19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas

CASE 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

CASE 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service

EVIDENTIARY HEARING

DATE: February 25, 2020

LOCATION: 19th Floor Hearing Room
Agency Building Three,
Empire State Plaza
Albany, New York

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ALJ MAUREEN F. LEARY, DPS

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(On the record 9:04 a.m.)

A.L.J. LEARY: Good morning.

A.L.J. COSTELLO: Good morning. This is the ninth day of the evidentiary hearings in cases 19-G-0309 and 19-G-0310. And we'll begin with appearances and I'll start with the companies.

MS. KRUEGER: Appearing on behalf of the companies Kara Krueger, Patrick O'Brien, Phil Decicco, and Ken Maloney of the law firm of Cullen and Dykman.

A.L.J. LEARY: Ms. Krueger, thank you.

A.L.J. COSTELLO: DPS staff.

MR. FORST: For DPS staff you have Brandon Goodrich, Nicholas Forst, and Raquel Parks will be joining us shortly.

A.L.J. COSTELLO: Okay. Department of State.

MS. KASON: New York Department of State Utility Intervention Unit, Jillian Kason and Katie O'Hare.

A.L.J. COSTELLO: Okay. Public Utility Law Project.

MS. WHEELOCK: Laurie Wheelock and Richard Berkley will be joining us shortly.

A.L.J. COSTELLO: Okay. City of New York.

MR. CONWAY: For the City of New York Adam Conway and Robert Loughney from the firm of Couch White.

A.L.J. COSTELLO: Okay. Do we have AARP? No? Okay. Sane Energy.

MS. DIESCHE: For the Sane Energy project we have Kim Fraczek and Lee Diesche.

A.L.J. LEARY: Can you speak into the microphone?

MS. DIESCHE: Yeah. It's on. For Sane Energy project Kim Fraczek and Lee Diesche.

A.L.J. COSTELLO: Is there anyone else who is making an appearance who I have not called?

MS. MURPHY: Yes. This is Erin Murphy on behalf of Environmental Defense Fund.

A.L.J. COSTELLO: Okay. Thank you. Anyone else? No? Okay. I believe we have a few preliminary matters before we bring the witness panel up.

A.L.J. LEARY: So Judge Costello and I learned last night that a part of the hearing, early day I believe of the hearing, was videotaped and

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audio -- audio/videotaped and then posted to a social media site, which was I believe Facebook. We did not recall and could not find any application seeking approval from us to videotape any part of the proceeding. So, you know, there are some issues associated with that. And just keep in mind we're first on kind of an inquiry here and I need to hear from potentially one or more parties.

But I want to just layout the ground rules, which many of you know or should know. Every party representing -- represented in this proceeding absolutely should know what I'm about to tell you, and I think many of you do.

So our regulations within the department 16NYCRR, Section 2.1A, indicates that a party may be represented by the party itself. So you don’t have to be a lawyer or through a representative who also need not be a lawyer. But if a party does either of those things, that party "Must conform to the standards of conduct required of attorneys appearing before the courts of the State of New York." And that is 16NYCRR, Section 2.1A. And that rule further provides that a person -- any person signing a pleading in a proceeding like this, many of

you have, or entering an appearance in the proceeding will be considered "To have agreed to conform to those standards, namely the ones that apply to attorneys appearing before the courts of the State of New York."

So those standards are what's at issue and, you know, in our view most significantly that rule states that failure to conform to those standards will be grounds for exclusion from the proceeding in which the party appears and any later proceeding that is before the department. So there's a pretty hefty sanction associated with not conforming to those standards, but none of this is necessarily easy for people who are not lawyers. But, as to lawyers, it's a little bit different view that -- that is taken.

So it's pretty clear across the state that the courts of New York do not allow the filming of judicial proceedings by any party, particularly without the approval -- prior written expressed approval of the -- of the judge and here of the presiding examiners, Judge Costello and I.

So in -- in posting an audiovisual representation of this hearing in which there was

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas live testimony by a witness panel also recorded and a colloquy that resulted from inquires that we -- that I personally had. This was a violation of our rule 2.1A. So that's our starting point.

So we're going to provide the party posting the video today and other parties with an opportunity to address what is an apparent violation and, you know, I don’t want to take up a lot of time since our time is -- is going to be limited, but any party wishing to be heard can -- can be heard. I only ask that you make it as brief as possible, briefer than I have been.

But I wanted to lay out on the record what the rules of the road are and -- and I also want to ask if any other party during the proceeding was aware of the filming that was happening or the video that was being taken. So let's hear from -- let's hear from Sane first. Who wants to be heard, Ms. Diesche or --

MS. FRACZEK: Yeah. I can -- I can answer that.

A.L.J. LEARY: Miss, is it Fraczek?

MS. FRACZEK: Fraczek.

A.L.J. LEARY: Fraczek?

MS. FRACZEK: Yeah. We represent a number of people that live in North Brooklyn in our neighborhood that were not able to be here. And this is a public proceeding. There was a person from the press here in the back of the room during that proceeding that was recording. So we assumed that it was fine to also record since it was open to the press. And people that are living on the -- the path of the MRI pipeline were very concerned and are unable to make it to Albany for these things and wanted to know what was happening. And we apologize for breaking any law that we didn't know that we were breaking.

A.L.J. LEARY: So that was Ms. Diesche, correct?

MS. FRACZEK: Ms. Fraczek.

A.L.J. LEARY: I'm sorry, Ms. Fraczek.

MS. FRACZEK: That's okay.

A.L.J. LEARY: And you're the director of the Sane Energy Project?

MS. FRACZEK: Yes.

A.L.J. LEARY: And who did the filming?

MS. FRACZEK: I did.

A.L.J. LEARY: Okay. And Ms. Diesche, were you aware that this filming was happening?

MS. DIESCHE: Yes. I knew it was being filmed. I didn't oppose.

A.L.J. LEARY: Okay. Anyone else in your organization here that day -- that was here -- was anyone else in your organization here that day?

MS. FRACZEK: No.

A.L.J. LEARY: No?

MS. FRACZEK: No.

A.L.J. LEARY: Are you aware whether anyone else knew about the filming?

A.L.J. COSTELLO: Any -- any of the other parties in the room. Do -- did you inform anybody that this was going on?

MS. FRACZEK: No.

A.L.J. COSTELLO: Okay.

MS. FRACZEK: But I was just filming my colleague Lee.

A.L.J. COSTELLO: Your colleague and it also captured a few other people that were -- were in the room as well as the testimony on the stand. It's not that it's -- it would've -- if you had requested it, you know, we would've had discretion to

allow it. It's the fact that nobody had the opportunity to know that this was going on and nobody had the opportunity to raise any questions about it. That's basically the issue.

MS. FRACZEK: I understand. We just saw that there was press here and notified us that were was press so --

A.L.J. COSTELLO: We -- we -- yes. We -- we were aware that the press was here.

MS. FRACZEK: Okay.

A.L.J. COSTELLO: So it -- it's just again this -- this process is -- it -- just try to keep it -- you know, you want it open and it is open to the public. But the -- the thing is it also has to be a fair process for everybody involved. So people have to know that they are being videotaped or they are being recorded and -- and the proper procedure would've been to request it beforehand.

MS. FRACZEK: Sure.

A.L.J. COSTELLO: Request that -- that’s all.

A.L.J. LEARY: So I don’t -- I did not see the press videotaping the hearing. Is -- were -- was the member of the press videotaping the hearing?

MS. FRACZEK: He was audio recording.

A.L.J. LEARY: He was audio recording.

A.L.J. COSTELLO: That's okay.

A.L.J. LEARY: That -- well, they're -- they're still supposed to seek approval. Okay.

A.L.J. COSTELLO: Well…

A.L.J. LEARY: So yeah, it's…

A.L.J. COSTELLO: And -- and they did seek approval through our public information officer that -- so we were aware that -- that was going on.

A.L.J. LEARY: So are -- are either of you lawyers?

MS. FRACZEK: No.

A.L.J. LEARY: Okay. That's fine. People make mistakes. We're taking this seriously because of the rules and we are required to take it seriously frankly because of the potential, you know, prejudice to others. I am going to request that you remove from Facebook the video at your earliest convenience and no later than 6:00 p.m. today.

MS. FRACZEK: No problem.

A.L.J. LEARY: Okay, great. Thanks. Anybody else care to be heard on this? Great. Let's -- do you want to cover the exhibit Mr. Goodrich that

you discussed offline?

MR. GOODRICH: Yes, please. So Exhibit 652, which has already been marked is -- was proffered by the company and it is the consensus -- Return on Equity Consensus document in proceeding 91-M-0509, which is the -- commonly known as the generic financing proceeding. It's from 1993, and because of the way our -- the departments and commissions recordkeeping system works it is not available on the -- the public case file. And so that's why I believe Mr. Maloney -- the company asked for it to be admitted and staff is -- is okay with that.

But we have asked National Grid and they have consented to also admit the recommended decision issued in the generic financing proceeding, which is also sadly not -- not on our public casefile. And the generic financing proceeding recommended decision is a 117 page document and it was issued on July 19, 1994. And I emailed it around to the parties just a few minutes ago for people to -- to review. But I would ask that that be given an exhibit number and, if need be, I can print off a copy or at least print off first pages for Your Honors.

A.L.J. LEARY: So I have to understand. I thought you said something about an RD first.

MR. GOODRICH: Yeah, recommended decision.

A.L.J. LEARY: So that will be an exhibit?

MR. GOODRICH: That's what I -- I'm asking be marked as an exhibit.

A.L.J. LEARY: Okay. And you just indicated that there was also an order.

MR. GOODRICH: No.

A.L.J. LEARY: No? Okay.

MR. GOODRICH: There was a consensus document, which was I guess something that came to a joint proposal from -- from 1993. That's -- that's the item that's already marked as 652.

A.L.J. LEARY: I thought -- okay. That's the consensus document. I viewed this as comments from the joint utilities.

MR. MALONEY: It -- it -- it came under a cover page that says comments, but there is a consensus document behind it. We took it the way it came.

A.L.J. LEARY: This consensus document is this also a document that is signed by this -- representatives of this agency and the answer is yes. I see it on page 29.

MR. MALONEY: That's correct.

A.L.J. LEARY: Okay, great. It's a consensus document among the joint utilities and counsel -- the department.

A.L.J. COSTELLO: And we'll mark the recommended decision as Exhibit 676.

MR. GOODRICH: Thank you. And did you want any sort of paper copy or paper first page or anything like that?

A.L.J. LEARY: We would like a copy.

A.L.J. COSTELLO: I can -- if you can email -- can you email it?

MR. GOODRICH: Well it's already emailed.

A.L.J. LEARY: To us?

MR. GOODRICH: But I'm saying if you want a paper copy, yes.

A.L.J. COSTELLO: We don’t need a paper copy.

MR. GOODRICH: Okay.

A.L.J. LEARY: We don’t need a paper copy.

MR. GOODRICH: Did you -- Your Honors, you had mentioned wanting an update about the -- that audit. Do you want to do that now or later?

A.L.J. LEARY: No. About Mr. Holst's, H-O-L-S-T, testimony on the audit that's currently pending.

MR. GOODRICH: Yeah.

A.L.J. LEARY: I inquired -- this is for the record -- I inquired of the company and Mr. Goodrich about whether there was any update to that pending audit and Mr. Goodrich, your response, if you would put that on the record.

MR. GOODRICH: So, as stated in Mr. Holst's testimony, at the time of assessment it was still ongoing. And I believe at that time it was already known that is was going to be concluded by department staff as opposed to the consultant who was then conducting the audit. And, if it wasn’t stated that way in his testimony, it -- it -- that occurred shortly thereafter and so I'm informing the -- Your Honors in the record that -- that that is the case that the consultant was removed from the case. And

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas department staff is concluding the audit; however, that -- the audit is still ongoing and so there is no report yet having been -- I can't talk this morning -- A report has not yet been issued and I do not anticipate one will be issued at least in the -- the near future, at least in the next I would say couple months at the very least. I don't honestly have a -- a known date for an expected release of that report.

A.L.J. LEARY: That's okay. Okay. We have a couple other outstanding issues, one of which is an update. I -- I -- I'm sorry Mr. Decicco. I don’t remember whether this was actually covered on the record or not. So the updating of the Eco Study and the discussion UIU and the company had about updating I believe a discovery response with the cost of service study, can you just address that on the record?

MR. DECICCO: Sure and actually if -- if UI wouldn’t mind maybe they could just put on the record exactly what they’ve requested and we can comment on our willingness to provide that.

MS. O'HARE: Sure. In -- in a request from UIU, the company -- UIU requested the companies to update Exhibit RDP12. It's premarked as Hearing

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas Exhibit 138 for KEDNY and RDP12 for KEDLI, which is premarked as Hearing Exhibit 154. These are the embedded cost of service studies that classify means as 100% demand that were submitted with the company's initial testimony and the companies will be updating these to reflect the most recent new requirement they have requested.

MR. DECICCO: And Phil Decicco for the company. Your Honors, we have agreed to update those and our expectation is we'll have them by this week. So I would propose that we provide those exhibits with new exhibit numbers and then we can file them in the proceeding when they're available.

MS. O'HARE: And just to clarify, they also include the work papers as well as part of this exhibit.

A.L.J. LEARY: So is this -- does this sound like for Friday -- this coming Friday would be a -- a good bet?

MR. DECICCO: I was told we'd have it on Thursday, but I don’t want to overpromise. So Friday, yes, would be fair. Yeah.

A.L.J. LEARY: Go with Friday then. Okay, great. Anybody else want to be heard on this?

And obviously this will be shared with all the parties. Finally, the one request I made early in the hearing, which I haven't covered again yet was related to what I'll call the 1990 baseline emission levels that in the early part of the hearing my understanding was was that the company would be providing that.

There was a spreadsheet referenced by the witness panel for the company that was in response to inquiry -- questioning on cross examination about 60% reduction in carbon by 2035, 80% by 2050. And I think the question was 60% reduction from what baseline and I believe it's the 1990 levels that were being referred to. So I didn’t want to lose that -- that thread and I'm just going to ask the company what -- you probably forgot about it like I did or Mr. O'Brien is shaking his head. No I didn’t. So, yeah, you're much younger than me so.

MR. DECICCO: Your Honor, a lot has happened in the last two weeks, but I believe we circulated that to the parties that day. But we're just going to confirm that. Maybe at a -- at a break we could get back on the record on that one in particular, but I -- I believe we had circulated that

the day that it was requested.

A.L.J. LEARY: And you did not circulate it to Judge Costello and I?

MR. DECICCO: Can we just get this back to you Judge? That would be helpful. I think it may have just gone to other parties.

A.L.J. LEARY: I don’t think we saw it, did we?

A.L.J. COSTELLO: No. I -- I haven't seen it.

A.L.J. LEARY: I -- I don’t think we have it, but we could be wrong. So let's -- can you report out at the break on this and just get Judge Costello and I -- and I'm going to assume nobody else needs to address this issue but speak now or forever hold your piece. Okay. Let's go with the panel. Anything else?

A.L.J. COSTELLO: No.

A.L.J. LEARY: Let's bring in the company's panel on gas infrastructure and operations.

A.L.J. COSTELLO: Would you one at a time state your names and business addresses for the record? And we'll start at the very end. Thank you.

MS. MAD: Sriridya Mad, 40 Sylvan

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas Road, Waltham, Massachusetts.

MS. HON: Caroline Hon, 40 Sylvan Road, Waltham, Massachusetts.

MR. TURRINI: Ross Turrini, 25 Hub Drive, Melville, New York.

MR. GRAHAM: Tim Graham, 300 Erie Boulevard, Syracuse, New York.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

PANEL: I do.

SRIRIDYA MAD; SWORN

CAROLINE HON; SWORN

ROSS TURRINI; SWORN

TIM GRAHAM; SWORN

A.L.J. COSTELLO: Okay. You may be seated.

DIRECT EXAMINATION

BY MR. DECICCO:

Q. Good morning, Panel. Panel, you have before you a document entitled the Direct Testimony of the Gas Infrastructure and Operations Panel on behalf of the Brooklyn Union Gas Company or KEDNY dated April 2019, consisting of 120 pages, a

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coversheet and a table of contents. Is that correct?

A. (Panel) Yes.

Q. You also have before you a document entitled the Supplemental Direct Testimony of the Gas Infrastructure and Operations Panel on behalf of KEDNY dated June 11, 2019, consisting of eight pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You also have before you a document entitled the Corrections and Updates Testimony of the Gas Infrastructure and Operations Panel on behalf of KEDNY dated July 3, 2019, consisting of 16 pages, a coversheet and a table of contents. Correct?

A. Yes.

Q. Switching to KEDLI. You have before you a document entitled the Direct Testimony of the Gas Infrastructure and Operations Panel on behalf of the KeySpan Gas East Corporation or KEDLI dated April 2019, consisting of 110 pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You have before you a document

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas entitled the Supplemental Direct Testimony of the Gas Infrastructure and Operations Panel on behalf of the KEDLI dated June 11, 2019, consisting of eight pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You also have before you a document entitled the Corrections and Updates Testimony of the Gas Infrastructure and Operations Panel on behalf of the KEDLI dated July 3, 2019, consisting of 15 pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You also have before you a document entitled the Rebuttal Testimony of the Gas Infrastructure and Operations Panel on behalf of both KEDNY and KEDLI dated September 18, 2019, consisting of 39 pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You have -- you also have before you a document entitled the Second Supplemental Testimony of the Gas Infrastructure and Operations Panel on behalf of both KEDNY and KEDLI dated

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December 13, 2019, consisting of 20 pages, a coversheet and a table of contents. Is that correct?

A. Yes.

Q. You have been very busy. Was the testimony prepared by you -- or was this testimony prepared by you or under your direction and supervision?

A. Yes.

Q. Is the information contained in the testimony true and correct to the best of your knowledge?

A. Yes.

Q. If the questions contained in these documents were asked today, would your answers be the same?

A. Yes.

Q. Do you adopt these documents as your sworn testimony in these proceedings?

A. Yes.

MR. DECICCO: Your Honors, I ask that the prefiled Direct First Supplemental and Corrections and Updates Testimony on behalf of KEDNY, the Direct First Supplemental and Corrections Testimony on behalf of KEDLI and the Rebuttal and

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas Second Supplemental Testimony on behalf of both KEDNY and KEDLI be entered into the record as if orally given in these proceedings.

A.L.J. COSTELLO: That request is granted and the court reporter should insert the following files at this point: KEDNY Gas Infrastructure and Operations Panel Direct Testimony, KEDNY Gas Infrastructure and Operations Panel Supplemental Direct Testimony, KEDNY Gas Infrastructure and Operations Panel Corrections and Updates Testimony, KEDLI Gas Infrastructure and Operations Panel Direct Testimony, KEDLI Gas Infrastructure and Operations Panel Supplemental Direct Testimony, KEDLI Gas Infrastructure and Operations Panel Corrections and Updates Testimony, KEDNY-KEDLI Gas Infrastructure and Operations Panel Rebuttal Testimony and KEDNY-KEDLI Gas Infrastructure and Operations Panel Second Supplemental Testimony. #

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Direct Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Date: April 2019

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I. Introduction and Qualifications

Q. Please introduce the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudhan.

Q. Mr. Turrini, please state your name and business address.

A. My name is Ross W. Turrini. My business address is 25 Hub Drive, Melville, New York 11747.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company"), a subsidiary of National Grid USA ("National Grid"), as the Group and US Chief Gas Engineer. I oversee approximately 800 employees and more than $9.5 billion of gas infrastructure assets serving over 3.6 million customers in New York, Massachusetts, and Rhode Island.

National Grid owns and operates three gas distribution companies in New York that provide retail gas service to more than 2.4 million customers in three service territories: Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk") serves areas of eastern and central New York; The

Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or "Company") serves Brooklyn, Staten Island, and parts of Queens in New York City; and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") serves customers on Long Island and the Rockaway Peninsula in Queens. I am responsible for all aspects of the performance of National Grid's New York gas networks, including emergency response, gas engineering, construction activities, operation and maintenance of gas transmission and distribution facilities, gas network strategy, including the gas capital investment strategy, and gas pipeline safety management.

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Civil Engineering from the United States Military Academy at West Point in 1985. I have worked for National Grid and its predecessor companies (the Long Island Lighting Company and KeySpan Corporation) for 25 years in various roles in engineering, operations, and procurement. Prior to joining National Grid, I spent five years as an Officer in the United States Army Corps of Engineers and three years in engineering and construction roles at Brown & Root Services Corporation, an international engineering, procurement, and construction company.

Q. Have you previously testified before the New York Public Service Commission ("Commission")?

A. Yes. I testified before the Commission on behalf of KEDNY and KEDLI in Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases"). I also testified most recently on behalf of Niagara Mohawk in Case 17-G-0239 (the "2017 NMPC Rate Case").

Q. Mr. Graham, please state your full name and business address.

A. My name is Timothy S. Graham. My business address is 300 Erie Boulevard West, Syracuse, New York 13212.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President for Gas Field Operations. I oversee approximately 2,300 employees and am responsible for the maintenance of KEDNY, KEDLI, and Niagara Mohawk's gas distribution systems. This includes gas metering, construction, and field activities to carry out mandated safety programs, leak response, leak repair, instrumentation and regulation, damage prevention, and inspection programs.

Q. Please describe your educational background and experience.

A. I received a Bachelor of Engineering in Electrical Engineering from the State University of New York at Buffalo. I also have a Master's in Business Administration from Lemoyne College in Syracuse, New York. I have worked for National Grid and its predecessor companies (Niagara Mohawk) for 32 years in various roles in engineering and operations.

Q. Ms. Hon, please state your full name and business address.

A. My name is Caroline Hon. My business address is 40 Sylvan Road, Waltham, Massachusetts, 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President for Gas Resource Management. I oversee approximately 400 employees and am responsible for the planning, scheduling, dispatch, dependency management, and clerical support for the capital, maintenance, and repair work across National Grid's gas network.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Science in Chemical Engineering from the Massachusetts Institute of Technology in 2001. I have worked for National

Grid for nine years in various roles in operations, finance, and shared services. Prior to joining National Grid, I worked as a consultant for eight years in Accenture's Utilities Transmission and Distribution practice, focused on work management process optimization and system implementation.

Q. Ms. Madhusudhan, please state your name and business address.

A. My name is Srividya Madhusudhan. My business address is 40 Sylvan Road, Waltham, Massachusetts, 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President of Property and Fleet. I oversee all the Facilities, Fleet and Property activity in support of National Grid Operating companies including KEDNY, KEDLI, and Niagara Mohawk. This includes buying, selling, leasing, licensing, administration of rights of way, Management of Rights of way and maintenance of fleet. I also oversee development of the facilities and fleet plans to support KEDNY and KEDLI's gas businesses and related capital forecasts.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Law from Chennai, India in 1991. I also have a Masters in Environmental Management from Rensselear Polytechnic Institute and an

MBA from the University of Connecticut. I have worked at National Grid since May of 2017 prior to which I worked at Eversource Energy for 18 years in various roles in environmental, Operations and Real Estate and Facilities.

II. Purpose of Testimony

Q. What is the purpose of the Gas Infrastructure and Operations Panel's testimony?

A. The purpose of the Panel's testimony is to provide a forecast of the capital investments of the Company for the twelve-month period ending March 31, 2021 ("Rate Year"), and the three subsequent twelve-month periods ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (Data Year 1, Data Year 2, and Data Year 3 are collectively referred to as the "Data Years"). The Panel also presents a forecast for the fifteen-month period ("Gap Period") between December 31, 2018, which is the end of the Company's Historic Test Year, and the beginning of the Rate Year.

The Panel discusses direct capital expenditures that will (i) increase the safety and reliability of the Company's gas network, (ii) modernize the Company's gas transmission and distribution infrastructure to reduce emissions while continuing to meet customer needs, and (iii) enhance resiliency and the

Company's ability to respond to future weather events. The Panel will also discuss the Company's practices and policies for maximizing the efficiency of its capital delivery program from planning and budgeting through the completion of construction.

The Panel's testimony provides an overview of the significant projects in the Company's capital plan, including retirement of leak prone pipe ("LPP") and new safety programs to identify and address system risks. The Panel's testimony also presents an overview of the Company's pipeline integrity and reliability programs that will improve the overall safety and reliability of the Company's gas system, and will also address recently enacted, as well as pending, pipeline safety regulations administered by the U.S. Department of Transportation ("DOT"), Pipeline and Hazardous Materials Safety Administration ("PHMSA"). The Panel discusses the Company's plans and infrastructure projects necessary to ensure that the Company can reliably serve its existing customers as well as forecast demand for new customer connections within the Company's operating territory.

The Panel also presents the Company's indirect capital investments, including supply chain, inventory management/investment recovery ("IM/IR"), and fleet investments. The Company is also proposing significant investments in

facilities during the Rate Year and Data Years that are necessary to: support the Company's O&M and gas safety programs; make needed upgrades to aging facilities; and create more modern and efficient customer support and employee work spaces.

Q. Does the Panel's testimony also address the Company's operations and maintenance ("O&M") programs?

A. Yes. In addition to capital investments in gas infrastructure, the Panel describes incremental labor (full time equivalent positions or "FTEs") and non-labor O&M expenses that the Company proposes in the Rate Year and Data Years, the costs of which are not fully reflected in the twelve-month period beginning January 1, 2018 and ending December 31, 2018 ("Historic Test Year"). These expenses represent known and measurable changes from Historic Test Year expenses that are necessary to (i) improve system safety and reliability, (ii) address new and emerging safety regulations, (iii) enhance customer service, and (iv) support the Company's capital investments. The Panel will also discuss the Company's staffing plan for the proposed new FTEs.

Q. Does the Panel address any other topics?

A. Yes. The Panel discusses the Williams Transco Northeast Supply Enhancement ("NESE") project and the importance of NESE in enabling KEDNY's proposed capital investment plan.

Q. Does the Panel sponsor any exhibits as part of its testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared under its direction and supervision:

Exhibit __ (GIOP-1): Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3.

Exhibit __ (GIOP-2): Charts Demonstrating Components of Unit Cost Increases

Exhibit __ (GIOP-3): Gas Safety and Reliability Surcharge Including Unit Cost Incentive Proposals

Exhibit __ (GIOP-4): Chart Summarizing Projected Leak Rates for LPP for Various Main Replacement Strategies.

Exhibit __ (GIOP-5): Data Sheets for Significant Capital Programs. This exhibit includes summaries of the Company's significant capital projects/programs.

Exhibit __ (GIOP-6): Demonstration of Variability in City/State Construction O&M.

Exhibit __ (GIOP-7): Projected Indirect Capital Expenditures (Supply Chain, Fleet, and Facilities) for the Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3, and Data Sheets for Significant Facilities Investments.

Exhibit __ (GIOP-8): Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3.

Exhibit __ (GIOP-9): Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3.

The capital expenditures presented throughout the testimony and in the exhibits include cost of removal ("COR"), as applicable.

Q. How is the Panel's testimony organized?

A. The testimony is organized into the following sections:

Sections I and II are introductory sections outlining the Panel's testimony.

Section III provides an overview of the Company's capital investment and O&M program priorities and objectives, strategy to address increasing costs, and significant capital programs, such as the retirement of leak prone

mains and services, and other key capital investments in gas pipeline safety and reliability.

Section IV provides details on the Company's proposed direct capital investment program for the Rate and Data Years, including the Company's spending rationales, categories of capital investment, and specific work activities within each category.

Section V describes the Company's indirect capital investment plan, which includes facilities, fleet, and supply chain (IM/IR) investments.

Section VI describes the Company's O&M programs, including those necessary to carry out the Company's proposed capital programs and those targeted at current and emerging safety regulations and implementation of gas pipeline-safety best practices. Section VI also describes the status of the Company's Local Law 30 compliance program, which concludes at the end of the Rate Year. With respect to gas pipeline safety, several of the O&M expenditures presented in the Panel's testimony and set forth in Exhibit __ (GIOP-8) and Exhibit __ (GIOP-9) are presented in more detail in KEDNY and KEDLI's Gas Safety Panel testimony.

Section VII describes the NESE project, its relationship to KEDNY's capital investment plan, and the important safety and reliability projects and programs that could not proceed in the absence of NESE.

III. Capital and O&M Plan Objectives and Priorities

A. Plan Objectives

Q. Please describe the Company's overall objective of its infrastructure and operations programs.

A. The Company designs its gas infrastructure and operations programs to provide safe and reliable gas delivery service to customers at reasonable costs. Over the last several years, the Company has significantly increased investment to modernize and enhance the resiliency of its gas assets. Significant capital investment over the next several years is required to ensure that the gas system continues to operate safely while meeting the demands of customers and to enable further modernization of the Company's facilities and operations in a manner that achieves meaningful methane emissions reductions in furtherance of the Company's Future of Heat strategy. The proposed plan includes capital and O&M spending to meet both customer needs and satisfy state and federal regulatory requirements and goals, including retirement of LPP, gas pipeline safety enhancements, and deployment of non-pipeline alternatives. While significant investments are required to meet these needs and objectives, in developing its capital and O&M plans, the Company balanced the need for spending to achieve safety and service objectives with the need to manage costs and minimize impacts on customer rates.

Q. What are the Company's projected direct and indirect capital forecasts?

A. A summary of the Company's total capital plan forecast for the Rate Year and Data Years is provided in Table 1.

Table 1: Indirect and Direct Capital Plan (Includes COR)

($000)	FY 2021	FY 2022	FY 2023	FY 2024
Direct CapEx	$952,053	$934,990	$931,812	$1,008,451
Indirect CapEx	$9,884	$7,459	$6,659	$5,350
Total Capital Plan	**$961,937**	**$942,449**	**$938,471**	**$1,013,801**

Q. Why is the scale of the Company's capital investment plan increasing?

A. Several developments have required that KEDNY and other natural gas distribution utilities increase their annual capital expenditures. Notably, pipeline safety incidents, such as the tragic events in San Bruno, California, Allentown, Pennsylvania, East Harlem, New York, and the more recent incident in Merrimack Valley, Massachusetts, have appropriately increased focus on pipeline safety and the need to carefully monitor and replace aging pipeline infrastructure. Weather events, such as Superstorm Sandy and the Polar Vortex, and the expectation that similar events will continue to occur, require the Company to find ways to protect its facilities from severe weather.

In response to these events, the industry has adopted new safety requirements and is implementing best practices to modernize and protect gas facilities. For example, since the 2016 KEDNY and KEDLI Rate Cases, (i) new Commission safety regulations and directives now require additional safety inspections of inside service lines and plastic fusions; (ii) PHMSA is implementing new safety requirements related to transmission system integrity management and verification this year; and (iii) the New York State Legislature is currently considering a bill that would require licensed professional engineers ("PEs") to review and certify gas facility design plans. The Commission and utility industry review of recent incidents has also resulted in recommendations for safety best practices, including overpressure protection improvements and enhanced contractor oversight. These mandated and recommended improvements require significant additional investments and compliance resources.

Q. How will the Company support the increased level of capital investment?

A. As the Company developed plans to modernize its gas assets, it also began to build and enhance its operations, engineering, resource planning, work management, and quality control organizations and capabilities to deliver increasing levels of capital investment. The Company will continue to develop these capabilities in the Rate Year and Data Years by adding incremental labor

resources to execute the capital plan and support the increased operations workload (discussed in Section VI). The capital investment plan includes additional field training resources, facilities, and tools to enhance the development of the internal work force.

Regarding contractor resources, over the past several years, the Company developed a procurement strategy that supports sustainable growth in qualified contractors to meet the work plan increases. To ensure adequate levels of qualified, skilled labor, and to address the challenges around developing qualified contractors, the Company has worked to:

- Establish longer term contracts to enable contractors to plan and invest in hiring, training, facilities, and equipment to meet the Company's construction needs
- Provide greater work plan visibility to contractors on forecast crew requirements, which will enable them to develop the required capacity
- Manage the work plan to limit seasonal variability to support a stable contractor workforce and promote worker retention
- Continue partnerships with community colleges, trade schools, and veterans group to attract and develop new sources of skilled labor to build the workforce

Q. Does the Company require additional personnel in the Rate Year and Data Years to execute its capital and O&M programs?

A. Yes. The Company forecasts the need for an additional 346.7 FTE positions by Data Year 3 to support the additional capital investment, increasing O&M workload due to new and pending regulatory requirements, new programs discussed below, and the gas safety programs presented in the Gas Safety Panel's testimony. These FTEs include positions in contractor oversight, gas pipeline safety management, corrosion control, field operations, instrumentation and regulation ("I&R"), gas control, gas system engineering, and gas system management. The cost of these FTEs will be charged to both capital and O&M programs based on the job function and nature of the work. Exhibit __ (GIOP-9) identifies the incremental FTE positions by function and the need for each position. KEDNY's Revenue Requirements Panel presents the labor O&M associated with these FTEs.

B. <u>Cost Pressures and Cost Mitigation</u>

Q. What explains the significant increases reflected in the investment plan?

A. The Company's investment plan reflects increases in both the scope of work (units) and the costs elements to perform this work (unit costs). As discussed above, there are several factors that are collectively increasing the size of the Company's investment plan over the next several years, including incremental

work driven by federal and state regulatory mandates and the need to upgrade aged infrastructure and replace dated facilities. At the same time, the Company is experiencing increasing unit costs resulting from new work requirements and enhanced safety practices, and the rising costs of materials and contractors, in part due to increasing demands for these resources across the industry

Indeed, a primary driver of increases in the capital plan is significantly higher unit costs for main installation work across multiple mandated programs, including the proactive and reactive main replacement, customer connections, corrosion, and city/state construction ("CSC") programs. The unit cost increases are explained in more detail, below, and unit cost component increases for main replacement are shown on Exhibit __ (GIOP-2).

Another driver of the capital investment plan is a significant increase in the volume of work in the Company's CSC program, which involves the relocation and/or protection of gas facilities that conflict with municipal public works projects. As described in detail below, the CSC program scope is continuing to grow as New York City is investing billions of dollars to improve its water and sewer systems. As a result of the sharp increase in scope and scale of the City's public works programs, the CSC program now represents more than a third of

the Company's total capital investment plan, and the work requirements and timing of this program are not within the Company's control.

Q. What factors are contributing to the Company's unit cost increases?

A. Exhibit __ (GIOP- 2) shows the components that have contributed to the main installation unit cost increase. The primary source of increased costs is New York City's new restoration and paving requirements that became effective in April 2017. Specifically, changes to Highway Rules §§ 2-01 through 2-09, 2-11 through 2-14, and 2-20 ("Paving Rules") now require significantly more restoration work, including:

- Increasing the size of the area that must be re-paved following an excavation. Under the new Paving Rules, the final restoration area is based on the largest dimension of the excavation. Under the prior regulations, KEDNY was authorized to pave the area of individual bell holes. The Paving Rules require KEDNY to pave the bell holes in addition to the area in between those holes. In some cases, this requirement doubles or triples the area needing to be repaved.

- Requiring that base material be replaced in kind or better, which means that concrete roadways must be replaced with concrete instead of the less expensive binder material that was previously utilized. In that regard, approximately 65 percent of New York City streets are concrete; and

- Requiring more saw cutting of roadway excavations instead of more efficient and cost-effective jackhammering.

The Historic Test Year was the first full year in which these new paving costs were effective, during which the Company performed tens of thousands of individual paving jobs in the City at a considerably increased cost. Indeed, in calendar year ("CY") 2017 to CY 2018, paving restoration costs accounted for approximately 46 percent ($526) of the total main replacement unit cost ($1,147).

Other factors contributing to an increase in the Company's main replacement unit cost, as shown on Exhibit __ (GIOP-2), include:

- Higher base labor (contractor costs): recently renegotiated long-term contractor agreements resulted in higher base labor contractor costs
- Mains Size: work scope includes more large-diameter main at higher replacement costs
- Increased time and materials ("T&M"): increasingly restrictive municipal stipulations that limit work time in streets and increasing costs to comply with arborist regulations (protection of trees on sidewalks proximate to work sites)

Q. What is the Company doing to moderate the size of its investment plan?

A. As set forth in more detail in Section IV, the Company develops its proposed capital and O&M investment plans through close coordination among the Asset Management, Field Operations, and Investment and Resource Planning teams. Major programs and investments undergo a robust options analysis to identify the most cost-effective solution. The alternatives considered and project benefit information for major capital investments are set forth in the data sheets included in Exhibit __ (GIOP-5). Additionally, the Company recently created a Capital Delivery organization tasked with enhancing the Company's ability to develop and execute its capital plan more efficiently for complex projects (Phase 1) and eventually for the entire capital program (Phase 2). As part of this effort, which is currently in the first year of Phase 1, Capital Delivery is working to identify efficiencies and opportunities for better ways of working that may reduce costs. Capital Delivery is also in the process of identifying and implementing targeted efficiency initiatives in support of National Grid's aspirational savings program, known as "Accelerate." The Revenue Requirements Panel describes the Accelerate Program.

Q. Please describe how the Company reflected Capital Delivery initiatives in the capital investment plan.

A. In recognition that efficiencies are expected over the term of the Rate Year and Data Years, the Company is making reasonable budget adjustments to the capital plan to reflect savings that have been identified to date, while maintaining a mechanism to return to customers the benefits of additional efficiencies achieved over the term of the rate plan. To that end, the Company proposes to reflect capital efficiencies in two ways:

- Targeted Efficiencies Adjustment: The Company included a line-item reduction to the overall capital investment plan in the amount of $0.6 million in the Rate Year, $2.7 million in Data Year 1, $1.4 million in Data Year 2, and $1.8 million in Data Year 3 to reflect savings that the Company believes are reasonably achievable across a portfolio of capital projects. The line-item reduction is shown on Exhibit __ (GIOP-1) as "Complex Capital Delivery Initiative – Savings." The calculation of this adjustment is discussed in more detail below.

- Net Plant Capital Tracker: The Company proposes to continue the net plant capital tracker mechanism that was approved in the 2016 KEDNY and KEDLI Rate Cases as part of a multi-year rate plan. Any capital efficiencies realized through the Company's Accelerate Program will be

returned to customers through the operation of the downward-only capital tracker.

Q. Please explain how the Company calculated the Complex Capital Delivery Initiative – Savings line item reduction to the capital plan.

A. The Company estimated potential savings that the Company reasonably expects to achieve across a portfolio of complex capital projects that are currently moving through Capital Delivery's project stages. The projects are pressure regulation station re-builds, LNG projects, IMP, IVP, and transmission station integrity projects, and system reinforcements. Estimated savings are not attributable to specific projects but are representative based on recent experience across the portfolio. The Company applied a five percent reduction to the five-year forecasts for these projects to derive the savings estimate for each year.

Q. Is the Company proposing unit cost productivity incentives to encourage additional efficiencies?

A. Yes. The Company is proposing to continue the unit cost productivity incentive for the Proactive Main Replacement Program, as adopted in the 2016 KEDNY and KEDLI Rate Cases. The Company is also proposing a similar incentive for the Customer Connections program, which is also a unit-cost driven program.

The Company proposes to recover both incentives through the existing Gas Safety and Reliability Surcharge, consistent with the current unit cost incentive. The GSRS, including the proposed incentive mechanisms, is presented in Exhibit __ (GIOP- 3). These productivity incentives align with the Company's efforts to identify and realize efficiency benefits through its Accelerate Program.

Q. How are the unit cost targets for the productivity incentives derived?

A. The unit cost targets are based on the Company's total costs for main and service replacements, including the costs of related safety programs that will be charged to the work orders for main replacements. For purposes of presentation of the capital investment plan in the rate case, Exhibit __ (GIOP-1) presents the costs of these additional programs, such as plastic fusion inspections, as separate line items for clarity of costs attributed to each program. However, in the interest of setting productivity incentives that reflect all elements of unit costs and are easily tracked and reported using program work orders, unit cost targets must include a full allocation of costs for the related safety programs to reflect the total costs for main and service replacements.

As shown on page 11 of Exhibit __ (GIOP-3), unit costs for the LPP productivity incentive include the costs for the Proactive Main Replacement

Program, plus allocated costs for plastic fusion inspections, low pressure main valve installations, replacement of high-density polyethylene ("HDPE") services, and contractor safety inspections. As shown on page 12 of Exhibit __ (GIOP-3), unit costs for the Customer Connections – Mains productivity incentive include the costs of the Customer Connections – Install Main program, plus allocated costs for plastic fusion inspections and contractor safety inspections.

Q. Are there emerging safety and regulatory changes that are not included in the Company's investment plan that may further increase capital and O&M requirements during the Rate Year and Data Years?

A. Yes. The Company's investment plan does not include costs to comply with: (i) the anticipated legislation to require professional engineering ("PE") certification of engineering designs for gas facilities, and (ii) the Commission's proposed new requirements for gas company operator qualification programs (see "Notice of Proposed Rulemaking," issued March 27, 2018 in Cases 14-G-0212 and 17-G-0318). There is uncertainty regarding when these proposals might be adopted as final, whether there will be changes to the scope and requirements prior to final adoption, and the timing of implementation following final adoption. Even if these new requirements are adopted without substantive changes to the current proposals, these items were introduced too

recently to allow the Company to complete the thorough analysis of estimated implementation costs necessary for inclusion in this rate filing.

Q. What is the Company's proposal regarding the anticipated PE and operator qualification requirements?

A. To the extent the PE and operator qualification requirements are finalized during the pendency of this rate proceeding, the Company intends to update its filing to include the costs of these mandated programs. If these new requirements are finalized after rates are established in this case, the Company proposes to recover prudent costs to implement the PE and operator qualification requirements through the Company's GSRS. Under either scenario, the unit cost targets for the productivity incentive would need to be updated to address these new costs.

C. <u>Significant Capital Programs</u>

Q. Please describe some of the significant and notable programs included in the capital plan.

A. The capital plan represents the investments required to provide safe and reliable service to the Company's customers. KEDNY's major capital programs and projects include the following:

- Proactive Main Replacement (LPP) Program

- Integrity Management Program ("IMP") and Integrity Verification Process Program ("IVP")
- Transmission Station Integrity Program
- CSC
- Completion of the Company's ongoing Metropolitan Reliability Infrastructure ("MRI") Project

The Company also proposes an innovative program to enable renewable natural gas projects to more cost effectively interconnect to the Company's system. These programs and projects are described in detail below and are included in Exhibit __ (GIOP-5).

i. <u>Proactive Main Replacement (LPP) Program</u>

Q. Please describe the inventory of LPP existing on the Company's system.

A. LPP in KEDNY's service territory is comprised of twelve-inch and smaller main and associated services that are (i) unprotected (*i.e.*, non-cathodically protected) steel pipe whether bare or coated and (ii) cast and wrought iron pipe, (iii) unprotected steel/wrought iron services, and (iv) copper and pre-1974 HDPE services. As of the end of CY 2018, the Company has approximately 1,458 miles of LPP in its remaining inventory.

Q. What is the Company's proposal regarding retirement of LPP?

A. To reduce the risk of leaks and breaks, improve system safety, performance, and reliability, meet the Company's commitment to enhance customer satisfaction, and reduce methane emissions, the Company has prioritized the retirement of poor performing and higher-risk gas infrastructure – specifically, LPP and associated services – that disproportionally contributes to leaks on KEDNY's system. The Company's current rate plan funds LPP replacements of 55 miles in CY 2017, 60 miles in CY 2018, and 65 miles in CY 2019.

The Company's proposal is to continue accelerating LPP retirement by five miles per year from the current, CY 2019 target of 65 miles, as depicted in Table 2. The performance metrics for LPP replacement, including minimum mileage targets, are set forth in more detail in the Companies' Gas Safety Panel testimony.

Table 2: LPP Retirement Rate Allowance

Miles	CY 2020	CY 2021	CY 2022	CY 2023
Target	70	75	80	85

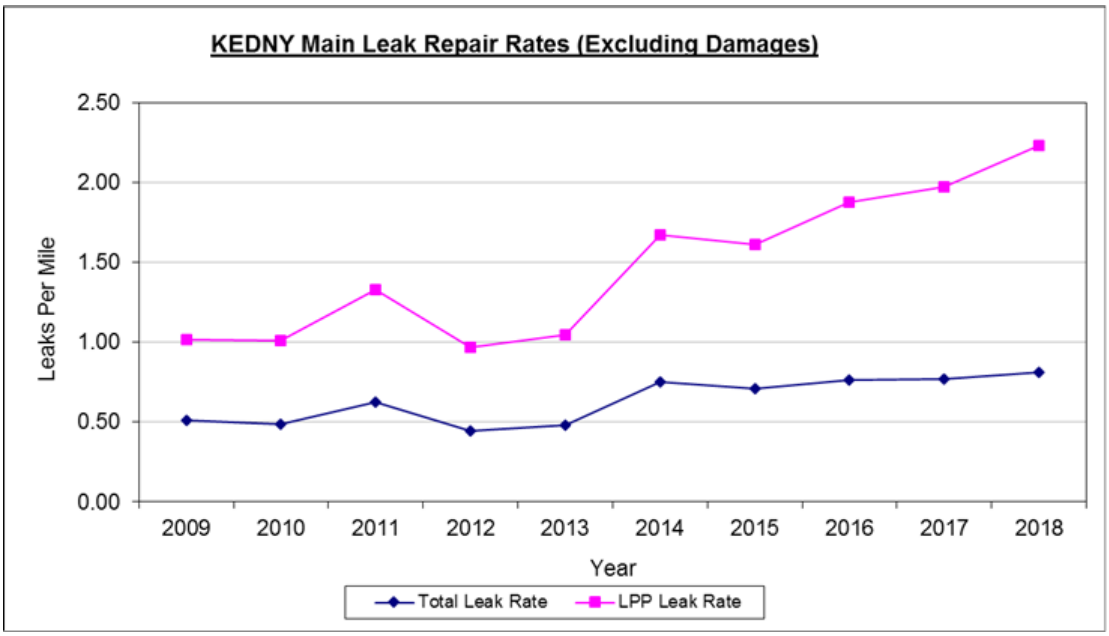
The proposed LPP retirement rate will position the Company to fully retire its remaining LPP inventory in 20 years.

Q. Why is the Company proposing to continue accelerating LPP retirement by five additional miles per year?

A. As would be expected, the Company has observed a significantly higher leak rate on its LPP inventory as compared to all other distribution facilities. While the current LPP inventory represents only 35 percent of KEDNY's distribution system, LPP is responsible for 96 percent of leak repairs, excluding excavation damages. The current leak rate for all distribution piping is 0.81 leaks per mile. The 2018 leak rate for LPP was 2.23 leaks per mile, representing an increase from the 2017 leak rate for LPP of 1.97.

Significantly, in KEDNY's service territory, the leak rate for LPP has continued to increase over the past several years, as shown in Table 3. In fact, the 2018 LPP leak rate was the highest in the past ten years. Thus, KEDNY's proposal to continue accelerating LPP retirement is warranted. Exhibit __ (GIOP-4) provides projected leak rates for LPP for various main replacement strategies.

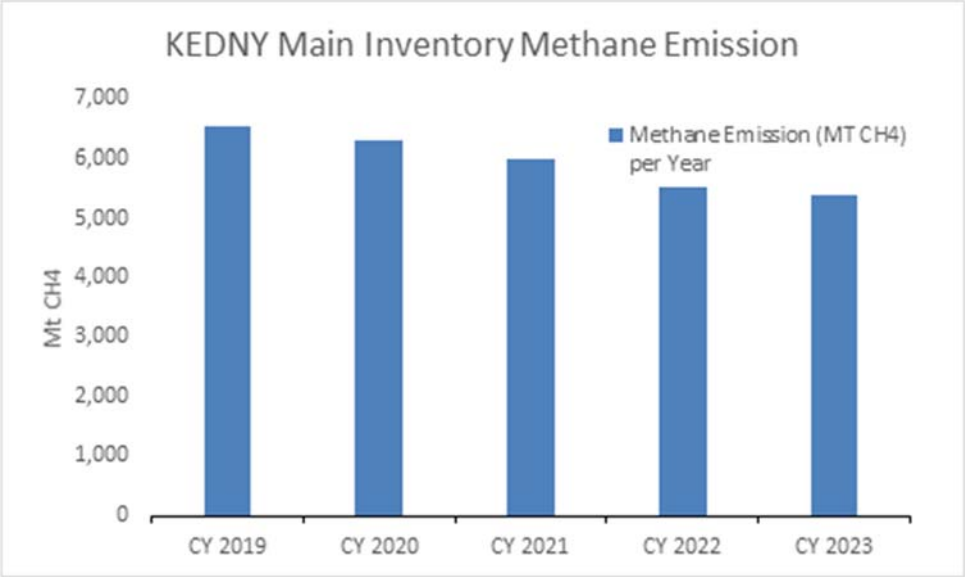
Table 3: KEDNY Historic Leak Rate



Q. Is there an environmental benefit associated with the retirement of LPP?

A. Yes. As is noted in the Company's Future of Heat Panel testimony, KEDNY and KEDLI are establishing a target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035, and retirement of LPP is a major component toward reaching this ambitious goal. Specifically, retirement of LPP reduces gas losses and fugitive emissions of methane. Table 4 provides a high-level estimate of potential methane emissions reductions over the next several years based on the Company's current facilities assuming the retirement of LPP pursuant to KEDNY's proposed program.

Table 4: Estimated Methane Emissions Reduction Through LPP Retirement



Additionally, the Company's LPP retirement algorithm prioritizes retirement of pipe segments with Type 3 leaks and service leaks when other risk factors are equal, thereby enhancing the emissions reduction benefits of its Proactive Main and Service Replacement Program.

Q. Do other capital programs contribute to retirement of LPP to meet the planned targets?

A. Yes. Several miles of LPP are retired through the CSC and Gas System Reliability Programs, but the bulk of LPP retirement occurs through the Company's Proactive Main Replacement Program. The Proactive Main Replacement Program also includes retirement of the associated services.

Q. How does the Company prioritize the retirement of main segments for the Proactive Main Replacement Program?

A. Each year, the Company prioritizes retirement of LPP segments by using a risk-ranking algorithm that is part of the Company's Distribution Integrity Management Plan ("DIMP") and the Company's Gas Operating Procedure for the Identification, Evaluation and Prioritization of Distribution Main Segments for Replacement (ENG04030). The Company's risk model calculates a relative risk score for each LPP segment based on specific performance data and localized incident probabilities and consequences, combined with calculated risk factors for the asset classes under evaluation. This risk-based algorithm, along with the Company's engineering judgment, which takes all factors and risks into consideration in each case, form the foundation of the LPP retirement strategy.

The Company is proposing to target more cast iron main for retirement where possible in consideration of other risk factors, due to increased age and risk of this material. Additionally, as a result of the Storm Hardening Collaborative following the 2016 KEDNY and KEDLI Rate Cases, the Company added flood zone factors to its risk ranking algorithm such that, where other risk factors are equal, the Company will target LPP retirements in Federal Emergency Management Agency ("FEMA") 100 and 500-year flood zones.

Q. What level of investment in the Proactive Main Replacement Program is required in the Rate Year and Data Years to achieve the Company's LPP retirement goals?

A. Annual program spending for the Proactive Main Replacement Program is $250.1 million in the Rate Year, $304.8 million in Data Year 1, $347.9 million in Data Year 2, and $407.6 million in Data Year 3. The capital cost of the Proactive Main Replacement Program is based on a forecast LPP replacement unit cost of approximately $1,280 per foot in the Rate Year. As shown on Exhibit __ (GIOP-1) and Exhibit __ (GIOP-3), costs for safety programs that impact main replacements are presented separately in the capital investment plan, but ultimately will be charged to main replacements.

Q. How does the Company manage the costs of its Proactive Main and Service Replacement Program?

A. To mitigate costs, retirement of LPP is coordinated with other programs (such as the public works, reinforcement, and reliability programs) to capture efficiency savings and cost avoidance. KEDNY will look for more of these opportunities to deploy construction resources more efficiently and will identify areas of the gas network where entire LPP systems can be retired efficiently and cost effectively.

Q. What is the Company's proposal to recover the cost of incremental proactive replacement of LPP?

A. The Company currently utilizes the GSRS adopted by the Commission in the 2016 KEDNY and KEDLI Rates case to recover a return on investment and depreciation expense associated with incremental investment in proactive LPP replacements. The Company proposes to continue use of the GSRS to recover costs of incremental proactive LPP replacements, as shown on Exhibit __ (GIOP- 3). The recovery of costs for incremental proactive LPP replacements will work in conjunction with the Company's net plant tracker. The surcharge mechanism ensures that KEDNY will recover LPP replacement costs only to the extent it is successful in delivering its program and encourages the Company to replace more than the mileage targets.

Q. Is the Company planning to change the depreciation of existing LPP assets?

A. Yes. As described in the testimony of Company Witness Paul M. Normand, the Company is proposing to depreciate existing LPP assets over the remaining 20-year term of the Proactive Main and Services Replacement Programs.

Q. Does the Company propose to continue reporting on LPP retirement?

A. Yes. The Company will continue to provide Department of Public Service Staff ("Staff") with visibility to the status of LPP retirement. The Company proposes to report to Staff on a quarterly basis, including main retired (feet, location), cost data, opportunistic retirements, and the status of the Company's LPP retirement work plan.

ii. Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs

Q. What is the Company's IMP?

A. The Company's transmission pipeline IMP is a safety program mandated by the Pipeline Safety Improvement Act of 2002 and corresponding DOT regulations. The IMP identifies and addresses potential issues affecting the physical soundness of Company facilities before they become safety or performance issues. The Company conducts baseline and periodic reassessments of transmission facilities to identify and evaluate potential threats to "Covered Segments" of pipelines, *i.e.*, transmission pipelines that could affect High Consequence Areas (areas where a pipeline failure could have significant adverse consequences), as well as remediation of significant defects discovered during such assessments. In regions of the United States where older gas

distribution systems are common, IMPs have become a key component of gas pipeline safety through improving pipeline integrity.

Q. Please describe the IMP capital investments.

A. Currently, IMP assessments may be accomplished through either external corrosion direct assessment ("ECDA") or in-line inspection ("ILI") technology, which uses a robotic device known as a "pig" to evaluate the condition of the inside of the pipeline. ILI is considered superior to ECDA because of the ability to identify condition problems originating internally. IMP investments are necessary to: (i) support in-line inspections (*e.g.*, installation of pig launchers and receivers, and pipe reconfiguration/replacement), and (ii) to resolve issues discovered during pipeline inspections conducted through either method.

The construction activities associated with these expenditures involve the installation of "hot tap" fittings, the reconfiguration of such fittings to allow ILI passage, the construction of access points to allow tethered in-line inspection and, in some cases, the replacement of pipeline segments.

Q. Please describe the IMP capital investment needs for the Rate Year and Data Years.

Currently, 21 percent of the Company's DOT pipeline is ILI enabled, increased from 17 percent when the 2016 KEDNY and KEDLI Rate Cases was filed. Over the long term, the Company's plan is to reach 80 percent ILI enablement of the Company's DOT piping. The Company's IMP capital investments in the Rate Year and Data Years will drive toward this goal and will significantly improve the Company's ability to identify integrity issues.

Q. How are the IMP workplan and associated IMP capital investment forecast determined?

A. The workplan began in 2002 based on initial assessments using the Company's existing risk model. Segments were prioritized for ILI enablement over a multi-year workplan based on a combination of their risk score and other relevant factors, such as facility characteristics or geography, to determine what work to do when. Some segments must be re-assessed in certain years according to when they were initially assessed. The workplan is updated annually and as required assessments are completed. Additionally, ILI projects typically consist of two years of design and procurement work followed by construction in the third year.

IMP spending can fluctuate significantly year to year depending on the projects identified for inclusion in the workplan. Because the work must be done by segment, the length, geographic location, and characteristics of the segments that need to be addressed each year contribute to different costs such that historic program costs are not indicative of future program needs. Once segments to be addressed by the workplan are identified, cost forecasts are level one project estimates based on prior projects of similar size and scope adjusted for known variables. Additionally, some larger projects in the IMP are separately budgeted on the capital investment plan.

The IMP workplan for the Rate Year and Data Years is described in more detail in the IMP data sheet in Exhibit __ (GIOP-5). The IMP capital investments required to address known risks identified via assessments and to increase ILI enablement are approximately $5.5 million in the Rate Year, $21.5 million in Data Year 1, $20.0 million in Data Year 2, and $28.0 million in Data Year 3.

Q. What is the status of the pending federal regulations in this area?

A. In May 2016, PHMSA issued a Notice of Proposed Rulemaking ("NPRM") that proposes new pipeline safety regulations that include a requirement for increased inspection of IMP-covered pipelines utilizing ILI technology and an expanded definition of High Consequence Areas, as well as a host of other

requirements. To meet these requirements, transmission pipelines must be ILI enabled.

The Company expects that regulations will become effective in the fall of CY 2019, and an implementation schedule will be included. Because the Company's IMP workplan is risk based, and in light of PHMSA's new regulations expanding IMP, the Company believes that its proposed IMP program is a reasonable and conservative approach to managing pipeline integrity during the Rate Year and Data Years.

Q. What is covered in the Company's IVP Program?

A. The Pipeline Safety Act of 2011 also mandates that PHMSA establish rules requiring operators to demonstrate their pipelines are "fit for service" by reviewing construction records for each pipeline segment to confirm it is operating within design parameters. The May 2016 NPRM also proposes new rules regarding the maximum allowable operating pressure ("MAOP") and pressure testing requirements for existing pipelines, including (i) eliminating the exemption for establishing the MAOP of pre-1970 "grandfathered" pipe segments; (ii) mandating additional pressure testing or replacement for pipelines without adequate pressure test records; and (iii) requiring operators who lack certain records to establish material properties using approved

methods (*e.g.*, cutting and testing pipe samples). In advance of a final rulemaking, PHMSA issued an advisory bulletin (ADB-11-01) directing operators to perform a detailed threat and risk analysis that includes a records review of their systems.

The Company's IVP program began in 2011 and includes thorough record reviews, pipeline replacement, and retirement of non-essential pipeline segments. As with the IMP, the IVP is primarily based on the Company's assessment of system risks, while also incorporating PHMSA's proposed rulemaking.

Q What is the status of the Company's IVP records review and its IVP Program proposal for the Rate Year and Data Years?

A. Through its IVP to date, KEDNY has completed the MAOP records review on 100 percent of its DOT-jurisdictional pipelines. Going forward, the IVP addresses transmission main replacements and testing necessitated by incomplete records identified by the review and pressure testing and preparing to comply with additional testing that will be required by the new rules. Where the Company has identified incomplete records, pipelines will be replaced, or records will be recreated through testing. This work is necessary to ensure system integrity regardless of PHMSA's proposed requirements. The IVP work

plan is described in more detail in the IVP data sheet in Exhibit __ (GIOP-5) and is levelized by year to manage spending at between approximately $4 and $6 million in the Rate Year and each subsequent year.

iii. Transmission Station Integrity Program

Q. Please describe the Transmission Station Integrity Program.

A. This program is similar to the Company's IVP, but the program addresses transmission-connected pressure regulating stations and take stations ("transmission stations"). PHMSA's 2016 NPRM, related advisory bulletins, and discussions between PHMSA and industry stakeholders during the NPRM process, have indicated that the proposed rules will apply to transmission stations. Therefore, the Company is proposing a program for IVP records review and, where records are inadequate to demonstrate that transmission station facilities are fit for service, capital investments for station rehabilitation or partial or full station replacements. Similar to the IMP and IVP Programs for transmission pipelines, these investments are prudent in advance of the final rulemaking to address risk presented by older pressure regulating facilities for which there are inadequate records. The Company is beginning to ramp-up this program in FY 2020, and reaches full implementation in Data Year 1, and the spending is expected to levalize year to year after Data Year 1. The forecast for this program is based on the number and vintage of transmission stations the

program is expected to address and level one estimates of similar station project rehabilitation and replacement projects historically. The capital investments required for this program are $3 million in the Rate Year and over $17 million in each of the Data Years.

iv. CSC

Q. Please describe what is included in the CSC Program.

A. The CSC Program is a mandated program that represents 34 percent of KEDNY's direct capital investment plan in the Rate Year. This program consists of construction work performed to accommodate third-party, municipal construction activity that could impact the integrity of the Company's natural gas facilities. Typical third-party construction activities that impact gas facilities include work on water, sewer, and drainage infrastructure, street reconstruction, road realignment, and bridge replacement. State regulations and Company procedures require the replacement of eight inch and smaller cast iron gas mains if roadway or underground construction is being performed in such a way that would impact the integrity of the Company's mains. Non-cast iron gas mains (*i.e.*, steel and plastic) are not subject to the same replacement regulations and are typically supported and protected if not in direct conflict with third-party construction. Direct conflicts are addressed through relocation regardless of material type.

Aging municipal infrastructure (buildings, schools, bridges, roadways, transportation systems, water mains and sewer facilities) will require significant upgrades in the coming years. Over the last several years, there has been an increase in the level of municipal construction activity, as Superstorm Sandy and recent gas incidents focused attention on the state of municipal infrastructure. Going forward, New York City will invest billions of dollars to upgrade its infrastructure, and many of these projects will directly impact KEDNY's gas system. Thus, the CSC Program is one of the largest programs in the mandated category in terms of costs. The forecast cost for this program is approximately $324 million in the Rate Year, comprised of reimbursable, non-reimbursable, and a large project that is separately budgeted and described in more detail below.

Q. Are there opportunities to retire LPP during CSC projects?

A. Yes. As part of the CSC construction program, the Company looks to identify cost-effective opportunities to retire LPP when main replacements are required to accommodate municipal construction. CSC projects present opportunities to perform safety and reliability upgrades on the Company's infrastructure, the costs of which can be offset by coordinating construction activities (shared trenching and paving) and securing third party reimbursements. Of the

approximately 381,686 linear feet (72.3 miles) of CSC main replacements forecast for FY 2020, approximately 163,680 linear feet (31 miles) of LPP is expected to be retired.

Q. How does the Company forecast CSC expenditures?

A. KEDNY forecasts its CSC expenditures by reviewing the known and planned work identified by municipalities, historical work volumes, and unit information. In large part, the forecast is based on New York State and New York City's current five-year construction plans. In CY 2018, approximately 317,948 linear feet (approximately 60.2 miles) of main replacement was required to address municipal infrastructure improvements. Thirty percent of this footage for this replacement work (equating to approximately $108 million) was subject to some level of reimbursement from third parties, while the remaining 70 percent of the footage (approximately $156 million) was non-reimbursable main and service replacements. Large, complex construction projects, such as the SE856 Phase 2 Transmission Offset Project at Sheffield and New Jersey Avenue ("SE856 Project"), described in more detail below, may be project managed by the Company's Capital Delivery organization and are budgeted separately from the CSC Blanket Program.

While the Company's CSC forecasts are reasonable based on available information, capital and O&M expenditures in this area are subject to a high degree of variability because the scope and scheduling of municipal construction projects are constantly revised. The New York City Department of Design and Construction's ("DDC") work plan is dynamic and is subject to change even within a fiscal year ("FY"). Because the Company does not control the scope or timing of projects, large walk-in projects can require the Company to undertake substantial unplanned capital work in any given year. For example, in CY 2018, the Flatlands Transmission Offset project was added to the DDC's workplan. This was not a known project that was included in the Company's CSC program during the 2016 KEDNY and KEDLI Rate Cases. The project was subsequently added to the FY 2020 work plan at an estimated cost of approximately $69.4 million. The project is expected to be complete prior to the Rate Year.

There is similar volatility with respect to CSC Program O&M expenditures, some of which are non-reimbursable "Support and Protect" expenditures. As shown in Exhibit __ (GIOP- 6), the Company's actual spending on non-reimbursable O&M varied from expected O&M expenditures based on DDC's work portfolio by 75 percent between FY 2016 and FY 2017, and by 118 percent between FY 2017 and FY 2018. The dramatic year-to-year fluctuation

of the CSC Program's O&M requirements are a significant impediment to creation of a reliable multiyear financial O&M forecast.

Q. Does the Company's forecast reflect potential reimbursement from municipalities and other third parties?

A. Yes. KEDNY's forecast reflects reimbursable amounts for expenditures pursuant to the Gas Facility Cost Allocation Act and other cost sharing arrangements with municipalities. KEDNY forecasts expected reimbursements total approximately $57.8 million in the Rate Year, $60.8 million in Data Year 1, $63.1 million in Data Year 2, and $64.9 million in Data Year 3. However, typically there is a one-and-a-half-year lag in payment of invoices by the City.

Q. Please describe New York City's joint bidding process and its potential impact on CSC costs.

A. In 2014, New York State adopted a "joint bidding" requirement for public works projects in cities with populations of one million or more (N.Y.S. Assembly Bill A10021B) that will impact KEDNY's CSC projects in New York City. The City has not yet fully implemented this process. Currently, utilities impacted by New York City public works projects are individually responsible for performing their own relocation work, as well as negotiating all support and protect work directly with the City's contractor. Under the "joint

bidding" model, the City will bid and manage the entire project. If the City's contractor is qualified by the Company to work on its gas facilities, the contractor may perform the gas relocation work. If the City's contractor is not gas qualified, KEDNY will perform its own relocation work ahead of the planned project. Each impacted utility will also be responsible for a portion of the "shared costs" of work performed by the City's contractors, including, for example, maintaining the construction site, establishing field offices, setting up transportation and managing contracts and expenses. It is unclear when the City will fully implement the joint bidding process, but in recent months, the City has increased discussions regarding implementation. It is likely that the process will be fully implemented within the next few years, but the scope of its application is currently unknown.

The City's workplan for FY 2019 and FY 2020 includes approximately 18 projects, across multiple boroughs and utilities, using the joint bidding process. Incremental costs of the joint bidding process for FY 2019 and FY 2020 appear to be about $0.5 million for each FY; however, the projects the City is jointly bidding are not necessarily representative of future project scopes that may use this process. Because the true cost implications on utilities of working through the City's selected contractors, as well as the impact of the shared costs

contribution, remains to be seen, the Company has not included additional costs in its CSC forecast.

Q. How does KEDNY address increasing costs in the CSC workplan and reimbursements?

A. KEDNY works closely with DDC's engineers, consultants, and contractors to minimize direct conflicts to the Company's existing facilities to reduce support and protect and replacement work as much as practicable and to address changes to DDC's workplan. Regarding invoicing and payments, KEDNY recently adopted process improvements to improve the timeliness of Company invoices to the City and is actively negotiating with the City regarding the payment backlog and other process issues related to administration of the CSC program.

Q. What changes is KEDNY proposing to the recovery mechanisms for the CSC Program to address the variability in CSC forecasting and reimbursements?

A. In light of the increasing volatility and scope of unplanned CSC projects, such as the Flatlands and SE865 projects, the Company is proposing to change the existing CSC tracker mechanism to a full reconciliation of the difference between forecast and actual spending (from the current 90 percent deferral

mechanism). The Company is not proposing any other changes to the mechanism or reporting requirements. As set forth in the Joint Proposal adopted in the 2016 KEDNY and KEDLI Rate Cases, the CSC capital reconciliation will work in conjunction with the net utility plant tracker. A demonstration of the CSC capital tracker mechanism is provided in the Revenue Requirements Panel's Exhibit __ (RRP-9), Schedule 2.

Moreover, to address the similar variability in O&M spending demonstrated in Exhibit __ (GIOP- 6), the Company is proposing to add an O&M tracker that would fully reconcile the difference between forecast and actual spending in the non-reimbursable "Support & Protect" O&M expense category. This tracker will provide assurance that the Company's customers are not overcharged to the extent expenses are lower than expected in any given year.

Q. Please describe the status of the LaGuardia Airport Redevelopment Project and the other significant CSC projects that are included in the Company's capital investment plan.

A. The Port Authority of New York and New Jersey's LaGuardia Airport upgrade project, requiring KEDNY to relocate its gate and governor stations and install a new gas distribution system, is ongoing and expected to be completed by the

end of the Rate Year. There is a small amount of capital funding included in the Rate Year to complete this project.

The SE856 Project was recently added to the DDC workplan and involves significant transmission and distribution work. The DDC will be installing new water main, high level storm sewer and sanitary sewer on Flatlands Ave and the vicinity in Brooklyn. This project could potentially require KEDNY to replace 4", 6", 8", 12" and 24" low pressure and 60psi gas mains. KEDNY could also have to offset the 350-psi transmission gas main at three locations within the scope of this project. The Company has not yet received from DDC the preliminary drawings for the entire scope of the project. The transmission work will be managed by the Capital Delivery organization and is budgeted separately from the CSC blanket program. The distribution work is included in the CSC blanket program but is shown as a separate line item on Exhibit __ (GIOP- 1) to clearly identify the capital requirements of this significant project. The capital investment plan includes approximately $101.9 million in the Rate Year and $2.0 million in Data Year 1 for this project. The cost estimate for the SE856 Project is preliminary, and the Company has not yet received the full project scope for SE856 from the DDC. The Company is working closely with DDC to attempt to minimize costs to the extent possible.

v. MRI

Q. Please describe the Metropolitan Reliability Infrastructure ("MRI") Project.

A. The MRI project was included in the 2016 KEDNY and KEDLI Rate Cases as a major gas system reliability project to provide increased supply diversity, pressure support, outage contingency, operational flexibility, and operational autonomy. The project will maximize existing capacity and also will enable proposed upstream capacity (*e.g.,* from Williams' NESE Project) that will better position the Company to meet customers' long-term supply needs over the next 15 to 20 years. Specifically, the MRI project will provide an operational loop to the Brooklyn backbone system that will:

- Immediately improve reliability by increasing supply sources. Sources from any directly connected supply point will be accessible and gas may be moved to any point across the Company's system. This will significantly reduce the reliance on any individual gate deliveries, and specifically dependence on gas from Consolidated Edison Company of New York ("Con Edison") at Newtown Creek.
- Allow the Brooklyn backbone system to be taken out of service for system integrity work, while still maintaining gas operations.
- Enhance the effectiveness of remote control valves on the transmission system by allowing increased utilization of remote valves in the event

of a system disturbance that requires a remote shut down (without disrupting service to potentially thousands of customers).

- Provide for the movement of incremental supplies from both the Narrows and the Lower New York Bay Lateral to any point across the system, including transfer points with Con Edison.

The MRI Project will enable KEDNY to move an additional 850 dekatherms each day by 2021.

Q. What is the current status of the MRI Project, and what is required to complete the project?

A. The MRI Project is on target to complete the first two phases by the end of CY 2019. The main portion of the project, phases one through four, involves installation of approximately 33,100 feet of transmission main and a pig launcher/receiver and a significant sewer crossing at Linden Boulevard. Phases one and two are nearing completion with approximately 20,000 feet of main installed to date. Phases three and four are pending construction and are expected to be in service before the 2020-2021 winter heating season. The remaining work on the project, phase five, entails installation of approximately 7,000 feet of transmission main and a permanent pig launcher/receiver on the Greenpoint facility. Phase five is expected to be in service before the 2021-

2022 winter heating season. The forecast to complete the MRI project is approximately $63.9 million in FY 2020, $100.0 million in the Rate Year and $25.6 million in Data Year 1. There is no projected spending for this project in Data Years 2 and 3.

vi. Renewable Natural Gas ("RNG") Interconnections Program

Q. Please describe KEDNY's new RNG Interconnections Program.

A. This is an innovative program that is part of the Company's overall strategy to encourage interconnection of third-party RNG projects to KEDNY's natural gas distribution system. High interconnection costs have proven to be a significant impediment to the development and interconnection of new RNG projects in KEDNY's service territory. Through this program, the Company will install and maintain some of the necessary interconnection equipment, specifically meters, analyzers, and odorization equipment, that currently is the sole financial and operational responsibility of project developers, thus lowering interconnection costs. This program complements the Company's development of a standard RNG project interconnection agreement that resulted from the 2016 KEDNY and KEDLI Rate Cases, as well as the Company's proposals toward decarbonizing the gas network (*e.g.*, a new green gas tariff offering), as discussed in the Future of Heat Panel's testimony.

Q. What is the basis for the scope and forecast for the RNG Interconnections Program?

A. Based on the number of inquiries KEDNY has received from RNG project developers over the past several years, and RNG projects currently under development, the Company forecasts approximately two RNG projects per year are likely to mature to the interconnection phase during the Rate Year and Data Years. The capital forecast for this program of $0.9 million per year based on the cost of meters, analyzers, and otherization equipment for two projects per year. A detailed description of the project and project costs is provided in the data sheet in Exhibit __ (GIOP-5).

IV. Gas Infrastructure Capital Investment

Q. How much is the Company planning to invest in its gas system assets in the Rate Year?

A. The Company plans to invest approximately $952.1 million in its gas infrastructure and other direct capital investments in the Rate Year, including cost of removal. Exhibit __ (GIOP-1), which provides the actual or budgeted direct capital investments for the Historic Test Year, Gap Period, Rate Year, and Data Years, is segmented into four primary spending rationales (programs): "Customer Connections," "Mandated," "Reliability," and "Non-

Infrastructure." Table 5 summarizes the planned direct capital investments for the Rate Year, and Data Years in each of these programs:

Table 5: Direct Capital by Spending Rationale

Spending Rationale	FY 2021 ($000)	FY 2022 ($000)	FY 2023 ($000)	FY 2024 ($000)
Customer Connections	80,612	114,263	138,471	144,024
Mandated	661,030	657,802	714,313	762,245
Reliability	200,469	154,881	70,881	93,880
Non-Infrastructure / Misc.	9,942	8,045	8,147	8,302
Total	952,053	934,990	931,812	1,008,451

*Inclusive of cost of removal

Each spending rationale is broken down further into sub-categories that identify specific programs.

Q. How were the projected capital estimates derived?

A. In accordance with the Company's budgeting policies and procedures, capital budgets are prepared annually with a five year forward look. Budget projections are based on historical work levels and unit cost performance for

ongoing mandated and routine work and programs, plus any identified new requirements, programs and projects. Projects that fall outside of routine work, such as safety-driven programs (*e.g.*, the Company's LPP replacement program in the Mandated Category), are developed by Engineering based on the most recent material, labor and overhead costs.

Q. What are the primary drivers of the difference in the Company's planned capital spending in the Rate Year compared to historic capital spending?

A. As Exhibit __ (GIOP-1) shows, the primary driver of the increase in planned capital investment in the Rate Year compared to the Historic Test Year is increased investment in the Mandated and Reliability programs. Investments for Mandated programs in the Rate Year are approximately $140 million higher than Historic Test Year spending in this category. Investments in the Reliability category in the Rate Year are approximately $88 million higher than the Historic Test Year in this category. The specific drivers for these increases are discussed below.

Q. Does the Company's revenue requirement in this case also include cost of removal associated with the capital investment plan?

A. Yes. Included in the capital costs discussed below, there is a level of cost of removal required to implement the Company's infrastructure investment plan.

As reflected in Exhibit __ (GIOP-1), the Company is forecasting costs of removal as follows: approximately $95.1 million in the Rate Year, $93.4 million in Data Year 1, $93.1 million in Data Year 2, and $100.7 million in Data Year 3. The capital forecasts for each program presented below are inclusive of cost of removal.

Q. What types of activities are associated with cost of removal?

The Company defines cost of removal as any work on an asset that results in it being removed from the asset inventory, whether or not a different asset is added in its place. This type of work would include, but is not limited to, the activities associated with disconnection, removal and disposal (or retirement in place) of gas mains, gas services, and related facilities.

Q. What information is presented in Exhibit __ (GIOP-5)?

A. Exhibit __ (GIOP-5) provides additional information for each of the significant gas capital projects and programs expected to be performed during the Rate Year. This additional information includes:

- Project or Program name
- Spending rationale
- Project or Program description
- Project or Program justification

- Estimated costs
- Customer benefits discussion
- Alternatives/Options Analysis
- Studies/references that support the program

Q. Please describe some of the technologies and practices the Company uses to reduce the total cost of its capital expenditures.

A. As discussed in the testimony of Company Witness Christopher J. Connolly, National Grid has undertaken a comprehensive effort to evaluate the technology and business processes that support its gas businesses (referred to as the Gas Business Enablement ("GBE") Program). Among the core investments of the GBE Program are new asset and work management, scheduling, geographic information system, and field mobility solutions designed to improve the Companies' gas safety and operational performance. While GBE is in process, the Company continues to employ a number of technologies and best practices designed to increase the efficiency and reduce the cost of its capital expenditures. These practices include:

- Increasing the amount of planned capital work (versus reactive work).
- Increasing coordination among capital programs to increase efficiencies (*e.g.*, leveraging LPP opportunities).

- Installing more small-diameter, high-pressure facilities that can be installed at lower cost.
- Using smaller excavating equipment, increasing operating efficiency and reducing instances of damage (because of decreased size and weight of equipment).
- Using "coring and keyhole" technology to repair existing mains.
- Enhancing contractor management.
- On-site reporting for work crews in many large construction projects.
- Continuing to deploy CISBOT and CIP lining (as described in more detail below).

In addition, as noted above, the Company's Capital Delivery organization is in the beginning stages of identifying opportunities for efficiencies that are expected to reduce costs.

A. <u>Capital Planning, Budgeting, and Sanctioning Process</u>

Q. Please describe the annual development of the Company's capital plan.

Each year, the Company develops a ten-year capital plan to achieve its performance objectives of delivering safe and reliable service. Investment Planning compiles proposed spending for programs and individual capital projects. Programs and projects are categorized into one of four spending

rationales (Mandated, Customer Connections, Reliability, and Non-Infrastructure).

The Company currently uses a tool, C55 (known internally as "Copperleaf"), for the prioritization and optimization of capital investments through a risk scoring and prioritization methodology. This tool uses criteria aligned to a common scale to compare dissimilar inputs (*e.g.*, financial and non-financial inputs) to determine the set of investments, alternatives, and the timing of projects that have the highest potential value.

After the Companies rank the investments in C55, meetings are scheduled with budget sponsors to review value models, project rankings, and various scenarios produced through the optimization of investments to meet capital planning targets. Once plan targets have been approved by the Gas Business Unit, they are submitted to Finance for review.

In late fall, the capital plan is reviewed by the New York Jurisdictional President (Company Witness John Bruckner) and the Vice President, Finance, New York (Company Witness David Doxsee). The New York Jurisdictional President reviews the overall customer, service quality, and financial impacts

of the investment plan and may request changes to the level or mix of investments.

In early winter, the capital plan is presented to the National Grid USA Board of Directors and the National Grid plc Executive Committee and, in early spring, the capital portfolio is presented to the National Grid plc Board of Directors for review and approval.

Q. Are there additional approvals needed before a project in the annual capital plan may proceed?

A. Yes. Aside from the capital planning and budgeting process, specific delegation of authority ("DOA") approval must be obtained for any project in the ten-year capital plan to proceed. This process includes the sanctioning documentation and review for projects over $1 million and other levels of review for smaller projects. Presently, all projects greater than $1 million require some level of sanctioning documentation and review. The U.S. Sanctioning Committee ("USSC") was established by the National Grid USA Board of Directors specifically for this purpose. Projects between $8 million and $25 million are reviewed and approved by the USSC. Projects between $25 million to $203 million are reviewed by the USSC and then are forwarded to a Senior Executive Sanctioning Committee ("SESC") for review and approval. For projects

between $1 million and $8 million, the USSC has delegated review to an informal committee led by the Investment Planning group and including, but not limited to, managers and directors from the Regulatory, Estimating, Asset Management, and Procurement groups. The committee reviews and finalizes sanctioning papers for these projects at a weekly meeting, and the committee then forwards the final sanction documents to the executive sponsor of the project for approval and signature. Projects less than $1 million do not require sanctioning and are approved through a supervisory delegation of authority hierarchy based on certain established thresholds.

Q. Please explain the difference between the DOA review and approval (sanctioning) process and the approved five-year capital plan used to forecast the Rate Year and Data Years.

A. The timing of the sanctioning process is not aligned with the capital planning process used to forecast the Rate Year and Data Years for purposes of the rate filing. As described above, the Company develops a long-term investment plan that is used as the basis for the forecast for the Rate Year and Data Year proposals. Project sanctioning, however, normally occurs immediately prior to the FY for which the investment is planned. For example, the FY 2020 capital plan will be formally sanctioned in early 2019. Thus, the Company's currently-

sanctioned or partially-sanctioned projects do not yet represent the full capital forecast proposed in the Rate Year and Data Years.

Q. Please describe how the Company's DIMP impacts its capital investment planning.

A. The DIMP involves a risk-based assessment of the Company's distribution system to identify threats in seven categories: corrosion, natural forces, excavation damage, other outside force damage, material and weld failure, equipment failure/malfunction and inappropriate operation. The DIMP requires evaluation and prioritization of the risks that these threats pose, and the implementation of measures to address the highest risks with an emphasis on leak management, enhanced damage prevention, operator qualification to reduce human error and system replacement. Consistent with the DIMP, the Company prioritizes asset replacements in its investment plan based on a risk ranking that considers, among other things, leak repair history, types of leak, pipe material, surrounding geography, segment length, nearby construction activity, field conditions, customer issues, open leaks and engineering judgment. The risk ranking factors are carefully designed to consider known differences in the performance of asset subclasses, extensive experience with asset failures, current performance data for the asset subclasses for various

threat categories, and subject matter experts' analysis and opinions on the future performance of the assets.

B. Customer Connections Category of Spending

Q. Please describe what is included in the Customer Connections spending category.

A. The Customer Connections category of work accounts for approximately eight percent ($80.6 million) of the total planned capital investment in the Rate Year.

KEDNY is required by the Public Service Law, the Commission's regulations, and tariff to extend service to new customers in its service territory in proximity to the Company's facilities, and to install new distribution main, up to a 100-foot tariff allowance per customer, to do so. Expanding the availability of natural gas in KEDNY's service territory can bring significant economic benefits in the form of energy cost savings for customers, job creation, and increased local tax revenues. Moreover, there are environmental benefits associated with conversion to natural gas from other, higher emission fuels such as fuel oil. To enable new customer connections, the Company must make significant capital investments in mains, services, and system reinforcements. Customer Connection programs are designed to support forecast customer demand and add new load by increasing system utilization in a cost-effective

way. Customer Connection programs involve the installation of new mains, services, and meters and include system reinforcement.

The Customer Connections – Install Main, Install Services, Meter Purchases, Install Meter, and Automatic Meter Reading ("AMR") Programs and costs are presented in Exhibit __ (GIOP-5).

Q. Please describe recent customer demand trends in the Company's service territory.

A. The updated customer demand forecast shows steady demand in the multifamily and commercial conversion market, steady growth in the multi-family new construction market, and an increase in the commercial new construction market. In the residential sector, the forecast shows an expected increase in demand in residential new construction, but a dip in residential conversions due to saturation in the market. The completion of reliability projects, including the Northern Queens Expansion Program and Brooklyn Navy Yard projects, will enable new connections and are expected to increase demand in previously under-served areas, subject to approval of the NESE project, as discussed below.

Q. If demand for new customer connections is expected to remain relatively flat, what explains the increases in the Customer Connections forecasts?

A. The increase is driven by higher unit costs for main and service installation due the factors discussed in Section III, above. While units are not expected to increase dramatically, the cost of main installation and reinforcements to serve new customers increased as compared to prior years.

Q. Please describe the Customer Connections – AMR Program.

A. The AMR Program includes the purchase and installation of AMR equipment for new customer connection installations. AMR equipment is not included in the Company's Customer Connections Purchase Meters budget. Based on the Company's forecast, the Company will need to install between 9,000 and 10,000 AMRs per year during the Rate Year and Data Years at costs of approximately $1.1 million each year.

Q. Please describe the Gas System Reinforcement Program in the Customer Connections category of spending.

A. The Gas System Reinforcements Program contains projects intended to ensure that minimum system pressures are maintained throughout the gas network during periods of peak demand. The Company models peak demand based on the sendout forecasts developed by Analytics, Modeling and Forecasting

(Company Witness Theodore Poe). As a result of demand for gas usage in its service territory, KEDNY has determined that it is necessary to complete a number of projects to ensure its ability to meet peak requirements. These reinforcement projects are essential to ensure the Company can continue to serve growing gas usage load. Without such reinforcements, the Company estimates that more than 42,000 existing customers could experience loss of service.

KEDNY experienced new records for sendouts during the last two winter heating seasons. During the winter of 2017/2018, KEDNY recorded four of its top ten sendout records, including a firm load record of 1,417,131, dekatherms on January 6, 2018, when the average temperature was eight degrees Fahrenheit. During the winter of 2018/2019, KEDNY recorded two of its top ten sendout records, including a second highest firm load of 1387,550 dekatherms on January 21, 2019 when the average temperature was thirteen degrees Fahrenheit.

As is discussed in Section VII, below, and in the testimony of Company Witness Elizabeth A. Arangio, additional capacity expected from Williams' NESE project is critical to meeting this level of increasing demand. Moreover, the recent growth in peak sendout underscores the need to ensure that the

Company's gas distribution facilities are capable of maintaining minimum system design pressures during periods of peak demand.

Q. Please provide examples of System Reinforcement projects.

A. Examples of System Reinforcement projects include:

- Replacing undersized mains with larger diameter mains. LPP is targeted whenever practical during this work.
- Looping or connecting system endpoints by installing new main.
- System pressure uprates (*e.g.*, 15 pounds per square inch ("psi") to 60 psi).
- Installing new district regulators and replacing existing undersized district regulators.
- Transferring existing low-pressure customers to an adjacent high-pressure main (*i.e.*, load shedding).

Q. What explains the fluctuation in the Gas System Reinforcement forecast from year-to-year?

A. The workplan for the Gas Systems Reinforcements Program is based on where reinforcements are needed and, thus, is zero-based each year. Estimates are derived based on the types of projects needed, creating uneven spending year-to-year.

Q. Please describe the Kew Gardens Gate and Belmont Gate Station projects in the Customer Connections category of spending.

A. These are major reinforcement projects that are separately budgeted and are project managed by the Capital Delivery organization because they are complex construction projects. The Kew Gardens Gate Station will install a new regulator station in Kew Gardens, Queens to address system pressure issues and reduce dependency on deliveries from Con Edison's transmission main. The project began in CY 2018 and is scheduled to be completed during the Rate Year and includes a total forecast of approximately $22.7 million. The Belmont Gate Station Project will rebuild an existing station that is undersized to increase capacity and address integrity issues. The total cost of this project in the Rate Years and Data Years is approximately $26.4 million, but there are no costs included in the Rate Year forecast because the project is scheduled to begin in Data Year 1 with the bulk of spending occurring in Data Year 3.

C. <u>Mandated Category of Capital Spending</u>

Q. What portion of the Company's capital investment plan is Mandated?

A. The Mandated category of work accounts for approximately 69 percent ($661 million) of the total planned capital investment in the Rate Year.

Q. Please describe what is included in the Mandated spending category.

A. Projects covered by the Mandated spending rationale are those needed to comply with regulatory obligations and rate plan commitments, including: the CSC Program (described above), the remaining Local Law 30 compliance, code-required corrosion testing and mitigation or other pipeline integrity related activity, proactive and reactive capital main and service replacement, plastic fusion inspections, and required meter replacements. Exhibit __ (GIOP-5) includes a summary description of each of the significant projects included in the Company's Mandated spending rationale/category, along with the estimated annual funding during the Rate Year and Data Years for each.

Q. Is the Company proposing to continue its CISBOT and Large Diameter Main Rehabilitation Programs for joint sealing and cast iron lining?

A. Yes. The Company's capital plan includes continuation of two programs, included in the 2016 KEDNY and KEDLI Rate Cases, that utilize a joint sealing robot or ("CISBOT") and cured-in-place pipe lining ("CIP") to cost effectively improve the safety and reliability of its large diameter mains. These technologies are proven to extend the life of the main, improve system performance, and reduce future joint leaks on treated pipeline segments for years. As agreed in the Joint Proposal in the 2016 KEDNY and KEDLI Rate Cases, the Company will continue to include main segments reconditioned through CISBOT and CIP lining in the inventory of LPP for eventual

retirement. The Company intends to address approximately two and a half miles of large diameter cast iron main in the Rate Year and each of the Data Years at a cost of approximately $5 million each year, adjusted for inflation. The Large Diameter Main Rehabilitation Program will address two miles in the Rate Year and each of the Data Years using CIP lining at a cost of approximately $14 million per year, adjusted for inflation.

Q. Why is the Company proposing to line fewer miles each year at a higher unit cost than reflected in the current rate plan?

A. Issues arose in the execution of CIP lining program over the past three years where an entire segment of main could not be continuously lined due to configuration issues. In these instances, work had to be stopped and small sections of main had to be replaced between segments that could be lined. As a lesson learned, the Company has adjusted it strategy to plan for replacement of small segments that cannot be lined. The proposed plan to line two miles is more realistic and includes the time and funding to replace segments that cannot be lined.

Q. Please explain KEDNY's Cross Bore Remediation, Latent Damages, and Atmospheric Corrosion Inside Inspections Programs.

A. The Cross Bore Remediation and Latent Damages Programs are existing programs involving safety enhancements undertaken when the Company is performing other mandated work, such as main replacement. In the 2016 KEDNY and KEDLI Rate Cases, the Company included a program to perform inspections on older installations for cross bores, instances when plastic main inadvertently bores through a sewer lateral as a result of horizontal directional drilling. The Company will complete inspections of historic HDD installations by the end of CY 2020. Going forward, the costs to inspect for cross bores on new installations is embedded in main installation unit costs; however, a small budget is necessary to mitigate cross bores that may be revealed through routine inspections. The capital investment plan includes $0.2 million in the Rate Year and each of the Data Years, adjusted for inflation to remediate cross bores. The Latent Damages Program is a capital program to remediate third party damages discovered during the course of normal operations. The Company seeks to continue this program at a funding level based on historic damage rates. The capital investment plan includes $0.4 million in the Rate Year and each of the Data Years, adjusted for inflation.

The Atmospheric Corrosion Inside Inspections Program is a similar proposed program to address remediation required when inside service inspections result in findings of substandard conditions. As a result of the Commission's 2017

Inside Service Line Inspections Order in Case 15-G-0244, baseline inspections result in additional remediation work that must be completed within a year of discovery. Although the majority of remediation work is O&M expense, some remediation requires capital service replacements. The Company estimates that capital funding of approximately $0.7 million in the Rate Year (to address a spike in remediation work anticipated from completion of baseline inspections), and $0.1 million per year in each of the Data Years, adjusted for inflation.

Q. Please describe what is included in the Reactive Main and Reactive Service Replacement Programs.

A. The Reactive Main and Reactive Service Replacement Programs provide for the replacement of gas mains and services and leak repair during urgent or emergency situations that fall outside the normal scope of integrity, reinforcement, reliability and public works programs. These replacements are performed in lieu of repair in instances when repairing damaged facilities is not possible, or where the pipeline segment is too short be covered by the Proactive Program.

The Main Replacements (Reactive) – Maintenance Program provides Field Operations the ability to quickly replace segments of main when emergent conditions warrant. The forecast for this program is based on Historic Test Year

costs, plus an adjustment to include replacement of segments under 50 feet that were previously expensed. This adjustment is made pursuant to a change in capitalization policy that is discussed in detail in Section IV(F), below.

The Service Replacement (Reactive) - Leaks Program addresses unanticipated service replacements due to leaks reported by the public and customers or detected during programmed leak surveys. The capital investment plan includes approximately $5.1 million in the Rate Year and each of the Data Years, adjusted for inflation.

The Service Replacement (Reactive) - Non-Leaks/Other Program addresses primarily customer-driven service replacements necessitated by damages, service abandonments, demolition requests, and service relocations. The capital investment plan includes approximately $5.2 million in the Rate Year and each of the Data Years, adjusted for inflation.

Q. In addition to the LPP replacement program, does the Company also have a program to proactively replace gas services?

A. Yes. KEDNY's Service Replacements - Proactive is the Company's existing program to proactively replace its inventory of inside, high pressure, unprotected (bare) steel services. Proactive replacement of these services is

warranted based on the Company's engineering risk assessment that suggested high pressure, unprotected steel services present a higher risk of failure, particularly at the "wall piece" where the service piping penetrates the foundation wall of the building. The accelerated replacement of high pressure bare steel services with meters/regulators located inside the building is also supported by the risk assessment conducted pursuant to the Company's DIMP. The current inventory is approximately 7,500 services, and KEDNY proposes to continue to replace 250 services per year, maintaining the current pace of this program. The cost of this program is approximately $2.1 million in the Rate Year and each of the Data Years, adjusted for inflation.

Q. What is the status of the Company's New York City Local Law 30 Program?

A. Local Law 30 is a New York City Building Code that requires every gas service line in the City to have a gas service valve or other emergency shut-off device installed outside for all gas services. Over the past three years, the Company incurred tens of millions of dollars above the current rate plan allowance to achieve significant completion of this work. The program is now in its final stage and will be complete by the end of the Rate Year. Because this program is ramping down in FY 2021, the O&M required for this program in the Rate Year is significantly lower ($16.3 million) as compared to the Historic Test

Year ($63.9 million), and the capital forecast decreases to approximately $11.4 million dollars in the Rate Year. There are no capital or O&M requirements for this program in the Data Years.

Q. Please describe what is included in the Corrosion Control Program.

A. This program funds work consisting of field testing, monitoring, upgrades, and repairs to existing corrosion control systems as mandated by Federal and State Code requirements. Part of this program addresses above ground gas mains at bridge locations, which includes complete recoating of existing aged, dis-bonded, deteriorated or uncoated gas mains, as well as retirement of LPP where it extends underground near these crossings. In addition, this program addresses the installation and testing of cathodic protection systems on buried piping. The Company's forecast for the Corrosion Control is based on the Historic Test Year program unit cost applied to a three-year historic average of units (CY 2016 – CY 2018), adjusted for inflation in each year. The Historic Test year is more indicative of future costs because it was the first full year that includes New York City's new restoration and paving requirements. The capital investment plan includes approximately $1.0 million in the Rate Year and each of the Data Years.

Q. Please describe what is included in the Valve Installation and Replacement Program.

A. Federal and state regulations require installation, inspection, operation, and maintenance of critical pipeline valves on all gas distribution systems. The purpose of these valves is to facilitate the rapid shutdown of distribution piping during gas emergencies such as third-party damage or water intrusion. A secondary purpose of these valves is to facilitate maintenance and pipe replacement on associated distribution piping.

This program will strengthen the Company's emergency response capabilities by improving the level at which Field Operations personnel can safely and efficiently isolate sections of the distribution system while mitigating customer impacts (*e.g.*, reducing the duration of future outages). Ensuring all critical valves are properly maintained and operable is a key public safety function and is essential to the effective operation of the Company's gas distribution system. The Rate Year forecast of approximately $0.1 million each year, adjusted for inflation in each of the Data Years.

Q. Please describe what is included in the Meter Changes Program.

A. The Meter Changes program involves the labor to replace gas meters that are retired from service due to required periodic testing, damage, failure, or any other reason.

Q. Please describe what is included in the Purchase Meters Program.

A. This program includes the purchase, testing, processing and delivery of gas meters and associated instrumentation needed to support the Meter Change program, Customer Connections, and Customer Meter Service ("CMS") operations.

Q. Why is there an increase in costs for the Purchase Meters Program beginning in FY 2020?

A. The Company has become aware that a certain population of meters installed prior to 2003 contain cadmium, which can cause problems for testing lab equipment. Therefore, the Company is no longer refurbishing any pre-2003 meters. This has increased the number of meters that may need to be replaced each year going forward.

D. <u>Reliability Category of Capital Spending</u>

Q. What portion of the Company's capital investment plan is Reliability?

A. The Reliability category accounts for approximately 21 percent ($200 million) of the total planned capital investment in the Rate Year.

Q. Please describe what is included in the Reliability category.

A. Investments in this category are intended to maintain safe, reliable service to customers by ensuring that all facilities on the gas system are operating efficiently and reliably. The Reliability category includes programs related to gas control, heaters, reactive Instrument & Regulation ("I&R"), pressure regulating facilities, valve installation/replacement, remote-controlled valves, gas planning, system reliability, water intrusion, system automation and control line integrity, special station projects (including over-pressure protection), storm hardening (remote control valves), and reliability and storm hardening upgrades to the Company's LNG facilities. Exhibit __ (GIOP-5) includes a summary description of significant projects included in the Reliability spending rationale/category, along with the estimated cost during the Rate Year and Data Years for each project. Additionally, two of the larger reliability projects are described below: the Marine Park Regulator Station Project and the Elmhurst Reliability Projects.

Q. Please describe what is included in the I&R Reactive Program.

A. The reactive I&R budget provides funding for capital investment in pressure regulating and control stations. Typical projects in this category include unplanned capital work resulting from emergency conditions, including the replacement of station valves, regulators and relief valves, as well as related capital work on station equipment.

In addition to the I&R Reactive program, the Company is proposing several I&R-related capital and O&M programs to address lessons learned from the Merrick Valley over pressurization issue. These programs are presented in KEDNY and KEDLI's Gas Safety Panel Testimony.

Q. Please describe the Gas Control Training Simulator Project.

A. This program is for the purchase, design, and implementation of a real-time, gas control system-modeled training simulator to meet Federal control room management regulatory requirements for scenario based situational training of gas system operators. The Company currently uses paper table-top exercises to meet this requirement. Use of a training simulator will improve the effectiveness of the training by enabling operators to mimic actual system specifications while drilling actions under normal, abnormal, and emergency operating conditions. The training simulator will benefit all National Grid's gas

local distribution companies; thus, KEDNY's forecast for this project represents KEDNY's allocation of the total cost.

Q. Please describe what is included in the Heater Installation and Pressure Regulating Facilities Programs.

A. There are 40 natural gas heaters currently operating on the Company's system. Because high-pressure gas cools when reduced to a lower pressure, heaters are required at pressure regulating stations to prevent freeze-ups that can impact flow control devices. In addition, cold gas temperatures can lead to reduced pipe toughness and increased potential for frost heave and cold temperature-induced stresses. The heater program adds new heaters (where required) and replaces existing heaters that have reached the end of their useful lives. The Pressure Regulating Facilities category provides funding for replacement and/or rebuilding and reconditioning of existing regulating and control stations.

The Pressure Regulating Facilities Program provides funding for replacement and/or rebuilding and reconditioning of existing regulating and control stations. Pressure regulating facilities (or stations) are designed to control system pressures and maintain continuity of supply during normal operating conditions and during periods of peak gas demand.

KEDNY has assessed regulating stations on its system, evaluating factors such as pressure, location, including whether the station is located in a FEMA 100 year or 500-year flood zone, and the number of dependent customers for each station. In addition, the assessment considered station condition, including pipe corrosion, location and type of overpressure protection, automation, condition of vaults, vault covers, wall sleeves, piping vents and ladders. The results of the assessment were used to create an overall risk rating for each station that serves as the basis for prioritizing projects in this program. The program includes full or partial replacement of existing stations, and storm hardening upgrades where necessary.

The investments in the Pressure Regulating Facilities Program do not cover the special project capital improvements to specific stations that are separately set forth in the investment plan and described below.

Q. Please describe the pressure regulation special projects.

A. Pressure Regulation Special Projects are capital investments to address reliability issues at specific stations that are separately budgeted and are not included in any other blanket reliability programs. These projects are described in detail in the "Pressure Regulation Special Projects" data sheet included in Exhibit __ (GIOP-5). The projects target facilities that have the highest

potential customer impact. Depending on the asset, these projects include station replacement or rebuild, incorporation of odorization, gas quality validation, pressure regulation, overpressure protection, storm hardening improvements, and addition of process pre-heating equipment.

Q. Please describe what is included in the System Automation category.

A. This program will install Remote Terminal Units ("RTUs") at multiple city gate and regulator stations. RTUs provide temperature, pressure and flow data back to the Gas Control Room. RTUs can also monitor gas detectors and intrusion alarms and allow Gas Control Operators to adjust flow and pressure set points at regulator stations. The benefits include enhanced calibration of network models from automation and telemetry data, improved accuracy of network analysis, and enhanced ability to forecast the need for capital reinforcements, which will lead to more efficient capital planning. Automation allows Gas Control Operators to selectively close valves, raise or lower pressures, and shut down take stations. System alarms also alert Gas Control Operators to system issues and allow quick pinpointing of the source.

PHMSA regulations regarding Control Room Management require Operators to ensure that "practices and procedures within their control rooms are adequate to maintain pipeline safety and integrity." These rules indicate that Operators

should have telemetry to monitor pipelines, as it would increase system awareness and enable a proactive response to abnormal operating conditions. The System Automation program complies with these regulations by providing for increased deployment of telemetry on the Company's system.

Q. How is system performance monitored currently?

A. Currently, 67 percent of the pressure regulation stations are equipped with some form of telemetry, while the 33 percent of the system relies on paper chart recorders. KEDNY's System Automation program adds automation capabilities to low pressure regulator stations. RTUs installed under the System Automation program will provide enhanced ability to monitor system performance and remotely adjust pressures on the gas system. The program will also replace aging and obsolete telemetry equipment.

Q. Please describe what is included in the Gas System Reliability – Gas Planning/Remote Controlled Valve ("RCV") Program.

A. This program includes capital projects required to maintain system minimum pressures on the gas network in the event of an abnormal operating condition (failure involving a regulator station, gate station, critical main or other major pressure facility on the system). Gas System Planning ensures that customers

continue to have reliable service and that no customers experience interruptions as a result of an unplanned outage of a facility under normal winter conditions. The program also involves the installation of additional RCVs on transmission pipelines to improve emergency response capability and reduce the risk of gas releases. In the event of a pipeline failure, RCVs allow control room operators to stop the flow of gas and to remotely isolate and shut down a portion of the system. Currently, most transmission pipelines can only be shut down using manually-controlled isolation valves, which can take longer to close and result in a larger customer impact. Improving response time through the expanded deployment of RCVs reduces the quantity of gas released and can limit the harm to the public and property.

Q. Please provide examples of Gas Planning Reliability projects.

A. Examples of Gas Planning Reliability projects include: eliminating distribution systems fed by a single district regulator or main, integrating distribution systems with the same operating pressures through pipeline connections, expanding supply diversity, and projects targeting areas of the system where large numbers of customers would experience a service interruption if a single gas facility became inoperable when the average daily temperature is five degrees Fahrenheit.

Q. Why does the Company propose to include installation of RCVs in the capital investment plan?

A. The PHMSA regulations promulgated in response to the Pipeline Safety Act of 2011 will mandate the installation of additional RCVs. But even in the absence of the PHMSA regulations, investment in RCVs is prudent given the safety and reliability benefits. As highlighted by recent industry events, there are significant operational benefits associated with the increased deployment of RCVs, such as enhanced pipeline shutdown capabilities.

Q. Please describe what is included in the Water Intrusion Program.

A. The Water Intrusion program is designed to address water entering the gas distribution system and protects against main obstructions, freeze and thaw issues, poor pressure and/or freezing customer services. This program targets the retirement of LPP that is susceptible to water intrusion but is not prioritized for replacement under the Main Replacement programs because of the absence of leaks and/or historical leak repair activity.

Q. Please describe the Storm Hardening – Remote Service Shutoff Valves Program.

A. In the 2016 KEDNY and KEDLI Rate Cases, the Company included a proposal for a Storm Hardening Program to install remote service shutoff valves on

approximately 25,000 gas services within the FEMA's designated flood zones over five years beginning in CY 2017. Because there was no viable shutoff valve mechanism commercially available for use on the system, the Company was unable to timely launch this program. The devices became available from the manufacturer during February of 2019, and the Company plans to begin installations in FY 2020. Because of the significant benefits of these devices, the Company is renewing its proposal for this program to address an increased scope of approximately 29,000 gas services from FY 2020 and FY 2024.

Q. What are the benefits and costs of this program?

A. The remote shutoff valves add a layer of protection for services that are especially susceptible to storm surge and flooding that could cause over-pressurization of the gas facilities connecting customers' premises. The impacts of recent severe storms (*i.e.*, Superstorm Sandy and Hurricane Irene) demonstrate the need for the Company to harden its infrastructure to provide greater protection from future storms. These valves will operate on a fixed communication network that will allow for remote operation and monitoring. Automated valves stop the flow of gas as soon as flooding is detected. This will prevent regulator over-pressurization and stop gas from flowing to premises with damaged equipment and/or extinguished pilot lights, mitigating the risk of a potential incident. Automated valves also provide a real-time count

of services impacted by flooding to inform the Company's storm response about the resources needed to restore the affected customers expeditiously. Lastly, in areas where flooding prevents physical access to valves and regulators, remote shut-off valves will allow the Company to interrupt only those services impacted by flooding, which could spare entire neighborhoods or larger areas from losing gas service because of access issues. Installation of the communication networks through which the valves operate can also be leveraged in the future for additional smart devices, such as methane detectors that are being developed.

The Company is proposing to ramp-up the program in FY 2020 and the Rate Year, installing approximately 500 valves in FY 2020 and approximately 4,000 valves in the Rate Year, with the bulk of the communications network systems in the Rate Year as well. The Company intends to install approximately 8,000 valves per year in each of the Data Years. Program costs in the Rate Year are approximately $7.4 million in the Rate Year.

Q. Please describe the Marine Park Regulator Station Project and the Elmhurst Reliability Project.

A. These are two of the larger project-managed reliability projects that are included in KEDNY's capital investment plan. The Marine Park Regulator Station

Project includes installation of a new regulator to enable increased supply delivery at the Floyd Bennet Field supply point that was constructed as part of the Brooklyn Queens Interconnection project. This project is necessary to ensure adequate service to firm customers and to avoid stranded supply at Floyd Bennet Field. The project is scheduled to commence in FY 2020, and the capital plan includes approximately $1.0 million in the Rate Year and $22.8 million in Data Year 1.

The Elmhurst Reinforcement and Reliability Project is required to support an increase in forecast demand in the 2nd Ward of Queens, including at LaGuardia Airport. The project involves creation of an interconnect that can be used to prevent outages in the event of a system emergency and will reduce dependence on supply from Con Edison. The project is scheduled to begin in Data Year 2, and the capital plan includes approximately $1.0 million in Data Year 2 and $35.0 million in Data Year 3.

Q. Please describe the compressed natural gas ("CNG") investments included in the KEDNY's capital plan.

A. The CNG investments included in the capital plan support the Company's fleet of CNG vehicles. These investments include upgrades necessary to maintain safe and reliable operation of the Company's CNG fueling stations, including

partial or full replacement of existing stations, replacement of station equipment, and/or installation of telemetry, safety design, and reliability improvements.

Q. Please describe the Company's LNG programs.

A. KEDNY maintains on-system supply through its LNG facility at the Greenpoint LNG Plant. This plant is a critical component of KEDNY's gas supply portfolio and gas operating network and plays a critical role in meeting peak demand. The Greenpoint LNG Plant is capable of supplying 290 million cubic feet of gas per day, which is critical for meeting KEDNY's peak day demand. Refilling the tanks is accomplished through liquefaction during the summer period when gas supplies are available and less expensive. The liquefaction system can refill at a rate of about 7 to 8.5 million cubic feet of gas per day and it takes between 150 and 200 days to refill both tanks. The Greenpoint LNG Plant is a cost-effective means of meeting peak demand, because gas is liquified during the summer months when prices are lower and stored for use in the winter during peak demand conditions. KEDNY's various LNG capital programs are necessary to ensure the continued safe, reliable operation of the Greenpoint LNG Plant.

Q. Please describe the Company's proposed capital investments for the LNG programs during the Rate Year and Data Years.

A. The Greenpoint LNG Plant has been in service for more than 50 years and requires significant investment to support continued safe and reliable operation. Over the past three years, the Company developed major projects to upgrade certain facilities at the Plant that were approved in the 2016 KEDNY & KEDLI Rate Cases, namely LNG tank modernization project and a new Salt Water Pump House. The Company's proposed capital plan includes funding to complete these ongoing projects, blanket funding to replace aging equipment and unexpected needs, and proposals for a number of continued and new special projects necessary to upgrade and storm harden critical facilities and equipment at the Plant, and to support safety systems.

Q. What is covered in the LNG Blanket Program?

A. The LNG Blanket Program provides funding for near-term and emergent capital projects needed to maintain safety and reliability at the Greenpoint LNG facility. Projects typically address equipment that is at or near the end of useful life. Examples of projects in this category include: upgrades to mechanical equipment and systems; upgrades and replacement of electrical and control systems; structural improvements of plant and facilities; procurement of capital tools and equipment; preliminary engineering and design of capital projects;

and retirement and decommissioning of equipment, plant and facilities. These projects will extend the service life of the facility and improve operational performance of plant equipment.

Q. Please describe the LNG Special Projects.

A. In addition to the LNG Blanket Program work, KEDNY has identified a number of significant, discrete capital projects to be completed in the Rate Year and Data Years. These projects are discussed in more detail in the "LNG Special Projects" data sheet that is included in Exhibit __ (GIOP-5). Several of these projects were identified during the Company's Storm Hardening Collaborative that resulted from the 2016 KEDNY and KEDLI Rate Cases and involve projects to address identified flood and climate change related risks to critical components or systems at the plant. Several projects are carry-overs from the current rate plan that are in various stages of completion, including projects deferred due to re-prioritization, changes in project scope, or delays in permitting approvals from municipal entities.

Q. Please described the status of the Salt Water Pump House Project that was included in the 2016 KEDNY and KEDLI Rate Cases.

A. The Salt Water Pump House Project will provide a dedicated source of salt water for two major systems, the deluge system and the fire hydrant system.

The Company began this project in 2016, but experienced delays and scope changes resulting from unexpected Fire Department of New York permitting requirements and stipulations. The bulk of construction work is anticipated to occur during the Rate Year. The capital investment plan includes approximately $35.1 million in the Rate Year and $15.3 million in Data Year 1 to complete this important reliability project.

Q. What is the status of the Tank Modernization Project?

A. The Greenpoint LNG Tank #1 (0.6 billion cubic feet ("bcf")) was put into service in 1968 and Tank #2 (one bcf) was put into service in 1971. In the 2016 KEDNY and KEDLI Rate Cases, the Company proposed a project to temporarily take Tank #2 out of service to allow entry into the tank to perform a major tank upgrade. However, execution of this project was dependent on the NESE project. Delays in Williams' NESE project resulted in delays to the planned Tank 1 and 2 upgrades. The Company's proposed capital investment plan includes funding to deliver this project in Data Years 1 through 3, based on the assumption that the NESE project will move forward in the next two years.

Q. Please explain the LNG Vaporizers 7 and 8 Refurbishment Project.

A. The Greenpoint LNG Plant includes six submersible combustion vaporizers that vaporize the liquified gas into the Company's gas transmission system. Two of the vaporizers, known as vaporizer numbers 7 and 8, have been in service for nearly 40 years and despite modifications that have been made to improve and maintain these units, conditions are deteriorating such that replacement is necessary. This project funds replacement of the vaporizers to ensure continued operability of the plant at existing levels. The capital investment plan includes approximately $10.2 million in Rate Year, $10.1 million in Data Year 1, and $3.0 million in Data Year 2.

E. <u>Non-Infrastructure and Other Capital Spending</u>

Q. What portion of the Company's capital investment plan is Non-Infrastructure and Other?

A. The Non-Infrastructure and Other category of work accounts for approximately one percent ($10 million) of the total planned capital investment in the Rate Year. Other Capital includes special projects not included in the Company's other investment programs, most notably KEDNY's investment in automated meter reading. The Non-Infrastructure budget also includes funds for the purchase of tools and equipment that meet the criteria for capitalization.

Q. Please describe the Company's Telecommunications Program.

A. The Company operates a two-way radio system that provides communications support to Gas Field Operations, CMS, Security, and other departments. This system also serves to support communications during emergencies when commercial communications systems are not operational. These systems contain ageing components and must be repaired or replaced upon failure. The Telecommunications Programs provide reactive funding for smaller replacement projects in three categories that collectively cover funding needs for towers at remote sites, radios installed in vehicles, and for general damages and failures.

Q. What is included in the Tools and Equipment program?

A. The Tools and Equipment program captures the items that are not used for specific projects but support the safe, efficient and on-going day-to-day operations of the gas business. Examples include tools (hand, power, pneumatic, hydraulic), specialty equipment, PPE, office machines, electronic data processing equipment and software applications, shop and garage equipment, and communications devices.

Q. What is included in the Company's Learning and Development ("L&D") Materials Tools and Equipment Program?

A. Historically, L&D has a limited stock of tools for use in training and has relied upon tools borrowed from Field Operations to supplement its supply and conduct required training of in-house and contractor employees. With workload and training requirements increasing, Field Operations can no longer spare tools and equipment sufficient for training purposes, and the limited stock of available tools is beginning to deteriorate. The proposed L&D Materials Tools and Equipment Program will purchase new tools and equipment for training purposes. The program will follow an innovative approach whereby the new tools will be used for training during the annual training sessions at the beginning of each year, and then the new tools will be sent into the field for use by the newly-trained employees. Following annual training, Field Operations will send its in-use tools back to L&D for refurbishment and use in supplemental training throughout the year. This tool swap enables better training because training will occur on the equipment that will actually be used in the field, enhances the quality of tools used in the field, and ensures that L&D has a sufficient inventory of tools to conduct supplemental training during the year after annual training has concluded. The program also increases the useful life of field tools that are swapped back into L&D for refurbishment that otherwise may have failed earlier in the field due to continuous use.

Q. Please describe the Company's Non-Infrastructure AMR Installation and AMR Replacement Programs.

A. The AMR Installation Program will complete KEDNY's five-year AMR deployment program that began in CY 2017. As of the beginning of FY 2020, the Company will have roughly 110,000 encoder receiver transmitters ("ERTs") to install to complete the program. The Company anticipates that AMR deployment in KEDNY's service territory will be complete by the end of the Rate Year.

The AMR Replacement Program funds the purchase of the ERT units to support the Company's mandated meter changes program and to replace existing units nearing the end of useful life or that have failed. The ERT units are not included in the Purchase Meters Program. KEDNY currently has an installed AMR population of approximately 1.4 million units and estimates that approximately 23,000 AMR units will require replacement during the Rate Year and each Data Year. This program does not include funding for ERTS to support meter installations for new customers; those are funded in the Customer Gas Connections – AMR Program.

Q. What is included in the Meter Testing Equipment Program?

A. The Meter Testing Equipment Program replaces meter test lab equipment that has reached its useful life that is needed to carry out the mandated meter testing program.

F. Capitalization Policy Change – Main Installation

Q. Is the Company proposing any changes to its capitalization policies for main installation?

A. Yes. Accounting for installation of segments of main under 50 feet in length is currently expensed by the Company. This threshold is arbitrary; moreover, there is no similar threshold for main replacement in Niagara Mohawk's service territory. Applicable accounting principles and regulations permit capitalization of new main installations, and there is no minimum threshold. The Company is proposing to remove the 50-foot threshold and capitalize all main installation activities in accordance with the FERC Chart of Accounts – Gas – Operating Expense Instructions, Accounts 367 Mains and 376 Mains. This change will standardize the accounting treatment of these work activities between National Grid's New York gas distribution companies. As discussed in the testimony of the Revenue Requirements Panel, the Company has normalized its Historic Test Year to reflect the change from expense to capital

for distribution and transmission main installation, and the Company's capital investment forecasts also reflect this capitalization change.

V. Indirect Capital

Q. Is the Company allocated indirect capital costs?

A. Yes. KEDNY is allocated a portion of indirect costs such as facilities, fleet services, and supply chain. The Company's indirect capital investment plan is presented in Exhibit __ (GIOP-7).

Q. Please describe the Fleet and Supply Chain investments included in the Rate Year and Data Years.

A. The proposed fleet and supply chain expenditures will provide required upgrades to the Company's critical operating facilities in support of its capital and O&M programs. Fleet requirements in the Rate Year and Data Year include construction of fleet bays in Springfield, required re-certification of a fuel tank system in Staten Island to meet environmental regulations; replacement of truck lifts that have reached their useful lives in Greenpoint, Canarsie, and Staten Island; a fleet offices refurbishment in Canarsie; installation of electric vehicle charging stations; and as-needed replacements of tools, equipment, and miscellaneous items required for upkeep of the

Company's fleet, such as shop equipment, tire lifts, *et cetera*. The fleet forecast includes approximately $0.7 million in the Rate Year.

Supply Chain investments support the service centers that are used to stage and deploy the Company's field workforce, store vehicles, equipment and materials, and provide various utility functions (*e.g.*, meter shop) in support of the operation and maintenance of KEDNY's system, including inventory management and warehousing. Supply chain needs during the Rate Year include paving, security cameras, and inventory racking. The supply chain forecast includes approximately $0.6 million in the Rate Year.

Q. Please describe the Company's proposed Facilities investments for the Gap Period, Rate Year, and Data Years.

A. The Company's Facilities investments are set forth in Exhibit __ (GIOP-7) and include a baseline facilities plan of approximately $5.7 million in the Gap Period, between $3.0 million and $3.6 million in the Rate Year and Data Years 1 and 2, and $4.6 million in Data Year 3 for general property improvements and necessary upgrades, and several larger projects that are described below and in data sheets included in Exhibit __ (GIOP-7). The Company is making significant investments in facilities required to support the Company's capital and O&M programs, enhance the customer experience, and ensure optimized

and efficient office workspaces. The total facilities investment plan for the Rate Year and Data Years for KEDNY is approximately $8.6 million in the Rate Year, $5.9 million in each of Data Years 1 and 2, and $4.6 million in Data Year 3.

Q. Please briefly describe the Company's significant Facilities projects.

A. There are five large projects included in the Company's Facilities capital plan for the Gap Period, Rate Year, and Data Years as follows:

- Pitkin Customer Office Remodeling: The Company will remodel the Pitkin Avenue Customer Office and Outreach Center to improve the work area for customer-facing payment center employees and walk-in customers, and including carpet and ceiling tile replacement, full painting, and general wear and tear repairs. The Company forecasts $0.5 million in capital expenditures in the Gap Period for this project. This project is discussed in the testimony of the Shared Services Panel.
- NYC Training Centers (Queens/Brooklyn and Staten Island): KEDNY will develop two 3,000 square foot Company-owned training facilities, one in Queens/Brooklyn and one in Staten Island, to accommodate growth in the amount of technical training required to ensure the competence of National Grid's gas workers and to validate the training/competency of the contractor workforce in KEDNY's service

territory. The Training Centers will also support New York State regulatory commitments for implementing First Responder training in New York City, and will better enable the Company to educate its workforce concerning revised PHMSA pipeline safety standards and enhanced engineering oversight standards. The new facility will permit the Company to deliver enhanced classes and hands on training to both employees and contractors. The Training Centers will include classrooms, lab-bench technical, and office facilities designed to meet increased training and evaluation demands. Forecast capital expenditures for this project are $13.6 million in the Gap Period and $2.4 million in the Rate Year.

- Greenpoint Electrical Project: The Company will perform multiple upgrades and modifications to the existing outdated and aged electrical distribution systems feeding the various buildings on the Greenpoint Gas Operations and LNG site, including reducing the number of electrical services to the site, removal of unnecessary feeders, installation of up-to-date switchgear and sub-metering systems, and modifying systems for redundancy and emergency power protection. These upgrades will improve the safety and efficiency of the site's electrical system. Projected capital expenditures are $0.5 million in the

Gap Period, $0.6 million in the Rate Year, and $0.8 million in Data Year 1.

- Canarsie Roof and Facades Project: KEDNY will replace the roofs and perform significant façade improvements, including window replacements, on all buildings at the Canarsie Operations Site to maintain the integrity of the buildings. The Company had the roofs and facades at the site inspected by a professional engineering firm, which recommended that repairs and replacements be conducted in the 2015-2020 timeframe. Forecast capital expenditures for this project are $1.9 million in the Gap Period, $2.3 million in the Rate Year, $1.5 million in Data Year 1, and $2.3 million in Data Year 2.
- Canarsie Parking/Paving Project: The Company will remove and reconstruct the employee parking lot adjacent to its Canarsie Operations Center. The reconstruction of the East 83rd Street lot will consist of removal of all existing asphalt and fencing, re-grading the entire lot, and the installation of a new drainage system, pavement, curbing, fencing, and landscaping. The Company will also remove and replace deteriorated asphalt in the interior lot. These improvements are needed to remedy the severe state of disrepair in the lots, which can cause vehicle damage and creates unsafe walking hazards. The drainage system repairs are required by New York City to address hazardous

impacts to the public. Forecast capital expenditures for this project are $0.8 million in the Gap Period and $0.3 million in the Rate Year.

Q. Are additional Facilities investments planned for the Gap Period, Rate Year, and Data Years?

A. Yes. National Grid is proposing a plan of facilities replacement for general facilities improvements and upgrades of $34.1 million in the Gap Period, $38.8 million in the Rate Year, $11.8 million in Data Year 1, $6.8 million in Data Year 2 and $6.8 million in Data Year 3. These facilities are National Grid Service Company investments and the costs are set forth in KEDNY's Revenue Requirements Panel's Exhibit __ (RRP-11), Workpapers to Exhibit RRP-3, Schedule 9, Workpapers 4, 7, 10, and 13. The large Facilities investment projects include:

- Replacement of the Hicksville Office and Operating Site: National Grid currently occupies 237,000 square feet of the 515,000 square feet available at this site. PSEG-LI occupies the remainder of the site. National Grid plans to relocate its operations to a 100,000 square foot site and, in so doing, achieve savings through a reduction in the total footprint and the elimination of inefficiencies that arise from the joint occupation of the current site. Forecast capital expenditures associated with the relocation for National Grid Service Company total

approximately $3.1 million in the Gap Period, $19 million in the Rate Year, $0.5 million in Data Year 1, $0.5 million in Data Year 2, and $0.5 million in Data Year 3.

National Grid has been sharing this site with PSEG-LI since it assumed responsibility for the administration of the Long Island Power Authority's transmission and distribution systems in 2014. Sharing major operations facilities is not optimal for the Company from an efficiency, safety or security perspective. Moreover, sharing a major operations site makes it more difficult to promote and brand National Grid to the public. In addition, it creates issues with respect to recruiting, training and retaining employees. Relocating National Grid's operations to a more efficient, Company-only site is clearly beneficial from a company, employee, and customer perspective.

- Brooklyn, NY Office Relocation: National Grid will relocate existing office personnel in the MetroTech office to another facility in Brooklyn, reducing the Company's footprint. National Grid is examining potential locations for this relocation at this time. However, a move to a smaller footprint will reduce costs and enable the Company to make more efficient use of space. As part of this plan, KEDNY's call center, customer office, and Sustainability Hub will remain at the current location and the Company will sublease the remainder of the space until

the end of the current lease term in February 2025. National Grid forecasts capital expenditures of $8.0 million in the Gap Period and $13.2 million in the Rate Year for this project. The Revenue Requirements and Shared Services Panels further discuss the planned MetroTech office relocation.

- IM Warehouse Project: – National Grid will consolidate and integrate the Gas Distribution and Gas Transmission supply chain by moving certain warehouse facilities from Greenpoint in Brooklyn to a new location on Long Island. This integration supports a comprehensive overall supply and inventory management program for the GBE Program in the U.S. The facility will help to improve services provided to Gas Transmission and Gas Distribution by enabling increased delivery frequency of needed materials and permitting National Grid to better optimize delivery routes. Forecast capital expenditures for this project are approximately $14.0 million in the Gap Period and $1.1 million in the Rate Year.
- Reservoir Woods Office Project: – National Grid is reconfiguring the corporate offices in Reservoir Woods, Massachusetts, which is utilized by employees who provide services to KEDNY and other National Grid affiliates, to create a more efficient space. Forecast capital expenditures are $6.0 million in the Gap Period.

VI. Gas O&M Expenses

Q. Please summarize the Panel's testimony regarding the costs of operating the gas system.

A. The Panel addresses major expenses associated with operating the Company's gas delivery system, and incremental O&M expenses the Company expects to incur in the Rate Year and Data Years.

Q. Please generally describe the nature of the Company's gas system O&M expenses.

A. O&M expenses relate to work performed to provide customer support, respond to emergencies, perform safety inspections and other compliance activities, restore service, and maintain the life of capital assets. The Company has a significant maintenance program to ensure that system assets are utilized to their fullest potential life expectancy. As gas facilities age, maintenance costs increase. These costs include more frequent inspection and testing required by regulatory changes, increased volume of repairs, more significant repair work, increased emergency work, and gas pipeline safety enhancements. These expenditures are required to prevent failure and maintain the life of the assets until replacement occurs. The Company's O&M programs are also designed to maintain the service commitments in its gas safety performance metrics, which cover various aspects of its performance in the areas of reliability and safety,

including metrics measuring pipeline replacement, emergency response, leak management, and damage prevention.

Q. How does the projected Rate Year non-labor expense level compare to the Historic Test Year non-labor expenses for operating the gas system?

A. There is incremental non-labor O&M for individual projects and programs; however, there is a net decrease in total non-labor O&M in the Rate Year and in each of the Data Years as compared to the Historic Test Year. This is due to the ramp down of the Local Law 30 Program that significantly decreases non-labor O&M for this program in the Rate Year and Data Years as compared to the Historic Test Year. Specifically, compared to the Historic Test Year, LL 30 Program O&M expense decreases by approximately $47.6 million in the Rate Year. Thus, considering the LL 30 offset, the Company projects its Rate Year O&M associated with gas operations to be approximately $24.7 million less than its adjusted O&M non-labor expense for the Historic Test Year.

Q. What are the projected incremental Rate Year O&M non-labor and labor expenses for gas operations and gas safety programs?

A. As shown on Exhibit __ (GIOP-8), the Company projects its incremental Rate Year non-labor O&M expense to be approximately $22.878 million. As shown in Exhibit __ (GIOP-9), the Company also proposed to hire 159.7 total

incremental FTEs in the Rate Year. The costs for these FTEs are presented in KEDNY's Revenue Requirements Panel testimony and exhibits.

Q. Please summarize the projected incremental O&M expenditures.

A. Increases in O&M expense are primarily driven by (i) the Company's increasing capital investments and increased costs for executing the Company's capital plan, (ii) changes to Company's O&M workload, and (iii) initiatives the Company is undertaking in the Rate Year to address new or expanding safety requirements and to implement lessons learned from recent industry gas system incidents.

The New York City restoration/paving cost increases also impact O&M costs of construction, leak repairs, and disconnects and reconnects. These costs are reflected in the Historic Test Year, but generally have increased costs as compared to prior years.

Q. What is the Company doing to manage its O&M costs?

A. The Company has implemented various initiatives to reduce its O&M expenses, including:

- Increasing the use of scheduled O&M work appointments to reduce multiple unproductive field visits to complete work.

- Coordinating O&M activities required at each premise so that multiple maintenance requirements can be completed during a single visit.
- Increasing the use of coring and low-dig technology, reducing debris removal and paving restoration costs associated with smaller roadway excavations.
- Modifying shift schedules to more efficiently respond to higher leak volumes.
- Exploring new work management systems to optimize dispatch of resources.

Notably, during CY 2018, the Company undertook a re-organization of its Gas Business Unit to optimize the level of managers and FTEs to arrive at a more efficient balance of personnel.

A. <u>Incremental O&M Costs Associated With Capital Investment</u>

Q. Please describe the Company's need for incremental O&M costs associated with its planned capital investments.

A. As discussed above, the Company plans to increase its capital investment program during the Rate Year. As shown on Table 6 and in Exhibit ___ (GIOP-8), the Company estimates incremental O&M non-labor expense of approximately $8.009 million in the Rate Year directly related to the Company's capital investments.

Table 6: Rate Year Incremental O&M and FTEs

Related to Capital Investments

Category ($000)	FY 2021 Non-Labor	FTEs
Capital Support – General	--	18
Disconnects & Reconnects	$53	--
IMP/IVP	$7,178	4.5
Transmission Station Integrity	$146	2
Storm Hardening	$521	0.6
RNG Program	$111	1.9*

* Included in new Future of Heat Engineering Group

i. Capital Support - General

Q. What O&M services will the various construction support functions provide to support the Company's increased capital investments?

A. Construction support functions include internal groups providing contract administration, project management, budgeting and resource planning. While the majority of costs from these functions are directly charged to capital projects, the Company incurs O&M expenses for items such as training, travel, conferences, licensing, new employee on-boarding, and costs for administering O&M contracts. The Company estimates that approximately 10 percent of construction support employees' time is O&M expense.

As KEDNY increases its capital expenditures, the Company will require additional capital support resources, including gas system engineering (estimators, designers, engineers), investment planning (clerks, inspectors, program managers), operations support (permit clerks) and resource planning (analysts, coordinators).

Q. What are the incremental support requirements in the Rate Year?

A. The Company forecasts approximately 18 FTEs for these support functions in the Rate Year. A breakdown of the positions is provided in Exhibit __ (GIOP-9).

ii. Disconnects & Reconnects

Q. Please describe the O&M costs associated with service line disconnects and reconnects.

A. Main replacements require the Company to disconnect gas service lines from the main being removed, and then reconnect the service to the new main. A 2,000-foot main replacement can require dozens of disconnects and reconnects, especially in densely populated, urban areas. The Company's capital plan involves significant investment in main and service replacements, including the accelerated retirement of LPP. This work will increase O&M costs for disconnection and reconnection of gas services by $0.053 million in the Rate

Year, $1.891 million in Data Year 1, $3.799 million in Data Year 2, and $5.983 million in Data Year 3, as shown in Exhibit___(GIOP-8).

iii. IMP/IVP

Q. Please describe the O&M requirements of the IMP and IVP Programs.

A. Incremental non-labor O&M in the amount of $7.178 million in the Rate Year, $2.097 million in Data Year 1, $0.719 million in Data Year 2, and $6.055 million in Data Year 3, as shown in Exhibit___(GIOP-8) and an additional 4.5 FTEs in the Rate Year and Data Years are required to conduct the required inspections via ECDA on pipelines that are not ILI-enabled and to conduct ILI on those that are already enabled. These costs include excavation and support for ECDA inspections, evaluation of testing data and the costs of non-capital repairs such as repair sleeves and on-site material testing. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

iv. Transmission Station Integrity

Q. Please describe the O&M requirements of the Transmission Station Integrity Program.

A. Similar to the IVP and IMP program needs for non-station assets, incremental non-labor O&M in the amount of $0.146 million in the Rate Year, $0.796

million in Data Year 1, $0.771 million in Data Year 2, and $0.787 million in Data Year 3, as shown in Exhibit___(GIOP-8) and two incremental FTEs in the Rate Year and Data Years are required to conduct records reviews and testing of stations. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

v. Storm Hardening – Remote Shutoff Valves

Q. Please describe the O&M requirements of the Storm Hardening – Remote Shutoff Valves Program.

A. The installation of storm hardening remote shutoff valves requires incremental non-labor O&M in the amount of $0.521 million in the Rate Year, $0.709 million in Data Year 1, $0.741 million in Data Year 2, and $0.903 million in Data Year 3, as shown in Exhibit___(GIOP-8), for pole rentals and information systems running costs for communication equipment to operate the valves. Additionally, the Company requires 0.6 incremental FTEs in the Rate Years and Data Years to oversee system monitoring and valve location support, investigate alarms, and replace valve components, as necessary.

vi. RNG Interconnection Program

Q. Please describe the O&M requirements of the Company's RNG Interconnection Program.

A. The Company's RNG Interconnection program requires non-labor O&M for ongoing maintenance of the interconnection equipment that the Company installs in the amount of approximately $0.111 million in the Rate Year, $0.221 million in Data Year 1, $0.332 million in Data Year 2, and $0.442 million in Data Year 3. The O&M for this program is shown in Exhibit___(RRP-3), Schedule 27. Additionally, 2.4 incremental FTEs are needed in the Rate Year and Data Years for the new Future of Heat Engineering Group that is described in the Future of Heat Panel's testimony. Approximately 1.9 of the Future of Heat FTEs will be dedicated to supporting administration of the Company's RNG Interconnection Program.

B. <u>Changes to O&M Workload</u>

Q. Please describe the non-safety program incremental O&M workload expenses.

A. There are two significant O&M programs where the Company projects incremental expenses in the Rate Year that are not related to safety requirements and initiatives: (i) the Traditional Research and Development ("R&D") Program, and (ii) the Fixed Factor Meter Inspection Program.

Q. Please describe the need for incremental O&M for the Company's Traditional R&D Program.

A. The Traditional Gas R&D Program is for short-term research associated with gas operations, end use, natural gas appliances, supply related storage, safety, and related measures that do not qualify for funding under the Millennium Program. The Historic Test Year reflects minimal spending for Traditional R&D in part due to departmental, budgeting, and personnel changes that resulted in an inadvertent gap in Traditional R&D spending. Traditional R&D may result in modernization of utility operations, reduced costs, improvements to public, customer, and employee safety, and/or environmental benefits; thus, the Company is committed to reinstating its Traditional R&D Program. KEDNY proposes one incremental FTE in the Rate Year and Data Years, and incremental non-labor O&M for Traditional R&D of approximately $1.768 million in the Rate Year, $1.460 million in Data Year 1, $1.346 million in Data Year 2, and $1.374 million in Data Year 3, as shown in Exhibit___(GIOP-8), for R&D projects to address:

- Advanced leak/methane detection equipment
- Auto/remote shut-off valves
- Communications
- Composite materials, liners, and related technologies
- Damage prevention and location technologies

- Gas venting
- Global navigation satellite system (GNSS); geospatial information system (GIS), and asset tracking and traceability
- IMP/IVP technologies
- RMD technologies
- Virtual reality and augmented reality
- Large diameter plastic
- Live main insertion
- Main installation via plow

Q. Please describe the Fixed Factor Meter Inspection Program and the incremental O&M required for this program.

A. Fixed factor meters are used in conjunction with pressure regulated installations and that employ a correction method system, based on a billing multiplier, to adjust the measured volume of gas delivered to standard pressure conditions. KEDNY's Fixed Factor Meter Inspection Program conducts mandated inspections to verify measurement accuracy for billing purposes. Pursuant to a pilot program for extended inspection intervals, KEDNY and KEDLI currently conduct fixed factor inspections annually of approximately one-fifth of the total fixed factor meter population, such that the entire population is inspected over five years. Niagara Mohawk conducts inspections of its entire population of

fixed factor meters annually. To better align fixed factor meter processes and resources across the three New York gas LDCs, KEDNY and KEDLI propose to conduct annual inspections of all fixed factor meters. KEDNY requires three incremental FTEs in the Rate Year and Data Years, specifically union field mechanics, to conduct the incremental inspections.

C. O&M Costs Related to Safety and Reliability Programs

Q. Please briefly describe the incremental O&M requirements for the Company's gas safety programs and initiatives.

A. The Company's safety programs address both new mandated safety inspection requirements (inside service line and plastic fusion inspections), and implementation of gas pipeline safety enhancements. These programs are described in detail in the Gas Safety Panel testimony. The O&M requirements of these programs are summarized in Table 7.

Table 7: Rate Year Gas Safety Programs Incremental O&M and FTEs

Category ($000)	FY 2021 Non-Labor	FTEs
Plastic Fusions In-Process Inspections	$102	0
Inside Service Line Inspections (mandated)	$5,900	3.5
Contractor Safety Inspections	$228	25
Inactive Accounts	--	78
I&R Training Lab	$75	0
I&R Control Line Survey	$1,496	2.5

Pipeline Safety Management	--	10.4
Damage Prevention	$2,384	1
New Materials Test Lab	--	0.5
Single Meter Regulator Inspections	$780	3
Gas Control SOP Training	--	2.3
First Responder Training	$50	2
Field Training	$50	2
Residential Methane Detection	$1,896	--
Enhanced High Emitter Methane Detection (Google Vehicle)	$250	0.4

VII. Impact of Williams' NESE Project on KEDNY's Investment Plan

Q. Please briefly describe the Transco Northeast Supply Enhancement ("NESE") Project.

A. As discussed in the testimony of Company Witness Elizabeth D. Arangio, Williams' NESE Project will expand the existing Transco interstate pipeline to provide an additional 400 million cubic feet of natural gas capacity per day to the New York area by the 2020 winter heating season. National Grid has contracted for 100 percent of the gas pipeline capacity that will be created by the NESE Project. This capacity is critical to meeting increasing demand in KEDNY and KEDLI's service areas. The NESE Project is currently awaiting permit approval by the New York Department of Environmental Conservation ("NYDEC").

Q. What is the impact on the Company's capital investment plan if NESE is not approved?

A. The Company's capital investment plan assumes that the NESE Project will be approved and placed in service. As stated in the testimony of Ms. Arangio, if NESE is not approved, KEDNY will have no choice but to impose a moratorium on new and additional gas service in affected areas of the Company's service territories to maintain system reliability. Moreover, several of the Company's planned Rate Year and Data Years capital programs and projects will be drastically reduced, deferred, or removed altogether. Specifically, the Company's customer connections programs would be suspended, and planned system reinforcements needed to support increasing demand would be reduced by approximately half. Notably, without additional capacity from NESE, the Company will be unable to take its LNG tanks out of service for sufficient time to complete needed repairs; thus, the capital project described above to make critical repairs to the LNG tanks could not proceed.

The MRI project will be completed according to the Company's current plans regardless of whether NESE is approved and in service. While a significant benefit of MRI is to enable optimization of the additional capacity NESE will provide, the MRI project is nearly complete and includes other important reliability benefits independent of NESE, such as increased design day pressure

capability and redundancy to the Brooklyn Backbone. MRI will be completed but will not provide all expected system benefits if NESE is not approved.

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Supplemental Direct Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Date: June 11, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudham.

Q. Is this the same Panel that previously submitted direct testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or "Company") in this proceeding?

A. Yes. The terms defined in the Panel's direct testimony have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of the Panel's supplemental direct testimony?

A. The purpose of the Panel's supplemental direct testimony is to discuss the potential impacts to the gas capital programs described in KEDNY's April 2019 rate filing if the Company is unable to obtain incremental natural gas supplies from Transcontinental Gas Pipe Line, LLC's Northeast Supply Enhancement ("NESE") project, which would provide 400,000 dekatherms/day of incremental gas supply to KEDNY and KEDLI's gas distribution systems to meet forecast demand for natural gas and enhance

reliability for existing customers. In our direct testimony, the Panel identified certain proposals that would be affected if the NESE project were not placed in service by the winter of 2020/2021. As discussed more fully by Company Witness Elizabeth D. Arangio, recent decisions by the New York State Department of Environmental Conservation ("DEC") and the New Jersey Department of Environmental Protection regarding required water permits create uncertainty as to whether and when the NESE project may proceed with permitting and construction. While the Company remains cautiously optimistic the NESE project will go forward, and therefore is not proposing any changes to its April 2019 filing at this time, the Panel addresses the impacts that gas supply constraints would have on KEDNY's forecast capital additions and certain gas operations and safety programs should the NESE project not be available during the proposed rate plan period.

Q. Does the Panel's supplemental direct testimony include any exhibits?

A. Yes. The Panel's supplemental direct testimony includes the following exhibits that were prepared or compiled under our direction and supervision:

(i) Exhibit ___ (GIOP-1S): Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

(ii) Exhibit ___ (GIOP-8S): Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

(iii) Exhibit __ (GIOP-9S): Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

III. Potential Impacts of the NESE Project and Gas Supply Constraints

Q. Did the Panel's direct testimony discuss the possibility that the NESE project might not be available and that this could affect the Company's filing?

A. Yes. While the Company's filing was prepared assuming the NESE project would be completed by the winter of 2020/2021, the Panel's direct testimony (at 118-120) identified the possibility that the NESE project would not be available as planned and explained that KEDNY's capital investment plan would be impacted in that scenario. Specifically, the Panel explained that several planned capital programs and projects could be eliminated, modified, or deferred, including the Customer Connections and Gas System Reinforcement programs. Further, without the planned

capacity from the NESE project, the Company would be unable to proceed with the planned maintenance and upgrades to the Greenpoint LNG facility.

Q. Please explain the impact to the Company's capital investment plan if the NESE project is not approved.

A. Exhibit ___ (GIOP-1S) is an illustrative capital investment plan reflecting the impact in the Rate Year and Data Years should the NESE project not be completed on time, including reductions to the Company's Customer Connections, Reinforcements, and LNG programs. As is stated by Company Witness Arangio, without the incremental capacity expected from the NESE project, KEDNY will not be able to procure sufficient gas supplies to meet the forecast demand from new firm customers, such that connecting additional firm load would jeopardize the Company's ability to provide safe and reliable gas service to existing customers. Therefore, investments in the Customer Connections programs associated with serving new load in these areas would not be necessary, including the installation of new main and the purchase and installation of new meters and associated AMR devices, as well as investments in the low-to moderate income gas conversion program that was proposed by the Company's Future of Heat Panel. Some investments in the Customer Connections programs would still be required to serve non-firm load and to serve new firm load on Staten

Island where capacity is available. Similarly, the Gas System Reinforcements blanket program would be reduced to remove projects no longer needed to serve incremental firm load in KEDNY's service territory.

Notwithstanding the reductions to the blanket Gas System Reinforcement program, one reinforcement and reliability project, the Jamaica Inlet Project would be added to the investment plan to resolve an existing low inlet pressure situation that the incremental NESE capacity was expected to resolve. Without the NESE project, the Company needs this project to maintain reliability to existing gas customers served by the Jamaica Station.

Finally, without the NESE project, the Company is unable to take the Greenpoint LNG Tank 2 out of service to perform planned maintenance and upgrades. The Company intends to perform some engineering work for this project for the purpose of conducting studies and design work for potential limited upgrades to the tank that would not require taking it out of service.

Q. What is the total impact of the capital forecast adjustments described above?

A. The above-described adjustments result in a total reduction (including cost of removal) to the proposed capital investment plan of approximately $46.1

million in the Rate Year, $78.6 million in Data Year 1, $90.2 million in Data Year 2, and $92.5 million in Data Year 3. Table 1 shows the adjustments to each impacted program. Exhibit __ (GIOP-1S) provides a revised version of the total capital forecast showing the adjusted programs.

Table 1: Summary of No-NESE KEDNY Capital Program Adjustments

($ in thousand)	FY20	FY21	FY22	FY23	FY24
Customer Connections					
Install Main	(14,161)	(17,837)	(18,239)	(18,183)	(18,015)
Install Services	(10,096)	(16,437)	(18,592)	(18,861)	(19,133)
Customer Contributions	-	(2,089)	(2,276)	(2,835)	(3,525)
Meter Purchases	-	(1,598)	(1,629)	(1,662)	(1,695)
Install Meter/Regulator	(517)	(992)	(1,065)	(1,085)	(1,106)
Automatic Meter Reading (AMR)	-	(902)	(920)	(938)	(957)
Gas System Reinforcement	(4,538)	(6,746)	(45,837)	(46,591)	(48,023)
LTNYXXXXX - Jamaica Inlet - PM	100	520	9,913	-	-
	(29,212)	**(46,080)**	**(78,646)**	**(90,156)**	**(92,455)**
Mandated					
Contractor Safety Inspections	-	(88)	(267)	(454)	(461)
	-	**(88)**	**(267)**	**(454)**	**(461)**
Reliability					
LTNY13231 - Marine Park Regulator Station - PM	-	(999)	(22,769)	-	-
LNG - Tank 2 Upgrade	-	-	(100)	(1,400)	-

LNG - Tank 1 Upgrade	-	-	(50)	(1,500)	(500)
	-	**(999)**	**(22,919)**	**(2,900)**	**(500)**
Total Capital Including Cost of Removal	**(29,212)**	**(47,167)**	**(101,832)**	**(93,510)**	**(93,416)**
Cost of Removal	**(2,918)**	**(4,712)**	**(10,173)**	**(9,342)**	**(9,332)**
Total Capital (Net of Removal)	**(26,294)**	**(42,455)**	**(91,659)**	**(84,168)**	**(84,083)**

Q. Would the supply constraints affect KEDNY's forecast of capital and O&M expenses for any of the proposed gas safety programs?

A. Yes. The Company's Contractor Inspector Safety Program, which will add contractor inspectors over three years to reach a one-to-one ratio of inspectors to crews, would also be adjusted to reflect reductions to the capital work plan. KEDNY would reduce its total incremental FTE proposal by two inspectors. The revised FTE forecast is provided in Exhibit __ (GIOP-9S). Associated reductions to the capital forecast and non-labor O&M forecast for the Contractor Inspector Program are set forth in Exhibit __ (GIOP-1S) and Exhibit (GIOP-8S), respectively.

Q. Does KEDNY plan to continue to provide updates in this proceeding concerning the status of the NESE project?

A. Yes. KEDNY is cautiously optimistic that the NESE project will be approved. As the project's status is uncertain, the Company will update Staff and the other parties throughout this proceeding and will supplement the record as further developments occur.

IV. Conclusion

Q. Does that conclude the Panel's supplemental direct testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Corrections and Updates Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Dated: July 3, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel ("GIOP").

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudhan. This is the same Panel that provided direct testimony as part of the Company's April 30, 2019 filing. The terms defined in the Panel's direct testimony have the same meanings here.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The purpose of the Panel's testimony is to identify and explain the following corrections and updates:

- Updates to the direct capital plan – The Panel identifies the following updates and corrections to the Company's gas capital programs and projects: (i) updates to the City/State Construction ("CSC") programs to reflect changes in NYCDDC's projected budgets and projects and to provide recently available FY 2019 O&M data to further support the Company's CSC Program O&M reconciliation mechanism proposal; (ii) updates to the Metropolitan Reliability Infrastructure ("MRI") project schedule and phase estimates, including an overall decrease in remaining project costs; (iii) updates to the Greenpoint LNG Salt Water Pump House

Project and Vaporizers 7 & 8 schedules that defer certain of the projects' costs from the Rate Year to the Data Years; and (iv) a correction to properly reflect the Gas Control Training Simulator Training Project as a Service Company asset and removing these costs from KEDNY's direct capital investment plan.

- Operator Qualification Program ("OQ Program") – The Panel briefly summarizes the Company's plans to implement the recommendations of Staff's "Operator Qualification White Paper" (dated February 2019) that was noticed for comment by the Commission in March of 2019 in Cases 14-G-0212 and 17-G-0318. The Panel updates the Company's capital investment and O&M plans to include implementation costs that are expected to be incurred during the Rate Year and Data Years.

- Updates to the indirect capital plan – The Panel describes general updates to the Company's facilities forecasts and to the forecasts for Service Company facilities projects that were identified in the Company's direct testimony.

- Updates to the O&M forecasts – The Panel updates its O&M forecast to reflect (i) a decrease in Rate Year costs to complete the Local Law 30

("LL30") Compliance Program and (ii) updates to the Company's incremental FTE hiring plans and position titles.

Q. Does the Panel's testimony also address the June 11, 2019 supplemental filing?

A. No. The supplemental filing was made to illustrate the potential impacts to the April 30 filing if KEDNY is unable to obtain incremental natural gas supplies from the NESE project. At this time, the Company remains optimistic that the NESE project will be approved. Therefore, all corrections and/or updates pertain to the capital forecast set forth in the April 30 base rate filing. The Company will continue to update the parties on the status of the NESE project and implications, if any, to this rate filing.

Q. Is the Panel sponsoring any exhibits as part of its corrections and updates filing?

A. Yes. The Panel sponsors the following exhibits that were prepared by one or more members of the Panel or under their supervision and direction:

- Exhibit ___ (GIOP-1CU): Revised Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3

- Exhibit __ (GIOP-6CU): Revised Demonstration of Variability in City/State Construction O&M
- Exhibit __ (GIOP-7CU): Revised Projected Indirect Capital Expenditures (Supply Chain, Fleet, and Facilities) for the Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3
- Exhibit ___ (GIOP-8CU): Revised Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3.
- Exhibit __ (GIOP-9CU): Revised Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3.

Additionally, the Company's responses to information requests ("IRs") that are relevant to the corrections and updates discussed below are included in Appendix A.

II. Corrections and Updates

A. Updates to Direct Capital Plan

Q. Please summarize the updates to the Company's direct capital investment plan and Exhibit __ (GIOP-1).

A. KEDNY's capital plan has been updated to reflect scheduling and/or budget changes in the Rate Year and Data Years for the following capital programs:

(i) the CSC Reimbursable, Non-Reimbursable, Reimbursements, and SE856 Phase 2 (Transmission and Distribution) Programs; (ii) the MRI Project; and (iii) the Greenpoint LNG Saltwater Pump House and Vaporizers 7 & 8 Projects. The Company is also removing allocated costs of the Gas Control Training Simulator Project from the capital plan because those costs will instead be reflected in the revenue requirement as a Service Company asset. The revised capital forecast is provided as Exhibit __ (GIOP-1CU), which highlights the changes from the plan included with the Panel's direct testimony. The update to each program is discussed in more detail below.

Q. What is the impact of these changes to the Company's forecast capital expenditures in the Rate Year and Data Years?

A. The Company has updated its capital forecast to reflect a net decrease in the Rate Year of approximately $96.2 million, a net increase in Data Year 1 of approximately $83 million, a net increase in Data Year 2 of approximately $62.7 million, and a net increase in Data Year 3 of approximately $36 million. These figures are inclusive of cost of removal ("COR").

Table 1: Direct Capital Forecast Updates Rate Year to Data Year 3 (including COR)

($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	Total
Original Forecast	849,153	952,053	934,990	931,812	1,008,451	4,676,459
Revised Forecast	861,801	855,873	1,017,970	994,495	1,044,442	4,774,582
Variance	12,648	(96,180)	82,980	62,683	35,991	98,123

The cumulative impact of these changes over the five-year period is a $98 million increase to KEDNY's capital forecast. The largest increases to KEDNY's direct capital forecast result from a reclassification of approximately $522 million of reimbursable CSC program work to non-reimbursable work, and the associated decrease in CSC reimbursements, driven by updated information from DDC regarding its currently-expected project designs and schedules.

The Revenue Requirement Panel's Corrections and Updates testimony addresses the revenue requirement impact of the Company's revised capital forecast. Additionally, the impacts of revised in-service dates for some of the major projects, including facilities projects, are included in the revised revenue requirements presented in Exhibit ___ (RRP-7CU) Schedule 1.

i. *CSC Program Changes*

Q. Please describe the changes to the CSC/Public Works-Non-Reimbursable, CSC/Public Works-Reimbursable, and CSC/Public Works-Reimbursements line items shown on Exhibit __ (GIOP-1CU).

A. The changes to the CSC Reimbursable, Non-Reimbursable, and Reimbursements items are described in the Company's responses to IR Nos. DPS-352 and DPS-402, which are included in Appendix A. In summary, the DDC has provided additional information to the Company regarding expected expenditures, designs, and plans for DDC projects that are anticipated to occur during the Rate Year and Data Years. As is typical, DDC's infrastructure project portfolio is dynamic, and there is currently no established project list or final work plan for the Rate Year and Data Years; however, information is exchanged monthly and the most recent update reflects a non-reimbursable to reimbursable split that is more aligned with actual historic work. These updates require a shift in the Company's forecasts of reimbursable category of CSC work to the non-reimbursable category, as shown on Exhibit __ (GIOP-1CU) and summarized in Table 2, below.

Table 2: CSC Reimbursable to Non-Reimbursable and Reimbursements Revisions (FY 2020 to Data Year 3)

Non-Reimbursable ($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024
Original Estimate	23,226	23,959	24,353	24,793	22,922
Revised Estimate	122,011	125,898	132,501	134,915	125,596
Reimbursable ($000)					
Original Estimate	247,909	255,813	270,094	275,018	256,181
Revised Estimate	149,125	153,875	161,946	164,896	153,507
Reimbursements ($000)					
Original Estimate	(43,380)	(57,842)	(60,807)	(63,063)	(64,903)
Revised Estimate	(42,285)	(33,400)	(34,103)	(35,789)	(36,453)

Q. Please describe the changes to the CSC SE856 Phase 2 Project.

A. As is also described in the Company's response to IR No. DPS-352, the Company received an update from DDC that changes the project design and schedule for the SE856 Phase 2 Project, for both the transmission and distribution work. The updated project scope and timeline: (i) increases the Company's work scope and extends the transmission work's project schedule from a November 2020 in-service date to a November of 2023 in-service date, and (ii) and extends the distribution work's project schedule in-service date one year from March 2021 to March 2022. The Company's revised forecasts reflect these changes and are summarized in Table 3, below. The DDC's design and schedule for this project is still preliminary and is subject to change.

Table 3: CSC Updated SE856 Phase 2 Forecast (FY 2020 to Data Year 3)

SE856 Phase 2 Transmission ($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024
Original Estimate	10,550	60,000	-	-	-
Revised Estimate	1,609	26,590	26,980	27,590	4,000
SE856 Phase 2 Distribution ($000)					
Original Estimate	2,500	41,900	2,000	-	-
Revised Estimate	-	14,400	14,400	-	-

ii. *MRI Project Updates*

Q. Please describe the changes to the MRI Project.

A. The Company's response to IR No. DPS-556 (Amended) describes the updates to the MRI project. In summary, refinements in the project estimates for the remaining forecasts and a re-evaluation of capital expenditures to date and work needed to achieve operation at transmission pressure by the in-service date of October 2020, and a recently-completed re-routing design change resulted in the following updates:

- The project schedule has been accelerated, resulting in reallocation of $25 million from the Rate Year to FY 2020.
- The remaining project forecast has been reduced by approximately $4.9 million; the reduction is to the Phase 5 (FY 2022) forecast.

- Refinement of Phases 4 and 5 of the project such that $75 million in costs originally included in Exhibit __ (GIOP-1) in Phase 4 (FY 2021) are now split into $35.426 million for Phase 4 and $39.574 million for Phase 5 in FY 2021.

iii. ***Greenpoint LNG Project Updates***

Q. Please describe the changes to Greenpoint LNG projects.

A. Certain work for two Greenpoint LNG projects, the Salt Water Pump House Project and the Vaporizers 7 & 8 Project have been deferred from the Rate Year resulting from a re-evaluation of project schedules and based in part on contractor resource availability. The Salt Water Pump House Rate Year forecast is decreased by approximately $25.4 million, and the Vaporizers 7 & 8 Project Rate Year forecast is decreased by approximately $9.7 million.

iv. ***Gas Control Training Simulator Program***

Q. Please describe the correction to the Gas System Reliability – Gas Control (Training Simulator) forecast.

A. KEDNY's portion of costs for the Gas Control Training Simulator Program was included in the capital investment plan; however, this program benefits all National Grid gas distribution companies in New York and New England, with

total projects costs shared. Therefore, the program is properly reflected as a Service Company asset in the revenue requirement instead of in KEDNY's capital investment plan.

B. OQ Program

Q. Please summarize the Company's plan to implement Staff's proposed OQ requirements.

A. As stated in the Panel's direct testimony at pages 24 to 25, the Company did not include in the April 30, 2019 rate filing costs to implement Staff's proposed new requirements for gas company operator qualification programs (*see* "Notice of Proposed Rulemaking," issued March 27, 2018 in Cases 14-G-0212 and 17-G-0318) due to uncertainty regarding the ultimate outcome of the rulemaking proceeding and due to the timing of the proposal that did not allow the Company to complete its analysis and develop implementation costs in time for inclusion in the April 30 filing.

While the Commission has yet to issue an Order, the Company has developed a plan to proactively begin implementation of enhanced operator qualification best practices that are recommended by Staff's proposal and are likely to become compliance requirements during the term of the rate plan. Proactive

implementation provides safety and reliability benefits to customers in the nearer term.

Q. What are the implementation costs?

A. The Company adjusted the direct capital investment plan by approximately $0.9 million in the Rate Year and $0.5 million in each of Data Years 1, 2, and 3, to reflect costs of Industrial Training Services ("ITS") systems, records maintenance, equipment such as eWallet cards used to verify OQ, and OQ test security. The Company also adjusted the O&M plan to include non-labor implementation costs of approximately $0.6 million in the Rate Year and $0.7 million in each of Data Years 1, 2, and 3, as well as 11 incremental FTE Technical Inspectors to maintain OQ records, conduct verifications including field inspections, conduct compliance reviews, and implement other program improvements, as required.

Q. Could the Commission's final OQ rules necessitate implementation costs that have not been included in the Company's investment plan?

A. Yes. Until the rulemaking is final, the full extent of new OQ requirements remains unknown. To the extent the final order in the rulemaking proceeding requires additional incremental investments, the Company reiterates the

proposal set forth in its direct testimony to recover prudent additional costs through the Company's GSRS.

C. Updates to Indirect Capital Plan

Q. Please describe the updates to the indirect capital investment plan.

A. KEDNY's facilities program forecasts have been updated to reflect more refined cost estimates and re-evaluation of project schedules. Updates are shown in Exhibit __ (GIOP-7CU), in Table 4, below and are summarized as follows:

- Facilities Base Spend: The FY 2020 forecast has been increased by $0.7 million to reflect a more refined cost estimate and unanticipated replacements of various assets (*e.g.,* HVAC equipment, sidewalks, lighting) that were inadvertently excluded from the original forecast.
- Pitkin Customer Office Remodeling: The project has been shifted from FY 2020 to the Rate Year.
- NYC Training Centers: The project has been shifted from FY 2020 to the Rate Year and Data Year 1.
- Canarsie Parking: The total project cost is reduced by $0.3 million and is re-allocated between FY 2020 and the Rate Year due to more work completed than was planned in FY 2019.

Table 4: KEDNY Total Facilities Program Forecast Updates

FY 2020 to Data Year 3 (including COR)

($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	Total
Original Forecast	21,427	8,575	5,850	5,850	4,600	46,302
Revised Forecast	7,657	18,425	10,250	5,850	4,600	46,782
Variance	(13,770)	9,850	4,400	0	0	480

Q. Please describe the updates to the Service Company facilities programs that were described in the Panel's direct testimony.

A. Updates to the Service Company facilities projects are shown in Exhibit___ (RRP-11CU), Workpapers to Exhibit __ (RRP-3CU), Schedule 9, Workpapers 4, 7, 10, and 13. Updated forecasts result in overall increases to the Service Company facilities projects, but certain projects have been deferred to the Data Years to more evenly spread costs over the period of the rate plan. The overall capital investment increases by approximately $7.8 million in the Rate Year, $5.0 million in Data Year 1, $8.4 million in Data Year 2, and $0.3 million in Data Year 3. This is mainly due to more refined cost estimates for the Brooklyn Office relocation (approximately $1.9 million), Hicksville Office relocation (approximately $1.7 million), and the Reservoir Woods Office project ($4.6 million in FY 2020). Project spending for the IM Warehouse has been deferred to Data Year 1 and certain costs related to the Long Island Office have been deferred to Data Year 2. An adjustment was made to remove costs

to maintain the existing Hicksville Office site in Data Years 2 and 3, following the office relocation. These costs were inadvertently left in the forecasts for those years in the Companies' April 2019 filing.

D. Updates to O&M forecasts

Q. Please describe the updates to the O&M forecasts.

A. The corrections and updates to the Company's O&M forecast are shown in Exhibit __ (GIOP-8CU), and Exhibit __ (GIOP-9CU) and are summarized as follows:

- LL30 Compliance Program: The costs to complete LL30 compliance in the Rate Year have been reduced by approximately $7 million because the Company accelerated work and was able to address more installations than planned in FY 2019. The correction to the LL30 forecast is also described in the Company's response to IR DPS-236 Supplemental.
- D&Rs Related to Capital: The forecast has been updated to correct an inadvertent double-count of costs associated with replacement of HDPE services. The correction is described in the Company's responses to IR Nos. DPS-499 and DPS-500.
- Incremental FTEs: Updates have been made to the hire dates for certain incremental FTEs to correct mis-stated hire dates and to reflect a more realistic hiring schedule. Additionally, corrections have been made to FTE

position titles that were inadvertently mis-stated. These corrections have been made in accordance with the response to IR No. DPS-374. These changes are reflected in Exhibit __ (GIOP-9CU) and in Exhibit__(RRP-11CU) Workpapers to Exhibit (RRP-3CU), Schedule 27, Workpaper 2

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Date: April 2019

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I. Introduction and Qualifications

Q. Please introduce the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudhan.

Q. Mr. Turrini, please state your name and business address.

A. My name is Ross W. Turrini. My business address is 25 Hub Drive, Melville, New York 11747.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company"), a subsidiary of National Grid USA ("National Grid"), as the Group and US Chief Gas Engineer. I oversee approximately 800 employees and more than $9.5 billion of gas infrastructure assets serving over 3.6 million customers in New York, Massachusetts, and Rhode Island.

National Grid owns and operates three gas distribution companies in New York that provide retail gas service to more than 2.4 million customers in three service territories: Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk") serves areas of eastern and central New York; The

Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") serves Brooklyn, Staten Island, and parts of Queens in New York City; and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI" or "Company) serves customers on Long Island and the Rockaway Peninsula in Queens. I am responsible for all aspects of the performance of National Grid's New York gas networks, including emergency response, gas engineering, construction activities, operation and maintenance of gas transmission and distribution facilities, gas network strategy, including the gas capital investment strategy, and gas pipeline safety management.

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Civil Engineering from the United States Military Academy at West Point in 1985. I have worked for National Grid and its predecessor companies (the Long Island Lighting Company and KeySpan Corporation) for 25 years in various roles in engineering, operations, and procurement. Prior to joining National Grid, I spent five years as an Officer in the United States Army Corps of Engineers and three years in engineering and construction roles at Brown & Root Services Corporation, an international engineering, procurement, and construction company.

Q. Have you previously testified before the New York Public Service Commission ("Commission")?

A. Yes. I testified before the Commission on behalf of KEDNY and KEDLI in Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases"). I also testified most recently on behalf of Niagara Mohawk in Case 17-G-0239 (the "2017 NMPC Rate Case").

Q. Mr. Graham, please state your full name and business address.

A. My name is Timothy S. Graham. My business address is 300 Erie Boulevard West, Syracuse, New York 13212.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President for Gas Field Operations. I oversee approximately 2,300 employees and am responsible for the maintenance of KEDNY, KEDLI, and Niagara Mohawk's gas distribution systems. This includes gas metering, construction, and field activities to carry out mandated safety programs, leak response, leak repair, instrumentation and regulation, damage prevention, and inspection programs.

Q. Please describe your educational background and experience.

A. I received a Bachelor of Engineering in Electrical Engineering from the State University of New York at Buffalo. I also have a Master's in Business Administration from Lemoyne College in Syracuse, New York. I have worked for National Grid and its predecessor companies (Niagara Mohawk) for 32 years in various roles in engineering and operations.

Q. Ms. Hon, please state your full name and business address.

A. My name is Caroline Hon. My business address is 40 Sylvan Road, Waltham, Massachusetts, 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President for Gas Resource Management. I oversee approximately 400 employees and am responsible for the planning, scheduling, dispatch, dependency management, and clerical support for the capital, maintenance, and repair work across National Grid's gas network.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Science in Chemical Engineering from the Massachusetts Institute of Technology in 2001. I have worked for National Grid for nine years in various roles in operations, finance, and shared services. Prior to joining National Grid, I worked as a consultant for eight years in Accenture's Utilities Transmission and Distribution practice, focused on work management process optimization and system implementation.

Q. Ms. Madhusudhan, please state your name and business address.

A. My name is Srividya Madhusudhan. My business address is 40 Sylvan Road, Waltham, Massachusetts, 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President of Property and Fleet. I oversee all the Facilities, Fleet and Property activity in support of National Grid Operating companies including KEDNY, KEDLI, and Niagara Mohawk. This includes buying, selling, leasing, licensing, administration of rights of way, management of rights of way and maintenance of fleet. I also oversee development of the facilities and fleet

plans to support KEDNY and KEDLI's gas businesses and related capital forecasts.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Law from Chennai, India in 1991. I also have a Masters in Environmental Management from Rensselear Polytechnic Institute and an MBA from the University of Connecticut. I have worked at National Grid since May of 2017 prior to which I worked at Eversource Energy for 18 years in various roles in environmental, Operations and Real Estate and Facilities.

II. Purpose of Testimony

Q. What is the purpose of the Gas Infrastructure and Operations Panel's testimony?

A. The purpose of the Panel's testimony is to provide a forecast of the capital investments of the Company for the twelve-month period ending March 31, 2021 ("Rate Year"), and the three subsequent twelve-month periods ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (Data Year 1, Data Year 2, and Data Year 3 are collectively referred to as the "Data Years"). The Panel also presents a

forecast for the fifteen-month period ("Gap Period") between December 31, 2018, which is the end of the Company's Historic Test Year, and the beginning of the Rate Year.

The Panel discusses direct capital expenditures that will (i) increase the safety and reliability of the Company's gas network, (ii) modernize the Company's gas transmission and distribution infrastructure to reduce emissions while continuing to meet customer needs, and (iii) enhance resiliency and the Company's ability to respond to future weather events. The Panel will also discuss the Company's practices and policies for maximizing the efficiency of its capital delivery program from planning and budgeting through the completion of construction.

The Panel's testimony provides an overview of the significant projects in the Company's capital plan, including retirement of leak prone pipe ("LPP") and new safety programs to identify and address system risks. The Panel's testimony also presents an overview of the Company's pipeline integrity and reliability programs that will improve the overall safety and reliability of the Company's gas system, and will also address recently enacted, as well as pending, pipeline safety regulations administered by the U.S. Department of Transportation ("DOT"), Pipeline and Hazardous Materials Safety Administration ("PHMSA"). The Panel discusses the Company's plans and

infrastructure projects necessary to ensure that the Company can reliably serve its existing customers as well as forecast demand for new customer connections within the Company's operating territory.

The Panel also presents the Company's indirect capital investments, including supply chain, inventory management/investment recovery ("IM/IR"), and fleet investments. The Company is also proposing significant investments in facilities during the Rate Year and Data Years that are necessary to: support the Company's O&M and gas safety programs; make needed upgrades to aging facilities; and create more modern and efficient customer support and employee work spaces.

Q. Does the Panel's testimony also address the Company's operations and maintenance ("O&M") programs?

A. Yes. In addition to capital investments in gas infrastructure, the Panel describes incremental labor (full time equivalent positions or "FTEs") and non-labor O&M expenses that the Company proposes in the Rate Year and Data Years, the costs of which are not fully reflected in the twelve-month period beginning January 1, 2018 and ending December 31, 2018 ("Historic Test Year"). These expenses represent known and measurable changes from Historic Test Year expenses that are necessary to (i) improve system safety

and reliability, (ii) address new and emerging safety regulations, (iii) enhance customer service, and (iv) support the Company's capital investments. The Panel will also discuss the Company's staffing plan for the proposed new FTEs.

Q. Does the Panel address any other topics?

A. Yes. The Panel discusses the Williams Transco Northeast Supply Enhancement ("NESE") project and the importance of NESE in enabling KEDLI's proposed capital investment plan.

Q. Does the Panel sponsor any exhibits as part of its testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared under its direction and supervision:

Exhibit __ (GIOP-1): Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3.

Exhibit __ (GIOP-2): Charts Demonstrating Components of Unit Cost Increases

Exhibit __ (GIOP-3): Gas Safety and Reliability Surcharge Including Unit Cost Incentive Proposals

Exhibit __ (GIOP-4): Chart Summarizing Projected Leak Rates for LPP for Various Main Replacement Strategies.

Exhibit __ (GIOP-5): Data Sheets for Significant Capital Programs. This exhibit includes summaries of the Company's significant capital projects/programs.

Exhibit __ (GIOP-6): Projected Indirect Capital Expenditures (Supply Chain, Fleet, and Facilities) for the Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3, and Data Sheets for Significant Facilities Investments.

Exhibit __ (GIOP-7): Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3.

Exhibit __ (GIOP-8): Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3.

The capital expenditures presented throughout the testimony and in the exhibits include cost of removal ("COR"), as applicable.

Q. How is the Panel's testimony organized?

A. The testimony is organized into the following sections:

Sections I and II are introductory sections outlining the Panel's testimony.

Section III provides an overview of the Company's capital investment and O&M program priorities and objectives, strategy to address increasing costs, and significant capital programs, such as the retirement of leak prone mains and services, and other key capital investments in gas pipeline safety and reliability.

Section IV provides details on the Company's proposed direct capital investment program for the Rate and Data Years, including the Company's spending rationales, categories of capital investment, and specific work activities within each category.

Section V describes the Company's indirect capital investment plan, which includes facilities, fleet, and supply chain (IM/IR) investments.

Section VI describes the Company's O&M programs, including those necessary to carry out the Company's proposed capital programs and those targeted at current and emerging safety regulations and implementation of gas pipeline-safety best practices. With respect to gas pipeline safety, several of the O&M expenditures presented in the Panel's testimony and set forth in Exhibit __ (GIOP-7) and Exhibit __ (GIOP-8) are discussed in more detail in KEDNY and KEDLI's Gas Safety Panel testimony.

Section VII describes the NESE gas supply project, its relationship to KEDLI's capital investment plan, and the important safety and reliability projects and programs that will not proceed in the absence of NESE.

III. Capital and O&M Plan Objectives and Priorities

A. Plan Objectives

Q. Please describe the Company's overall objective of its infrastructure and operations programs.

A. The Company designs its gas infrastructure and operations programs to provide safe and reliable gas delivery service to customers at reasonable costs. Over the last several years, the Company has significantly increased investment to modernize and enhance the resiliency of its gas assets.

Significant capital investment over the next several years is required to ensure that the gas system continues to operate safely while meeting the demands of customers and to enable further modernization of the Company's facilities and operations in a manner that achieves meaningful methane emissions reductions in furtherance of the Company's Future of Heat strategy. The proposed plan includes capital and O&M spending to meet both customer needs and satisfy state and federal regulatory requirements and goals, including retirement of LPP, gas pipeline safety enhancements, and deployment of non-pipeline alternatives. While significant investments are required to meet these needs and objectives, in developing its capital and O&M plans, the Company balanced the need for spending to achieve safety

and service objectives with the need to manage costs and minimize impacts on customer rates.

Q. What are the Company's projected direct and indirect capital forecasts?

A. A summary of the Company's total capital plan forecast for the Rate Year and Data Years is provided in Table 1.

Table 1: Indirect and Direct Capital Plan (Includes COR)

($000)	FY 2021	FY 2022	FY 2023	FY 2024
Direct CapEx	540,023	549,970	578,938	623,162
Indirect CapEx	46,111	7,098	6,587	3,860
Total Capital Plan	**586,134**	**557,068**	**585,524**	**627,022**

Q. Why is the scale of Company's capital investment plan increasing?

A. Several developments have required that KEDLI and other natural gas distribution utilities increase their annual capital expenditures. Notably, pipeline safety incidents, such as the tragic events in San Bruno, California, Allentown, Pennsylvania, and East Harlem, New York, and the more recent incident in Merrimack Valley, Massachusetts, have appropriately increased focus on pipeline safety and the need to carefully monitor and replace aging pipeline infrastructure. Weather events such as Superstorm Sandy and the Polar Vortex, and the expectation that similar events will occur in the future, require the Company to find ways to protect its facilities from severe weather.

In response to these events, the industry has adopted new safety requirements and is implementing best practices to modernize and protect gas facilities. For example, since the 2016 KEDNY and KEDLI Rate Cases, (i) new Commission safety regulations and directives now require additional safety inspections of inside service lines and plastic fusions; (ii) PHMSA is implementing new safety requirements related to transmission system integrity management and verification this year; (iii) and the New York State Legislature is currently considering a bill that would require licensed professional engineers ("PEs") to review and certify gas facility design plans. The Commission and utility industry review of recent incidents has also resulted in recommendations for safety best practices, including overpressure protection improvements and enhanced contractor oversight. These mandated and recommended improvements require significant additional investments and compliance resources.

Q. How will the Company support the increased level of capital investment?

A. As the Company developed plans to modernize its gas assets, it also began to build and enhance its operations, engineering, resource planning, work management, and quality control organizations and capabilities to deliver increasing levels of capital investment. The Company will continue to develop these capabilities in the Rate Year and Data Years by adding

incremental labor resources to execute the capital plan and support the increased operations workload (discussed in Section VI). The capital investment plan includes additional field training resources, facilities, and tools to enhance the development of the internal work force.

Regarding contractor resources, over the past several years, the Company developed a procurement strategy that supports sustainable growth in qualified contractors to meet the work plan increases. To ensure adequate levels of qualified, skilled labor and the challenges around developing qualified contractors, the Company has worked to:

- Establish longer term contracts to enable contractors to plan and invest in hiring, training, facilities, and equipment to meet the Company's construction needs
- Provide greater work plan visibility to contractors on forecast crew requirements which will enable them to develop the required capacity
- Manage the work plan to limit seasonal variability to support a stable contractor workforce and promote worker retention
- Continue partnerships with community colleges, trade schools, and veterans group to attract and develop new sources of skilled labor to build the workforce

Q. Does the Company require additional personnel in the Rate Year and Data Years to execute its capital and O&M programs?

A. Yes. The Company forecasts the need for an additional 135.3 FTE positions by Data Year 3 to support the additional capital investment, increasing O&M workload due to new and pending regulatory requirements, new programs discussed below, and the gas safety programs presented in the Gas Safety Panel Testimony. These FTEs include positions in contractor oversight, gas pipeline safety management, corrosion control, field operations, instrumentation and regulation ("I&R"), gas control, gas system engineering, and gas asset management. The cost of these FTEs will be charged to both capital and O&M programs based on the job function and nature of the work. Exhibit __ (GIOP-8) identifies the incremental FTE positions by function. KEDLI's Revenue Requirements Panel presents the labor O&M associated with these FTEs.

B. <u>Cost Pressures and Cost Mitigation</u>

Q. What explains the significant increases reflected in the investment plan?

A. The Company's investment plan reflects increases in both the scope of work (units) and the costs elements to perform this work (unit costs). As discussed above, there are several factors that are collectively increasing the size of the Company's investment plan over the next several years, including incremental

work driven by federal and state regulatory mandates and the need to upgrade aged infrastructure and replace dated facilities. At the same time, the Company is experiencing increasing unit costs resulting from new work requirements and enhanced safety practices, and the rising costs of materials and contractors, in part due to increasing demands for these resources across the industry.

Indeed, a primary driver of increases in the capital plan is significantly higher unit costs for main installation work across multiple mandated programs, including the proactive and reactive main replacement, customer connections, corrosion, and city/state construction ("CSC") programs.

Q. What factors are contributing to the Company's unit cost increases?

A. Exhibit __ (GIOP-2) shows the components that have contributed to the main installation unit cost increases. The factors contributing to an increase in the Company's main replacement unit cost include:

- Restoration Costs: increasing municipal restoration requirements and significantly increasing costs for dumping excavation spoils (more stringent environmental regulations has led to a reduction in dumping locations and higher costs)

- Services: work scope includes mains with higher service density (more services per mile of main) than in prior years
- Test Holes: process safety improvements requiring stricter procedures related to test holes
- Cross Bores: process safety improvements requiring more camera scoping during HDD in sewer districts

Q. What is the Company doing to moderate the size of its investment plan?

A. As set forth in more detail in Section IV, the Company develops its proposed capital and O&M investment plans through close coordination among the Asset Management, Field Operations, and Investment and Resource Planning teams. Major programs and investments undergo a robust options analysis to identify the most cost-effective solution. The alternatives considered and project benefit information for major capital investments are set forth in the data sheets included in Exhibit __ (GIOP-5). Additionally, the Company recently created a Capital Delivery organization tasked with enhancing the Company's ability to develop and execute its capital plan more efficiently for complex projects (Phase 1) and eventually for the entire capital program (Phase 2). As part of this effort, which is currently in the first year of Phase 1, Capital Delivery is working to identify efficiencies and opportunities for better ways of working that may reduce costs. Capital Delivery is also in the

process of identifying and implementing targeted efficiency initiatives in support of National Grid's aspirational savings program, known as "Accelerate." The Revenue Requirements Panel describes the Accelerate Program.

Q. Please describe how the Company reflected Capital Delivery initiatives in the capital investment plans.

A. In recognition that efficiencies are expected over the term of the Rate Year and Data Years, the Company is making reasonable budget adjustments to the capital plan to reflect savings that have been identified to date, while maintaining a mechanism to return to customers the benefits of additional efficiencies achieved over the term of the rate plan. To that end, the Company proposes to reflect capital efficiencies in three ways:

- Targeted Efficiencies Adjustment: The Company included a line-item reduction to the overall capital investment plan in the amount of $0.9 million in the Rate Year, $1.2 million in Data Year 1, $1.0 million in Data Year 2, and $1.0 million in Data Year 3 to reflect savings the Company believes are reasonably achievable across a portfolio of capital projects. The line-item reduction is shown on Exhibit __ (GIOP-1) as "Complex Capital Delivery Initiative – Savings"). The calculation of this adjustment is discussed in more detail below.

- Complex Project Forecast Adjustments: The Company identified specific savings opportunities for two complex construction projects that are proceeding through the Capital Delivery process, the Southeast Suffolk Infrastructure Project ("Southeast Suffolk Project") and Phase 3 of the Northwest Nassau Transmission Main and Control Valve Project ("Northwest Nassau Project"). The capital investment plan reflects savings of approximately $20.0 million for the Southeast Suffolk Project and approximately $6.0 million for the Northwest Nassau Project.

- Net Plant Capital Tracker: The Company proposes to continue the net plant capital tracker mechanism that was adopted in the 2016 KEDNY and KEDLI Rate Cases as part of a multi-year rate plan. Any capital efficiencies realized through the Company's Accelerate Program will be returned to customers through the operation of the downward-only capital tracker.

Q. Please explain how the Company calculated the Complex Capital Delivery Initiative – Savings line item reduction to the capital plan.

A. The Company identified more than $4 million of potential savings that it reasonably expects to achieve across a portfolio of complex capital projects that are currently moving through Capital Delivery's project stages. The projects are pressure regulation station re-builds, LNG projects, IMP, IVP, and transmission station integrity projects and system reinforcements. Estimated savings are not attributable to specific projects but are representative based on recent experience across the portfolio. The Company applied a five percent reduction to the five-year forecasts for these projects to derive the savings estimate for each year.

Q. What is the basis for the complex project forecast adjustments to the Southeast Suffolk and Northwest Nassau Projects?

A. For the Southeast Suffolk Project, Capital Delivery identified an alternate project design and route that resulted in elimination of one phase of the project (reducing the project from three to two phases) and savings of approximately $20.0 million. For Phase 3 of the Northwest Nassau Project, Capital Delivery identified changes in work procedures and methods that resulted in savings of approximately $6.0 million, including eliminating sheeting for some sections of trenching, use of a flowable (cement and grout) backfill method that is more cost effective than traditional backfilling, and utilizing pipe staging and prefabrication.

Q. Is the Company proposing unit cost productivity incentives to encourage additional efficiencies?

A. Yes. The Company proposes to continue the unit cost productivity incentive for the Proactive Main Replacement Program that was adopted in the 2016 KEDNY and KEDLI Rate Cases. The Company proposes a similar incentive for the Customer Connections program, which is also a unit-cost driven program. The Company proposes to recover both incentives through the existing Gas Safety and Reliability Surcharge ("GSRS"), consistent with the current unit cost incentive. The GSRS, including the proposed incentive mechanisms, is presented in Exhibit __ (GIOP- 3). These productivity incentives align with the Company's efforts to identify and realize efficiency benefits through its Accelerate Program.

Q. How are the unit cost targets for the productivity incentives derived?

A. The unit cost targets are based on the Company's total costs for main and service replacements, including the costs of related safety programs that will be charged to the work orders for main replacements. For purposes of presentation of the capital investment plan in the rate case, Exhibit __ (GIOP-1) presents the costs of these additional programs, such as plastic fusion inspections, as separate line items for clarity of costs attributed to each program. However, in the interest of setting productivity incentives that

reflect all elements of unit costs and are easily tracked and reported using program work orders, unit cost targets must include a full allocation of costs for the related safety programs to reflect the total costs for main and service replacements.

As shown on page 11 of Exhibit __ (GIOP-3), unit costs for the LPP productivity incentive include the costs for the Proactive Main Replacement Program, plus allocated costs for plastic fusion inspections, low pressure main valve installations, and contractor safety inspections. As shown on page 12 of Exhibit __ (GIOP-3), unit costs for the Customer Connections – Mains productivity incentive include the costs of the Customer Connections – Install Main program, plus allocated costs for plastic fusion inspections and contractor safety inspections.

Q. Are there emerging safety and regulatory changes that are not included in the Company's investment plan that may further increase capital and O&M requirements during the Rate Year and Data Years?

A. Yes. The Company's investment plan does not include costs to comply with: (i) the anticipated legislation to require professional engineering ("PE") certification of engineering designs for gas facilities, and (ii) the Commission's proposed new requirements for gas company operator

qualification programs (see "Notice of Proposed Rulemaking," issued March 27, 2018 in Cases 14-G-0212 and 17-G-0318). There is uncertainty regarding when these proposals might be adopted as final, whether there will be changes to the scope and requirements prior to final adoption, and the timing of implementation following final adoption. Even if these new requirements are adopted without substantive changes to the current proposals, these items were introduced too recently to allow the Company to complete the thorough analysis of estimated implementation costs necessary for inclusion in this rate filing.

Q. What is the Company's proposal regarding the anticipated PE and operator qualification requirements?

A. To the extent the PE and operator qualifications requirements are finalized during the pendency of this rate proceeding, the Company intends to update its filing to include the costs of these mandated programs. If these new requirements are finalized after rates are established in this case, the Company proposes to recover prudent costs to implement the PE and operator qualification requirements through the Company's GSRS. Under either scenario, the unit cost targets for the productivity incentive would need to be updated to address these new costs.

C. Significant Capital Programs

Q. Please describe some of the significant and notable programs included in the capital plan.

A. The capital plan represents the investments required to provide safe and reliable service to the Company's customers. KEDLI's major capital programs and projects include the following:

- Proactive Main Replacement (LPP) Program
- Integrity Management Program ("IMP") and Integrity Verification Process Program ("IVP")
- Transmission Station Integrity Program
- Southeast Suffolk Project
- Completion of the Northwest Nassau Project

The Company is also proposing an innovative program to enable renewable natural gas projects to more cost effectively interconnect to the Company's system. These programs and projects are described in detail below and are included in Exhibit __ (GIOP-5).

i. Proactive Main Replacement (LPP) Program

Q. Please describe the inventory of LPP existing on the Company's system.

A. LPP in KEDLI's service territory is comprised of twelve inch and smaller main and associated services that are: (i) unprotected (*i.e.*, non-cathodically

protected) steel pipe (whether bare or coated), (ii) cast and wrought iron pipe, (iii) pre-1985 vintage Aldyl-A plastic pipe, (iv) unprotected steel/wrought iron, copper services, and (v) pre-1974 High Density Polyethylene ("HDPE") and pre-1985 Aldyl-A plastic services. As of the end of CY 2018, the Company has approximately 3,455 miles of LPP in its remaining inventory.

Q. What is the Company's proposal regarding retirement of LPP?

A. To reduce the risk of leaks and breaks, improve system safety, performance and reliability, meet the Company's commitment to enhance customer satisfaction and reduce methane emissions, the Company has prioritized the retirement of older and higher-risk gas infrastructure – specifically, LPP and associated services – that disproportionally contributes to leaks on KEDLI's system.

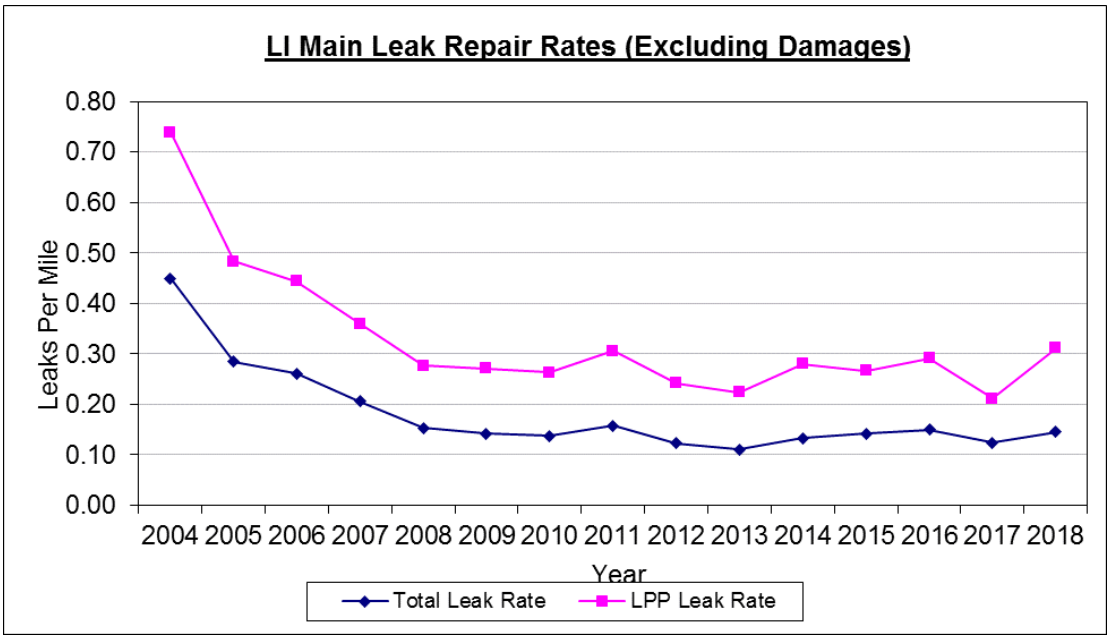
The Company's current plan funds LPP replacements of 115 miles in CY 2017, 135 miles in CY 2018, and 155 miles in CY 2019. The Company proposes to continue its current pace of LPP retirement of 155 miles per year beginning in CY 2020, with a cumulative total of at least 620 miles from CY20 to CY23. The performance metrics and targets for LPP replacements are presented in detail in KEDNY and KEDLI's Gas Safety Panel testimony.

The proposed LPP retirement rate will position the Company to fully retire its remaining LPP inventory in 20 years.

Q. Why is the Company proposing to continue the current pace of LPP retirement through Data Year 3?

A. As the Company has worked over the last several years to retire LPP, the Company observed a leveling-off of the overall leak rate, as shown in Table 2. While leak rates on LPP still outpace the leak rate for overall distribution piping, the leak rates have not continued to rise significantly over the past several years. The current leak rate for all distribution piping is 0.15 leaks per mile. The 2018 leak rate for LPP was 0.31 leaks per mile.

Table 2: KEDLI Historic Leak Rate



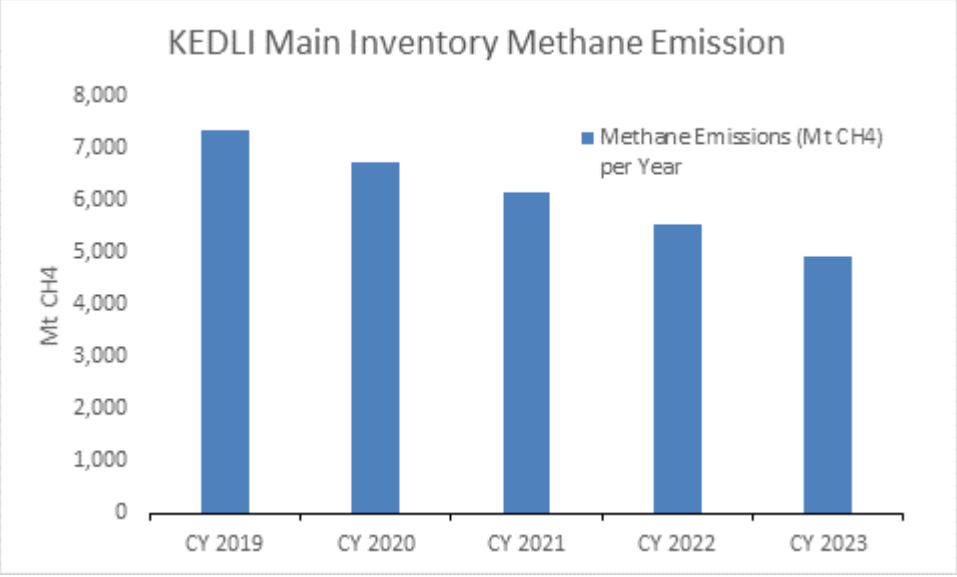
Because the leak rates have stabilized to a lower range for nearly a decade, KEDLI's proposal enables the Company to moderate the overall cost of the LPP retirement program in the near term while still reaching the goals of the program over the next 20 years. Exhibit __ (GIOP-4) provides projected leak rates for LPP for various main replacement strategies.

Q. Is there an environmental benefit associated with the retirement of LPP?

A. Yes. As is noted in the Company's Future of Heat Panel testimony, KEDNY and KEDLI are establishing a target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035, and retirement of LPP is a major component toward reaching this ambitious goal. Specifically,

retirement of LPP reduces gas losses and fugitive emissions of methane. Table 3 provides a high-level estimate of potential methane emissions reductions over the next several years based on the Company's current facilities assuming the retirement of LPP pursuant to KEDLI's proposed program.

Table 3: Estimated Methane Emissions Reduction Through LPP Retirement



Additionally, the Company's LPP retirement algorithm prioritizes retirement of pipe segments with Type 3 leaks and service leaks when other risk factors are equal, thereby enhancing the emissions reduction benefits of its Proactive Main and Service Replacement Program.

Q. Do other capital programs contribute to retirement of LPP to meet the planned targets?

A. Yes. Several miles of LPP are retired through the CSC and Gas System Reliability Programs, but the bulk of LPP retirement occurs through the Company's Proactive Main Replacement Program. The Company forecasts that of the 155 replacement miles per year, the Proactive Main Replacement Program will replace approximately 145.5 miles per year. The Proactive Main Replacement Program also includes retirement of the associated services.

Q. How does the Company prioritize the retirement of main segments for the Proactive Main Replacement Program?

A. Each year, the Company prioritizes retirement of LPP segments by using a risk ranking algorithm that is part of the Company's Distribution Integrity Management Plan ("DIMP") and the Company's Gas Operating Procedure for

the Identification, Evaluation and Prioritization of Distribution Main Segments for Replacement (ENG04030). The Company's risk model calculates a relative risk score for each LPP segment based on specific performance data and localized incident probabilities and consequences, combined with calculated risk factors for the asset classes under evaluation. This risk-based algorithm, along with the Company's engineering judgment, which takes all factors and risks into consideration in each case, form the foundation of the LPP retirement strategy. The Company is proposing to target more cast iron main for retirement, where possible, in consideration of other risk factors, due to increased age and risk of this material. Additionally, as a result of the Storm Hardening Collaborative following the 2016 KEDNY and KEDLI Rate Cases, the Company added flood zone factors to its risk ranking algorithm such that where other risk factors are equal, the Company will target LPP retirements in Federal Emergency Management Agency ("FEMA") 100 and 500-year flood zones.

Q. What level of investment in the Proactive Main Replacement Program is required in the Rate Year and Data Years to achieve the Company's LPP retirement goals?

A. As shown in Exhibit __ (GIOP-1), annual program spending for the Proactive Main Replacement Program is $228.9 million in the Rate Year, $234.7 million

in Data Year 1, $240.5 million in Data Year 2, and $246.5 million in Data Year 3. The total capital cost of the Proactive Main Replacement Program is based on a forecast LPP replacement unit cost of approximately $298 per foot in the Rate Year, exclusive of costs for plastic fusion inspections, contractor safety inspections, and low pressure valve installations, which are set forth separately for rate case presentation purposes, as depicted in Exhibit __ (GIOP-1) and Exhibit __ (GIOP-3).

Q. How does the Company manage the costs of its Proactive Main and Service Replacement Program?

A. The Company's proposal to continue the current pace of retirements over the next several years allows the Company to levelize the program costs during the Rate Year and Data Years. Additionally, to mitigate costs, retirement of LPP is coordinated with other programs (such as the public works, reinforcement, and reliability programs) to capture efficiency savings and cost avoidance. KEDLI will look for more of these opportunities to deploy construction resources more efficiently and will identify areas of the gas network where entire LPP systems can be retired efficiently and cost effectively.

Q. What is the Company's proposal to recover the cost of incremental proactive replacement of LPP?

A. The Company utilizes the GSRS adopted by the Commission in the 2016 KEDNY and KEDLI Rate Cases to recover a return on investment and depreciation expense associated with incremental investment in proactive LPP replacement. The Company proposes to continue use of the GSRS to recover costs of incremental proactive LPP replacements, as shown on Exhibit __ (GIOP-3). The recovery of costs for incremental proactive LPP replacements will continue to work in conjunction with the Company's net plant tracker. The GSRS ensures that KEDLI will recover proactive LPP replacement costs only to the extent it is successful in delivering its program and encourages the Company to replace more than the mileage targets.

Q. Is the Company planning to change the depreciation of existing LPP assets?

A. Yes. As described in the testimony of Company Witness Paul M. Normand, the Company is proposing to depreciate existing LPP assets over the remaining 20-year term of the Proactive Main and Services Replacement Programs.

Q. Does the Company propose to continue reporting on LPP retirement?

A. Yes. The Company will continue to provide Department of Public Service Staff ("Staff") with visibility to the status of LPP retirement. The Company proposes to report to Staff on a quarterly basis, including main retired (feet, location), cost data, opportunistic retirements, and the status of the Company's LPP retirement work plan.

ii. Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs

Q. What is the Company's IMP?

A. The Company's transmission pipeline IMP is a safety program mandated by the Pipeline Safety Improvement Act of 2002 and corresponding DOT regulations. The IMP identifies and addresses potential issues affecting the physical soundness of Company facilities before they become safety or performance issues. The Company conducts baseline and periodic reassessments of transmission facilities to identify and evaluate potential threats to "Covered Segments" of pipelines, *i.e.*, transmission pipelines that could affect High Consequence Areas (areas where a pipeline failure could have significant adverse consequences), as well as remediation of significant defects discovered during such assessments. In regions of the United States

where older gas distribution systems are common, IMPs have become a key component of gas pipeline safety through improving pipeline integrity.

Q. Please describe the IMP capital investments.

A. Currently, IMP assessments may be accomplished through either external corrosion direct assessment ("ECDA") or in-line inspection ("ILI") technology, which uses a robotic device known as a "pig" to evaluate the condition of the inside of the pipeline. ILI is considered superior to ECDA because of the ability to identify condition problems originating internally. IMP investments are necessary to: (i) support in-line inspections (*e.g.*, installation of pig launchers and receivers, and pipe reconfiguration/replacement), and (ii) resolve issues discovered during pipeline inspections conducted through either method.

The construction activities associated with these expenditures involve the installation of "hot tap" fittings, the reconfiguration of such fittings to allow ILI passage, the construction of access points to allow tethered in-line inspection and, in some cases, the replacement of pipeline segments.

Q. Please describe the IMP capital investment needs for the Rate Year and Data Years.

Currently, 21 percent of the Company's DOT pipeline is ILI enabled, increased from 17 percent when the 2016 KEDNY and KEDLI Rate Cases was filed. Over the long term, the Company's plan is to reach 80 percent ILI enablement of the Company's DOT piping. The Company's IMP capital investments in the Rate Year and Data Years will drive toward this goal and will significantly improve the Company's ability to identify integrity issues.

Q. How are the IMP workplan and associated IMP capital investment forecast determined?

A. The workplan began in 2002 based on initial assessments using the Company's existing risk model. Segments were prioritized for ILI enablement over a multi-year workplan based on a combination of their risk score and other relevant factors, such as facility characteristics or geography, to determine what work to do when. Some segments must be re-assessed in certain years according to when they were initially assessed. The workplan is updated annually and as required assessments are completed. Additionally, ILI projects typically consist of two years of design and procurement work followed by construction in the third year.

IMP spending can fluctuate significantly year to year depending on the projects identified for inclusion in the workplan. Because the work must be

done by segment, the length, geographic location, and characteristics of the segments that need to be addressed each year contribute to different costs such that historic program costs are not indicative of future program needs. Once segments to be addressed by the workplan are identified, cost forecasts are level one project estimates based on prior projects of similar size and scope adjusted for known variables. Additionally, some larger projects in the IMP are separately budgeted on the capital investment plan.

The IMP workplan for the Rate Year and Data Years is described in more detail in the IMP data sheet in Exhibit __ (GIOP-5). The IMP capital investments required to address known risks identified via assessments and to increase ILI enablement are approximately $7.4 million in the Rate Year, $7.4 million in Data Year 1, $7.0 million in Data Year 2, and $7.0 million in Data Year 3.

Q. What is the status of the pending federal regulations in this area?

A. In May 2016, PHMSA issued a Notice of Proposed Rulemaking ("NPRM") that proposes new pipeline safety regulations that include a requirement for increased inspection of IMP-covered pipelines utilizing ILI technology and an expanded definition of High Consequence Areas, as well as a host of other

requirements. To meet these requirements, transmission pipelines must be ILI enabled.

The Company expects that regulations will become effective in the fall of CY 2019, and an implementation schedule will be included. Because the Company's IMP workplan is risk based, and in light of PHMSA's new regulations expanding IMP, the Company believes that its proposed IMP program is a reasonable and conservative approach to managing pipeline integrity during the Rate Year and Data Years.

Q. What is covered in the Company's IVP Program?

A. The Pipeline Safety Act of 2011 also mandates that PHMSA establish rules requiring operators to demonstrate their pipelines are "fit for service" by reviewing construction records for each pipeline segment to confirm it is operating within design parameters. The May 2016 NPRM also proposes new rules regarding the maximum allowable operating pressure ("MAOP") and pressure testing requirements for existing pipelines, including (i) eliminating the exemption for establishing the MAOP of pre-1970 "grandfathered" pipe segments; (ii) mandating additional pressure testing or replacement for pipelines without adequate pressure test records; and (iii) requiring operators who lack certain records to establish material properties using approved

methods (*e.g.*, cutting and testing pipe samples). In advance of a final rulemaking, PHMSA issued an advisory bulletin (ADB-11-01) directing operators to perform a detailed threat and risk analysis that includes a records review of their systems.

The Company's IVP program began in 2011 and includes thorough record reviews, pipeline replacement, and retirement of non-essential pipeline segments. As with the IMP, the IVP is primarily based on the Company's assessment of system risks, while also incorporating PHMSA's proposed rulemaking.

Q What is the status of the Company's IVP records review and its IVP Program proposal for the Rate Year and Data Years?

A. Through its IVP to date, KEDLI has completed the MAOP records review on 100 percent of its DOT-jurisdictional pipelines. Going forward, the IVP addresses transmission main replacements and testing necessitated by incomplete records identified by the review and pressure testing and preparing to comply with additional testing that will be required by the new rules. Where the Company has identified incomplete records, pipelines will be replaced, or records will be recreated through testing. This work is necessary to ensure system integrity regardless of PHMSA's proposed requirements.

The IVP work plan is described in more detail in the IVP data sheet in Exhibit __ (GIOP-5) and is approximately $0.8 million in the Rate Year, $3.0 million in Data Year 1, $6.5M in Data Year 2, and $29.5 million in Data Year 3.

iii. Transmission Station Integrity Program

Q. Please describe the Transmission Station Integrity Program.

A. This program is similar to the Company's IVP, but the program addresses transmission-connected pressure regulating stations and take stations ("transmission stations"). PHMSA's 2016 NPRM, related advisory bulletins, and discussions between PHMSA and industry stakeholders during the NPRM process have indicated that the proposed rules will apply to transmission stations. Therefore, the Company is proposing a program for IVP records review and, where records are inadequate to demonstrate that transmission station facilities are fit for service, capital investments for station rehabilitation or partial or full station replacements. Similar to the IMP and IVP Programs for transmission pipelines, these investments are prudent in advance of the final rulemaking to address risk presented by older pressure regulating facilities for which there are inadequate records. The Company is beginning to ramp-up this program in FY 2020, and reaches full implementation in Data Year 2, and the spending is expected will levelize year to year after Data Year 2. The forecast for this program is based on the

number and vintage of transmission stations the program is expected to address and level one estimates of similar station project rehabilitation and replacement projects historically. The capital investments required for this program are $3.0 million in the Rate Year, $15.0 million in Data Year 1, $19.4 million in Data Year 2, and $19.8 million in Data Year 3.

iv. Southeast Suffolk Project

Q. Please describe the Southeast Suffolk Project.

A. The Southeast Suffolk Project is a reliability project to address system needs in the eastern region of KEDLI's service area east and south of Riverhead, New York. The Company's system in that area is operating at close to capacity, and the Company expects increasing demand in the area over the next five years. Absent this project, the Company may need to implement a moratorium for gas sales in the limited area east and south of Riverhead, New York. This project is required to address system needs and avoid a limited moratorium regardless of whether the NESE project is approved. The project will install approximately 8,500 feet of 16-inch, 350 psig transmission main, a new regulator station, and 5,800 feet of 12-inch steel distribution main. The Company's capital investment plan includes approximately $20 million in the Rate Year, $21.6 million in Data Year 1, $0.1 million in Data Year 2, and $0.3 million in Data Year 3.

v. ***Northwest Nassau Project***

Q. Please describe the Northwest Nassau Transmission Project.

A. This three-phase transmission reliability project was included in the 2016 KEDNY and KEDLI Rate Cases to address three aging natural gas transmission lines in northwestern Nassau County. In response to the events of San Bruno and PHMSA's IVP recommendations, the Company undertook a review of historical records to substantiate the MAOP of its gas transmission mains, as well as a review of each pipe's operating history. This review indicated that replacement and down rating of these three transmission pipelines will mitigate system risks, increase reliability, resolve low pressure issues, and position the gas transmission system to accommodate a future supply point in that area. When all phases are complete, the project will replace 8.8 miles of transmission main, install 5.6 miles of new pipe, down rate 9.1 miles of pipe and construct two new regulator stations. Phase One of the project has been completed. As originally proposed in 2016, the entire project was expected to be complete by the end of Fiscal year 2023, but the project schedule has slipped into Fiscal Year 2024 due to municipal permitting delays. To complete Phases Two and Three, the Company's investment plan includes approximately $84.5 million in the Rate Year, $75.0

million in Data Year 1, $80.0 million in Data Year 2, and $49.0 million in Data Year 3.

vi. Renewable Natural Gas ("RNG") Interconnections Program

Q. Please describe KEDLI's new RNG Interconnections Program.

A. This is an innovative program that is part of the Company's overall strategy to encourage interconnection of third-party RNG projects to KEDLI's natural gas distribution system. High interconnection costs have proven to be a primary impediment to the development and interconnection of new RNG projects in KEDLI's service territory. Through this program, the Company will install and maintain some of the necessary interconnection equipment, specifically meters, analyzers, and odorization equipment, that currently is the sole financial and operational responsibility of project developers, thus lowering interconnection costs. This program complements the Company's development of a standard RNG project interconnection agreement that resulted from the 2016 KEDNY and KEDLI Rate Cases, as well as the Company's proposals toward decarbonizing the gas network (*e.g.*, a new green gas tariff offering), as discussed in the Future of Heat Panel's testimony.

Q. What is the basis for the scope and forecast for the RNG Interconnections Program?

A. Based on the number of inquiries KEDLI has received from RNG project developers over the past several years, and RNG projects currently under development, the Company forecasts approximately one RNG project per year is likely to reach maturity to the interconnection phase during the Rate Year and Data Years. The capital forecast for this program of $0.5 million per year based on the cost of meters, analyzers, and odorization equipment for one project per year. A detailed description of the project and project costs is provided in the data sheet in Exhibit __ (GIOP-5).

IV. Gas Infrastructure Capital Investment

Q. How much is the Company planning to invest in its gas system assets in the Rate Year?

A. The Company plans to invest approximately $540 million in its gas infrastructure and other direct capital investments in the Rate Year, including cost of removal. Exhibit __ (GIOP-1), which provides the actual or budgeted direct capital investments for the Historic Test Year, Gap Period, Rate Year, and Data Years, is segmented into four primary spending rationales (programs): "Customer Connections," "Mandated," "Reliability," and "Non-

Infrastructure." Table 4 summarizes the planned direct capital investments for the Rate Year, and Data Years in each of these programs.

Table 4: Direct Capital by Spending Rationale

Spending Rationale	FY 2021 ($000)	FY 2022 ($000)	FY 2023 ($000)	FY 2024 ($000)
Customer Connections	113,890	111,055	95,363	101,610
Mandated	280,457	308,551	330,683	361,011
Reliability	141,115	125,905	148,397	155,955
Non-Infrastructure/ Misc.	4,561	4,459	4,495	4,586
Total	540,023	549,970	578,938	623,162

*Inclusive of cost of removal

Each spending rationale is broken down further into sub-categories that identify specific programs.

Q. How were the projected capital estimates derived?

A. In accordance with the Company's budgeting policies and procedures, capital budgets are prepared annually with a five year forward look. Budget projections are based on historical work levels and unit cost performance for ongoing mandated and routine work and programs, plus any identified new

requirements, programs and projects. Projects that fall outside of routine work, such as safety-driven programs (*e.g.*, the Company's LPP replacement program in the Mandated Category), are developed by Engineering based on the most recent material, labor, and overhead costs.

Q. Does the Company's revenue requirement in this case also include cost of removal associated with the capital investment plan?

A. Yes. Included in the capital costs discussed below, there is a level of cost of removal required to implement the Company's infrastructure investment plan. As reflected in Exhibit __ (GIOP-1), the Company is forecasting costs of removal as follows: approximately $36.7 million in the Rate Year, $37.4 million in Data Year 1, $39.4 million in Data Year 2, and $42.3 million in Data Year 3. The capital forecasts for each program presented below are inclusive of cost of removal.

Q. What types of activities are associated with cost of removal?

The Company defines removal as any work on an asset that results in it being removed from the asset inventory, whether a different asset is added in its place. This type of work would include, but is not limited to, the activities associated with disconnection, removal and disposal (or retirement in place) of gas mains, gas services and related facilities.

Q. What information is presented in Exhibit __ (GIOP-5)?

A. Exhibit __ (GIOP-5) provides additional information for each of the significant gas capital projects and programs expected to be performed during the Rate Year. This additional information includes:

- Project or Program name
- Spending rationale
- Project or Program description
- Project or Program justification
- Estimated costs
- Customer benefits discussion
- Alternatives/Options Analysis
- Studies/references that support the program

Q. Please describe some of the technologies and practices the Company uses to reduce the total cost of its capital expenditures.

A. As discussed in the testimony of Company Witness Christopher J. Connolly, National Grid has undertaken a comprehensive effort to evaluate the technology and business processes that support its gas businesses (referred to as the Gas Business Enablement ("GBE") Program). Among the core investments of the GBE Program are new asset and work management, scheduling, geographic information system, and field mobility solutions

designed to improve the Companies' gas safety and operational performance. While GBE is in process, the Company continues to employ a number of technologies and best practices designed to increase the efficiency and reduce the cost of its capital expenditures. These practices include:

- Increasing the amount of planned capital work (versus reactive work).
- Increasing coordination among capital programs to increase efficiencies (*e.g.*, leveraging LPP opportunities).
- Installing more small-diameter, high-pressure facilities that can be installed at lower cost.
- Using smaller excavating equipment, increasing operating efficiency and reducing instances of damage (because of decreased size and weight of equipment).
- Using "coring and keyhole" technology to repair existing mains.
- Enhancing contractor management.
- On-site reporting for work crews in many large construction projects.
- Continuing to deploy CIP lining (as described in more detail below).

In addition, as noted above, the Company's Capital Delivery organization is in the beginning stages of identifying opportunities for efficiencies that are expected to reduce costs.

A. Capital Planning, Budgeting, and Sanctioning Process

Q. Please describe the annual development of the Company's capital plan.

Each year, the Company develops a ten-year capital plan to achieve its performance objectives of delivering safe and reliable service. Investment Planning compiles proposed spending for programs and individual capital projects. Programs and projects are categorized into one of four spending rationales (Mandated, Customer Connections, Reliability and Non-Infrastructure).

The Company currently uses a tool, C55 (known internally as "Copperleaf"), for the prioritization and optimization of capital investments through a risk scoring and prioritization methodology. This tool uses criteria aligned to a common scale to compare dissimilar inputs (*e.g.*, financial and non-financial inputs) to determine the set of investments, alternatives, and the timing of projects that have the highest potential value.

After the Companies rank the investments in C55, meetings are scheduled with budget sponsors to review value models, project rankings, and various scenarios produced through the optimization of investments to meet capital planning targets. Once plan targets have been approved by the Gas Business Unit, they are submitted to Finance for review.

In late fall, the capital plan is reviewed by the New York Jurisdictional President (Company Witness John Bruckner) and the Vice President, Finance, New York (Company Witness David Doxsee). The New York Jurisdictional President reviews the overall customer, service quality and financial impacts of the investment plan and may request changes to the level or mix of investments.

In early winter, the capital plan is presented to the National Grid USA Board of Directors and the National Grid plc Executive Committee and, in early spring, the capital portfolio is presented to the National Grid plc Board of Directors for review and approval.

Q. Are there additional approvals needed before a project in the annual capital plan may proceed?

A. Yes. Aside from the capital planning and budgeting process, specific delegation of authority ("DOA") approval must be obtained for any project in the ten-year capital plan to proceed. This process includes the sanctioning documentation and review for projects over $1 million and other levels of review for smaller projects. Presently, all projects greater than $1 million require some level of sanctioning documentation and review. The U.S. Sanctioning Committee ("USSC") was established by the National Grid USA

Board of Directors specifically for this purpose. Projects between $8 million and $25 million are reviewed and approved by the USSC. Projects between $25 million to $203 million are reviewed by the USSC and then are forwarded to a Senior Executive Sanctioning Committee ("SESC") for review and approval. For projects between $1 million and $8 million, the USSC has delegated review to an informal committee led by the Investment Planning group and including, but not limited to, managers and directors from the Regulatory, Estimating, Asset Management, and Procurement groups. The committee reviews and finalizes sanctioning papers for these projects at a weekly meeting, and the committee then forwards the final sanction documents to the executive sponsor of the project for approval and signature. Projects less than $1 million do not require sanctioning and are approved through a supervisory delegation of authority hierarchy based on certain established thresholds.

Q. Please explain the difference between the DOA review and approval (sanctioning) process and the approved five-year capital plan used to forecast the Rate Year and Data Years.

A. The timing of the sanctioning process is not aligned with the capital planning process used to forecast the Rate Year and Data Years for purposes of the rate filing. As described above, the Company develops a long-term investment

plan that is used as the basis for the forecast for the Rate Year and Data Year proposals. Project sanctioning, however, normally occurs immediately prior to the FY for which the investment is planned. For example, the FY 2020 capital plan will be formally sanctioned in early 2019. Thus, the Company's currently-sanctioned or partially-sanctioned projects do not yet represent the full capital forecast proposed in the Rate Year and Data Years.

Q. Please describe how the Company's DIMP impacts its capital investment planning.

A. The DIMP involves a risk-based assessment of the Company's distribution system to identify threats in seven categories: corrosion, natural forces, excavation damage, other outside force damage, material and weld failure, equipment failure/malfunction and inappropriate operation. The DIMP requires evaluation and prioritization of the risks that these threats pose, and the implementation of measures to address the highest risks with an emphasis on leak management, enhanced damage prevention, operator qualification to reduce human error and system replacement. Consistent with the DIMP, the Company prioritizes asset replacements in its investment plan based on a risk ranking that considers, among other things, leak repair history, types of leak, pipe material, surrounding geography, segment length, nearby construction activity, field conditions, customer issues, open leaks and engineering

judgment. The risk ranking factors are carefully designed to consider known differences in the performance of asset subclasses, extensive experience with asset failures, current performance data for the asset subclasses for various threat categories, and subject matter experts' analysis and opinions on the future performance of the assets.

B. Customer Connections Category of Spending

Q. Please describe what is included in the Customer Connections spending category.

A. The Customer Connections category of work accounts for approximately 21 percent ($113.9 million) of the total planned capital investment in the Rate Year.

KEDLI is required by the Public Service Law, the Commission's regulations, and tariff to extend service to new customers in its service territory in proximity to the Company's facilities, and to install new distribution main, up to a 100-foot tariff allowance per customer, to do so. Expanding the availability of natural gas in KEDLI's service territory can bring significant economic benefits in the form of energy cost savings for customers, job creation, and increased local tax revenues. Moreover, there are environmental benefits associated with conversion to natural gas from other, higher emission

fuels such as fuel oil. To enable new customer connections, the Company must make significant capital investments in mains, services, and system reinforcements. Customer Connection programs are designed to support forecast customer demand and add new load by increasing system utilization in a cost-effective way. Customer Connection programs involve the installation of new mains, services and meters and include system reinforcement.

The Customer Connections – Install Main, Install Services (including the Clean Choice Program), Meter Purchases, Install Meter, and Automatic Meter Reading ("AMR") Programs and costs are presented in Exhibit __ (GIOP-5).

Q. Please describe recent customer demand trends in the Company's service territory.

A. Gas price advantage is expected to increase between CY 2020 and 2023 and is expected to drive continued demand for gas, particularly in the new construction market. The updated customer demand forecast shows steady demand in the residential conversion market. The commercial conversion market is forecast to decrease slightly due to market saturation. The residential new construction market is expected to increase sharply, and multi-

family new construction is expected to begin rising in CY 2020. Non-residential construction is expected to decrease.

Q. What explains the increases in the Customer Connections forecasts?

A. The increases are driven by higher unit costs of main and service installation due the factors discussed in Section III, above. While units are not expected to increase dramatically, the cost of main installation and reinforcements to serve new customers increased as compared to prior years. Additionally, the anticipated change in sales mix in market to mostly residential projects requires more capital spending because costs per customer connection are higher for residential installations.

Q. Please describe the Customer Connections – Clean Choice Program.

A. The Clean Choice program provides alternatives to customers who want to convert to a cleaner fuel in areas that meet customer density and minimum customer connections requirements set by the program. The program will focus on more geothermal installations, and where geothermal is not an option, customers will still have the option to make an environmentally beneficial decision by conversion from oil to natural gas. This program will incorporate the web-based fuel switching calculator that is discussed in KEDNY and KEDLI's Future of Heat Panel testimony.

Q. Please describe the Customer Connections – AMR Program.

A. The Customer Connections-AMR Program includes the purchase and installation of AMR equipment for new customer connection installations. AMR equipment is not included in the Company's Purchase Meters budget. Based on the Company's forecast, the Company will need to install between 7,000 and 8,000 AMRs per year during the Rate Year and Data Years at costs of approximately $1.0 million per year.

Q. Please describe the Gas System Reinforcement Program in the Customer Connections category of spending.

A. The Gas System Reinforcements Program contains projects intended to ensure that minimum system pressures are maintained throughout the gas network during periods of peak demand. The Company models peak demand based on the sendout forecasts developed by Analytics, Modeling and Forecasting (Company Witness Theodore Poe). In response to demand for gas usage in its service territory, KEDLI determined that it is necessary to complete several projects to ensure its ability to meet peak requirements. These reinforcement projects are essential to ensure the Company can continue to serve growing gas usage load. Without such reinforcements, the Company estimates that more than 32,000 existing customers could experience loss of service.

KEDLI experienced new records for sendouts during the last two winter heating seasons. During the winter of 2017/2018, KEDLI recorded four of its top ten sendout records, including a firm load record of 1,014,810 dekatherms on January 6, 2018, when the average temperature was eight degrees Fahrenheit. During the winter of 2018/2019, KEDLI recorded two of its top ten sendout records, including a firm load record of 1,029,616 dekatherms on January 21, 2019, when the average temperature was thirteen degrees Fahrenheit.

As discussed in Section VII, below, and in the testimony of Company Witness Elizabeth A. Arangio, additional capacity that is expected from the Williams' NESE project is critical to meeting this level of increasing demand. Moreover, the recent growth in peak sendout underscores the need to ensure that the Company's gas distribution facilities can maintain minimum system design pressures during periods of peak demand.

Q. Please provide examples of System Reinforcement projects.

A. Examples of System Reinforcement projects include:

- Replacing undersized mains with larger diameter mains. LPP is targeted whenever practical during this work.

- Looping or connecting system endpoints by installing new main.
- System pressure uprates (*e.g.*, 15 pounds per square inch ("psi") to 60 psi).
- Installing new district regulators and replacing existing undersized district regulators.
- Transferring existing low-pressure customers to an adjacent high-pressure main (*i.e.*, load shedding).

Q. What explains the fluctuation in the Gas System Reinforcement forecast from year-to-year?

A. The workplan for the Gas Systems Reinforcements Program is based on where reinforcements are needed and, thus, is zero-based each year. Estimates are derived based on the types of projects needed, creating uneven spending year-to-year.

Q. Please describe the Riverhead Transmission Main Project.

A. The Riverhead Transmission Main Project is a reinforcement project that will install approximately 8,200 feet of 12-inch, 350 psig transmission main to address system reliability needs in the "North Fork" area of KEDLI's system that is east and north of Riverhead, New York. The existing 124 psig system serving the area is not sufficient to meet increasing demand in the area. The

project will begin design phase in Data Year 1 and is phased over several years to match the expected increase in demand over time. The project requires an investment of approximately $0.2 million in Data Year 1, $1.0 million in Data Year 2, and $23.7 million in Data Year 3.

C. Mandated Category of Capital Spending

Q. What portion of the Company's capital investment plan is Mandated?

A. The Mandated category of work accounts for approximately 52 percent ($280.5 million) of the total planned capital investment in the Rate Year.

Q. Please describe what is included in the Mandated spending category.

A. Projects covered by the Mandated spending rationale are those needed to comply with regulatory obligations and rate plan commitments, including: the CSC Program that requires the Company to relocate facilities, code-required corrosion testing and mitigation or other pipeline integrity related activity, proactive and reactive capital main and service replacement, plastic fusion inspections, and required meter replacements. Exhibit __ (GIOP-5) includes a summary description of each of the significant projects included in the Company's Mandated spending rationale/category, along with the estimated annual funding during the Rate Year and Data Years for each.

Q. Please describe what is included in the CSC Program.

A. This program consists of construction work performed to accommodate third-party, municipal construction activity that could impact the integrity of the Company's natural gas facilities. Typical third-party construction activities that impact gas facilities include work on water, sewer, and drainage infrastructure, street reconstruction, road realignment and bridge replacement. State regulations and Company procedures require the replacement of eight inch and smaller cast iron gas mains if roadway or underground construction is being performed in such a way that would impact the integrity of the Company's mains. Non-cast iron gas mains (*i.e.*, steel and plastic) are not subject to the same replacement regulations and are typically supported and protected if not in direct conflict with third-party construction. Direct conflicts are addressed through relocation regardless of material type.

Aging municipal infrastructure (buildings, schools, bridges, roadways, transportation systems, water mains and sewer facilities) will require significant upgrades in the coming years. Over the last several years, there had been an increase in the level of municipal construction activity, as Superstorm Sandy and recent gas incidents focused attention on the state of municipal infrastructure, but this work in KEDLI's service area seems to have plateaued. The forecast cost for this program is approximately $9.8 million in

the Rate Year, $10.0 million in Data Year 1, $10.5 million in Data Year 2, and $10.5 million in Data Year 3.

Q. Are there opportunities to retire LPP during CSC projects?

A. Yes. As part of the CSC Program, the Company looks to identify cost-effective opportunities to retire LPP when main replacements are required to accommodate municipal construction. CSC projects present opportunities to perform safety and reliability upgrades on the Company's infrastructure, the costs of which can be offset by coordinating construction activities (shared trenching and paving) and securing third party reimbursements. Of the approximately 15,096 linear feet (2.9 miles) of CSC main replacements forecasted for FY 2020, approximately 10,560 linear feet (2 miles) of LPP is expected to be retired.

Q. How does the Company forecast CSC expenditures?

A. KEDLI forecasts its CSC expenditures by reviewing the known and planned work identified by municipalities, historical work volumes, and unit information. In CY 2018, approximately 9,692 linear feet (approximately 1.8 miles) of main replacement was required to address municipal infrastructure improvements. Approximately 41 percent of this footage for this replacement work (equating to approximately $5.0 million) was subject to some level of

reimbursement from third parties, while the remaining approximately 59 percent of this footage ($1.9 million) was non-reimbursable main and service replacements.

Q. Does the Company's forecast reflect potential reimbursement from municipalities and other third parties?

A. Yes. KEDLI's forecast reflects reimbursable amounts for expenditures pursuant to the Gas Facility Cost Allocation Act and other cost sharing arrangements with municipalities. KEDLI forecasts expected reimbursements total approximately $1.1 million in the Rate Year, $1.1 million in Data Year 1, $0.9 million in Data Year 2, and $0.9 million in Data Year 3.

Q. Please describe New York City's joint bidding process and its potential impact on CSC costs.

A. In 2014, New York State adopted a "joint bidding" requirement for public works projects in cities with populations of one million or more (N.Y.S. Assembly Bill A10021B) that will impact KEDLI's CSC projects in New York City (on the Rockaway Peninsula). The City has not yet fully implemented this process. Currently, utilities impacted by New York City public works projects are individually responsible for performing their own relocation work, as well as negotiating all support and protect work directly

with the City's contractor. Under the "joint bidding" model, the City will bid and manage the entire project. If the City's contractor is qualified by the Company to work on its gas facilities, the contractor may perform the gas relocation work. If the City's contractor is not gas qualified, KEDLI will perform its own relocation work ahead of the planned project. Each impacted utility will also be responsible for a portion of the "shared costs" of work performed by the City's contractors, including, for example, maintaining the construction site, establishing field offices, setting up transportation and managing contracts and expenses. It is unclear when the City will fully implement the joint bidding process, but in recent months, the City has increased discussions regarding implementation. It is likely that the process will be fully implemented within the next few years, but the scope of its application is currently unknown. The City's workplan for FY 2019 and FY 2020 includes approximately 18 projects, across multiple boroughs and utilities, using the joint bidding process. Incremental costs of the joint bidding process for FY 2019 and FY 2020 appear to be about $0.5 million for each FY; however, the projects the City is jointly biding are not necessarily representative of future project scopes that may use this process. Because the true cost implications on utilities of working through the City's selected contractors, as well as the impact of the shared costs contribution, remains to be seen, the Company has not included additional costs in its CSC forecast.

Q. What changes is KEDLI proposing to the recovery mechanisms for the CSC Program to address the variability in CSC forecasting and reimbursements?

A. To address the uncertainty in municipal public works scheduling and project scope from year-to-year, KEDLI is proposing to change the existing CSC tracker mechanism to a full reconciliation of the difference between forecast and actual spending (from the current 90 percent deferral mechanism). The Company is not proposing any other changes to the mechanism or reporting requirements. As set forth in the Joint Proposal adopted in the 2016 KEDNY and KEDLI Rate Cases, the CSC capital reconciliation will work in conjunction with the net utility plant tracker. The CSC capital tracker mechanism is presented in Exhibit __ (RRP-9), Schedule 2.

Q. Is the Company proposing to continue its Large Diameter Main Rehabilitation Program for cast iron lining?

A. Yes. The Company's capital plan includes continuation of its Large Diameter Main Rehabilitation Program, included in the 2016 KEDNY and KEDLI Rate Cases, that utilizes cured-in-place pipe lining ("CIP") to cost effectively improve the safety and reliability of its large diameter mains. This technology is proven to extending the life of the main, improve system performance and

reduce future joint leaks on treated pipeline segments for years. As agreed in the Joint Proposal in the 2016 KEDNY and KEDLI Rate Cases, the Company will continue to include main segments reconditioned through CIP lining in the inventory of LPP for eventual retirement. The Large Diameter Main Rehabilitation Program will address one mile in the Rate Year and each of the Data Years using CIP lining at a cost of approximately $6.5 million per year.

Q. Why has the cost to CIP line one mile of pipe increased?

A. Issues arose in the execution of CIP lining program over the past three years where an entire segment of main could not be continuously lined due to configuration issues. In these instances, work had to be stopped and small sections of main had to be replaced between segments that could be lined. As a lesson learned, the Company adjusted it strategy to plan for replacement of small segments that cannot be lined. The proposed plan to line one mile per year is more realistic and includes the time and funding to replace small sections that cannot be lined.

Q. Please explain KEDLI's Cross Bore Remediation, Latent Damages, and Atmospheric Corrosion Inside Inspections Programs.

A. The Cross Bore Remediation and Latent Damages Programs are existing programs involving safety enhancements undertaken when the Company is

performing other mandated work, such as main replacement. In the 2016 KEDNY and KEDLI Rate Cases, the Company included a program to perform inspections on older installations for cross bores, instances when plastic main inadvertently bores through a sewer lateral as a result of horizontal directional drilling. The Company will complete inspections of historic HDD installations by the end of CY 2020. Going forward, the costs to inspect for cross bores on new installations is embedded in main installation unit costs; however, a small budget is necessary to mitigate cross bores that may be revealed through routine inspections. The capital investment plan includes $0.1 million in the Rate Year and each of the Data Years, adjusted for inflation to remediate cross bores. The Latent Damages Program is a capital program to remediate third party damages discovered during normal operations. The Company seeks to continue this program at a funding level based on historic damage rates. The capital investment plan includes $0.5 million in the Rate Year and each of the Data Years, adjusted for inflation.

The Atmospheric Corrosion Inside Inspections Program is a similar proposed program to address remediation required when inside service inspections result in findings of substandard conditions. As a result of the Commission's 2017 Inside Service Line Inspections Order in Case 15-G-0244, baseline inspections result in additional remediation work that must be completed

within a year of discovery. Although the majority of remediation work is O&M expense, some remediation requires capital service replacements. The Company estimates that capital funding of approximately $0.7 million in the Rate Year (to address a spike in remediation work anticipated from completion of baseline inspections), and $0.1 million per year in each of the Data Years, adjusted for inflation.

Q. Please describe what is included in the Reactive Main and Reactive Service Replacement Programs.

A. The Reactive Main and Reactive Service Replacement Programs provide for the replacement of gas mains and services and leak repair during urgent or emergency situations that fall outside the normal scope of integrity, reinforcement, reliability, and public works programs. These replacements are performed in lieu of repair in instances when repairing damaged facilities is not possible, or where the pipeline segment is too short be covered by the Proactive Program.

The Main Replacements (Reactive) – Maintenance Program provides Field Operations the ability to quickly replace segments of main when emergent conditions warrant. The forecast for this program is based on Historic Test Year costs, plus an adjustment to include replacement of segments under 50

feet that were previously expensed. This adjustment is made pursuant to a change in capitalization policy that is discussed in detail in Section IV(F), below.

The Service Replacement (Reactive) - Leaks Program addresses unanticipated service replacements due to leaks reported by the public and customers or detected during programmed leak surveys. The capital investment plan includes approximately $1.9 million in the Rate Year and each of the Data Years, adjusted for inflation.

The Service Replacement (Reactive) - Non-Leaks/Other Program addresses primarily customer-driven service replacements necessitated by damages, service abandonments, demolition requests, and customer requests to relocate existing services. The capital investment plan includes approximately $4.7 million in the Rate Year and each of the Data Years, adjusted for inflation.

Q. Please describe what is included in the Corrosion Control Program.

A. This program funds work consisting of field testing, monitoring, upgrades, and repairs to existing corrosion control systems as mandated by Federal and State Code requirements. Part of this program addresses above ground gas mains at bridge locations, which includes complete recoating of existing aged,

dis-bonded, deteriorated or uncoated gas mains, as well as retirement of LPP where it extends underground near these crossings. In addition, this program addresses the installation and testing of cathodic protection systems on buried piping. The Company's forecast for the Corrosion Control is based on the Historic Test Year program unit cost applied to a three-year historic average of units (CY 2016 – CY 2018), adjusted for inflation in each year. The capital investment plan includes approximately $1.0 million in the Rate Year and each of the Data Years.

Q. Please describe what is included in the Valve Installation and Replacement Program.

A. Federal and state regulations require installation, inspection, operation, and maintenance of critical pipeline valves on all gas distribution systems. The purpose of these valves is to facilitate the rapid shutdown of distribution piping during gas emergencies such as third-party damage or water intrusion. A secondary purpose of these valves is to facilitate maintenance and pipe replacement on associated distribution piping.

This program will strengthen the Company's emergency response capabilities by improving the level at which Field Operations personnel can safely and efficiently isolate sections of the distribution system while mitigating

customer impacts (*e.g.*, reducing the duration of future outages). Ensuring all critical valves are properly maintained and operable is a key public safety function and is essential to the effective operation of the Company's gas distribution system. The Rate Year forecast of approximately $0.1 million each year, adjusted for inflation in each of the Data Years.

Q. Please describe what is included in the Meter Changes Program.

A. The Meter Changes program involves the labor to replace gas meters that are retired from service due to required periodic testing, damage, failure, or any other reason.

Q. Please describe what is included in the Purchase Meters Program.

A. This program includes the purchase, testing, processing, and delivery of gas meters and associated instrumentation needed to support the Meter Change program, new customer connections, and Customer Meter Service ("CMS") operations.

D. <u>Reliability Category of Capital Spending</u>

Q. What portion of the Company's capital investment plan is Reliability?

A. The Reliability category accounts for approximately 26 percent ($141.1 million) of the total planned capital investment in the Rate Year.

Q. Please describe what is included in the Reliability category.

A. Investments in this category are intended to maintain safe, reliable service to customers by ensuring that all facilities on the gas system are operating efficiently and reliably. The Reliability category includes programs related to gas control, heaters, reactive Instrument & Regulation ("I&R"), pressure regulating facilities, valve installation/replacement, remote-controlled valves, gas planning, system reliability, water intrusion, system automation and control line integrity, special station projects (including over-pressure protection), storm hardening (remote control valves), and reliability and storm hardening upgrades to the Company's LNG facilities. Exhibit __ (GIOP-5) includes a summary description of significant projects included in the Reliability spending rationale/category, along with the estimated cost during the Rate Year and Data Years for each project.

Q. Please describe what is included in the I&R Reactive Program.

A. The reactive I&R budget provides funding for capital investment in pressure regulating and control stations. Typical projects in this category include unplanned capital work resulting from emergency conditions, including the replacement of station valves, regulators and relief valves, as well as related capital work on station equipment.

In addition to the I&R Reactive program, the Company is proposing several I&R-related capital and O&M programs to address lessons learned from the Merrimack Valley over pressurization issue. These programs are presented in KEDNY and KEDLI's Gas Safety Panel Testimony.

Q. Please describe the Gas Control Training Simulator Project.

A. This program is for the purchase, design, and implementation of a real-time, gas control system-modeled training simulator to meet Federal control room management regulatory requirements for scenario based situational training of gas system operators. The Company currently uses paper table-top exercises to meet this requirement. Use of a training simulator will improve the effectiveness of the training by enabling operators to mimic actual system specifications while drilling actions under normal, abnormal, and emergency operating conditions. The training simulator will benefit all National Grid's gas local distribution companies ("LDCs"); thus, KEDLI's forecast for this project represents KEDLI's allocation of the total cost.

Q. Please describe what is included in the Heater Installation and Pressure Regulating Facilities Programs.

A. There are 28 natural gas heaters currently operating on the Company's system. Because high-pressure gas cools when reduced to a lower pressure, heaters are

required at pressure regulating stations to prevent freeze-ups that can impact flow control devices. In addition, cold gas temperatures can lead to reduced pipe toughness and increased potential for frost heave and cold temperature-induced stresses. The heater program adds new heaters (where required) and replaces existing heaters that have reached the end of their useful lives. The Pressure Regulating Facilities category provides funding for replacement and/or rebuilding and reconditioning of existing regulating and control stations.

The Pressure Regulating Facilities Program provides funding for replacement and/or rebuilding and reconditioning of existing regulating and control stations. Pressure regulating facilities (or stations) are designed to control system pressures and maintain continuity of supply during normal operating conditions and during periods of peak gas demand.

KEDLI has assessed regulating stations on its system, evaluating factors such as pressure, location, including whether the station is in a FEMA 100 year or 500-year flood zone, and the number of dependent customers for each station. In addition, the assessment considered station condition, including pipe corrosion, location and type of overpressure protection, automation, condition of vaults, vault covers, wall sleeves, piping vents and ladders. The results of

the assessment were used to create an overall risk rating for each station that serves as the basis for prioritizing projects in this program. The program includes full or partial replacement of existing stations, and storm hardening upgrades where necessary.

The investments in the Pressure Regulating Facilities Program do not cover the special project capital improvements to specific stations that are separately set forth in the investment plan and described below.

Q. Please describe the pressure regulation special projects.

A. Pressure Regulation Special Projects are capital investments to address reliability issues at specific stations that are separately budgeted and are not included in any other blanket reliability programs. These projects are described in detail in the "Pressure Regulation Special Projects" data sheet included in Exhibit __ (GIOP-5). The projects target facilities that have the highest potential customer impact. Depending on the asset, these projects include station replacement or rebuild, incorporation of odorization, gas quality validation, pressure regulation, overpressure protection, storm hardening improvements, and addition of process pre-heating equipment.

Q. Please describe what is included in the System Automation category.

A. This program will install updated Remote Terminal Units ("RTUs") at multiple city gate and regulator stations. RTUs provide temperature, pressure and flow data back to the Gas Control Room. RTUs can also monitor gas detectors and intrusion alarms and allow Gas Control Operators to adjust flow and pressure set points at regulator stations. The benefits include enhanced calibration of network models from automation and telemetry data, improved accuracy of network analysis, and enhanced ability to forecast the need for capital reinforcements, which will lead to more efficient capital planning. Automation allows Gas Control Operators to selectively close valves, raise or lower pressures, and shut down take stations. System alarms also alert Gas Control Operators to system issues and allow quick pinpointing of the source.

PHMSA regulations regarding Control Room Management require Operators to ensure that "practices and procedures within their control rooms are adequate to maintain pipeline safety and integrity." These rules indicate that Operators should have telemetry to monitor pipelines, as it would increase system awareness and enable a proactive response to abnormal operating conditions. The System Automation program complies with these regulations by providing for increased deployment of telemetry on the Company's system.

Currently, KEDLI's pressure regulation stations are equipped with some form of telemetry, but only 96 percent of the system has full remote control of the regulators. Additionally, some of the telemetry equipment was installed years ago and the technology has become obsolete. KEDLI's System Automation program adds automation capabilities to low pressure regulator stations and updates RTU equipment and communications capabilities. This program will provide enhanced ability to monitor system performance and remotely adjust pressures on the gas system.

Q. Please describe what is included in the Gas System Reliability – Gas Planning/Remote Controlled Valve ("RCV") Program.

A. This program includes capital projects required to maintain system minimum pressures on the gas network in the event of an abnormal operating condition (failure involving a regulator station, gate station, critical main or other major pressure facility on the system). Gas System Planning ensures that customers continue to have reliable service and that no customers experience interruptions because of an unplanned outage of a facility under normal winter conditions. The program also involves the installation of additional RCVs on transmission pipelines to improve emergency response capability and reduce the risk of gas releases. In the event of a pipeline failure, RCVs allow control room operators to stop the flow of gas and to remotely isolate and shut down a

portion of the system. Currently, most transmission pipelines can only be shut down using manually-controlled isolation valves, which can take longer to close and result in a larger customer impact. Improving response time through the expanded deployment of RCVs reduces the quantity of gas released and can limit the harm to the public and property.

Q. Please provide examples of Gas Planning Reliability projects.

A. Examples of Gas Planning Reliability projects include: eliminating distribution systems fed by a single district regulator or main, integrating distribution systems with the same operating pressures through pipeline connections, expanding supply diversity, and projects targeting areas of the system where large numbers of customers would experience a service interruption if a single gas facility became inoperable when the average daily temperature is five degrees Fahrenheit.

Q. Why does the Company propose to include installation of RCVs in the capital investment plan?

A. The PHMSA regulations promulgated in response to the Pipeline Safety Act of 2011 will mandate the installation of additional RCVs. But even in the absence of the PHMSA regulations, investment in RCVs is prudent given the safety and reliability benefits. As highlighted by recent industry events, there

are significant operational benefits associated with the increased deployment of RCVs, such as enhanced pipeline shutdown capabilities.

Q. Please describe what is included in the Water Intrusion Program.

A. The Water Intrusion program is designed to address water entering the gas distribution system, resulting in main obstructions, poor pressure and/or freezing customer services. This program targets the retirement of LPP that is susceptible to water intrusion but is not prioritized for replacement under the Main Replacement programs because of the absence of leaks and/or historical leak repair activity.

Q. Please describe the Storm Hardening – Remote Service Shutoff Valves Program.

A. In the 2016 KEDNY and KEDLI Rate Cases, the Company included a proposal for a Storm Hardening Program to begin installing remote service shutoff valves on KEDLI's approximately 60,000 gas services within the FEMA's designated flood zones over five years beginning in CY 2017. Because there was no viable shutoff valve mechanism commercially available for use on the system, the Company was unable to timely launch this program. The devices became available from the manufacturer during February of 2019, and the Company plans to begin installations in FY 2020. Because of

the significant benefits of these devices, the Company is renewing its proposal for this program from FY 2020 through FY 2024.

Q. What are the benefits and costs of this program?

A. The remote shutoff valves add a layer of protection for services that are especially susceptible to storm surge and flooding that could cause over-pressurization of the gas facilities connecting customers' premises. The impacts of recent severe storms (*i.e.*, Superstorm Sandy) demonstrate the need for the Company to harden its infrastructure to provide greater protection from future storms. These valves will operate on a fixed communication network that will allow for remote operation and monitoring. Automated valves stop the flow of gas as soon as flooding is detected. This will prevent regulator over-pressurization and stop gas from flowing to premises with damaged equipment and/or extinguished pilot lights, mitigating the risk of a potential incident. Automated valves also provide a real-time count of services impacted by flooding to inform the Company's storm response about the resources needed to restore the affected customers expeditiously. Lastly, in areas where flooding prevents physical access to valves and regulators, remote shut-off valves will allow the Company to interrupt only those services impacted by flooding, which could spare entire neighborhoods or larger areas from losing gas service because of access issues. Installation of the

communication networks through which the valves operate can also be leveraged in the future for additional smart devices, such as methane detectors that are being developed.

The Company is proposing to ramp-up the program in FY 2020 and the Rate Year, installing approximately 1,000 valves in FY 2020 and approximately 9,000 valves in the Rate Year, with the bulk of the communications network systems in the Rate Year as well. The Company intends to install approximately 16,500 valves per year in each of the Data Years. Program costs in the Rate Year are approximately $15.6 million.

Q. Please describe the compressed natural gas ("CNG") investments included in the KEDLI's capital plan.

A. The CNG investments included in the capital plan support the Company's fleet of CNG vehicles. These investments include upgrades necessary to maintain safe and reliable operation of the Company's CNG fueling stations, including partial or full replacement of existing stations, replacement of station equipment, and/or installation of telemetry, safety design, and reliability improvements.

Q. Please describe the Company's LNG programs.

A. KEDLI maintains on-system supply through its LNG facility at the Holtsville LNG Plant. This plant is a critical component of KEDLI's gas supply portfolio and gas operating network and plays a critical role in meeting peak demand. The Holtsville LNG Plant has one storage tank and is capable of supplying 100 million cubic feet of gas per day. Refilling the tank is accomplished through liquefaction during the summer period when gas supplies are available and less expensive. The liquefaction system can refill at a rate of about 6 million cubic feet of gas per day and it can take up to100 days to refill the tank. The Holtsville LNG Plant is a cost-effective means of meeting peak demand, because gas is liquified during the summer months when prices are lower and stored for use in the winter during peak demand conditions. KEDLI's various LNG capital programs are necessary to ensure the continued safe, reliable operation of the Holtsville LNG Plant.

Q. Please describe the Company's proposed capital investments for the LNG programs during the Rate Year and Data Years.

A. The Holtsville LNG Plant has been in service since 1971 and requires significant investment to support continued safe and reliable operation. The Company's proposed capital plan includes funding to complete a major tank upgrade, blanket funding to replace aging equipment and unexpected needs,

and proposals for a number of new special projects necessary to upgrade critical facilities and equipment at the plant and to support safety systems.

Q. What is covered in the LNG Blanket Program?

A. The LNG Blanket Program provides funding for near-term and emergent capital projects needed to maintain safety and reliability at the Holtsville LNG facility. Projects typically address equipment that is at or near the end of useful life. Examples of projects in this category include: upgrades to mechanical equipment and systems; upgrades and replacement of electrical and control systems; structural improvements of plant and facilities; procurement of capital tools and equipment; preliminary engineering and design of capital projects; and retirement and decommissioning of equipment, plant and facilities. These projects will extend the service life of the facility and improve operational performance of plant equipment.

Q. Please describe the LNG Special Projects category.

A. In addition to the LNG Blanket Program work, KEDLI has identified several significant, discrete capital projects to be completed in the Rate Year and Data Years. These projects are discussed in more detail in the "LNG Special Projects" data sheet that is included in Exhibit __ (GIOP-5). Several projects are carry-overs from the current rate plan that are in various stages of

completion, including projects deferred due to re-prioritization, changes in project scope, or delays in permitting approvals from municipal entities.

Q. What is the status of the Tank Modernization Project?

A. The Holtsville LNG tank (0.6 billion cubic feet ("bcf")) was put into service in 1971. In the 2016 KEDNY and KEDLI Rate Cases, the Company proposed a project to temporarily take the tank out of service to allow entry into the tank to perform a major upgrade. However, execution of this project was dependent on the NESE project. Delays in Williams' NESE project resulted in delays to the planned tank upgrades (NESE's original permit request was denied, and the project is at least one year past its original schedule). The Company's proposed capital investment plan includes funding to deliver this project, based on the assumption that the NESE project will move forward in the next two years.

E. <u>Non-Infrastructure and Other Capital Spending</u>

Q. What portion of the Company's capital investment plan is Non-Infrastructure and Other?

A. The Non-Infrastructure and Other category of work accounts for approximately 1 percent ($4.6 million) of the total planned capital investment in the Rate Year. Other Capital includes special projects not included in the

Company's other investment programs, including AMR replacements. The Non-Infrastructure budget also includes funds for the purchase of tools and equipment that meet the criteria for capitalization.

Q. Please describe the Company's Telecommunications Program.

A. The Company operates a two-way radio system that provides communications support to Gas Field Operations, CMS, Security, and other departments. This system also serves to support communications during emergencies when commercial communications systems are not operational. These systems contain ageing components and must be repaired or replaced upon failure. The Telecommunications Programs provide reactive funding for smaller replacement projects in three categories that collectively cover funding needs for towers at remote sites, radios installed in vehicles, and for general damages and failures.

Q. What is included in the Tools and Equipment program?

A. The Tools and Equipment program captures the items that are not used for specific projects but support the safe, efficient and on-going day-to-day operations of the gas business. Examples include tools (hand, power, pneumatic, hydraulic), specialty equipment, PPE, office machines, electronic

data processing equipment and software applications, shop and garage equipment, and communications devices.

Q. What is included in the Company's Learning and Development ("L&D") Materials Tools and Equipment Program?

A. L&D has a limited stock of tools for use in training and has relied on tools borrowed from Field Operations to supplement its supply and conduct required training of in-house and contractor employees. With workload and training requirements increasing, Field Operations can no longer spare tools and equipment sufficient for training purposes, and the limited stock of available tools is beginning to deteriorate. The proposed L&D Materials Tools and Equipment Program will purchase new tools and equipment for training purposes. The program will follow an innovative approach whereby employees will train on new tools during the annual training sessions at the beginning of each year, and then the new tools will be sent into the field for use by the newly-trained employees. Following annual training, Field Operations will send its in-use tools back to L&D for refurbishment and use in supplemental training throughout the year. This tool swap (i) enables more effective training on the equipment that will be used in the field, (ii) enhances the quality of tools used in the field, and (iii) ensures that L&D has a sufficient inventory of tools to conduct supplemental training during the year

after annual training has concluded. The program also increases the useful life of field tools that are swapped back into L&D for refurbishment that otherwise may have failed earlier in the field due to continuous use.

Q. Please describe the Company's Non-Infrastructure AMR Replacement Program.

A. The AMR Replacement Program funds the purchase of the ERT units to support the Company's mandated meter changes program and to replace existing units nearing the end of useful life or that have failed. The ERT units are not included in the Purchase Meters Program. KEDLI currently has an installed AMR population of approximately 610 thousand units and estimates that approximately 18,000 AMR units will require replacement during the Rate Year and each Data Year. This program does not include funding for ERTS to support meter installations for new customers; those are funded in the Customer Gas Connections – AMR Program.

Q. What is included in the Meter Testing Equipment Program?

A. The Meter Testing Equipment Program replaces meter test lab equipment that has reached its useful life that is needed to carry out the mandated meter testing program.

F. Capitalization Policy Change – Main Installation

Q. Is the Company proposing any changes to its capitalization policies for main installation?

A. Yes. Accounting for installation of segments of main under 50 feet in length is currently expensed by the Company. However, this threshold is arbitrary; moreover, there is no similar threshold for main replacement in Niagara Mohawk's service territory. Applicable accounting principles and regulations permit capitalization of new main installations, and there is no minimum threshold. The Company is proposing to remove the 50-foot threshold and capitalize all main installation activities in accordance with the FERC Chart of Accounts – Gas – Operating Expense Instructions, Accounts 367 Mains and 376 Mains. This change will standardize the accounting treatment of these work activities between National Grid's New York gas distribution companies. As discussed in the testimony of the Revenue Requirements Panel, the Company has normalized its Historic Test Year to reflect the change from expense to capital for distribution and transmission main installation, and the Company's capital investment forecasts also reflect this capitalization change.

V. Indirect Capital

Q. Is the Company allocated indirect capital costs?

A. Yes. KEDLI is allocated a portion of indirect costs such as facilities, fleet services, and supply chain (IM/IR). The Company's indirect capital investment plan is presented in Exhibit __ (GIOP-6).

Q. Please describe the Fleet and Supply Chain investments included in the Rate Year and Data Years.

A. The proposed fleet and supply chain expenditures will provide required upgrades to the Company's critical operating facilities in support of its capital and O&M programs. Fleet requirements in the Rate Year and Data Year include a gasoline tank replacement and fuel tank recertification to meet environmental regulations at Bellmore; remediation of aqueous film forming foam ("AFFF") at fuel islands; repair of portable lifts at Roslyn, paint hazard remediation in Hicksville, Hewlett, and Brentwood, construction of a garage in Brentwood to accommodate indoor on-site repairs of the 55 trucks assigned to this yard; and as-needed replacements of tools, equipment, and miscellaneous items required for upkeep of the Company's fleet (*e.g.*, shop equipment, tire lifts). The fleet forecast includes approximately $1.2 million in the Rate Year.

Supply Chain investments support the service centers that are used to stage and deploy the Company's field workforce, store vehicles, equipment and materials, and provide various utility functions (*e.g.*, meter shop) in support of the operation and maintenance of KEDLI's system, including inventory management and warehousing. Supply chain needs during the Rate Year include paving, security cameras, and inventory racking. The supply chain forecast includes approximately $0.4 million in the Rate Year.

Q. Please describe the Company's proposed Facilities investments for the Gap Period, Rate Year and Data Years.

A. The Company's Facilities investments are set forth in Exhibit __ (GIOP-6) and include a baseline facilities plan of $0.3 million in the Gap Period, $0.3 million in the Rate Year, $2.5 million in Data Year 1, $2.5 million in Data Year 2, and $3.1 million in Data Year 3 for general property improvements and upgrades. KEDLI is also planning to complete several large facilities projects described below and in data sheets included in Exhibit __ (GIOP-6). The Company is making significant investments in facilities required to support the Company's capital and O&M programs, enhance the customer experience, and ensure optimized and efficient office workspaces. The total facilities investment plan for the Rate Year and Data Years for KEDLI is

approximately $44.5 million in the Rate Year, $5.8 million in Data Year 1, $5.6 million in Data Year 2, and $3.1 million in Data Year 3.

Q. Please briefly describe the large Facilities projects.

A. The large projects are as follows:

- Bayshore: The Company plans to construct a new welding shop and other improvements at its facility in Bay Shore, New York. The new facility will include an improved work area for the fleet organization. Improvements would include upgrades to the building's structure, doors, plumbing, electrical and heating, ventilation and air conditioning system. The forecast capital expenditures for this project total $5.7 million in the Gap Period.
- Materials Testing Lab & Equipment: The Company currently houses its materials testing laboratory at its Hicksville site. The Company plans to relocate this facility to its existing property located at 25 Hub Drive in Melville, New York. The Company plans to construct and operate a 6,500 square foot modern facility capable of meeting existing and future business requirements. The increased capacity and capability created by the new lab will enable the Company to (i) to better comply with increasing safety inspection and testing requirements, including the testing of plastic fusion joints, (ii) perform

testing of larger diameter pipes and thus provide support for the "cured in place" lining and CISBOT large diameter programs. Forecast capital expenditures for this project total $0.2 million in the Gap Period and $4.3 million in the Rate Year.

- Melville Hub Expansion (Training Academy): The Company plans to enlarge and improve its existing training academy located at 25 Hub Drive in Melville, New York. The expanded facility will accommodate enhanced training initiatives to ensure the competence of gas workers and to validate the training competency of the contractor workforce in KEDLI's service territory. The facility will also support first responder training on Long Island and will better permit the Company to educate its workforce about revised PHMSA pipeline safety standards and enhanced engineering oversight standards. The new facility will permit the Company to deliver enhanced classes and hands on training to both employees and contractors. The new space will include a mix of classrooms, lab facilities, technical and office space designed to meet increased training and evaluation demands. The forecast capital expenditures for this project total $1.6 million in the Gap Period and $3.4 million in the Rate Year.

- Additional Operations Sites: KEDLI's gas operations currently use 10 operating sites. Eight of those sites are shared with PSEG-LI. KEDLI proposes to reduce the total number of sites from ten to eight, to reduce the number of shared sites from six to two, and to develop more efficient sites that will serve as modern, energy efficient gas operations sites. Forecast capital expenditures for National Grid Service Company associated with the project total $2.3 million in the Gap Period, $2.5 million in the Rate Year, $9.7 million in Data Year 1, $1.6 million in Data Year 2, and $1.3 million in Data Year 3. Forecast capital expenditures for KEDLI for this project total $26 million in the Gap Period, $36.6 million in the Rate Year, $3.4 million in Data Year 1 and $3.2 million in Data Year 2.

  The relocation of National Grid's operations sites will enhance its ability to recruit and retain employees by enabling the Company to more clearly distinguish itself from PSEG-LI. The relocation will also enable the Company to use space more efficiently and improve its environmental footprint. Sharing operations sites with PSEG-LI creates the same issues as sharing the much larger Hicksville facility and does not make long term business sense in most circumstances. In addition, over the ten-year period beginning in FY 2020 by relocating

operations sites, the Company projects that it will save approximately $16.5 million compared to maintaining the status quo.

The process of relocating National Grid's operations sites will cause it to incur incremental O&M expenses in the first four years of the project that will be more than offset by savings once the project nears completion.

The Company has not adjusted its revenue requirement in this case to account for either incremental O&M expense or incremental O&M savings associated with the relocation of its operations centers because the Company forecasts that it will incur incremental O&M costs for this project of $2.0 million in the Rate Year and $2.4 million in Data Year 1 that will be offset by incremental O&M savings of $1.8 million in Data Year 2, and $2.9 million in Data Year 3. Thus, over the four-year period reflected in the Company's rate filing, the incremental savings associated with this project slightly exceed the incremental costs by approximately $0.4 million. The Company believes that offsetting the backloaded incremental savings in Data Years 2 and 3 against the upfront incremental costs in the Rate Year and Data Year 1 is a reasonable ratemaking approach.

Q. Are additional Facilities investments planned for the Gap Period, Rate Year and Data Years?

A. Yes. National Grid is proposing a plan of facilities replacement for general facilities improvements and upgrades of $34.1 million in the Gap Period, $38.8 million in the Rate Year, $11.9 million in Data Year 1, $6.8 million in Data Year 2 and $6.8 million in Data Year 3. These facilities are National Grid Service Company investments and the costs are set forth in the Revenue Requirements Panel's Exhibit __ (RRP-11), Workpapers to Exhibit RRP-3, Schedule 9, Workpapers 4, 7, 10, and 13. The large facilities investment projects include:

- Replacement of the Hicksville Office and Operating Site: National Grid currently occupies 237,000 square feet of the 515,000 square feet available at this site. PSEG-LI occupies the remainder of the site. National Grid plans to relocate its operations to a 100,000 square foot site and, in so doing, achieve savings through a reduction in the total footprint and the elimination of inefficiencies that arise from the joint occupation of the current site. Forecast capital expenditures associated with the relocation for National Grid Service Company total approximately $3.1 million in the Gap Period, $19 million in the Rate

Year, $0.5 million in Data Year 1, $0.5 million in Data Year 2, and $0.5 million in Data Year 3.

National Grid has been sharing this site with PSEG-LI since it assumed responsibility for the administration of the Long Island Power Authority's transmission and distribution systems in 2014. Sharing major operations facilities is not optimal for the Company from an efficiency, safety or security perspective. Moreover, sharing a major operations site makes it more difficult to promote and brand National Grid to the public. In addition, it creates issues with respect to recruiting, training and retaining employees. Relocating National Grid's operations to a more efficient, Company-only site is clearly beneficial from a company, employee, and customer perspective.

- Brooklyn, NY Office Relocation: National Grid will relocate existing personnel in the MetroTech office to another facility in Brooklyn and reduce the existing footprint. National Grid is examining potential locations for this relocation at this time. However, a move to a smaller footprint will reduce costs and enable the Company to make more efficient use of space. As part of this plan, KEDNY's call center, customer office, and Sustainability Hub will remain at the current location, and the Company will sublease the remainder of the space until the end of the current lease term in February 2025. National Grid

forecasts capital expenditures of $8.0 million in the Gap Period and $13.2 million in the Rate Year for this project. KEDLI will receive only its properly allocated share of responsibility for the costs of this investment. The Revenue Requirements and Shared Services Panels further discuss the planned MetroTech office relocation.

- IM Warehouse Project – National Grid will consolidate and integrate the Gas Distribution and Gas Transmission supply chain by moving certain warehouse facilities from Greenpoint in Brooklyn to a new location on Long Island. This integration supports a comprehensive overall supply and inventory management program for the GBE Program in the U.S. The facility will help to improve services provided to Gas Transmission and Gas Distribution by enabling increased delivery frequency of needed materials and permitting National Grid to better optimize delivery routes. Forecast capital expenditures for this project are approximately $14.0 million in the Gap Period and $1.1 million in the Rate Year.
- Reservoir Woods Office Project – National Grid is reconfiguring the corporate offices in Reservoir Woods, Massachusetts, which is utilized by employees who provide services to KEDLI and other National Grid affiliates, to create a more efficient space. Forecast capital expenditures are $6.0 million in the Gap Period.

VI. Gas O&M Expenses

Q. Please summarize the Panel's testimony regarding the costs of operating the gas system.

A. The Panel addresses major expenses associated with operating the Company's gas delivery system, and incremental O&M expenses the Company expects to incur in the Rate Year and Data Years.

Q. Please generally describe the nature of the Company's gas system O&M expenses.

A. O&M expenses relate to work performed to provide customer support, respond to emergencies, perform safety inspections and other compliance activities, restore service and maintain the life of capital assets. The Company has a significant maintenance program to ensure that system assets are utilized to their fullest potential life expectancy. As gas facilities age, maintenance costs increase. These costs include increased volume of repairs, more significant repair work, and increased emergency work. These expenditures are required to prevent failure and maintain the life of the assets until replacement occurs. The O&M workload is also increasing to keep pace with expanding inspection and testing requirements from new regulatory and pipeline safety mandates. Finally, the Company's O&M programs are designed to maintain the service commitments in its gas safety performance

metrics, which cover various aspects of its performance in the areas of reliability and safety, including metrics measuring pipeline replacement, emergency response, leak management, and damage prevention.

Q. How does the projected Rate Year non-labor expense level compare to the Historic Test Year non-labor expenses for operating the gas system?

A. As shown on Exhibit __ (GIOP-7), the Company projects its incremental Rate Year non-labor O&M expense to be approximately $15.496 million for gas operations and gas safety programs, which is approximately 13.7 percent greater than its adjusted O&M expense for the Historic Test Year.

Q. What are the projected Rate Year O&M incremental labor requirements?

A. As shown in Exhibit __ (GIOP-8), the Company also proposed to hire 75.3 total incremental FTEs in the Rate Year to support its O&M and gas safety program requirements. The costs for these FTEs are supported by the Revenue Requirements Panel and presented in Exhibit___(RRP-3), Schedule 27.

Q. Please summarize the projected incremental O&M expenditures.

A. Increases in O&M expense are primarily driven by (i) the Company's increasing capital investments and increased costs for executing the Company's capital plan, (ii) changes to Company's O&M workload, and (iii) initiatives the Company is undertaking in the Rate Year to address new or expanding safety requirements and to implement lessons learned from recent industry gas system incidents.

Q. What is the Company doing to manage its O&M costs?

A. The Company has implemented various initiatives to reduce its O&M expenses, including:

- Increasing the use of scheduled O&M work appointments to reduce multiple unproductive field visits to complete work.
- Coordinating O&M activities required at each premise so that multiple maintenance requirements can be completed during a single visit.
- Increasing the use of coring and low-dig technology, reducing debris removal and paving restoration costs associated with smaller roadway excavations.
- Modifying shift schedules to more efficiently respond to higher leak volumes.

- Exploring new work management systems to optimize dispatch of resources.

Notably, during CY 2018, the Company undertook a re-organization of its Gas Business Unit to optimize the level of managers and FTEs to arrive at a more efficient balance of personnel.

A. Incremental O&M Costs Associated With Capital Investment

Q. Please describe the Company's need for incremental O&M costs associated with its planned capital investments.

A. As discussed above, the Company plans to increase its capital investment program during the Rate Year. As shown in Exhibit ___ (GIOP-7), the Company estimates incremental O&M non-labor expense of approximately $6.343 million in the Rate Year directly related to the Company's capital investments.

Table 5: Incremental O&M Costs Related to Capital Investments

Category ($000)	FY 2021 Non-Labor	FTEs
Capital Support – General	0	5
Disconnects & Reconnects	$596	0
IMP/IVP	$4,353	5.5
Transmission Station Integrity	$517	2
Storm Hardening	$877	1.4
RNG Program	0	0

* Included in new Future of Heat Engineering Group

i. Capital Support - General

Q. What O&M services will the various construction support functions provide to support the Company's increased capital investments?

A. Construction support functions include internal groups providing contract administration, project management, budgeting and resource planning. While the majority of costs from these functions are directly charged to capital projects, the Company incurs O&M expenses for items such as training, travel, conferences, licensing, new employee on-boarding, and costs for administering O&M contracts. The Company estimates that approximately 10 percent of construction support employees' time is O&M expense.

As KEDLI increases its capital expenditures, the Company will require additional capital support resources, including gas system engineering (estimators, designers, engineers), investment planning (clerks, inspectors, program managers), operations support (permit clerks) and resource planning (analysts, coordinators).

Q. What are the incremental support requirements in the Rate Year?

A. The Company forecasts approximately 13.9 FTEs for these support functions in the Rate Year. A breakdown of the positions is provided in Exhibit __ (GIOP-8).

ii. Disconnects & Reconnects

Q. Please describe the O&M costs associated with service line disconnects and reconnects.

A. Main replacements require the Company to disconnect gas service lines from the main being removed, and then reconnect the service to the new main. A 2,000-foot main replacement can require dozens of disconnects and reconnects, especially in more densely populated areas. The Company's capital plan involves significant investment in main and service replacements, including the retirement of LPP. This work will increase O&M costs for disconnection and reconnection of gas services by $0.596 million in the Rate Year, $0.529 million in Data Year 1, $0.574 million in Data Year 2, and $0.674 million in Data Year 3, as shown in Exhibit___(GIOP-7).

iii. IMP/IVP

Q. Please describe the O&M requirements of the IMP and IVP Programs.

A. Incremental non-labor O&M in the amount of $4.353 million in the Rate Year, $2.946 million in Data Year 1, $3.728 million in Data Year 2, and $3.719 million in Data Year 3, as shown in Exhibit___(GIOP-7) and an additional 5.5 FTEs in the Rate Year and Data Years are required to conduct the required inspections via ECDA on pipelines that are not ILI-enabled and to conduct ILI on those that are already enabled. These costs include excavation and support for ECDA inspections, evaluation of testing data and the costs of non-capital repairs such as repair sleeves and on-site material testing. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

iv. Transmission Station Integrity

Q. Please describe the O&M requirements of the Transmission Station Integrity Program.

A. Similar to the IVP and IMP program needs for non-station assets, incremental non-labor O&M in the amount of $0.517 million in the Rate Year, $0.505 million in Data Year 1, $1.057 million in Data Year 2, and $1.078 million in Data Year 3, as shown in Exhibit___(GIOP-7), and two incremental FTEs in

the Rate Year and Data Years are required to conduct records reviews and testing of stations. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

v. Storm Hardening – Remote Shutoff Valves

Q. Please describe the O&M requirements of the Storm Hardening – Remote Shutoff Valves Program.

A. The installation of storm hardening remote shutoff valves requires incremental non-labor O&M in the amount of $0.877 million in the Rate Year, $1.309 million in Data Year 1, $1.382 million in Data Year 2, and $1.822 million in Data Year 3, as shown in Exhibit___(GIOP-7) for pole rentals and information systems running costs for communication equipment to operate the valves. Additionally, the Company requires 1.4 incremental FTEs in the Rate Years and Data Years to oversee system monitoring and valve location support, investigate alarms, and replace valve components as necessary. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

vi. RNG Interconnection Program

Q. Please describe the O&M requirements of the Company's RNG Interconnection Program.

A. The Company's RNG Interconnection program requires non-labor O&M for ongoing maintenance of the interconnection equipment that the Company installs in the amount of $0.055 million in the Rate Year, $0.110 million in Data Year 1, $0.166 million in Data Year 2, and $0.221 million in Data Year 3. The O&M for this program is shown in Exhibit __ (RRP-3), Schedule 27. Additionally, 2.2 incremental FTEs are needed in the Rate Year and Data Years for the new Future of Heat Engineering Group that is described in KEDNY and KEDLI's Future of Heat Panel's testimony. These FTEs will, in part, support administration of the Company's RNG Interconnection Program. The costs for FTEs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel.

B. <u>Changes to O&M Workload</u>

Q. Please describe the non-safety program incremental O&M workload expenses.

A. There are two significant O&M programs where the Company projects incremental expenses in the Rate Year that are not related to safety requirements and initiatives: (i) the Traditional Research and Development ("R&D") Program, and (ii) the Fixed Factor Meter Inspection Program.

Q. Please describe the need for incremental O&M for the Company's Traditional R&D Program.

A. The Traditional Gas R&D Program is for short-term research associated with gas operations, end use, natural gas appliances, supply related storage, safety and related measures that do not qualify for funding under the Millennium Program. The Historic Test Year reflects minimal spending for Traditional R&D, in part due to departmental, budgeting, and personnel changes that resulted in an inadvertent gap in Traditional R&D spending. Traditional R&D may result in modernization of utility operations, reduced costs, improvements to public, customer, and employee safety, and/or environmental benefits; thus, the Company is committed to reinstating its Traditional R&D Program. KEDLI proposes one incremental FTE in the Rate Year and Data Years, and incremental non-labor O&M for Traditional R&D of approximately $0.956 million in the Rate Year, $0.802 million in Data Year 1, $0.751 million in Data Year 2, and $0.708 million in Data Year 3, as shown in Exhibit___(GIOP-7), for R&D projects to address:

- Advanced leak/methane detection equipment
- Auto/remote shut-off valves
- Communications/IT
- Composite materials, liners, and related technologies
- Damage prevention and location technologies

- Gas venting
- Global navigation satellite system (GNSS); geospatial information system (GIS), and asset tracking and traceability
- IMP/IVP technologies
- RMD technologies
- Virtual reality and augmented reality
- Large diameter plastic
- Live main insertion
- Main installation via plow

Q. Please describe the Fixed Factor Meter Inspection Program and the incremental O&M required for this program.

A. Fixed factor meters are used in conjunction with pressure regulated installations and that employ a correction method system, based on a billing multiplier, to adjust the measured volume of gas delivered to standard pressure conditions. KEDLI's Fixed Factor Meter Inspection Program conducts mandated inspections to verify measurement accuracy for billing purposes. Pursuant to a pilot program for extended inspection intervals, KEDNY and KEDLI currently conduct fixed factor inspections annually of approximately one-fifth of the total fixed factor meter population, such that the entire population is inspected over five years. Niagara Mohawk conducts

inspections of its entire population of fixed factor meters annually. To better align fixed factor meter processes and resources across the three New York gas LDCs, KEDNY and KEDLI propose to conduct annual inspections of all fixed factor meters. KEDLI requires one incremental FTE in the Rate Year and Data Years, specifically union field mechanics, to conduct the incremental inspections.

C. O&M Costs Related to Safety and Reliability Programs

Q. Please briefly describe the incremental O&M requirements for the Company's gas safety programs and initiatives.

A. The Company's safety programs address both new mandated safety inspection requirements (inside service line and plastic fusion inspections), and implementation of gas pipeline safety enhancements. These programs are described in detail in KEDNY and KEDLI's Gas Safety Panel testimony. The O&M requirements of these programs are summarized in Table 6.

Table 6: Rate Year Gas Safety Programs Incremental O&M and FTEs

Category ($000)	FY 2021 Non-Labor	FTEs
Plastic Fusions In-Process Inspections	203	0

Inside Service Line Inspections (mandated)	135	1.5
Contractor Safety Inspections	123	15
Inactive Accounts	0	2
I&R Training Lab	75	0
I&R Control Line Survey	500	2.5
Pipeline Safety Management	442	5.1
Damage Prevention	1,004	1
New Materials Test Lab	0	0.5
Single Meter Regulator Inspections	4,000	25
Gas Control SOP Training	0	2.2
First Responder Training	50	2
Field Training	50	2
Residential Methane Detection	1,264	0
Enhanced High Emitter Methane Detection (Google Vehicle)	350	0.6

VII. Impact of Williams' NESE Project on KEDLI's Investment Plan

Q. Please briefly describe the Transco Northeast Supply Enhancement ("NESE") Project.

A. As discussed in the testimony of Company Witness Elizabeth D. Arangio, Williams' NESE Project will expand the existing Transco interstate pipeline to provide an additional 400 million cubic feet of natural gas capacity per day to the New York area by the 2020 winter heating season. National Grid has contracted for 100 percent of the gas pipeline capacity that will be created by the NESE Project. This capacity is critical to meeting increasing demand in KEDNY's and KEDLI's service areas. The NESE Project is currently

awaiting permit approval by the New York Department of Environmental Conservation ("NYDEC").

Q. What is the impact on the Company's capital investment plan if NESE is not approved?

A. The Company's capital investment plan assumes that the NESE Project will be approved and placed in service. As stated in the testimony of Ms. Arangio, if NESE is not approved, KEDLI will have no choice but to impose a moratorium on new and additional gas service in affected areas of the Company's service territories to maintain system reliability. Moreover, several of the Company's planned Rate Year and Data Years capital programs and projects will be drastically reduced, deferred, or removed altogether. Specifically, the Company's customer connections programs would be suspended, and planned system reinforcements needed to support increasing demand would be reduced by approximately half. Notably, without additional capacity from NESE, the Company will be unable to take its LNG tank out of service for sufficient time to complete needed repairs; thus, the capital project described above to make critical repairs to the LNG tank could not proceed.

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Supplemental Direct Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Date: June 11, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudham.

Q. Is this the same Panel that previously submitted direct testimony on behalf of KeySpan Gas East Corporation d/b/a National Grid NY ("KEDLI" or "Company") in this proceeding?

A. Yes. The terms defined in the Panel's direct testimony have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of the Panel's supplemental direct testimony?

A. The purpose of the Panel's supplemental direct testimony is to discuss the potential impacts to the gas capital programs described in KEDLI's April 2019 rate filing if the Company is unable to obtain incremental natural gas supplies from Transcontinental Gas Pipe Line, LLC's Northeast Supply Enhancement ("NESE") project, which would provide 400,000 dekatherms/day of incremental gas supply to KEDNY and KEDLI's gas distribution systems to meet forecast demand for natural gas and enhance

reliability for existing customers. In our direct testimony, the Panel identified certain proposals that would be affected if the NESE project were not placed in service by the winter of 2020/2021. As discussed more fully by Company Witness Elizabeth D. Arangio, recent decisions by the New York State Department of Environmental Conservation ("DEC") and the New Jersey Department of Environmental Protection regarding required water permits create uncertainty as to whether and when the NESE project may proceed with permitting and construction. While the Company remains cautiously optimistic the NESE project will go forward, and therefore is not proposing any changes to its April 2019 filing at this time, the Panel addresses the impacts that gas supply constraints would have on KEDLI's forecast capital additions and certain gas operations and safety programs should the NESE project not be available during the proposed rate plan period.

Q. Does the Panel's supplemental direct testimony include any exhibits?

A. Yes. The Panel's supplemental direct testimony includes the following exhibits that were prepared or compiled under our direction and supervision:

(i) Exhibit ___ (GIOP-1S): Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

(ii) Exhibit ___ (GIOP-7S): Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

(iii) Exhibit __ (GIOP-8S): Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3 (without NESE).

III. Potential Impacts of the NESE Project and Gas Supply Constraints

Q. Did the Panel's direct testimony discuss the possibility that the NESE project might not be available and that this could affect the Company's filing?

A. Yes. While the Company's filing was prepared assuming the NESE project would be completed by the winter of 2020/2021, the Panel's direct testimony (at 118-120) identified the possibility that the NESE project would not be available as planned and explained that KEDLI's capital investment plan would be impacted in that scenario. Specifically, the Panel explained that several planned capital programs and projects could be eliminated, modified, or deferred, including the Customer Connections and Gas System Reinforcement programs. Further, without the planned

capacity from the NESE project, the Company would be unable to proceed with the planned maintenance and upgrades to the Holtsville LNG facility.

Q. Please explain the impact to the Company's capital investment plan if the NESE project is not approved.

A. Exhibit ___ (GIOP-1S) is an illustrative capital investment plan reflecting the impact in the Rate Year and Data Years should the NESE project not be completed on time, including reductions to the Company's Customer Connections, Reinforcements, and LNG programs. As stated by Company Witness Arangio, without the incremental capacity from the NESE project, KEDLI will not be able to procure sufficient gas supplies to meet the forecast demand from new firm customers, such that connecting additional firm load would jeopardize the Company's ability to provide safe and reliable gas service to existing customers. Therefore, investments in the Customer Connections programs associated with serving new firm load would not be necessary, including the installation of new main and the purchase and installation of new meters and associated AMR devices, as well as investments in the low-to-moderate income gas conversion program that was proposed by the Company's Future of Heat Panel. Similarly, the Gas System Reinforcements program would be reduced to remove projects

no longer needed to serve incremental firm load in KEDLI's service territory.

Finally, without the NESE project, the Company is unable to take the Holtsville LNG Tank out of service to perform planned maintenance and upgrades. The Company intends to perform some engineering work for this project for the purpose of conducting studies and design work for potential limited upgrades to the tank that would not require taking it out of service.

Q. What is the total impact of the capital forecast adjustments described above?

A. The above-described adjustments result in a total reduction (including cost of removal) to the proposed capital investment plan of approximately $85.5 million in the Rate Year, $92.0 million in Data Year 1, $115.0 million in Data Year 2, and $137.4 million in Data Year 3. Table 1 on the following page shows the adjustments to each impacted program. Exhibit __ (GIOP-1S) provides a revised version of the total capital forecast showing the adjusted programs.

Table 1: Summary of No- NESE KEDLI Capital Program Adjustments

($ in thousand)	FY20	FY21	FY22	FY23	FY24
Customer Connections					
Install Main	(4,831)	(21,235)	(18,272)	(14,671)	(14,964)
Install Services	(12,893)	(21,620)	(22,003)	(22,405)	(22,853)
Clean Choice Program - Main	(17,205)	(20,790)	(18,314)	(15,731)	(13,037)
Clean Choice Program - Services	(2,481)	(4,769)	(4,198)	(3,602)	(2,982)
Customer Contributions	-	(619)	(2,557)	(2,698)	(2,803)
Meter Purchases	-	(1,429)	(1,579)	(1,611)	(1,643)
Install Meter/Regulator	-	(799)	(1,003)	(1,037)	(1,074)
Automatic Meter Reading (AMR)	-	(953)	(990)	(1,010)	(1,030)
Gas System Reinforcement	(2,499)	(12,662)	(16,850)	(25,856)	(13,294)
LTLI10860 Riverhead Transmission Main - PM	-	-	(195)	(1,000)	(23,700)
LTLI10985- Southeast Suffolk Infrastructure - Phase 2	-	-	-	(100)	(300)
	(39,909)	**(84,876)**	**(85,961)**	**(89,721)**	**(97,679)**
Mandated					
Contractor Safety Inspections	-	(628)	(1,914)	(3,253)	(3,200)
	-	**(628)**	**(1,914)**	**(3,253)**	**(3,200)**
Reliability					
LNG - Tank Upgrade			(4,113)	(22,039)	(36,483)
	-	**-**	**(4,113)**	**(22,039)**	**(36,483)**

Total Capital Including Cost of Removal	**(39,909)**	**(85,504)**	**(91,989)**	**(115,013)**	**(137,362)**
Cost of Removal	**(2,714)**	**(5,814)**	**(6,255)**	**(7,821)**	**(9,341)**
Total Capital (Net of Removal)	**(37,195)**	**(79,690)**	**(85,733)**	**(107,192)**	**(128,022)**

Q. Would the supply constraints affect KEDLI's forecast of capital and O&M expenses for any of the proposed gas safety programs?

A. Yes. The Company's Contractor Inspector Safety Program, which will add contractor inspectors over three years to reach a one-to-one ratio of inspectors to crews, would also be adjusted to reflect reductions to the capital work plan. KEDLI would reduce its total incremental FTE proposal by 13 inspectors and one supervisor. The revised FTE forecast is provided in Exhibit __ (GIOP-8S). Associated reductions to the capital forecast and non-labor O&M forecast for the Contractor Inspector Program are set forth in Exhibit __ (GIOP-1S) and Exhibit (GIOP-7S), respectively.

Q. Does KEDLI plan to continue to provide updates in this proceeding concerning the status of the NESE project?

A. Yes. KEDNY is cautiously optimistic that the NESE project will be approved. As the project's status is uncertain, the Company will update

Staff and the other parties throughout this proceeding and will supplement the record as further developments occur.

IV. Conclusion

Q. Does that conclude the Panel's supplemental direct testimony?

A. Yes.

Before the Public Service Commission

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Dated: July 3, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel ("GIOP").

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudhan. This is the same Panel that provided direct testimony as part of the Company's April 30, 2019 filing. The terms defined in the Panel's direct testimony have the same meanings here.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The purpose of the Panel's testimony is to identify and explain the following corrections and updates:

- Updates to the direct capital plan – The Panel identifies the following updates and corrections to the Company's capital programs and projects: (i) a correction to the unit costs for proactive leak prone pipe ("LPP") replacement; (ii) updates to Phases 1 and 2 of the Northwest Nassau Transmission Main Project ("Northwest Nassau Project") project schedule and estimates; (iii) a correction to the Customer Connections main installation and customer contributions forecasts for FY 2020; and (iv) a correction to properly reflect the Gas Control Training Simulator Training

Project as a Service Company asset and removing these costs from KEDLI's direct capital investment plan.

- Operator Qualification Program ("OQ Program") – The Panel briefly summarizes the Company's plans to implement the recommendations of Staff's "Operator Qualification White Paper" (dated February 2019) that was noticed for comment by the Commission in March of 2019 in Cases 14-G-0212 and 17-G-0318. The Panel updates the Company's capital investment and O&M plans to include implementation costs that are expected to be incurred during the Rate Year and Data Years.

- Updates to the indirect capital plan – The Panel describes general updates to the Company's facilities forecasts.

- Updates to the O&M forecasts – The Panel updates its O&M forecast to reflect updates to the Company's incremental FTE forecasts, hiring plans, and position titles.

Q. Does the Panel's testimony also address the June 11, 2019 supplemental filing?

A. No. The supplemental filing was made to illustrate the potential impacts to the April 30 filing if KEDLI is unable to obtain incremental natural gas supplies from the NESE project. At this time, the Company remains optimistic that the NESE project will be approved. Therefore, all corrections and/or updates pertain to the capital forecast set forth in the April 30 base rate filing. The Company will continue to update the parties on the status of the NESE project and implications, if any, to this rate filing.

Q. Is the Panel sponsoring any exhibits as part of its corrections and updates filing?

A. Yes. The Panel sponsors the following exhibits that were prepared by one or more members of the Panel or under their supervision and direction:

- Exhibit ___ (GIOP-1CU): Revised Actual and Projected Direct Capital Expenditures: Historic Test Year, Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3
- Exhibit __ (GIOP-3CU): Revised Gas Safety and Reliability Surcharge Including Unit Cost Incentive Proposals
- Exhibit __ (GIOP-6CU): Revised Projected Indirect Capital Expenditures (Supply Chain, Fleet, and Facilities) for the Gap Period, Rate Year, Data Year 1, Data Year 2, and Data Year 3

- Exhibit ___ (GIOP-7CU): Revised Incremental O&M Expenditures: Rate Year, Data Year 1, Data Year 2, and Data Year 3
- Exhibit __ (GIOP-8CU): Revised Incremental FTE Positions by Function in the Rate Year, Data Year 1, Data Year 2, and Data Year 3

Additionally, the Company's responses to information requests ("IRs") that are relevant to the corrections and updates discussed below are included in Appendix A.

II. Corrections and Updates

A. Updates to Direct Capital Plan

Q. Please summarize the updates to the Company's direct capital investment plan and Exhibit __ (GIOP-1).

A. KEDLI's capital plan has been updated to correct errors and reflect scheduling and/or budget updates in the Gap Period, Rate Year, and Data Years for the following capital programs: (i) the Proactive Main Replacements (LPP) Program, (ii) the Northwest Nassau Project, and (iii) the Customer Connections – Install Main and Customer Contributions. The update to each program is discussed in more detail below. The Company is also removing allocated cost of the Gas Control Training Simulator Project from the capital

plan because those costs will instead be reflected in the revenue requirement as a Service Company asset. The revised capital forecast is provided as Exhibit __ (GIOP-1CU), which highlights the updates to the capital plan, and includes the updated capital forecast for the Distribution Station Over Pressure Protection Program that is discussed in the Company's Gas Safety Panel Testimony.

Q. What is the impact of these updates to the Company's forecast capital expenditures in the Rate Year and Data Years?

A. The Company's updated capital forecast reflects a net increase in the Rate Year of approximately $27.6 million, a net increase in Data Year 1 of approximately $41.1 million, a net increase in Data Year 2 of approximately $10.8 million, and a net increase in Data Year 3 of approximately $7.2 million. These figures are inclusive of cost of removal ("COR").

Table 1: Direct Capital Forecast Updates FY 2020 to Data Year 3 (including COR)

($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	Total
Original Forecast	446,698	540,023	549,970	578,938	623,162	2,738,791
Revised Forecast	432,391	567,574	591,080	589,712	630,379	2,811,136
Variance	(14,307)	27,551	41,110	10,774	7,217	72,345

The cumulative impact of these updates over the five-year period is a $72.3 million increase to KEDLI's direct capital forecast in FY 2020, the Rate Year, and Data Years. The Revenue Requirement Panel's Corrections and Updates testimony addresses the revenue requirement impact of the Company's revised capital forecast. Additionally, revised in-service dates for some of the major projects, including facilities projects, are presented in Exhibit __ (RRP-7CU), Schedule 1.

i. *Proactive Main Replacement (LPP)*

Q. Please describe the correction to KEDLI's Proactive Main Replacement (LPP) Program forecast.

A. A correction to the forecast is required because the LPP unit costs were understated due to a formula error in the workpaper used to develop the unit costs. Specifically, an incorrect formula was used to sum the components of unit costs that inadvertently excluded a contractor cost component from the total. The error resulted in the unit costs for the Rate Year and Data Years being understated as set forth in Table 2, below. The Company's response to IRs DPS-597 and DPS-602 discuss the unit cost correction and are attached in Appendix A.

Table 2: KEDLI LPP Replacement Unit Cost Correction (Rate Year to Data Year 3)

Unit Cost Per Foot	FY 2021	FY 2022	FY 2023	FY 2024
April 30 Filing	$298	$305	$313	$321
Corrected Unit Cost	$306	$314	$322	$330

Based on this correction to the unit costs, the forecast for the Proactive Main Replacement Program has been increased by approximately $6.3 million in the Rate Year, $6.4 million in Data Year 1, $6.6 million in Data Year 2, and $6.7 million in Data Year 3. The revised forecast is included Exhibit __ (GIOP-1CU). Exhibit __ (GIOP-3CU) provides the revised LPP incremental cost recovery mechanism and the LPP productivity incentive mechanism updated for the correct unit costs.

ii. *Northwest Nassau Project*

Q. Please describe the updates to the Northwest Nassau Project.

A. The project schedules for Phases 1 and 2 have been revised due to a contractor employee safety accident that occurred at the current project construction site. The Company ceased work from February 2019 until June 2019 to conduct a "safety stand-down," which involved an incident investigation and process safety reviews with crews. The project cost estimate has increased due in part to the delay associated with the safety stand down and a request by one of the

municipalities in which the project is located to re-route part of the project. The revised forecasts for Northwest Nassau are included in Exhibit __ (GIOP-1CU) and are summarized in Table 3, below. The Company's response to IR DPS-566 also discusses the updates to the project.

Table 3: Updates to Northwest Nassau Project Forecasts (FY 2020 to Data Year 3)

Phase 1 ($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024
Original Estimate	-	-	-	-	-
Revised Estimate	4,504	-	-	-	-
Phase 2 ($000)					
Original Estimate	57,000	59,500	5,000	-	-
Revised Estimate	30,705	79,239	38,000	2,500	-

iii. Customer Connections – Install Main and Customer Contributions

Q. Please describe the correction to the Customer Connections main installation and customer contributions forecasts.

A. The costs of main installation of approximately $6.7 million and an estimated customer contribution of approximately $5 million for a large fuel cell generator project were inadvertently excluded from the FY 2020 forecasts. Exhibit __ (GIOP-1) includes these costs, which net out to an increase to the FY 2020 capital plan of approximately $1.7 million.

iv. *Gas Control Training Simulator Program*

Q. Please describe the correction to the Gas System Reliability – Gas Control (Training Simulator) forecast.

A. KEDLI's portion of costs for the Gas Control Training Simulator Program was included in the direct capital investment plan; however, this program benefits all National Grid gas distribution companies in New York and New England, with total project costs shared. Therefore, the program is properly reflected as a Service Company asset in the revenue requirement instead of in KEDLI's direct capital investment plan.

B. OQ Program

Q. Please summarize the Company's plan to implement Staff's proposed OQ requirements.

A. As stated in the Panel's direct testimony at pages 24 to 25, the Company did not include in the April 30, 2019 filing costs to implement Staff's proposed new requirements for gas company operator qualification programs (*see* "Notice of Proposed Rulemaking," issued March 27, 2018 in Cases 14-G-0212 and 17-G-0318) due to the timing of the proposal that did not allow the Company to complete its analysis and develop implementation costs in time for inclusion in the April 30 filing.

The Company has developed a plan to proactively implement enhanced operator qualification best practices that are recommended by Staff's proposal and are likely to become compliance requirements during the term of the rate plan. Proactive implementation provides safety and reliability benefits to customers in the nearer term.

Q. What are the implementation costs?

A. The Company adjusted the direct capital investment plan by approximately $0.7 million in the Rate Year, $0.5 million in each of Data Years 1 through 3, to reflect costs of Industrial Training Services ("ITS") systems, records maintenance, equipment such as eWallet cards used to verify OQ, and OQ test security. The Company also adjusted the O&M plan to include non-labor implementation costs of approximately $0.3 million in the Rate Year, $0.4 million in each of Data Years 1, 2, and 3, as well as 10 incremental FTE Technical Inspectors to maintain OQ records, conduct verifications including field inspections, conduct compliance reviews, and implement other program improvements as required.

Q. Could the Commission's final OQ rules necessitate implementation costs that have not been included in the Company's investment plan?

A. Yes. Until the rulemaking is final, the full extent of new OQ requirements remains unknown. To the extent the final order in the rulemaking proceeding requires additional incremental investments, the Company reiterates the proposal set forth in its direct testimony to recover prudent additional costs through the Company's GSRS.

C. Updates to Indirect Capital Plan

Q. Please describe the updates to KEDLI's indirect capital investment plan.

A. KEDLI's facilities program forecasts have been updated to reflect more refined cost estimates and re-evaluation of project schedules that move some projects out of the Rate Year and lengthen the total design/construction periods. Specifically:

- Facilities Base Spend forecast increased by approximately $0.7 million in FY 2020 due to more refined cost estimates and replacements of various assets (*e.g.,* HVAC equipment, sidewalks, lighting) that were inadvertently excluded from the original forecast
- The Bayshore New Building Completion forecast decreased by approximately $0.1 million in FY 2020 because more work was completed during FY 2019 than had been anticipated when developing the forecast

- The Materials Test Lab forecast increased by approximately $0.03 million due to a more refined estimate, and the project was deferred to Data Year 1
- The Melville HUB Expansion forecast increased by approximately $0.5 million in FY 2020 due to a more refined estimate, and the project schedule was changed for design to occur in FY 2020 and construction to occur in Data Year 1
- The LI Operations Sites project forecasts (New Large Ops Site and Other New LI Ops Sites) increased in total by approximately $0.005 million due to more refined estimates, and portions of the project schedules were deferred to Data Years 1, 2, and 3, which shifted approximately $42.3 million from FY 2020 and the Rate Year to the Data Years.

Updates to KEDLI's facilities projects are shown in Exhibit __ (GIOP-6CU). In total, the KEDLI facilities program updates decrease FY 2020 forecast by $25.1 million and the Rate Year forecast by approximately $22.4 million, due to the project deferrals noted above, and net to a total increase from FY 2020 to the end of Data Year 3 of approximately $1.1 million.

Table 4: KEDLI Facilities Program Forecast Updates FY 2020 to Data Year 3 (including COR)

($000)	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	Total
Original Forecast	31,042	44,533	5,821	5,609	3,060	90,065
Revised Forecast	5,958	22,085	38,125	18,821	6,209	91,198
Variance	(25,084)	(22,448)	32,304	13,212	3,149	1,133

Q. Please describe the updates to the Service Company facilities programs that were described in the Panel's direct testimony.

A. Updates to the Service Company facilities projects are shown in Exhibit___ (RRP-11CU), Workpapers to Exhibit __ (RRP-3CU), Schedule 9, Workpapers 4, 7, 10, and 13. Updated forecasts result in overall increases to the Service Company facilities projects, but certain projects have been deferred to the Data Years to more evenly spread costs over the period of the rate plan. The overall capital investment increases by approximately $7.8 million in the Rate Year, $5.0 million in Data Year 1, $8.4 million in Data Year 2, and $0.3 million in Data Year 3. This is mainly due to more refined cost estimates for the Brooklyn Office relocation (approximately $1.9 million), Hicksville Office relocation (approximately $1.7 million), and the Reservoir Woods Office project ($4.6 million in FY 2020). Project spending for the IM Warehouse has been deferred to Data Year 1 and certain costs related to the Long Island Office have been deferred to Data Year 2. An adjustment was made to remove costs

to maintain the existing Hicksville Office site in Data Years 2 and 3, following the office relocation. These costs were inadvertently left in the forecasts for those years in the Companies' April 2019 filing.

D. Updates to O&M forecasts

Q. Please describe the updates to the O&M forecasts.

A. The corrections and updates to the Company's O&M forecasts are shown in Exhibit __ (GIOP-8CU), and Exhibit __ (GIOP-9CU). The O&M updates include updates to gas safety programs that are described in the Gas Safety Panel's Corrections and Updates testimony, as well as the O&M for the Operator Qualification Program that is described above.

Additionally, updates have been made to the hire dates for certain incremental FTEs to correct mis-stated dates and reflect a more realistic hiring schedule, and corrections have been made to FTE position titles that were inadvertently mis-stated. The hire date corrections have been made in accordance with the Company's response to IR No. DPS-374. Finally, the labor costs for all FTEs has decreased due to a change in capital overheads as discussed in the testimony of the Revenue Requirements Panel. These updates are reflected in Exhibit __ (GIOP-9CU) and in Exhibit__(RRP-11CU) Workpapers to Exhibit (RRP-3CU), Schedule 27, Workpaper 2.

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Case 19-G-0309
Case 19-G-0310

September 18, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel.

A. KEDNY and KEDLI's ("Companies") Gas Infrastructure and Operations Panels (collectively, "Panel") consist of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudhan.

Q. Is this the same Panel that testified previously in this proceeding?

A. Yes. Capitalized terms defined in the Panel's direct, supplemental, and corrections and updates testimony have the same meanings here.

Q. What is the purpose of the Panel's rebuttal testimony?

A. The purpose of the Panel's rebuttal testimony is to respond to certain recommendations set forth in the prepared testimony of the Department of Public Service Staff ("Staff") Gas Infrastructure and Operations Panel ("SGIOP"), the Staff Pipeline Safety Panel ("SPSP"), the Staff Policy Panel ("SPP"), the Staff Efficiency and Sustainability Panel ("SESP") (collectively, "Staff"), as well as the New York City Gas Infrastructure and Safety Panel ("NYCGISP") and New York City Policy Panel ("NYCPP") (collectively, "NYC"), the Environmental Defense Fund's witness Joseph Von Fischer ("EDF"), and the New York State Laborers Organizing Fund's witness John Hutchings ("NYSLOF").

Regarding recommendations and proposed adjustments to the Companies' forecast incremental gas capital and operations and maintenance ("O&M") expenditures, the Panel's rebuttal testimony will address:

- The overall presentation of the Companies' capital investment forecast in light of Staff's position that the forecast should assume the Williams' NESE Project will not be in service during the Rate Year;
- Staff's and other parties' proposed adjustments to the Companies' Mandated, Reliability, Customer Connections, and Non-Infrastructure programs;
- Staff's and other parties proposed adjustments to the Companies' O&M programs and incremental FTEs; and
- Other recommendations by Staff and other parties regarding: LNG tank upgrade plans, capital reporting, NYC's proposals for storm hardening and its green infrastructure assets, and hiring practices for contractor labor to support the Companies' capital and O&M programs.

The Companies' Gas Safety Panel addresses the SGIOP's recommendations and proposed adjustments to the LPP metrics and incentives, the Companies' Low Pressure Valves Program, and the Companies' Enhanced Contractor Inspections Program.

Q. Does the Panel sponsor any exhibits as part of its rebuttal testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared under its direction and supervision:

- Exhibit __ (GIOP-1R): KEDNY and KEDLI's capital investment plans with and without the NESE Project in service
- Exhibit __ (GIOP-2R): KEDNY and KEDLI's O&M plans with and without the NESE Project in service

- Exhibit __ (GIOP-3R): KEDNY and KEDLI's proposed incremental FTEs with the NESE Project in service and without the NESE Project in service
- Exhibit __ (GIOP-4R): Corrections and updates to the Companies' No-NESE adjustments to the capital plan
- Exhibit __ (GIOP-5R): Corrections and updates to the Companies' No-NESE adjustments to the O&M and incremental FTE plans
- Exhibit __ (GIOP-6R): Relevant IR responses

II. NESE Impacts and Blanket Adjustments

A. Assumption of Absence of NESE Project

Q. Please describe the Companies' approach to presenting their capital and O&M investment plans for the Rate Year and Data Years considering the current status of the Williams' NESE Project.

A. In its direct testimony, Staff states its position that the Companies' capital and investment plans should be adjusted to assume the NESE Project is not in service because permitting approvals have not yet been granted. As of the date of this filing, final decisions on the NESE Project permits are still pending in New York and New Jersey. While the Companies remain cautiously optimistic that the project will be approved, in consideration of Staff's position and the continued uncertainty, the Companies are presenting their rebuttal testimony assuming that the NESE Project will not be in service in the Rate Year and Data Years. However, the Companies agree with the SPP that the record contains sufficient information on the capital investment plan under either scenario to allow the

Commission to set rates based on any developments with the NESE Project during these proceedings.

Q. What information have the Companies provided in this proceeding regarding the impact on their capital and O&M investment plans should the NESE Project not be available?

A. On June 10, 2019, the Companies submitted Supplemental Testimony that provided the Companies' projected adjustments to their capital plans in the Rate Year and Data Years if the NESE Project was not in service. These adjustments were described in the Supplemental Testimony and presented in the Companies' respective Exhibits __ (GIOP-1S) (direct capital investment plans without NESE), KEDNY Exhibit __ (GIOP-8S) (incremental O&M without NESE), KEDNY Exhibit ___ (GIOP- 9S) (Incremental FTEs without NESE), KEDLI Exhibit __ (GIOP-7S) (incremental O&M without NESE), and Exhibit __ (GIOP-8S) (incremental FTEs without NESE).

Q. Is the Panel presenting any corrections and updates to the Supplemental Testimony regarding the absence of NESE?

A. Yes. The Companies' have identified errors in its Supplemental Testimony that should be corrected as well as an update to labor overhead rates, as provided in the Companies' response to IR DPS-877.

First, as is stated in the Companies' Supplemental Testimony, funding is required in the Rate Year and Data Years to conduct engineering for the LNG Tank Upgrade Projects. For

KEDNY's Tank 2 Upgrade project, the engineering forecast was inadvertently omitted from Data Years 1 and 2, and the construction costs were inadvertently included in Data Year 3. KEDNY is correcting its forecast to add the engineering costs and remove the construction costs.

Second, for the Enhanced Contractor Inspector Program, the Supplemental Testimony sets forth the number of FTEs and the associated capital labor that would be removed because of less capital work in the absence of NESE. KEDNY and KEDLI's Supplemental Testimony mistakenly applied rounding that resulted in imprecise FTE adjustments to remove O&M labor in the Rate Year and Data Years.

Finally, the Supplemental Testimony was prepared in reaction to the denial of the NESE Project's permits in late May, and the Companies have since continued to evaluate the impacts on customer connections. To that end, the Companies have determined that the Customer Connections Install Main, Install Services, and Install Meter/Regulator Programs budgets do not accurately reflect the costs to connect all of the customers who were approved to received service prior to the May 15, 2019, and who are expected to connect to the system over the next three years. The Companies note that all customers expected to connect were included in the revised sales and demand forecasts.

Q. Please describe the updated and corrected exhibits the Panel is presenting regarding the Companies' capital investment forecasts.

A. Exhibit __ (GIOP-1R) restates the Companies' Corrections and Updates capital plans to include two versions of the capital plan for each company – one assuming NESE is in service, and one assuming NESE is not in service for each Company. Similarly, Exhibit __ (GIOP-2R) restates Corrections and Updates O&M forecast with and without NESE in service during the Rate Year and Data Years, and Exhibit __ (GIOP-3R) restates the Companies' Corrections and Updates incremental FTE proposals with and without NESE. The NESE impact adjustments are highlighted in the exhibits and include the above-described corrections and updates to the no-NESE adjustments.

The corrections and updates to the Companies' no-NESE capital adjustments are shown in Exhibit __ (GIOP-4R) as variances to the Supplemental Testimony's projected adjustments. The adjustments to the Enhanced Contractor Inspector Program O&M are shown in Exhibit __ (GIOP-5R).

There are three additional corrections shown in Exhibit __ (GIOP-1R), Exhibit __ (GIOP-2R), and Exhibit __ (GIOP-3R). The first correction is to address an error in KEDNY's IVP program Rate Year forecast (the Data Years are correct) that was identified in Staff's direct filing (discussed in more detail below). The second correction is to include the current estimate of the Newtown Creek project as discussed in the Companies' Future of Heat Panel's rebuttal testimony. The third correction is to accept a reduction of six incremental FTEs to support capital program work that were inadvertently duplicated with FTEs proposed for the Enhanced Contractor Inspector Program.

With the exception of the corrections and updates noted above, all of the information presented in Exhibit __ (GIOP-1R), Exhibit __ (GIOP-2R), and Exhibit __ (GIOP-3R) was previously provided in the Companies' filings in this proceeding. The rebuttal exhibits are intended to consolidate and clarify the Companies' proposals with and without the NESE Project.

Q. How do the Companies' propose to adjust the capital and O&M investment plans in the event the NESE Project is approved after rates are set in this proceeding?

A. The Companies recommend that a final order or rate settlement in this proceeding include a re-opener mechanism to adjust rates in the event that the NESE Project is approved and expected to be in service during either the Rate Year or Data Years. The filed testimony of both Staff and the Companies include capital and O&M investment plan recommendations and sufficient information to support adjustment to the Companies' capital and incremental O&M requirements if the NESE Project is approved.

B. Staff's Blanket Adjustments to Capital Programs

Q. Please summarize the SGIOP's general approach and basis for proposed adjustments to the Companies' capital forecasts.

A. The SGIOP generally supports the Companies capital plans (SGIOP at 32) but proposes adjustments for certain programs in the mandated, non-infrastructure, and customer connections categories, and proposes an overall blanket reduction to the entire Reliability category of spending. The SGIOP offers the following justifications for its proposed adjustments:

- The SGIOP rejects the Companies' higher inflation rates and other known factors that are expected to increase Rate Year and Data Year costs for the CSC and Proactive Main Replacement Programs (*e.g.* increasing contract costs). (SGIOP at 46-52).
- Where the Companies' forecasts were derived using historic averages of spending in FY 2017 and FY 2018, the SGIOP believes that forecasts should be based on "updated" two-year averages or three-year averages to include FY 2019. (SGIOP at 38, 58, 74).
- The SGIOP lacks confidence in the Companies' Reliability Category forecasts because there have been historic variances between the Companies' budgets and actual spending in this category. (SGIOP at 61-65).
- Because of the timing of the Companies' sanctioning process, which occurs closer to the Rate Year, sanction papers are generally not yet available for the Companies' major investments. The SGIOP claims that without sanction papers, it is unable to determine whether proposed projects are truly needed, whether forecasts are reasonable, and whether alternatives were considered. (SGIOP at 28-29).

Q. Have the Companies provided adequate information to the SGIOP to enable a thorough review of the proposed capital investment plan?

A. Yes. The Companies' direct testimony explains the capital budgeting planning and sanction processes including the timing of sanctioning that, as the SGIOP notes, occurs closer to the spending year. Although it is true that sanctioning documentation is not yet available for the Rate Year projects and programs, the Companies' direct testimony and

exhibits provided extensive information for each program and project, including data sheets (Exhibits __ (GIOP-5) that contain the same information that is typically included in sanctioning documentation such as detailed project descriptions, projects needs and justification, forecasting methodologies, cost drivers, applicable regulatory requirements, and alternatives analyses. The Companies also responded to extensive discovery in this proceeding regarding the details of the capital plan.

Q. Does the Panel agree with the SGIOP's methods and justifications for its proposed adjustments?

A. No, the SGIOP's proposed blanket adjustments are not supported by sufficient data and analysis. As discussed below, the SGIOP ignores known, significant cost drivers that will increase costs of CSC and Proactive Main Replacement Programs. Removal of these costs from the budgets for these programs will impair the Companies' ability to deliver high priority capital programs that are needed to meet important reliability, safety, and policy goals, including work to enable New York City and municipal infrastructure projects and to improve safety and reduce methane emissions through replacement of LPP.

Q. Please describe how the Companies determined their proposed capital forecasts.

A. In contrast to the SGIOP's blanket approach, the Companies determined the appropriate basis and methodology for forecasting Rate Year and Data Year investments on a program-by-program basis. Following a detailed analysis of each program, including examination of historic spending, the Companies determined for each program whether an average of historic costs or the HTY represented the best predictor of Rate Year costs. The Companies

also determined the projects for which a project estimate, rather than a forecast based solely on historic costs, was more appropriate. In this way, the Companies carefully considered any anomalies prior to setting its forecasts for each program. The Companies' methodologies for deriving program forecasts are provided in the Companies' testimony and exhibits and in response to IR DPS-761.

Q. How will the SGIOP's proposals impact the Companies' ability to provide safe and reliable service?

A. In general, the SGIOP's attempt to more closely align the Rate Year forecasts with the Companies' average historic costs and spending levels shifts funds away from the Companies' highest priority mandated, safety, and reliability programs. The SGIOP's recommendations will hinder the Companies' ability to deliver on many of the Companies' and the Commission's important policy, environmental, and safety objectives.

More specifically, proposed downward adjustments in the mandated category will challenge the Companies' ability to deliver on aggressive goals for LPP replacement, methane reduction, and programs and projects discussed below that are necessary to reduce overall system risks. Proposed adjustments in the Reliability category will compromise the Companies' ability to replace aging facilities and equipment that are critical to system operations and to add new equipment that enables the Companies to remotely operate and control the gas system. Hampering the Companies' ability to address reliability and gas system reinforcement work is especially concerning considering the possibility that the NESE Project may not be available to deliver additional supply.

The Companies appreciate the SGIOP's concerns for managing increasing costs; however, the reality is that the SGIOP's proposed adjustments will result in the reprioritization of capital investments in a manner that fails to achieve the Companies' and the Commission's goals. This is not the right result from a public policy or safety perspective.

The Panel discusses in more detail the impacts of Staff's and intervenors' recommendations and proposed adjustments to individual capital programs in the sections that follow.

III. Proposed Adjustments to Mandated Programs

A. City State Construction ("CSC") Program

i. The SGIOP's Recommendations

Q. Please explain the SGIOP's adjustments to KEDNY's CSC/Public Works Reimbursable and Non-Reimbursable forecasts.

A. Staff recommends removal of approximately $11 million and $14 million from KEDNY's CSC Reimbursable and Non-Reimbursable forecasts, respectively. Staff explains that it agrees with KEDNY's general methodology for forecasting the CSC program needs based on 20 percent of New York City's expected construction budget but disagrees with KEDNY's additions to the forecast of incremental funding to cover increased restoration and paving costs that are anticipated in the Rate Year and Data Years as "unnecessary." Staff notes that, in FY 2018, KEDNY spent approximately 17 percent of the City's construction budget. Staff also notes that KEDNY has in place a true-up mechanism, which

Staff suggests will make KEDNY whole if actual spending in the Rate Year exceed the rate allowance for this program. (SGIOP at 48-48).

Q. Does KEDNY agree with Staff's adjustments to its CSC Program?

A. No. KEDNY believes that its CSC program forecast appropriately estimate program needs in the Rate Year and Data Years based on recent trends and known cost drivers for this program.

Q. Does the true-up mechanism Staff mentions address KEDNY's concern that the program will be under-funded if Staff's recommendations are accepted?

A. No. Pursuant to the Joint Proposal adopted in the 2016 KEDNY and KEDLI Rate Cases, KEDNY and KEDLI have a reconciliation mechanism for CSC capital expenditures. To the extent that Companies' actual capital spending for CSC, net of reimbursements, differs from the forecast amount in a Rate Year, the Companies can defer the revenue requirement effect (excluding O&M expenses) associated with 90 percent of the difference for future recovery from or return to customers. This mechanism provides some protection against the volatility in the CSC process due to changing New York City plans and the timing lag between rate setting and the City's determinations of project that will be constructed year-to-year. As was stated in KEDNY's direct testimony, however, a partial deferral mechanism is not a substitute for appropriate rate recovery. The Companies must be permitted to recover sufficient amounts in rates to fully fund the necessary work of this mandated program. The Companies' forecasts to set rates are based on known factors that increase costs in the Rate Year; whereas the reconciliation is intended to cover for unknown

cost increases. Therefore, having a reconciliation in place does not blunt the detrimental effects of removing nearly $25 million from KEDNY's CSC program forecasts.

Q. Does Staff agree with the Companies' proposal to modify their current reconciliation mechanism to allow deferral of 100 percent of the difference between forecast and actual expenditures and to add a reconciliation mechanism for non-reimbursable O&M?

A. No. The SGIOP recommends that the capital deferral mechanism be changed from the current 90 percent/10 percent reconciliation to an 80 percent/20 percent reconciliation based on the SGIOP's belief that the Companies need an incentive to effectively manage program aspects and costs that are in the Companies' control. (SGIOP at 89-90). Staff's reasoning is flawed. As stated above, the reconciliation mechanism is intended to manage the aspects of CSC that are *not* within the Companies' control, including the unpredictability of the work that will ultimately be required each year. Not only is the SGIOP's recommendation unjustified, but it is egregious when coupled with the significant reduction to KEDNY's CSC program budget. There is simply no basis for the mechanism to provide less than a full reconciliation, let alone a reduction to the mechanism that already exists.

Moreover, the SGIOP states no justification for disallowing an O&M reconciliation mechanism for KEDNY considering that the Company has demonstrated significant volatility in O&M spending year-on-year. (See Exhibit __ (GIOP-6CU)). Even if KEDNY manages this program to maximize efficiencies and controllable costs to the highest

standard, the inability to predict future program needs would still justify the need for a reconciliation mechanism.

In short, KEDNY's CSC program is unique in its challenges to accurately forecast, and, for that reason, capital and O&M reconciliation mechanisms are appropriate to protect both customers and the Companies.

ii. NYC's Recommendations

Q. What are NYC's recommendations regarding KEDNY's CSC Program?

A. NYC does not support the proposed change to the capital reconciliation mechanism or the addition of an O&M reconciliation mechanism for similar reasons as the SGIOP. Moreover, NYC notes past challenges regarding coordination of the CSC program and requests that KEDNY dedicate executive-level personnel to coordinate with NYC agencies and that the Commission implement a reporting metric to track KEDNY's progress in addressing NYC requests.

Q. How does KEDNY respond to these recommendations?

A. KEDNY's response regarding the reconciliation mechanisms is the same as stated above. Regarding assignment of an executive to coordinate communications with NYC agencies, KEDNY supports opportunities to further improve communications and coordination and is willing to discuss with the City how best to accomplish this.

KEDNY disagrees, however, with NYC's characterization of prior challenges in administration of the CSC program and, in particular, that these issues resulted solely from KEDNY's actions. KEDNY acknowledges that there is room to improve, but NYC's own actions–for example, the extended time NYC takes to process, review, and pay bills–have frustrated the smooth administration of this program. Also, NYC fails to note the significant progress that has been made over the last few years to address many of these issues, including invoicing, as stated in the Companies' responses to IRs DPS-884 and CNY-21. These process enhancements demonstrate KEDNY's willingness to improve coordination and, therefore, Commission involvement and additional reporting is unnecessary.

B. <u>Proactive Main Replacement Program (Leak Prone Pipe)</u>

Q. Please describe the SGIOP's adjustments to the Companies' forecasts for their Proactive Main Replacement Programs.

A. The SGIOP generally agrees with the Companies' forecasting methodology for LPP replacement but recommends adjustments to remove approximately $6 million from KEDNY's program forecast and approximately $10 million from KEDLI's program forecast. The SGIOP disagrees with certain assumptions the Companies made in calculating the unit costs. Specifically, the SGIOP contests the inflation factors the Companies used for the contractor expense components of the unit costs. (SGIOP at 51-52).

Q. Do the Companies agree with these adjustments?

A. No. The Companies' contractor cost inflation factors for LPP replacement work reflect price increases that the Company knows will occur during the Rate Year and Data Years due to the expirations of contractor agreements during that period and changes in terms and conditions that will significantly increase costs (more significantly in KEDLI's service territory) since the existing agreements were executed. This information was provided in the Companies' responses to IRs DPS-494, part 3, DPS-955, part 3, DPS-597, part 3, and DPS-956, part 3. For example, all of KEDLI's mains and services contractor agreements expire in the Rate Year, and KEDLI entered into a new agreement with one of its contractors at the beginning of FY 2020 at significantly increased pricing as compared to the last several years. The Company validated the increased pricing, which is due to a combination of external cost drivers such as increases in dumping costs and changes to the Company's ways of working/process improvements that drive overall safety and customer benefits. Therefore, the Companies' unit cost calculations reasonably account for known cost drivers in the Rate Year and Data Years that differ from the prior period and should be accepted. The SGIOP's adjustments will underfund these programs and jeopardize the Companies' ability to keep pace with targets and goals for LPP removal, and the associated methane reduction.

Q. Are there other recommendations regarding the Proactive Main Replacement Programs?

A. Yes. Staff and NYC provide recommendations regarding the LPP replacement targets, metrics, incentives, and reporting. These recommendations are addressed by the Companies' Gas Safety Panel's Rebuttal Testimony.

C. <u>Transmission Station Integrity Program</u>

Q. Please describe the SGIOP's adjustments to the Companies' Transmission Station Integrity Programs.

A. Staff's adjustments to these programs remove approximately 94 percent of the programs' Rate Year budgets, effectively gutting the programs and compromising the Companies' ability to achieve timely compliance with the PHMSA regulations that will be in place in the Rate Year. The SGIOP's only basis for the reduction is that the Companies have not identified specific station projects, so the design phase funding should be removed until the Companies have completed records evaluation and project identification. (SGIOP at 57).

Q. Why do the Companies need funding for design work in the Rate Year if specific station projects have not yet been identified?

A. As is stated in the Companies' direct testimony, the Transmission Station Integrity Program is similar to the Companies' Integrity Verification Program and is intended to enable timely compliance with PHMSA regulations that will be in place in the Rate Year. The regulations will require records review and verification for transmission station facilities and capital station rehabilitation or replacement projects where records are inadequate or the

Companies are unable to demonstrate that the facilities meet PHMSA's fit for purpose standards. The SGIOP correctly notes that this program is just ramping up; however, based on the initial records review results, the Companies' experience with the transmission pipeline IVP program, and the Companies' knowledge of the vintages, characteristics, and past record keeping systems and practices for its existing transmission station assets, the Companies have reasonably concluded that approximately two stations per year will need either refurbishment or replacement in each year. Based on the pace of records review, design phase will need to begin in the Rate Year, or the Companies will not be positioned to timely complete needed projects. The Companies' forecasts were generically based on conservative estimates for replacing stations but waiting to identify specific projects is not an option considering the pace that will need to be maintained to ensure compliance. Therefore, it is vital that the design costs are included in the Rate Year.

D. Reactive Main and Proactive and Reactive Service Replacement Programs

Q. What adjustments to the Companies' Reactive Main Replacement and Proactive and Reactive Service Replacements does the SGIOP recommend?

A. The SGIOP notes that the Companies' response to IR DPS-761 shows that the forecasts for these programs are based on the average spending over two years (FY 2017 and FY 2018), plus inflation. The SGIOP recommends recalculating the budgets for these programs based on updated two-year averages (FY 2018 and FY 2019) and using a different inflation factor than the Companies applied.

Q. Please comment on the SGIOP's recommendation.

A. The Companies' explanation in DPS-761 of the basis for the forecasts of these programs was incomplete. In addition to averaging two years of historic costs and inflation, these forecasts also reflect the Companies' proposal to capitalize costs for replacements of segments under 50 feet that previously were expensed. Staffs' adjustments do not properly reflect this capitalization change. Additionally, the Companies maintain that FY 2017 and FY 2018 are appropriate proxies for Rate Year costs.

E. KEDLI's Corrosion Program

Q. Is the SGIOP recommending an adjustment to KEDLI's Corrosion Program?

A. Exhibit __ (SGIOP-4) shows removal of $0.73 million from KEDLI's Corrosion Program, but the adjustment is not discussed in the SGIOP's testimony, and therefore is not supported.

F. KEDNY's IVP

Q. Please explain the SGIOP's adjustment to KEDNY's capital budget for the IVP program.

A. Exhibit __ (SGIOP-4) indicates a reduction to KEDNY's IVP program of approximately $0.17 million. KEDNY accepts the correction of this apparent error in the Company's original filing. This correction is included in Exhibit __ (GIOP-1R).

IV. Proposed Adjustments to Reliability Programs

A. Blanket Adjustment to All Reliability Programs

Q. Please explain the SGIOP's blanket reduction to the entire Reliability spending category.

A. The SGIOP determined that KEDNY and KEDLI have typically spent approximately 74 percent and 71 percent, respectively, of their total budgets in the Reliability category. The SGIOP reasons that the Companies' spending in the Rate Year and Data Years for the entire category should be limited to this trend and, therefore, removes 26 percent and 29 percent from the Reliability category total forecast for KEDNY and KEDLI, respectively. Staff performed no program or project specific analysis for any of the forecasts included in the Reliability category of spending, except for the Storm Hardening – Remote Service Shutoff Valves Program and the RNG Interconnections Program.

Q. Do the Companies agree with the SGIOP's recommended blanket adjustment?

A. No. The SGIOP's methodology is flawed, adjustments to the projects and programs included in the category are unsupported, and the adjustment will hinder the Companies' ability to deliver important programs that are needed to enable continued safe and reliable service, including, but not limited to, (i) the Companies' Northwest Nassau and MRI Projects, I&R and Pressure Regulation Programs that are vital to address overpressure risks and to implement best practices in light of the Columbia Gas Merrimack Valley incident; (ii) Heater Installations that are required due to assets reaching the ends of their useful lives, (iii) LNG facility programs and projects, (iv) needed reliability upgrades; and (v) the installation of remote control valves.

A blanket adjustment to a spending category, rather than consideration of the unique factors that influence historic spending variances in each program and factors considered by the Companies when projecting forecasts for each program, is inappropriate. Variances may be experienced for a host of different reasons in each project, and factors outside the Companies' control can also influence variances. For example, the MRI Project experienced delays due to a design change required by a municipality, and, as is stated in KEDNY's GIOP Panel Direct Testimony, the LNG Salt Water Pump House Project experienced delays resulting from unanticipated permit requirements and stipulations imposed by the FDNY.

Moreover, the forecasts for the various projects and programs within the Reliability Category are based on different needs and forecasting methodologies. Historic spending trends do not necessarily reflect Rate Year and Data Year needs. For example, in the Heaters program, heaters must be replaced as they reach the end of their useful lives, so that the program needs can vary significantly depending on the age of assets. Additionally, station refurbishments or replacements are risk-driven, and the work plan is in part dependent on the conditions of the assets as they continue to age. These are the types of factors the Companies consider when setting future project and program budgets, and that are wholly ignored by the SGIOP's attempt to adjust an entire category of spending based on historic trends without sufficient analysis or support.

Q. How would the proposed 26 percent and 29 percent reductions impact the projects and programs in the Reliability Program?

A. There would be a substantial negative impact. The SGIOP does not suggest how the Companies should apply the total reduction among the various programs, but there are no good alternatives. There are programs the Companies cannot deliver if the reduction is applied equally against all programs (reducing each program forecast by 26 percent or 29 percent). For example, the Companies cannot complete the NWN or MRI projects for approximately 70 percent of the budgets. If the Companies divert funding to higher priority projects and programs in the spending category, other projects and programs may need to be deferred altogether, increasing overall safety and reliability risk on the system.

Additionally, the SPSP's proposed reduction fails to consider that forecasts in the Reliability spending category are higher in the Rate Year and Data Years compared to historic spending levels due to incremental work and new programs (*i.e.,* the Distribution Station Over Pressure Protection) that address the identified best practices as well as the recommendations of Staff in the wake of the Columbia Gas Merrimack Valley incident. The SPSP's proposal jeopardizes the Companies' ability to implement industry best practices and needed improvements to I&R, gas system control, and pressure regulation.

B. <u>Storm Hardening – Remote Shutoff Valves</u>

Q. Please explain the SGIOP's and NYC's recommendations for the Companies' Storm Hardening – Remote Shutoff Valves program.

A. The SGIOP recommends deferring the program for one year considering that the remote valve technology is new and recommends that the Companies conduct additional, third-party testing. NYC requests coordination and access to data collected by the devices.

Q. Are the Companies able to defer the program from the Rate Year to Data Year 1?

A. No. The program is in progress, and the Companies are on track to begin installations this year. Contracts are in place with vendors for the valves and for support systems such that capital and O&M spending cannot easily be deferred, and there is work required during the Rate Year on communications network, IT changes to the Companies' customer systems, data security, and other supporting systems that also cannot easily be deferred. For example, installation of communication network routers and pole attachment agreements with other utilities and municipalities take a long time to negotiate and execute. O&M expenses budgeted for the Rate Year largely consist of pole rentals, electric costs, cellular network costs, cloud data storage costs, and network monitoring/maintenance that needs to be in place. The Companies cannot simply cease this ongoing work for one year in the Rate Year and then start it up again. Moreover, the Companies and the manufacturer are conducting extensive testing on the valves that is expected to be complete by the end of December of 2019. While contractual obligations may not allow for complete deferral of the program out of the Rate Year, the Companies are evaluating the possibility of extending this program to phase installations more slowly and allow for additional in-field testing and trouble-shooting and possibly third-party testing.

Q. Does the SGIOP clearly set forth its recommended adjustments to the program budget in support of their recommendation to defer the project out of the Rate Year?

A. No. The SGIOP's testimony does not set forth specific reductions and simply states that the program should be deferred by one year. Exhibit __ (SGIOP-7) and Exhibit __ (SGIOP-8) show removal of the entire O&M budget and FTE to support this program.

On Monday, September 16, 2019, in response to a discovery request, the SGIOP provided a corrected version of Exhibit __ (SGIOP-4) and corrected Reliability category workpapers showing that the SGIOP does not intend to remove all capital spending, but rather applies the blanket Reliability Category percentage reduction to the Companies' FY 2020 budgets for Storm Hardening Remote Valves Program. The corrected Exhibit __ (SGIOP-4) and Reliability category workpapers are provided as Exhibit __ (RRP-5R). There is no justification provided for the amount of the adjustment, and it is unclear why the SGIOP would propose a reduced capex budget for this program in the Rate Year but remove all O&M and the FTEs.

Q. Should any adjustment be made to the forecasts for this program?

A. Not at this time. Considering the uncertainty and lack of support for the SGIOP's proposed adjustments and to allow the Companies to perform contractual work that is needed in the Rate Year to support future installations of the valves, the Rate Year capital and O&M budgets and incremental FTEs should remain as proposed by the Companies.

Q. Please respond to NYC's requests regarding the Storm Hardening – Remote Shutoff Valves Program.

A. The Companies are willing to better coordinate with NYC regarding this program. The Companies are evaluating NYC's request for data sharing and is open to further discussions with NYC regarding whether data sharing is feasible and possible parameters such as uses, frequency of data sharing, and non-disclosure requirements.

C. <u>RNG Interconnection Project</u>

Q. Does Staff support KEDNY's proposed RNG Interconnection Program?

A. Staff supports the program but recommends deferral of KEDNY's program by one year. (SESP at 56-57). KEDNY prefers to begin this program in the Rate Year, as there are several projects currently in development that may benefit from this program. Smoothing the path to bring additional RNG into the Companies' systems supports the Companies' longer-term methane reduction goals.

Additionally, the SESP recommends removal of incremental FTEs proposed for the Future of Heat Engineering group that would support this program. The SESP suggests that removal of the Clean Conversion Program would allow for a reallocation of FTEs intended for that program to support Future of Heat Projects. (SESP at 57). Removal of the proposed FTEs is inappropriate, however, as it would represent a double-count of the Companies' no-NESE adjustments. The Companies' no-NESE adjustments remove all costs, including labor and non-labor capital and O&M expense. Moreover, even if the RNG Interconnections Program were to be deferred by a year, the FTEs proposed as the

Future of Heat Engineering group are being added to broadly cover integration of Future of Heat needs including enhanced integration of non-pipeline alternatives, in addition to the workload increase resulting from ongoing RNG projects in the development pipeline. These positions are needed in the Rate Year even if the capital RNG Interconnection program is delayed. For these reasons, the Companies' Future of Heat Engineering FTEs should remain.

V. Proposed Adjustments to Customer Connections Programs

A. Gas System Reinforcements

Q. Please address the SGIOP's recommendation that the Companies' Gas System Reinforcement Programs in the Rate Year should be adjusted downward by 75 percent in the event that the NESE Project is not approved, rather than by 50 percent as the Companies recommend.

A. The SGIOP's recommendation is not supported and may jeopardize the Companies' ability to ensure reliable service and meet its obligations to serve increasing demand from existing customers. The Companies acknowledge that their proposed 50 percent reduction to this program in reaction to the absence of the NESE Project is a high-level estimate, as the Companies are continuing to evaluate the impacts of capacity constraints against the expected increase in demand from existing customers. The SGIOP's recommendation, however, is not informed by any specific analysis, but by simply assuming that the Companies' needs will be less than the Companies predict due to the moratorium on new connections if the NESE Project is not approved. The Companies are better positioned to estimate their own gas system reinforcement needs, even at a high level. The need for

reinforcements will be increased in some areas where the Companies expected the NESE Project to provide capacity relief, and the risk in accepting the SGIOP's recommendation is that this program will be underfunded, and the effects of the lack of the NESE Project will begin to negatively impact the reliability of firm service to existing customers. The Companies' proposed reduction was a conservative estimate; the SGIOP's reduction simply cuts too deep.

B. Customer Connections Unit Costs

Q. Please summarize the SGIOP's recommended adjustments to the Companies' unit costs for Customer Connections Programs.

A. The SGIOP generally agrees with the Companies' methodology for deriving the forecasts for Customer Connections installations but recommends that the unit costs be calculated based on an updated three-year average (FY 2017 to FY 2019) rather than the two-year average the Companies used (FY 2017 to FY 2018) for the Install Main, Install Services, Install Meter/Regulator, and Automatic Meter Reading programs.

Q. What is the Companies' response to this recommendation?

A. The Companies maintain that for a one-year rate plan, the two-year average of FY 2017 to FY 2018 is an appropriate proxy for setting the forecast for the Rate Year.

The Companies also note that, as stated in Section II (A) above, the units should be adjusted to properly reflect the program forecasts for new connections in the event that the NESE Project is not approved. The correction to the forecasts that are shown in Exhibit __ (GIOP-

1R) reflect the adjusted units and the Companies' methodology for deriving unit costs in the Rate Year and Data Years.

VI. Proposed Adjustments to Non-Infrastructure Programs

Q. Please comment on the proposed adjustments to the Non-Infrastructure Programs.

A. The SGIOP proposes to adjust the forecasts for several Non-Infrastructure programs based on updated or expanded historic averages. Generally, the Companies believe that averages based on FY 2017 and FY 2018 are an appropriate proxy for costs in the Rate Year. Additionally, for the Meter Testing Equipment Programs and KEDLI's Tools & Equipment Program, increases in costs follow the trend of increased capital workload and should not be reduced. For the Telecomm programs, the forecasts for the Rate Year reflect the need to replace facilities that are at or near the end of useful life.

VII. Proposed O&M Adjustments

A. IMP

Q. Please address the SGIOP's proposed downward adjustment to the O&M budget for IMP Inspections.

A. The SGIOP's recommendation is for a 100 percent reduction to the Companies' IMP non-labor O&M budget. The SGIOP's rationale for totally eliminating the incremental funding is that the SGIOP believes there has not been a significant increase to the IMP capital budget. The SGIOP's assumption that IMP O&M is proportionally tied to the capital budget is incorrect. In fact, the O&M budget is not directly tied to the capital IMP program in any given year. Rather, the O&M expense is directed toward conducting mandated inspections

and performing records reviews, not making IMP capital improvements. Each year's O&M budget for the IMP program is zero-based and is highly variable from year to year because different assets of varied size and characteristics might be in the work plan in any given year. By way of illustration, Exhibit __ (GIOP-4R) provides examples of the workplan variability, and the workplan for the Rate Year. The SGIOP's proposed reduction in O&M for IMP inspections and records reviews will not allow the Company to meet its regulatory requirements in the Rate Year.

Q. Is the SGIOP also proposing to reduce incremental FTEs for the IMP/IVP program?

A. Yes. For the same reasons stated above, the FTEs to support IMP/IVP are needed to support the increased O&M workload that is expected in the Rate Years and Data Years.

B. <u>O&M FTEs Supporting CapEx Workload</u>

Q. Does the SGIOP recommend removal of incremental FTEs the Companies proposed to support O&M work driven by increased capital workload?

A. Yes. The SGIOP notes that the Companies have proposed incremental FTEs for O&M in support of the increased capital workload, including additional inspectors, and the Companies have also proposed an enhanced Contractor Inspector program to add FTEs to bring the ratio of inspectors to crews to one-to-one. The SGIOP perceives that there is an overlap between FTEs in these two categories and recommends reduction of O&M FTEs supporting capital workload by nine FTEs for KEDNY and by three FTEs for KEDLI.

Q. Do the Companies agree with these adjustments?

A. In part. Upon further review, the Companies have determined that there is an overlap between six FTEs for KEDNY (five field inspectors and one supervisor) between O&M supporting capital workload and the Enhanced Contractor Inspector Program. Therefore, the Companies accept the reduction of six FTEs for KEDNY. This reduction is reflected in Exhibit __ (GIOP-2R) and Exhibit __ (GIOP-3R); however, the other FTEs are required to support increasing capital workload and do not overlap with the Contractor Inspector Program. The Companies oppose removal of these positions.

C. <u>Research and Development FTE</u>

Q. Does the SGIOP recommend an adjustment to the Companies' proposal of one FTE to be split between KEDNY and KEDLI for the Companies' Research and Development Program?

A. No, the SGIOP does not include this recommendation in its testimony or exhibits; however, removal of the 0.5 FTE for KEDLI is reflected in Staff's Revenue Requirements Panel Exhibit ____ (SRRP-1 and 2), Schedule 7(c). This adjustment is unsupported and unjustified. The Companies have proposed incremental funding for the Research and Development Program that Staff has not opposed. The addition of 0.5 FTE for each Company is reasonable for management of the additional programming. Moreover, the SPSP recommends *adding* certain items to the Companies' Research and Development Plan: (1) management of the Companies' proposed Enhanced High Emitter Methane Detection Program; (2) a plan for development of advanced RMDs; and (3) enhanced tracking and reporting for the Companies' Expanded Residential Methane Detector

Program. In light of these recommendations and increased work in this program, removal of the FTE is not appropriate.

VIII. Other Recommendations

A. LNG Tank Upgrade Plans Without the NESE Project

Q. Please explain the Companies' LNG Tank Upgrade projects and the SGIOP's recommendation to address the Companies' LNG tanks in the event that the NESE Project is not approved?

A. As is stated in the Companies' Supplemental Testimony, the Companies cannot undertake the planned upgrades of the LNG facility tanks if the NESE Project is not approved because the Companies cannot take the tanks out of service absent the additional capacity the NESE Project will provide. The Supplemental Testimony also notes that in the event the NESE Project is not approved, the Companies intend to conduct engineering design work to explore possible alternative projects to make improvements to the tanks without taking them out of service. Engineering and design work can also be done so that the tank upgrades can move forward to construction expediently whenever sufficient capacity is available to allow the tanks to be removed from service. The SGIOP recommends that the Companies file a proposal for addressing "needed tank repairs" by April of 2021. (SGIOP at 71).

Q. What is the Companies' response to this proposal?

A. The Companies are considering acceleration of engineering and design analysis to determine if alternative projects can address external LNG tank condition issues; some analysis work has already begun. To be clear, there is no way to address all needed repairs

to the LNG tanks without being able to access the insides of the tanks to assess the conditions. If the NESE Project is not approved, the Companies will not be able to take the tanks out of service without imposing curtailments of firm service for the duration of time it takes to complete the tank upgrades. Notwithstanding, the Companies believe that exploring external repairs may provide some degree of mitigation of the risk that the tanks will fail while the Companies continue to explore capacity solutions. This engineering and design work, even if accelerated, will take more time than a few months to complete, and the SGIOP's deadline of April of 2020 is unreasonable. The Companies will continue to engage with Staff on this issue and the timing of plans to address the LNG tanks.

B. Capital Reporting

Q. Do the Companies support the SGIOP's recommendations for capital reporting?

A. For purposes of a one-year rate plan, the Companies support capital reporting that is aligned with the reporting structure adopted in the 2016 KEDNY and KEDLI Rate Cases. The Companies are open to discussing modifications to reporting requirements that are not overly burdensome in the event of a multi-year rate plan.

C. New York City's Storm Hardening and Green Infrastructure Proposals

i. Storm Hardening in NYC

Q. What are NYC's other recommendations related to storm hardening of the Companies' gas facilities located in the New York City?

A. NYC makes the following recommendations related to storm hardening:

- Mini-gate stations: accelerate storm hardening of two mini-gate stations that are located in the flood zones, the Clifton Gate and the Citizens Gate stations, to the Companies' five-year capital plan (address by CY 2025);
- Meter set elevations: develop guidelines for standard elevations for new meter set installations in flood zones; and
- Greenpoint LNG: commission a third-party flood vulnerability study of the Greenpoint LNG facility to be completed by the end of the Rate Year.

Q. Have the Companies worked with NYC to address their concerns regarding storm hardening of the Companies' facilities?

A. Yes. As is noted by NYC, the Companies recently hosted a Storm Hardening Collaborative with NYC and other interested parties to develop storm hardening recommendations that are detailed in a report filed on April 26, 2018 in Cases 16-G-0058 and 16-G-0059. The Companies have implemented all of the recommendations in the report and met with NYC and the other collaborative parties to review implementation status early in 2019. Indeed, the Companies' proposed investment plans in this proceeding include specific investments (*e.g.*. LNG projects) that directly resulted from the collaborative.

Q. How do the Companies approach storm hardening of their gas systems?

A. The Companies manage their gas facilities holistically considering all system risks. As is demonstrated in the Companies' rate filing, and in particular the Future of Heat Panel's testimony, the Companies recognize the effects of climate change as a significant factor that increases system risks and is committed to continuously address these risks. However,

climate change impacts are not the only risk the Companies must contend with, and the Companies must allocate available resources in any given year toward projects and programs that meet various needs to ensure safe and reliable service to its customers. The Companies' capital and O&M investment plans filed in this proceeding reflect this balance and include significant projects to address storm hardening including the installation of remote shutoff valves and various LNG facility projects.

Q. Please address each of NYC's recommendations for additional storm hardening investments beyond what the Companies have proposed.

A. The Companies do not believe that additional investments are warranted in the Rate Year and Data Years to ensure system reliability.

- Mini-gate stations: The Companies' 10-year station work plan is risk-based and considers multiple significant factors including the age and current condition of each station. As a result of the Storm Hardening Collaborative, the Companies now specifically consider location in flood zone as a factor in this analysis. However, as NYC states, the Clifton Gate and the Citizens Gate stations, and an additional seven stations located within FEMA flood zones are included in the 10-year plan but are not currently scheduled to be addressed until after year five. This is due to the need to address other stations located outside of the flood zone that score higher from a total overall risk standpoint. To accelerate work on the stations as NYC suggests, either other stations that are riskier would need to be deferred, or the Companies would require significant additional resources to convert its 10-year plan into a 5-year plan for all stations. Even with significant additional funding, however, it is

not clear that such an accelerated plan is even possible given overall levels of qualified resources available to conduct the work. Additionally, the Companies and NYC note that regarding the Clifton Gate and Citizens Gate Stations, there are other system redundancies to mitigate flood risk. The Companies also re-assess all stations and the work plan every three years – so that if total conditions at stations within flood zones warrant, these stations may move up in priority in future iterations of the 10-year plan. In short, the Companies' approach to mini gate station refurbishment and replacements is reasonable and should not be revised based solely on flood zone locations.

- Meter set elevations: In accordance with the recommendations of the Storm Hardening Collaborative, the Companies conducted an analysis of the feasibility of standardizing guidance for meter set elevations in flood zones and do not recommend increasing the standard height design. The Company conducted outreach among other utilities via a survey regarding the feasibility and practices for increasing standard height design for meter sets and regulators among its peers. A majority of the respondents were not increasing the height of the meter sets or regulators. The Company also considered the geography of its service territory and found that the flood baseline varies drastically, which creates a significant challenge for determining an alternate standard design height. The Company is not recommending increasing the standard height design for regulators and meters sets for new installations. The Company is actively pursuing Remotely Operated Service Shutoff Valves for the services in the flood plains that will automatically shut off the gas service in the event of the flooding to mitigate risk for customers

and communities located within flood plains. This information was provided in the Companies' response to IR CNY-14.

- Greenpoint LNG: As NYC notes in its testimony, the Companies conducted a flood study in response to the Storm Hardening Collaborative. The study identified the critical components and facilities at the Greenpoint LNG site and used an overlay map that included flood risk data NYC provided. Several of the projects planned for the LNG sites in the Rate Year and Data Years address the risks identified. In short, the Companies are aware that the Greenpoint LNG site is subject to flooding and climate change impacts and are now targeting specific critical assets to address flooding risk. The Companies will continue to evaluate their 10-year capital investment plans for opportunities to make further improvements that address climate change impacts at this site. NYC's recommendation to conduct a third-party study that is more detailed could be considered in the later years of a multi-year rate plan (for example, in Data Year 3) to better inform future investments but would have limited value for Rate Year planning.

ii. NYC's Green Infrastructure Assets

Q. What are NYC's recommendations regarding their Green Infrastructure Assets?

A. NYC states concerns that the Companies' construction work is performed without adequate consideration of NYC's Green Infrastructure Assets. NYC recommends that the Companies install their facilities six feet from the curb rather than three feet, that the Companies provide notice to NYC agencies prior to street openings, and that the Companies attend additional training related to the Green Infrastructure Assets.

Q. What is the Companies' response to these recommendations?

A. The Companies are willing to coordinate with NYC on this issue and can discuss additional training that the Companies assume would be provided by NYC. Regarding installation set-backs, the Companies are unable to commit to a standard six-foot clearance for installations. The Companies' construction work is varied, and the Companies must operate within existing rights-of-ways, avoid conflicts with other infrastructure, and comply with other permit stipulations and municipal code requirements. The Companies are open to further discussions with NYC on this issue.

D. NYSLOF's Union Hiring Recommendation

Q. What is NYSLOF's recommendation regarding hiring of contractors?

A. NYSLOF recommends that the Companies hire only union labor from construction trade locals affiliated with the Building and Construction Trades Council of Greater New York.

Q. Please comment on NYSLOF's recommendation.

A. The Companies value the expertise and dedication that their large number of contracted employees bring to their jobs, the Companies' customers, and the communities in which they work, including the vast number of employees represented by organized labor. However, the Companies note that NYSLOF's recommendation is self-interested. NYSLOF states that it is affiliated with the Laborers International Union of North America and that, in New York, its members primarily work in the construction industry and are organized into more than 24 local unions and five district councils. (NYSLOF at 1).

Substantively, the Companies disagree with a proposal to limit their contractor work force to *only* union laborers. Such a restriction would remove market competition for labor resources, impairing the Companies' ability to obtain qualified contractor services at the least cost, and risking cost increases that customers would ultimately bear.

As NYSLOF notes, the Companies already employ many of its union members; indeed, most of the Companies' contracted labor are unionized. Notwithstanding, the Companies have a long and productive partnership with both organized labor and non-union contractors and vendors, as well as employment terms and approaches that balance the needs of employees, are best for customers, and meet the standards for quality, performance, safety and cost competitiveness. Although the majority of the Companies' contract work is performed by union labor – quality, safety and compliance are not mutually exclusive to either a union or non-union workforce.

The Companies' union and non-union labor force is heavily vetted, as NYSLOF acknowledges. (NYSLOF at 4). The Companies endeavor to contract only with companies that conduct their operations lawfully, safely and ethically and in compliance with all applicable laws and regulations, irrespective of whether they are represented by organized labor associations. All contractors that bid on the Companies' projects must meet ISN safety standards whether they are union or not. In addition, all contractors performing work on gas pipelines must have the associated Operator Qualifications for the tasks they are performing. Operator Qualifications are required per 49 CFR 192 & 195 to work on

gas pipelines and are not exclusive to union contractors. In addition to maintaining Operator Qualifications, welders are required to perform challenge testing to qualify to weld on the Companies' pipelines. Prior to being placed on the Companies' bid lists contractor vetting includes a review of their safety and technical training.

Additionally, as noted in the Panel's Direct and Corrections and Updates testimony, the Companies are implementing improvements to their contractor inspection, quality control and quality assurance, and operator qualification and training programs. Therefore, restricting hiring to union-only labor is not necessary and may lead to increased costs.

Q. Does that conclude your testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Second Supplemental Testimony

of

Gas Infrastructure and Operations Panel

Ross W. Turrini
Timothy S. Graham
Caroline Hon
Srividya Madhusudhan

Case 19-G-0309
Case 19-G-0310

Date: December 13, 2019

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudham.

Q. Is this the same Panel that previously submitted testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies") in these proceedings?

A. Yes. Previously-defined terms have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of the Panel's second supplemental testimony?

A. The purpose of the Panel's second supplemental testimony is to (i) update the Companies' capital and O&M expense forecasts to incorporate changes necessary to implement the recent settlement agreement between the Companies and DPS Staff in Case 19-G-0678 concerning the gas supply constraints in downstate New York and related moratorium on new customers connections, and (ii) update the Companies' cost of service to reflect the impact of contract negotiations with certain contractors.

Q. Does the Panel's second supplemental testimony include any exhibits?

A. Yes. The Panel's second supplemental testimony includes the following exhibits that were prepared or compiled under our direction and supervision:

- Exhibit ___ (GIOP-1SS): KEDNY and KEDLI's Revised Capital Investment Plans Reflecting New Customer Connections
- Exhibit ___ (GIOP-2SS): KEDNY and KEDLI's Revised O&M Plans Reflecting New Customer Connections
- Exhibit ___ (GIOP-3SS): KEDNY and KEDLI's Revised Proposed Incremental FTEs
- Exhibit ___ (GIOP-4SS): KEDLI's Revised Proactive Main Replacement (LPP) Unit Costs for FY2020-FY2024
- Exhibit ___ (GIOP-5SS): Worksheets Summarizing Changes to Capital Plan
- Exhibit ___ (GIOP-6SS): Worksheets Summarizing Changes to O&M Plan

III. <u>Updates to Incorporate the Settlement in Case 19-G-0678</u>

Q. Please summarize how the Companies presented their capital and O&M investment plans in these filings.

A. The Companies direct and corrections and updates testimony and exhibits were prepared under the assumption the NESE project would be in-service by the winter of 2020/2021, thereby allowing the Companies to continue connecting new customers. To provide the Commission with a full and complete evidentiary record in these proceedings, the Companies filed supplemental testimony on June 11, 2019, reflecting a cost of service under a no-NESE project scenario where the Companies would not be connecting new customers during the Rate Year and Data Years.

Staff, in its direct testimony, took the position that the Companies' capital and investment plans should be adjusted to assume the unavailability of the NESE project. Accordingly, in rebuttal, the Companies revised their capital and O&M investment plans to assume the NESE project would not be in service and that new customers would not be connected in the Rate Year or Data Years.

Q. Please describe the recent settlement in Case 19-G-0678 and how it impacts these rate filings.

A. On November 26, 2019, the Commission approved a settlement agreement between the Companies and DPS Staff in Case 19-G-0678 ("Settlement")

where the Companies agreed to lift the moratorium for the next two winters (through September 1, 2021). To enable new customer connections, the Companies have developed an operational plan and supply solution that provides for the expanded use of portable compressed natural gas ("CNG") injections to maintain system pressure during peak conditions, as well as various demand-side measures and the continued reliance on city gate peaking supplies. The Panel is supplementing its testimony and exhibits to reflect the impacts of the Settlement on various programs and cost components in the Companies' capital and O&M expense forecasts.

Specifically, the Companies are updating the capital forecasts for the Customer Connections programs to reflect the costs of new customer additions (installing main and services, meters, and AMR requirements) and the Gas System Reinforcements and Reliability projects necessary to maintain system pressures and serve additional load (*e.g.*, transmission upgrades). To a large extent, these costs were included in the Companies' initial filing on April 30, 2019 and updated in the Companies' July 3, 2019 corrections and updates filing.

The Companies are also updating their capital and O&M plans to reflect the costs of projects that are required to provide the additional on-system supply

necessary to meet peak demand, including new CNG and liquified natural gas ("LNG") facilities and upgrades. These investments are critical to enabling new customer connections and are expressly contemplated in the Settlement.

The updates are reflected in Exhibit __ (GIOP-1SS) and Exhibit __ (GIOP-2SS) and are described in detail below.

Q. Please describe the Companies' updates to the Customer Connections forecasts.

A. KEDNY and KEDLI's capital forecasts for the Install Main, Install Services, Meter Purchase, Install Meter/Regulator, and AMR line items have been updated to reflect the costs to connect new customers as agreed in the Settlement. These forecasts largely are consistent with the Customer Connections budgets presented in the Companies' corrections and updates filings.

Additionally, KEDLI has updated its Meter Purchases, Install Meter/Regulator, and AMR forecasts to remove costs associated with the Clean Choice Program because it is no longer proposing this gas conversion program. As updated in Exhibit __ (GIOP-1SS), the Meter Purchases,

Install Meter/Regulator and AMR forecasts now reflect meter and AMR requirements for new customer connections during the Rate Year and Data Years.

Q. Please describe the updates required to KEDNY's Gas System Reinforcement and Reliability programs to enable new customer connections.

A. The Companies' rebuttal filing reflected adjustments to KEDNY's Gas System Reliability Program due to the anticipated moratorium on new customer connections. The rebuttal filing also contemplated the addition of the Jamaica Inlet Project that was required to address low system pressures in the absence of the additional supply from NESE.

Now that the moratorium has been lifted, KEDNY's Gas System Reinforcement Program has been updated to remove the moratorium-related adjustments, thus restoring the forecast costs for this program to the amount reflected in the corrections and updates filing.

However, KEDNY's MRI Reliability project must be accelerated by one year to provide the hydraulic system support that is critical for enabling new customer connections over the next few winters. Exhibit __ (GIOP-1SS)

reflects the acceleration of the MRI project. Acceleration of the MRI project avoids the need for the Jamaica Inlet Project, which has been removed from KEDNY's capital forecast. Additionally, the Marine Park Regulator Station Program has been removed from the forecast because it was only needed to support the NESE project.

Additionally, over the longer term KEDNY requires one new system reliability project to support new customer connections. The Clove Lakes Transmission Loop Project has been added to the FY 2021 and FY 2022 capital plans to enable KEDNY to conduct a feasibility study and engineering analyses. Looping of the Clove Lakes Transmission Main with a new 30-inch transmission main across the borough of Staten Island will facilitate KEDNY's ability to take more gas through the TETCO Goethals Take Station and ultimately move this gas across KEDNY's system to supply constrained areas. This project will increase supply diversity and delivery capacity for long term growth to meet design day conditions as well as improve overall system reliability. This project is essential to maintaining pressure on the transmission system as part of KEDNY's longer-term plans to support new customer connections but is not expected to be in service until FY 2025, at the earliest. The forecast estimates for

conducting the feasibility and engineering studies are $0.5 million in FY 2021 and $1.5 million in FY 2022.

Q. Please describe the updates required to KEDLI's Customer Connections – Gas System Reinforcement Program and Reliability category programs to enable new customer connections in the absence of the NESE project.

A. KEDLI has also updated its Gas System Reinforcement Program forecast to remove the moratorium-related adjustments that were presented in the Companies' rebuttal filing. The forecast is restored to the version that was included in the corrections and updates filing to reflect reinforcements required to connect new customers.

Additionally, KEDLI's updates include adding back the Riverhead Transmission and Southeast Suffolk Phase 2 projects that were removed in the rebuttal testimony. These projects are required to support new customer connections in eastern Long Island. Moreover, to enable new customer connections, KEDLI has accelerated Phase 1 of the Southeast Suffolk project to ensure it will be in service in FY 2022. Accordingly, Exhibit __ (GIOP-1SS) reflects a shift of $5 million of the project forecast from FY 2022 to FY 2021.

Q. Are the Companies updating the costs for the Enhanced Contractor Inspection Program?

A. Yes. In the Companies' rebuttal filing, the Companies reduced the number of contractor inspectors required for this program because there would be a reduction in contractor crews due to the moratorium on new connections, and thus fewer inspectors were needed to meet the Companies' goal of a one-to-one inspector to crew ratio. Now that the moratorium is no longer in place, the contractor crew reduction will not occur. Thus, the Companies have adjusted the capital and O&M forecasts to the same levels that were submitted in the corrections and updates filing.

Q. Please explain the investments in CNG facilities that are required to support service to new customers.

A. The Companies' interim plans to ensure the peak day supply needed to serve new customers rely on the expanded deployment of CNG. There are two existing CNG injection sites in KEDLI's service territory at Glenwood and Riverhead (available January 2020), which will be utilized to serve peak demand for the winter 2019/2020.

For the winter 2020/2021, to meet increased gas demand from new and existing customers, the Glenwood CNG facility will be expanded to double

its capacity. Two additional CNG injection facilities (a third and a fourth CNG facility) will be required in KEDLI's service territory at locations to be determined. At this point, five potential properties that are owned by National Grid in KEDLI's service territory are being considered. The capital costs for each new facility are estimated at approximately $1 million in FY 2020 and $10 million in FY 2021. These costs are reflected in Exhibit __ (GIOP-1SS). The new facilities will also require incremental O&M expense that is included in Exhibit __ (GIOP-2SS).

Finally, the Companies require a CNG Filling Station to be located on Staten Island where trucks will be filled with CNG for transportation to both KEDLI's and KEDNY's CNG injection sites. The total cost for the project is approximately $30 million, and the project is expected to be in service in March of 2022. The cost is being reflected in KEDNY's capital investment forecast in Exhibit ___ (GIOP-1SS) because the asset is being built in KEDNY's service territory; however, KEDLI will pay 50 percent of the cost to KEDNY as a rental because the asset equally benefits KEDNY and KEDLI. KEDLI's rental expense related to this project is reflected in Exhibit ___ (GIOP-6SS) and KEDNY's rental income related to this project is reflected in Exhibit __ (RRP-3SS) and is included in the revenue requirement accordingly.

Q. Are updates required to the Companies' LNG investments to support new customer connections?

A. Yes. The Companies' plans for interim solutions to the capacity constraints involve additional reliance on the Companies' LNG capabilities, including updates to the Companies' existing LNG Tank Upgrade programs and to KEDNY's LNG Truck Load/Unload Station Program. Additionally, new investments are required for KEDNY's Greenpoint LNG site to add: (i) a new portable LNG facility and (ii) two additional vaporizers.

Q. Please describe the updates to the LNG Tank Upgrade Programs.

A. The Companies' LNG Tank Upgrade Programs as submitted in the April 30 rate case filing were initially designed as full replacements of the tanks. This project would require the tanks to be taken out of service for an extended period, including the winter heating season. As has been stated in the Companies' prior submissions, it is not possible to take the tanks out of service during the winter absent the additional supply capacity. However, considering the age of the tanks and the importance of reliable operation of the Companies' LNG facilities, particularly now that the Companies will be adding new customers over the next several winters, the Companies have designed a more streamlined and targeted approach to performing inspections and repairs on the tanks. KEDNY's Greenpoint LNG Tank 2

will be prioritized over Tank 1 because of known condition issues related to the foundation heating system. Tank 2 will be taken out of service from April of 2024 to October of 2024 to perform inspection and any necessary repairs so that the tank will be in service for the winter heating season of November 2024 to March 2025. Tank 1 will be addressed beginning in FY 2026. This modified work scope will ensure that the tanks are in service for the winter heating seasons.

KEDLI's Holtsville LNG Tank will be taken out of service during the Spring of 2023 and will be back in service by November of 2023.

Updates to the forecasts for the Greenpoint Tank 2 Upgrade Program, removal costs of the Greenpoint Tank 1 Upgrade Program, and updates to the Holtsville Tank Upgrade Program are reflected in Exhibit __ (GIOP-1SS).

Q. Please describe the update to KEDNY's LNG Truck Load/Unload Program.

A. KEDNY's Truck Load/Unload Program is a special project that is described in the April 30 filing on page 101 of Exhibit __ (GIOP-5). The program provides capability to transfer portable LNG in the event of an unplanned

outage at the Greenpoint liquefaction unit. The project was planned to begin in FY 2020; however, the project has been delayed because KEDNY and the City of New York have not yet executed a Memorandum of Understanding ("MOU") that would allow KEDNY to transport LNG via trucks through New York City. In light of the increased role LNG capabilities will play in enabling KEDNY to deliver reliable service to new and existing customers, the Company anticipates that the MOU will be executed in the near future. Thus, KEDNY's updated capital plan reflects delaying the investment in this project from FY 2020 to FY 2021 and compressing the remaining work plan to ensure that the station is in service by FY 2023 to remain on schedule to ensure reliability. Compression of the work plan requires an additional $2.44 million in project costs to cover expediting fees for faster equipment manufacturing.

Q. Please explain the new investments required for KEDNY's Greenpoint LNG site.

A. KEDNY's plan to ensure the continued ability to support customer load includes the addition of portable LNG capabilities at the Greenpoint site that will allow LNG delivered via truck to on-system injection points. The project is anticipated to be in service in November of 2022 and requires a capital investment of $0.5 million in FY 2020 and $4.5 million in FY 2021

for permitting, engineering and site work, and investments of $7.5 million in FY 2022 and $15 million in FY 2024 for construction.

Additionally, KEDNY's original rate case filing included a project to replace vaporizers 3 and 4 that will be complete in the Spring of 2020 and that will enable delivery of increased peak hour and peak day supply from the Greenpoint plant for the winter of 2020/2021 (the new vaporizers replacing 3 and 4 will be named 11 and 12 to avoid confusion on system schematics and drawings). To support increased customer demand beyond the 2020/2021 winter, KEDNY needs to add two additional vaporizers to the Greenpoint site, which will be Vaporizers 13 and 14. The LNG facility has enough pump capacity for Vaporizers 13 and 14. Each vaporizer is capable of outputting 2,500 Dth/hr of natural gas. Over a twenty-four hour period, this equates to 60,000 Dth per day (they are not run simultaneously). By comparison, a similar CNG project proposed in Greenpoint would decompress CNG at a rate of 2,200 Dth/hr up to 6,400 Dth per day. For a CNG injection facility to match the daily output of the vaporizers, it would require 143 CNG trailers. Vaporizers 13 and 14 are expected to be in service in FY 2023, and the costs of $1 million in FY 2020, $13 million in FY 2021, $21 million in FY 22, and $20 million in FY 23 are included in Exhibit __ (GIOP-1SS).

IV. Impact of Recent Contractor Agreements

Q. Please explain why the Companies are updating contractor costs.

A. The Companies have completed contract renegotiations with certain contractors that will provide necessary services in the Rate Year and Data Years. These negotiations have resulted in known and measurable decrease to the unit costs of KEDLI's Proactive Main Replacement (LPP) Program and an increase to the damage prevention mark out costs for both Companies, and, therefore, require updates to the capital and O&M expense forecasts.

Q. Please describe the update to the forecast for KEDLI's Proactive Main Replacement (LPP) Program.

A. KEDLI renegotiated its agreement with a contractor that provides services for the Proactive Main Replacement (LPP) Program. A copy of the renegotiated agreement was provided in the Company's response to IR No. DPS-1043.

Q. How does the agreement impact KEDLI's costs for this program?

A. The agreement results in reduction of the program's unit costs from $306 per foot to $304 per foot, as shown in Exhibit ___ (GIOP-4SS). Revising

the cost per foot reduces KEDLI's capital expense forecast by $1.929 million in the Rate Year, as shown in Exhibit ___ (GIOP-1SS).

Q. Please explain damage prevention mark outs and how the costs of these mark outs are included in the Companies' rate filings.

A. The Companies' Damage Prevention group performs mark outs of the Companies' underground utilities to allow for safe excavation and prevent damage to existing facilities, as required by the Commission's regulations. Damage prevention mark outs are performed when required to support one of the Companies' projects or when customers, third parties, and municipalities contact New York State's Dig Safe One Call Center. KEDNY uses internal employees to locate facilities on Company-related jobs and a contractor for all third-party Dig Safe mark out requests. Conversely, KEDLI sends all Dig Safe mark out requests, whether internal or external, to a contractor to mark out facilities.

The costs associated with damage prevention mark outs are included in KEDNY's O&M expense and KEDLI's O&M and capital expense. To prepare the rate case forecasts, the Companies increased these expense components from the Historic Test Year by standard annual inflation to arrive at the Rate Year forecast.

Q. Please explain recent developments that will affect the contractor cost component of damage prevention mark out costs.

A. On August 19, 2019, the Companies solicited contractor bids for their damage prevention mark out work. Bids were received from contractors on September 16, 2019. Thereafter, bidders were evaluated both commercially and on their ability to execute technical requirements. Extensive negotiations were also conducted with bidders between September 16 and November 8.

Q. Have the Companies selected contractors based on the bids received?

A. Yes. The Companies' are in the process of executing agreements with two contractors to provide damage prevention mark out services from February 2020 to December 2022. The agreements are expected to be executed on or about January 1, 2020. The Companies' are therefore updating the capital and O&M forecasts to include these increased costs.

Q. What effect will the updated agreements have on the Companies' contractor costs?

A. The contracts will raise the labor rates for these contractors in excess of the standard annual inflation rate used to derive the Companies' Rate Year forecasts.

Q. What are the drivers of the higher contractor bids?

A. The increase is attributable to three factors: (i) the prior contractor labor rates were based on long-term, dated agreements; (ii) recent unionization of the contractor labor force; and (iii) an increased scope of work to incorporate locator assurance technology. The Companies have not bid this work in over a decade. While the prior agreements provided for minimal price increases, the rates paid to those contractors were generally lower than market prices at the time. Contractors were no longer willing to accept below-market rates during the recent negotiations. Additionally, the contractors providing services under the selected agreements unionized during the late Spring and early Summer of 2019. While the Companies were generally aware that these labor forces were unionizing, the increase in labor rates experienced was significantly higher than anticipated or reflected in the Companies' initial filing. In addition, the new agreements require that the contractors use updated technology (locator assurance technology that enables real-time tracking of technician performance against industry standards to improve overall effectiveness of locating field work) when marking out facilities. This requirement expands the scope of work required in prior agreements and will require additional training and inventory for these contractors.

Q. What are the overall impacts of the changes to the contracts for damage prevention mark-out services to the Companies' capital and O&M forecasts?

A. Tables 1 and 2 below present summaries of the impacts of these agreements on the Companies' capital and O&M forecasts. These impacts are also reflected in the Damage Prevention line item in the updated Exhibit ___ (GIOP-1SS) and Exhibit ___ (GIOP-2SS).

Table 1: KEDNY Damage Prevention Mark Out O&M Updates FY2020 to Data Year 3

($000)	FY2021	FY2022	FY2023	FY2024
Original Forecast	7,588	7,979	8,445	9,005
Revised Forecast	14,417	14,755	15,102	15,460
Variance	6,829	6,776	6,657	6,455

Table 2: KEDLI Damage Prevention Mark Out Capital and O&M Updates FY2020 to Data Year 3

($000)	FY2021		FY2022		FY2023		FY2024	
	Capital	O&M	Capital	O&M	Capital	O&M	Capital	O&M
Original Forecast	3,315	1,637	3,384	1,766	3,455	1,914	3,528	2,083

Revised Forecast	8,339	3,574	8,339	3,574	8,339	3,574	8,339	3,574
Variance	5,024	1,937	4,955	1,808	4,884	1,660	4,811	1,491

Q. Did the Companies consider alternatives to use of contractors given the increase in costs for damage prevention mark out work?

A. Yes. The Companies conducted an analysis on whether it would be less expensive if this work was performed in-house. The results of that analysis determined that outsourcing the work was the least cost option. Procurement also performed an analysis that compared all bids received based on various criteria and determined that the contractors selected were the best overall option.

V. Conclusion

Q. Does this conclude the Panel's second supplemental testimony?

A. Yes.

MR. DECICCO: Thank you, Your Honor.

BY MR. DECICCO: (Cont.)

Q. Panel, with regard to the exhibits for your KEDNY Direct Testimony, did you sponsor nine exhibits that were identified as GIOP1 through GIOP9, which have been premarked for identification as Exhibits 58 through 66?

A. Yes.

Q. For your KEDLI Direct Testimony, did you sponsor eight exhibits that were identified as GIOP through GIO -- GIOP1, excuse me, through GIOP8, which have been premarked for identification as Exhibits 67 through 74?

A. Yes.

Q. For your KEDNY First Supplemental Testimony, did you sponsor three exhibits that were identified as GIOP1S, GIOP8S and GIOP9S, which have been premarked for identification as Exhibits 206 through 208?

A. Yes.

Q. For your KEDLI First Supplemental Testimony, did you sponsor three exhibits that were identified as GIOP1S, GIOP7S and GIOP8S, which have been premarked for identification as Exhibits 209

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through 211?

A. Yes.

Q. For you KEDNY Corrections and Updates Testimony, did you sponsor an appendix and five exhibits that were identified as GIOP1CU, GIOP6CU, GIOP7CU, GIOP8CU and GIOP9CU, which have been premarked for identification as Exhibits 266 through 271?

A. Yes.

Q. For your KEDLI Corrections and Updates Testimony, did you sponsor an appendix and five exhibits that were identified as GIOP1CU, GIOP3CU, GIOP6CU, GIOP7CU and GIOP8CU, which have been premarked for identification as Exhibits 272 through 277?

A. Yes.

Q. We're getting there. For KEDNY and KEDLI Rebuttal Testimony, did you sponsor six exhibits that were identified as GIPOR1 through GIOPR6, which have been premarked for identification as Exhibits 330 through 335?

A. Yes.

Q. And finally, for your KEDNY and KEDLI Second Supplemental Testimony, did you sponsor

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six exhibits that were identified GIOP1SS through GIOP6SS, which have been premarked for identification as Exhibits 369 through 374?

A. Yes.

Q. Were these exhibits prepared by you or under your direct supervision and control?

A. Yes.

Q. Are these exhibits complete and accurate to the best of your knowledge?

A. Yes.

MR. DECICCO: Your Honor, the panel is available for cross examination.

A.L.J. COSTELLO: Department of Public Service staff.

CROSS EXAMINATION

BY MR. FORST:

Q. Good morning.

A. (Panel) Good morning.

Q. I'm about to pass around a document, so just bear with us one second. So the document being passed around right now is a response to DPS request 1048. It's a three page response document, including attachments for a total of 15 pages.

A.L.J. COSTELLO: That will be marked as Exhibit 677.

BY MR. FORST: (Cont.)

Q. Panel, are you familiar with this document?

A. Yes.

Q. And was that document prepared by you or under your direct supervision?

A. Yes.

Q. And is the information contained in that document true and accurate to the best of your knowledge?

A. Yes.

Q. Excellent. Thank you. We're passing out another document, if you'll bear with us. Thank you. The document being passed around is a response document to information request DPS1048 supplemental. It's a two page response document that was submitted on December 3, 2019.

A.L.J. COSTELLO: That will be marked as Exhibit 678.

BY MR. FORST: (Cont.)

Q. Panel, are you familiar with this document?

A. Yes.

Q. And was this document prepared by you under your direct supervision?

A. Yes.

Q. And is the information contained in the document true and accurate to the best of your knowledge?

A. Yes.

Q. Excellent. Thank you. Panel, we're passing out a third document, if you'll bear with us.

MR. FORST: So Your Honors, the document that's being passed around currently is a response to DPS1088. What's being passed around currently is a two page response document submitted on February 13, 2020. It is a redacted version of a confidential request, which we also have copies of.

A.L.J. COSTELLO: That will be marked as Exhibit 679.

A.L.J. LEARY: Do we have the confidential?

MR. FORST: Yes. We'll be passing that around in one second. And Your Honors, we'd ask that the confidential version be given its own

number.

A.L.J. COSTELLO: And the confidential version we'll mark as 680.

BY MR. FORST: (Cont.)

Q. Panel, are you familiar with the documents, both the redacted and confidential versions?

A. Yes.

A.L.J. LEARY: Your mic is not on.

MR. FORST: My apologies.

BY MR. FORST: (Cont.)

Q. Panel, are you familiar with the documents before you, both the redacted and confidential versions?

A. Yes.

Q. And were these documents prepared by you or under your direct supervision?

A. Yes.

Q. And is the information contained therein true and accurate to the best of your knowledge?

A. Yes.

MR. FORST: No further questions, Your Honor.

A.L.J. COSTELLO: Okay. Thank you. Sane Energy.

MS. DIESCHE: So we're going to start by addressing the Metropolitan Reliability Infrastructure project and I'd like to direct your attention to Sane IR4, which you responded to on October 28th and was prefiled as Exhibit 581.

MR. DECICCO: Does the party have a copy of the exhibit?

MS. DIESCHE: I thought we didn't have to if it had been prefiled. We have copies of all the new ones, but I thought the thing was -- we didn't have to bring it if it was prefiled.

MR. DECICCO: I'm not sure what you mean by prefiled.

A.L.J. COSTELLO: Is there any -- does anyone -- yeah. Do we have access to this?

A.L.J. LEARY: Does the panel have this?

MR. TURRINI: We have a lot of stuff. We don't know if we have this.

MS. HON: I don't think -- I don't think I have that one.

A.L.J. LEARY: Okay. So take a look.

MS. DIESCHE: Can we -- can we look it up?

UNIDENTIFIED SPEAKER: We can look.

MS. DIESCHE: I'll look electronically and see.

A.L.J. LEARY: Can you give the identifying IR so potentially the panel can find it? They may not be able to find it as a hearing exhibit.

MS. DIESCHE: Okay. Sane IR4. The response was on October 28th, and it was prefiled as Exhibit 581.

A.L.J. LEARY: And do you have a copy for -- just one copy for circulation?

MS. DIESCHE: I don’t have a printed copy. No.

A.L.J. LEARY: Okay.

MR. GOODRICH: I do have it electronically. I can email it around if you would like.

A.L.J. LEARY: That would be great Mr. Goodrich. Thank you.

BY MS. DIESCHE:

Q. Was this response prepared by you or under your supervision?

A. (Turrini) Yes.

Q. Is it a -- do you have a true and correct copy in front of you?

A. Yes.

A.L.J. LEARY: Your mic is not on.

BY MS. DIESCHE: (Cont.)

A. Oh, I'm sorry. Yes.

Q. So in that response, the company said they have not done a greenhouse gas emissions evaluation of the Metropolitan Reliability Infrastructure project. Is that still true?

A. Yes.

Q. And why has an evaluation not been completed?

A. There's no regulatory requirement for us to recomplete that type of a study.

Q. Okay. And we're now going to pass out Sane IR8. We just put them all together, every document we're going to have so Kim doesn’t have to go around five times.

A.L.J. COSTELLO: That's -- no. That's fine.

MS. DIESCHE: But I'll - they're in order. I'll address them.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Thank you.

BY MS. DIESCHE: (Cont.)

Q. All right, so the document folks have in front of them is Sane IR8. The company responded on November 4th. It is a six page document. Was this response prepared by you or under your supervision?

A. Yes.

Q. Is this a true and correct copy?

A. Yes.

Q. In the response, the company said a pipe route between these points was then determined by studying surveys, completing test pits to identify utilities along the route and then ensuring that pipe can be installed on the selected route. Did any of the surveys look at demographics to determine if there are disadvantaged communities along the pipeline route?

A. No.

Q. Did any of the surveys look at co-pollutants in the communities along the pipeline route?

A. Could you define what you mean by

co-pollutants?

Q. I'm referring to it how it is defined in the CLCPA.

A. So could you just -- you're asking a question. Could you define it?

A.L.J. COSTELLO: Just for a moment. I just want to identify the document as Exhibit 681.

BY MS. DIESCHE: (Cont.)

Q. We can come back to that. In response to Sane R -- Sane IR8 question one, the company said "During these canvases, the team provided project information related to the project need, benefits and an estimated schedule verbally and in written -- in -- in writing to residents and businesses along and adjacent to the project route." Were residents informed about the cost of the pipeline and the impacts the rate hike would have on their monthly bills?

A. I don't know the answer to that.

Q. So I would now like to address the document…

A.L.J. LEARY: Mic.

BY MS. DIESCHE: (Cont.)

Q. Sorry. I would now like to

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address a document. It is titled Community Board Number One. It was filed as a public comment yesterday. It was a resolution passed on February 11th from Brooklyn Community Board One. It is three pages long.

A.L.J. COSTELLO: That will be marked for identification as Exhibit 682.

BY MS. DIESCHE: (Cont.)

Q. In this resolution, Community Board One opposes the Metropolitan Reliability Infrastructure project. Does the fact that the community opposes this pipeline change the company's position at all?

MR. DECICCO: Your Honor, can I just lay a little foundation as to whether the panel is familiar with this document or they've seen it before?

BY MS. DIESCHE: (Cont.)

Q. Have -- okay. Have you seen this document before?

A. No. We haven't.

A.L.J. LEARY: Do you want the panel to take the time now to review this document?

MS. DIESCHE: Yes, please. Can you

please become familiar with the document?

A.L.J. LEARY: Has the panel finished reviewing?

MR. TURRINI: Yes, we have.

BY MS. DIESCHE: (Cont.)

Q. Does the communities opposition to the pipeline impact the company's position at all to build it?

A. So having reviewed the document, it's actually riddled with inaccuracies and -- and a lot of falsehoods. It's a -- a lot of subjective opinions that are included in the document based on the facts of the requirements for the gas system in New York City and the -- our obligation to serve safely reliable the gas customers in New York City. No. It does not change our opinion about the need for the project.

Q. I'd like now to address the company's supplemental testimony that was filed in December on page six, lines 19 through 21.

A.L.J. COSTELLO: Is that KEDNY or KEDLI?

MS. DIESCHE: KEDNY.

BY MS. DIESCHE: (Cont.)

Q. So beginning line 19, the company said "However, KEDNY's MRI Reliability project must be accelerated by one year to provide the hydraulic system support that is critical for enabling new customer connections over the next few winters. How does the MRI project provide the hydraulic system support?

A.L.J. LEARY: I need a page number again.

MS. DIESCHE: Page six, lines 19 through 21.

MR. GOODRICH: I think this is the second supplemental, not the supplemental.

MS. DIESCHE: Oh, sorry, the one that was filed in December.

MR. DECICCO: Yep. For -- for the record, Your Honor, that's the second supplemental testimony that was filed on behalf of both KEDNY and KEDLI.

A.L.J. LEARY: Okay. Thank you.

MR. DECICCO: Page six.

BY MS. DIESCHE: (Cont.)

A. So -- oh, so I'll go ahead and answer the question. So the -- the reference in lines

19 through 21 of this testimony actually refer to the issue that's associated with the short-term settlement of NESE. So the issue that we had when it's -- implementing the moratorium in New York, as we testified yesterday, they talked about the -- the peak day and the seasonal supply. The actual issue that we've had for the last three years, which is projected out into the future is actually a peak hour issue. The peak hour issue is the four hours in the morning and the four hours in the evening when everybody gets up in the morning and everybody comes home from work when gas usage peaks. So those are the actual -- actual peak hours of the day. The Metropolitan Reliability Infrastructure project is a critical project that allows us to support that settlement and reliably move existing gas around our system in order to allow us to continue to connect new customers as part of that settlement.

Q. Are there particular areas within the company's service territory where this hydraulic support is necessary?

A. It's actually critical in what we call the Jamaica area of Queens. That is currently the low spot that we have on our system, and without

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the MR -- MRI Reliability project phase four for the winter of 21-22 we are projecting outages in that Jamaica area of Queens.

Q. I would now like -- yeah -- to look at Sane IR11.

A.L.J. COSTELLO: We'll mark that for identification as Exhibit 683.

BY MS. DIESCHE: (Cont.)

Q. So the company responded on January 9th. It is a two page document. Was this response provided by you or under your supervision?

A. Yes.

Q. Is this a true and correct copy?

A. Yes.

Q. In response to question four, the company said "Gas is currently stored in Greenpoint as CNG for use as a Queens transportation fuel as well as LNG. Those functions are unaffected by the construction of MRI." That was in response to a question asking if any gas from that pipeline will be stored in those LNG. So that -- I just want to make sure that I understand that that's saying that no gas will be stored from the pipeline as LNG or CNG.

A. So just to be clear on what that

pipeline does, that pipeline doesn't bring any incremental gas into New York. It allows us to reliably move our existing supplies of gas throughout New York City service territory. It loops the transmission -- main transmission main that runs through Brooklyn called the Brooklyn backbone. The criticality of looping that main allows us to do federally mandated inspection and maintenance work on that pipeline while keeping our system reliable. The Greenpoint LNG currently every year takes gas off of our gas distribution and transmission system and super-cools it and converts it and stores it in LNG form as opposed to the gases form. When you supercool it, it compresses over 600 times into a liquid, which allows us to efficiently store for use in the winter when the pipelines into New York City are constrained so when we can use it on the coldest days of the year to meet our peak day requirements.

Q. But will gas that will be transported by the pipeline be stored in the LNG -- as LNG or CNG?

A. Gas that we bring into New York City is getting converted currently into LNG and CNG and this pipeline gives us no incremental capability

to doing that. It just allows us to move gas efficiently around our service territory to balance the pressure and demand requirements.

Q. I'd like to I guess bring your attention then to the April filing where you say that additional capacity…

A.L.J. COSTELLO: Please refer to a page number.

MS. DIESCHE: Yeah. I'm trying to find a page number.

A.L.J. LEARY: And also identify direct supplemental rebuttal, second supplemental, etc.

MS. DIESCHE: Yes. Okay. So this -- the direct testimony filed in April, page 51.

A.L.J. COSTELLO: And it's for KEDNY?

MS. DIESCHE: KEDNY line seven.

BY MS. DIESCHE: (Cont.)

Q. The company says "The -- the MRI project will enable KEDNY to move an additional 850 dekatherms each day by 2021." I know that was in reference to the NESE pipeline being built. If the NESE pipeline is built, would this pipeline still allow the -- the company to move additional gas?

A. It's one of many ways that we would move the NESE gas around our system. Like I said, it's a reliability project that allows us to move gas around our system.

Q. And the company has applied to add trucking capabilities, LNG trucking capabilities, to bring gas in and out of the Greenpoint facility. If gas is brought in, would that be transported also by the MRI project?

A. So the company's applied and is negotiations with the City of New York from a memorandum of understanding to bring -- truck LNG through New York City on an emergency basis should there be an emergency need for incremental LNG supply during a very cold winter. That LNG would go into our -- be downloaded off of the trucks through a -- a trucking station, put into our LNG storage tanks in Greenpoint and then be vaporized then put into our system for use on the very coldest days. It could go anywhere in the system depending on how the demands and the draw on the system were that day.

Q. Thank you. I'd now like to bring your attention to Sane IR12. The company responded on January 31st and it is three pages long.

A.L.J. COSTELLO: That will be marked for identification as Exhibit 684.

BY MS. DIESCHE: (Cont.)

Q. In this response -- oh, sorry. Was this prepared by you or under your supervision?

A. Yes.

Q. Is this a true and accurate copy?

A. Yes.

Q. In that response, the company said "The company installs and maintains" -- when asked about the useful life of the MRI project, the company said "The company installs and maintains infrastructure to safely and reliably operate its system in a manner to provide the most benefit to the 1.3 million customers in New York City with minimal impact to the community served over the long-term. We estimate the assets would remain in service as long as needed to provide safe and reliable service to our gas customers in accordance with our regulatory obligations." Has the company done any assessments about what the actual like the years of the useful life of this investment would be?

A. Certainly, we understand the years of the useful life from an asset integrity

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standpoint.

Q. And how many years does the company?

A. I don’t have the specific timetable, but it -- when you put steel pipe in the ground, if it's ketotic, we protect it. In theory, it could last indefinitely.

A.L.J. LEARY: Excuse me. Would you pull the microphone closer?

MR. TURRINI: Sorry.

A.L.J. LEARY: No worries.

MR. TURRINI: Would you like me to repeat what I said?

A.L.J. LEARY: No. I could hear it. I just need everyone in the room to hear it.

MR. TURRINI: Okay.

BY MS. DIESCHE: (Cont.)

Q. Okay. I would like to now move onto liquified natural gas. Was a greenhouse gas emissions evaluation done for the two new LNG vaporizers the company is proposing to add at your Greenpoint facility?

A. No.

Q. Was a greenhouse gas emissions

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evaluation done of adding LNG trucking abilities at your Greenpoint facility?

A. No.

Q. Was a greenhouse gas emissions assessment done of the doubling the compressed natural gas capacities at the Glenwood facility?

A. No.

Q. What about adding two new compressed natural gas facilities somewhere within the service territory?

A. No.

Q. Were any of -- was a survey of the demographics done to determine if there were any disadvantaged communities near the Greenpoint LNG facility?

A. No.

Q. Was a study done to look at the demographics to determine if there were any disadvantaged communities near the Glenwood CNG facility?

A. No.

Q. Okay. I would now ask you to look at Sane IR14, which the company responded to on February 14th. It is one page long.

A.L.J. COSTELLO: That will be marked for identification as Exhibit 685.

BY MS. DIESCHE: (Cont.)

Q. Sane Energy asked that since a report that recently was published, a 20-month investigation, into radioactivity in oil and gas was published in Rolling Stone and it found that the fossil fuel industry has known since the early 70s that radioactive material is building up within pipes, pumps and compressors as scale. The company said that this does not change their position to not test for radon. Has the company, while replacing the leak-prone pipe, found scale within its pipelines?

A. So could you define what you mean by scale?

Q. It is a buildup of material within the pipeline, which has been found to be radioactive.

A. So we've found no radioactive material in our pipelines.

Q. You have tested for radioactive material in your pipelines?

A. So we are required to clean our pipelines under the mega rule. We test for the

required material that we're required to test for and we clean those pipelines using environmental contractors when we retire them.

Q. Is radioactivity, though, one of them?

A. I have not seen any reports of any radioactivity.

A.L.J. LEARY: So let the question be asked before you respond so that the record doesn't have both of you speaking at the same time.

BY MS. DIESCHE: (Cont.)

Q. You said that you test for what you're required. Are you required to test for radioactivity?

A. I do not believe we are required to test for radioactivity.

Q. But has the company found buildup of scale within your pipelines?

A. Periodically, we find mostly dust and other type debris in the pipelines.

Q. But not scale ever?

A. I'm not exactly sure what the definition of scale is in -- in whatever report was developed. I'd have to take a look at it.

Q. Thank you. That is the end of our questioning.

A.L.J. COSTELLO: Okay. Thank you. We received an email that AARP will be waiving its cross examination. So we're back to the companies if they have any redirect.

A.L.J. LEARY: I just have a question for Sane. Hold on. Let's see because you gave us what appears to be a letter.

MS. DIESCHE: Yeah. It was -- it --it hasn’t been filed as a comment yet, so I didn’t know if I could refer to it.

A.L.J. LEARY: Okay. So you're not asking that this be marked as an exhibit?

MS. DIESCHE: No.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Do you need a few moments or?

MR. DECICCO: No, Your Honors. No redirect by the company.

A.L.J. COSTELLO: Anything else? Judge Leary, do you have anything else? Okay. Okay, Panel, we thank you for your testimony. At this time, you're excused.

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(Off the record 10:04 a.m.)

(On the record 10:19 a.m.)

A.L.J. COSTELLO: I'm going to ask the panel starting with the gentleman that's furthest from me to state your names and business addresses for the record.

MR. FISHER: Brian Fisher, 3 Empire State Plaza, Albany, New York.

MR. SHI: Qin Shi, DPS Long Island, 125 East Bethpage, Plainview.

UNIDENTIFIED SPEAKER: This one.

MR. SHI: Sorry. Qin Shi, DPS Long Island Office, 125 East Bethpage Road, Plainview, New York 11725.

MR. WALTERS: Sean Walters, 125 East Bethpage Road, Plainview, New York.

MR. TINTERA: Mark Tintera, 125 East Bethpage Road, Plainview, New York.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

PANEL: I do.

BRIAN FISHER; SWORN

QIN SHI; SWORN

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SEAN WALTERS; SWORN

MARK TINTERA; SWORN

A.L.J. COSTELLO: Okay. Thank you. You may be seated. Counsel, you may proceed.

BY MR. FORST:

Q. Panel members, before you is a document entitled Prepared Testimony of Staff Gas Infrastructure and Operations Panel consisting of a cover page and 97 pages of questions and answers dated August 2019 and one exhibit submitted with your testimony labeled SGIOP1 Redacted, SGIOP1 Confidential. Is that correct?

A. (Panel) Yes.

Q. These exhibits have been premarked as Exhibits 463 and 464. Panel members, do you also have before you seven exhibits labeled as SGIOP2 through SGIOP8 marked as Hearing Exhibits 465 through 471. Is that correct?

A. Yes.

Q. Panel members, before you is a document entitled prepared Supplemental Testimony of Staff Gas Infrastructure and Operations Panel consisting of a cover page and 23 pages of questions and answers dated January 2020. Is that correct?

A. Yes.

Q. Do you have any changes or corrections to that supplemental testimony?

A. (Shi) Yes.

A. (Walters) No.

A. (Shi) Yes. On page eight, line six, the days should read 2016-2018. And on page eight, line eight, the days should read 2014-2018.

Q. Is this the only correction?

A. Yes.

Q. Panel, do you also have before you three exhibits labeled SGIOP9 Redacted, SGIOP9 Confidential and SGIOP9A marked as Hearing Exhibits 584, 585 and 586 respectively?

A. (Panel) Yes.

Q. Panel, do you also have three additional exhibits labeled SGIOP10, SGIOP11 and SGIOP12 premarked as Hearing Exhibits 526, 527 and 528?

A. Yes.

Q. Was this set of testimony and exhibits prepared by you or under your direct supervision?

A. Yes.

Q. If you were asked the same questions today under oath would you answer them the same way?

A. Yes.

Q. And do you affirm that the information contained in your testimony and exhibits to be true to the best of your knowledge, information and belief?

A. Yes.

Q. Your Honors, I would like to move that the prefiled Direct and Supplemental Testimony of the Staff Gas Infrastructure and Operations Panel be entered into the record as if given orally here today.

A.L.J. COSTELLO: That is granted. I assume that you will be providing the corrected testimony -- a version of the corrected testimony?

MR. FORST: Yes, we will Your Honor.

A.L.J. COSTELLO: Okay. So, at this point, the court reporter should insert DPS Staff Gas Infrastructure and Operations Panel Corrected Direct Testimony and DPS Staff Gas Infrastructure and Operations Panel Supplemental Testimony. #

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Exhibits of:
Staff Gas Infrastructure and
Operation Panel

Mark Tintera
Assistant Engineer (Electrical)

Qin Shi
Utility Engineer Specialist 2

Sean Walters
Professional Engineer 1

Office of Electric, Gas and
Water

125 East Bethpage Road,
Plainview, New York 11803

Brian Fisher
Utility Engineer Specialist 2

State of New York, Department of
Public Service, Office of
Electric, Gas and Water

Three Empire State Plaza,
Albany, New York 12223-1350

Introductions and Qualifications

Q. Members of the Staff Gas Infrastructure and Operation Panel (Panel), please state your names, employer, and business addresses.

A. Mark Tintera, Qin Shi, and Sean Walters. We are employed by the New York State Department of Public Service (Department). Our business address is 125 East Bethpage Road, Plainview, NY 11803. My name is Brian Fisher. I am also employed by the Department, however, my business address is Three Empire State Plaza, Albany, NY 12223.

Q. Mr. Tintera, what is your position at the Department?

A. I am employed as an Assistant Engineer, Electrical, in the Electric Distribution Section in the Office of Electric, Gas, and Water.

Q. Please summarize your educational and professional background.

A. I graduated from the State University of New York at Farmingdale in 2008 with a Bachelor of Science degree in Mechanical Engineering Technology. My professional experience includes six years designing drive components at

Designatronics Inc., one year designing mechanical systems at Genesys Engineering, and a total of two years at the Town of Huntington as a Building Permit Examiner, and Suffolk County as a Public Health Sanitarian overseeing and inspecting the installation of hazardous material storage systems. I have been employed by the Department since July 2014 and my responsibilities have included various aspects of electric and gas utility regulation. This includes reviewing utilities' emergency response plans, overseeing emergency drills, participating in rate cases, overseeing Federal Emergency Management Agency work, performing safety inspections on electric facilities, conducting electric load forecasting reviews, performing engineering analyses and reviews of rates and tariff filings, performing reliability analyses of electric service, performing interconnection reviews and providing recommendations, reviewing the utilities' substations, distribution circuits and other facilities construction documents, and analyzing the operation, maintenance, and construction

cost of electric and gas facilities.

Q. Have you previously testified in utility rate proceedings or other ratemaking proceedings?

A. Yes. I previously testified in Matter No. 15-00262, Long Island Power Authority – Three Year Rate Proposal, regarding the Delivery Service Adjustment and on the Storm Reserve Panel, as well as the Energy Efficiency and REV Panel. Most recently, I testified in Cases 16-G-0058 and 16-G-0059 that established gas rates for The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid (KEDLI) (collectively the Companies). My testimony was included in Staff's Gas Infrastructure and Operations Panel (GIOP) primarily relating to the subject area of reviewing growth capital expenditures.

Q. Mr. Shi, what is your position at the Department?

A. I am employed as a Utility Engineer Specialist 2.

Q. Please describe your educational background.

A. I graduated from University of Buffalo with a Bachelor of Science Degree in Civil Engineering

and a Bachelor of Arts Degree in Mathematics in 2005.

Q. Please describe your responsibilities and professional experience with the Department.

A. I joined the Department in February 2007. I started in the Electric Distribution Section where my responsibilities included monitoring and conducting observation of electric utilities' field activities in order to ensure utility compliance with the Public Service Law and with electric service and safety standards; investigating customer complaints associated with reliability issues; and responding to electric emergency events. In 2009, I joined the Water Rates Section. My responsibilities in that Section included performing cost and depreciation studies, developing rates and rate structures, analyzing operation and maintenance (O&M) expense accounts, and reviewing water companies' escrow accounts. In 2012, I joined the Major Utility Rates Section. My duties included the review and evaluation of electric and gas utility capital expenditure (CapEx) and O&M budgets and the engineering analyses of

electric and gas utility rate, statement, and tariff proposals. In 2014, I joined the Department's Long Island Office. My duties include the review of electric capital and O&M expense, emergency response plans, and performance metrics.

Q. Have you previously testified in utility rate proceedings or other administrative proceedings?

A. Yes. I have testified in several proceedings such as Cases 08-E-0539, 11-W-0082, 13-E-0030, 13-G-0031, 13-S-0032, and Matter 15-00262. Most recently, I have testified in Cases 16-G-0058 and 16-G-0059 that established gas rates for KEDNY and KEDLI. My testimony was primarily in the subject area of the Companies' sales forecasts.

Q. Mr. Walters, what is your position at the Department?

A. I am employed as a Professional Engineer.

Q. Please briefly describe your educational background and professional experience.

A. I received a Bachelor of Engineering in Mechanical Engineering from Stony Brook University in 2008. I then worked with Padilla

Construction Services, Inc. as a civil engineer from 2008 until 2014. My key responsibilities included quality control, contract compliance, testing and inspections, and budget oversight for construction contracts that included structural steel, concrete, earthwork, electrical, and mechanical systems.

Q. Please describe your responsibilities and professional experience with the Department.

A. I joined the Department in 2014 as a Junior Engineer. In 2015, I was promoted to Utility Engineer 1. I received my Professional Engineer license in 2016 and was subsequently promoted to Professional Engineer. My responsibilities with the Department have covered a wide range of New York State utility company operations. More specifically, this has included the review of capital expenses, O&M expenses, performance metrics, management audits, rate design and tariff structures, renewable energy policies, safety performance, and emergency response plans for conformance with New York State and industry-wide best practices.

Q. Have you previously testified before the

Commission?

A. Yes. I have previously testified before the Commission in other proceedings including Cases 16-G-0058 and 16-G-0059 that established gas rates for KEDNY and KELDI. My testimony was primarily in the subject area of capital expenditures.

Q. Mr. Fisher, what is your position at the Department?

A. I am employed as a Utility Engineering Specialist 2 assigned to the Gas and Water Rates and Policy Section in the Office of Electric, Gas and Water.

Q. Please summarize your education and professional experience.

A. I received a Bachelor of Science degree in Civil Engineering from Clarkson University in 2006. I also received a Master of Science degree in Environmental Engineering from the University of Notre Dame in 2009 and a Ph.D. in Civil Engineering from Southern Methodist University in 2012. I began my employment with the Department as a Junior Engineer in July 2012. While employed at the Department, I have

prepared, analyzed, and reviewed reports and studies involving operating revenues, sales forecasts, O&M expenses, embedded costs, revenue allocation and rate design. My current duties include the engineering analyses of gas and water utility rates, pricing, and tariff filings.

Q. Have you testified before the Commission in other proceedings?

A. Yes, I testified in Cases 13-G-0031 and 16-E-0060, Consolidated Edison Company of New York, Inc., regarding the embedded cost of service study, revenue allocation and rate design, and various proposed tariff changes. I also testified in Case 14-E-0318, Central Hudson Gas & Electric Corporation, regarding the ECOS study, revenue allocation and rate design, and proposed tariff changes, and Cases 17-E-0459 and 17-G-0460, regarding the net plant model, common capital including information technology projects, and price out. I testified in Case 18-E-0067, Orange and Rockland Utilities, Inc., regarding electric and common capital expenditures, and the net plant model. Finally,

I testified in Case 19-G-0066, Consolidated Edison Company of New York, Inc., regarding gas sales forecast, depreciation, embedded cost of service study, revenue allocation, firm and interruptible rate design and price out.

Summary of Testimony

Q. What is the purpose of the Panel's testimony in these proceedings?

A. Our testimony will recommend an overall level of capital expenditures (CapEx) for KEDNY and KEDLI projected for the Rate Year, which is the twelve months ending March 31, 2021 and to recommend a level of net plant and depreciation expense to be reflected in the Companies' revenue requirements. It is our position that the Companies should spend at the levels they deem appropriate to provide safe and adequate service. We are, however, recommending adjustments to the capital programs, which will impact the amount of plant in service balances and depreciation expense. These adjustments reflect the level of capital additions we believe the Companies have justified in their

initial testimonies, corrections and updates testimonies, and responses to requests for information (IRs) provided during the discovery phase of the proceeding.

Q. Please define the HTY, Rate Year, and Data Years in these proceedings.

A. The HTY is the twelve-month period ending December 31, 2018. The Rate Year is the twelve-month period ending March 31, 2021. The Data Years, One, Two, and Three, are subsequent twelve-month periods ending March 31, 2022, March 31, 2023, and March 31, 2024, respectively.

Q. What are KEDNY's and KEDLI's fiscal years (FY)?

A. The Companies' FY runs from April 1 to March 31. They are identified by the calendar year in which the FY ends, such that FY 2021 begins on April 1, 2020, ends on March 31, 2021, and is identical to the Rate Year.

Q. What is the purpose of reviewing the Companies gas capital budget in this proceeding?

A. The intent of reviewing the Companies' CapEx budgets is to determine the appropriate level of gas capital expenditures to be used in setting

rates for the Rate Year. Our recommended adjustments reflect the level of capital additions we believe are reasonable, given the Companies' initial rate case presentations, responses to IRs during the discovery phase of these proceedings, and information shared during in-person meetings. As such, we believe that with our adjustments, we are recommending the level of plant in service that is most appropriate for the Commission to use when setting rates.

Q. If the Companies complete projects that they deem appropriate to provide safe and adequate service at higher spending levels than forecast in the Rate Year, will the higher cost be reflected in rates?

A. No. The rates that a customer pays through base delivery rates will be set in accordance with the level of forecast net plant, and other cost of service items, that the Commission adopts in these proceedings. If the Companies add plant at levels in excess of the levels forecast in base delivery rates, we recommend no provision that automatically adjusts rates upward to

recognize that increase. Conversely, however, if the Companies add plant at levels less than provided for in rates, we recommend that the Companies credit customers the associated revenue requirement impact of the shortfall as compared to the forecast levels. This reconciliation is addressed in our testimony.

Q. What other items will the Panel discuss?

A. We will discuss, among other things: (1) the scope of our review of the Companies' CapEx budgets; (2) the scope of our review of the Companies' historical CapEx; (3) the Companies' CapEx budgeting process; (4) an overview of the Companies' CapEx budgets;(5) the Companies' proposed CapEx by budget categories; (6) the Companies' capitalization policy; (7) our proposed Rate Year net plant in service and depreciation expense for KEDNY and KEDLI; (8) the Companies' proposed incremental O&M expenses; (9) the Companies' City State Construction True Ups; (10) the proposed incentives for Leak Prone Pipe and customer connections in the Gas Safety and Reliability Surcharge; (11) the Capital Investment

Reconciliation Mechanism; and (12) various reporting requirements.

Q. Please describe the Panel's overall CapEx recommendations and adjustments.

A. We recommend reducing KEDNY's Rate Year direct capital budget, as shown in the Company's Exhibit___(GIOP-1), by $124.3 million with the following adjustments based on our review: (1) for the Customer Connection Category, we recommend a downward adjustment of $50.5 million; (2) for the Mandated Category, we recommend a downward adjustment of $37.6 million; (3) for the Reliability Category, we recommend a downward adjustment of $36.4 million; and (4) for the Non-Infrastructure Category, we recommend an upward adjustment of $0.2 million. Additionally, we recommend adjusting KEDNY's indirect capital budget: (5) for facilities, we recommend an upward adjustment of $0.09 million.

We recommend reducing KEDLI's Rate Year direct capital budget, as shown in the Company's Exhibit___(GIOP-1), by $157.5 million with the following adjustments based on our review: (1)

for the Customer Connection Category, we recommend a downward adjustment of $92.5 million; (2) for the Mandated Category, we recommend a downward adjustment of $18.3 million; (3) for the Reliability Category, we recommend a downward adjustment of $46.3 million; and (4) for the Non-Infrastructure Category, we recommend an downward adjustment of $0.3 million. Additionally, we recommend adjusting KEDLI's indirect capital budget: (5) for facilities, we recommend a downward adjustment of $0.03 million.

Q. Please describe the Panel's overall O&M expense recommendations and adjustments.

A. We recommend reducing KEDNY's Rate Year incremental O&M expense budget, shown in Exhibit___(GIOP-8CU), by $4.806 million and we recommend reducing KEDLI's Rate Year incremental O&M expense budget, shown in Exhibit___(GIOP-7CU), by $3.010 million.

Q. Please describe the Panel's overall FTE employee adjustment.

A. We recommend reducing KEDNY's Rate Year FTE count, as shown in Exhibit___(GIOP-9CU), by 11.6

FTEs and we recommend reducing KEDLI's Rate Year FTE count, as shown in Exhibit___(GIOP-8CU), by 7.9 FTEs.

Q. Please describe the Panel's depreciation expense recommendations and adjustments.

A. We ran our capital budget adjustments through the Companies' plant models, which resulted in an overall decrease in net plant in service and an associated decrease to depreciation expense from the KEDNY and KEDLI's Exhibit___(RRP-7CU), Schedule 1, Page 4.

Q. Please summarize the Panel's remaining recommendations.

A. We have additional recommendations concerning the Capital Investment Reconciliation Mechanisms, and further Reporting Requirements, all of which we address in greater detail later on in our testimony.

Q. Did you provide your adjustments to the Staff Revenue Requirements Panel?

A. Yes. We provided the average net plant in service balance, the associated annual depreciation expense, our recommended O&M expenses and our FTE adjustments to the Staff

Revenue Requirements Panel.

Q. In your testimony will you refer to, or otherwise rely upon, any information obtained during the discovery phase of this proceeding?

A. Yes. We will refer to various IR responses throughout our testimony. All of the IR responses are included in Exhibit___(SGIOP-1).

Q. Are you sponsoring any additional exhibits?

A. Yes. We are sponsoring the following additional exhibits:

Exhibit___(SGIOP-2) – contains an analysis of the Gas CapEx Variances;

Exhibit___(SGIOP-3) - contains a comparison of the Companies' Actual, Budget and Forecast versus the Panel's Forecast Capital Investment Graphs;

Exhibit___(SGIOP-4) – contains a Summary of the Panels' CapEx Adjustments by Budget Category;

Exhibit___(SGIOP-5) – contains a Summary of the Monthly Net Plant Results;

Exhibit___(SGIOP-6) - contains the Plant Model Results Comparison;

Exhibit___(SGIOP-7) contains a Summary of the Panel's O&M adjustments; and,

Exhibit___(SGIOP-8) – contains a Summary of the Panel's FTE adjustments.

Review of the Capital Budget

Q. Please explain the scope of the Panel's review of the proposed CapEx programs and projects.

A. We recommend a level of capital spending for the Rate Year and, therefore, reviewed all of the plans and projects from the beginning of the historic test year (HTY) through FY 2021, or the Rate Year. We then incorporated those adjustments into the plant in service model. We also reviewed and make recommendations concerning the CapEx budgets for the Data Year or FYs 2022, 2023 and 2024.

Q. Please explain how Department Staff typically conducts a capital investment review during a rate proceeding.

A. Staff typically reviews how a company plans and controls its major capital expenditures by reviewing and evaluating the company's processes and procedures. We also evaluate whether the company is pursuing its budget priorities with sufficient oversight from the Board of

Directors, with executive management, and then measuring the project's progress during the year. We also analyze how load forecasting impacts capital planning by addressing current and future loads.

Q. Please describe how the capital budgeting process should be performed.

A. The capital budgeting process should be both a top-down and bottom-up process. The top-down perspective should represent the strategic goals and objectives for the utility and be incorporated into both the capital and O&M budgets. The company should incorporate these strategic goals and objectives in the planning process. The bottom-up budgeting consists of identifying capital requirements on a project-by-project basis at the departmental level. Ultimately, capital budget development should primarily consist of a bottom-up assessment of system needs giving due consideration to top-down financial considerations.

Q. Please explain how the approval process functions within the capital budgeting process.

A. The approval of the capital budget by the Board

of Directors should not constitute authorization to proceed on the individual projects. Project managers should complete design engineering and preparation work to be submitted for formal budget approval, which should then be followed by a funding authorization. Our review looks to determine whether this process is being followed. After the authorization is granted, the company should closely track, on a monthly basis, the actual expenditures for each project to the initial authorized planned expenditures. Variance reports should provide sufficient detail as to why projects are over- or under-spent, or if there exists a need for a project with a greater priority to supersede the need for spending on any other particular project. Management should closely track the company's expenditures and inform the Board of Directors on a timely basis of any issues that come to management's attention.

Q. What capital planning information did the Panel review in these proceedings?

A. We reviewed the processes and procedures used by the Companies, including historic expenditures,

forecast expenditures, and changes to O&M expenses related to changes to the forecast expenditures. We believe the Companies are required to justify each and every capital program or project, and related projected O&M change.

Q. Please explain why the Panel conducts a historical review of expenditures.

A. The purpose of a historic review is two-fold. It provides insight as to the company's capital spending performance and it is a final review of actual expenditures before they are incorporated into rate base.

Q. What does the Panel request from a company to conduct a historical review?

A. We expect that, on the day a company files its rate case, it will provide the actual and budgeted CapEx amounts for the last five historic years, including the HTY, in aggregate and, for mass activities, by blanket project grouping or otherwise by specific project. Historic budgeted expenditure levels should be the levels approved by the company's Board of Directors for each historic period. We expect

that the reporting format stay consistent from year to year. We also expect in a rate case that the company will be able to provide a full descriptive analysis to detail for each project line item, by year, whether project schedule slippage, scope change, cost variance, or a combination thereof was the basis for a deviation from budget.

Q. Please explain how the Panel typically reviews a company's forecast CapEx budget.

A. We review each CapEx project or blanket grouping by its associated line item. We review: (i) a fully detailed description, including all studies and alternative analyses, on a project- or program-specific basis; (ii) a justification of project expenditures and budgets; (iii) the current construction schedule, with major milestones and in-service dates included; (iv) the project- or program-specific cost/benefit analysis; and (v) the associated corporate management project authorization or sanction papers.

Q. Why is the review of this information important?

A. Review of the documentation provided by the

utility is how we verify that a company developed its plans to spend its capital budget in the best interest of customers.

Q. Does the Panel also review the related and potential O&M costs and savings associated with the forecast capital expenditures?

A. Yes. We typically review by program or project, either aggregated by blanket or individual project, each O&M program change in line with the capital plan. We also typically ask for and review a detailed description of the related O&M program; the related actual and budgeted O&M expense amounts for the last five historic years by program; the related O&M expense for both actual and planned units of work completed on annual basis for the last five years; a justification for the related O&M program change; a cost breakdown of the related O&M expense; and, the planned units of related O&M expense work to be completed in future annual periods.

Q. Why is documentation of O&M expenditures necessary and important?

A. Like the forecast CapEx documentation, there are

typically impacts to O&M expenses when changes to the capital budgets are made. Without this documentation, we cannot be sure that the associated forecast O&M costs are appropriate in the Rate Year.

Q. How will the Panel address the Companies' capital expenditures in their "five year forward look" capital plan?

A. We will discuss our historical review and our review of the forecast.

Historical Review

Q. Did the Panel review the Companies' historical CapEx variance reports?

A. Yes. We reviewed the Companies' responses to IR DPS-084, which provided the Companies' FY 2014 through FY 2018 history of planned and actual gas capital expenditures by grouping and in total. We also reviewed the CapEx quarterly reports filed with the Secretary to the Commission in Cases 16-G-0058 and 16-G-0059 that detailed the Companies' calendar year history of planned and actual capital expenditures by grouping and in total, and the Companies'

response to IR DPS-914, which shows the most recent CapEx variance reports for FY 2019.

Q. What did the Panel learn from its analysis of this data?

A. We made several observations. First, the Companies consistently sort their capital expenditures into categories depending on the overall objective of the project. The four primary categories include Customer Connections, Mandated, Reliability, and Non-Infrastructure. Second, the average annual increase in capital expenditures between FYs 2017 and 2019 was 23.8 percent and 22.2 percent for KEDNY and KEDLI, respectively. Lastly, we observed that there are always variances between the Companies' capital budgets and the amount expended in any given year.

Q. Can you briefly explain some of the variances that you observed in the variance reports?

A. Yes. As shown in Exhibit___(SGIOP-2), which was developed using the Companies' response to IR DPS-084 and DPS-914 , there were instances of significant variances between planned and actual spending. Our variance analysis shows that the

Companies consistently over-estimated their budgets for some accounts. An example of a budget category that is typically over-estimated, as shown graphically in Exhibit___(SGIOP-3), is the Gas Reliability budget where in FY 2017 through FY 2019 KEDNY had the following variances: -36 percent, -30 percent, and -18 percent, respectively; and KEDLI had the following variances: –23 percent, -21 percent, and –36 percent, respectively.

Capital Budgeting Process

Q. Explain how the Companies generally develop their annual CapEx budgets.

A. Each year, as explained in the Companies' Gas Infrastructure and Operations Panel Direct Testimony, the Companies develop a ten-year capital plan. The process begins with the Companies compiling proposed spending for programs and individual capital projects. These programs and projects are categorized into one of four spending rationales (Customer Connections, Mandated, Reliability, Non-infrastructure).

Q. How do the Companies determine which programs and projects to include in their capital plans?

A. The Companies are currently using a tool, C55, for the prioritization and optimization of capital investments. This tool uses a risk scoring and prioritization methodology to optimize the gas investment portfolio. After the Companies rank the investments in C55, meetings are scheduled with budget sponsors to review value models, project rankings, and various scenarios produced throughout the optimization of investments to meet capital planning targets.

Q. After the capital plan is complete, who approves it?

A. The New York Jurisdictional President – currently, Company Witness John Bruckner - and the Vice President, Finance, New York - currently, Company Witness David Doxsee – review the capital plan in late fall. The New York Jurisdictional President reviews the overall customer, service quality, and financial impacts of the investment plan as part of the business planning process and may request changes to the

level or mix of investments. In early winter, the capital plan is presented to the National Grid USA Board of Directors and the National Grid plc Executive Committee. In early spring, the capital portfolio is presented to the National Grid plc Board of Directors for review and approval.

Q. Are additional approvals needed before a project in the annual capital plan may proceed?

A. Yes. Specific delegation of authority approval must be obtained for any project in the capital plan to proceed. The delegation of authority approval process includes the sanctioning documentation and review for projects over $1 million. Smaller projects under $1 million do not require sanctioning and are approved through a supervisory delegation of authority hierarchy based on certain established thresholds. Projects with estimated costs between $8 million and $25 million are reviewed and approved by the U.S. Sanctioning Committee. Projects with estimated costs between $25 million and $203 million are reviewed by the U.S. Sanctioning Committee and then forwarded to the Senior

Executive Sanctioning Committee for review and approval. For projects between $1 million and $8 million, the U.S. Sanctioning Committee delegates review to an informal committee led by the Investment Planning group and includes, but is not limited to, managers and directors from the regulatory, estimating, asset management, and procurement groups.

Q. What is the sanctioning process?

A. The sanctioning process identifies the appropriate spending levels, by specific programs or projects. Sanction papers provide cost and project details, as well as potential alternatives and the ramifications of those alternatives so that the Companies can make informed decisions regarding capital forecasts and associated adjustments.

Q. How does the sanctioning process affect Staff's review of the capital plan?

A. Without sanction papers, it is difficult for Staff to appreciate whether the Companies' capital forecasts reflect its true need for investment, if alternatives were considered, if the most appropriate project was selected, and

whether cost estimates included in the budget are reasonable. The Companies state that the timing of the sanctioning process is not aligned with the capital planning process used to forecast the Rate Year forecast. Projects are normally sanctioned immediately prior to the FY in which the investment is planned, whereas capital budgets are developed on a longer term investment plan. Consequently, sanctioning papers have not yet been developed for all projects included in the capital plan. Due to this timing difference, there are cases where we support certain projects or programs but are skeptical of the cost estimates provided.

Overview of KEDNY's and KEDLI's Capital Budgets

Q. Summarize the Companies' historic capital investments for the most recent five FY period between 2015 and 2019.

A. According to the Companies' responses to IRs DPS-084 and DPS-914, KEDNY spent approximately $345.484 million, $450.042 million, $474.405 million, $662.196 million, and $715.456 million during FYs 2015 through 2019, respectively.

KEDLI spent approximately $226.309 million, $245.677 million, $252.397 million, $310.179 million, and $377.120 million during FYs 2015 through 2019, respectively.

Q. Describe the capital spending plan the Companies proposes for the period 2021 through 2024.

A. As shown in Exhibit___(GIOP-1CU), of the Companies' Corrections and Updates Testimony filed on July 3, 2019; KEDNY proposes spending levels of $770.371 million, $916.275 million, $895.145 million, and $940.102 million for FYs 2021 through 2024, respectively. KEDLI proposes spending levels of $528.979 million, $550.887 million, $549.611 million, and $587.513 million for FYs 2021 through 2024, respectively.

Q. Do the Companies assume that the North East Supply Enhancement (NESE) Project will be constructed?

A. Yes.

Q. Has the Staff Policy Panel instructed you to make recommendations assuming that the NESE Project will not be available?

A. Yes.

Q. Did the Companies provide additional testimony

explaining how the instant cases would be impacted if the NESE Project was further delayed or not constructed?

A. The Companies filed supplemental testimony on June 11, 2019 that detailed how their cases would change under a delayed or no NESE Project scenario.

Q. Describe the capital spending plan the Companies propose for the period FY 2021 through 2024 assuming the NESE Project is not constructed.

A. As shown in Company Exhibit___(GIOP-1S), KEDNY proposes spending levels of $814.488 million, $749.925 million, $754.556 million, and $823.624 million for FYs 2021 through 2024, respectively. KEDLI proposes spending levels of $423.612 million, $426.838 million, $432.378 million, and $452.766 million for FYs 2021 through 2024, respectively. The adjustments the Companies made to their budgets due to the NESE Project not being constructed are discussed in more detail later in our testimony.

Q. How are the Companies' CapEx investment plans categorized?

A. The Companies' forecast gas capital investments

are categorized into four spending rationales: (1) Growth; (2) Mandated; (3) Reliability; and (4) Non-Infrastructure.

Q. How did the Panel review the Companies' capital forecasts for the Rate Year, and FYs 2022 through 2024?

A. We reviewed the testimony, exhibits, and IR responses. We then held technical conferences, interviewed witnesses, and issued additional IRs.

Q. What are the Panel's recommendations regarding the Companies' forecast capital investment plans?

A. We generally support the Companies' plans, but recommend specific adjustments, including reductions associated with the NESE Project's unavailability during the Rate Year, to each of the budget categories, as explained further on in our testimony.

Customer Connections Category

Q. Please summarize the Companies' Rate Year forecasts for the Customer Connections category.

A. As shown in Exhibit___(GIOP-1CU), the Companies'

forecast for Customer Connections is the sum of the installation of main, services and meter/regulators, minus customer contributions, meter purchases, AMR, gas system reinforcement, and other specific projects associated with customer connections. The total Customer Connections category forecasts shown in the Companies' Corrections and Updates for the Rate Year are approximately $80.61 million and $113.89 million for KEDNY and KEDLI, respectively.

Q. Explain how the Companies developed their Customer Connection budgets.

A. The Companies' response to IR DPS-347 described how the Companies developed their capital budgets for Customer Connection category. The Companies stated that the budget for each line item under the Customer Connections category with the exception of "Contributions and Gas System Reinforcement," is the product of forecast units multiplied by unit costs. The unit costs are based on a two-year average escalated at 2.5 percent per year. The Companies' response to IR DPS-347, Attachments 1

and 2, provided the estimated units and unit cost for KEDNY and KEDLI from FY 2020 through FY 2024, respectively. Attachment 3 shows the development of unit costs using a two-year average of historic years actual spends from FY 2017 and FY 2018 as the basis then escalated by a 2.5 percent annual inflation rate for all cost types except contractors, which utilized a 3 percent annual inflation rate. This product was then divided by the two-year average of the units from FY 2017 and FY 2018 to develop unit costs as basis for forecasting. The forecast units for each line item under Customer Connection are based on various factors that impact the number of new connections including: (1) anticipated changes in various market segments, (2) project inventories, (3) any rate/regulatory changes, (4) system constraints, and (5) other external factors, such as fuel pricing, economic conditions, and new building forecasts.

Q. Have the Companies proposed to increase their Customer Connections category budget as compared to their historic investment plans?

A. KEDLI proposes an increase, but KEDNY does not.

Q. Please explain KEDLI's changes.

A. KEDLI proposes to increase its spending for many of the items in the Customer Connection category, which results in significant increases to the total customer connections budget, which we summarized in Exhibit___(SGIOP-3). KEDLI's annual average expenditures budgets for the Customer Connection category over the past five FYs, 2015 through 2019, was $101.850 million. For the past three FYs 2017 through 2019, the annual average actual expenditures was $92.359 million. KEDLI proposed a Rate Year Customer Connection category budget of $113.89 million, which is an approximate 12 percent increase over KEDLI's five-year annual average expenditure from FY 2015 through FY 2019, and an approximately 23 percent increase over the three-year actual annual average expenditures from FY 2017 through FY 2019.

Q. Did the Companies file supplemental testimonies to address the scenario that the NESE Project will not be available in the Rate Year?

A. Yes.

Q. Describe the Customer Connections CapEx plan the Companies propose for the Rate Year assuming the NESE Project is unavailable in the Rate Year.

A. As shown in Company Exhibit___(GIOP-1S), KEDNY proposes budgets of $34.531 million and KEDLI proposes $29.013 million. In addition, KEDNY requests $520,000 for the Jamaica Inlet Project in the Rate Year to resolve an existing low inlet pressure situation without the NESE Project. Originally, this situation could be resolved by the incremental NESE Project capacity.

Q. If the NESE Project is unavailable in the Rate Year, does the Panel propose any adjustments to the Customer Connections budget?

A. Yes. We recommend two adjustments: (1) the unit costs for the install main, install service, meter purchase, install meter/regulator and AMR should be developed using a three-year average, and (2) the Gas System reinforcements should be reduced by 75% for the level projected in corrections and updates. Based on our review, we recommend a downward adjustment of $50.506 million and $92.545 million for KEDNY and KEDLI,

respectively, in Rate Year, as shown in Exhibit___(SGIOP-4).

Q. Please explain why the Panel recommends using a three-year average to project unit costs.

A. For KEDNY, we found that the main unit cost increase for the most recent years was premised on the City of New York's (NYC) new restoration and paving requirements. Although we believe a two-year average unit costs seems reasonable, the new requirement only became effective in April of 2017, and thus cost increases associated with the new requirement could not be fully captured in 2017 results. In addition, based on the Companies' two-year average we found the unit costs (except Install Meter/Regulator) for KEDNY are too low but the unit costs for KEDLI appear high (except Install Main Commercial, Tariff Main, Install Meter, and Meter Purchase). More specifically, we believe the Companies did not accurately reflect the new costs associated with the NYC requirement by using a two-year average of FY 2017 and FY 2018.

Q. What does the Panel recommend for developing unit costs?

A. We recommend using a three-year average from FYs 2017 to 2019. We believe using a three-year average including 2019 data would more reasonably and accurately reflect the increase in unit costs due to NYC's new paving requirements.

Q. When the Panel used a three-year average cost, what were the results?

A. As shown in Exhibit___(SGIOP-4), we multiplied the three-year average unit costs with escalating inflation by the Companies forecast units for each of the budget line items, however, we also removed Clean Choice Program Main and Service from KEDLI's Capital budget under Customer Connection as recommended by Staff Policy Panel. This resulted in a decrease of $38.187 million and $75.847 million for KEDNY and KEDLI for the budget items projection be based on units and unit costs under Customer Connection category, respectively.

Q. Does the Panel agree with the Companies' Gas System Reinforcement budget under the Customer Connection Category assuming the NESE Project is not available in the Rate Year?

A. No.
Q. Please explain.
A. We reviewed the Companies' supplemental filings and their response to DPS-723, which reduced the budget of Gas System Reinforcement for both KEDNY and KEDLI by 25% in FY 2020, by 50% in FY 2021, and 75% in FY 2022 through FY 2024. The Companies' response to DPS-996 also indicated that the Companies' was applied a percentage reduction to the blanket reinforcement program based on assumed reductions in new customer connections without the NESE Project. The Companies could not provide supporting documentation or workpapers to support how these reductions or the percentage used to reduce the program budgets were developed.
Q. As of mid-May 2019, have the Companies attached customers to their distribution systems?
A. Except for KEDNY's Staten Island service area, the Companies have stopped adding new firm customers due to the NESE Project being unavailable, which means only those firm customers who submitted an application prior to May 16, 2019 have or will be added to the

system.

Q. Typically, how are reinforcement projects developed.

A. The Companies project a level of customer connections and if those connections result in pressure problems, the Companies develop projects to address the concerns.

Q. If the Companies are generally not attaching customers, should their reinforcement projects be reduced?

A. Yes. We believe the reinforcement projects that are needed to support these additional loads will most likely be curtailed; beyond the Companies estimate.

Q. What does the Panel recommend the Gas System Reinforcement budget assuming the NESE Project is not available in the Rate Year?

A. We recommend to reducing the Companies proposed Reinforcement budget for both KEDNY and KEDLI by 75% in the Rate Year. Given the situation just discussed, a 25 percent further reduction should enable the Companies to continue to provide safe and reliable service to existing firm customers in the Rate Year. This results in a decrease of

$10.231 million and $16.079 million for the Gas System Reinforcement budget for KEDNY and KEDLI, respectively.

Q. Summarize the Panel's adjustments assuming the NESE Project is unavailable in the Rate Year.

A. We used the three-year average unit costs with escalating inflation multiplied by the Companies forecast units considering the NESE Project is unavailable, which the Companies provided in their response to DPS-948. This would result in downward adjustments of $38.187 million for KEDNY and $75.847 million for KEDLI in Rate Year.also informed us that the Clean Choice Programs should be eliminated. These changes result in an upward adjustment of $8.773 million for KEDNY and a downward adjustment of $22.044 million for KEDLI, respectively, to the Companies Rate Year corrections and updates budgets, as shown in Exhibit___(SGIOP-4).

Mandated Category

Q. Please describe the Mandated category, and the Companies' spending forecasts for this category.

A. The Mandated spending category includes the

projects and programs needed to comply with regulatory obligations, to coordinate with third party projects, and to meet commitments of capital investments to further enable the safe operation of the system. The mandated section comprises the largest portion of the Companies' capital budgets, representing approximately three quarters of KEDNY's total budget, and half of KEDLI's total budget. Some of the projects and programs included are the City/State Construction (CSC) or Public Works projects, corrosion control program, and the proactive and reactive main and service replacement programs. The mandated category also includes pipeline integrity management program (IMP) and integrity verification program (IVP), numerous inspection and maintenance programs, the programs to manage the Companies' inventory of customer gas meters, among others.

Q. What levels of spending do the Companies propose for the mandated program?

A. As shown in the Companies' corrections and updates testimony Exhibits___(GIOP-1CU), the Companies' forecast total Mandated Rate Year

expenditure levels of $625.472 million and $287.365 million for KEDNY and KEDLI, respectively. The largest expenses in this category for KEDNY are the CSC/Public Works program, and the Main Replacements (Proactive)-Leak Prone Pipe programs (LPP Program). Together, these two programs account for 86 percent of the expenses in KEDNY's Mandated category. The largest expense for KEDLI is related to Main Replacements (Proactive) - Leak Prone Pipe (LPP program), which accounts for 82 percent of the expenses in KEDLI's Mandated category. Other significant programs for KEDLI include CSC/Public Works, the IMP, and the Contractor Safety Inspector Program.

Q. Have the Companies proposed to increase spending in the mandatory category from levels established in its previous capital investment plans?

A. As shown in Exhibit___(SGIOP-3), the Companies are proposing continuously increasing levels of spending for the mandated category. Compared to the three year average of actual expenditures in FYs 2017 through 2019, the Companies' Rate Year

proposals are 41 percent, and 47 percent higher for KEDNY and KEDLI, respectively. The major drivers of the increases are escalating unit costs associated with main installation and replacement, increases in work load, and new programs.

Q. Did the Companies provide any updates to their Mandatory capital expenditures for the Rate Year, considering the potential impacts of the NESE Project being unavailable?

A. Yes. In their updated Exhibit___(GIOP-1S), KEDNY and KEDLI made downward adjustments of approximately $0.09 million and $0.63 million, respectively, which represent minor reductions in the Contractor Safety Inspections programs in the Rate Year.

Q. Does the Panel recommend adjustments to the mandatory capital investments?

A. Yes.

Q. Does the Panel’s recommendations assume the construction of the NESE Project?

A. No. Our recommendations are based on other factors as discussed below.

Unit Cost Increases

Q. Please briefly explain the escalating unit costs KEDNY is experiencing.

A. KEDNY's cost to install gas main has been impacted by new paving and restoration regulations enacted by the New York City Department of Transportation (NYCDOT). As described in the Companies' response to IR DPS-244, the new requirements increase the scope of work KEDNY is required to perform anytime it performs excavations. The larger scope of work includes larger restoration areas, more expensive materials, and different removal methods such as saw-cutting roadways instead of less expensive methods like jackhammering. KEDNY implemented the new requirements on April 1, 2017.

Q. How have these new requirements impacted KEDNY's budgets?

A. As expected, the new requirements have led to increases in the unit costs to install and replace new gas main. For every linear foot of gas main the company installs or replaces, it now must perform additional work. This has led

to higher total budgets for KEDNY's programs which include main installation and replacement.

Q. Is KEDLI also affected by these regulations?

A. Not significantly. KEDLI is not affected as significantly by the NYCDOT Regulations because most of its service territory lies outside of the NYCDOT jurisdiction.

City and State Construction

Q. Please explain the CSC Program.

A. The CSC Program consists of main replacement work and support and protect work which is required to accommodate public work projects including work on water, sewer, and drainage infrastructure, street reconstruction, and bridge replacement projects, among others. These public work projects are implemented by outside parties including the City of New York (NYC) and various municipalities on Long Island. Company procedures require the replacement of any main that is in direct conflict with a public work project, as well as any 8" or smaller cast iron main which is not in direct conflict, but whose integrity may be compromised

by the project. Other pipe such as plastic pipe, which is not in direct conflict but may be compromised by a project is supported and protected during construction activities. The Companies do not dictate what projects are done by NYC and municipalities on Long Island, and s they do not have complete cost control over these program's budgets.

Q. How do the Companies develop their budgets for this program?

A. KEDNY and KEDLI take different approaches to develop the CSC budgets. As explained in the Companies' response to IR DPS-352, KEDNY uses the NYC's construction budget as a basis to estimate its CSC program budget. More specifically, KEDNY uses a 20% spending ratio of the NYC Department of Design and Construction's budget as its estimate for this program. KEDNY then adds expected incremental paving and restoration costs and allocates the total program budget between the Non-Reimbursable and Reimbursable budgets based on NYC's reimbursement projections. In addition, KEDNY separates any uniquely large projects it is

aware of and develops a cost estimate for these projects separate and apart from the general CSC budget. These specific project estimates are in addition to the general CSC budget. An example of this is the SE856 project which KEDNY has budgeted an additional $40 million for in the Rate Year as shown on Exhibit___(GIOP-1).

Q. Does the Panel agree with KEDNY's methodology?

A. Yes, we agree as this methodology is reasonable based on the nature of this program as supporting projects in NYC and Long Island, as well as, the level of historical expenditures.

Q. Does the Panel recommend any adjustments to KEDNY's proposed Rate Year budget for the CSC program?

A. Yes. We recommend a downward adjustment in the Rate Year of $24.773 million from $279.773 million to $255 million.

Q. Explain why the Panel proposes this adjustment.

A. While we agree with KEDNY's methodology of using NYC's construction budget to estimate the CSC budget, we do not agree with adding expected incremental paving and restoration costs to the budget.

Q. Explain why the incremental paving and restoration costs should be excluded.

A. KEDNY's new paving requirements were fully implemented on April 1, 2017 and so the costs experienced in FY 2018 are indicative of the level of spending incurred by the new requirements. KEDNY's actual CSC expenditures in FY 2018 were 17% of NYC's construction budget. We, therefore, propose a total CSC budget of $255 million, which is 20% of NYC's Rate Year construction budget of $1.275 Billion. We contend that the additional incremental funding on top of this is unnecessary. In addition, KEDNY currently utilizes a reconciliation mechanism which protects KEDNY from excessive expenditures above and beyond its budget. This true up mechanism is discussed in more detail later in our testimony.

Q. What is KEDLI's CSC budget proposal for the Rate Year?

A. KEDLI proposes a total CSC budget of $10.878 million in the Rate Year.

Q. Does the Panel recommend an adjustment to KEDLI's CSC budget in the Rate Year?

A. Yes. We propose a total downward adjustment of $3.746 million in the Rate Year for KEDLI's CSC budget.

Q. Explain why the Panel proposes this adjustment.

A. As previously mentioned, KEDLI takes a different approach to estimating the CSC budget than KEDNY. KEDLI states in the response to IR DPS-761 that it develops its CSC budget by taking a three-year average of historical expenditures and adds inflation to forecast the Rate Year budget. KEDLI takes this approach because their CSC expenditures have been relatively stable in recent years. The average total expenditures for KEDLI's CSC category for FY 2017 through FY 2019 is $6.860 million. Adding inflation using the same inflation factor the Companies use to inflate other expenses from the HTY to the Rate Year, as seen in Exhibit___(RRP-8CU), to inflate the majority of its expenses results in our Rate Year forecast of $7.131 million. This results in the downward adjustment of $3.746 million.

Leak Prone Pipe

Q. How have the Companies developed the Rate Year budget for their Proactive Main Replacement - Leak Prone Pipe Program?

A. As explained in the Companies' responses to IRs DPS-494 and DPS-597, the Companies used the unit costs experienced in FY 2018 and applied annual inflation factors to arrive at a projected Rate Year unit cost. The Companies then applied this unit cost to the projected mileage of Leak Prone Pipe abandonment.

Q. Does the Panel agree with this method?

A. Yes, this method is reasonable as the costs incurred in FY 2018 are indicative of the escalated unit costs associated with new paving requirements.

Q. Does the Panel agree with the Companies' proposed unit prices for the LPP program?

A. We do not agree with Companies' proposed unit prices for the LPP program because we disagree with assumptions made in the calculation of the unit prices in the Rate Year. More specifically, we disagree with the inflation factor the Companies use to arrive at the Rate

Year unit prices. KEDNY and KEDLI both propose an annual rate of 2.5 percent on all price components except for contractor costs. KEDNY proposes an annual rate of 3 percent for contractors. KEDLI proposes inflating 60 percent of the contractor costs by 10.5 percent, and the remaining 40 percent by 2.5 percent. Both proposals result in a significantly higher budget, as these rates are far greater than those the Companies use to inflate other expenses throughout their operations.

Q. What does the Panel propose?

A. We recommend using the same inflation factor, 3.957 percent, the Companies use to inflate other expenses from the HTY to the Rate Year as seen in Exhibit___(RRP-8CU). This results in a downward adjustment of $5.99 million in the Rate Year for KEDNY, and a downward adjustment of $10.027 million in the Rate Year for KEDLI.

New Programs

Q. Are the Companies proposing any new programs in their mandatory capital program budgets?

A. Yes, the Companies are proposing several new

programs. Some of these include the Contractor Safety Inspection program, the Plastic Fusion QA/QC Re-Digs, the Low Pressure Main Valve Installation, and the Transmission Station Integrity Programs.

Q. Please explain the Contractor Safety Inspection program?

A. Through the Contractor Safety Inspection program, the Companies propose to hire a significant number of new employees to support increased on-sight supervision of contractors.

Q. Did the Staff Pipeline Safety Panel recommend adjustments to the Companies' proposals that impact the new programs?

A. Yes. The Staff Pipeline Safety Panel made recommendations that impact the following programs: Contractor Safety Inspections, Low Pressure Main Valve Installation Program and the Buried Vent Line Program.

Q. Please explain the changes to these programs.

A. The Staff Pipeline and Safety Panel made the following adjustments: a reduction of two inspectors for KEDNY and a reduction of one inspector for KEDLI. We reduced the capital

budgets for these programs accordingly. Additionally, the Staff Pipeline Safety Panel recommended that the Low Pressure Main Valve Installation Program and Buried Vent Line Programs be removed from the Capital budget and funded through other means as explained in their testimony. In conjunction with the Staff Pipeline Safety Panel's recommendations, we reflected the removal of these programs from the Companies' CapEx budgets accordingly.

Q. Does the Panel propose any adjustments to the new programs in addition to those recommended by the Staff Pipeline Safety Panel?

A. Yes. We propose adjustments to the Plastic Fusion QA/QC Re-Digs program, and to the Transmission Station Integrity Program.

Q. Please explain the Plastic Fusion QA/QC Re-Digs program?

A. The Companies are implementing this program in response to the Commission's Order in Case 14-G-0212. The Commission required the Companies develop a QA/QC program which includes a statistically significant number of random checks on plastic fusion work that has been

completed.

Q. What is the scope of the Companies' proposed program?

A. The Companies propose an annual sample size of 200 inspections, or Re-Digs, where they will re-excavate completed work and perform a QA/QC inspection on the plastic fusion weld. The associated reporting requirements of this program are laid out in the Commission's Order.

Q. What are the proposed costs associated with this program?

A. The Companies propose capital budgets of $3.25 million and $0.974 million for KEDNY and KEDLI respectively. The Companies state the budgets are based on unit prices of $15,000 and $3,335 per re-excavation inspection for KEDNY and KEDLI, respectively. The unit costs are multiplied by the number of re-digs planned to develop the budget.

Q. Does the Panel propose an adjustment to these budgets?

A. Yes. We propose the budget be calculated using the Companies' unit prices and projected number of inspections. This results in program budgets

of $3 million and $0.667 million for KEDNY and KEDLI, respectively. Our recommendation results in downward adjustments of $0.250 million and $0.307 million for KEDNY and KEDLI, respectively. The Companies did not justify their proposed budget in excess of a total budget based off the unit price and number of inspections.

Q. Please describe the Transmission Station Integrity Program.

A. The Transmission Station Integrity Program is a new program the Companies are proposing to address anticipated regulations that are expected to be issued prior to the Rate Year by the Pipeline and Hazardous Materials Safety Administration (PHMSA). The program is similar to their IVP, but it addresses transmission connected Pressure Regulating Stations and Take Stations. The program will review records associated with these stations, and where records are inadequate to demonstrate stations are fit for service, the program will provide capital investments for station rehabilitation projects and station replacement projects. The

Companies budgets include funding for the IVP review of station records and funding for the design of capital projects for two transmission stations each year which will be completed once the program is fully implemented by Data Year 1.

Q. Does the Panel propose any adjustments to this program?

A. Yes. In the Companies' responses to IRs DPS-922 and DPS 923, KEDNY and KEDLI indicate that this program is just ramping up. The Companies are in the process of reviewing records associated with transmission stations and have not yet identified any specific projects that need to be rehabilitated or replaced. Consequently, we propose to delay the funding associated with the detailed design phase until specific projects are identified and prioritized appropriately. This results in a downward adjustment of $2.820 million for both KEDNY and KEDLI in the Rate Year.

Q. Does the Panel propose adjustments to any other mandated programs?

A. Yes. We propose adjustments to the following programs: Main Replacements (Reactive), Service

Replacements – Proactive, Service Replacements – Leaks, and Service Replacements – Non-Leaks.

Q. Please Explain these adjustments.

A. In their response to DPS-761, the Companies provided their methodology for developing the budget for each capital program. The Companies indicate these programs are developed using a two-year average of actual expenditures plus inflation. The panel calculated the most recent 2 year average of actual expenditures and inflated it using the same inflation factor of 3.957 percent which the Companies use to inflate other expenses from the HTY to the Rate Year as seen in Exhibit___(RRP-8CU). The resulting adjustments are reflected in staff Exhibit___(SGIOP-4).

Q. Please summarize your overall adjustments to the mandatory budget category.

A. Our proposed adjustments in the mandated category result in total downward adjustments in the Rate Year of $37.635 million to KEDNY and $18.318 million to KEDLI.

Reliability Category

Q. Summarize the Companies' proposed Reliability category forecast for the Rate Year.

A. As shown in Exhibit___(GIOP-1), the Companies' forecast total Reliability Rate Year CapEx of $200.47 million and $141.12 million for KEDNY and KEDLI, respectively. The reliability portion of the Companies' capital investment plan is comprised of several different categories, with the largest expenses for KEDNY related to Pressure Regulating Facilities, liquefied natural gas (LNG), Metropolitan Reliability Infrastructure (MRI), Storm Hardening-Remote Service Shutoff Valves, and remote-controlled valves. The largest expenditures for KEDLI are comprised of the Pressure Regulating Facilities, LNG, Northwest Nassau Transmission Main and Control Valve, Storm Hardening-Remote Service Shutoff Valves, and remote-controlled valves programs.

Q. Did the Companies update their Reliability capital expenditures for the Rate Year?

A. Yes. In their updated Exhibits___(GIOP-1CU), the Companies updated their total Reliability

expenditure levels to $139.85 million and $161.76 million for KEDNY and KEDLI, respectively. The observed variance between KEDNY's updated and initial filing reflects updates to: the MRI project, including accelerating the schedule, reallocation between the phases, and an approximate $4.9 million reduction in the remaining project forecast; the Greenpoint LNG projects, which entails deferring approximately $35.1 million for the Salt Water Pump House Project and the Vaporizers 7 & 8 Projects from the Rate Year to the Data Years; and, the Gas Control Training Simulator Training Project because the Companies believe this project should be a Service Company asset and KEDNY removed the costs from its direct capital investment plan. The observed variance between KEDLI's updated and initial filing reflects updates to: the Northwest Nassau Transmission Main Project, including delays in the project schedule and an increase in the project costs; and, the Gas Control Training Simulator Training Project because the Companies believe this project should be a Service Company asset and

KEDLI removed the costs from its direct capital investment plan.

Q. Does the Panel support the changes made in this correction?

A. Yes. However, we are concerned with the Companies' ability to accurately schedule projects, and contractor resources. KEDNY cited a re-evaluation of project schedules and contractor resource availability as reasons for deferring the Salt Water Pump House Project and the Vaporizers 7&8 Projects. These projects have been previously deferred and this is indicative of a trend in the reliability category of deferring projects and underspending the total reliability budget.

Q. Does the Panel believe it was appropriate to remove the Gas Control Training Simulator from the capital investment plans?

A. Yes. This program benefits all National Grid gas distribution companies in New York and New England and is more properly reflected as a Service Company asset.

Q. Did the Companies provide any updates to their Reliability capital expenditures for the Rate

Year, considering the potential impacts of the NESE Project being unavailable?

A. In their updated Exhibit___(GIOP-1S), KEDNY made a downward adjustment of approximately $1.0 million for the removal of the Marine Park Regulator Station Project from the Rate Year. KEDLI did not propose any adjustments to the Reliability capital expenditures for the Rate Year.

Q. Have the Companies proposed spending increases to the Reliability category, in their corrections and updates testimony, compared to its previous capital investment plans?

A. Yes. The Companies have proposed increasing their spending for many programs and projects in the Reliability category, which has resulted in significant increases to the total reliability budget for each Company. As can be seen from Exhibit___(SGIOP-2), the Companies' annual average actual reliability expenditures over the past three historical FYs, from 2017 through 2019, were $81.1 million and $41.7 million for KEDNY and KEDLI, respectively. The proposed budget for the Reliability categories in the

Rate Year, considering the no NESE scenario, are $138.8 million and $161.8 million for KEDNY and KEDLI, respectively. This is an increase of $57.7 million or 171 percent for KEDNY, and $120.1 million or 388 percent for KEDLI.

Q. Further explain the increases in the Reliability category.

A. The Reliability category for both Companies is comprised of multiple annual expenditure programs and numerous unique projects. Together, they form the comprehensive approach the Companies take to maintain safe and reliable service on the system. The Companies propose significant increases to a number of their annual programs including the Pressure Regulating Facilities Program, Heater Installation Program, Gas Planning/Remote Control Valves (RCV) Program, and System Automation Program.

Q. Explain how the Companies developed their Reliability budgets.

A. The Companies' response to IR DPS-761, Attachment 1, describes the methodology the Companies used to develop their capital budgets

for the Reliability category. For KEDNY, the majority of the programs and projects in the Reliability category, 81 out of 85, utilize program/project estimates to develop their Rate Year forecast, with the remaining four programs/projects utilizing historic spend plus inflation of unit costs. For KEDLI, the majority of the programs and projects in the Reliability category, 61 out of 64, utilize program/project estimates to develop their Rate Year forecast, with the remaining four programs/projects utilizing historic spend plus inflation.

Q. Does the Panel agree with the methodologies utilized by the Companies to develop their Rate Year forecasts?

A. Yes. The Panel generally agrees with the methodologies utilized by the Companies; however, we have found large historical variance in the Companies program and project estimates compared to actual dollars spent.

Q. Does the Panel recommend adjustments based on these large historical variances?

A. Yes.

Q. Please describe your recommended Reliability CapEx adjustments.

A. Based on our review of the projects and past spending levels, we recommend a reduction of $36.366 million and $46.344 million to the Rate Year Reliability budget for KEDNY and KEDLI, respectively. As stated in the historical review section of this testimony we found significant variances between planned and actual spending in the reliability category, with the budget for the category being historically over-estimated. Utilizing a three-year average, we calculated that KEDNY and KEDLI historically spent approximately 74 percent and 71 percent of their reliability budgets, respectively. We applied these percentages as a slippage adjustment to the Companies' proposed Rate Year reliability budgets.

Q. Does the Panel have any additional recommendations for the Reliability budget Projects or Programs?

A. Yes, we also have recommendations regarding the following Programs: (1) Storm Hardening – Install Remote Service Shutoff Valves Program;

and (2) LNG Tank Upgrades Projects.

Storm Hardening

Q. Briefly describe the Storm Hardening – Remote Service Shutoff Valves Program.

A. This program is explained in the Companies' Exhibit___(GIOP-5), page 77, and page 60, for KEDNY and KEDLI, respectively. Generally, this project would install remotely operated shutoff valves, flood sensors, and a fixed communication network within the FEMA defined flood zones across the Companies' operating regions. This would allow the Companies to monitor and control the shutoff valves, mitigating the risk of flood damage to the Companies' facilities.

Q. Did KEDNY and KEDLI address the Storm Hardening – Remote Service Shutoff Valves Program in their last rate Cases 16-G-0059 and 16-G-0058, respectively?

A. Yes. In the 2016 KEDNY and KEDLI Rate Cases, the Companies proposed a five-year, FY 2017 to FY 2021 Storm Hardening – Remote Service Shutoff Valve Program, which Staff supported.

Q. What is the current status of the program?

A. As this is a new technology, not used anywhere else in the country, the Companies were unable to obtain a viable shutoff valve mechanism commercially available for use on the system, delaying the launch of the program.

Q. Were the Companies able to find a manufacturer?

A. Yes. The devices became available from the manufacturer in February 2019.

Q. Have the Companies' proposals for this program changed since their 2016 rate case proposals.

A. Yes. The Companies' renewed proposal is for FY 2020 through FY 2024 with an increase in the scope from 25,000 to 29,000 gas services and 58,000 to 60,000 gas services for KEDNY and KEDLI, respectively. Additionally, now that the Companies were able to find a manufacturer, they have a more accurate forecast of the costs which has led to significant increases in the cost of the program.

Q. Does the Panel support the Companies' renewed proposal for this Program?

A. Yes. We support the increase in scope of this project as it relies on more accurate GIS data provided by the New York State Office of

Information Technology Services, according to KEDNY's and KEDLI's responses to IRs DPS-561, and DPS-568, respectively. Additionally, we support the increase in costs associated with this program, because the original costs were based on an estimate and the newly proposed costs are based on the manufactures price quotes.

Q. Does the Panel have any additional concerns with this program?

A. Yes. After attending a technical conference with the Companies and reviewing the Companies' responses to IRs DPS-561, DPS-568, DPS-832, and DPS-954, we have concerns with the Companies' ability to move forward with this program on the schedule that they have proposed. According to their response to IR DPS-954, we find that the Companies are still in the process of testing the valves and will not be completed until December of this year. At this time, several issues have been found with the valves and are currently being resolved.

Q. What is the Panel's recommendation for this program?

A. We recommend in consideration of the large cost and size of this deployment that the valves be tested by an independent laboratory. To accommodate the additional testing time, and time to resolve any further issues that may arise, we recommend deferring this program for one year.

LNG Tank Upgrades

Q. Describe the LNG Tank Upgrades Projects.

A. As explained in the Companies' Exhibits___(GIOP-5), page 105, and page 85, for KEDNY and KEDLI, respectively, the Companies have LNG tanks that have been in service for an extended amount of time and are in need of service. KEDNY has two LNG tanks totaling 1.6 billion cubic feet of storage, at their Greenpoint LNG facility and KEDLI has one, 0.6 billion cubic feet tank at their Holtsville facility. These LNG tanks have been in service for approximately fifty years and require a detailed inspection and overhaul of various tank related systems.

Q. Did KEDNY and KEDLI address the LNG tank upgrades projects in their last rate Cases 16-G-

0059 and 16-G-0058, respectively?

A. Yes. In the 2016 rate cases, the Companies proposed the detailed inspection and upgrade of tank number two at Greenpoint and the detailed inspection and upgrade of the tank at Holtsville, that was generally supported by Staff.

Q. Did the Companies provide a reason, if any, that these projects did not move forward on schedule?

A. Yes. As stated in the Companies' GIOP direct testimony, execution of these projects were dependent on the NESE Project. The unavailability of the NESE Project has resulted in delays to the planned tank upgrades.

Q. Did the Companies provide additional testimony or support explaining how they would move forward with upgrading and repairing these tanks if the NESE Project was further delayed or not constructed?

A. No. The Companies only stated in their direct testimony of the GIOP panel that without additional capacity from the NESE Project, they would be unable to take the LNG tanks out of service for sufficient time to complete needed

repairs.

Q. What is the Panel's recommendation for these projects?

A. We recommend that the Companies put forth a reasonable proposal of how they will address the needed tank repairs and upgrades if the NESE project doesn't move forward and file their proposal by the beginning of FY 2021.

Non-Infrastructure Category

Q. Summarize the Companies' Rate Year forecast for the Non-Infrastructure category.

A. In Exhibit___(GIOP-1CU), the Companies' forecast a total Non-Infrastructure Rate Year expense of $9.942 million and $4.560 million for KEDNY and KEDLI, respectively. The Non-Infrastructure portion of the Companies' capital investment plan are the expenses associated with the Automatic Meter Reading (AMR) Replacement program; Tools and Equipment; Telecommunication program (including Radio Capital Expenditures, Comm Site Upgrades, and Damage Failure); and Learning & Development for Materials, Tools and Equipment.

Q. Did the Companies provide any changes/updates to their Non-Infrastructure capital expenditures in their correction and supplemental filings?

A. No.

Q. Please describe how the Companies forecast the budgets of Non-infrastructure budget for Rate Year?

A. According to the Companies' response to IR DPS-766, the Companies propose the following: (1) for the Telecomm program, the Companies' budget forecast is based on FY 2019 budgets with escalating inflation; (2)for Learning and Development budget, the Companies' budget forecast is based on projected incremental employee need in FY 2021; (3) for Tools and Equipment, the Companies' forecast is based on the average of FY 2017 and FY 2018 cost adjusted for the expected rates of LPP replacement and other mandated programs as well as the cost of inflation at two percent each year; (4) for AMR, the Companies' forecast is based on number of employees needed to complete the target number of AMR installations in Rate Year; (5) for Meter Testing Equipment, the Companies' forecast is

based on FY 2018 historic spending plus two percent annual inflation; and (6) for AMR Replacement, the Companies' forecast is based on the FY 2018 historical spending, units completed, and plus two percent annual inflation.

Q. Does the Panel agree with the Companies' forecasted budget for Non-Infrastructure category

A. No, we do not agree with the Companies' forecasted budget.

Q. Please explain why the Panel does not agree.

A. The Companies forecast is not reasonable because they have typically underspent their historic budgets. For example, concerning the Telecomm program, KEDNY budgeted $108,000 in FY 2019 but only spent $51,000, or approximately 47 percent of its budget. For past three years, the actual average spending was approximately $38,000, or only approximately 36 percent of the total budget.

Q. What does the Panel recommend on Non-Infrastructure budget be for the Rate Year?

A. We recommend that Telecomm program, Tools &

Equipment, and Meter Testing Equipment budgets be based on a three-year average of actual historic spending, escalated by inflation. We have no adjustments to AMR Installation & Replacement, and Learning and Development for both KEDNY and KEDLI. This results in an increase of KEDNY's budget by $241,263 and a decrease of KEDLI's budget by $302,798 for the Non-Infrastructure in Rate Year.

Facilities

Q. Summarize the Companies' Rate Year forecast for the Facilities category.

A. In KEDNY's Exhibit___(GIOP-7CU) and KEDLI's Exhibit___(GIOP-6CU), the Companies' forecast a total Facilities Rate Year expense of $18.425 million and $22.085 million for KEDNY and KEDLI, respectively.

Q. Please describe the Companies' Facilities budget for Rate Year?

A. According to the Companies' response to DPS-992, the Facilities budget historically has three components including Baseline Spend, Major Facilities Projects, and Cost of Removal. For

the Rate Year, both KEDNY and KEDLI have proposed budgets including Base Spend and Major Facilities Projects. KEDNY proposed $2.975 million and $15.450 million for Baseline Spend and Major facilities projects, respectively. KEDLI proposed $0.300 million and $21.785 million for Baseline Spend and Major facilities projects, respectively.

Q. Describe your capital expense adjustments for the Facilities category.

A. We reviewed the Companies historic budget and actual spending level for Baseline spend and found that KEDNY and KEDLI spent on average $2.932 million and $0.254 million from FY 2017 through FY 2019, respectively. Based on our findings, we recommend an upward adjustment of $92,060 for KEDNY and a downward adjustment of $31,401 for KEDLI on Base Spend under the Facilities category to reflect a three-year historic average escalated by inflation.

Capitalization Policy

Q. Are the Companies proposing any changes to the existing capitalization policies for main

installation?

A. Yes. Currently, KEDNY and KEDLI charge any main installation less than 50 linear feet to their O&M budget as an expense. The Companies are proposing to remove this threshold and capitalize all main installation no matter the length. The Companies state this charge is in accordance with FERC accounting guidelines.

Q. What is the Panel's opinion on this change?

A. We believe this change is reasonable. Additionally, the Companies state that the 50 feet threshold is an arbitrary policy, however, the change would allow them to standardize the accounting treatment of these work activities between National Grid's New York gas distribution companies. It is important to note that this change will reduce both the O&M budget and revenue requirement if it is implemented. The Staff Revenue Requirements Panel has confirmed that the Companies reflected this change correctly.

Net Plant in Service and Depreciation Expense

Q. Did the Companies develop a model to forecast

the average net utility plant in service and depreciation expense for the Rate Year?

A. Yes. The Companies provided, in their response to IR DPS-087, the models they used to develop the forecast monthly components of net utility plant in service.

Q. What are the Companies' projections for net utility plant in service and depreciation expense for the Rate Year?

A. KEDNY projected $5.796 billion for gas net utility plant in service, and $159.3 million in annual gas depreciation expense. KEDLI projected $4.006 billion for gas net utility plant in service, and $107.8 million in annual gas depreciation expense. These projections are based on actual inputs as of March 31, 2019.

Q. Did the Panel verify the Companies' projections?

A. Yes. We reviewed the plant in service models' calculations to verify their accuracy.

Q. Did the Panel develop its own forecast?

A. Yes.

Q. How did the Panel develop its forecast of net utility plant in service and depreciation expense for the Companies?

A. We modified the Companies' models to reflect our recommended adjustments to the capital budgets, as shown in Exhibit___(SGIOP-4), and depreciation rates. The results are shown on Exhibit___(SGIOP-5).

Q. Please explain what is shown on page one of Exhibit___(SGIOP-5).

A. Exhibit___(SGIOP-5) summarizes the projected monthly plant in service balances for additions, non-interest bearing construction work in progress, reserve for depreciation, net utility plant in service, and depreciation expense. Exhibit___(SGIOP-5) shows the monthly balances and calculates the average of the monthly balances for each of the components. This yields the overall average net utility plant in service upon which the Companies should be allowed a return, as well as the Rate Year level of annual depreciation expense.

Q. What are the Panel's Rate Year projections for gas net utility plant in service and gas depreciation expense?

A. We project gas net utility plant in service of $84.014 billion for KEDNY and $57.700 billion

for KEDLI, and gas depreciation expense of $141.8 million for KEDNY and $106.8 million for KEDLI. These figures were provided to the Staff Revenue Requirements Panel.

Q. How do the Panel's recommendations compare to the Companies' Corrections and Updates filings?

A. As shown on page one of Exhibit___(SGIOP-6), we recommend adjustments that, in total, reduce KEDNY's Rate Year average gas net plant in service by $53.0 million, and reduce its annual gas depreciation expense by $17.5 million. We recommend adjustments that, in total, reduce KEDLI's Rate Year average gas net plant in service by $91.7 million, and reduce its annual gas depreciation expense by $11.7 million.

Incremental O&M Expenses

Q. Please summarize the Companies' incremental O&M expenses forecast in the Rate Year.

A. In the Companies' Corrections & Updates Testimony, KEDNY forecasts total incremental non-labor operating expenses of $10.866 million in the Rate year for GIOP programs, as shown in Exhibit___(GIOP-8CU). KEDLI forecasts total

incremental non-labor operating expenses of $7.299 million in the Rate year for GIOP programs, as shown in Exhibit___(GIOP-7CU). Additionally, the Companies propose hiring 29.1 FTEs for KEDNY and 15.9 FTEs for KEDLI in the Rate Year to support GIOP programs, as shown in KEDNY Exhibit___(GIOP-9CU) and KEDLI Exhibit___(GIOP-8CU). The proposed increases are utilized to support multiple programs in the companies' capital budgets. The Companies' Gas Infrastructure and Operations Panels explain, beginning on pages 108 and 99 of their direct testimonies for KEDNY and KEDLI, respectively, that the increases in O&M expense are driven by the Companies' increasing capital investments and increased costs for executing the capital plan, an increase in O&M workload, and expanding safety initiatives. Additionally, New York City restoration cost increases also impacts costs of construction, leak repairs, and disconnects and reconnects.

Non-Labor O&M Expenses

Q. Does the Panel recommend adjustments to the non-

labor Operating expenses the Companies propose in the Rate Year?

A. Yes. We recommend non-labor operating expense reductions of $4.806 million and $3.010 million for KEDNY and KEDLI, respectively. We recommend reductions to the following budget categories which are reflected in Exhibit___(SGIOP-7): Pipeline Integrity - IMP (PHMSA Rules), and the Storm Hardening Program.

Q. Describe the Pipeline Integrity - IMP program.

A. The Companies Gas Infrastructure and Operations Panels explain, beginning on page 34 of their Direct Testimony, that the program supports inspections and assessments of transmission pipelines.

Q. Does the Panel support the Pipeline Integrity - IMP program (PHMSA)?

A. Yes, however, we contend that the incremental non-labor operating expenses (OpEx) proposed by the companies are not necessary to complete the work in the Rate Year.

Q. Please explain.

A. In the Companies' responses to IRs DPS-499 and DPS-712, the Companies state they are proposing

increased OpEx to support the work included under this program. This includes external corrosion direct assessments (ECDA), In-line Inspections, test data analysis, and non-capital repairs. However, neither the capital budget, or the OpEx workload for the Rate Year is being increased as compared to the most recent FY or the HTY; therefore, the existing OpEx budget should be able to support the needs of this work in the Rate Year. This results in a downward adjustment of $4.285 million in the Yate Year.

Q. Please describe the Storm Hardening Program.

A. The Storm Hardening program will install automatic shut off valves in FEMA flood zones and is described in more detail previously.

Q. Please describe the Panel’s recommended adjustment to the incremental non-labor OpEx in the Rate Year for the Storm Hardening Program.

A. As stated earlier in our testimony, the Panel has concerns with the Companies’ ability to implement this program on the proposed schedule. The Panel proposes removing the increased OpEx budget associated with this program in the Rate Year consistent with our proposed capital budget

adjustments. This results in a downward adjustment of $0.521 million in the Yate Year.

Incremental FTE's

Q. Please summarize the Companies' proposal to hire additional FTEs in the Rate Year.

A. As discussed previously and as shown in Exhibits___(GIOP-9CU) and Exhibits___(GIOP-8CU), KEDNY and KEDLI plan to hire 29.1 and 15.9 FTEs, respectively, in the Rate Year to support GIOP programs. KEDNY and KEDLI propose to hire, in the Rate Year, an additional 29.1, and 15.9 FTEs, respectively to support its capital programs.

Q. How will the unavailability of NESE impact the Companies' FTE requirements?

A. Without the NESE Project, the Companies' capital budgets are decreased significantly which results in reduced work load for all FTEs associated with reduced capital programs. In the Companies' response to IR DPS-946, the Companies state they do not currently have a formal deployment or utilization plan for employees in this scenario. We recommend that

if the NESE Project is unavailable, the Companies should reassess staffing levels, and redeploy their current FTEs as appropriate, prior to hiring any additional FTEs.

Q. Does the Panel have additional adjustments to the incremental FTEs proposed in the Rate Year by KEDNY and KEDLI.

A. Yes. We recommend that the incremental Rate Year FTEs proposed by KEDNY and KEDLI be reduced by 11.6 and 7.4, respectively, as shown on Exhibit___(SGIOP-8). Our recommended adjustments impact the following categories: OpEx Support for Capital, IMP/IVP OpEx, and Storm Hardening.

Q. Please explain the basis for these recommendations below.

A. While the Companies should hire at the levels it deems appropriate to provide safe and adequate service, these FTE adjustments reflect the level of O&M expense we believe the Companies have justified in its initial testimonies, Corrections and Updates testimonies, and responses to requests for information during the discovery phase of the proceeding. Thus, we

believe that with our FTE adjustments, we are recommending the level of O&M expense that is most appropriate for the Commission to use when setting rates in the Rate Year.

Q. Please describe the IMP/IVP program and the Panel's proposed adjustments.

A. As discussed earlier, the IMP/IVP programs support inspections and assessments of transmission pipelines. As shown in our Exhibit__(SGIOP-8), we propose reducing the incremental FTEs proposed in the Rate Year by two FTEs and three FTEs for KEDNY and KEDLI, respectively.

Q. Why does the Panel propose these reductions?

A. We propose these reductions because neither the capital expenditures or the O&M work load are significantly increased in the Rate Year compared to the HTY or most recent FY. Therefore, the program should be sufficiently supported by the existing staffing levels. This is consistent with our recommendations for the non-labor OpEx adjustments in this program.

Q. Explain the FTEs included under the OpEx Support for Capital and the Panel's proposed adjustments

to this category.

A. The Companies are proposing to hire 18 new FTEs for KEDNY and five new FTEs for KEDLI under this category. The incremental FTEs will be utilized across multiple capital programs including the CSC, and Proactive Main Replacement Programs.

Q. Please describe the Panel's adjustments to these FTEs.

A. The Panel recommends reducing the proposed FTEs under this category by nine FTEs for KEDNY, and by three FTEs for KEDLI. These adjustments are reflected on our Exhibit__(SGIOP-8).

Q. Why does the Panel recommend these adjustments?

A. In the responses to IRs DPS-280 and DPS-281, the Companies provided explanations for each of the proposed incremental FTEs. Many of the explanations for additional inspectors, supervisors, and contract oversight analysts were related to maintaining appropriate oversight on contractors performing capital work. We appreciate the need for appropriate oversight on contractors for numerous reasons, however, as explained earlier in our testimony, the Companies have implemented a new contractor

safety inspection program to address this issue within the budget for that program. Consequently, we recommend removing these incremental FTEs.

Q. Does the Panel have another basis upon which they recommend an adjustment to the number of FTEs for this program?

A. Yes. The Companies plan to hire a significant number of FTEs through this contractor safety inspector program. The goal of the program is to achieve a one to one ratio of inspectors to work crews. With the number of work crews varying at any given time, the exact number of inspectors required is difficult to determine. Additionally, it is yet to be seen what ratio is required to achieve optimal oversight. We recommend that the Companies utilize the new FTEs in the contractor safety inspector program to assess and further define these parameters.

Q. Does the Panel recommend any other adjustments to the FTEs included under the OpEx Support for Capital category?

A. Yes. We recommend removing one Welder in KEDLI's forecast in the Yate Year. KEDLI

proposed hiring three Welders to account for increased work load and to begin the training process in anticipation of future attrition. We propose reducing this by one FTE to account for our reductions in the proposed capital budget as well as the uncertainty associated with actual attrition timelines of existing employees, which KEDLI did not provide.

Q. Please explain the Panel's FTE adjustments to the Storm Hardening Program.

A. Consistent with our proposed capital budget and non-labor OpEx adjustments for this program, we recommend removing the proposed incremental FTEs for this program in the Rate Year.

City State Construction True Ups

Q. Do the Companies currently have a true up mechanism for their City State Construction Program?

A. Yes. As provided for in recent rate cases, the Companies continue to utilize a reconciliation mechanism to recover the excess difference between actual expenditures and budgeted amounts for this program in the capital budget. The

current mechanism allows the Companies to recover 90 percent of the difference between the budgeted amount and the actual expenditure.

Q. Does the Panel support the true up mechanism?

A. We accept the true up because it has been in effect for some time.

Q. Do the Companies propose any changes to this mechanism?

A. Yes. The Companies proposes to change the true up from a 90 percent recovery to 100 percent recovery.

Q. Does the Panel support this change?

A. No. Although aspects of this program are beyond the Companies' control, KEDNY and KEDLI still manage the projects and contractors who ultimately complete the work associated with this program. This includes the full range of project management, contract oversight, quality control, material procurement, workforce control, and productivity, as well as many other factors which affect the total expenditures. Accordingly, we believe it is important to provide the Companies a strong incentive to manage the work efficiently and keep costs at a

minimum in order to benefit its customers. We, therefore, propose the true up mechanism at be adjusted to 80 percent to provide this incentive.

Q. Do the Companies propose any additional changes to the existing reconciliation mechanisms?

A. Yes. As stated on page 48 of KEDNY's testimony of the GIOP panel, KEDNY is also proposing to add an O&M Tracker that would fully reconcile the difference between forecast and actual spending in the non-reimbursable "Support and Protect" O&M expense category.

Q. Does the Panel support this new tracker?

A. No. As explained above, we believe it is important to provide the Companies with the necessary incentive to manage projects efficiently and cost effectively. In a one-year rate case, the Companies should be able to manage their O&M budgets and the true-up mechanism is not necessary.

Main Installation Incentives

Q. Please briefly describe the leak prone pipe incentives the Companies are proposing.

A. As shown in Exhibit___(GIOP-3), the Companies propose to recover the following costs and incentives under through the Gas Safety and Reliability Surcharge: Costs for incremental LPP replacement, Positive Revenue Adjustments (PRA) for LPP removal incentive, PRAs for the LPP unit cost incentive, costs for incremental leak repairs, PRAs for leak repair incentive, and PRAs for customer connections unit cost and productivity incentives.

Q. Did the Staff Policy Panel inform the Panel that these some of these incentives should be removed?

A. Yes.

Q. Is there a need to develop unit cost caps?

A. As the Staff Policy Panel proposes to remove the specific leak prone pipe incentives, there is no longer a need to develop the unit cost caps.

Capital Investment Reconciliation Mechanism

Q. Does the Panel propose a Capital Investment Reconciliation Mechanism?

A. Yes. As we stated earlier, based on our review of the Companies historical capital budgets and

expenditures, it is clear that there continuing variances between the capital budget and actual expenditures. To address this issue, we propose a Capital Investment Reconciliation Mechanism to protect customers from paying delivery rates that are too high because the Companies were not able to implement the entire capital plan, or conversely was able to do so more efficiently.

Q. Please briefly describe the proposed Capital Investment Reconciliation Mechanism.

A. We recommend that the actual net revenue requirement in the Rate Year be compared with the net revenue requirement approved by the Commission. When measuring the net plant in service, we recommend including depreciation expense. The mechanism is to be downward-adjusting only. The calculations should be made and filed with the Secretary on or before July 31st of the subsequent Rate Year. Any balance owed to customers would be deferred with carrying charges as calculated using the pre-tax rate of return approved by the Commission in this proceeding.

Q. Why does the Panel recommend that the mechanism

be a one-way, downward-only true-up mechanism?

A. Customers are paying a return on a forecast level of rate base. The one-way mechanism protects customers if the Company under-spends its capital budget, or if there is significant slippage, or delays, in closing projects to plant in service.

Q. Has the Commission previously adopted a one-way, downward-only, capital reconciliation mechanism in a litigated rate proceeding?

A. Yes. The Commission adopted a one-way downward only CapEx true-up mechanism in the Consolidated Edison Company of New York, Inc. electric rate case, Case 08-E-0539, and in Central Hudson Gas and Electric Corporation, Case 08-E-0887.

Q. Will the Gas Safety and Reliability Surcharge proposed by the Company impact the Capital Investment Reconciliation Mechanism?

A. Yes. It is important to understand how these mechanisms will function together.

Q. Please explain.

A. The Capital Investment Reconciliation Mechanism is an asymmetric reconciliation mechanism. If net plant additions are lower than projected,

the revenues collected to support the increment between actual and target net plant would be deferred for the benefit of customers. The Company would bear the risk of carrying costs resulting from net plant additions higher than projected.

Q. Please continue.

A. This reconciliation mechanism must work in concert with mechanisms approved in the Gas Safety Reliability Surcharge to prevent any unintended consequences.

Reporting Requirements

Q. What does the Panel recommend regarding CapEx and variance reporting requirements?

A. It is important for Department Staff and the Commission to monitor the Companies' capital investments. To that end, we recommend that the Companies be required to make regular filings. The filings should occur: (1) prior to the start of each FY; (2) quarterly during the FY; and (3) after the end of the FY.

Q. What information should the Companies be required to file before the FY, or shortly after

the Commission sets rates in this case?

A. Prior the beginning of the FY, the Companies should file with the Secretary to the Commission their LPP prioritization summary to identify the proposed projects and their estimated costs, an inventory of Type 3 leaks on their systems, and the approved five-year capital plans in the same format as Exhibit___(GIOP-1).

Q. What information would be filed quarterly?

A. The Companies should file quarterly variance reports to Staff in the same format as Exhibit___(GIOP-1) with explanations for any variances between the approved budget and actual expenditures. The Companies should also be required to report in detail the progress of LPP retirement mileage, Type 3 leaks repaired, a summary of the current Type 3 inventory, and new customers attached to the system, if the NESE Project is available.

Q. When should the quarterly reports be filed?

A. We recommend the Commission require the quarterly reports be filed within 45 days after the end of each of the first three calendar quarters of each FY.

Q. What information should be filed annually on a FY basis?

A. We recommend the Commission require the annual reports include the following information: (1) a final summary of the variance of capital expenditures (dollars and units) for all capital projects and programs including all on-going and active construction projects and programs in the same format as Exhibit___(GIOP-1); (2) a narrative explaining any cost or timeline deltas exceeding 10 percent; (3) a narrative on project design, permitting and or construction status (including a detailed construction schedule for each project) for any ongoing projects; (4) a description of any new projects or programs; and (5) capital project sanctioning documents for any projects exceeding $1 million that were authorized during the previous FY.

Q. When should the annual reports be filed?

A. We recommend the annual reports be filed not later than 60 days after the end of the last quarter in each FY.

Q. Should the reporting requirements continue beyond the FY?

A. Yes. It is important for the Commission to monitor the Companies' capital investment plans on an ongoing basis. Informational reports filed at regular intervals are critical to this oversight.

Q. Does this conclude the Panel's testimony?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY and
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

February 2020

Prepared Supplemental Testimony
of:
Staff Gas Infrastructure and
Operation Panel

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Introduction

Q. Members of the Staff Gas Infrastructure and Operation Panel (Panel), please state your names.

A. Our names are Mark Tintera, Qin Shi, Sean Walters, and Brian Fisher.

Q. Has the Panel previously sponsored testimony in these proceedings?

A. Yes. We submitted pre-filed testimony on August 30, 2019. Our testimony addressed the Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan East Gas Corporation d/b/a National Grid's (KEDLI), collectively the Companies, proposals related to capital expenditures (CapEx) and operation and maintenance (O&M) expenses.

Purpose of Testimony

Q. What is the purpose of this testimony?

A. The purpose of this testimony is two-fold. First, we will provide corrections to our pre-filed testimony on August 30, 2019. Second, we will respond to the second supplemental testimony filed in these proceedings by the Companies.

Q. In your testimony, will you refer to, or otherwise rely upon any information obtained from discovery request in

this proceeding?

A. Yes, our testimony will refer to Company responses to Staff's Information Requests (IRs). These responses are contained in Exhibit___(SGIOP-9). We will refer to these responses by the designation given when sent, e.g., DPS-123.

Q. Are you sponsoring any exhibits?

A. Yes. We have updated our exhibits from our original testimony to reflect our corrections, and to reflect staff's positions on the updates contained in the second supplemental testimony filed by the companies.

- Exhibit__(SGIOP-9): contains IR Responses which are relevant to our review of the Companies' second supplemental testimony;
- Exhibit__(SGIOP-10): contains an updated and corrected summary of the Panel's CapEx adjustments by Budget Category;
- Exhibit__(SGIOP-11): contains an updated summary of the Net Plant Model Results Comparison; and
- Exhibit__(SGIOP-12): contains an updated summary of O&M adjustments.

Q. What updates and adjustments did the Companies make in their second supplemental testimonies?

A. As discussed by the Staff Policy Panel, there has been a

recent settlement, a development that affects these cases. Following the Order Adopting and Approving Settlement, effective November 26, 2019, in Case 19-G-0678, the Companies lifted a moratorium on new gas customer connections that began in May 2019. To the effectuate the terms of the agreement, and to lift the moratorium, the Companies have revised their CapEx and O&M plans to enable new customer connections. This includes updates to the Customer Connections, Mandated and Reliability budget categories for the Rate Year, as well as updates to O&M program expenses in the Rate Year.

Q. Why does the Panel need to update its position and exhibits in response the second supplemental testimonies filed by the Companies?

A. Our original testimony and exhibits do not address the changes made to the Companies' CapEx budgets in their second supplemental testimonies. Therefore, our testimony is being supplemented, and our exhibits are being updated accordingly, to provide an accurate record of our review, analysis, and recommendations of the updated CapEX budgets. For each CapEx spending category, we will first explain our corrections, and then provide adjustments in response to the Companies' second supplemental testimony.

Customer Connections

Q. Does the Panel have any corrections to the testimony submitted on August 30, 2019 associated with Customer Connections budget category.

A. Yes, we have a correction for KEDNY. In Exhibit ___(SGIOP-4) in consideration of the North East Supply Enhancement (NESE) Project not moving forward, we incorrectly included $520,000 for the Jamaica Inlet project as a portion of KEDNY's Fiscal Year (FY) 2021 CapEx budget, it should be removed. Moreover, as we will discuss later, the Jamaica Inlet project is no longer relevant because the Companies' lifted the moratorium.

Q. Please describe the Companies' second supplemental filing update to the Customer Connections forecasts.

A. The Companies updated their CapEx forecasts for the Install Main/Services, Meter Purchase, Install Meter/Regulator, AMR, and Gas System Reinforcement budget items to reflect the costs for connecting new customers in response to the recent Settlement in Case 19-G-0678. These updated forecasts largely are consistent with the customer connections budgets in the Companies' corrections and updated (C&U) filings. In addition, KEDLI removed the Clean Choice program (CCP) as recommended by the Staff Policy Panel. Therefore, KEDLI

updated its forecasts to reflect the decrease on the number of units forecasted for Meter Purchases, Install Meter/Regulator, and AMR as a result of eliminating the CCP.

Q. Please describe the Companies' updates to the Gas System Reinforcement category under the Customer Connections forecasts.

A. KEDNY updated the Gas Reinforcement System program to remove the cost of the Jamaica Inlet project due to the moratorium having been lifted. This project was added in KEDNY's Rebuttal filing to address low system pressure for the no NESE scenario. The Companies also updated KEDLI's Gas Reinforcement Program by shifting $5 million from FY 2022 to FY 2021 to accelerate Phase 1 of the Southeast Suffolk project in order to support new customer connections.

Q. Does the Panel agree with the Companies' Customer Connection forecast updates for the Gas Reinforcement category?

A. While we generally agree, we do have concerns with specific budget line items.

Q. Please describe the budget line items under Customer Connections the Panel does not agree with.

A. We have concerns and do not agree with the following

budget line items: 1) KEDNY's Install Main Commercial (Comm) and Meter Purchase; 2) KEDLI's Install Main; 3) KEDLI's Meter Purchase, Install Meter/Regulator, and AMR to reflect eliminating the Clean Choice Program; and 4) KEDLI's Gas Reinforcement.

Q. Please describe the Panel's adjustments for KEDNY's Install Main.

A. KEDNY's Install Main is comprised of three groups: Install Main for Residential (Res), Commercial (Comm), and Tariff. The Company developed unit cost based on a two-year average from 2017 and 2018, escalated by 2.5 percent per year. We reviewed the Companies responses to IRs DPS-84, DPS-947, and DPS-1059, and found that the unit cost of Install Main (Comm) for FY 2017 and FY 2019 were higher due to installation of larger than average mains used during those two years, as per the Companies' responses. Based on the higher costs driven by the larger main in the historic period, we recommend using a three-year average (2016, 2018, & 2019).

Q. How did the Panel forecast Install Main (Comm) in the Rate Year?

A. We recommend normalizing the historic data for larger than average mains by excluding one of the FYs. We recommend using 2016 data instead of 2017 data to develop

our average unit cost. Our adjustment reduces the proposed unit cost in the Rate Year for KEDNY's Install Main (Comm) from $589 to $526. We also found that KEDNY's proposed number of units (15,000) of Tariff Main in the Rate Year is approximately two times higher than a three-year historic average (i.e., 6,214 for 2016-2018), and approximately three times higher than a five-year historic average (i.e., 4,870 for 2014-2018). We recommend reducing KEDNY's proposed units for Tariff main from 15,000 to 6,500, a level consistent with the three-year average. Moreover, Tariff Main is about 3 to 14 percent of the total cost for Install Main and the level of arborists need associated with this Tariff Main work can lead to a large variance in the unit cost. For example, the highest unit cost, which was $1,175 in 2017, was driven by the increased need for arborists, while the lowest unit cost, which was $266 in 2018, was due to the need for much fewer arborists. Therefore, we recommend using a three-year average (2015, 2016, & 2019) to normalize the unit cost, by excluding the highest unit cost in 2017 and the lowest unit cost in 2018. This results a downward adjustment of $3.66 million for Install Main, as shown in Exhibit___(SGIOP-10) for RY.

Q. Please describe the updated adjustment for KEDNY's Meter

Purchase line item.

A. For KEDNY's Meter Purchase, we recommend using a three-year average excluding 2019 data and including 2016 data. We recommend using 2016 data instead of 2019 data because the unit cost in 2019 was double the three-year average historical unit cost which, as stated in the Companies' response to DPS-1059, was due to two factors: 1)an increase in the higher cost of large Commercial and Industrial Meters purchased, and 2) a decrease in the total number of residential meters purchased. Our recommendation results in a decrease of $125K for Meter Purchase in the Rate Year.

Q. What groups make up KEDLI's Install Main category?

A. KEDLI forecasts four Install Main groups: Residential, Commercial, Tariff, and Residential Underground Distribution (RUD).

Q. Please explain the Panels' KEDLI Install Main commercial adjustments.

A. We recommend reducing KEDLI's proposed Install Main Commercial footage from 33,000 to 24,000 using a three-year average. Based on our review, we found a three-year average of KEDLI's Install Main (comm) is 23,692 feet. This is also consistent with our proposal for unit cost by using a three-year average.

Q. Please explain the Panels' KEDLI Install Main Tariff adjustments.
A. We decreased the footage of Tariff Main from 36,500 to 15,000 based on a three-year historic average. Based on our review, we found KEDLI's proposed unit was about 160% higher than a three-year historic average. Therefore, we recommended 15,000 feet of Tariff Main in the Rate Year.
Q. Did you review the unit costs associated with KEDLI's Tariff Main item?
A. Yes, we reviewed the historic unit costs. As shown in the Companies' response to DPS-1059, KEDLI's Tariff Main unit cost for 2019 was $369 as compared to $165 in 2017, and $246 in 2018. KEDLI claims the costs were higher in 2019 because of higher contractor costs associated with Department of Environmental Protection dumping cost increases.
Q. Should 2019 costs be normalized for forecasting purposes?
A. Yes. Therefore, we updated the unit cost of KEDLI's Tariff Main using a three-year average, by excluding 2019 data and including 2016 data because the 2019 unit cost was approximately 42 percent higher than a five-year historic average and 55 percent higher than a three-year historic average. Our recommended unit cost of $249 is higher than KEDLI's proposed unit cost of $223 for Tariff

Main.

Q. What is the overall impact of your recommendations for KEDLI's Install Main budget category?

A. Our recommendations result in a downward adjustment of $4.50 million for KEDLI's Install Main in the Rate Year.

Q. Please explain the Panels' KEDLI Meter Purchase and Install Meter/Regulator adjustments.

A. We reviewed the Companies' response to DPS-1058, found the number of units for Meter Purchase, Install Main, and AMR were decreased due to removing the CPP, and as a result, we updated the forecasts on Meter Purchases and Install Meter/Regulator to reflect removing costs associated with the Clean Choice Program. This results in an upward adjustment of 413,656 and 850,153 for Meter Purchase and Install Meter/Regulator, respectively, as shown in Exhibit___(SGIOP___10).

Q. Does the Panel agree with all of the Companies' Gas System Reinforcement updates?

A. No.

Q. What updates on the Companies' Gas System Reinforcement does the Panel not agree with?

A. We do not agree to shift $5 million from FY 2022 to FY 2021 for KEDLI's Southeast Suffolk Infrastructure - Phase 1. KEDLI's originally proposed $20 million in FY 2021

and $21.6 million in FY 2022 for this project.

Q. Why does KEDLI propose shifting $5 million for the Project from 2022 to 2021?

A. KEDLI proposed shifting $5 million for the Project from 2022-2021 to accelerate this project and to be in service by 2022.

Q. Does this proposed shift accelerate the completion date of the project?

A. No. The proposed shift of $5 million from FY 2022 to FY 2021 results in no change to the outcome to complete this project in FY 2022. Therefore, we reject KEDLI's proposal to shift $5 million from FY 2022 to FY 2021 for KEDLI's Southeast Suffolk Infrastructure - Phase 1.

Q. Please summarize the Panel's total adjustment for KEDNY and KEDLI's Customer Connections.

A. Our adjustments result in a downward adjustment of $545K and $9 million for KEDNY and KEDLI's Customer Connection Rate Year budgets in their Second Supplemental Filing, respectively, and as shown in Exhibit___(SGIOP-10).

Mandated

Q. Does the Panel have any corrections to the testimony submitted on August 30, 2019 associated with Mandated budget category?

A. No, we have no corrections associated with the Mandated Category.

Q. Please describe the Companies' changes to the Mandatory category in their second supplemental testimony.

A. The first change the Companies made affects the Contractor Safety Inspections program. In the first supplemental testimony the Companies reduced the budgets for the contractor inspection program to reflect a lower number of contractor inspectors associated with a no NESE scenario, where they expected a reduction in the customer connections work load. Now, as a result of the Companies' agreement to lift the moratorium, the customer connections workload levels are forecasted to resume at levels comparative to their original forecast. The Companies have, therefore, increased the CapEx and O&M forecasts to reflect the higher number of contractor inspectors proposed prior to the moratorium.

Q. Does the Panel agree with the Companies' update to the contractor inspection program?

A. Yes. Given that the budgets have been increased to represent the higher level of workload for customer connections, it is appropriate to increase the number of contractor inspectors to the original forecast. We also confirmed with the Staff Pipeline Safety Panel that the

adjustment for this program, decreasing the Companies' proposal by 5%, should still be applied to this originally forecasted figure. This adjustment is reflected in our Exhibit___(SGIOP-10).

Q. Please describe the other changes the Companies made to the Mandatory category in their Second Supplemental Testimony.

A. KEDLI revised their capital budget for Leak Prone Pipe (LPP) to reflect the terms of new contracts it recently finalized with one of their contractors. This resulted in a reduction to KEDLI's proposed Leak Prone Pipe budget of $1.929 million.

Q. Does the Panel agree with the updated LPP budget?

A. No. We maintain our adjustment proposed in our original testimony for this program.

Q. Please explain given that KEDLI claims the costs are based on a new contract?

A. In attachment 9 of the Companies' response to IR DPS-1043, KEDLI provided a budget impact summary for the new contract. The summary shows an estimated increase in total spends associated with the new contract when compared to the existing contract. Further, we requested the workpapers supporting this estimated budget impact in IR DPS-1080. In the Companies' response to IR DPS-1080,

KEDLI provided another workpaper which shows the total difference in the forecasted spend between the old contract and the new contract, but again, does not but support nor provides further insight into how the forecasted total spends were developed from the new contract.

Q. Please explain why this is not sufficient support.

A. The contract that has been negotiated is complex. It is not one lump sum price paid to the contractor for the Rate Year. Instead, each unit price has been updated. The Companies' support includes a myriad of assumptions to generate the estimated costs and does not justify any increases or decreases in total spend.

Q. What does the Panel recommend?

A. We recommend maintaining our original adjustment to this program. Since the assumptions KEDLI made in developing its forecasted cost increase are unknown, and the contract does not guarantee a certain amount of work, or payment to the contractor, it is unreasonable to assume the significant cost increase proposed by KEDLI at this time.

Q. Did KEDNY make changes to the programs in the mandatory category?

A. Yes. KEDNY updated its proposed budgets for Damage

Prevention. The Staff Policy Panel is addressing that proposed change.

Reliability

Q. Does the Panel have any corrections to the testimony submitted on August 30, 2019 associated with Reliability budget category.

A. Yes. In our initial testimony we proposed delaying the start of the Storm Hardening – Remote Service Shutoff Valves project by one year, however, our slippage adjustment excluded moving the FY 2020 Capital and O&M project expenses into FY 2021. KEDNY's and KEDLI's FY 2020 budgets for this program are $3.136 million and $7.199 million, respectively. Therefore, we removed the program from FY 2021 and applied our proposed blanket reliability category negative adjustment of approximately 26 percent and 29 percent for KEDNY and KEDLI, respectively, to the Companies' FY 2020 budgets resulted in a proposed FY 2021 budget of approximately $2.321 million, and $5.136 million. Performing the same adjustment for the associated O&M expenses resulted in a FY 2021 budget of $0.164 million and $0.289 million for KEDNY and KEDLI respectively.

Q. Did Staff make any corrections to the Panel's Original

testimony related to Newtown Creek?
A. Yes. Staff corrected an error in its plant in service model. Staff adjusted the Budget for the program upward to reflect the most recent forecast.
Q. Please describe KEDNY's updates to its Reliability programs.
A. KEDNY provided updates to its Metropolitan Reliability Infrastructure (MRI) project, Marine Park Regulator Station Program, Clove Lakes Transmission Loop Project, and its CNG and LNG programs.
Q. Describe the updates to the MRI project?
A. KEDNY proposes accelerating the Metropolitan Reliability Infrastructure (MRI) project by one year to provide the hydraulic system support needed to enable new customer connections. The Company proposes moving Phase 1-4 from FY 2021 to FY 2020 and Phase 5 from FY 2022 to FY 2021. According to the Companies' response to IR DPS-1076, KEDNY states that its contractor is able to accelerate the work schedule of this project with no additional costs.
Q. Does the Panel agree with the update?
A. Yes.
Q. Describe KEDNY's update to the Marine Park Regulator Station Program?

A. KEDNY removed approximately $1.0 million in FY 2021 associated with the Marine Park Regulator Station Program citing that the program was only needed to support the NESE project.

Q. Does the Panel agree with the update?

A. Yes.

Q. Explain KEDNY's updates concerning the Clove Lakes Transmission Loop project.

A. To support new customer connections, KEDNY proposes performing a feasibility study and engineering analyses for the Clove Lakes Transmission Loop Project, with an associated budget of $0.5 million in FY 2021.

Q. Does the Panel agree with the update to the Clove Lakes Project?

A. Yes.

Q. Please describe the Companies' proposals concerning CNG and LNG.

A. The Companies are proposing to construct a CNG Filling Station located on Staten Island to transport CNG to KEDNY and KEDLI injection sites. The project cost is estimated to be $0.2 million in FY 2020 and $18.8 million in FY 2021 and is reflected in KEDNY's capital investment forecast with KEDLI paying 50 percent of the cost as a rental. The Companies state that without the additional

supply of NESE they will be unable to take their LNG tanks out of service to inspect and rehabilitate them as originally planned. As a result, they have developed a streamlined approach to inspect and repair their existing LNG tanks. Using this new approach, KEDNY's two Greenpoint LNG tanks will be taken out of service, inspected and repaired during off-peak seasons and then placed back into service during the critical winter heating season. Additionally, KEDNY proposes to delay the start of the LNG Truck Load/Unload Program that is described in their April 30th filing, from FY 2020 to FY 2021. In order to remain on schedule, KEDNY proposes to compress the remaining work plan, at a projected cost $2.44 million, of which $1.55 million will be in FY 2021. On page 13 of 20 of the KEDNY's Gas Infrastructure and Operations Panel second supplemental testimony, it cites a failure to execute a Memorandum of Understanding (MOU) with the City of New York as a reason for the delay. Finally, KEDNY proposes several new investments for its Greenpoint LNG site. These plans include the addition of portable LNG capabilities at the site that will allow LNG delivered via truck to on-system injection points, requiring a capital investment of $0.5 million in FY 2020 and $4.5 million in FY 2021 for permitting, engineering,

and site work, and two additional vaporizers with a cost of $1 million in FY 2020 and $13 million in FY 2021.

Q. What is the total financial impact of these reliability upgrades and updates?

A. KEDNY's proposed reliability project and program updates account for capital forecast increases of approximately $25.5 million in FY 2020 and $29.5 million in FY 2021.

Q. Does the Panel agree with KEDNY's updated proposals?

A. After reviewing the Companies' response to information requests DPS-1075, DPS-1076, and DPS-1077 we believe the projects are necessary to continue to provide safe and adequate service.

Q. Please describe KEDLI's updates to its Reliability program.

A. In order to meet increased gas demand from new customers, KEDLI proposes doubling the capacity of its existing Glenwood CNG facility at a cost of $0.5 million in FY 2020 and $6.5 million in FY 2021, in addition to adding two CNG injection facilities at undetermined locations at an estimated cost of $0.98 million each in FY 2020 and $9.9 million each in FY 2021. KEDLI proposes taking its Holtsville LNG tank out of service, inspecting and repairing it during the off-peak seasons and then placing it back into service during the critical winter heating

season. The change in scope of work will add $0.3 million in FY 2020 and $3.658 million in FY 2021 to the Holtsville LNG tank upgrade project.

Q. What is the total financial impact of these reliability upgrades and updates?

A. KEDLI's proposed reliability project/program updates account for capital forecast increases of approximately $2.8 million in FY 2020 and $30 million in FY 2021.

Q. Does the Panel agree with KEDLI's updated proposals?

A. We find the updates reasonable and necessary to support the connection of new customers.

Q. Does the Panel have any adjustments to the Companies' reliability category forecast?

A. Yes. We recommend adjustments to the Companies' updates using the methodology consistent with our adjustments made to the reliability budget in our direct testimony.

Q. Please explain.

A. As stated in our direct testimony, we found significant variances resulting from over-estimating project/program costs between planned and actual spending in the reliability category. Based on these variances, we propose an application of the same negative adjustment of approximately 26 percent and 29 percent for KEDNY and KEDLI, respectively, to the Companies' proposed Rate Year

budgets and to all newly proposed or updated programs and projects within the reliability category.

Q. Why should the same adjustment be applied to the Companies supplemental filing?

A. We recommend that until the Companies can demonstrate the ability to accurately forecast reliability budgets without over-estimating that Staff's methodology to adjust the Companies forecast budgets still provides for a more accurate forecast.

Net Plant and Deprecation Expense

Q. Does the Panel have any corrections to the Net Plant proposed in the testimony filed on August 30, 2019?

A. Yes. In our original filing, we inadvertently double counted the reduction in net plant for the no NESE scenario. We subsequently provided a corrected Net Utility Plant model.

Q. What are the Panel's Rate Year projections for gas net utility plant in service and gas depreciation expense?

A. We project gas net utility plant in service of $5.805 billion for KEDNY and $3.988 billion for KEDLI, and gas depreciation expense of $145.8 million for KEDNY and $97.1 million for KEDLI. These figures were provided to the Staff Revenue Requirements Panel.

Q. How do the Panel's recommendations compare to the Companies' Second Supplemental filings?

A. As shown on page one of Exhibit___(SGIOP-11), we recommend adjustments that, in total, reduce KEDNY's Rate Year average gas net plant in service by $2.7 million, and reduce its annual gas depreciation expense by $13.8 million. We recommend adjustments that, in total, reduce KEDLI's Rate Year average gas net plant in service by $10.3 million, and reduce its annual gas depreciation expense by $10.8 million.

O&M

Q. Do you have any corrections to the O&M adjustments proposed in your testimony filed on August 30, 2019?

A. Yes. Our adjustment to the O&M expenses associated with the storm hardening program have been corrected as explained in the Reliability category above. This correction is reflected in our Exhibit___(SGIOP-12). This is the only correction.

Q. In the Companies' second supplemental testimony, did the Companies update the proposed O&M expenses associated with GIOP programs?

A. Yes. The Companies updated the O&M expenses to add incremental funding for the expansion work at the

Glenwood CNG site. They also propose incremental funding for the two new CNG sites that will be required at locations yet to be determined.

Q. Does the Panel agree with these incremental expenses?

A. Yes. Given the need to meet new and increased demand, these incremental expenses for the associated capital investments are appropriate.

Q. Does this conclude the Panel's testimony?

A. Yes

MR. FORST: Your Honors, I now proffer the panel for cross examination.

A.L.J. COSTELLO: Okay, companies.

CROSS EXAMINATION

BY MR. DECICCO:

Q. Good morning, Panel.

A. (Panel) Good morning.

Q. We are all that stands between the end of the hearing, so there's no pressure on us here to keep this going. But I'd like to discuss various adjustments the panel is recommending with regard to the company's capital investment forecast in the rate year. And I will talk about some specific adjustments in a minute, but I just wanted to hit one or two general concepts first. And does the panel agree that the companies as the operator of the gas networks must make the investments they deem necessary to maintain safe and adequate service?

A. Yes.

Q. When looking…

MR. FORST: Pardon me one second.

MR. DECICCO: Sure.

MR. FORST: Can I just ask the panel that they speak directly into the microphones?

MR. WALTERS: For the record, the answer was yes.

MR. FORST: Thank you.

BY MR. DECICCO: (Cont.)

Q. When looking at the total capital spending over the term of the current rate plan, is the panel aware whether the companies have underspent or overspent the capital budgets -- excuse me -- that are used to set rates?

A. (Walters) Yes. We are aware.

Q. Okay and I'm sorry. Is the panel aware that the companies have overspent or underspent their capital budgets during that period and we can break it out by company?

A. We're aware that a -- regarding the entire capital budget the company has variances with regard to its budget.

Q. Okay but why don’t we…

A.L.J. LEARY: I'm sorry. Let's break this down.

MR. DECICCO: Yeah. I'm -- I'm going to do that I think in a way that will be easier.

A.L.J. LEARY: So I missed one of the words you said and did -- you're aware that the

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas companies overspend or underspend?

MR. WALTERS: Can you clarify which year you're referring to?

MR. DECICCO: Yeah -- yeah. Why don't I ask this question in a way that's more productive? I'm going to turn your attention to the panel's exhibit SGIOP2, which has been premarked as Exhibit 465 for identification and I have copies -- can I just pass these out -- for anyone that doesn't have it in front of them.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Thank you.

BY MR. DECICCO: (Cont.)

Q. Does the panel have SGIOP2 in front of them?

A. (Walters) Yes.

Q. Okay.

A.L.J. LEARY: This is Hearing Exhibit 465 already marked.

BY MR. DECICCO: (Cont.)

Q. And is this an exhibit the panel -- this is an exhibit the panel prepared. Is that correct?

A. Correct.

Q. And is summarizes the panel's analysis of the company's capital budgets, actual investments and the variances between the budgets and -- and the actual expenditures. Is that correct?

A. Correct.

Q. Okay and starting with page one, which is for KEDNY. Looking at the four most recent year's depicted fiscal years 16-19, would you agree that KEDNY has overspent its total budget -- its total capital budget in each of those years?

A. We would agree.

Q. Okay. And -- and I don’t want to make you do math here, but a -- just looking at the -- at the numbers there, the percentage in particular on the bottom, would you agree that KEDNY has overspent on average about 14.275% over that four-year period?

A. I have not done the math, but if that's what these averages would equate to -- I can't say for sure -- but the percentages for each year are listed on the exhibit.

Q. Okay. And turning to page two of SGIOP2, again marked as Exhibit 465 for identification, this page is for KEDLI. Correct?

A. Correct.

Q. And looking at the four most recent years depicted, fiscal years 16 to 19, would you agree that KEDNY -- KEDLI overspent its rate allowance in three of those years?

A. Correct.

Q. Okay. And -- and again I'm not going to make you do math, but would you agree that subject to check it's about on average 6.37% on average over that four-year period?

A. Again, the variance is listed for each year.

Q. Okay. Thank you. I'd like to discuss now the concept of a capital investment reconciliation mechanism, which the panel discusses starting on page 91 of its direct testimony. And I'll just let you get there. Okay. Is this mechanism also commonly referred to as a net plant tracker?

A. It is.

Q. And staff proposes to continue the one-way downward only net plant tracker in the current rate plan. Is that correct?

A. That's correct.

Q. So under this net plant tracker,

if the rate allowance for capital investments is higher than the revenue requirement associated with the companies actual investments in that year meaning the companies effectively overspent -- effectively -- I'm sorry. Strike that question. Let's start over. So if the rate allowance for capital investment is higher than the revenue requirement associated with the companies actual investment in the rate year meaning the companies effectively spent less than the approved budget, the companies would defer the difference for return to customers. Is that correct?

A. That's correct.

Q. Okay. So and - and just to sort of how that would play out. So in the one year we identified that KEDLI underspent its budget to the extent that it underspent that budget the revenue -- associated revenue requirement would then flow back to customer -- be deferred for eventual return to customers. Is that correct?

A. Correct.

Q. Okay. But if -- if the rate allowance is lower than the revenue requirement for the companies capital investments, meaning the companies effectively overspend their rate allowance,

the companies are not made whole under your proposal. Is that correct?

A. Correct.

Q. So in the years that we just discussed where KEDNY overspent its budget according to your analysis the companies would not be able to collect additional recoveries from customers under this net plant tracker. Is that correct?

A. That's correct.

Q. I'd like now to discuss the Panel's approach to adjustments in general. And beginning on 30 -- page 32 of your direct testimony, the Panel testifies that you generally support the company's capital investment plans, but recommend specific adjustments to the -- to the -- to the programs in the capital's -- excuse me -- to the company's budget. Is that correct?

A. (Cont.) That's correct.

Q. Okay. And the first category is customer connections. And I'm going to now turn to your supplemental testimony, starting on page five. Is -- is the Panel there?

A. (Shi) Yes.

Q. Okay. Thank you. Is the Panel

familiar with the recent settlement between the companies and the Department of Public Service regarding the down state supply constraints and new gas connections?

A. Yes.

Q. And is the -- is it the Panel's understanding that the companies will be connecting new gas customers in the rate year?

A. Yes.

Q. And on that basis, does the Panel agree that the reasonable costs to connect new customers in the rate year should be reflected in the company's rate allowance?

A. Yes, that's correct.

Q. Okay. Starting on page seven of your testimony, you've proposed several adjustments in this category based on the company's forecast unit cost. Is that correct?

A. Yes. But for -- but that was a -- a breakdown component. I mean, installment, yes, based on the company and focus number of unit. But not for the tariff -- tariff main.

Q. Okay. Yeah. Let's -- let's -- let's just break that down. And -- and I appreciate

the clarification. And can you just explain what -- what you mean by unit cost, what -- what -- what a unit cost is in the context of your testimony?

A. Okay. The unit cost, I mean, the company basically, when the company project the unit cost, you know, I state in the direct testimony, they base it on the, you know, two years average, which is the 2017 and '18. Then escalate by the, you know, 2.5%, accept the contract -- contractor.

Q. And -- and is it fair to say that you looked at the companies' average unit costs in prior years and developed your own forecast of what the unit cost should be?

A. No. Staff or staff unit cost of oil based on the most recent three years.

Q. Okay. And did you use the most recent three years in every case?

A. Largely, you know, yeah, for most -- most the budget item.

Q. Okay. And in calculating a forecast for a line item, is it true that you take the unit costs and then multiply that by the forecast number of units to get at the budget for each line item?

A. That's correct.

Q. Okay. And would you say that staff's approach in this category was to make normalizing adjustments, in some cases, that eliminated some years based on staff's view of whether that was, for instance, an outlier, that year was an outlier, or -- or represented the companies' future costs?

A. Could you repeat it?

Q. Sure. And would you -- would you agree that staff's approach in this category was to make normalizing adjustments that eliminated years that the Panel viewed as outliers?

A. Could you -- could you specify which one?

Q. Sure. Why -- why -- we'll -- we'll -- we'll get into some specific examples. I was asking generally if that was your approach to making unit cost adjustments.

A. Yes.

Q. Okay. So in some cases, you've argued that more recent data is a better indicator of future costs. Is that correct?

A. It depends, you know, which

specific item you talking about.

Q. Okay. Fair enough. Yeah, let's get into the specific items, because I think that'd be more productive. But I'd just before we do, would you agree that historic data should be adjusted for known and measurable changes to the company's forecast when that information is available?

A. What do you mean by measurable?

Q. When -- when the companies know that they're going to incur additional costs, and that is measurable or capable of being forecast. Do you agree that historic data should be updated to reflect that type of information?

A. Could you give me some like example?

Q. Sure. So to the extent, for instance, the company's historic data did not reflect the impacts of new paving regulations, like as being experienced in New York City, would you agree that the historic data should be updated -- and this is a hypothetical -- to reflect sort of the known costs of that incremental work?

A. Yes. I think as state in the staff, you know, direct testimony, I think -- just

give me one second. Yes. I -- as state in the staff direct testimony, we believe that, you know, three years average, you know, will capture, you know, some of the most recent change, including the new paving regulation.

Q. Okay.

A. That's we believe that, you know, staff unit cost will be more reasonable compared to the -- the company's.

Q. So -- and -- and again, you're -- you're -- you're saying that you're using the three most recent years in order to make those calculations?

A. Yeah, for most the budget item.

Q. Okay. Why don't we look at some specifics, including -- we'll start with KEDNY's install main commercial, which the Panel discusses on page seven of their testimony beginning at line 20.

A.L.J. LEARY: Direct.

MR. DECICCO: I'm sorry. Yes, their supplemental testimony.

A.L.J. LEARY: Supplemental.

MR. DECICCO: Yeah, page seven, line 20.

BY MR. DECICCO: (Cont.)

Q. And the panel recommends uni -- using a unit cost based on a three-year average. But in this case, would you agree the Panel proposes to exclude one of the last three fiscal years under a theory that that year was an outlier?

A. (Cont.) Yes. But the major driver is not because the -- the paving. Because the large size of the main, you know, the company used in the past, you know, three years, which including one the official year, 2017 and 2019.

Q. Yeah.

A. That's why staff, you know, normalizing.

Q. So -- so in this example, staff removed fiscal year 2017, and is recommending using a three year average consisting of fiscal years 2016, 2018, and 2019. Is that correct?

A. That's correct.

Q. Okay. And that adjustment reduced the unit cost from $589 to $526. Is that correct?

A. Yes.

Q. Is it the Panel's view that costs in fiscal year 2016 are more indicative of the

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company's spending during the rate year fiscal '21 in this area?

A. Could you repeat it?

Q. Sure. In -- in normalizing out fiscal year 2017, is it the panel's view that costs in -- in 2016 are more indicative of the company's spending during the rate year fiscal year '21 in this area?

A. Yeah. It's close to it because staff, they escalate that by GDP in --.

Q. Okay. Does the Panel believe that fiscal year 2016 reflects the cost, for example, to perform the additional New York City paving requirements?

A. I -- we believe that it reflects it similar to fiscal year '17, because the paving requirements did not take effect until fiscal year '18.

Q. Okay. And does the Panel believe that the 2016 contractor costs are more indicative that -- than -- than 2017 costs, for example?

A. The company did not provide, you know, additional information for showing that. So we don't know.

Q. Okay. As the Panel sits here today, do you know the size of the main installations that the company will be performing in the rate year?

A. Could you re -- rephrase you question?

Q. Sure. As the Pa -- as the Panel sits here today in -- in the category that we're discussing, do you know the size of the main installations the company will be performing in the rate year?

A. That's different size. Yes, we are aware of the different size, but the -- the company did not provide the -- the detailed breakdown for the rate years.

Q. Did -- did the Panel request that information from the company?

A. We did request that how the company, you know, developed the unit cost. But the -- for the detailed breakdown, but the company did not provide that.

Q. Okay. And is the -- is the Panel in this category using the same three years to forecast the number of units in the rate year?

A. Say that again.

Q. Sure. Is the Panel using the same three years, 2016, 2018, 2 -- 2019, to forecast the number of units in the rate year?

A. It's for install commercial main, we use the company's -- the forecast number of the unit.

Q. You -- you adopted the company's forecast for the units in this category?

A. Yes.

Q. Okay.

A. For install main commercial.

Q. That's -- that's the Panel's testimony? Okay. We can move on then. With regard to -- we're going to move onto now KEDNY meter purchases starting on page nine of the supplemental testimony. The Panel proposes to exclude in this case the most recent year, fiscal year 2019, and instead use a three-year average unit cost that includes 2016, 2017, and 2018. Is that correct?

A. That's correct.

Q. Okay. And was 2019 the high -- the company's sort of highest costs over the last several years? Is that -- is that the basis for the exclusion in this -- in this case?

A. Yes, according to the company response that 2019 was double than three years average historical unit cost, because by the two factor. One was an increase on the high -- higher cost of the large commercial industry meter the company purchased in that year. And also, a decrease in the total number of the residential meter purchase.

Q. So -- so it's the Panel's position that in this -- in this instance, more recent data, 2019 data, is -- is not indicative of what the company will spend in -- in the rate year fiscal year '21?

A. I -- I mean, usually, in general speaking, you know, residential -- number of the residential meter usually that have the more large number. But in 2019, you know, when we found that looks like residential meter number is less than, you know, the commercial.

Q. Okay. And again, with respect to the units, is it -- is the Panel using the same three years for purposes of forecasting the number of units the company will -- will install in the -- in the --

A. No, we just try to normalize

that, what the general, you know, situation -- general situation will look like.

Q. And -- and -- and how did the Panel go about normalizing the number of units in this category?

A. That's -- that's the reason we exclude that 2019.

Q. Okay. But I'm -- I'm saying no -- with -- I -- I understand with respect to unit cost you excluded 2019. With respect to the number of units, did you also exclude 2019?

A. The number of the unit, we used that -- the company's forecast.

Q. Okay. And is that true in all these categories? Is it the Panel's position that they intended to use the company's forecasts in -- in -- in -- in forecasting the number of units?

MR. FORST: Objection, Your Honors. I think that's a vague question. There's a number of individual categories associated with all of these. And to ask generally that they accept, you know, the company's position, I don't think that's going to be clear for the record.

A.L.J. LEARY: Yeah, I'm a little bit

lost. So can you break it down, Mr. Decicco?

MR. DECICCO: Sure, yeah.

BY MR. DECICCO: (Cont.)

Q. Just with respect to -- to -- the -- the category we're talking about, KEDNY meter purchase, is it the Panel's testimony that they intended to use the company's forecasts of units in this category?

A. (Cont.) Yes, for meter purchase. For this category, yes.

Q. Okay. We can move on. So now -- and -- and I'm just going to move through a couple of these categories and the customer connections. Now talking about the KEDLI install main commercial. And KEDLI, starting on page nine of the supplemental testimony, line 17, "Staff updated the unit costs of KEDLI's tariff main using a three-year average by excluding 2019 data, and instead including 2016 data." Is that correct?

A. Could you say that again? Which -

-

Q. Sure. And again, I'm on page nine, starting this -- this discussion starts on -- on page seven, excuse me, line 17, and moves up to

the top of page 10. Is it true in this category, staff updated the unit cost of KEDLI's tariff main using a three-year average by excluding 2019 data and including 2016 data?

A. Yes. We -- as for tariff main, yes.

Q. Okay.

A. We excluded 2019 data.

Q. And -- and fiscal '19 was the highest year in terms of unit cost in the company's recent experience. Is that correct?

A. Yes. That was higher, because the higher contract cost associated with the Department of Environmental Protection cost.

Q. And -- and as the Panel sits here today, do they have any reason to believe that the company will not continue to incur those costs in the -- in the most recently available fiscal year, fiscal year '19?

A. There was not response from the company was showing that.

Q. So it's the Panel's view that that co -- that those costs will not continue to be recovered in -- in -- in subsequent years?

A. Based on the information we have, we don't know.

Q. Okay. And I should have said that the cost will continue to be incurred in -- in future years.

A. What do you mean by the cost incurred?

Q. So -- so I believe your testimony was that fiscal year '19, in your view, was -- was an outlier, because it incurred certain costs related to dumping, I believe, DOT requirements. Is it the Panel's view that those costs will not continue in the rate year?

A. We don't know. In the rate year, there's no data showing. We don't know what portion of the, you know -- that cost associated with DEP that increase caused will carry in the rate years.

Q. And thank you for correcting me. It's -- it's DEP requirements, not -- not DOT requirements. Thank you.

A. DEP.

A.L.J. LEARY: That's New York City Department of Environmental Protection. Is that the agency to which you are referring?

MR. SHI: No. I -- that was the EP.

A.L.J. LEARY: And I say that is the New York City Department of Environmental Protection. Is that correct, Mr. Decicco?

MR. SHI: New York State.

MR. WALTERS: We believe it's the New York State Department of Environmental Protection.

A.L.J. LEARY: Okay. So that would be New York State Department of Environmental Conservation, DEC. So I just want the record to be clear. What is it? It's not DOT.

BY MR. DECICCO: (Cont.)

Q. I believe it's the DEP. I believe it's the --

A. (Shi) DEP. That was state by the company's, you know, testimony.

Q. Okay. I believe, Your Honor, it's the city Department of Environmental Protection.

A.L.J. LEARY: So it is New York City Department of Environmental Protection, DEP.

MR. DECICCO: Yeah.

MR. SHI: That's correct.

A.L.J. LEARY: Thank you.

BY MR. DECICCO: (Cont.)

Q. Okay. And -- and again, in this category, for the number of units, is the Panel using the same three years in forecasting the number of units for -- for KEDLI install main?

A. (Cont.) Yes.

Q. It's inc -- it's including 2019?

A. That's correct.

Q. Okay. So the Panel believes that 2019 is a good measure of the rate year number of units, but not a corresponding unit cost. Is that -- is that correct?

A. For some items, yes.

Q. Okay. I'd like now to talk now about the reinforcement programs. And staring on page 11 of the Panel's supplemental testimony, line 20, "The Panel proposes a proposed adjustment to the Southeast Suffolk infrastructure phase one project."

A. Yes.

Q. Okay. And before discussing the specific adjustments, does the Panel agree generally that reinforcement programs are important to maintain system pressure, especially when adding additional gas load for new customers?

A. Yes.

Q. Okay. Specific to this project, the Southeast Suffolk infrastructure phase one project, does the Panel support the need for the South -- Southeast Suffolk infrastructure project to maintain system pressures on KEDLI's system?

A. Yes, we support the need.

Q. Okay. On page 12 of your supplemental testimony, line four, the Panel explains, "KEDLI's proposed shift of $5 million for the project from fiscal '22 to fiscal '21, which is the rate year." Is that correct?

A. Could you repeat that?

Q. Sure. One page 12 of your supplemental testimony, beginning on line four, the Panel discusses the proposed shift of $5 million of this project from fiscal '22 into fiscal '21, which is the rate year. Is that correct?

A. That's company proposed.

Q. Yes. I'm saying the Panel discusses that proposal, correct? And is it true that the panel opposes the shift because -- and -- and I'm quoting from the supplemental testimony, page 12, line nine. "The proposed shift of $5 million from fiscal 2 -- 2022 to fiscal year 2021 results in not

change to the outcome to complete this project in fiscal 2022. Therefore, we redirect -- we reject KEDLI's proposal to shift $5 million from fiscal year 2022 to fiscal year 2021 for the Southeast -- for KEDLI's Southeast Suffolk infrastructure phase one."

A. That's correct.

Q. Okay. So was it the Panel's understanding that KEDNY was proposing -- KEDLI was proposing to -- to accelerate the in-service date for this project?

A. There is no supporting document showing that they will move the service date. From my understanding, the project's still completely in 2022.

Q. Okay. And -- and I'm going to ask you if you're aware of -- the company testified in its second supplemental testimony, beginning on page eight, starting on line 17, "More -- moreover, to enable new customer connections, KEDLI has accelerated phase one of the Southeast Suffolk project to ensure that it will be in service in fiscal year '22." Are you aware of that testimony?

A. Yes --

Q. Okay.

A. -- fiscal year 2022.

Q. So does the Panel agree that KEDLI did not say it was accelerating the in-service date for the project into fiscal year 2021?

A. We don't have. The company did not provide us the detail about when is the actual service date.

Q. Okay. But -- but according to testimony I just read, the company said that it -- it wanted to ensure that the project would be in service in fiscal year 2022. Is that your understanding?

A.L.J. LEARY: Excuse me. I need to know a little bit more specifically the testimony to which you --

MR. DECICCO: Sure.

A.L.J. LEARY: -- are referring and I believe quoted.

MR. DECICCO: Yeah.

A.L.J. LEARY: Could you say that for the record again?

MR. DECICCO: Sure. I'll -- I'll read it -- I'll read it one more time.

A.L.J. LEARY: No. I don't need you to read it. I need you to tell me --

MR. DECICCO: Oh, I'm sorry.

A.L.J. LEARY: -- where you got it.

MR. DECICCO: It's the -- the Panel's -- the -- the company's Gas Infrastructure and Operation Panel, second supplemental testimony, page eight, starting on line 17. And actually, I have a copy of it if -- if you'd like to see it.

A.L.J. LEARY: I think that that was -- you're reading my mind, because I think the Panel would like to see that testimony for themselves, see the context in which it was stated, and so forth. I -- we don't need it. Oh, they do have it? Does the Panel already have this testimony in front of them? So it'd be the second supplemental testimony of the company's Gas -- I'm sorry. It's KEDLI's.

MR. DECICCO: It -- it's on behalf of both companies, Your Honor.

A.L.J. LEARY: It's on behalf of both.

MR. DECICCO: Correct.

A.L.J. LEARY: But --

MR. WALTERS: We do, Your Honor.

A.L.J. LEARY: Okay. Sorry about that. Yeah, I just --

MR. WALTERS: Just cite the page

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again.

BY MR. DECICCO: (Cont.)

Q. Sure. It is on page eight of 20, starting on line 17. And -- and -- and after you've had a chance to review -- oh, I'll -- I'll wait.

A. (Cont.) The company original budget also, I mean, original -- the filing also stated the completion also in the fiscal, you know, 2022. So I believe there was no change.

Q. Okay. And -- and -- and my question was that the company -- yeah. I think you answer, to understand, was not claiming that they were accelerating the in-service date of the project. Is that correct?

A. From the -- the information we received, yes.

Q. Okay. Has the panel done any analysis to determine the project can be delivered in fiscal year 2022 without accelerating the spend in the rate year, as the company has requested?

A. Yes, I believe so.

Q. Okay. And has -- and -- and was has the panel determined with respect to the company's ability to deliver the -- the project in

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2022 without accelerating the spend as -- as proposed?

A. Based on the company's budget in those two years. There was not budget on the fiscal year 2023 for the phase one.

Q. Okay. I -- I just want to -- the company has -- has testified that it -- it felt the need to accelerate the spend in order to deliver the project in fiscal year 2022. And I'm just asking whether the Panel has an independent view of whether that's necessary.

A. If there's no change in the completion date, I don't -- we don't believe that it's necessary.

Q. Okay. So you're basing that totally on what you see in the budget, not on the company's analysis of -- of the work that needs to be done?

A. The company has not provide us the detail about, you know, the service -- service date change.

Q. Okay. Let's talk next about city/state construction. Starting on page 46 of your direct testimony you discuss proposed adjustments to

the company's city/state construction program. Is that correct?

A. Bear with me a sec.

Q. Sure.

A. That's correct.

Q. Okay. And could you just describe generally what your understanding is of -- of city/state construction?

A. Our pro -- our testimony describes the program. In general, it consists of main replacement work and support and protect work, which is required to accommodate public work projects, including work on sewer, water, drainage infrastructure, st -- street reconstruction, and bridge replacement projects, among others.

Q. Okay. And looking at KEDNY's city/state construction project program in particular, does the Panel agree this work is reactive to New York City's construction requirements?

A. Yes, that's true.

Q. Okay. And does the Panel agree that the company's obligated to perform this work to accommodate New York City's construction activities?

A. Yes.

Q. Okay. On page 48, line eight, the Panel testifies that it supports the company's methodology for calculating city-state construction, but suggests the company's adjustment to add the cost for incremental paving is unnecessary. Is that correct?

A. That's correct.

Q. Does the panel agree that the new paving requirements in the city of New York are adding significant cost to the company's construction work?

MR. FORST: Objection. Could the counsel clarify what he means by "significant"?

BY MR. DECICCO: (Cont.)

Q. Turning your -- your attention to line -- to page 45, beginning on line 18, the question was, "How have these new requirements impacted KEDNY's budget?" And the -- the question's referring to -- to this -- to the city's paving requirements. Is that correct?

A. (Cont.) Yes.

Q. Okay. And the Panel testified, beginning on line 20. "As expected, the new

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requirements have led to increases in the unit cost to install and replace new gas main." So is it the Panel's testimony that these new paving regulations have increased the company's costs for main replacements?

A. Yes.

Q. Is the Panel aware whether KEDNY has historically underrun or overrun its city/state construction budget?

A. Bear with me one second. I don't have that data in front of me. But I believe in recent years there has been increased costs, and they have overrun budgets in recent historical years.

Q. Okay. So -- so why don't we just look at the most recent year. And I'd ask that you look at SGIOP-1, which is a DPS 9-14 -- and this has been marked for identification as Exhibit 463 -- beginning on page 722. Do you have that in front of you? And I have copies of it for anyone that will need it.

A. Can I have a copy of it, please?

Q. Sure.

MR. FORST: Your Honors, for anyone following along electronically, that's page 734 of

the PDF exhibit.

A.L.J. COSTELLO: Thank you.

MR. DECICCO: And -- and -- and thank you, Mr. Forst.

BY MR. DECICCO: (Cont.)

Q. Looking at page 724 -- well, first, could -- could the Panel explain generally what this IR is -- is -- is providing?

A. (Cont.) This IOR provides the most recent fiscal year 2019 company's budgeted and actual capital expenditures.

Q. Okay. And looking about a third of the way down the page, there's a number of categories entitled "city/state construction." Is that correct?

A. That's correct.

Q. Okay. And looking at t -- the three major categories there, city/state construction non-reimbursable, city/state construction reimbursable, and city/state construction reimbursements, would the Panel agree that in fiscal year 2019 the company overspent its capital budget for city/state construction?

MR. FORST: Objection. Could the

counsel specify what category? He just listed four separate ones, and now is referring to city/state construction in general.

MR. DECICCO: Okay. Well, it's -- we can go through each of them. I was referring to the three major ones, actually. It was three.

BY MR. DECICCO: (Cont.)

Q. But for the city/state construction non-reimbursable, could you -- what does this IR indicate the company did with respect to its -- its budget?

A. (Cont.) They overspent this budget on this line item.

Q. And -- and how much did they overspend by in that year?

A. Approximately $84 million.

Q. And to be fair, in the lines below, the line below, did the company underspend its budget in -- in that category?

A. Yes.

Q. And also true of the line below that too?

A. Yes.

Q. Okay. Would you agree, though,

between these three categories the company overspent its budget by about $20 million?

A. The three categories are -- can you clarify?

Q. Sure. C -- c -- city/state construction non-reimbursables, city/state construction reimbursables, city/state construction reimbursements.

A. Yes. It looks like they slightly overspent.

Q. By slightly --

A. I don't have the math in front of me, but it looks like under 10%.

Q. It -- it's -- well, would the Panel agree that, in round numbers, 84 million less 54 million less 10 million is $20 million?

A. Correct.

Q. Is it the Panel's view that $20 million is a slight overspend in this category?

A. No.

Q. Okay. Let's talk about the true-up mechanism. On page 88 of your direct testimony, line 17, you discuss the city/state construction true-up mechanisms. And I'll give us all a moment to

get there. Again, it's line 88, excuse me, page 88 of your direct testimony, line 17.

(Off the record discussion)

A.L.J. LEARY: The exhibit that was just discussed by the Panel was hearing Exhibit 463.

BY MR. DECICCO: (Cont.)

Q. So under the current true-up mechanism in the company's current rate plan, to the extent the company's -- and it's -- it's by company. But to the extent either company overspends the rate allowance for city/state construction, they confer -- can defer 90% of the revenue requirement impact and absorb 10 -- the other 10%. Is that correct?

A. (Cont.) That's correct.

Q. Okay. And it's vice versa on any underspend to the extent that they underspend their budget, they -- they will return 10 -- 90% to customers and -- and absorb the other 10% or retain the other 10%. Is that correct?

A. Correct.

Q. Okay. On page 90, line one, the Panel proposes to ad -- adjust the true-up mechanism from a 90% mechanism to an 80% mechanism. And -- and that effectively means the companies would only

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recover 80% of the revenue requirement associated with any overspend. Is that correct?

A. That's correct.

Q. And conversely, would only return to customers 80% of any underspend. Is that correct?

A. That's correct.

Q. Okay. So I -- I just want to -- and with respect to the -- to the rate year, has the -- has the Panel made a -- a pr -- has proposed an adjustment with respect to the rate year recovery for city/state construction?

A. Can you repeat the question?

Q. Sure. For the -- for city/state construction -- and -- and I want to be specific, the KEDNY -- has the Panel recommended an adjustment to the forecast budget for city/state construction?

A. Yes, we have.

Q. And what is the amount of the adjustment that this -- that the Panel is proposing?

A. Between the two line items, non-reimbursable and reimbursable, it's about $24 million.

Q. Okay. And so just to summarize, the Panel's proposing the -- a $24 million adjustment

and also proposing a change to the tracker from a 90% tracker to an 80% tracker. Is that fair?

A. Yes, that's fair.

Q. It is the Panel's view that the company's shareholders should absorb prudently incurred city/state construction costs?

MR. FORST: Objection. Could the counsel clarify what he means by "prudently incurred"? I think we've had numerous discussions about how that term is being used here.

A.L.J. LEARY: We -- we have to sort of stay firm on this prudent bar. So would you rephrase Mr. Decicco?

MR. DECICCO: Sure.

BY MR. DECICCO: (Cont.)

Q. Is it the Pan -- I'll take the word "prudent" out. I think we can ask the same question. Is it the Panel's view that the company shareholders -- and I'll say -- should fund the cost of city/state construction costs?

A. (Cont.) The Panel does not believe that its proposed adjustments will -- or leading to shareholders funding this program.

Q. So in -- in the example, we just

want the most recent year where we -- we said the company had overspent its -- its rate allowance for city/state construction by $20 million. Wouldn't it be true in that case that the company's shareholders would be funding a portion of that overspend?

A. Well, the company had proposed the way it budgeted this program, and staff accepted that methodology. And staff adjusted it for what it viewed was incremental cost that the company had added, which were in -- unnecessary in the rate year, because they had already been reflected in prior years.

Q. Which I appreciate. And -- and -- and -- and I -- thank you. But the question was, in the example that we just discussed in fiscal year '19 when the company overspent its city/state construction budget, under the company's -- under the Panel's proposal, wouldn't the company -- wouldn't the company shareholders be paying 20% of that -- of that overrun?

A. That is correct.

Q. Is it the Panel's view that a mechanism that puts the company at risk for the current 10% of city/state construction costs does not

provide sufficient incentive for it to manage costs?

A. Can you repeat the question?

Q. Sure. Is it the Panel's view that the current mechanisms that puts the company at risk for 10% of potential city/state construction overruns does not provide sufficient incentive to manage costs?

A. It is.

Q. On page 90, line 13, the Panel also testifies that it opposes the proposed tracker to reconcile operation and maintenance -- maintenance expense, excuse me, associated with the city's city/state construction, excuse me, associated with the city/state construction program. And that would be in -- the company had proposed in both direction. Is that correct?

A. Correct.

Q. Does the Panel agree that the O&M, or operation and maintenance expense, is also a function of the work required by the city?

A. It is.

Q. And similarly, does KEDNY, for -- for example, have an obligation to provide O&M expense to support the city's project, in some cases?

A. It does.

Q. And does the Panel agree that the level of O&M expense work is not within the company's complete control?

A. Yes.

Q. Thank you. And moving on. We're -- and we're more than halfway there. So getting on to leak-prone pipe. The company's direct testimony, page 51, the company, excuse me, the panel discusses adjustments to the leak-prone pipe program. And I'm just going to say -- strike that. Does the Panel agree that the leak-prone pipe replacement provides significant customer b -- benefits in terms of safety, reliability and reducing leaks and methane emissions?

A. Yes.

Q. Is the Panel aware of the commission's stated goal to reduce the statewide average replacement timeline for leak-prone pipe to 20 years?

A. Yes.

Q. Is the Panel aware that the Staff Pipeline Safety Panel is supporting an increase to the negative revenue adjustment associated with

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achieving in the company's leak-prone pipe replacement targets from eight basis points to 12 bases points?

A. Yes.

Q. In the Panel's direct testimony, beginning on line 50 -- beginning on line page 52, excuse me, line one, the Panel opposes certain inflation adjustments specifically to remove KEDLI's proposed inflation rate on contractor increases. Is that correct?

A. That's correct.

Q. As a general matter, does the Panel agree it is reasonable to make known and measurable adjustments to the company's forecast unit costs to reflect increasing contractor costs?

A. Known and measurable increases, correct. Yes. However, the Panel does not -- has not been provided enough information to determine whether these contractor costs are known and measurable.

Q. Okay. Fair enough. And -- and we'll get to those. But I -- yeah. I was asking you a general question. And I appreciate the answer. If the company's contractor costs are increasing in the rate year, would the Panel agree that would be an

appropriate adjustment to the forecast? And that's a hypothetical question.

A. Yes.

Q. Okay. And without accounting for any known contractor increases, and holding the company to replacement targets, does the Panel agree that these programs would be underfunded?

A. No.

Q. And -- and can you explain that?

A. Why we think that they would not be underfunded?

Q. Correct.

A. The Panel reviewed historic costs for this program, where the -- and similar to install main for customer connections, the company developed its budget for leak-prone pipe on a unit cost basis. And so the comp -- the staff reviewed historical unit costs, and applied appropriate inflation factors, and then developed the budget based on the amount of units to meet the targets.

Q. So -- so why don't we turn to your supplemental testimony on this topic. And -- and beginning on page 14, line 8, you explain that KEDLI in its second supplemental testimony, which was file

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in December of 2019, updated its forecast for leak-prone pipe to reflect the impacts of the final negotiated agreement with its contractor for main installations. Is that correct?

A. That's correct.

Q. And the final negotiations actually resulted in a decrease to the unit costs for KEDLI's proactive main replacement, or its leak-prone pipe program, as compared to the original inflation adjustment in its initial filing. Is that correct?

A. As compared to to their original adjust -- proposal, correct.

Q. Correct. So the supplemental inflation factor was lower than what the company initial filed for?

A. Correct.

Q. Okay. In discovery, did the companies provide copies of the updated contracts?

A. They did.

Q. Did they also provide a forecast of the impacts from the updated contracts on their leak-prone pipe budget?

A. They did.

Q. It is the Panel's position that

the company's contractor costs will not be going up in the rate year?

A. The Panel was unable to determine this with confidence based on the information that was provided in response to the information requests where the Panel requested this information.

Q. But -- but looking at the contract, is it -- is it the Panel's position that the contractor costs stated in the contract are not going up?

A. Looking at the contract, again, staff was unable to determine the cost impact of the new contract.

Q. And -- and among the reasons you cite in your testimony is that the new contract is complex. Is that -- is that fair?

A. That's fair.

Q. And did the company provide a forecast that included its analysis of the contract implications, and therefore, what the forecast would be in -- in the rate year?

A. The company provided some information of their forecast.

Q. Okay. Today is February 25, 2020.

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Staff has had to this point two months to analyze these adjustments and perform discovery. Is that correct?

MR. GOODRICH: Objection. The -- staff put in its testimony on January 29th. The fact that it's now February 25th is -- is irrelevant to that.

MR. DECICCO: Well, I'm -- I'm just saying discovery is still open. I'm just -- the question I'm just asking counsel is whether they've had two months to consider this.

MR. GOODRICH: Discovery --

A.L.J. LEARY: Well, -- okay. Hold on. This is the start of what I -- sounds like a fairly argumentative approach. And I want to just say, that in a rate case, everybody in this room knows who the burden is on. It's not on people to ask the company in discovery for information. I -- having said that, if there's missing information, certainly any party can ask in discovery. So I would like you to move beyond this question, because I find it argumentative, and ask the Panel -- and you can discuss at the break what or what -- what may or may not be requested that you're suggesting that DPS requests from the company. Or that you'll voluntarily

provide to this Panel and DPS staff. But let's move on to something more concrete in the Panel's testimony.

MR. DECICCO: Yeah, I -- I certainly don't want to be argumentative here in the last Panel. So I will -- I will withdraw that question.

BY MR. DECICCO: (Cont.)

Q. With regards to the -- the plastic fusion quality assurance/quality control re-dig program, in the Panel's direct testimony, page 54, line 14 -- let me get there myself -- the Panel proposes a -- as an -- an adjustment to the plastic fusion quality assurance/quality control, or QA/QC, re-dig program. Is that correct?

A. (Cont.) That's correct.

Q. And -- and is it the Panel's understanding that this program was implemented in response to a commission requirement?

A. That's correct.

Q. And the Panel supports this program?

A. Yes.

Q. The Panel proposes an adjustment of $250,000 for KEDLI, and $307,000 for KEDNY based

on its calculation of unit costs, and then multiplying that unit cost times the forecast units. Is that correct?

A. That's correct.

Q. Okay. In rebuttal testimony, the company's Gas Safety Panel, at page 45, testified that the company subsequently identified that their initial forecast failed to include the capital labor required to conduct the inspections, and that the capital labor is not included in the unit price. Is the Panel familiar with that testimony?

A. Yes.

Q. Does the Panel agree that the capital labor to perform the inspections is an appropriate component of the costs?

A. Yes.

Q. Pardon me. With regards to the transmission station integrity program, beginning on page 56 of the Panel's direct testimony, the Panel proposes to delay funding for the transmission station integrity program, because the program is new, and the companies had not yet identified specific projects. Is that correct?

A. That's correct.

Q. Does the Panel agree the company's required to follow federal gas safety regulations, including those promulgated by the Pipeline and Hazardous Materials Safety Administration or FHMSA?

A. Yes.

Q. Does the Panel agree that integrity management and integrity verification have been topics of considerable focus at PHMSA in recent years?

A. Yes.

Q. Even though this is a new program, does the Panel agree that based on the company's experience with inter -- integrity verification programs and the knowledge of the system assets, the companies would have a reasonable basis for forecasting work in this area?

A. The company explained in its testimony that it is currently undertaking a comprehensive review process in order to identify work in this program.

Q. So to the extent that that work produces projects that the company feels that it needs to do under the regulations, does the Panel

agree that the company should perform that work?

A. Yes.

Q. Okay. I'm going to move to the reliability category. And -- and could I just ask generally what the Panel's understanding of the reliability category of the company's investment plan is?

A. (Tintera) The reliability category is to -- programs and projects that are intended to provide safe and reliable service to customers within the service territory.

Q. Okay. And is it -- and -- and I don't want to characterize or -- or put words in your mouth, so feel free to disagree. But does -- does the Panel -- does the Panel generally support these programs that the company's proposing and is not -- and with a couple of exceptions, is not recommending the companies eliminate any reliability -- reliability programs. Is that correct?

A. Yeah, the Panel has not recommended elimination of any projects or programs from this category. And generally, we do support, you know, the company's ability to select, rank, and prioritize projects and programs within this

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category.

Q. Okay. Thank you. But in -- in -- in the Panel's analysis of this category, the Panel applies a slippage adjustment, which is a per -- percentage reduction to the budget based on prior years' actual spending against the entire budget category. Is -- is that correct?

A. That's correct.

Q. And I'm -- I'm at your -- your testimony on -- direct testimony on page 65. And for KEDNY, that slippage adjustment is 26%?

A. Yes.

Q. And for KEDLI, the slippage adjustment is 29%. Is that correct?

A. Yes.

Q. Okay. And -- and again, in arriving at those numbers, you just looked at -- at the prior spend in the tire -- entire budget category and then applied any variance to the -- to the rate year forecast. Is that correct?

A. We took a three-year average of the historic variances between the actual spend and budgets, and took an average of that, yes, and applied that to -- to make the slippage adjustment.

Q. Okay. Thanks. With regard to the basis for the slippage adjustment, do you agree that in a given year a reliability project may move in or out of the company's work plan for any number of reasons?

A. Yes.

Q. And would you agree that could include items like operational priorities, or permits, or damages that they have to respond to?

A. Yes, it could include that.

Q. Okay. And in some cases, projects are accelerated or deferred into another year. Is that -- is that your understanding?

A. Yes.

Q. So would you agree that an underspend in a -- in a line item in the reliability category in one year does not necessarily mean the companies will continue to underspend in subsequent years?

MR. FORST: Objection. I don't believe counsel's laid a foundation for that premise.

MR. DECICCO: Well, I -- I think the slippage adjustment is the foundation for that premise, because the -- the Panel is saying that they

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are applying a forward-looking adjustment based on prior years' variances. So I'm just trying to -- to confirm whether they -- the Panel understands that -- that a prior year underspend would automatically mean that that would continue going forward.

MR. FORST: And -- and I believe the Panel testified that they took a three-your -- three-year historic average of spending to forecast their adjustment. And I believe the premise that you're laying out is that one single year would somehow upend that.

MR. DECICCO: Well, I understand why -- yeah. I'm -- the foundation I laid was that the Panel acknowledged that in any year projects m -- may move in and out of a given year. And so I was just asking the question of whether in one year an underspend would nec -- would necessitate an underspend in a future, whether they could in fact overspend in a future year. I -- I think I did lay a foundation for this question.

A.L.J. LEARY: But this is hypothetically speaking, in one year, can you underspend and the next year overspend?

MR. DECICCO: Correct.

A.L.J. LEARY: Or in one year overspend and the next year underspend?

MR. DECICCO: Correct.

A.L.J. LEARY: I think the Panel can answer that. It's a hypothetical question, but I'm not sure how it comes back to this particular case. But --

MR. DECICCO: Let me just -- I wanted -- maybe I can just restate the question just to make sure that we're all on the same page in terms of what I asked.

BY MR. DECICCO: (Cont.)

Q. So I had asked you in preor -- prior questions whether or not work can move in and out of the budget for any number of reasons. And I believe the Panel said that it could. And so the question I asked was, would you agree than an underspend in a budget category in one year does not necessarily -- necessarily mean the companies will continue to underspend in subsequent years?

A. (Cont.) Well, yeah. One year does not necessarily make a trend. We do analyze trends. For example, looking at KEDNY, we didn't look at one year. We looked at three years. And the three-year

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underspend variance was in 30 -- in fiscal year '17, 36.1%, fiscal year '18, 29.8% underspend, and fiscal year '19, 18.1%. And similarly, for KEDLI.

Q. Okay. With respect to the -- to the slippage adjustment, the Panel is not suggesting that the company should do 74% or 71% of the programs in the portfolio. Is that correct?

A. That's correct.

Q. Okay. But it's the Panel's belief that the company can deliver these projects for -- for a budget that's 74% or 71% of the current forecast?

A. Yes, based on historic -- historic spending, the company has been able to, or has underspent, their budget. So we believe with that adjustment, they still will be able to perform the projects and have the ability to move money where it's needed.

Q. Okay. The Panel notes in its direct testimony at page 64 that 142 of the 149 projects in the reliability category are based on project-specific estimates. Is that accurate?

A. Yes.

Q. Sorry, I'm sniffling in the

microphone. And what is a project-specific estimate?

A. Company utilizes different methodologies for -- for forecasting their budgets. And a project program estimate is one in which they analyze previous projects. There's no unit cost or anything like that that they utilize, but they use a best guess estimate and use whatever data is available to forecast that project.

Q. Is it fair to say that budget is based on the particular project, not a program, as -- as compared to leak-prone pipe, but based on the actual project that the company is looking to achieve?

A. Yes.

Q. Okay. And in proposing your -- your slippage adjustment -- adjustments, did you consider -- and -- and applying those to the categories in the reliability area -- did you consider, for example, project complexity?

A. Yes. The Panel reviews every program and project, the complexity of it, the need for it, the cost benefits of it, the alternatives to it, and other aspects of the project. So they were considered, yes.

Q. And did you consider issues like safety and reliability, and the company's needs to get projects completed in order to -- to achieve safety and reliability issues?

A. Yes.

Q. Okay. However, when you apply that slippage adjustment, is -- is that just based on prior year spend or on the actual work that the company is -- is -- is proposing to do in the rate year?

A. It's based on a combination of the company's actual expenditures in prior years applied to the current forecast of work.

Q. Yeah. And -- and -- and I probably didn't ask the question well. But my -- my question was, are you just the -- the slippage adjustment is based on prior years. Is that correct?

A. Correct, which the Panel believes shows a representation of the company's ability to spend its forecasted budget.

Q. And -- and I guess my -- and -- and I appreciate that. My -- my -- I guess my question is, when applying the slippage adjustment based on prior years, you're just applying that to

the work in the rate year without considering the specific programs or the issues we just discussed in terms of reliability or safety?

A. No, we do consider the future need for programs. It's not just based on historicals. I mean, that is how we calculated the average, but the company has potentially more needs this year, different projects, and such. And in fact, the increase in budget over historicals for KEDNY is 155%. So that does represent a knowledge of the need for new -- a different budget.

And also for KEDLI, our -- our recommendation is a 314% increase. The company is seeking 460% increase, which is, you know, a -- a huge increase. So we do recognize that there are, you know, additional needs and things that may not be captured in historic -- in an historic variance.

Q. Okay. And -- and -- and the last item in the capital category, storm hardening. And I believe this is addressed on page 67 of your direct testimony. Oh, maybe not.

A.L.J. LEARY: Page 66.

MR. DECICCO: I'm sorry, 66. Thank you, Your Honor.

BY MR. DECICCO: (Cont.)

Q. Does the Panel agree that storm hardening is a priority?

A. (Cont.) Yes.

Q. And does the panel support the company's proposed storm hardening program, at least the concept of the report, if not the cost, excuse me, the -- the concept of the program, if not the cost in the rate year?

A. As I -- as I stated before, the Panel supports the company's ability to prioritize and -- projects and programs that seems best fit to reliably and safely provide service with in its ser -- service territory.

Q. Okay. With regard to the storm hardening program, is it -- is it the company's, excuse me, the Panel's proposal that the companies defer the cost of this program one year from the rate year into fiscal year '22 to conduct more testing?

A. Generally, yes.

Q. Okay. Has the company expressed a need to conduct more testing for its storm hardening program?

MR. FORST: Your Honors, point of

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order. Could counsel point to the specific section of the page 66 he's referring to. I -- you know, he's referred to storm hardening in general. And I believe page 66 refers to the storm hardening remote service shut-off values program. I just want to verify.

MR. DECICCO: Correct. And that's -- and -- and I apologize if I was not clear. That's -- that's the program that -- that I'm discussing.

BY MR. DECICCO: (Cont.)

Q. And I'm happy to re-ask -- re-ask the question with that if that wasn't the Panel's understanding. Thank you.

A. (Cont.) Well, from data requests that we've -- information requests that we've received from the company, and analyzed, and discussions, well, we do -- we do feel that further testing the company -- further testing is required.

Q. And -- and I guess my only question was, is that the -- had the company expressed the need to conduct more testing? Or is that the Panel's view?

A. At certain points, the company has expressed the need. I -- I can't point to exactly where in testimony. But at the time, they were still

doing testing, which was -- I believe some testing was completed by the end of this year. But they -- they -- they were still testing the -- the valves at -- at the time of -- at the time of their filing and somewhere in between.

Q. Okay. And -- as is the Panel aware whether the company has -- has since indicated that the program is -- is ready for deployment, if you're aware?

A. (Walters) I believe that maybe the Panel's -- it is the Panel's view that the testing -- immediately after testing, the program is not immediately ready for full deployment, that there would be testing and field testing required to verify these valves' correct operation.

Q. Okay. So let's just move on to one or two O&M adjustments, and then we are done. On page 81 of the Panel's direct testimony, the Panel proposes to completely eliminate the incremental non-labor operating expense for the pipeline integrity management program in the rate year. Is that correct?

A. Can you repeat the question?

Q. Sure. On page 81 of -- of the Panel's direct testimony, the Panel proposes to

completely eliminate the incremental non-labor operating expense for the pipeline integrity management program in the rate year. Is that correct?

A. That's correct.

Q. On page 85, the Panel also proposes to eliminate the incremental positions, or FTEs, full-time equivalents, to perform this work. Is that also correct?

A. Give me a second. I'm reading the page.

Q. Sure.

A. That's correct.

Q. Okay. And -- and what is the Panel's understanding of the integrity management and verification programs?

A. The Panel describes its understanding of the program on page 81 of its direct testimony. And it states that this -- generally speaking, this program performs testing to verify the integrity of existing pipeline, transmission pipeline, in line with federal regulations.

Q. And -- and is it true that the Panel supports the company's integrity management program generally?

A. Yes.

Q. And would the Panel agree the companies must adhere to federal regulations in this area, including the specific requirements of the Pipeline Hazardous Materials and Safety Administration, or PHMSA, in this area?

A. Yes, as it has -- had to do so in the past as well.

Q. Is the Panel of the opinion that the operating expense in this area, which is the adjustments we're talking about, is tied to the level of capital investments?

A. That is not an appropriate characterization of the Panel's understanding, as stated in the response to an IR that the company put in, National Grid 46.

Q. Would the Panel agree, though, among the reasons that's cited for eliminating this -- this item for the OPX budget, or operating expense budget -- and I'm looking on page 82, starting on line five. It says, "However, neither the capital budget or the OPX workload for the rate year is being increased as compared to the most recent fiscal year or the historic test year." Could you explain what

you meant by referencing the capital budget in that sentence in your testimony?

A. So in that statement, the staff was making an indirect statement that the capital budget can impact projects in the operations and maintenance budget of this program for possibly the current year and future years due to the fact that the capital projects will prepare the pipeline for inspection, which is then expensed. And so that is what we're referring to there.

Q. Okay. But would you agree that the -- the work in this category in -- in -- that you're -- that you're proposing to adjust is in fact expense work, not capital work?

A. Yes.

Q. Okay. Do you understand what a zero-based budget is?

A. Yes.

Q. Is it your understanding that the company uses a zero-based budget for its integrity management program?

A. Yes.

Q. And could you explain what a zero-based budget is?

A. It's a budget where the company, or whoever it is who's developing the budget, will build the budget up bottom-up and -- starting at zero. And then develop the budget based on the cost of each component that's required.

Q. Okay. And -- and that's as opposed to just taking last year's budget and inflating it, as an example?

A. Correct.

Q. Okay.

Does the panel agree that in a zero-based budget, there can be variabil -- variability year to year, depending on the projects identified for the particular year?

A. That is correct. However, in recent historical years, there has been much less variability than the company's proposing in the rate year. And I would also like to note that the federal regulations require inspections to occur on a multi-year cycle.

Q. So the -- the last category of the O&M is updated contractor costs. And in your supplemental testimony on page 15 -- well, it's -- it's -- it's starting on page 15 and going into page

16. You describe the company's adjustments, or KEDNY's update, to proposed -- strike that. On page 15, line 24, you describe an update to the KEDNY's proposed budget for damage prevention. Is that correct?

A. That's correct.

Q. Have the -- the members of this Panel analyzed the company's proposal with regards to its update to its damage prevention expenditures?

A. We have, together with the Staff Policy Panel.

Q. And have -- has this Panel many any determination with regards to the company spending in the damage prevention area?

A. This adjustment was discussed in the Policy Panel's testimony, and reflects it in our exhibit.

Q. Okay. And -- and -- and it is your testimony that you were not involved in -- in recommending that adjustment?

A. That is correct.

Q. Okay. Then we will move on. The last item I'd like to discuss on page 28 of your direct testimony, the Panel makes a point that

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sanctioning papers would enable the Panel's review of the proposed capital investments in the rate year. Is that correct?

A. Can you repeat the question?

Q. Sure. And -- and -- and on page 28 of the -- of the Panel's testimony, the Panel makes the point that sanction papers would enable the Panel's review of the proposed capital investments in the rate year. Is that correct?

A. That's correct, together with other information.

Q. Sure. And -- and just in terms of enabling your view of the -- of the company's capital programs, is it true that the company provided the sworn testimony of the company's chief engineer and other experts regarding the needs for its proposed capital investments?

A. That's correct.

Q. And did the -- did the testimony include white papers that provided information typically found in sanctioning papers, including background justification, cost estimates, and alternatives considered?

A. Yes.

Q. Did the companies facilitate technical sessions to present their proposals in more detail?

A. Yes.

Q. Have the companies made themselves available to answer any questions regarding the capital investment programs?

A. Generally, yes.

Q. And -- and through the nearly 10 months of discovery to this point, have the companies provided the Panel with information requested on capital programs through numerous discovery requests?

A. Yes.

Q. Okay. I have no further questions. Thank you. Thank you, Panel, I appreciate your patience.

A. Thank you.

A.L.J. LEARY: I just want to clarify just so I understand that this last line of questioning was about sanctioning papers that were not provided to the Panel. Is that correct?

MR. DECICCO: Yeah. And I don't want to characterize the testimony, but -- but the Panel makes a point that because of the timing of when

sanctioning papers were available that -- that they could have enabled their review. And -- and obviously, I asked the questions that I asked.

A.L.J. LEARY: Okay. Thanks.

MR. DECICCO: Thank you, again.

A.L.J. COSTELLO: Okay. Staff, do you need -- do you need a moment?

MR. FORST: If we could take about 10 minutes, that would be perfect.

A.L.J. COSTELLO: Okay. 10 minutes. We'll go off the record.

(Off the record 11:44 a.m.)

(On the record 12:04 p.m.)

A.L.J. COSTELLO: Staff, any redirect?

MR. FORST: We have one redirect question, Your Honor.

REDIRECT

BY MR. FORST:

Q. Panel, earlier you testified about the company's rate allowance for city/state construction. Could you clarify the Panel's recommendations regarding city/state construction rate allowance?

A. (Walters) Yes. So the Panel

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recommends its rate allowance based on the company's proposed methodology of using 20% of the city's construction budget. And we believe this is appropriate, because based on information we received in information requests, we saw that in fiscal year '18 the company had spent 17% of the city's budget. And this year is representative of the costs which will be incurred, because the paving regulations were in place in this year.

Q. Just to follow-up, Your Honors. So based on the Panel's analysis, it also recommends a true-up mechanism. Can you explain why that true-up mechanism is appropriate based on your analysis?

A. Yes. We believe the proposed 80% true-up is appropriate, because as time goes on, the company should becoming more efficient at managing its workload, and also with these regulations in place.

MR. FORST: No further questions, Your Honors.

A.L.J. COSTELLO: Any of you recross?

RECROSS

BY MR. DECICCO:

Q. I -- I'd just ask the Panel, does

the Panel have any reason to believe that the city will be doing less work in the future?

A. (Walters) The Panel did not investigate that. The -- the company provided the city's construction budget, which is how it developed its budget.

Q. And is -- is the Panel aware of any year where the company has underspent its proposed city/state construction over the term of the current rate plan? And -- and I'm talking about KEDNY now, just to be clear.

A. No, it has not.

Q. Okay. Thank you.

MR. GOODRICH: One moment, Your Honor.

A.L.J. COSTELLO: Sure.

MR. FORST: No further questions, Your Honors.

A.L.J. COSTELLO: Okay. Thank you, Panel. Thank you very much for your testimony, and you are excused.

MR. TURRINI: Thank you.

A.L.J. COSTELLO: We'll just go off the record for a moment.

MR. BERKLEY: Your Honors, my

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A.L.J. COSTELLO: Yes, certainly.

MR. BERKLEY: Thank you. I'd like to note my appearance for the Public Utility Law Project. This is Richard Berkley. I returned to the hearing room at 10:10. Thank you.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: We apologize, I --

A.L.J. COSTELLO: This will go off the record, just --

(Off the record 12:08 p.m.)

(On the record 12:08 p.m.)

A.L.J. COSTELLO: I just wanted to correct something that I stated earlier with respect to the testimony that was filed by the Staff Gas Infrastructure and Operations Panel with respect to notifying the court reporter what files to insert. I believe I referred to corrected direct testimony by that Panel. And that -- that was incorrect. They are -- they made corrections to their supplemental testimony. So I just wanted to note that for the record.

Okay. And what -- what we plan on doing now is going through the affidavits, just

marking them as exhibits, and admitting the testimony that is being sponsored by the affidavits. And we're going to start with the -- the companies. And what I'm going to do is just mark the affidavits all at one time for the companies, and then for each party, and admit the testimony at that time after I've done marked all of them.

So for the companies, we have an affidavit by Phillip A. Decicco, two pages, sworn to on February 12, 2020. That will be marked as Exhibit 686. We have a two-page affidavit by John Bruckner, sworn to on February 7, 2020. That will be marked as Exhibit 687. We have a two-page affidavit by Maureen P. Heaphy, sworn to on February 13, 2020. And that will be marked as Exhibit 688. We have an affidavit, two-page, by Anne E. Bulkley, sworn to on February 18, 2020. That will be marked as Exhibit 689. We have an affidavit by Theo -- Theodore Poe, Jr., two pages, sworn to on February 13, 2020. That will be marked as Exhibit 690. We have a two-page affidavit by Jonathan Cohen, sworn to on February 13, 2020, which will be marked at Exhibit 691. We have a two-page affidavit by David B. Doxey, sworn to on February 13, 2020. That will be marked as Exhibit 692. And finally, for

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the companies, we have an affidavit by James M. -- and I -- is it M -- Molloy? Is that correct? Okay. Molloy, M-O-L-L-O-Y -- two pages, sworn to on February 18, 2020. And that will be marked as Exhibit 693.

And at this point in the record, the court reporter should insert the following files: KEDNY-KEDLI Phillip A. Decicco direct testimony; KEDNY-KEDLI John Bruckner direct testimony; KEDNY-KEDLI Maureen P. Heaphy direct testimony, corrections and updates testimony, and rebuttal testimony; KEDNY-KEDLI Anne E. Bulkley direct testimony and rebuttal testimony; KEDNY-KEDLI Theodore Poe, Jr. direct testimony, supplemental direct testimony, corrections and updates testimony, second supplemental testimony, and third supplemental testimony, KEDNY Jonathan Cohen direct testimony, corrections and updates testimony; KEDLI Jonathan Cohen direct testimony and corrections and updates testimony; and KEDNY-KEDLI Jonathan Cohen rebuttal testimony. Finally, for the companies, we have KEDNY-KEDLI David B. Doxey and J -- James M. Molloy, supplemental direct testimony.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Philip A. DeCicco

Dated: April 2019

Testimony of Philip A. DeCicco

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Testimony of Philip A. DeCicco

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Philip A. DeCicco. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid" or the "Company"), as the Vice President of New York Regulation and Pricing. My responsibilities include overseeing regulatory matters for National Grid's New York businesses, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI").

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Finance from Fairfield University in 1995 and a Juris Doctorate from Fordham Law School in 1998. I have worked for National Grid for over 20 years, primarily as an attorney holding various roles of increasing responsibility within the Legal department. I assumed my current role in 2018.

II. Purpose of the Testimony

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to describe the status of the recently initiated New York State Public Service Commission ("Commission") Comprehensive Management and Operations Audit of National Grid's New York electric and gas businesses in Case 18-M-0196 ("Management Audit"), conducted by Saleeby Consulting Group ("SCG"). My testimony also discusses the status of three other Commission audits: the Comprehensive Management and Operations Audit of National Grid's New York gas business in Case 13-G-0009 ("Gas Management Audit"); the Operations Audit of the Accuracy of New York State Utilities' Self-Reported Data in Case 13-M-0314 ("Data Audit"); and the Focused Operations Audit of Major Utility Internal Staffing Levels and Use of Contractors for Selected Core Utility Functions at Major New York Energy Utilities in Case 13-M-0449 ("Staffing Audit").

III. Management Audit Update

Q. Please discuss the status of the Management Audit.

A. In May 2018, the Commission issued Request for Proposals for an independent consultant to conduct an audit of National Grid's New York electric and gas businesses in accordance with Section 66(19) of the Public

Service Law ("PSL"). In September 2018, the Commission selected SCG to conduct the audit under the supervision of Department of Public Service Staff ("Staff"). The Management Audit formally commenced in November 2018 with a full-day presentation by National Grid senior management on its electric and gas operations to SCG and Staff.

The Management Audit is currently ongoing, with a final report due by September 30, 2019. To date, National Grid has found the audit to be very professional and marked by open and productive discussions among the parties. The Company has endeavored to provide SCG full access to its personnel, records, and systems in the six months since the audit commenced, during which the Company has answered more than 1,200 data requests, produced over 250 management employees for interviews, and allowed SCG to participate in numerous senior level meetings across the electric and gas businesses. National Grid is fully committed to supporting the Management Audit and looks forward to the continued opportunity to work collaboratively with Staff and SCG to obtain objective feedback on the management and operations of the New York electric and gas businesses. Pursuant to PSL Section 66(19), once the report is issued, the Company will file an implementation plan with the Commission.

IV. Other Commission Audits

Q. Have there been any other Commission-initiated management or operations audits of National Grid in the last several years?

A. Yes. The Commission initiated the Gas Management Audit in 2013 along with two statewide operations audits of New York utilities, the Data Audit (2013) and Staffing Audit (2014).

Q. Please provide a summary of the Gas Management Audit and its status.

A. In February 2013, the Commission commenced the Gas Management Audit pursuant to PSL Section 66(19). The Commission selected NorthStar Consulting Group ("NorthStar") to perform the audit. In August 2014, NorthStar released its audit report. The report, which contained 31 recommendations, described generally strong performance by National Grid's gas business in a number of key functional areas, including gas operations, load forecasting, supply procurement, and customer information systems.

On November 3, 2014, National Grid filed its implementation plan with the Commission detailing plans to implement the audit recommendations. The implementation plan was approved by the Commission by Order dated May 14, 2015. In the ensuing months, the Company worked collaboratively with

Staff to implement the recommendations. By letter dated September 28, 2018, Staff confirmed that all 31 recommendations had been implemented. A copy of the close-out letter is on file with the Commission in Case 13-G-0009.

Q. Please discuss the status of the Data Audit.

A. In August 2013, the Commission instituted an operations audit of the accuracy of the New York utilities' self-reported data regarding electric reliability, gas safety, and customer service. The Commission selected Overland Consulting ("Overland") to conduct the audit. In April 2016, the Commission issued an order publishing the final report and directing the utilities to file implementation plans.

National Grid filed its implementation plan with the Commission in May 2016. The implementation plan, which addressed 106 recommendations, was approved by the Commission in March 2017. By letter dated March 13, 2018, Staff confirmed the Company implemented all recommendations. A copy of the close-out letter is on file with the Commission in Case 13-M-0314.

Q. Please discuss the status of the Staffing Audit.

A. In January 2014, the Commission instituted an audit of the internal staffing levels and use of contractors for selected core utility functions at the major New York utilities. The Commission selected The Liberty Consulting Group ("Liberty") to conduct the audit. In February 2017, the Commission authorized the public release of Liberty's audit report. Liberty identified a total of 27 recommendations aimed at enhancing National Grid's staffing and workforce management processes. In March 2017, National Grid filed its implementation plan, which was approved by the Commission in December 2017. In its latest implementation plan update filed with the Commission, National Grid advised that all 27 recommendations had been implemented and were pending Staff's review and approval.

Q. Does that conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

John Bruckner

Date: April 2019

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I. Introduction and Overview

Q. Please state your name and business address.

A. My name is John Bruckner, and my business address is 25 Hub Drive, Melville, New York 11747.

Q. Please explain your role and principal responsibilities at National Grid.

A. I am the Jurisdictional President for National Grid USA's ("National Grid") New York businesses. I am responsible for the performance of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY"), KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (together, the "Companies"), and Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk"). I also serve on the boards of directors for KEDNY, KEDLI, and Niagara Mohawk.

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Marine Engineering from U.S. Merchant Marine Academy in 1985. I received a Master of Business Administration degree from New York Institute of Technology in 1992. I joined the Long Island Lighting Company ("LILCO") in 1985 as a substation maintenance engineer in the electric division. Over the years, I have held positions of increasing responsibility in LILCO's, and later KeySpan Corporation's ("KeySpan"), electric transmission and distribution business, including Director of Transmission and Distribution Project

Management and President of Long Island Electric Transmission and Distribution. In 2015, I was named Senior Vice President, Gas and Electric Engineering, responsible for engineering operations for National Grid. In 2017, I was named the Executive Vice President of Network Operations for National Grid. I was named President of the New York Jurisdiction in April 2018. I also serve on the boards of directors for several New York energy, business, academic, and non-profit organizations.

Q. What is the purpose of your testimony?

A. My testimony provides an overview of the Companies' gas rate filings and introduces the witnesses that will provide testimony in support of those filings. I will provide background on KEDNY and KEDLI's businesses and explain the Companies' goals in making these filings. I will address the factors driving the Companies' need for rate relief and describe the important steps taken to prepare these filings. Finally, I will discuss the Companies' views on the future of heat, highlight the key issues in these cases, and explain how the Public Service Commission's ("Commission") granting the relief sought will further the objectives shared by the Companies, our customers, and the State of New York.

Q. Please provide an overview of the Companies' rate filings.

A. The current energy landscape is evolving beyond the traditional utility construct of years past. Technology and innovation are driving change in the energy industry, and utilities must provide more dynamic, efficient, and interactive services to meet heightened customer expectations. At the same time, the confluence of an industry focus on gas safety and the imperative to reduce carbon emissions demands that utilities rise to the challenge of delivering a safe and sustainable energy future. This rate filing will enable us to achieve step changes in our performance in these key areas and lead the transition to New York's energy future. Accordingly, investment is required not only to enhance the Companies' core operations, but also to support energy policy objectives. These rate filings present comprehensive, measured proposals designed to align with the Companies' priorities of (i) maintaining affordability and mitigating customer bill impacts; (ii) funding capital programs necessary to modernize our gas infrastructure and replace aging and outdated information technology systems; (iii) delivering the Future of Heat; (iv) improving the customer experience; and (v) preserving credit quality and allowing the Companies the opportunity to earn a reasonable rate of return.

KEDNY and KEDLI seek to increase revenues by $236.8 million and $49.4 million, respectively, to eliminate revenue deficiencies in the twelve months ending March 31, 2021 ("Rate Year"). For typical residential heating customers, the

proposed base rate increases would result in delivery increases of 17.8 percent (11.99 percent on the total bill) for KEDNY customers and 6.9 percent (5.15 percent on the total bill) for KEDLI customers.

While these rate increases are significant, they are needed to deliver safe and reliable service in a rising cost environment. The Companies are focused on managing controllable costs – and to mitigate the impact of the proposed rate increases, the Companies have implemented various measures to offset the Companies' need for rate relief. Nevertheless, we are experiencing upward pressure on operating expenses. This is especially true for KEDNY, where the costs of working in the heavily urban environment of New York City are amplified and there is need to recover significant site investigation and remediation ("SIR") costs. While KEDLI is similarly experiencing cost increases, the cost pressures are far less pronounced in its Long Island service territory. The difference between the types and magnitude of costs incurred by the Companies ultimately results in significantly different revenue requirements and bill impacts for customers; however, in these filings both Companies strive to deliver on the same overall service objectives and common goals of providing safe, reliable, and sustainable energy to our customers.

While the Companies' filings propose new rates for the Rate Year only, the Companies are interested in exploring a multi-year rate plan settlement that would allow for a phase in of these revenue increases and the ability to manage the impact on customers' bills. To facilitate such discussions, the Companies have included cost projections for three years (the "Data Years") beyond the Rate Year.

Q. Please describe the Companies' need to earn a reasonable rate of return.

A. Without the rate increase proposed in these filings, current rate levels will not permit the Companies to recover their costs of providing safe and reliable service. Current rate levels would also not permit the Companies to earn a reasonable rate of return on our investments. The Companies' proposal brings their respective rates of return to reasonable levels, while, as noted above, mitigating the impact on customers.

The Companies' ability to access capital markets on terms favorable to our customers will be directly affected by the rates authorized by the Commission in this proceeding. For decades, the combination of stable earnings, generally manageable business risks, and declining or stable market interest rates have allowed us to access capital markets on favorable terms and reduce borrowing costs for the benefit of customers. As our portfolio of capital investments grows to meet the demands of today's business, it is increasingly important that we maintain the

ability to cost-effectively finance our investments. However, due, in part, to recent federal tax reform, and the resulting impacts on cash flows, the Companies' credit ratings were recently downgraded by Moody's from A2 to A3. Without a reasonable return on equity, funding for incremental capital investments, and support for timely recovery of operating expenses such as SIR costs, the Companies are at risk of further downgrades, which could impact the Companies' credit quality and raise costs for customers as the cost of borrowing increases.

Q. Please discuss the Companies' need for capital investment and how it is presented in these rate filings.

A. Continued funding for capital investments and improvements is needed to modernize the gas system, enhance pipeline safety, mitigate the effect of future severe weather incidents, and reduce methane emissions. These investments will also allow the Companies to meet mandated federal, state, and local regulatory requirements, which now account for more than 60 percent of total capital expenditures. The capital investment proposals presented in this filing include a suite of projects, programs, and initiatives designed to address the needs identified above while also providing safe and reliable service to our customers, strengthening our core business, and providing a foundation for future energy developments.

Q. Please describe the differences between the capital investments proposed by KEDNY and KEDLI.

A. As noted above, there is a significant difference between the revenue deficiencies of KEDNY and KEDLI, and that difference is largely attributable to several capital cost drivers that arise in KEDNY's service territory. New York City is undertaking significant investment in public works projects. As a result, the volume and scale of KEDNY's city/state construction work ("CSC"), which involves the relocation and/or protection of gas facilities that conflict with municipal public works projects, continues to grow and now represents approximately one-third of KEDNY's total capital investment plan. The timing and scope of CSC projects are outside of KEDNY's control, and these projects are expensive and prone to a high degree of cost variability. At the same time, KEDNY is experiencing higher unit costs resulting from new work requirements, including recent changes to paving and restoration requirements in New York City, as well as increased costs for contractors due to demand. These increased costs impact many of the mandated programs in KEDNY's capital investment plan, including CSC and main replacement projects. These factors, which are further compounded by the increased volume and scope of construction projects in New York City, have contributed to the differences in capital requirements and costs between the Companies as these cost drivers are not currently experienced to the same degree in KEDLI's service territory.

Q. Please discuss how the Companies have incorporated their Future of Heat strategy in these filings.

A. Combating climate change and reducing carbon emissions are significant challenges. National Grid is committed to leading on these issues and carving a path for natural gas in delivering the future of heat and energy in New York. State and federal energy policy objectives, including the Commission's Reforming the Energy Vision ("REV") proceeding, are focused on providing green, efficient solutions for current and future consumer energy needs. Natural gas is a sustainable, reliable, and cost-effective form of energy that can be used to lower carbon emissions and reduce customers' energy burdens. As discussed more below, the Companies are proposing various projects and programs to improve system performance and safety, aggressively reduce methane emissions (with a targeted goal of reducing total methane emissions 60 percent by 2035), increase customer choice in meeting their energy needs through the expanded deployment of geothermal heating and clean conversions, modify and expand the Companies' energy efficiency programs, and promote development of renewable natural gas ("RNG") and other emerging technologies to reduce carbon emissions in the gas supply. The Companies also propose a package of non-pipe alternative ("NPA") solutions to defer or offset investment in traditional natural gas infrastructure. These proposals are consistent with New York State's policy objectives and provide a progressive future for natural gas in the Companies' service territories.

The Companies' desire to maintain a space for natural gas in the heat and energy landscape is countered by the potential unavailability of natural gas resources. The Companies are facing serious gas supply constraints that, if not timely addressed, will prevent the Companies from serving new customers in their service territories. The inability to serve new customers will negatively impact carbon emissions and economic development in the communities we serve as those customers will continue to rely on dirtier and more expensive forms of energy, such as oil, to meet their energy demands. In these filings, the Companies will discuss our efforts to overcome these supply constraints with prudent pipeline solutions and highlight the consequences should these solutions not be approved. The Companies also propose a package of additional energy efficiency programs, non-pipe solutions, and other projects to reduce environmental impacts, support customer service requests, reduce natural gas consumption to potentially relieve system constraints, and provide alternative service options for our customers.

Q. Please discuss the Companies' approach to mitigating customer bills.

A. The Companies fully understand the impact of increased energy costs and take seriously our obligations to mitigate customers' bills. Indeed, beginning January 1, 2019, the Companies' rates reflected the impacts of recent federal tax reform laws, returning more than $70 million to customers through bill offsets, and the Companies plan to reflect further savings in these filings to help further reduce bill

impacts. These filings were developed with a keen focus on balancing the need for continued investment with the need to maintain affordability and protect our most vulnerable customers. National Grid is committed to continuously identifying opportunities to work more efficiently and productively for the benefit of customers – and has reduced its revenue requirements in prior rate filings by hundreds of millions of dollars because of various efficiency efforts.

In this case, the Companies have implemented several aggressive efficiency measures to reduce the overall costs of providing service. National Grid recently restructured its U.S. business to obtain efficiencies and leverage best practices, achieve economies of scale, and ultimately better serve our customers. In connection with the restructuring, National Grid launched the Accelerate Program, which sets out a very ambitious, aspirational target of achieving 20 percent efficiency improvements in its operating and capital expenses by fiscal year 2021. The Accelerate Program is focused on identifying ways to work more efficiently, improve customer service, and reduce costs across many segments of the Companies' operations. Because of National Grid's work through the Accelerate Program, the Companies have reduced their revenue requirements by approximately $43 million in the Rate Year and made an additional adjustment of more than $6 million to reflect the Commission's typical one percent productivity adjustment, for a total of approximately $49 million in savings in the Rate Year

alone. While the Accelerate Program ends in the Rate Year, as a way of further mitigating the rate increase, the Company carried this same level of savings forward in each of the Data Years and applied inflation. In doing so, the Company assumed the risk that this significant level of savings can be sustained in future years and not offset by new work requirements. The total efficiency savings over the four-year term of the proposed rate plan for the Accelerate Program alone is approximately $124 million for KEDNY and $48 million for KEDLI. While we do not know if this level of savings can be achieved, we have nonetheless reflected the savings in our filing to help mitigate bill impacts to customers. Further discussion regarding the Companies' efficiency measures is provided in the testimony of the Revenue Requirements Panels.

Q. Please describe the Companies' proposal to improve the customer experience.

A. The Companies' filings are focused on improving various aspects of the overall customer experience. Specifically, the Companies propose to develop and implement digital solutions for certain customer interactions, enhance customer service offices, provide further support for economic development and growth in the Companies' service territories, and enhance the affordability of the Companies' energy efficiency and low income programs.

Significantly, the Companies are implementing a new Customer Information System ("CIS") that will replace the decades-old systems in use today with a modern solution capable of providing greater ease of use, data transparency, and additional functionality to meet customer expectations, handle an increasingly complex billing environment, and incorporate marketplace and regulatory objectives. The CIS program is a pivotal investment as it will support and enable existing and future programs that will benefit both the Companies and our customers.

Q. Are the Companies' filings consistent with the goals and objectives outlined in New York State's Energy Plan?

A. Yes. As I discussed above, the Companies strongly support New York State's energy policies. The filings reflect the Companies' commitment to modernizing gas infrastructure to promote resiliency and reliability, to deploying new technologies to enhance safety, reliability and customer engagement, to promoting market solutions that drive efficiencies and enhance customer service options, and to assisting customers with energy management. The Companies look forward to working with all stakeholders to promote these policies in a manner that is beneficial to our customers and communities.

Q. Please introduce the other witnesses who provide testimony in the Companies' filings.

A. In addition to my testimony, the Companies' rate case filings are supported by the testimony of 16 witnesses or witness panels. The witnesses and the subject they address are as follows:

- The Gas Infrastructure and Operations Panels consists of Ross Turrini, Group and US Chief Engineer Gas, Timothy Graham, Vice President – Gas Field Operations, Caroline Hon, Vice President – Gas Resource Management, and Srividya Madhusudhan, Vice President – Facilities and Property Services. The panel's testimony discusses the Companies' plans to deliver necessary investments in gas infrastructure, including the continued replacement of leak prone pipe, investments to promote the safe and reliable operation of our gas networks, and investments to upgrade aging facilities and create more modern and efficient customer support and employee work spaces. The panel also discusses the Companies' operations and maintenance costs to provide service to customers, as well as the incremental work that is driving cost increases, particularly in New York City.
- The Gas Safety Panel consists of Ross Turrini, Timothy Graham, and Annette Saxman, Director – Gas Pipeline Safety and

Compliance. The panel presents the Companies' proposals for new gas safety programs and gas safety performance metrics, including enhancements to training, inspection, contractor oversight, and quality control capabilities.

- Ann E. Bulkley of Concentric Advisors addresses the Companies' cost of equity capital.
- Jonathan Cohen, Director – U.S. Planning and Strategy, addresses the Companies' overall cost of capital and capital structure.
- Maureen Heaphy, Vice President – U.S. Compensation, Benefits and Pensions, describes the Companies' compensation and benefits programs and efforts to control the costs of those programs.
- Paul M. Normand of Management Applications Consulting, Inc. presents the Companies' proposed gas depreciation rates for the Companies' leak prone mains and services.
- Charles F. Willard, Director – Site Investigation and Remediation, discusses the Companies' SIR programs, efforts to control and mitigate SIR expense, and forecast SIR costs.
- The Shared Services Panel consists of Paula Leaverton, Manager – Real Estate and Property Tax, Jody Allison, Vice President – Revenue Cycle Management, Jeffrey P. Martin, Director – Billing and Revenue Strategy, Arlene Gans, Director – Customer Contact

Centers, New York, and Lisa Tallet, Director – New York Customer Energy Management. The panel's testimony sets forth the Companies' proposals relating to property tax, uncollectible expense, customer offices, customer service quality metrics, investment in a new CIS, and low income programs.

- The Future of Heat Panel consists of Don Chahbazpour, Director – Gas Utility of the Future, Lisa Tallet, Arthur Hamlin, Manager – Economic Development, and Owen Brady, Product Management Specialist – Emerging Products. The panel outlines the Companies' proposals to reduce carbon emissions in the gas supply, programs to promote customers' energy choices that align with the Commission's energy goals, and proposals regarding economic development and outreach and education.
- Philip DeCicco, Vice President – New York Regulation and Pricing, provides an overview of recent and ongoing Commission management audits.
- Theodore E. Poe, Manager – Gas Load Forecasting and Analysis, presents the Companies' gas sales forecasts.
- Elizabeth D. Arangio, Director – Gas Supply Planning, discusses the Companies' efforts to purchase natural gas supplies on a reliable,

cost-efficient basis, as well as the current gas supply constraints in downstate New York.

- Christopher J. Connolly, Business Lead for the Gas Business Enablement ("GBE") Program, discusses this critical program, which is needed to build a modern platform that supports customer demands.
- The Information Technology ("IT") Panel consists of Stephen Olive, Senior Vice President and US Chief Information Officer, Daniel J. DeMauro, Director – IT Regulatory, Mukund Ravipaty, Director – Security Services, Design, and Architecture, and Jeffrey P. Martin. The panel describe the major IT investments and initiatives during the Rate Year and Data Years.
- The Revenue Requirements Panels consists of David B. Doxsee, Vice President – Finance, New York, Stephanie Briggs, Director – Revenue Requirements, New York, and James M. Molloy, Director – Revenue Requirements. The panel's testimony sets forth the revenue requirements for the Rate Year and describes the Companies' efforts to review the historic test year costs to ensure they were accurately charged. The panel also discusses the Companies' efforts to mitigate customer bill impacts through various efficiency programs. The historic test year is the twelve

months ended December 31, 2018 ("Historic Test Year" or "Test Year").

- The Rate Design Panels consists of Dawn M. Herrity, Principal Analyst – Gas Pricing New York, Kellie I. Smith, Manager – New York Gas Pricing, and Howard Gorman of HSG Group, Inc. The panel's testimony addresses the marginal and embedded cost of service studies, the revenue forecast, revenue allocation, rate design, and bill impacts.

II. Background and Rate History

Q. Please describe the Companies.

A. KEDNY serves 1.2 million customers in the Boroughs of Brooklyn, Queens, and Staten Island in New York City with approximately 4,100 miles of gas distribution main. KEDLI serves 590,000 customers in Nassau and Suffolk Counties on Long Island and the Rockaways in New York City with approximately 8,200 miles of gas distribution main. Combined, KEDNY and KEDLI have $6 billion in rate base. Both Companies (and their predecessors) have served customers in these territories for more than 100 years and operate gas distribution systems that are among the oldest in the United States.

While the Companies' priority has always been the safe and reliable delivery of natural gas, we take great pride in a long history of community engagement. The Companies have demonstrated a commitment to improving the quality of life in the communities we serve through support for economic development, charitable, and educational programs. The combination of KEDNY's Cinderella program and the Companies' other economic development programs have awarded millions of dollars in funding to non-profits, community groups, developers, and customers in our service territories over the years. This continued investment has not only helped to spur economic revitalization and regional growth, but also to strengthen our partnership with the communities and customers we serve. In addition to the Companies' direct economic investment, our employees actively engage in numerous volunteer activities within communities throughout our New York service territories with the support of National Grid's "Power to Serve" program. National Grid has partnered with educators and students, from kindergarten to college, to promote STEM education and attract students to STEM careers, and we intend to continue to these academic partnerships in the future.

Q. Please describe the Companies' efforts to enhance the quality of service provided to customers.

A. Since the Commission's approval of Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases"), the Companies have invested more than $2

billion to modernize and enhance our gas networks. Those investments have included aggressive efforts to remove aging and leak prone infrastructure as well as proactively identify areas on the network in need of upgrade to increase system reliability. The Companies have also made substantial investment and progress in improving system resiliency. The past decade has seen increasingly severe weather events in our service territories that have significantly affected our customers and our networks. These investments, coupled with our talented and dedicated work force, have allowed us to maintain strong operational performance and service levels despite increasingly severe weather conditions. Indeed, in the past winter, the Companies safely and reliably delivered unprecedented volumes of gas to our customers, establishing three of the top ten daily delivery records for KEDNY and four of KEDLI's top ten daily delivery records, and setting peak throughput records of 1,461,285 dekatherms and 1,115,728 dekatherms for KEDNY and KEDLI, respectively, on January 21, 2019.

Safety remains a top priority for the Companies, and that priority is echoed through all aspects of our business. In the 2016 KEDNY and KEDLI Rate Cases, the Companies' gas safety metrics were reset to provide stringent safety targets and currently measure the Companies performance in the areas the leak prone pipe ("LPP") removal, leak management, damage prevention, emergency response, and gas safety regulations compliance. These metrics include negative revenue

adjustments to penalize poor performance and provide incentives for the Companies to exceed the specified targets. The Companies have also implemented various safety and training programs and participated in numerous regulatory proceedings and utility industry collaboratives over the past few years. These efforts have not only helped to improve our safety performance, but also positioned us at the forefront of the safety discussion so that we are better equipped to identify emerging safety issues.

While we are proud of our track record of providing safe and reliable gas operations in New York, recent events in the gas industry coupled with increased regulatory requirements demand continued improvements not only to remain compliant, but also to meet the expectations of the customers we serve. The Companies are committed to working with our regulators to make the necessary improvements. Indeed, in these rate filings we propose several modifications and additions to our current safety practices and programs, including enhancements to the frequency and types of safety inspections and contractor oversight in the field, the addition of incremental safety programs to address issues such as pressurization and methane detection, further standardization of operating procedures for quality assurance and control, and new training and testing labs to provide additional employee competency. The Companies are confident that these changes will deliver immediate and continued improvements in this crucial area.

Q. What have the Companies done to help customers manage their energy usage and pay their bills?

A. Service affordability is major concern for all customers in our service territories. Customers want stability and predictability in their bills, so they can appropriately budget for their energy needs. They also want the ability to monitor and manage their energy consumption to further control costs and maximize the efficiency of their usage.

For the past few years, customers have benefitted from low commodity costs, which have created consistency and affordability in the commodity portion of customer bills. The Companies have also applied a variety of pricing tools, products, and services to achieve stability in the delivery portion of customer bills and make our services more affordable for all customers. For example, the current rate plans for KEDNY and KEDLI include rate mitigators and levelization, which provides annual consistency in the delivery charges. Additionally, as noted above, customer bills were recently further reduced by the return of nearly $70 million dollars to reflect the impacts of federal tax reform.

The Companies presently offer a series of programs to assist our low income customers with bill payment and affordability. Eligible low income customers receive residential rate discounts, service reconnect fee waivers, and rebate

programs to help reduce their bills and the costs of other energy-related services. To maximize the efficacy of these programs, the Companies employ trained consumer advocates who work directly with customers and local agencies, such as the Human Resource Administration in New York City, to identify customers who may be eligible for our programs, provide program materials and customer support, and ultimately connect these customers to the necessary programs and services to assist with their bills. These programs have helped to offset millions of dollars in costs for low income customers over the past three years in the Companies' service territories.

The Companies also offer numerous efficiency products and services to keep services affordable and encourage cleaner and more efficient energy usage.

These stability and affordability efforts are directly reflected in customer bills. Tables 1 and 2 below depict the average annual residential heating bills for KEDNY and KEDLI, respectively. The tables illustrate that residential bills have remained relatively stable in recent years, even with the 2017 rate increase.

Table 1 – KEDNY Average Annual Residential Heating Bill
(1,000 Therm Annual Usage)

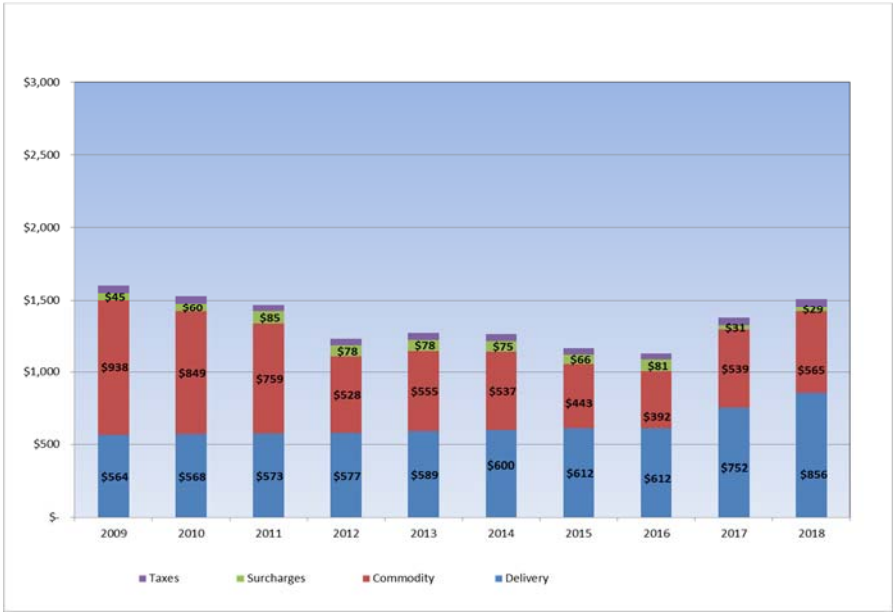
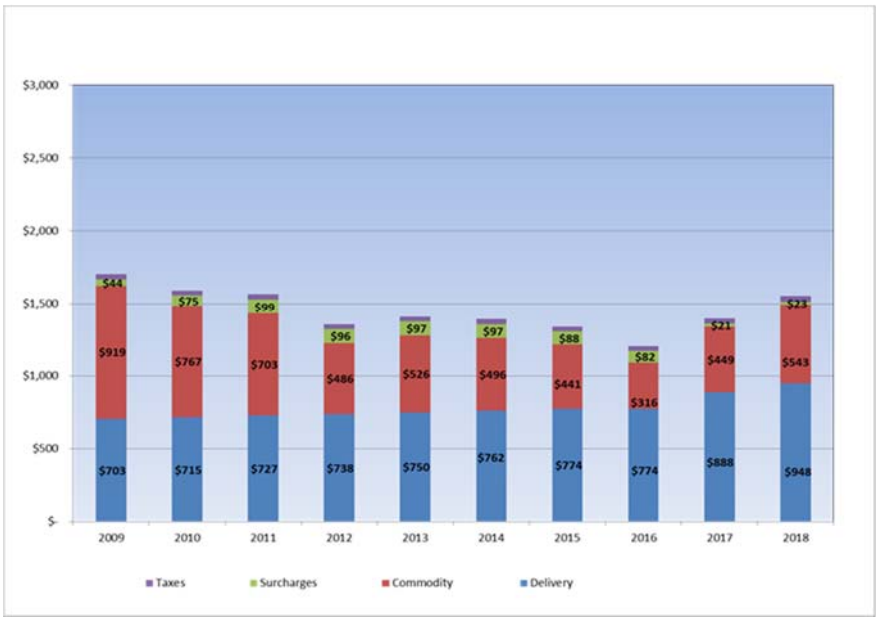


Table 2 – KEDLI Average Annual Residential Heating Bill
(1,000 Therm Annual Usage)



Q. Please provide the recent ratemaking history for KEDNY and KEDLI.

A. The Companies last filed for base rate increases in the 2016 KEDNY and KEDLI Rate Cases. The Joint Proposal adopted in those proceedings provided for three-year rate plans for the Companies that commenced in calendar year ("CY") 2017. Prior to the 2016 KEDNY and KEDLI Rate Cases, customer annual base delivery rates remained relatively flat and consistent for KEDNY and reflected only modest increases for KEDLI during the period from 1995-2016 due to the adoption of long-term rate plans, which, in some cases, incorporated base rate freezes or decreases. Accordingly, customers have generally benefitted from rates that remained relatively stable over the past two decades.

Q. Please explain the Companies' need for rate relief.

A. Increased capital expenditures and operation and maintenance ("O&M") expenses are the principal drivers of the Companies' revenue deficiencies. The Companies have engaged in numerous efforts to operate as efficiently as possible and control costs to minimize the effects on customers, as demonstrated in this proposal. However, current rate allowances are simply inadequate to cover the Companies' costs of providing safe and reliable service to our customers, especially in New York City. Capital expenditures required to maintain and modernize gas infrastructure have increased over the years and will continue to rise in the future.

Increases in property tax costs for both Companies and KEDNY's growing SIR expense have further exacerbated the forecast revenue deficiencies.

III. Objectives for the Rate Filing

Q. What are the overall objectives of the Companies' filing?

A. As outlined above, the Companies have several objectives in this filing that are also important to customers and the Commission. First and foremost, KEDNY and KEDLI must adjust base rates and establish compensatory rate plans to recover the costs of providing safe and reliable service to customers, to fund necessary investment in gas infrastructure, to modernize our networks, and enhance service to customers. Recovery of costs associated with necessary investments will provide the Companies with a fair opportunity to earn a reasonable return and must be effectuated in a balanced manner that mitigates the impact on customer bills.

Q. Did the Companies conduct any customer and stakeholder outreach prior to making these rate filings?

A. Yes. The Companies recognize the impact of these cases on our customers. We undertook significant outreach to our customers and communities to educate them about the Companies' need for rate relief and to hear from them as to their priorities and expectations of us. We found that our overall objectives align with

those of our customers. In developing these cases, we have reflected much of the feedback we received from customers.

Q. Please describe the efforts the Companies made to hear customers' priorities and what you learned.

A. The Companies conducted more than 200 outreach meetings and contacts to date with customers, various state agencies, local governments, school districts, hospitals, economic and community partners and elected officials throughout KEDNY and KEDLI's service territories. Our outreach was welcomed and well received. We heard directly customers' expectations, what they believe we do well, and where we need to improve.

The top priorities are the reliability, safety, and the affordability of gas service. All constituents are concerned about bill impacts, as the affordability of gas service is important to all our customers. A number of these stakeholders raised phasing in increases over multiple years to soften rate impacts for customers. Customers praised the hard work of our consumer advocates and would like to see us to do more to assist customers struggling to pay their bills.

The Companies confirmed that customers want us to play a lead role in gas safety education and energy cost management. Customers want to be able to

communicate with the Companies more easily and effectively. While many praised the contacts they had with the Companies, they seek more direct and real-time communications with us on issues impacting their energy usage, gas safety, and energy efficiency offerings and incentives. The Companies' educational partners strongly supported our STEM grants, mentoring, career panels and pipeline training programs that are helping to create the engineers of tomorrow and supporting the Companies' workforce development needs.

We also heard various areas of concern and where improvement is needed. The Companies heard fair criticism and frustration on issues such as gas construction impacts. Customers want more transparency and efficiency in our construction process and the impact it will have on their daily lives. Our customers continue to raise concerns with billing issues, including bill accuracy, and challenge the Companies to improve our billing processes. Customers are also concerned with the adequacy of gas supply and how supply issues may affect the future of their service.

While we heard appreciation for our legacy Cinderella program and economic development programs, our communities are looking for more economic development support and programs better tailored to current market conditions to

continue the revitalization and beautification of neighborhoods in the communities we serve.

This feedback from our customers and constituents was very valuable and has guided our development of these filings.

A. Investing in Infrastructure and Developing Our Workforce

Q. Please explain the Companies' objectives and proposals with respect to gas infrastructure investment.

A. KEDNY and KEDLI have made substantial infrastructure investments in their systems in the past several years. However, significant work remains to maintain the required level of safe and reliable service as well as modernize our facilities to keep pace with the emerging gas issues. These filings present the Companies' capital investment programs, which are designed to enable the Companies to continue to maintain safe and reliable service, remove leak prone pipe, reinforce the gas distribution systems, implement new safety requirements and safety best practices, make property improvements to facilities that support capital and O&M programs, and meet the expectations of our regulators and customers. The Companies' capital spending programs are described in the testimony of the Gas Infrastructure and Operations Panels and Gas Safety Panel. The Companies

propose the following levels of indirect and direct capital spending, including cost of removal, for their gas businesses during the Rate Year and Data Years:

$M	Rate Year	Data Year 1	Data Year 2	Data Year 3
KEDNY	$961,937	$942,449	$938,471	$1,013,801
KEDLI	$586,134	$557,068	$585,524	$627,022

Q. Are the Companies actively engaged in efforts to ensure that they are managing their gas networks in a cost-effective manner?

A. Yes. National Grid recently created the Capital Delivery organization, which is intended to systematically improve our ability to develop and execute the capital plan more efficiently for complex projects and the overall capital program. The organization will identify efficiencies and opportunities for better ways of working, with the objective of reducing costs. While the organization is still relatively new and in the process of development, its initial efforts have yielded some efficiencies that are reflected in the Companies' proposed capital investment plans.

While modernizing our infrastructure is critically important, the Companies are focused on more than installing new mains and service. As discussed in the Companies' Gas Infrastructure and Operations Panel testimonies, the Companies are moving toward more integration of NPAs and energy efficiency solutions, where possible, to offset traditional capital investments. However, at the same

time, the Companies must continue robust capital investments to ensure that we can safely and reliably serve existing load and the expected demand of our customers.

Q. Is there a focus to the Companies' investments in utility infrastructure?

A. Yes. Gas safety remains the Companies' top priority and the proposals in this filing reflect our increased focus in this crucial area. Indeed, most of the capital investments in this filing fall under the overarching umbrella of gas safety. Investments to modernize gas distribution and transmission infrastructure, monitor and replace aging infrastructure, and retire LPP will help the Companies operate the gas system in a safe and efficient manner. For KEDNY, substantial investment in the City State Construction program will provide continued support for mandated municipal projects that impact our facilities, while also providing the opportunity to address potential LPP that may exist in those areas during the process. These investments also have the added benefit of aiding the Companies in the delivery of reliable service and meeting, or exceeding, the expectations of regulators and customers.

The Companies are fully dedicated to removing LPP to maximize the safety of New York gas distribution networks. In the 2016 KEDNY and KEDLI Rate Cases, the Companies established an aggressive pace of LPP replacement, replacing minimum cumulative three-year targets of 180 and 405 miles of leak prone pipe for KEDNY

and KEDLI, respectively. To further challenge the Companies, we propose to increase LPP replacement to four-year minimum cumulative targets of 310 and 620 miles for KEDNY and KEDLI, respectively. The Companies also propose limited incentives to encourage additional main replacements and productivity improvements to reduce the costs of this program. The Gas Infrastructure and Operations Panels and Gas Safety Panel discuss the Companies' leak prone pipe proposals, metrics, and incentives.

The Companies' gas capital investment plans also include several other programs, ranging from targeted proactive repairs to supply initiatives, that will enhance the safety and reliability of its gas distribution system. These programs and their projected costs and benefits are outlined in the testimony of the Gas Infrastructure and Operations Panels.

Q. Will these investments yield any environmental benefits?

A. Yes. As part of our Future of Heat strategy, the Companies are focused on continuing a leadership role in initiatives aimed at reducing emissions. National Grid recently announced an aspirational target to reduce 60 percent of methane emissions on its gas systems by 2035. Investments in aggressive main replacement programs, leak repair, and enhanced work practices will reduce gas losses, methane emissions, and greenhouse gas emissions, all which will substantially benefit the

environment and contribute to National Grid and State clean energy targets. The testimony of the Future of Heat Panel proposes several products and services to further support these important environmental objectives, including:

- A green gas tariff that will allow customers to elect to have a portion of their gas sourced from renewable/green supplies;
- Incentives for investments in RNG projects and development of renewable supplies; and
- Geothermal and Customer Demand Demonstration Projects to promote customer choice and encourage alternative and efficient energy usage.

Q. What are the Companies' proposals with respect to gas safety and compliance?

A. The Companies have improved their performance and focus on gas safety over the past few years. However, recent events in the gas industry demonstrate the need for constant vigilance as well as continued improvements in this area. In this proposal, we have included a variety of programs and investments, both capital and O&M, to comply with new and emerging gas safety regulatory requirements and to adopt industry recommendations that drive further gas safety improvements and respond to lessons learned from recent industry gas safety incidents. These programs and investments are targeted to improve the Companies' field

performance as well as our internal processes, training, testing, and internal resources to provide holistic approach to gas safety improvement.

The Companies recently implemented a process safety program that adopts the American Petroleum Institute's recommended pipeline safety management system standards (Recommended Practice 1173). The program provides a framework for identifying hazards, controlling potential risks, and addressing safety and maintenance requirements throughout a pipeline's life cycle to reduce the likelihood of safety incidents. The Companies will build upon the progress of this program by enhancing and adding incremental gas safety and compliance programs.

We will also enhance our customer education and outreach programs to continue to educate the public on the importance of recognizing and reporting gas odors and deploying additional damage prevention resources to protect our underground facilities. Building upon the success of the program implemented in the 2016 KEDNY and KEDLI rate cases, the Companies propose to expand the scope of KEDNY's Residential Methane Detection Program and to incorporate this program in KEDLI's service territory, targeting the installation of approximately 90,000 residential methane detectors between the two service territories over the term of the rate plan.

The Companies propose to modify our gas safety performance metrics and incentives. The proposed modifications will adjust some of the targets established in the 2016 KEDNY and KEDLI Rate Cases and incent the Companies to continue to aggressively address our leak backlog and LPP replacements. The metrics proposed remain stringent, but the Companies believe that these adjustments will more meaningfully reflect the conditions we are seeing in the field and we believe to be reasonable based on recent experience.

Detailed discussions of the proposals highlighted above are provided in the testimony of the Gas Infrastructure and Operations Panels and the Gas Safety Panel.

Q. Are the Companies proposing to hire additional employees to support gas operations?

A. Yes. These rate plans present substantial increases to work in our capital investment plan and increased O&M workload as well as increases in our gas safety activities for the benefit of the Companies and our customers. Current staffing levels are not adequate for the workload projected, and therefore incremental resources are required to ensure the successful and safe delivery of the plans. Specifically, KEDNY and KEDLI will need approximately 250 incremental positions before the end of the Rate Year, and 500 incremental positions by the end of Data Year 3. The vast majority of these positions (95

percent) are required to support incremental safety programs such as contractor oversight, inspections, and inactive accounts.

In the coming months, National Grid will look to recruit and train employees needed to support capital and O&M programs, including jobs in field operations, pipeline safety management, corrosion control, and gas system engineering. Further justification for these positions is provided in the testimonies of the Gas Infrastructure and Operations Panels, Gas Safety Panel, Future of Heat Panel, and Revenue Requirements Panels. These positions will provide the opportunity to develop coveted STEM work skills and experience and offer the prospect of long-term careers in the energy industry.

Q. What are the Companies doing to hire and train the next generation of utility workers?

A. National Grid has thousands of highly skilled, dedicated employees that support our businesses and customers in New York. However, it is well known that the broader utility industry is facing a challenge because of the aging demographic of the utility workforce. National Grid is not immune to this challenge, and, in the coming years, many of our most experienced and skilled utility professionals will be eligible for retirement. Accordingly, it is imperative that we attract and retain the next generation of utility workers to support our business and operate and

maintain our gas infrastructure. The Companies continue to utilize our Strategic Workforce Planning Center of Excellence to analyze workforce demographics and future business requirements and develop plans to address short and long-term planning needs. Additionally, the Companies will continue to partner with vocational schools, community colleges and veterans' associations to create a job "pipeline."

Q. How are the Companies addressing potential gas supply issues in their service territories?

A. The Companies have signed a precedent agreement with Transcontinental Gas Pipe Line Company LLC for gas transportation service on an upstream pipeline project, known as Northeast Supply Enhancement project ("NESE"), that will provide access to additional gas supplies necessary to relieve the current supply constraints in the Companies' service territory. The constraint relief provided by NESE will allow for continued, reliable service delivery to our existing firm service customers as well as provide the potential to serve additional gas load. The Companies' filings in these proceedings reflect supply levels assuming that NESE is placed in service by the winter 2020/2021 season. However, if NESE is unable to bring new supply to our territories, the Companies will not be able to add new or additional gas load without jeopardizing the Companies' ability to provide safe, reliable service to its existing firm customers. Company Witness Elizabeth D. Arangio and the Gas

Infrastructure and Operations Panels discuss the impacts to the Companies if the NESE project is not completed within its current projected timeline.

B. Information Technology

Q. What are the Companies' proposals and strategy for upgrading IT?

A. Modern energy networks rely on state-of the-art information systems to monitor, manage, and optimize system performance, support state and federal energy goals, address emerging utility and market trends, meet evolving customer and business demands, and stay ahead of emerging cyber-security threats. Many of National Grid's legacy systems, however, are at or near the end of their useful lives and rely on outdated technology that is insufficient to support evolving business requirements. Aged infrastructure is more prone to outages and extended down time, which can negatively impact network reliability and resiliency. The current systems lack the flexibility needed to drive performance improvements and efficiencies for the Companies and our customers in the future.

KEDNY and KEDLI currently utilize outdated and disparate customer systems. CSS (supporting KEDLI and Niagara Mohawk) is 20 years old and CRIS (supporting KEDNY) is 30 years old. Neither system can readily support escalating customer expectations or increasingly sophisticated billing needs (*e.g.,* distributed energy resources). The CIS Replacement Program will replace these

outdated systems with a modern, flexible, and more capable customer system that will meet the requirements of new customer program and rate options, incorporate digital and emerging technology solutions, improve customer choice, and reduce billing issues and certain costs associated with operating our aged systems.

The testimony of the IT Panel discusses in more detail the Companies' investment levels, our strategy to deliver on these information systems investments, and how we propose to report on these investments to the Commission.

C. Gas Business Enablement Program

Q. Please describe the GBE Program.

A. The GBE Program is a comprehensive business transformational program focused on strengthening and improving the performance of National Grid's U.S. gas business with regard to managing assets, delivering work, and serving customers. The GBE Program will replace a number of aged computer systems, support continuous improvement in gas safety performance, facilitate more effective and efficient delivery of the overall work plan, and progress the satisfaction of evolving customer expectations. Certain aspects of the program have already been deployed in New York.

As the GBE Program will support all of National Grid's U.S. operating companies, each operating company will bear its allocated share of the total program cost, which is currently estimated at approximately $539.3 million. Niagara Mohawk received funding for the GBE Program in the 2017 NMPC Rate Case. To date, KEDNY and KEDLI have borne $67.4 million in connection with the GBE Program. These costs were not reflected in the 2016 KEDNY and KEDLI Rate Cases, and therefore were borne by shareholders, for the benefit of customers, to progress the project. Funding is requested in these rate filings for KEDNY and KEDLI's allocated share of future program costs. The funding request is imperative to support future development and deployment of the program in those service territories.

Further information regarding the program and the Companies' requested investment levels is provided in the testimony of Company Witness Christopher J. Connolly.

D. <u>Enhancing the Customer Experience</u>

Q. Please describe the Companies' philosophy regarding the customer experience.

A. Customer service is central to National Grid's goal of an effortless customer experience with increased choice and flexibility. Customer service is a core

function of our business and is at the forefront of our decisions. The Companies believe customer service is a multi-faceted experience that begins at the time of service application and continues throughout the life of the service relationship. It encompasses office and field employees through a series of interactions, such as a service calls, bill payments, and meetings with consumer advocates. The Companies are committed to continually improving customer service to enhance the experience of the residents and businesses we serve. These rate filings reflect the Companies' steadfast focus on and commitment to our customers and are designed to facilitate various investments needed to enhance the overall customer experience.

Q. What are the Companies' proposals with respect to customer service?

A. The Companies are proposing to refine the ways in which we interact with our customers, both in the field and through digital media. CIS and GBE are core investments that will update and enhance our information technology systems to improve customer service levels and the Companies' overall performance. Beyond these core investments, the Companies are proposing several service enhancements, products, and programs designed to improve the experience of our customers. As outlined in the testimony of the Shared Services Panel, the proposals include implementation of Electronic Deferred Payment Agreements, improvement of our Pitkin Avenue customer office, socializing credit card transaction fees, and

implementation of automated building energy usage data uploads to the Environment Protection Agency's ("EPA") ENERGY STAR Portfolio Manager web site.

Q. Please provide an overview of the Companies' proposed products, services, and new technologies to engage customers and promote new energy solutions.

A. As discussed by the Shared Services Panel, Gas Infrastructure and Operations Panels, and the Future of Heat Panel, the Companies have developed a suite of products and services to facilitate energy efficiency, empower and enable customers to better manage their energy usage, and to advance clean energy goals in our service territories. The Companies believe these offerings will help provide a sustainable and affordable path toward a low-carbon future for customers and the environment.

The Companies have had success with previous energy efficiency programs and believe we can build upon those successes by expanding existing and adding incremental programs to our Energy Efficiency Transition Implementation Plan ("ETIP") portfolio. The portfolio proposals will further encourage energy savings across various market sectors for the benefit of customers. The Companies propose to maintain existing ETIP budgets but change the way we fund our energy

efficiency programs to better measure and track ETIP costs and align our recovery of energy efficiency costs with Commission policy.

The current rate filing presents an opportunity to offer new products and services in the Companies' territories that align with the State's clean energy goals, encourage sustainability and affordability, reduce consumption, and promote low-carbon energy solutions.

Achieving these goals will require a multi-faceted effort, including increased customer participation, reducing methane emissions, and decarbonizing the gas network. National Grid has developed a variety of products and services, such as a green gas tariff and revised natural gas vehicles rates, to support affordable implementation of green technologies using natural gas; a fuel-switching calculator to aid customers in their evaluation of sustainable choices; and a program to incent developers and non-residential customers to invest in renewable natural gas alternatives when developing their projects. Additionally, the Companies believe that continued research and development is necessary to leverage available technology and develop innovative solutions to meet our customers energy needs. The Companies will continue our partnership with research institutes to evaluate clean energy technologies low-carbon gas alternatives.

Q. Please describe the proposals for demonstration projects that support the Companies' future of heat objectives.

A. The Companies are committed to testing scalable technologies and programs that create cleaner, more resilient energy options, expand customer choice, and animate the market for clean energy technologies. KEDNY and KEDLI propose expansion of two existing demonstration programs to further explore how those projects can be used to scale the low-carbon gas network of the future. In addition, the Companies will continue their work to reduce carbon in the gas network.

The Geothermal Demonstration Project approved by the Commission in the 2016 KEDNY and KEDLI Rate Cases authorized KEDLI to test shared-loop ground-source heat pump ("GSHP") systems as a cost-effective, NPA to extending traditional natural gas infrastructure. Use of geothermal technology allows for the decarbonization of traditional heating and cooling sources by converting customers who may be using oil as an energy source and provides homeowners with potential energy cost savings. The Companies seek to build upon the initial success of the project as part of the Sustainable Heat Initiative, which is aimed at sustainably meeting customers' heating requirements. The expanded project would target low-to-moderate income customers, as well as commercial and industrial customers, and develop partnerships with geothermal developers to facilitate market animation.

As part of the Sustainable Heat Initiative, the Companies also propose to implement the Clean Conversion Program that will provide customer education and a feasibility analysis to determine if conversion from oil heat to another energy source, including a transition to geothermal resources, is the most viable solution for their energy needs.

In the 2016 KEDNY and KEDLI Rate Cases, the Commission also approved a commercial gas demand response program that allowed the Companies to provide commercial firm gas customers with credits for gas demand reduced during specified periods. The primary objective of the gas demand response project was to evaluate the effectiveness of voluntary peak reductions and determine whether the provision of credits would ultimately affect customer consumption. The project was successful during its first few years, resulting in gas curtailments that positively impacted gas system pressure, and reduced customer supply costs. The Companies propose to expand the project to firm gas delivery customers who meet the eligibility requirements.

The Companies will also advance our efforts to enhance the diversity of supply and present viable alternatives to natural gas and oil. KEDNY will continue its Newtown Creek Demonstration Project, which converts the biogas from wastewater and food waste to pipeline-quality RNG that can be added to the natural

gas system. Similarly, KEDLI is proposing to scope a Power-to-Gas ("P2G") Demonstration Project to convert excess renewable electricity to pipeline-quality gas and connect those resources to the distribution system to alleviate constraints and reduce carbon emissions.

Q. Please discuss the earnings adjustment mechanisms ("EAMs") proposed by the Companies.

A. In these filings, the Companies propose investments, products, and services that are focused on system efficiency, energy efficiency and supporting the State's clean energy goals. These investments, products, and services are designed to further build on existing programs, and in doing so, delivering outcomes that benefit customers while at the same time achieving the Commission's objectives of promoting energy efficiency, facilitating alternative energy solutions, and reducing carbon emissions.

To emphasize the value of these outcomes for customers, the Companies have included EAMS to incent performance in the areas of system efficiency, energy efficiency, and customer carbon reduction. Achievement of the ambitious targets set out in the EAMs will require the Companies not only to leverage the programs and services that will be created through the investments contained in these filings, but also challenge the Companies and our customers to increase our focus on

providing more efficient and innovative ways to power homes, businesses, and vehicles. The Companies also propose separate incentive mechanisms tied to activities that further support REV objectives, like adoption of an NPA incentive mechanism and an E-Commerce Platform Service Revenue, to drive additional performance in these areas. The Future of Heat Panel describes the proposed EAMs and incentive mechanisms.

E. Low Income and Economic Development Proposals

Q. Please explain the Companies' low income proposals.

A. The Companies recognize that many of our customers experience financial difficulties and struggle to make ends meet. The Companies have a long history of providing much needed assistance to our most vulnerable customers through community partnerships and rebate and discount programs to offset customer bills. The Companies propose to continue existing programs, such as the Energy Affordability Program ("EAP") and Low-Income Conversion Rebate Program, as well as modify aspects of existing programs to increase customer participation in these programs.

A further description of the Companies' low income customer initiatives and efforts, including the operation of the highly successful Consumer Advocate

program, is provided in the testimony of the Shared Services Panel and Future of Heat Panel.

Q. Please explain the Companies' economic development proposals.

A. Economic development is incredibly important to the communities we serve. Investment in economic development can attract and retain businesses, create job stability, build the local tax base to fund government programs, diversify the economy, and promote a cleaner energy future in a given region. These benefits all increase the wellbeing and quality of life within a community.

For decades, the Cinderella Community Development Program has helped to revitalize local communities in the KEDNY and KEDLI service territories. Through this program, we have provided awards for rehabilitation of abandoned buildings, to provide new affordable housing and retail stores, and to support the development of green technology and innovative building technologies. The rate plans approved in the 2016 KEDNY and KEDLI Rate Cases included additional economic development discount and grant programs to assist customers in our service territories. We are extremely proud of our economic development programs and wish to continue the revitalization of the communities we serve.

In developing this filing, the Companies solicited feedback from customers, program participants, and other stakeholders to evaluate the efficacy of existing programs and identify whether additional enhancements or modifications were required to make the programs more responsive to customer needs. Based the feedback received, the Companies developed changes to our comprehensive suite of economic development programs that will add new economic development programs and modify existing programs to more accurately reflect the customers and communities we serve. Specifically, we propose to expand the eligibility for certain programs, including the Cinderella program, broaden business incentive rates, increase the annual level of discounts provided, and modify areas targeted for discount programs to better reflect economic development priorities. Collectively, these modified programs will help offset customer costs for natural gas infrastructure upgrades to accommodate business expansion, incorporate new sustainable gas technologies, and support overall regional development and economic growth in our service territories. These programs are addressed in the testimony of the Future of Heat Panel.

F. Site Investigation and Remediation Costs

Q. What do the Companies propose relative to recovery of their SIR costs?

A. SIR programs are government-mandated programs that require the management, investigation, and remediation of manufactures gas plant ("MGP") sites that have

been impacted by the past release of substances by property owners. Both Companies are responsible for MGP sites in their service territories. KEDNY is also responsible for two federal Superfund sites associated with waterways adjacent to former MGP sites. The Companies are presently engaged in clean-up activities at these sites, which will continue well beyond the Rate Year and Data Years. While the Companies actively try to manage and control SIR costs, including taking action to reduce MGP site cost responsibility, the scope and timing of activities at these sites are ultimately subject to the purview of the EPA and New York State Department of Environmental Conversation. The Companies propose to recover future SIR costs in a manner consistent with present recovery methods and maintain the existing SIR Recovery Surcharge and annual reconciliation mechanisms to defer costs above the allowance for future refund or recovery from customers.

Company Witness Charles Willard addresses our SIR programs, cost mitigation efforts, proposed funding levels, and the status of work at major MGP sites, including the Gowanus Canal and Newtown Creek Superfund sites. The remediation of these superfund sites is likely to have a significant impact on KEDNY's cash flows and thus it is particularly important for the Commission to approve KEDNY's SIR cost recovery proposals.

IV. The Companies' Gas Rate Filings

Q. Please address the review and presentation of the cost of service data in the Historic Test Year and the Rate Year.

A. The Revenue Requirements Panels describes how the cost data included in the Companies' filings was reviewed and prepared. To facilitate Department of Public Service Staff's ("Staff") review of Historic Test Year costs, National Grid reviewed the accounting for costs charged to the Companies in the Historic Test Year. This detailed review included a comprehensive review of service company and operating company costs in the Historic Test Year, including labor allocations, vendor charges and employee expenses, among other items. The review was focused on verifying that the Historic Test Year costs were charged correctly in accordance with National Grid's cost allocation methodologies and were appropriate to include in KEDNY and KEDLI's cost of service. The Companies' review should facilitate Staff's timely audit of costs included in KEDNY and KEDLI's revenue requirements.

Q. What rate of return on equity and capital structure do the Companies propose?

A. We are proposing a return on equity of 9.65 percent for the Rate Year and a capital structure with a 48 percent equity component. For a multi-year rate plan, the Companies propose that their returns on equity reflect a premium of 50 basis points

and that the equity component of their capital structures should be increased gradually from 48 percent to 50 percent, as discussed in the testimony of Company Witnesses Ann E. Bulkley and Jonathan Cohen.

The interests of the Companies and our customers will be served if these rate filings establish rates that compensate KEDNY and KEDLI's cost of providing service to customers and afford the Companies a reasonable opportunity to earn allowed returns on equity that are commensurate with enterprises of similar risk. Investors use the return on equity as a key benchmark in assessing investment opportunities in public utilities. A return on equity that is below what investors believe they can earn on investments in companies with similar risk would impair our ability to attract capital, both debt and equity, on reasonable terms.

Q. What are the rate impacts of the Companies' filing?

A. Both Companies require a modification to existing rates to continue the provision of safe and reliable service to our customers. KEDNY's revenue deficiencies are significant, and thus the rate impacts are significant, as well. In contrast, KEDLI requires a relatively modest revenue increase to achieve required service levels and the proposed rate increase will have less of an impact on customer bills.

As discussed in the testimony of our Rate Design Panels, the Companies proposal results in total bill increases for typical usage residential customers of 11.99 percent and 5.15 percent for KEDNY and KEDLI, respectively. The Company would be open to levelizing the rate increase over the term of a multi-year rate plan to mitigate bill impacts to customers. To that end, the Companies look forward to working with Staff and the other parties to arrive at a multi-year rate plan that permits recovery of the Companies' cost of service and mitigates the impacts for our customers.

Q. Does that conclude your testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Maureen P. Heaphy

Dated: April 2019

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I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Maureen P. Heaphy. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("Service Company"), a subsidiary of National Grid USA ("National Grid"), and currently hold the position of Vice President of U.S. Compensation, Benefits, and Pensions. My responsibilities include overseeing compensation, pension and benefit strategies, policies, and programs for all National Grid U.S. operations, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies").

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Accounting and Computer Applications and Information Systems from New York University in

1983. In 1991, I received a Master of Business Administration in Finance from St. John's University. I joined KeySpan Corporation in 1983 and held several professional and managerial positions in Treasury and Accounting. In 1991, I joined the Human Resources organization where my focus has been on design, strategy, administration, and implementation of compensation, benefit, and pension programs for employees and retirees.

Q. Have you previously testified before the New York State Public Service Commission ("Commission") or any other regulatory commissions?

A. Yes. I testified before the Commission on behalf of the Companies in Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases"). I also testified on behalf of Niagara Mohawk Power Corporation before the Commission in Cases 17-E-0238 and 17-G-0239 (the "2017 NMPC Rate Case"), Cases 12-E-0201 and 12-G-0202, and Case 10-E-0050. I have also submitted testimony concerning matters related to employee compensation and benefits before: (i) the Massachusetts Department of Public Utilities on behalf of Massachusetts Electric Company, Nantucket Electric Company, Boston Gas Company, and Colonial Gas Company; (ii) the Rhode Island Public Utilities Commission

on behalf of The Narragansett Electric Company; and (iii) the Federal Energy Regulatory Commission on behalf of National Grid Generation, LLC.

Q. What is the purpose of your testimony?

A. My testimony addresses and supports the Companies' employee compensation, benefit, and pension costs, including those associated with medical, dental, life insurance, and other post-employment benefit ("OPEB") plans, for the twelve months ended December 31, 2018 ("Historic Test Year"), as well as the forecast costs for the twelve months ending March 31, 2021 ("Rate Year") and the twelve months ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2") and March 31, 2024 ("Data Year 3") (collectively, the "Data Years"). The information concerning projected cost changes has been provided to the Revenue Requirements Panel and was used to develop the Companies' proposed revenue requirements. I will address changes in National Grid's workforce that affect the Companies' revenue requirements as well as the overall cost of the Companies' compensation and benefit programs.

I will demonstrate that the costs of the total compensation and benefit programs included in the Companies' revenue requirements, which

include the costs of base salary, performance-based variable pay and various other benefits, are reasonable and necessary and must be incurred by KEDNY and KEDLI to meet their obligations to provide safe and reliable utility service to customers. To that end, I will explain and support a detailed study of the Companies' total compensation and benefit programs for their management workforce, as well as analyses of the compensation and benefits provided to the Companies' union workforce. The results of these studies demonstrate that KEDNY and KEDLI's total overall compensation and benefit programs are reasonable and market competitive. I will also explain specifically why the costs of the variable pay plan should continue to be included in the revenue requirements and how that plan is structured to align the interests of the Companies with their customers.

In addition, I will discuss National Grid's ongoing efforts to monitor and control the costs of various elements of the Companies' employee compensation, benefit, and pension package.

Q. Do you sponsor any exhibits as part of your testimony?

A. Yes. I am sponsoring the following exhibits that were prepared or compiled under my direction and supervision:

(i) Exhibit __ (MPH-1) sets forth a list of incremental positions that National Grid forecasts to be filled prior to the end of the Rate Year and the Data Years;

(ii) Exhibit __ (MPH-2) sets forth Willis Towers Watson's comprehensive evaluation of National Grid's compensation and benefits for management (non-union) employees. This exhibit consists of the following schedules:

Schedule 1 – Competitive Assessment of National Grid's Total Compensation and Benefits Package;

Schedule 2 – Target Compensation and Target Variable Pay as a Percent of Market Assessment; and

Schedule 3 – BENVAL Analysis for Management Employees: 2018 Benefit Plans

(iii) Exhibit __ (MPH-3) sets forth information provided by Willis Towers Watson concerning increases in management base salary costs during the relevant periods, as well as historical information concerning increases in cash compensation provided by National Grid since 2009. This exhibit consists of the following schedules:

Schedule 1 – Market Merit Increases; and

Schedule 2 – 10 Year Wage Increase History

(iv) Exhibit __ (MPH-4) sets forth evaluations of KEDNY and KEDLI's compensation and benefits for union employees. This exhibit consists of the following schedules:

Schedule 1 – General Wage Increases for Union Employees;

Schedule 2 – Union Wage Comparison for KEDNY;

Schedule 3 – Union Wage Comparison for KEDLI;

Schedule 4 – BENVAL Analysis for KEDNY Union Employees – 2018 Benefit Plans for Local 101; and

Schedule 5 – BENVAL Analysis for KEDLI Union Employees – 2018 Benefit Plans for Local 1049

Q. How is the oversight of the Human Resources function conducted at National Grid?

A. The oversight of National Grid's Human Resources function is performed on a centralized basis and, with certain limited exceptions, uniform compensation, benefit, and pension packages for management employees have been instituted for all National Grid U.S. based operations, including KEDNY and KEDLI.

Q. Is a portion of the Companies' workforce unionized?

A. Yes. The majority of the Companies' employees are unionized and members of collective bargaining units. The union employees in KEDNY are members of the Transport Workers Union of America, AFL-CIO, Local 101 ("TWU Local 101") and the International Brotherhood of Electric Workers Local 3 ("IBEW Local 3"). The union employees in KEDLI are members of the International Brotherhood of Electric Workers Local 1049 ("IBEW Local 1049"). The total compensation and benefits for all represented workers is determined by collective bargaining.

II. Workforce Adjustments

Q. Do the Companies' revenue requirements include any costs associated with incremental positions that are forecast to be filled through the end of the Rate Year and Data Years?

A. Yes. National Grid expects to fill 252.4 incremental positions before the end of the Rate Year, and forecasts the need for 503.4 incremental positions by the end of Data Year 3. Exhibit __ (MPH-1) sets forth a list of these positions, over 95 percent of which (482 positions) are required to support incremental safety programs such as contractor oversight, inspections, and inactive accounts. The Company will update this list as the case proceeds. The need for these employees is discussed in the

testimony of the following witness panels: Gas Infrastructure and Operations ("GIOP"), Gas Safety ("GSP"), Future of Heat ("FHP"), and Revenue Requirements. The revenue requirement impacts associated with these additional positions are discussed in the testimony of the Revenue Requirements Panels.

Q. Are there any other adjustments to National Grid's management and union workforces as of December 31, 2018 that should be made to arrive at a representative workforce level in the Rate Year?

A. Yes. Additional adjustments were made to eliminate 57 management employees and 121 union employees who were on long-term leave as of December 31, 2018 and are not expected to return. In addition, adjustments were made to add 75 management interns and nine seasonal union employees to the Service Company, and three management interns to KEDNY/KEDLI to bring the Rate Year count of temporary employees equal to the Historic Test Year average. The adjustments to the revenue requirements associated with these employees are discussed by the Revenue Requirements Panels.

III. Review of Compensation and Benefits

Q. Please describe National Grid's philosophy concerning employee wages and benefits.

A. National Grid's overall approach to compensation is to ensure that (i) a significant portion of employee compensation is tied to the attainment of performance goals that create benefits for customers and are consistent with the policy goals established by National Grid's regulators, (ii) employees' total compensation is comparable to median compensation for comparable positions in both the utility industry and general industry and is reasonable after considering base and variable pay and benefits in total, and (iii) variable pay is based on both the overall performance of National Grid and the performance of each individual in achieving goals that are tied to the attainment of customer satisfaction, safety, and reliability objectives. To provide safe, reliable, and efficient utility service to its customers, National Grid must attract, retain and engage high performing, qualified personnel. To accomplish this, National Grid provides a total compensation package that recognizes and rewards excellence, maintains fair and competitive market pay and benefits for employees, and encourages employees to improve skills while providing a safe working environment. Doing so under the cost containment pressures faced by all companies is a critical challenge. To meet this challenge, National Grid

has developed a "Total Rewards Program" to provide employees with an overall compensation, benefits, and pension package that is market competitive, offers flexibility and choice, and supports a high-performance culture by directly linking performance to rewards. By maintaining a comprehensive and competitive approach to total rewards that establishes appropriate levels of pay and benefits, National Grid can attract and retain a high-quality workforce and motivate employees to perform at high levels.

Q. What are the elements of the total reward package?

A. The compensation elements of the total reward package are cash compensation, which includes both fixed and variable pay, and a number of benefits, which include medical and dental plans, life insurance, long-term disability, a 401(k) savings plan, retirement plans, other post-employment benefits, vacations, and holidays. It is National Grid's philosophy to require employees to share the costs of certain benefits, consistent with market practice.

Q. Please describe National Grid's efforts to ensure the reasonableness of the cost of the wages and benefits it offers to employees.

A. As part of its effort to provide a market competitive package while controlling the cost of compensation, benefit, and pension programs, National Grid monitors the marketplace to ensure that its cash compensation and benefit programs are both cost-effective and sufficient to enable it to attract, retain, and engage the highly skilled workforce needed to deliver excellent customer service and achieve the financial success required by the capital markets. Specifically, National Grid utilizes the services of Willis Towers Watson to provide information concerning the overall competitiveness of its compensation and benefits package.

Q. Did Willis Towers Watson conduct a study of the competitiveness of the overall compensation package offered by National Grid to its management employees?

A. Yes. Willis Towers Watson recently conducted a study of the competitiveness of the overall compensation package offered by National Grid to its management workforce. This study involved analyses of the total cash compensation (fixed and variable) and employee benefits package offered by National Grid to its management employees. On an

overall basis, this study concluded that the value of National Grid's overall compensation, benefits, and pension package is slightly below the median value of the peer group's package, but generally within the 10 percent corridor that is considered the zone of reasonableness. The results of the management study are set forth on Exhibit __ (MPH-2).

Q. Please explain how Willis Towers Watson performed its study of the competitiveness of National Grid's management compensation.

A. Willis Towers Watson compared the total management compensation and benefits package provided by National Grid to the total management compensation and benefits packages provided by 38 peer companies. These peer companies include utility companies (*e.g.*, Consolidated Edison Company of New York, Inc., Eversource Energy Services Company), and general businesses that have significant workforces in the United States (*e.g.*, American International Group, Inc. (AIG), International Business Machines Corporation (IBM), United States Steel Corporation). While many of these companies are comparable in size to National Grid in terms of revenues, some are smaller, and some are larger.

Q. Does the peer group provide a representative sample of the types of entities that National Grid competes with to attract management employees?

A. Yes. On an overall basis, it is reasonable to compare National Grid's total management compensation package to the companies set forth in the peer group used by Willis Towers Watson. National Grid believes that if its total compensation package falls within a 10 percent corridor (90-110 percent) of the median level of these companies, then the overall package is reasonably designed to enable National Grid to attract and retain highly qualified management employees. While the results of the study show that National Grid's total compensation package is slightly below median, National Grid is proposing certain adjustments, discussed more fully later in my testimony that will ensure that National Grid's compensation will continue to approach the median level.

Q. Please explain how Willis Towers Watson performed its comparison of the total compensation package provided by National Grid to that provided by the peer group companies.

A. This comparison was performed by analyzing (i) the total cash compensation, including both fixed and variable pay, provided to employees in approximately 92 percent of the management positions at

National Grid to the total compensation, including both fixed and variable pay, provided by the peer group companies, and (ii) the relative value of the benefits provided to National Grid employees in comparison to the relative value of the benefits provided by the peer group. The positions evaluated by Willis Towers Watson represent approximately 5,800 employees of National Grid's total management workforce of approximately 6,300 employees as of January 2019. The results of the cash compensation and benefits comparisons were added to arrive at the total compensation and benefits value. The result of the total compensation study is that National Grid's total package of management compensation is generally close to the median level of the peer group, as shown on Schedule 1 of Exhibit __ (MPH-2).

Q. Please explain why Willis Towers Watson's study compares the relative value of National Grid's total compensation to the relative value of the total compensation provided by the peer group.

A. The purpose of performing an overall compensation study is to determine whether the total level of cash compensation and benefits is sufficient to attract a highly qualified workforce. From this perspective, cash compensation – fixed and variable – is best analyzed by comparing the cash compensation provided to various positions at National Grid to cash

compensation provided by other companies with comparable positions. In this case, Willis Towers Watson analyzed the cash compensation provided by members of the peer group to management positions that fall within various salary bands to the equivalent cash compensation provided to the equivalent positions at National Grid. The comparison of total compensation for various benchmarked positions produces an overall comparison of cash compensation that permits National Grid to evaluate whether management positions within various salary bands are receiving a reasonable (*i.e.* within the 10 percent corridor of the median) level of cash compensation that will ensure that National Grid remains competitive in the labor market. The results of these analyses are set forth in Schedule 2 of Exhibit __ (MPH-2).

In contrast, acquiring information concerning peer companies' costs of various employee benefits and comparing those costs is very difficult if not impossible. The reason for this is that the cost of various benefits can vary dramatically from year to year and from company to company. For example, the cost of pension and other post-employment benefits ("OPEBs") can change because of changes in the funded status of the plans established to provide those benefits, the investment performance of the plans, and the mortality rate of those participating in the plan.

Similarly, the cost of health insurance can vary because of the claims experience of individual companies and the age and health of individual employees in the workforce and their dependents. Changes in costs associated with these factors do not determine whether the benefits being provided by a particular company are competitive. Instead, it is the relative value of the benefits that provides the proper measurement of whether a benefits package is sufficient to attract, retain and engage highly qualified personnel. The analyses performed by Willis Towers Watson properly consider the relative value of the benefits being provided by National Grid and the peer group.

Q. Are the analyses of compensation and benefits being presented in this case comparable to the analyses that were presented in the 2016 KEDNY and KEDLI Rate Case and the 2017 NMPC Rate Case?

A. Yes. The analyses follow the same basic methodology that was presented in the 2016 KEDNY and KEDLI Rate Cases and the 2017 NMPC Rate Case.

Q. How did Willis Towers Watson determine the relative value of the benefits package provided by National Grid and the peer group?

A. The relative value of the various benefits is established based on a common set of actuarial assumptions and a single employee population. This establishes a controlled environment in which differences in value are exclusively a function of the differences in benefit plan provisions. The relative value of various benefits is determined by dividing the value of the particular benefit for individual companies by the average value of all the companies in the peer group. Relative values are developed for each benefit and for the entire benefit plan.

Q. Are all elements of National Grid's compensation package – fixed compensation, variable compensation, and benefits – necessary elements of its overall compensation package?

A. Yes. National Grid's overall management compensation package is designed to be competitive with the marketplace. It is the total compensation package – fixed cash compensation, variable pay, benefits and pensions – that permits National Grid to be competitive with the marketplace. The fixed component of National Grid's cash compensation package along with the benefits package is insufficient to be competitive with the marketplace. If National Grid did not provide variable pay, it

likely would not be able to attract the caliber of employees it seeks to hire because its compensation package would not be competitive. Indeed, if National Grid did not have variable pay as a component of its compensation package, its package would only be 84 percent of the median compensation, as noted in Schedule 1 of Exhibit __ (MPH-2).

Q. In addition to the analysis of total compensation being presented in these cases, does National Grid also undertake analyses of various individual elements of its compensation to ensure that these individual elements remain competitive with the marketplace?

A. Yes. While it is important that National Grid's total compensation package remains competitive with the marketplace, it is also important that individual elements of the compensation package remain competitive, as well. Thus, National Grid also undertakes efforts to ensure that: (i) its total cash compensation (fixed and variable pay) and (ii) various benefits, such as healthcare and retirement benefits, individually track the market. National Grid needs to stay well-informed of what workers demand in today's labor market. To that end, in addition to ensuring that its overall compensation is reasonable, National Grid utilizes the services of Willis Towers Watson to ensure that its total cash compensation (fixed and variable) is at or near median levels as compared to the peer group, and

that its various key benefits remain at that level, as well. Analyses of individual elements of National Grid's compensation and benefits package are included in Schedules 2 and 3 of Exhibit __ (MPH-2). As can be seen from the data provided in these schedules, National Grid is currently in a competitive position with respect to total cash compensation and total benefits.

IV. Management and Union Compensation

Q. Previously in your testimony, you mentioned that the total compensation of employees is intended, on average, to approximate median compensation levels. Do individual employee salaries deviate from these levels?

A. Yes. Individual salaries deviate above and below the median levels based on experience, individual performance, and length of time in a position.

Q. Is the officer compensation program similar in philosophy and operation to the management cash compensation program?

A. Yes. The approach taken with respect to officer compensation is basically the same as described for the management employee program. Overall compensation for officers, including National Grid's top officers, is benchmarked to median compensation of an appropriate peer group and

includes a variable pay component. For purposes of preparing the revenue requirements, the Companies have included payroll expense for only the fixed compensation (base pay) and the benefit component of the compensation package for National Grid's top officers, referred to as Band A employees. The variable pay component for National Grid's Band A employees is not included in the revenue requirements.

Q. Please describe the "banded" approach to overall compensation.

A. National Grid's compensation structure comprises six bands. Band A consists of the top officers such as executive and senior vice presidents. Band B consists of less senior officers (vice presidents). Band C is primarily for directors who report directly to an officer. Band D is for managers who have at least one direct report or may directly report to an officer. Band E is for supervisors who must have at least one direct report and who report to a director or manager. Band F is for general administrative staff. Bands D, E, and F are also for individual employees with unique career paths. Each band has a salary range that consists of a minimum and a maximum salary as well as multiple market reference points within the band. The market reference points track the varying level of compensation for positions within the bands. The structure is reviewed annually by the Human Resources function. The band structure

employed by National Grid generally corresponds to the market study prepared by Willis Towers Watson in analyzing National Grid's total compensation.

Q. Please describe National Grid's variable pay program.

A. National Grid's variable pay program is known as the Annual Performance Plan (the "Plan"). As discussed above, the Plan is a critical element of National Grid's total compensation package. The purpose of the Plan is to ensure that employees are working toward common goals that incorporate the perspective of customers and regulators and that high performance is recognized. The Plan is intended to motivate employees to achieve the highest possible level of individual performance while ensuring that, at all times, all safety, health, and environmental requirements are adhered to and standards of customer service are achieved. The Plan is designed to ensure that variable pay is based on both the overall performance of National Grid and the performance of the individual. It is imperative that individual performance goals reflect objectives that provide employees with a greater line of sight as to how they support the overall corporate objectives and create value for customers.

Q. Do all management employees participate in the Plan?

A. No. There are a few management employees who are covered by a sales commission plan. These commissioned employees do not participate in the Plan and are not eligible for any variable pay awards under the Plan. Seasonal and temporary employees are also ineligible for participation in the Plan due to their temporary employment status.

Q. Please elaborate on the goals established under the Plan.

A. The Plan's goals include clear linkage to objectives that are consistent with the interests of customers and Commission policies. The corporate goals in the Plan are linked directly to the U.S. business strategy and are built around three key areas of focus – customers, communities, and people – that will foster the enhancement of National Grid's connection with its customers and stakeholders, delivering energy with greater safety and reliability, and investing in infrastructure and networks across the U.S. footprint. Within these areas of focus, the key performance metrics are concentrated on safety, reliability, customer ease, customer trust, and controllable costs and are critical to how National Grid runs its business.

The Plan has been structured to establish the following percentages for payouts: (i) Bands C through F – 50 percent based on customers,

communities and people objectives and 50 percent based on individual objectives, and (ii) Band B – 40 percent based on individual objectives and 60 percent based on overall financial objectives. These payout percentages have been in effect since April 1, 2017, the start of the 2017/2018 performance period.

The payout percentages for Bands C, D, E, and F comprise the majority of National Grid's management employees with the greatest amount of contact with customers. With 50 percent of the payout for these employees tied to the achievement of safety, reliability, and customer metrics, the variable pay plan is specifically designed to focus this group on objectives that benefit customers.

Q. Please describe the metrics that National Grid will evaluate to determine whether it has met its customer, communities, and people objectives.

A. For the customer objectives, National Grid has developed performance metrics that directly measure how customers perceive the Company, experience service, and trust that the Company will provide them with the advice that they need to make good energy decisions. In addition, performance metrics measure and evaluate controllable costs and

efficiencies through complex capital construction in a manner that is intended to focus the workforce on providing better, more efficient service to customers. To evaluate performance for the communities' objectives, metrics will focus on reliability. National Grid measures its performance against a number of reliability metrics applicable in the jurisdictions in which it operates, including those adopted by the Commission. Finally, regarding the people objectives, the key performance metrics will focus on safety. Similar to the reliability metrics, National Grid will measure its performance against a number of safety metrics applicable to the jurisdictions in which it operates, including those adopted by the Commission.

Q. How are individual objectives established under the Plan?

A. Individual objectives are established for management employees by individuals in consultation with their managers and supervisors. These objectives are individually designed to ensure that employees are enhancing their expertise and performing their assigned tasks at the highest possible levels. In addition, individual objectives are designed to focus employees on common goals. For example, the individual objectives for a manager charged with completing regulatory reports might include: (i) completing all required reports in a timely fashion; (ii)

taking continuing education classes to ensure that the manager is aware of recent changes in reporting requirements and regulatory policies that may affect these reports; and (iii) leading and developing his or her team. Objectives are established for individual employees shortly after the start of the Plan year.

Q. Is employee performance under the variable pay plan considered in evaluating candidates for promotion?

A. Yes. Employees' success in achieving the goals established each year under the variable pay plan is a significant factor that is considered in evaluating employees for promotional opportunities. The individual goals established for employees are designed to encourage their professional growth and to put them in a position to qualify for promotions when opportunities arise.

Q. Please explain why National Grid includes a variable pay component as part of total cash compensation for its employees.

A. Variable cash compensation provides direct and specific incentives to employees to achieve or exceed certain operating performance goals of importance to National Grid, its customers, and its regulators, including the customer service, safety, and reliability metrics the Commission has

approved for KEDNY and KEDLI. Accordingly, the variable pay component of National Grid's overall employee compensation package aligns the interests of National Grid and KEDNY and KEDLI with the interests of its customers and regulators and assists KEDNY and KEDLI in meeting its public policy objectives. Moreover, as noted above, under National Grid's total compensation structure, base pay and benefits alone are not an adequate level of compensation to allow National Grid to meet its goal of attracting and retaining highly qualified employees to provide safe, reliable, and efficient service to customers. Today's marketplace dictates that variable pay is a fundamental component of any private sector entity's efforts to attract qualified employees, as demonstrated by the fact that 95 percent of the positions in the peer group have variable pay included in their compensation packages. Top talent in the market demands compensation that is directly linked to performance. For National Grid to compete in this marketplace and be viewed as an employer of choice, variable pay must be a part of the overall compensation package. If National Grid did not have a variable pay component, it would be out of step with the market and likely would not be able to attract the highly motivated employees necessary for KEDNY and KEDLI to deliver outstanding customer service in a reliable, efficient, and safe manner.

Q Is it correct that the variable pay plan at National Grid is pay-at-risk?

A. Yes, that is correct. National Grid's variable pay plan is designed to be part of a total compensation program; it is a pay-at-risk plan, not extra or bonus pay. This conclusion is borne out by the analysis of total compensation prepared by Willis Towers Watson. The variable pay component is labeled "pay-at-risk" because if performance measures are not achieved, employees will not receive their variable pay. This "at-risk" compensation allows National Grid to align pay with performance that is consistent with the goals of customers and regulators and to allocate compensation to those employees who are most deserving. Thus, variable compensation is not "bonus" or "optional" compensation that National Grid provides to employees, but a required element in the total compensation program and a necessary and prudent cost of doing business.

Q. Do the goals of National Grid's variable pay plan promote employee conduct that is contrary to Commission policy or the interests of National Grid's customers?

A. No. The goals of the Plan have been designed to ensure that they promote safety, reliability, and the delivery of a high level of service to our customers. This includes the individual objectives, which are typically

directly tied to established service quality measures such as safety, reliability, and customer satisfaction and further support the corporate objectives, which are all focused on our customers. The goals of the Plan are fully consistent with Commission policy and the interests of National Grid's customers.

Q. How did KEDNY and KEDLI determine the level of variable pay expense included in their proposed revenue requirements?

A. Similar to the approach followed by National Grid in the 2016 KEDNY and KEDLI Rate Cases and the 2017 NMPC Rate Case, the level of variable pay expense included in the proposed revenue requirements does not reflect any variable pay component for National Grid's employees in Band A or the variable pay component for National Grid's employees in Band B that is tied to the achievement of overall financial objectives. In addition, the Companies propose to continue the current reconciliation whereby any variable compensation amounts reflected in rates that are not paid to employees for any reason are deferred with appropriate carrying costs and returned to customers.

KEDNY forecasts the following levels of variable compensation expense: $9.686 million for the Rate Year; $10.088 million for Data Year 1;

$10.443 million for Data Year 2; and $10.749 million for Data Year 3. KEDLI forecasts the following levels of variable compensation expense: $5.466 million for the Rate Year; $5.657 million for Data Year 1; $5.855 million for Data Year 2; and $6.025 million for Data Year 3.

The adjustments to the amounts included in the Data Years reflect the application of a labor inflation factor, as discussed in the testimony of the Revenue Requirements Panels. In determining the variable pay included in the Rate Year for both KEDNY and KEDLI, the Companies have assumed that they will attain the target levels of performance and payout.

Q. Do the variable pay amounts the Companies are proposing to include in the revenue requirements include any payments to union employees?

A. Yes, they do. I discuss variable pay for union employees later in my testimony.

Q. Do they include any payments to employees who are covered by the sales commission plan?

A. No, they do not.

Q. What do you mean by "target levels?"

A. "Target levels" represent a level of performance that is judged to be 50 percent of stated maximums. Target performance is best described as an above average level of performance that is consistent with overall expectations. Achieving target performance requires employees to perform at a high, but nonetheless expected level. High performers would exceed "target levels" while average or below average performers would fall below "target levels."

Q. Please discuss the Commission policy concerning variable pay.

A. KEDNY and KEDLI's current rates, as adopted by the Commission in the 2016 KEDNY and KEDLI Rate Cases, reflect an amount of variable compensation expense and, most recently, the rates that were adopted by the Commission and took effect on April 1, 2018 in the 2017 NMPC Rate Case also reflect an amount of variable compensation expense.

In addition, in a decision involving Orange and Rockland Utilities, Inc. in Case 10-E-0362 ("O&R"), the Commission indicated that there were two ways a utility could substantiate that variable pay expense should be included in the revenue requirement. First, a utility could attempt to justify the inclusion of variable pay expense by demonstrating that: (i)

total compensation, including variable compensation for a class or classes of employees, is reasonable by reference to a study that compares total compensation to that provided by similarly situated companies; and (ii) that the variable compensation plan supports the provision of safe and adequate service and does not promote employee behavior that would be contrary to customer interests or Commission policies. Second, to the extent a utility seeks funding in rates for a variable compensation program that would award payments above what would be considered reasonable total compensation for a particular class of employees, then the utility must demonstrate that such a program will return quantifiable or demonstrated benefits to customers in a financial sense or in terms of reliability, environmental impact or customer service that justify the variable pay expense.

Q. Do the Companies believe that their rate filings in these cases justify the inclusion of variable pay expense under either of the tests adopted by the Commission?

A. Yes. KEDNY and KEDLI believe that this testimony and the associated exhibits support findings by the Commission that: (i) the Companies' total compensation expense, including variable pay expense, is reasonable compared to the similarly situated companies included in the

compensation study; (ii) the Companies' variable compensation plan does not promote employee behavior that would be contrary to customer interests or Commission policies; and (iii) the Company's variable pay program is designed to encourage employees to attain objectives that support the provision of safe and adequate service and are consistent with the interests of customers and the Commission's policy concerning variable pay. The analyses of total compensation presented in this testimony are consistent with the types of analyses that the Commission determined would be necessary to justify inclusion of variable pay costs in rates.

Q. Did National Grid increase management wages in the Historic Test Year?

A. Yes. During the Historic Test Year, National Grid increased management wages by 3.25 percent (3.00 percent took effect on July 1, 2018 and 0.25 percent took effect on October 1, 2018).

Q. Is National Grid projecting any management wage increases in the period covered by the Companies' rate filings in these proceedings?

A. Yes. National Grid is projecting the following increases:

Effective Date	Percentage
July 1, 2019	3.35 percent
July 1, 2020	3.10 percent
July 1, 2021	3.10 percent
July 1, 2022	3.10 percent
July 1, 2023	3.10 percent

The management wage increase that will take effect on July 1, 2019 was approved by senior leadership in February 2019. The forecast of proposed increases in management wages is based on market studies that are currently available to National Grid, including the information set forth in Schedule 1 of Exhibit __ (MPH-3). These increases are also in line with the average wage increases provided over the last ten years, as set forth in Schedule 2 of Exhibit __ (MPH-3). Nonetheless, National Grid will continue to monitor market information and may revise its projections if market conditions require such adjustments.

Q. Please explain what is included in the 3.35 percent management salary increase approved for July 1, 2019.

A. The 3.35 percent includes a 3.1 percent average increase in base pay provided to National Grid's entire management workforce. The remaining 0.25 percent represents increases associated with career path progressions.

Q. Does National Grid follow the same compensation philosophy for its union employees?

A. Yes. However, the compensation provided to union employees is the result of collective bargaining. Pursuant to their respective union contracts, members of TWU Local 101 (KEDNY) will receive a base wage increase of 1.5 percent on April 16, 2019; members of IBEW Local 3 (KEDNY) will receive a base wage increase of 1.5 percent on April 16, 2019, October 16, 2019 and April 16, 2020; and members of IBEW Local 1049 (KEDLI) received a base wage increase of 3.0 percent on February 14, 2019 and will receive 3.0 percent on February 14, 2020. These base wage increases are also set forth in Schedule 1 of Exhibit __(MPH-4). Because all of these collective bargaining agreements are scheduled to expire during the course of these rate proceedings (the TWU Local 101 contract expires on October 15, 2019, the IBEW Local 3 contract expires on October 15, 2020 and the IBEW Local 1049 contract expires on

February 13, 2021), the Companies have included projected base pay increases in the revenue requirements, assuming new contracts are negotiated upon the expiration of the existing contracts. These projections are not intended to guarantee that increases will be included in the contract negotiations.

Q. Does compensation for union employees also include a variable pay component?

A. Yes. Variable pay gives union employees a stake in National Grid's performance and provides direct incentives for employees to strive to meet or exceed metrics tied to safe, reliable, and efficient performance, which, in turn, results in better service for customers. As with the management program, the variable compensation component is part of the total compensation package designed to link rewards and results. Because this compensation philosophy is an important part of National Grid's effort to deliver value to customers, including variable pay as a component of compensation for union employees has been one of National Grid's priorities in labor negotiations. Under the current collective bargaining agreements, union workers participate in the same variable pay plan as management employees. For KEDNY, members of TWU Local 101 and IBEW Local 3 have a target payment of 2.5 percent of base pay. For

KEDLI, members of IBEW Local 1049 have a target payment of 1.75 percent of base pay. The payout for union employees is tied to the same corporate performance metrics that are applicable to management employees. As previously discussed, these metrics are focused on enhancing safety, reliability and the customer experience.

Q. Has National Grid attempted to evaluate whether the cash compensation and benefits provided to union employees are reasonable?

A. Yes. National Grid recently reviewed analyses of the compensation and benefits provided to its union workforce. Willis Towers Watson assisted National Grid in preparing the benefits portion of this study. Willis Towers Watson also provided data concerning the cash compensation paid to certain positions that are held by National Grid's union workers. The results of the analyses are set forth on Exhibit __ (MPH-4) and they demonstrate that National Grid's union total cash compensation is within market competitive levels.

Q. Please explain how Willis Towers Watson analyzed the competitiveness of KEDNY and KEDLI's union benefits package.

A. Willis Towers Watson compared the total benefits package provided to KEDNY and KEDLI union workers to the total benefits package provided by 15 peer companies with unionized workforces. The analyses were performed in a manner similar to the analysis of management benefits and were accomplished by analyzing the relative value of benefits provided by National Grid to its union workforce to the relative value of the benefits provided by the peer group to their equivalent workforces. In total, these analyses showed that the employer value of the benefits provided by National Grid to KEDNY's unionized workforce was slightly below the median level of the peer group and the value of the benefits provided by National Grid to KEDLI's unionized workforce was above the median level of the peer group. The results of the analyses are set forth in Schedule 4 and Schedule 5 on Exhibit __ (MPH-4).

Q. Did Willis Towers Watson attempt to analyze the cash compensation provided to union workers in the union peer group?

A. Willis Towers Watson does not have information available that specifically addresses cash compensation paid to unionized workforces of the peer group used to compare union benefits. Moreover, the Bureau of

Labor Statistics ("BLS") does not provide such information. However, the BLS does provide general information on differences between union and management wages, and Willis Towers Watson did have information for certain positions that are filled by union workers at KEDNY and KEDLI. That data is included in Schedules 2 and 3 of Exhibit __ (MPH-4).

Q. Please describe the compensation data that was provided by Willis Towers Watson and used by National Grid to evaluate its union compensation.

A. The data was obtained from a survey of 138 utility and energy companies conducted by Willis Towers Watson. The participants ranged in size from companies having revenues of less than $1 billion annually (41 participants) to participants having revenues of more than $6 billion annually (34 participants). The data was provided for non-union positions that could be compared to some of KEDNY and KEDLI's most common union positions. This cash compensation data was adjusted upward by 9 percent to account for the wage premiums that are typically reflected in the collective bargaining agreements of unionized workforces based on information provided by the BLS concerning the difference between union and management compensation in the utility industry. KEDNY and KEDLI's hourly wages for equivalent positions were generally higher than

the hourly wages identified by Willis Towers Watson. This comparison is set forth on Schedules 2 and 3 of Exhibit __ (MPH-4).

Q. Does the analysis of cash compensation provided to union workers consider the impact of variable pay?

A. Yes. The level of variable pay provided to the Companies' union workforce at target increases the workers' cash compensation by 2.5 percent for members of TWU Local 101 and IBEW Local 3 and 1.75 percent for members of IBEW Local 1049. Thus, on Schedules 2 and 3 of Exhibit __ (MPH-4), KEDNY and KEDLI's union wages are adjusted upward by these percentages to compare them to the cash compensation, including variable pay, provided to the comparison group. This comparison shows that the total cash compensation, including variable pay, provided to the Companies' union workforce is within market competitive levels when compared to the comparison group.

Q. What are the performance objectives of the union variable pay plan?

A. As previously mentioned, the performance objectives are tied to corporate goals with performance metrics for safety, reliability, and the customer experience. These goals are consistent with the Commission's policies and customer interests. The union variable pay plan is designed to

encourage union workers to assist the Companies in providing safe and reliable utility service to customers.

V. Cost Control Measures

Q. What efforts has National Grid undertaken to control costs associated with its health and welfare programs?

A. National Grid has taken many steps to control the costs associated with health and welfare benefit programs. These steps include:

(i) The majority of National Grid's health and welfare benefit plans are self-insured, which affords National Grid a greater ability to control costs than it would have under third-party insurance programs. Because these plans are self-insured, the program costs are directly linked to the utilization of benefits. National Grid's medical plans also provide extensive health and wellness programs designed to reduce health risks and thereby help mitigate costs. These programs, combined with coverage for preventive check-ups and screenings, provide a benefit structure that, over time, should help to mitigate costs and improve employee wellness.

(ii) National Grid aggressively negotiates with vendors and periodically conducts a competitive bidding process to achieve the lowest administrative service fees and premiums and to obtain maximum discounts when rolling out a new program or upon the expiration of an

existing contract. For example, in late 2017, National Grid negotiated the renewal of the contract with its national Pharmacy Benefits Manager ("PBM"), CVS Caremark, which resulted in projected savings over the term of the contract. These savings are embedded in the Historic Test Year level of expense.

(iii) The prescription drug program is carved out of each medical plan and is administered in its entirety with a CVS Caremark plan. Because CVS Caremark is National Grid's PBM for its entire U.S. business, National Grid has been able to gain economies of scale and incurs lower expenses by leveraging a volume discount.

(iv) Generic step therapy, specialty preferred drug therapy, an exclusive specialty pharmacy program and mandatory mail order for maintenance drugs were introduced into the prescription drug programs to lower prescription drug costs.

(v) Prior authorization for compounded drugs costing more than $300 is required in the prescription drug programs. In addition, bulk powders were excluded from coverage for compound drugs. Compounded drugs may contain three to five drugs along with expensive bases and solvents in the preparation that dramatically increase the cost.

(vi) A standard formulary was implemented in addition to an advanced control specialty formulary to help ensure clinically appropriate utilization and cost effectiveness of specialty therapies.

(vii) Consumerism has been introduced into the medical plans to drive greater consumer attention to costs and benefits, a culture of health, and employee education by improving cost control, quality, patient compliance, and healthy outcomes through consumer directed health plans, advanced health strategies, condition-specific performance comparisons of providers and treatment options and wellness programs to support a healthy workforce.

(viii) Employees who participate in a consumer driven health plan are provided with a tax-advantaged health savings account ("HSA") with some funding from National Grid that allows them to save for the payment of qualified medical expenses.

(ix) A wellbeing program was introduced in collaboration with Redbrick Health. The program allows employees and their families to connect with resources and tools that will help them achieve their personal wellness goals (*i.e.*, improving physical fitness, managing a chronic condition, or making better nutritional choices).

(x) Telehealth Services were implemented to allow immediate, around-the-clock access to physicians, specialists, and other health care providers

through remote monitoring. Telehealth services can be less expensive, may provide more convenient care options, and ultimately improve care outcomes.

(xi) National Grid participates in both industry groups and benefits councils to learn best practices and stay abreast of market developments and seeks new ideas and suggestions from its vendors.

Q. Please describe any other changes that have been made to the management health and welfare plans since the Companies' last rate filing through the end of the Historic Test Year.

A. To maintain market competitiveness while controlling costs, National Grid made a number of changes in healthcare benefit plan designs that lowered the cost of healthcare benefits while maintaining market competitiveness. These changes included:

<u>Select Provider Plan (formerly the Exclusive Provider Organization Plan)</u>

- An increase in office visit co-pays for a primary care physician/specialist to $25/$50;
- Implementation of 90 percent coinsurance (which eliminated first dollar coverage) as well as a $400/$800 annual deductible for individual/family coverage;

- Implementation of 90 percent coinsurance for diagnostic lab, x-ray and advanced radiology;
- Moved in-patient hospital services and emergency room visits from co-pays to 90 percent coinsurance;
- Increases in co-pays for prescription drug benefits to $10/$40/$80 for retail and $20/$80/$160 for mail order.

Preferred Provider Organization Plans ("PPO") – the national PPO plan and all regional PPO plans were eliminated as of January 1, 2018.

Consumer Driven Health Plans ("CDHP") – two consumer-driven health plans were introduced coupled with an HSA. The plans offer comprehensive coverage with preventive care covered at 100 percent. The plans have higher deductibles and out-of-pocket expenses, but lower employee contributions. National Grid makes an annual contribution into the HSA in the amount of $375 for individual coverage and $750 for individual with dependents coverage:

- The Health Savings Plan has an annual deductible for in-network services for individual/family coverage of $1,800/$3,600 with 90 percent coinsurance; and
- The Health Investment Plan has an annual deductible for in-network services for individual/family coverage of $3,000/$6,000 with 80 percent coinsurance.

In addition, the following changes were implemented and apply to all medical plan options:

- An additional coverage level option for medical and dental plans. The new four tier structure includes employee only, employee plus spouse, employee plus child(ren) and family;
- A spousal buy-up of $100 per month was introduced for those employees who are covering a spouse under the National Grid medical plan that have access to employer sponsored subsidized group coverage through their company;
- A wellbeing program was implemented that will reduce employee medical plan contributions and allow employees to get an additional HSA contribution from National Grid of $375 for individual coverage and $750 for coverage with dependents if they complete certain wellness incentive activities;
- An additional dental plan option was introduced that has no coverage for orthodontia, a lower annual plan maximum, and lower employee contributions;
- Implementation of several prescription drug utilization management programs previously discussed in my testimony;

- Exclusion of bulk powders on compound drugs and the introduction of an advanced control specialty formulary to help ensure clinically appropriate utilization of specialty therapies; and
- Implementation of a "dispense as written" penalty where employees pay more for a brand name prescription drug when a generic alternative is available.

Any savings associated with these initiatives are embedded in the Historic Test Year and reflected in the Companies' revenue requirements.

Q. Are there any healthcare changes to the management plans that took effect on January 1, 2019?

A. There were no design changes to the healthcare plans except for an enhancement to the Consumer Driven health plans with the introduction of reduced coinsurance for in-patient services rendered at Blue Distinction Centers; however, National Grid introduced two new programs to assist employees and their family members in improving their health. One program will focus on diabetes care and the other will provide enhanced care management by providing access to on demand concierge support to those participating in a Blue Cross Blue Shield medical plan.

Q. Were there other changes made to the benefit plans offered to management employees other than healthcare?

A. Yes. In 2018, National Grid implemented two new innovative benefit programs that support a multi-generational workforce and are designed to provide employees what they need to stay healthy, support their work/life needs and maintain engagement and productivity to meet the needs of the business. A description of the two programs follows:

Student Loan Repayment Program – This program was introduced in April 2018 to support the professional development and financial wellbeing of employees and assist those who have obtained, or are in the process of obtaining, a degree and have outstanding student loans or have taken out parent loans to help fund their child's education. Under this program, National Grid will make monthly payments toward the principal portion of student loans the employees have taken to pay for their own education or their child's education to an accredited institution. The repayment starts at $50 per month with an escalation factor of $25 per year, up to $150 per month, the longer an employee remains in the program. The maximum lifetime benefit under this program of $6,000 per employee.

Caregiver Program – This program was introduced in July 2018 to help employees keep their work and life in balance and provides them with

access to high-quality child and elder care, along with programs and resources (*i.e.,* educational advice for finances and college admissions, special needs parenting and support in finding caregivers for child care, elder care, and household help) to support employees and their families. The program is employee paid with some services being partially subsidized by National Grid.

Q. Do the Companies' revenue requirements include any incremental costs associated with these two programs?

A. Yes. As discussed in the testimony of the Revenue Requirements Panels, an adjustment was made to Historic Test Year costs associated with the Student Loan Repayment Program to reflect a full twelve months of costs, and to reflect an escalation in the monthly reimbursement amount provided under the program. As a result, the revenue requirement reflects Rate Year expense of $0.118 million for KEDNY and $0.070 million for KEDLI, inflated through Data Year 3 at the rate of general inflation. Similarly, an adjustment was made to Historic Test Year costs associated with the Caregiver Program to reflect a full twelve months of expense. As a result, the revenue requirement reflects Rate Year expense of $0.051 million for KEDNY and $0.025 million for KEDLI, inflated through Data Year 3 at the rate of general inflation.

Q. Please describe any other changes that have been made to the union health and welfare plans since the Companies' last rate filing through the end of the Historic Test Year.

A. National Grid negotiated some medical benefit plan design changes with each of the unions that were discussed in the Companies' last rate filing. The changes that took effect after the filing in 2016 through the end of the Historic Test Year are as follows:

TWU Local 101 and IBEW Local 3

- Annual deductible for in-network services for individual/family coverage increased to $250/$500;
- The level of coinsurance decreased from 100% to 95%;
- Increases in office visit co-pays for a primary care physician/specialist to $30/$40 and $40/$60;
- Increases in co-pays for prescription drug benefits to $10/$35/$60; and
- Emergency room visits, and in-patient hospital stays subject to 95% coinsurance; co-pays and coverage at 100% was eliminated.

IBEW Local 1049

- Annual deductible for in-network services for individual/family coverage increased to $200/$400;
- The level of coinsurance decreased from 100% to 95%;

- Increase in office visit co-pays for a primary care physician/specialist to $30/$30;
- Increases in co-pays for prescription drug benefits to $10/$25/$50; and
- Emergency room visits, and in-patient hospital stays subject to 95% coinsurance; co-pays and coverage at 100% was eliminated.

Q. Are there any healthcare plan design changes to the union plans that took effect on January 1, 2019?

A. No. There were no healthcare plan design changes to the union plans that took effect on January 1, 2019.

Q. Have any new collective bargaining agreements been negotiated that will take effect after the end of the Historic Test Year?

A. Yes. In November 2018, National Grid and IBEW Local 1049 negotiated a new two-year agreement which commences after the expiration of the current contract term on February 14, 2019. The contract, which expires on February 13, 2021, includes a few changes to medical plan benefits in the Preferred Provider Organization plan that will be implemented over the course of the contract term: (a) an increase in the in-network deductible from $200/$400 for individual/family coverage to $250/$500

on January 1, 2020 and $275/$550 on January 1, 2021, (b) an increase in the copay for a specialist office visit from $30 to $35 on January 1, 2020, (c) an increase in the out of pocket maximum from $1,750/$3,500 for individual/family coverage to $1,800/$3,600 on January 1, 2020 and $1,850/$3,700 on January 1, 2021, and (d) introduction of a standard formulary and a dispense as written requirement for prescription drugs. The projected savings associated with these changes total $2.567 million in the Rate Year - KEDLI's allocated O&M portion of the savings is $0.608 million. These savings have been reflected in the Companies' revenue requirements.

Q. Does existing law impose any mandates on employers that may have an impact on the Companies' costs in the near future?

A. Yes. The Patient Protection and Affordable Care Act will impose an excise tax on plans that are termed "Cadillac" medical plans. The tax is scheduled to go into effect in 2022.

Q. How will the excise tax be determined?

A. The excise tax is a 40 percent tax on high cost "Cadillac" medical plans that will be imposed on medical plans that exceed predetermined cost thresholds. The thresholds will be based on the $10,200

individual/$27,500 family limits that were originally set for the 2018 implementation and indexed to 2022 and thereafter. Notably, the excise tax thresholds are based on U.S. national averages so that companies in the Northeast region are more likely to incur the tax because medical costs are historically higher.

Q. Does National Grid expect to trigger the excise tax in 2022?

A. National Grid expects that some of its plans would be above the thresholds and would trigger the excise tax in 2022.

Q. What steps are being taken by National Grid to reduce the probability of triggering the excise tax?

A. National Grid's benefit strategies are focused on attempting to avoid the excise tax in 2022 (and beyond) and controlling costs while maintaining market competitiveness for plans provided to management employees. This is also a critical component and key focus during all union negotiations. National Grid's comprehensive strategy will drive greater consumer attention to costs and benefits, a culture of health and employee education by improving cost control, quality, patient compliance and healthy outcomes through consumer directed health plans, advanced health strategies, condition-specific performance comparisons of providers

and treatment options and wellness programs to support a healthy workforce. As discussed above, National Grid has already implemented some of the key elements of the strategy and is attempting to manage plan costs in an effort to come below the tax thresholds for both management and union employees, both active and retired. However, at this time, it is not possible to say with certainty whether and to what extent National Grid will be successful in avoiding the excise tax. National Grid's efforts to avoid the tax must be balanced against its need to continue to offer market-competitive medical benefits to its management and union workforces.

Q. Are the Companies seeking recovery of the excise tax and have any projected levels of excise tax been reflected in the revenue requirements for the Data Years?

A. Consistent with the deferral treatment approved by the Commission in the 2017 NMPC Rate Case, the Companies are requesting authority to defer for future recovery the costs associated with the excise tax that may be assessed in 2022 and beyond. The Companies are proposing deferral treatment for the actual level of excise tax to be incurred because of the uncertainty surrounding the law as well as the many other variables that may affect the Companies' efforts to avoid the excise tax. As such, there

is no projected level of excise tax included in the revenue requirements for the Data Years.

VI. Employee Benefits

Q. Please describe any adjustments to the Historic Test Year expense for employee medical and dental benefit plans that the Companies are including in the proposed revenue requirements.

A. National Grid is projecting annual increases of approximately 7.5 percent in the costs associated with medical benefits and annual increases of approximately 4 percent in the costs associated with dental benefits over the three years ending March 31, 2022. These projected increases are generally consistent with national projections for health care trends and the projections gathered by National Grid's benefit consultant. Despite the fact that health care costs have increased significantly above the general rate of inflation for many years, the Companies applied a general inflation rate to the Historic Test Year level of expense to determine the level of medical and dental claims expenses included in the Rate Year revenue requirements consistent with Commission precedent. Although the Companies chose to follow Commission precedent, the use of a general inflation factor to project health insurance increases likely will deprive KEDNY and KEDLI of the ability to fully recover anticipated

health care costs in the Rate Year unless savings in other areas beyond the productivity adjustments reflected in the Companies' proposed rates are achieved to offset the expected increase in health care costs.

Q. Please describe any adjustments to the Historic Test Year expense for other employee benefit plans that the Companies are including in their proposed revenue requirements.

A. The forecast Rate Year revenue requirements for other employee benefit plan expenses (life insurance, and other active benefits) were also developed by applying the general inflation rate to the Historic Test Year level of expense, consistent with Commission precedent.

Q. Please describe National Grid's efforts to manage OPEB benefits.

A. The process of ensuring fair, competitive, and efficient benefit programs for employees and retirees is one of the core activities of National Grid's U.S. Compensation, Benefits, and Pensions department. Compensation, Benefits, and Pensions staff continually endeavor to ensure proper vendor performance and compliance with federal and state laws and regulations. Trends in compensation, benefit, and pension plan design are continually monitored and compared to National Grid's plans to ensure that they continue to be fair, reasonable, and competitive.

On an annual basis, National Grid analyzes the funding, recording, and administration of OPEBs. Less frequently, but on a consistent basis, National Grid considers more significant changes designed to reduce the overall cost of the OPEB plans.

Q. What challenges does National Grid face in considering fundamental changes to retirement health and life insurance benefits on a frequent basis?

A. Retiree programs are unique among the collection of employee benefits. Unlike health care programs for which employees derive current value, retiree programs represent benefits to be provided in the future. As such, their value tends to accrue over a longer period of time. Because of these long accrual periods, significant changes in retiree programs are difficult to adjust to when they occur close to or during retirement. Preparation for retirement involves assessing one's financial preparedness provided through benefit plans, Social Security, and individual savings. Given the fixed benefits payable under National Grid's pension plan, and the potential for significant health care expense as participants' age, retiree health insurance benefits are one of the key factors in reaching the conclusion that one can retire. Therefore, and as further outlined below, National Grid has reached the conclusion that changes to OPEB benefit

levels are only feasible in connection with major benefit alignments either following corporate transactions, in connection with labor negotiations, or when major legislative changes are made. Any OPEB benefit change will usually be done in connection with other benefit alignments and will generally be carefully constructed to affect only those employees who have sufficient time to adapt to the changes before reaching retirement age.

Another reason retiree programs are unique is the wide geographic dispersion of the participants. Unlike active employees, who live geographically proximate to the Companies' business operations, retirees move to other parts of the United States and other countries. This makes communication with them and implementation of changes to their benefits administratively burdensome and costly.

Q. Are there particular challenges involved in modifying OPEB benefits for union employees?

A. With respect to OPEBs for union employees, National Grid is bound by law to negotiate with TWU Local 101, IBEW Local 3 and IBEW Local 1049 for these benefits, and having reached agreement on them, the Companies are bound to maintain those benefit levels until the current

collective bargaining agreement expires, unless they are renegotiated earlier by mutual consent.

Q. Is there a relationship between National Grid's ability to make changes to the OPEB benefits provided to management employees and the benefits provided to union employees?

A. Management employee benefits can be changed at any time. However, the existence and substantial size of the union workforce and the need to recruit future managers from the union ranks creates a significant connection. It would be shortsighted to design or change benefits without fully evaluating the impact on this connection. In the case of OPEB benefits in particular, a union employee would be unlikely to accept, let alone seek, a management job that provides either significantly fewer or less valuable benefits than he or she is eligible for under the union contract. Many of the best union to management recruits are rank-and-file employees who have worked many years in a physical field work position and seek to move into supervision. These employees are highly valued by the management team, as they can ably and immediately contribute to meeting National Grid's objective of providing safe and reliable service at a reasonable cost. Requiring union employees to accept benefit reductions to move to management would be a significant impediment to any such

moves. National Grid's desire to facilitate the promotion of union employees manifests itself in a three to five-year cycle of OPEB benefit level reviews for both union and management employees, absent a major triggering event such as a merger. This cycle generally corresponds to the terms of the collective bargaining agreements.

Q. Please explain the OPEB analyses performed on an annual basis.

A. The annual process to examine OPEBs focuses on the following three categories: (i) actuarial assumptions around plan expense and funding; (ii) general vendor performance reviews; and (iii) Medicare prescription drug program refund administration.

Actuarial assumptions for expense and funding: Each year, working with its actuary, Aon, National Grid evaluates the proper level of several actuarial factors used to determine plan expense and funding, including discount rate, medical trend rate, and mortality rate. Depending on economic information and trends at the time, rates are either continued or modified as appropriate. In addition, National Grid evaluates various options to ensure optimal funding of these benefits.

General vendor performance reviews: All active and retiree benefit plan administration is consolidated with a single vendor. In addition, National Grid's benefits staff periodically meets with the third-party vendors who

are responsible for claims payments to review their performance. National Grid's arrangements with these vendors are for administrative services only, so the focus is on transaction performance and participant satisfaction. National Grid continuously reviews the delivery and administration of the benefits under the retiree medical plan with the ultimate objective of reducing administration costs and improving the delivery of services.

Medicare prescription drug refund administration: National Grid obtains sizeable subsidies from the government in return for providing retiree drug benefits to retirees who are 65 years of age or older ("Post 65 retirees"). In 2013, National Grid implemented an Employer Group Waiver Plan for Post 65 retiree prescription drug benefits. This plan was more financially advantageous to National Grid primarily because of the large pharmaceutical manufacturer discounts on brand drugs as well as federal catastrophic reinsurance payments.

Q. Is the OPEB plan the same for all employees?

A. No. National Grid has two OPEB plans – the OPEB plan applicable to union employees and the OPEB plan applicable to management employees. In terms of the overall cost of these plans, the cost of the union plan is greater because the union workforce is larger.

In terms of the OPEB benefits offered to each workforce, the broad categories of benefits are the same. Both union and management employees receive retiree medical benefits and may receive life insurance benefits depending on their hire date.

Q. What has National Grid done to control the cost of management retirement benefits since the Companies' last rate filing?

A. Because the plan offerings for pre-65 retirees are the same as those offered to the active management employees, the healthcare changes that were made to active management plan benefits that were discussed previously in my testimony are incorporated into the pre-65 retiree plans.

Q. Were there any changes to the union post-employment benefit costs since the Companies' last rate filing?

A. No. However, the changes that will be made to active employee medical plan benefits under the new contract for IBEW Local 1049 will be incorporated into the pre-65 retiree plans.

Q. Is National Grid considering any other changes that would impact post-employment benefit costs?

A. Yes. National Grid is considering moving the post 65 retiree management and union population to a private exchange. The Company is looking at the existing medical plan offerings and structures and evaluating the benefits and feasibility of providing medical coverage for these groups through a private exchange.

Q. What has National Grid done to control pension costs since the Companies' last rate filing?

A. Consistent with the practice for OPEBs, National Grid continually reviews the pension plan benefits it offers to employees to control plan costs and ensure that the benefit offerings are in line with market practices. National Grid evaluates the proper level of several actuarial assumptions used to determine pension plan expense and funding.

Q. Has National Grid made other changes to its pension plans that affect the cost of those plans since the Companies' last rate filing?

No. There have been no changes to the pension plan benefits since the Companies' last rate filing. National Grid closed the defined benefit pension plans to new entrants for both management and union employees

in KEDNY and KEDLI. The management plans were closed as of December 31, 2010 and the union plans for KEDNY's Local 101 and Local 3 were closed as of October 15, 2010 and December 31, 2011, respectively. The defined benefit pension plan for KEDLI's Local 1049 was closed as of February 13, 2015. Employees in management and the respective unions hired after the dates noted above are provided with a defined contribution plan.

Q. What other steps has National Grid taken with respect to the management of its retirement plans since the Companies last rate filing?

A. National Grid continues to encourage greater participation and investment management in its 401(k) plans so that employees will take greater ownership and responsibility for funding their own retirement by saving more and investing wisely. Management employees continue to receive a company match equal to 50 percent of the first eight percent of compensation they contribute to the plan. Members of TWU Local 101 receive a company match equal to 25 percent of the first six percent of base pay they contribute to the plan and members of IBEW Local 3 and IBEW Local 1049 receive a company match equal to 50 percent of the first six percent of base pay they contribute to the plan.

Q. Does this conclude your direct testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Maureen P. Heaphy

Dated: July 3, 2019

Q. Please state your name and business address.

A. My name is Maureen P. Heaphy. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. Are you the same Maureen P. Heaphy who previously submitted direct testimony in these proceedings?

A. Yes, I am. The terms defined in my direct testimony have the same meanings here.

Q. What is the purpose of your corrections and updates testimony?

A. The purpose of my corrections and updates testimony is to: (i) adjust the employee headcount figures used in determining the revenue requirements to reflect developments since the case was filed; and (ii) update base pay wage increases and health care benefit costs included in the revenue requirements in these proceedings to reflect changes resulting from the recently negotiated collective bargaining agreement between National Grid and the Transport Workers Union of America, AFL-CIO, Local 101 ("TWU Local 101").

Q. Do you sponsor any exhibits to your corrections and updates testimony?

A. Yes. Exhibit ___ (MPH-1CU) sets forth an updated list of the positions that National Grid forecasts to be filled prior to the end of the Rate Year and Data Years. This exhibit was prepared under my direction and supervision. In addition, Appendix A includes the pertinent portion of KEDNY's response to Information Request No. DPS-388.

Q. Please describe the basis for the change in employee headcount since the case was filed.

A. In my direct testimony (at pages 7 and 8), I indicated that National Grid expected to fill 252.4 incremental positions before the end of the Rate Year and forecast the need for 503.4 incremental positions by the end of Data Year 3. Exhibit ___ (MPH-1CU) sets forth an updated list of the positions that National Grid forecasts to be filled. National Grid forecasts that 251.7 and 494.7 incremental positions will be filled before the end of the Rate Year and Data Years, respectively. These updates are explained in the corrections and updates testimony of the Gas Infrastructure and Operations Panels, the Future of Heat Panel, the Gas Safety Panel, and the Revenue Requirements Panels.

Q. Please describe the recently negotiated collective bargaining agreement with TWU Local 101.

A. TWU Local 101 represents employees providing gas distribution services in KEDNY's service territory pursuant to a collective bargaining agreement, which was set to expire on October 15, 2019. In anticipation of the agreement's expiration, National Grid and TWU Local 101 participated in negotiations to ensure the continuity of services as well as the continued provision of safe and reliable gas service to customers.

As discussed in the response to Information Request No. 388, on April 27, 2019, three days before the Companies filed their direct testimony in these proceedings, National Grid and TWU Local 101 reached a tentative three-year agreement to replace the existing contract, which was set to expire at midnight on October 15, 2019. The contract was ratified by the TWU Local 101 membership on May 31, 2019.

Q. Does the new contract include base pay increases and if so, please describe them?

A. Yes. Pursuant to the terms of the new contract, members of TWU Local 101 will receive a base pay wage increase of 2.75 percent on October 16, 2019, a 3.00 percent base pay wage increase on October 16, 2020, and a

3.00 percent base pay wage increase on October 16, 2021. The corrections and updates testimony of the Revenue Requirements Panels discuss the adjustment made to the revenue requirements to reflect the base pay increase.

Q. Please describe the changes in healthcare benefits pursuant to the provisions of the new contract with TWU Local 101.

A. The contract includes several changes to healthcare benefits including changes in employee contributions and other changes that result in lower healthcare costs for the Companies. These changes include:

- New Healthcare Administrator: effective January 1, 2020, the administration for the two Preferred Provider Organization ("PPO") plans will move from G.H.I. to Aetna, Inc. This change will significantly increase the utilization of in-network services provided, which will result in lower claims costs.
- Premier PPO Plan: (i) an increase in the in-network deductible from $250/$500 for individual/family coverage to $275/$550 on January 1, 2021 and $300/$600 on January 1, 2022; (ii) an increase in the co-pay for primary/specialist office visits from $30/$40 to $35/45 on January 1, 2022; (iii) an increase in the out of pocket maximum from $1,900/$3,800 for individual/family coverage to $2,000/$4,000 on

January 1, 2022 and $2,100/$4,200 on January 1, 2022; (iv) a decrease in co-insurance from 95% to 90% on January 1, 2022; (v) elimination of first dollar coverage for diagnostic x-rays and labs to 95% co-insurance on January 1, 2021 and 90% co-insurance on January 1, 2022; and (vi) an increase in brand and preferred brand prescription drug co-pays from $35/$60 to $40/$80 on January 1, 2022.

- Standard PPO Plan: (i) an increase in the in-network deductible from $400/$800 for individual/family coverage to $450/$900 on January 1, 2021 and $500/$1,000 on January 1, 2022; (ii) an increase in the out of pocket maximum from $2,400/$4,800 for individual/family coverage to $2,500/$5,000 on January 1, 2022; and (iii) an increase in brand and preferred brand prescription drug co-pays from $35/$60 to $40/$80 on January 1, 2022.
- Consumer Driven Health Plan: the Company will continue to make an annual contribution into an employee's health savings account, the total amount the employee will be eligible to receive is not changing ($750/$1,500 for individual/family coverage); however, as of January 1, 2020, the Company contribution will be $300/$600 and the employee will get an additional contribution from National Grid of $450/$900 if they complete certain wellness incentive activities.

- Prescription Drugs: implementation of several prescription drug utilization management programs as of January 1, 2020: (i) introduction of a standard formulary; (ii) exclusion of bulk powders on compound drugs; (iii) the introduction of an advanced control specialty formulary to help ensure clinically appropriate utilization of specialty therapies; and (iv) implementation of a "dispense as written" penalty where employees pay more for a brand name prescription drug when a generic alternative is available.
- Dental Plan: introduction of employee contributions for dental coverage as of January 1, 2020.
- Vision Plan: introduction of a fully insured, voluntary, company paid vision plan through EyeMed Vision Care.

Q. Does the change in healthcare benefits for TWU Local 101 impact the Companies' revenue requirements?

A. Yes. The effects of these changes are discussed in the corrections and updates testimony of the Revenue Requirements Panels.

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Maureen P. Heaphy

Case 19-G-0309
Case 19-G-0310

September 18, 2019

Rebuttal Testimony of Maureen P. Heaphy

I. Introduction

Q. Please state your name and business address.

A. My name is Maureen P. Heaphy. My business address is One Metrotech Center, Brooklyn, New York 11201.

Q. Are you the same Maureen P. Heaphy who previously submitted direct and corrections and updates testimony in these proceedings?

A. Yes, I am. The terms defined in my direct and corrections and updates testimony have the same definitions here.

II. Purpose of the Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my rebuttal testimony is to respond to certain recommendations of Department of Public Service Staff ("Staff") witness Daniel S. Gadomski and the Staff Revenue Requirements Panel ("SRRP"). Specifically, I will address Mr. Gadomski's recommendation that $0.270 million of operating expenses for KEDNY and $0.167 million of operating expenses for KEDLI should be disallowed based on his claim that National Grid's July 1, 2019 management wage increase of 3.35 percent and wage increases in subsequent years of 3.10 percent should be reduced to 3.0

percent. I will also briefly address the SRRP's recommendation that a vacancy rate be applied to the Companies' headcounts.

III. Management Wage Increases

Q. Why is Mr. Gadomski recommending that a portion of the Companies' July 1, 2019 management wage increase as well as portions of subsequent increases in the Rate Year and Data Years be disallowed?

A. Mr. Gadomski asserts (at 16) that the WorldatWork survey that shows a forecast of "Total Salary Budget Increases" of 3.0 percent for the actual 2019 median and projected 2020 median is more reasonable to include in the Companies' revenue requirements than the actual 3.35 percent that was implemented by the Companies on July 2019 and the 3.10 percent annual wage increases projected on July 1st of subsequent years.

Q. Do you agree with Mr. Gadomski's recommendation?

A. No, I do not. In my direct testimony, I presented a compensation study that showed that National Grid's management compensation is below the market median of a representative peer group of utility and non-utility companies' total compensation. Mr. Gadomski performed an independent assessment of the results of the study. Although his final calculation slightly differed from what I presented, he concluded (at 11-12) that National Grid's total

compensation was within the reasonable range of plus or minus 10 percent of median total compensation of its similarly situated peers, which Mr. Gadomski acknowledged is the standard for determining whether a company's compensation is market competitive and has been used by Staff to evaluate compensation benchmarking results in numerous prior rate cases (at 9). The 3.35 percent increase implemented this past July, which is comprised of a 3.1 percent average increase in base pay and a 0.25 percent increase for career path progressions, will not cause National Grid's total management compensation to exceed the median compensation of the peer group. The 3.1 percent annual wage increases requested in subsequent years will similarly not cause National Grid's total management compensation to exceed the median compensation of the peer group. For this reason alone, Mr. Gadomski's recommendation should be rejected.

Q. Are there additional reasons why you believe that the Companies' wage increases are justified and reasonable?

A. Yes. In retroactively applying a 3.0 percent wage increase to the July 1, 2019 wage increase already paid by National Grid to its employees, Mr. Gadomski attempts to adjust a known and measurable change reflected in the Companies' filings. Further, Mr. Gadomski's use of a single data point, the WorldatWork survey, is incredibly limited and fails to incorporate a review of

a representative set of data sources. Prior to approving the salary increases, National Grid performed a holistic review of numerous, available market studies, including the WorldatWork survey, and used the average of the median merit increases from those studies for its wage increases, as shown in the following table.

Data Source	Forecast Increase
Aon Hewitt	3.10%
Culpepper	3.09%
Mercer	3.30%
Wordat Work	3.00%
Salary Com	3.00%
CRI	3.00%
ERI	3.20%
Pay Factors	3.10%
Average	**3.1%**

Therefore, the Companies' data is more reflective of actual market conditions and is also reasonably conservative when compared to the wage increase spectrum of its market peers. The Companies provided Staff with the comprehensive documents reviewed by senior management prior to approving the salary increases in its response to IR No. DPS-803 (the relevant portions of which are included as Exhibit ___ (MPH-1R)). Finally, as indicated in my direct testimony and shown in Schedule 2 of Exhibit ___ (MPH-3) thereto, the

increases included in the Companies' filings are in line with the prior wage increases paid by National Grid.

Q. Do you have any additional comments?

A. Yes. Mr. Gadomski did not address that included within the 3.35 percent increase was a 0.25 percent increase for career path progressions. These costs are a legitimate business expense required to retain high performing employees as they advance to more senior levels across the organization. This is another reason why Mr. Gadomski's proposed adjustment is unreasonable and should be denied.

IV. Vacancy Rate

Q. Please summarize Staff's recommendation to apply a vacancy rate to the labor forecast.

A. The SRRP (at 24) applied vacancy rates of 3.16 percent and 8.20 to the forecast of labor expense for KEDNY and KEDLI, respectively, to account for future vacancies that Staff believes will be experienced by the Companies for various reasons "as a normal part of doing business." Application of Staff's recommended vacancy rates reduces KEDNY's labor expense by $2.759 million and KEDLI's labor expense by $3.229 million.

Q. Do you agree with this recommendation?

A. No. The Companies' headcounts at the end of December 2018 (the end of the Historic Test Year) used to develop the Companies' labor forecasts only reflected positions actually filled by employees at that time. The Companies' Historic Test Year headcount did not include positions that were vacant at that time. Therefore, the SRRP's proposed adjustment is incorrect. A more detailed description of the Companies' objections to this recommendation is presented in the rebuttal testimony of the Revenue Requirements Panels.

Q. Does this conclude your rebuttal testimony?

A. Yes, it does.

Before the Public Service Commission

The Brooklyn Union Gas Company d/b/a National Grid NY

and

KeySpan Gas East Corporation d/b/a/ National Grid

Direct Testimony

of

Ann E. Bulkley

Testimony of Ann E. Bulkley

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I. INTRODUCTION AND QUALIFICATIONS

Q. Please state your name, affiliation, and business address.

A. My name is Ann E. Bulkley. I am a Senior Vice President of Concentric Energy Advisors, Inc. ("Concentric"), located at 293 Boston Post Road West, Suite 500, Marlborough, Massachusetts 01752.

Q. On whose behalf are you submitting this Direct Testimony?

A. I am submitting this Direct Testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"), collectively referred to as "the Companies," wholly-owned subsidiaries of National Grid USA ("National Grid"). My Direct Testimony is part of the Companies' rate case filings before the New York State Public Service Commission ("PSC" or "Commission").

Q. Please describe your experience in the energy and utility industries.

A. I hold a Bachelor's degree in Economics and Finance from Simmons College and a Master's degree in Economics from Boston University, and have more than 20 years of experience consulting to the energy industry. I have advised numerous energy and utility clients on a wide range of financial and economic issues with primary concentrations in valuation and utility rate matters. Many of these assignments have included the determination of the cost of capital for valuation and ratemaking purposes.

I have included my resume and a summary of testimony that I have filed in other proceedings as Attachment A.

Q. Please describe Concentric's activities in energy and utility engagements.

A. Concentric provides regulatory, financial, and economic advisory services to a large number of energy and utility clients across North America. Our regulatory, economic, and market analysis services include utility ratemaking and regulatory advisory services; energy market assessments; market entry and exit analysis; corporate and business unit strategy development; demand forecasting; resource planning; and energy contract negotiations. Our financial advisory activities include buy and sell-side merger, acquisition and divestiture assignments; due diligence and valuation assignments; project and corporate finance services; and transaction support services. In addition, we provide litigation support services on a wide range of financial and economic issues for clients throughout North America.

II. PURPOSE AND OVERVIEW OF TESTIMONY

Q. What is the purpose of your Direct Testimony?

A. The purpose of my Direct Testimony in this proceeding is to present evidence and provide a recommended range for the Companies' cost of equity (sometimes referred to as the Return on Equity or "ROE" for rate-

setting purposes) and capital structure for their gas distribution utility operations. My analysis and recommendations are supported by the data presented in Exhibits __ (AEB-1) through __ (AEB-12).

Q. Please provide a brief overview of the analyses that led to your ROE recommendation.

A. As discussed in more detail in the remainder of my Direct Testimony, it is important to consider the results of several analytical approaches in determining a reasonable recommendation for the Companies' ROE. To develop my ROE recommendation, I developed two proxy groups – the Combined Utility Proxy Group and the Natural Gas Utility Proxy Group – that consist of companies that face risk generally comparable to that faced by the Companies. The Combined Utility Proxy Group includes both electric utilities and natural gas distribution utilities. Recognizing that the ROE that is being established in these proceedings is for natural gas distribution companies, the Natural Gas Utility Proxy Group consists entirely of natural gas distribution companies. I developed a multi-stage Discounted Cash Flow ("DCF") model and two forms of the Capital Asset Pricing Model ("CAPM"). I have presented the DCF and CAPM results weighted equally, and with the Commission's conventional 2/3 weighting of the DCF and 1/3 weighting of the CAPM. I have considered the range of results established using both the Natural Gas Proxy Group and the Combined Utility Proxy Group because the Natural Gas Proxy Group,

although limited in size, is more risk comparable to the Companies than the Combined Utility Proxy Group.

The use of a multi-stage DCF model and two forms of the CAPM is consistent with the approach employed by the Commission in prior cases. While my equal weighting of the DCF and CAPM results does not conform to the weighting typically employed in proceedings before the Commission in the past, I explain in my Direct Testimony why placing less emphasis on the DCF model at this time is consistent with the goals of the Recommended Decision ("RD") issued in the Generic Finance Proceeding ("GFP"), Case 91-M-0509, which is the docket that has been relied on by the Commission to establish the ROE formula.

Q. Please summarize the results of the ROE estimation models that you considered in your analyses.

A. As noted above, I considered the results of the multi-stage form of the DCF model and two versions of the CAPM. The results of my analyses are summarized in Figure 1 (below).

Figure 1: Summary of Analytical Results

Combined Utility Proxy Group			
	Low	**Mean**	**High**
DCF	8.92%	9.20%	9.67%
Mean CAPM	10.66%	10.72%	10.90%
50%/50% DCF/CAPM	9.79%	9.96%	10.29%
67%/33% DCF/CAPM	9.50%	9.71%	10.08%
Natural Gas Proxy Group			
	Low	**Mean**	**High**
DCF	8.59%	9.04%	10.02%
Mean CAPM	10.89%	10.95%	11.12%
50%/50% DCF/CAPM	9.74%	10.00%	10.57%
67%/33% DCF/CAPM	9.35%	9.68%	10.39%

Q. What are your conclusions regarding the appropriate cost of equity for the Companies?

A. Based on the quantitative and qualitative analyses discussed throughout my Direct Testimony and the weighting of the DCF and CAPM results presented in Figure 1, and based on my assessment of the business and financial risk environment of KEDLI and KEDNY relative to the proxy groups, I conclude that the appropriate ROE for the Companies is within the range of 9.65 percent and 10.55 percent. The Companies are requesting an ROE of 9.65 percent, which is at the low end of the range of reasonableness and is a conservative estimate of the investor-required ROE.

Q. Please summarize your analysis of the appropriate ratemaking capital structure for the companies.

A. The analysis presented in Section VIII of my Direct Testimony demonstrates that the Companies' requested equity ratio of 48 percent to 50 percent is below the mean equity ratio of 56.65 percent for the operating utility companies in Combined Utility Proxy Group, and the 57.94 percent for the operating utility companies in the Natural Gas Proxy Group over the last four years. Therefore, I conclude that the Companies' requested equity ratio is reasonable, and in fact, conservative.

Q. How is the remainder of your Direct Testimony organized?

A. The remainder of my Direct Testimony is organized as follows:

Section III – Discusses the regulatory guidelines and financial considerations pertinent to the development of the Cost of Capital;

Section IV – Briefly discusses the current capital market conditions and the effect of those conditions on the Companies' cost of equity;

Section V – Explains my selection of the proxy groups of electric and gas distribution utilities used to develop my analytical results;

Section VI – Explains my analyses and the analytical bases for my ROE recommendation;

Section VII – Summarizes the specific regulatory and business risks that have a direct bearing on the Companies' cost of equity;

Section VIII – Provides an assessment of the Companies' proposed capital structure;

Section IX – Provides an assessment of the effect of a Multi-Year Rate Plan on the ROE; and

Section X – Summarizes my conclusions and recommendations.

III. REGULATORY GUIDELINES AND FINANCIAL CONSIDERATIONS

Q. Please describe the guiding principles to be used in establishing the cost of capital for a regulated utility.

A. The United States Supreme Court's precedent-setting *Hope* and *Bluefield* cases established the standards for determining the reasonableness of a utility's allowed ROE. Among the standards established by the Court in those cases are: (1) consistency with the returns on equity investments in other businesses having similar or comparable risks; (2) adequacy of the return to support credit quality and access to capital; and (3) an understanding that the means of arriving at a fair return are not controlling, only that the end result leads to just and reasonable rates.[1]

Based on those standards, the Commission's order in these cases should provide the Companies with the opportunity to earn a ROE that is: (1) adequate to attract capital at reasonable terms, thereby enabling them to continue to provide safe, reliable service; (2) sufficient to support the

[1] Federal Power Commission v. Hope Natural Gas Co., 320 U.S. 591 (1944) ("Hope"); Bluefield Waterworks & Improvement Co., v. Public Service Commission of West Virginia, 262 U.S. 679 (1923) ("Bluefield").

financial soundness of the Companies' operations; and (3) commensurate with returns on equity investments in enterprises having comparable risks. The authorized ROE should enable the Companies to finance capital expenditures at reasonable rates and maintain their financial flexibility over the period during which rates are expected to remain in effect.

Q. Has the Commission conducted a proceeding to review the standard for estimating the Cost of Capital for a regulated utility?

A. Yes. On August 21, 1991, the Commission established the GFP to review the Commission's then-current methodology for estimating the cost of equity and to examine various alternatives.[2]

Q. Please provide a brief summary of the purpose of the GFP.

A. The GFP was initiated because the Commission recognized that the DCF method was particularly sensitive to interest rate fluctuations and was producing returns far below the returns produced by other methodologies.[3] The Commission's goal in opening the GFP was to eliminate controversy around ROE calculations and attempt to find common ground on contentious issues by developing a consensus approach for setting utility equity returns. Among other things, the Commission examined whether there should be greater consistency in rate of return determinations from

[2] Case 91-M-0509, Proceeding on Motion of the Commission to Consider Financial and Regulatory Policies for New York State Utilities, Recommended Decision, (issued July 19, 1994) ("Generic Finance RD"), at 2.

[3] *Id*.

company to company, such that differences in returns could be directly attributed to differences in risk between companies, and whether the Commission's historical primary reliance on DCF-based ROE determinations continued to provide fair returns.[4] The Commission's inquiry considered the merits of a generic process to reduce redundancy in litigating equity returns, and sought a robust, but standardized, approach to setting ROE such that ROE results were commensurate with the risk of the individual company and would not be skewed by the shortcomings of a single methodology.

Q. Please describe the conclusions outlined in the RD of the Generic Finance Proceeding.

A. Ultimately, the RD concluded that the Commission should implement a generic process for setting returns, based on proxy groups (not company-specific data), and that reliance on the DCF method should be replaced with a combination of the DCF and CAPM methodologies. The RD proposed to use as a preferred convention that gives a respective 2/3rd to 1/3rd weighting to the results of the DCF and CAPM analyses. The RD recognized that the CAPM "should figure prominently in the analysis" because this methodology provides fundamental information on interest rates and the returns required by stocks as a result of changes in interest rates. At that time, the CAPM was not accorded the same level of prominence as the DCF

[4] *Id.* at 13-14.

analysis, given that the former had previously only been used as a check.[5] However, while the RD recognized a benefit to establishing an "operating norm" with respect to setting the ROE, it also recognized that there may be good reason to adjust either the weightings of the DCF and CAPM models or to rely on different ROE estimation models. Specifically, the RD provides the following guidance:

> In either an annual-proceeding to determine a rate of return or in individual proceedings, the 2/3 DCF and 1/3 CAPM convention should be the presumption, but as Multiple Intervenors suggests, parties would not be barred from introducing new methods or different weightings. Such parties, however, would have the burden of convincing other parties and the Commission of the relevance or superiority of their proposals. [6]

To establish the "operating norm," the RD recommended specific forms of the ROE estimation models – a two-stage DCF approach and a Traditional and Zero Beta CAPM. In the DCF model, the first-stage growth was determined by the implied growth rate in Value Line dividend forecasts for four- to six-years in the future. The second growth rate began with the end of the four- to six-year period of the first stage and extended infinitely. The second stage included what is termed an SV adjustment for external growth through additional equity issuances.[7] The CAPM result was proposed to be based on the average of the Traditional and Zero-Beta forms of the model.[8]

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5 *Id.* at 27.
6 *Id*.
7 *Id.* at 21.
8 *Id.* at 24.

Dividend yields in the DCF analysis and risk-free bond yields in the CAPM analyses were based on six months of yield data.[9] Equity ratios were capped at the upper end of the levels necessary to maintain an "A" bond rating.[10] Though the GFP RD was never formally adopted by the Commission, it has served as a touchstone for the Commission's ROE determinations for more than 25 years.

Q. Does the analysis presented in the remainder of your Direct Testimony meet the intentions of the GFP RD?

A. Yes, it does. As discussed in greater detail in Section V, the methodologies that I have applied to estimate the return on equity are consistent with Commission precedent since the RD in the GFP. Moreover, the models used in my analysis extend the principles advanced in the RD in the GFP to best practices in financial analysis and current capital market conditions, as was contemplated in the RD.

Specifically, I rely on the weighted results of DCF and CAPM analyses. In developing these ROE estimation models, I rely on proxy groups of risk-comparable companies as discussed in Section IV. I have used both the DCF and CAPM methodologies to estimate the return on equity. The multi-stage DCF model that I relied on is consistent with the methodology that the Commission has relied on in that it allows growth rates to vary over

[9] *Id.* at 26.
[10] *Id.* at 43.

time. Consistent with the fundamental principles upheld by the Commission, I have applied two versions of the CAPM: Traditional and Zero Beta. Finally, consistent with the principles of the GFP, to reduce the volatility associated with the reliance on any one model, I have considered the DCF and CAPM results weighted equally, and with the RD's proposed 2/3 weighting on the DCF and 1/3 weighting of the CAPM.

Q. Do the principles and intentions of the RD in the GFP require adherence to a static formula?

A. No. The GFP and RD did not require rote adherence to a static formula, rather, they promoted some basic principles and afforded parties the flexibility to investigate approaches to address changing financial market conditions. The RD recognized the benefit of using multiple approaches for setting ROE and although it found benefits to a preferred convention for setting ROE, it did not bar parties from introducing new cost of capital estimation methods or weightings and specifically recognized that there may be circumstances where this would be superior. Capital market conditions vary widely over time and each ROE methodology (DCF and CAPM) may be impacted differently by identical conditions. The impact of these conditions on ROE must be assessed and interpreted by the practitioner to determine if their effects are directionally appropriate and are of a reasonable magnitude. Accordingly, it is incumbent on the practitioner to review the results of the analyses and exercise judgment as to how to

weight those results in the overall ROE determination. A close read of the RD reveals that the Commission expressed some uncertainty around the correct level of weighting and certainly indicated a willingness to revisit its proposed weightings in the future. It is particularly fitting that the Commission, which is seeking to update the traditional utility regulatory model with new, innovative approaches suitable to current industry circumstances in the New York Reforming the Energy Vision ("NY REV") efforts, consider the integrity of the intent and principles of the RD and demonstrate the flexibility to adapt the weightings of each methodology to the applicable capital market conditions.

Q. Is flexibility of approach and judgment important to ROE determination?

A. Yes, it is. When faced with the task of estimating the cost of equity, analysts are inclined to gather and evaluate as much relevant data (both quantitative and qualitative) as can be reasonably analyzed. Analysts and academics understand that ROE models are tools to be used in the ROE estimation process, and that strict adherence to any single approach, or the specific results of any single approach, can lead to flawed conclusions. No model can exactly pinpoint the correct return on equity; rather, each model brings its own perspective and set of inputs that inform the estimate of ROE. That position is consistent with the *Hope* finding that "[u]nder the statutory

standard of 'just and reasonable,' it is the result reached, not the method employed, which is controlling."[11]

Though each model brings a different perspective and adds depth to the analysis, each model also has its own set of inherent weaknesses and should not be relied upon individually without corroboration from other approaches. Changes to inputs as a result of changes in economic conditions could have widely different effects on the results of the various analyses.

Regardless of which analyses are performed to estimate the investor's required return on equity, the analyst must apply judgment to assess the reasonableness of results and to determine the best weighting to apply to results under prevailing capital market conditions. No one model can reliably and consistently estimate the cost of capital that meets the fairness standard of *Hope* and *Bluefield* in all market conditions.

IV. CAPITAL MARKET CONDITIONS

Q. Why is it important to analyze capital market conditions?

A. The ROE estimation models rely on market data that are either specific to the proxy group, in the case of the DCF model, or to the expectations of market risk, in the case of the CAPM. The results of the ROE estimation

[11] *Hope*, 320 U.S. at 602.

models can be affected by prevailing market conditions at the time the analysis is performed. While the ROE that is established in a rate proceeding is intended to be forward-looking, the analyst uses current and projected market data, specifically stock prices, dividends, growth rates and interest rates in the ROE estimation models to estimate the required return for the subject company.

As is discussed in the remainder of this section, analysts and regulatory commissions have concluded that current market conditions have affected the results of the ROE estimation models. As a result, it is important to consider the effect of these conditions on the ROE estimation models when determining the appropriate range and recommended ROE for a future period. If investors do not expect current market conditions to be sustained in the future, it is possible that the ROE estimation models will not provide an accurate estimate of investors' required return during that rate period. Therefore, it is important to consider projected market data to estimate the return for that forward-looking period.

Q. What factors are affecting the cost of equity for regulated utilities in the current and projected capital markets?

A. The cost of equity for regulated utility companies is being affected by several factors in the current and prospective capital markets, including: (1) the current low interest rate environment and the corresponding effect on

valuations and dividend yields of utility stocks relative to historical levels; (2) the market's expectation for interest rates; and (3) recent Federal tax reform. In this section, I discuss each of these factors and how it affects the models used to estimate the cost of equity for regulated utilities.

A. The Effect Of Market Conditions On Valuations

Q. How has the Federal Reserve's monetary policy affected capital markets in recent years?

A. Extraordinary and persistent federal intervention in capital markets artificially lowered government bond yields after the Great Recession of 2008-2009, as the Federal Open Market Committee ("FOMC") used monetary policy (both reductions in short-term interest rates and purchases of Treasury bonds and mortgage-backed securities) to stimulate the U.S. economy. As a result of very low or zero returns on short-term government bonds, yield-seeking investors have been forced into longer-term instruments, bidding up prices and reducing yields on those investments. As investors have moved along the risk spectrum in search of yields that meet their return requirements, there has been increased demand for dividend-paying equities, such as natural gas and electric utility stocks.

Q. How has the period of abnormally low interest rates affected the valuations and dividend yields of utility shares?

A. The Federal Reserve's accommodative monetary policy has caused investors to seek alternatives to the historically low interest rates

available on Treasury bonds. A result of this search for higher yield is that the share prices for many common stocks, especially dividend-paying stocks such as utilities, have been driven higher while the dividend yields (which are computed by dividing the dividend payment by the stock price) have decreased to levels well below the historical average.

Q. How have higher stock valuations and lower dividend yields for utility companies affected the results of the DCF model?

A. During periods of general economic and capital market stability, the DCF model may adequately reflect market conditions and investor expectations. However, in the current market environment, the DCF model results are distorted by the historically low level of interest rates and the higher valuation of utility stocks. Value Line recently commented on the high valuations of electric utilities:

> Even after a pullback in late 2018, most stocks in the Electric Utility Industry are still priced expensively, in our view. Many of the equities are still trading within our 2021-2023 Target Price Range. The industry's average dividend yield is 3.5%, and some stocks have yields that aren't significantly higher than the median of all stocks under our coverage. For the 3- to 5-year period, the group's average total return potential is just 5%.[12]

This is further supported by a recent Edward Jones report on the utility sector:

[12] Value Line Investment Survey, Electric Utility (West) Industry, January 25, 2019, at 2217.

> Utility valuations have come down as 10-year Treasury bond rates have climbed back over 3%. On a price-to-earnings basis, they do remain significantly above their historical average, but have declined to less unreasonable levels. We have seen utility valuations moving in line with interest rate movements, although there have been exceptions to this. Overall, however, we believe the low-interest rate environment has been the biggest factor in pushing utilities higher since many investors buy them for their dividend yield.

> Utilities have declined from their all-time highs reached late in 2017, but are still trading significantly above their average price-to-earnings ratio over the past decade. The premium valuation continues to reflect not only the low interest rate environment, but also the stable and predominantly regulated earnings growth we foresee.[13]

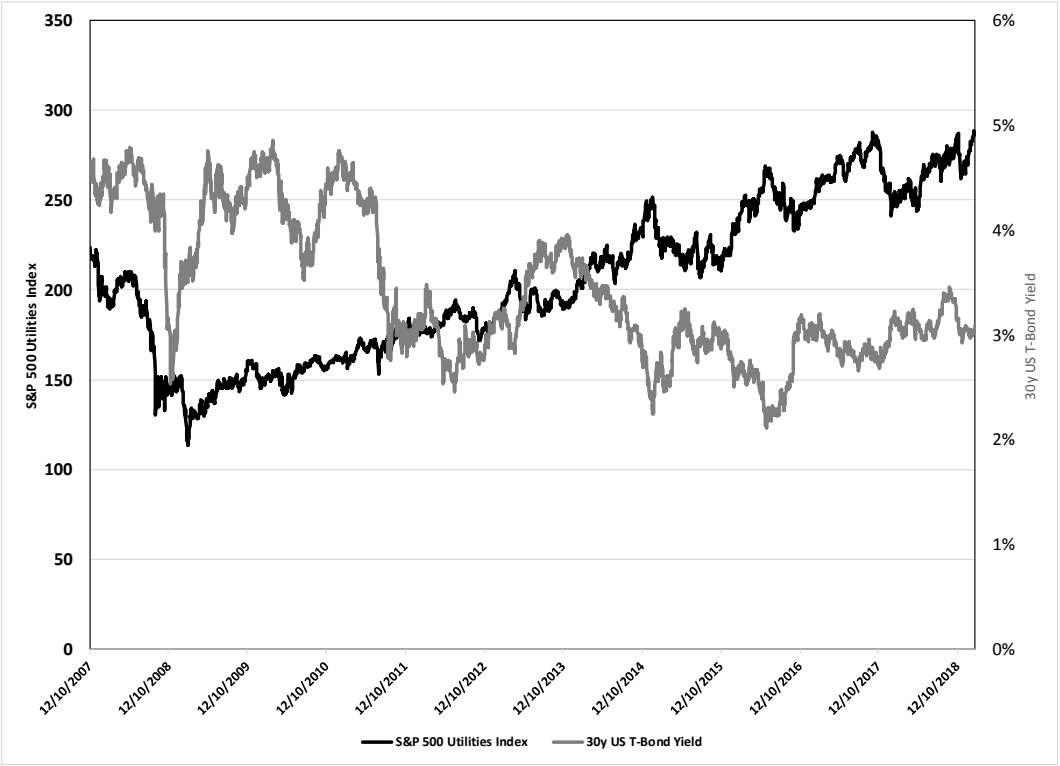
As noted by Value Line and Edward Jones, over the last few years, utility stocks have experienced high valuations and low dividend yields; driven by investors moving into dividend paying stocks from bonds due to the low interest rates in the bond market, however, those dynamics are changing. Value Line and Edward Jones recognize that as interest rates increase, bonds become a substitute for utility stocks. As utility stock prices decline, the dividend yields will increase. This change in market conditions implies that the ROE calculated using historical market data in the DCF model may understate the forward-looking cost of equity.

[13] Andy Pusateri and Andy Smith. Edward Jones, Utilities Sector Outlook (January 16, 2019), at 2-3. [Reference to figure omitted.]

Q. How did the Standard & Poor's ("S&P") Utilities Index respond to the market conditions that existed following the Great Recession of 2008-2009?

A. Figure 2**Error! Reference source not found.**, demonstrates market conditions from 2007-2019 as measured by the S&P Utilities index and the yield on 30-year Treasury bonds. As shown in Figure 2, the S&P Utilities index increased steadily from the beginning of 2009 through early November 2017, as yields on 30-year Treasury bonds declined in response to accommodative federal monetary policy.

Figure 2: S&P Utilities Index and U.S. Treasury Bond Yields (2007-2019)



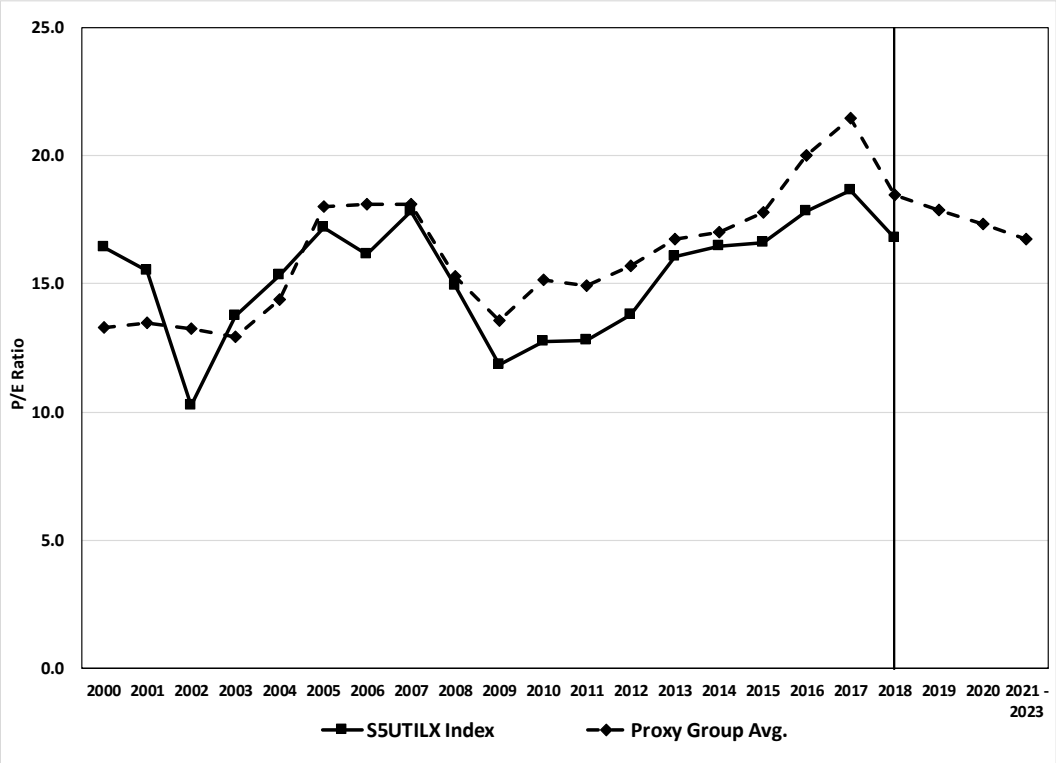
Source: Bloomberg Professional

Q. How do the valuations of public utilities compare to the historical average?

A. Figure 3 summarizes the average historical and projected price-to-earnings ("P/E") ratios for the proxy companies calculated using data from Bloomberg Professional and Value Line.[14] As shown in Figure 3, the average P/E ratio for the Combined Utility Proxy Group companies was higher in 2017 than at any other time in the last seventeen years and is significantly higher than the average projected P/E ratio for the group for the period from 2021-2023. In 2018 however, the average P/E ratio for the proxy companies has decreased slightly to 18.45 from the high in 2017 of 21.44. All else equal, if P/E ratios for the proxy companies continue to decline, as Value Line projects, the ROE results from the DCF model would be higher. Therefore, the DCF model using historical market data is likely understating the forward-looking cost of equity for the proxy group companies.

[14] Selection of the Proxy Companies is discussed in detail in Section IV of my Direct Testimony.

Figure 3: Average Historical Combined Utility Proxy Group P/E Ratios



Note: Figure includes data through February 28, 2019.

Sources: Bloomberg Professional, Value Line

Q. How do equity investors view the utilities sector based on these recent market conditions?

A. Investment advisors have suggested that utility stocks may underperform as a result of market conditions. Barron's recently published its seventh annual review of income-producing investments in which Barron's ranked eleven different sectors based on projected performance in 2019. The utility sector ranked ninth out of the eleven sectors with Barron's noting that:

> Utilities, however, aren't cheap; they are valued at an average of 17 times projected 2019 earnings, a premium to the S&P 500, at about 14. That may make it hard for utilities to best the index in 2019, barring a market collapse. Earnings growth is running at a mid-single-digits yearly pace.[15]

Similarly, a recent report on the market outlook for 2019 from J.P. Morgan Asset Management noted:

> As prospects for slower economic growth become clearer in the middle of next year, the Fed may signal it will pause. Such a signal, or a trade agreement with China, could lead multiples to expand, pushing the stock market higher and potentially adding years to this already old bull market. However, even if the bull market does end in the next few years, it is important to remember that late-cycle returns have typically been quite strong.
>
> This leaves investors in a tough spot – should they focus on a fundamental story that is softening, or invest with an expectation that multiples will expand as the bull market runs its course? The best answer is probably a little bit of each. We are comfortable holding stocks as long as earnings growth is positive, but do not want to be over-exposed given an expectation for higher volatility. As such, higher-income sectors like financials and energy look more attractive than technology and consumer discretionary, and we would lump the new communication services sector in with the latter names, rather than the former. However, given our expectation of still some further interest rate increases, it does not yet seem appropriate to fully rotate into defensive sectors like utilities and consumer staples. Rather, a focus on cyclical value should allow investors to optimize their upside/downside capture as this bull market continues to age.[16]

[15] Bary, Andrew. "Best Income Investments for 2019." Barron's, Barron's, 4 Jan. 2019, www.barrons.com/articles/the-best-income-ideas-for-2019-51546632171.

[16] J.P. Morgan Asset Management, "The investment outlook for 2019: Late-cycle risks and opportunities", November 30, 2018, at 5.

This view was further supported by UBS who noted that:

> Our underweight views on consumer staples and utilities sectors reflect our preference for sectors that are more leveraged to continued favorable economic growth than these two defensive sectors. In addition, consumer staples are contending with sluggish organic growth. High dividend yields for the utilities sector makes it most negatively exposed to higher interest rates. Our industrials underweight is a bit of a hedge against a potential increase in trade frictions.[17]

Q. Have regulators recently responded to the historically low dividend yields for utility companies and the corresponding effect on the DCF model?

A. Yes. The Federal Energy Regulatory Commission ("FERC") has determined that current capital market conditions have caused the DCF model to understate equity costs for regulated utilities at this time.[18] The FERC recently proposed a methodology that reflects the commission's current view that investors rely on multiple ROE estimation models. The proposed methodology includes an equal weighting of the DCF, CAPM,

[17] UBS, "2019 outlook: Aging gracefully", December 5, 2018, at 7.

[18] FERC Docket No. EL11-66-001, Opinion No. 531 (June 19, 2014), footnote 286. While Opinion No. 531 was recently remanded to the FERC by the D.C. Circuit Court on other grounds, that decision did not question the finding by the FERC that capital market conditions were anomalous. *See, Emera Maine v. FERC* 854 F.3d 9 (D.C. Dir. 2017). Additionally, the methodologies that were relied on by FERC to establish the range have not been challenged. *See also* FERC, Docket No. EL 11-66-001, et al., Order Directing Briefs, issued October 16, 2018, at para. 32. This Order develops a proposed methodology to address the issues that were remanded to FERC. The proposed methodology includes an equal weighting of the DCF, CAPM, Expected Earnings and Risk Premium models to better reflect investor behavior and capital market conditions.

Expected Earnings and Risk Premium models to better reflect investor behavior and capital market conditions.[19]

In addition, the Illinois Commerce Commission ("ICC"), the Pennsylvania Public Utility Commission ("PPUC") and the Missouri Public Service Commission ("Missouri PSC") have all considered this in recent decisions.

B. THE CURRENT AND EXPECTED INTEREST RATE ENVIRONMENT

Q. Please provide a brief summary of the recent monetary policy actions of the Federal Reserve.

A. Based on stronger conditions in employment markets, a relatively stable inflation rate, steady economic growth, and increased household spending, the Federal Reserve raised the short-term borrowing rate by 25 basis points on four occasions in 2018. Since December 2015, the Federal Reserve has increased interest rates nine times, bringing the federal funds rate to the range of 2.25 percent to 2.50 percent. While the Federal Reserve recently indicated at the March 2019 meeting that going forward it will be patient in determining future adjustments to the federal funds rate due to recent global economic and financial developments and low inflationary pressures, they have not indicated that they will not raise interest rates over the coming year. In fact, Bloomberg recently noted that some officials saw higher rates

[19] FERC, Docket No. EL 11-66-001, et al., Order Directing Briefs, issued October 16, 2018, at para. 32.

as appropriate later this year if economic growth continued above its longer-run trend rate, according to the minutes.[20]

Additionally, in October 2017, the FOMC started reducing the size of the Fed's $4.5 trillion bond portfolio by no longer reinvesting the proceeds of the bonds it holds. In response to the Great Recession, the Fed pursued a policy known as "Quantitative Easing," in which it systematically purchased mortgage-backed securities and long-term Treasury bonds to provide liquidity in financial markets and drive down yields on long-term government bonds. Although the Federal Reserve discontinued the Quantitative Easing program in October 2014, it continued to reinvest the proceeds from the bonds it holds. Under the initial balance sheet normalization policy, the FOMC gradually reduced the Federal Reserve's securities holdings by $10 billion per month initially, ramping up to $50 billion per month by the end of the first twelve months.[21] However, at the March 2019 meeting, the FOMC announced that it intends to slow the reduction of its holdings of Treasury Securities starting in May 2019 and ultimately conclude the program in September 2019.[22]

[20] FOMC, Federal Reserve press release, March 20, 2019. *See also*, Torres, Craig. "Fed Minutes Show Some Rate Flexibility During Year of Patience." Bloomberg.com, Bloomberg, 10 Apr. 2019, www.bloomberg.com/news/articles/2019-04-10/fed-minutes-show-some-rate-flexibility-during-year-of-patience.

[21] Federal Reserve press release, Addendum to the Policy Normalization Principles and Plans, June 14, 2017, implemented at FOMC meeting, September 20, 2017.

[22] Federal Reserve press release, Balance Sheet Normalization Principles and Plans, March 20, 2019.

Q. How does the recent change in the Federal Reserve's policy affect the yields on long-term government bonds?

A. While the Federal Reserve has recently indicated to that it will be patient in determining future adjustments to the federal funds rate, this is not unusual as monetary policy has a lagged effect on the economy. As Federal Reserve Bank of San Francisco notes:

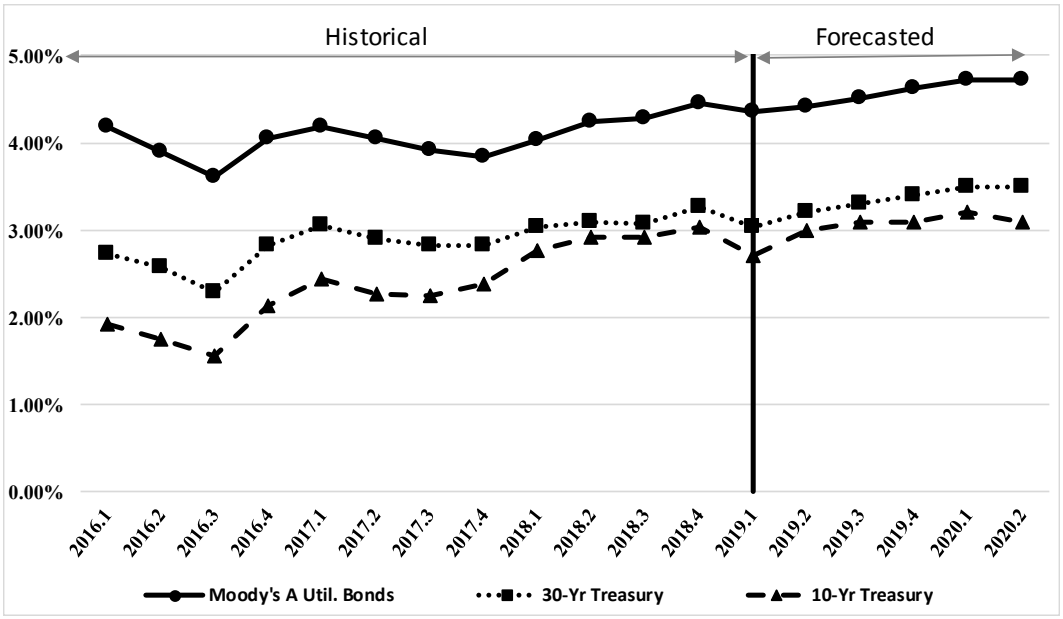
> It can take a fairly long time for a monetary policy action to affect the economy and inflation. And the lags can vary a lot, too. For example, the major effects on output can take anywhere from three months to two years. And the effects on inflation tend to involve even longer lags, perhaps one to three years, or more.[23]

Since December 2015, the Federal Reserves has increased the federal funds rate nine times, four of which occurred in 2018 and three in 2017. Therefore, given recent market volatility and lagged effect that monetary policy has on the economy, it is reasonable to expect the Federal Reserve to be patient with future increases. However, it is important to note, that the Federal Reserve is continuing to reduce the size of its balance sheet by no longer reinvesting the proceeds of the bonds it holds over the near-term. This policy in conjunction with the lagged effect of past increases in the federal funds rate suggests that the yields on long-term government bonds should continue to increase over the near-term which is consistent with investors'

[23] Federal Reserve Bank of San Francisco, "U.S. Monetary Policy: An Introduction - How does monetary policy affect the U.S. economy?", February 6, 2004. https://www.frbsf.org/education/teacher-resources/us-monetary-policy-introduction/real-interest-rates-economy/

expectations. As shown in Figure 4, investors are expecting continued increases in interest rates on both government and corporate/utility bonds over the next few years.

Figure 4: Interest Rate Conditions[24]



Q. Have you examined the effect of the Federal Reserve's monetary policy on the yields of long-term government bonds over the past few years?

A. Yes. As shown in Figure 4, yields on long-term government bonds have increased since the Federal Reserve started to raise the federal funds rate in 2016. However, the increase in long-term government bond yields has not been as pronounced as the rise in short-term interest rates. This is due to a shift in the supply and demand of long-term government bonds that has

[24] Source: Historical data from Bloomberg Professional. Forecast data from Blue Chip Financial Forecasts, Volume. 38, No. 2, February 1, 2019, at 2.

occurred since 2009. For example, since the Great Recession of 2008-2009, federal debt has increased significantly which has resulted in an increase in the supply of Treasury bonds in the market. In general, an increase in supply should result in a decrease in the price of Treasury bonds and an increase in yield. However, long-term government bonds yields have not increased as fast as expected given the increase in supply. This is because the demand for Treasury bonds has also increased since 2009. As noted in a recent article published by the St. Louis Federal Reserve, the demand for government bonds increased for a number of reasons some of which included increased holdings by foreign governments as countries in Europe and Asia faced their own economic uncertainty, and increased holdings from commercial banks due to new regulations that required banks to hold a larger portion of high-quality liquid assets.[25] This has resulted in a more gradual increase in the yields on long-term government bonds over the past few years.

Q. Is the demand for long-term government bonds expected to continue to increase?

A. No, it is not. As noted in the Federal Reserve article:

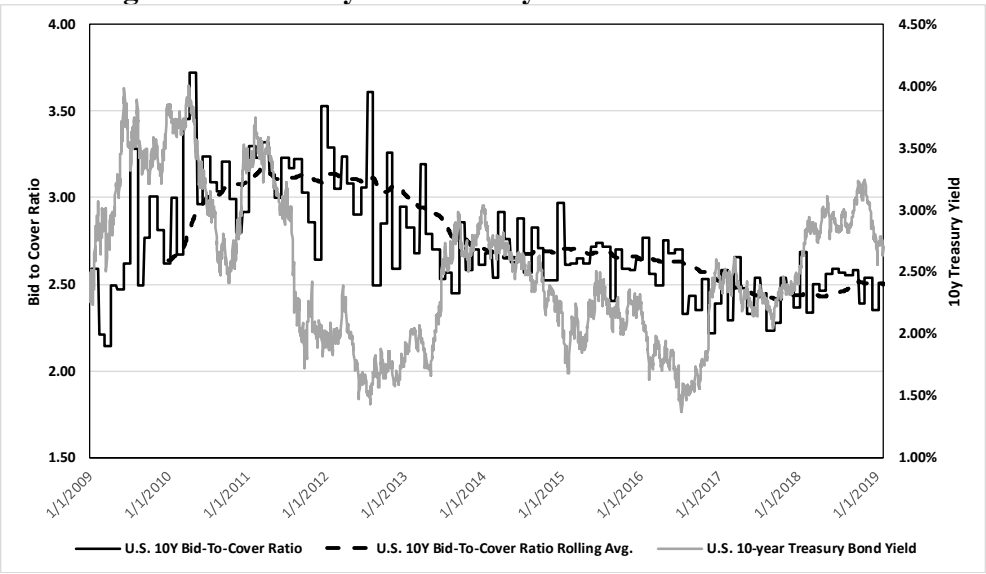
[25] David Andolfatto and Andrew Spewak, Federal Reserve Bank of St. Louis, "On the Supply of, and Demand for, U.S. Treasury Debt," Economic Synopses, No. 5, 2018. https://doi.org/10.20955/es.2018.5.

> Some evidence suggests that the growth in demand for Treasuries has already begun to soften. …[F]oreign holdings have remained more or less constant since 2014, largely because of declining holdings in Japan and China. Likewise, regulation and policy changes such as the Dodd-Frank Act and new rules for prime money market funds may have only transitory effects on the demand for Treasuries. For example, the pace of growth of the ratio of commercial bank Treasury security holdings to private loans has slowed since 2014 (see Figure 3), as has the growth of investment in government money market funds since 2017 (Figure 4).[26]

Furthermore, another indicator of the demand for Treasury bonds is the bid to cover ratio which represents the dollar amount of bids received versus the dollar amount sold in a Treasury security auction. Therefore, a higher bid-to-cover ratio is indicative of an increase in the demand for government bonds. As shown in Figure 5, the bid-to-cover ratio for the 10-year U.S. Treasury bond is currently at its lowest point since 2009 which indicates that the demand for long-term government bonds has declined. The decline in demand is occurring at a time when the supply of Treasury bonds is expected to increase as the Federal Reserve continues its balance sheet unwind and the federal government issues bonds to offset the reduced tax revenue associated with the implementation of the Tax Cuts and Jobs Act ("TCJA"). As a result, yields on long-term government bonds are expected to continue to increase over the near-term which is consistent with investors' expectations shown in Figure 4.

[26] *Id.*

Figure 5: U.S. 10-year Treasury Bond Bid-to-Cover-Ratio



Q. What effect do rising interest rates have on the cost of equity?

A. As interest rates continue to increase, the cost of equity for the proxy companies using the DCF model is likely to be an overly conservative estimate of investors' required returns because the proxy groups' average dividend yields reflects the increase in stock prices that resulted from substantially lower interest rates. However, my CAPM analysis includes estimated returns based on near-term projected interest rates, reflecting investors' expectations of market conditions over the period that the rates that are determined in this case will be set. As discussed in Section VI, the CAPM may be a more reliable model in current capital market conditions and therefore it would be reasonable to place greater emphasis on the results of this model.

C. Effect of Tax Reform on the Return on Equity and Capital Structure

Q. Are there other factors that should be considered in determining the cost of equity for the Companies?

A. Yes. The effect of the TCJA should also be considered in the determination of the cost of equity. The credit rating agencies have commented on the effect of the TCJA on regulated utilities. In summary, the TCJA is expected to reduce utility revenues due to the lower federal income taxes and the requirement to return excess accumulated deferred income taxes. This change in revenue is expected to reduce Funds From Operations ("FFO") metrics across the sector, and absent regulatory mitigation strategies, is expected to lead to weaker credit metrics and negative ratings actions for some utilities.[27]

Q. Have credit or equity analysts commented on the effect of the TCJA on utilities?

A. Yes. Moody's Investors Services ("Moody's") indicated that while the TCJA was credit positive for many sectors, it has an overall negative credit impact on regulated operating companies of utilities and their holding companies due to the reduction in cash flow metrics that results from the change in the federal tax rate and the loss of bonus depreciation.

[27] FitchRatings, Special Report, What Investors Want to Know, "Tax Reform Impact on the U.S. Utilities, Power & Gas Sector", January 24, 2018.

Moody's noted that the rates that regulators allow utilities to charge customers are based on a cost-plus model, with tax expense being one of the pass-through items. Utilities will collect less taxes at the lower rate, reducing revenue. While the taxes are ultimately paid out as an expense, under the new law utilities lose the timing benefit, reducing cash that may have been carried over a number of years. The lower tax rate combined with the loss of bonus depreciation will have a negative effect on utility cash flows and will ultimately negatively impact the utilities' ability to fund ongoing operations and capital improvement programs with internally generated cash.

Q. How has Moody's responded to the increased risk for utilities resulting from the TCJA?

A. In January 2018, Moody's issued a report changing the rating outlook for several regulated utilities from Stable to Negative, including KEDNY and KEDLI.[28] At that time, Moody's noted that the ratings change affected companies with limited cushion in their ratings for deterioration in financial performance. In June 2018, Moody's downgraded the outlook for the entire regulated utility industry from stable to negative for the first time ever. Moody's cites ongoing concerns about the negative effect of the TCJA on cash flows of regulated utilities. While noting that "[r]egulatory

[28] Moody's Investor Service, Global Credit Research, Rating Action: Moody's changes outlooks on 25 US regulated utilities primarily impacted by tax reform, January 19, 2018.

commissions and utility management teams are taking important first steps"[29] and that "we have seen some credit positive developments in some states in response to tax reform,"[30] Moody's concludes that "we believe that it will take longer than 12-18 months for the majority of the sector to show any material financial improvement from such efforts."[31]

Q. Has Moody's changed its outlook for utilities in 2019?

A. No. Consistent with the prior reports issued by Moody's in January and June of 2018, Moody's is maintaining its negative outlook for regulated utilities in 2019 as a result of continued concerns over the effect of the TCJA on cash flows as well as increasing debt.[32] Moody's notes that "[t]he combination of financial pressures is expected to keep the sector's ratio of funds from operations to debt down around 15% in the year ahead".[33]

Q. What does it mean for Moody's to downgrade a credit outlook?

A. A Moody's rating outlook is an opinion regarding the likely rating direction over what it refers to as "the medium term." A Stable outlook indicates a low likelihood of a rating change in the medium term. A Negative outlook indicates a higher likelihood of a rating change over the medium term.

[29] Moody's Investors Service, "Regulated utilities – US: 2019 outlook shifts to negative due to weaker cash flows, continued high leverage", June 18, 2018, at 3.
[30] *Id*.
[31] *Id.*
[32] Moody's Investors Service, Research Announcement: Moody's: US regulated utilities sector outlook for 2019 remains negative, November 8, 2018.
[33] *Id.*

While Moody's indicates that the time period for changing a rating subsequent to a change in the outlook from Stable will vary, on average Moody's indicates that a rating change will follow within a year of a change in outlook.[34]

Q. Have any utilities experienced a downgrade related to cash flow metrics resulting from the TCJA?

A. Yes. Recently, Moody's downgraded the senior unsecured rating and issuer rating of KEDNY and KEDLI, noting the persistent weakness of key credit metrics with limited likelihood of a recovery in the medium term. As discussed in more detail in Section VII, Moody's rationale includes the expectation of reduced revenues for the Companies due to the TCJA.[35] In addition, there have been several other downgrades. Figure 6 summarizes credit rating downgrades for utilities that have resulted from tax reform.

[34] Moody's Investors Service, Rating Symbols and Definitions, July 2017, at 27.

[35] Moody's Investors Service, Rating Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019 and Moody's Investor Services, Ratings Action: Moody's takes action on certain National Grid subsidiaries and affirms other ratings, March 29, 2019..

Figure 6: Credit Rating Downgrades Resulting from TCJA

Utility	Rating Agency	Credit Rating before TCJA	Credit Rating after TCJA	Downgrade Date
American Water Works	Moody's	A3	Baa1	4/1/2019
KeySpan Gas East Corporation (KEDLI)	Moody's	A2	A3	3/29/2019
Xcel Energy	Moody's	A3	Baa1	3/28/2019
ALLETE, Inc.	Moody's	A3	Baa1	3/26/2019
Brooklyn Union Gas Company (KEDNY)	Moody's	A2	A3	2/22/2019
Avista Corp.	Moody's	Baa1	Baa2	12/30/2018
Consolidated Edison Company of New York	Moody's	A2	A3	10/30/2018
Consolidated Edison, Inc.	Moody's	A3	Baa1	10/30/2018
Orange and Rockland Utilities	Moody's	A3	Baa1	10/30/2018
Southwestern Public Service Company	Moody's	Baa1	Baa2	10/19/2018
Dominion Energy Gas Holdings	Moody's	A2	A3	9/20/2018
Piedmont Natural Gas Company, Inc.	Moody's	A2	A3	8/1/2018
WEC Energy Group, Inc.	Moody's	A3	Baa1	7/12/2018
Integrys Holdings Inc.	Moody's	A3	Baa1	7/12/2018
OGE Energy Corp.	Moody's	A3	Baa1	7/5/2018
Oklahoma Gas & Electric Company	Moody's	A1	A2	7/5/2018

Q. Have other rating agencies commented on the effect of the TCJA on ratings?

A. Yes. S&P and Fitch have also commented on the implications of the TCJA on utilities. S&P published a report on January 24, 2018 entitled "U.S. Tax Reform: For Utilities' Credit Quality, Challenges Abound" in which S&P concludes:

The impact of tax reform on utilities is likely to be negative to varying degrees depending on a company's tax position going into 2018, how its regulators react, and how the company reacts in return. It is negative for credit quality because the combination of a lower tax rate and the loss of stimulus provisions related to bonus depreciation or full expensing of capital spending will create headwinds in operating cash-flow generation capabilities as customer rates are lowered in response to the new tax code. The impact could be sharpened or softened by regulators depending on how much they want to lower utility rates immediately instead of using some of the lower revenue requirement from tax reform to allow the utility to retain the cash for infrastructure investment or other expenses. Regulators must also recognize that tax reform is a strain on utility credit quality, and we expect companies to request stronger capital structures and other means to offset some of the negative impact.

> Finally, if the regulatory response does not adequately compensate for the lower cash flows, we will look to the issuers, especially at the holding company level, to take steps to protect credit metrics if necessary. Some deterioration in the ability to deduct interest expense could occur at the parent, making debt there relatively more expensive. More equity may make sense and be necessary to protect ratings if financial metrics are already under pressure and regulators are aggressive in lowering customer rates. It will probably take the remainder of this year to fully assess the financial impact on each issuer from the change in tax liabilities, the regulatory response, and the company's ultimate response. We have already witnessed differing responses. We revised our outlook to negative on PNM Resources Inc. and its subsidiaries on Jan. 16 after a Public Service Co. of New Mexico rate case decision incorporated tax savings with no offsetting measures taken to alleviate the weaker cash flows. It remains to be seen whether PNM will eventually do so, especially as it is facing other regulatory headwinds. On the other hand, FirstEnergy Corp. issued $1.62 billion of mandatory convertible stock and $850 million of common equity on Jan. 22 and explicitly referenced the need to support its credit metrics in the face of the new tax code in announcing the move. That is exactly the kind of proactive financial management that we will be looking for to fortify credit quality and promote ratings stability.[36]

In S&P's 2019 trends report, the rating agency notes that the utility industry's financial measures weakened in 2018 and attributed that to tax reform, capital spending and negative load growth. In addition, S&P expects that weaker credit metrics will continue into 2019 for those utilities operating with minimal financial cushion. S&P further expects that these utilities will look to offset the revenue reductions from tax reform with

[36] Standard and Poor's Global Ratings, "U.S. Tax Reform: For Utilities' Credit Quality, Challenges Abound", January 24, 2018.

equity issuances. The rating agency reported that in 2018 regulated utilities issued nearly $35 billion in equity, which is more than twice the equity issuances in 2016 and 2017.[37]

Finally, FitchRatings recognized the implications of tax reform but indicated that any ratings actions will be guided by the response of regulators and the management of the utilities. Fitch notes that the solution will depend on the ability of utility management to manage the cash flow implications of the TCJA. Fitch offers several solutions to provide rate stability and to moderate changes to cash flow in the near term, including increasing the authorized ROE and/or equity ratio as measures that can be implemented.[38]

Q. What conclusions do you draw from your analysis of capital market conditions?

A. The important conclusions resulting from capital market conditions are:

- The assumptions used in the ROE estimation models have been affected by recent historical capital market conditions.
- Recent market conditions are not expected to persist as the Federal Reserve continues to normalize monetary policy. As a result, the

[37] Standard & Poor's Ratings, "Industry Top Trends 2019, North America Regulated Utilities", November 8, 2018.

[38] FitchRatings, Special Report, What Investors Want to Know, "Tax Reform Impact on the U.S. Utilities, Power & Gas Sector", January 24, 2018.

recent historical market conditions are not reflective of the market conditions that will be present when the rates for the Companies will be in effect.

- It is important to consider the results of a variety of ROE estimation models, using forward-looking assumptions to estimate the cost of equity.

- Without adequate regulatory support, the TCJA will have a negative effect on utility cash flows, which increases investor risk expectations for utilities.

V. PROXY GROUP SELECTION

Q. Please explain why you have used a group of proxy companies to determine the cost of equity for the Companies.

A. In these proceedings, we are focused on estimating the cost of equity for the Companies' rate-regulated, natural gas distribution utility operations in New York. Because ROE is a market-based concept and the Companies are not publicly traded, it is necessary to establish a group of companies that are both publicly traded and comparable to the Companies in certain fundamental business and financial respects to serve as their "proxy" in the ROE determination process. As discussed later in my Direct Testimony, the proxy companies used in my analyses all possess a set of operating and risk characteristics that are substantially comparable to the Companies and

thus provide a reasonable basis for the derivation and assessment of the Companies' ROE.

In utility rate proceedings before the Commission over the past 25 years (since the RD in the GFP),[39] the Commission has endorsed the use of proxy groups for the purpose of determining utility ROEs. Because proxy companies are now commonly used as the basis for estimating the utility cost of equity, the primary objective of the screening process is to establish a group of companies that are as comparable as possible to the Companies with respect to fundamental financial and business risks. As a practical matter, while the determination of an appropriate ROE necessarily requires a degree of informed judgment, the careful selection of a risk-appropriate comparison group serves to mitigate the extent to which subjective assessments must be applied.

Q. Please provide a summary profile of the Companies.

A. KEDNY's business consists of its regulated natural gas transportation and distribution operations in New York State. KEDNY distributes natural gas to retail customers in the New York City boroughs of Brooklyn, Staten Island, and Queens.[40] KEDNY's long-term issuer ratings are A3 (Moody's, Outlook: Stable), and A- (S&P, Outlook: Stable).[41]

[39] Generic Finance RD at 133-134.
[40] KEDNY, Consolidated Financial Statements, 2017 at 11.
[41] Source: SNL Financial, accessed February 25, 2019.

KEDLI's business consists of its regulated natural gas transportation and distribution operations in New York State. KEDLI distributes natural gas to retail customers in Long Island, New York and Queens, New York.[42] KEDLI's long-term issuer ratings are A3 (Moody's, Outlook: Stable), and A- (S&P, Outlook: Stable).[43]

Q. How did you select the companies included in your proxy groups?

A. The Commission has historically relied on proxy groups generally comprised of electric utilities even for the purposes of establishing the ROE for a natural gas distribution utility. In recognition of that practice, I began with the companies that Value Line classifies as "Electric Utilities" and "Natural Gas Distribution Companies." That combined group includes 49 domestic U.S. utilities. I simultaneously applied the following screening criteria to establish a risk-comparable Combined Utility Proxy Group that includes electric utility companies and natural gas distribution companies:

- To ensure that information regarding the proxy group companies is consensus-based, I eliminated the companies that are not covered by at least two utility industry equity analysts;
- I eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to

[42] KEDLI, Consolidated Financial Statements, 2017 at 10.
[43] Source: SNL Financial, accessed February 25, 2019.

S&P and Moody's because such companies do not have a similar financial risk profile to that of the Companies;

- I eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source because such characteristics are incompatible with the DCF model;
- To ensure that the proxy group consists of companies that are primarily transmission and distribution regulated utilities, I eliminated companies that have owned generation comprising greater than 60 percent of the Company's MWh sales to ultimate customers;
- To ensure that the proxy group consists of companies that are primarily regulated utilities, I eliminated companies with less than 70 percent of total operating income derived from regulated utility operations; and
- I eliminated companies known to be party to a merger, acquisition, or other transformational transaction as such activities may have a temporary effect on such companies' stock prices and projections unrelated to the overall cost of capital.

Q. What is the composition of your Combined Utility Proxy Group?

A. My Combined Utility Proxy Group consists of the 20 companies presented in Figure 7.

Figure 7: Combined Utility Proxy Group

Company	Ticker
ALLETE, Inc.	ALE
Alliant Energy Corporation	LNT
Atmos Energy Corporation	ATO
Black Hills Corporation	BKH
CMS Energy Corporation	CMS
Consolidated Edison, Inc.	ED
Eversource Energy	ES
FirstEnergy Corporation	FE
Hawaiian Electric Industries, Inc.	HE
New Jersey Resources Corporation	NJR
Northwest Natural Gas Company	NWN
NorthWestern Corporation	NWE
ONE Gas, Inc.	OGS
Portland General Electric Company	POR
PPL Corporation	PPL
Public Service Enterprise Group Inc.	PEG
Sempra Energy	SRE
South Jersey Industries, Inc.	SJI
Southwest Gas Corporation	SWX
Spire, Inc.	SR

Q. Why do you believe that net operating income is an appropriate screening criterion?

A. In establishing my proxy group, I relied on the percentage of net operating income derived from regulated operations instead of the percentage of total revenue derived from regulated operations because net operating income is more representative of the contribution of that business segment to earnings and the corporation's overall financial position. Specifically, a significant portion of gas and electric utility company revenue is derived from the costs of purchased gas, purchased fuel, and purchased power, which, in most

cases, are recoverable through tracking mechanisms and do not, therefore, contribute to earnings. Furthermore, this portion of total revenue can fluctuate considerably based on the cost of gas and other inputs. Therefore, relying exclusively on a revenue screen does not provide a clear or necessarily consistent indicator of the contribution of the regulated utility operations to a company's earnings. Net operating income excludes the cost of purchased commodity and therefore more closely represents the contribution of the business segment to earnings.

Q. Please provide an example of a company that has been included in the proxy group because net operating income was used instead of total revenue as a screening criterion.

A. New Jersey Resources ("NJR") would have been excluded from the Combined Utilities Proxy Group if the percentage of total revenue derived from regulated operations were used as a screening criterion instead of the percentage of net operating income derived from regulated operations. NJR has an Energy Service segment that provides unregulated, wholesale natural gas to customers that include natural gas distribution companies, industrial companies, electric generators natural gas/liquids processors, retail aggregators, wholesale marketers, and natural gas producers across the US Coast and Canada.[44] In 2017, the Energy Service segment had operating revenues of approximately $1.46 billion.[45] When compared to NJR's total

[44] New Jersey Resource Corporation 2017 Form 10-K, page 11.
[45] New Jersey Resource Corporation 2017 Form 10-K, page 52.

operating revenue of approximately $2.27 billion, it is clear that NJR's percentage of revenue derived from regulated operations would not meet the revenue screening criterion.[46] However, Energy Service's 2017 operating revenue consisted of $1.44 billion in natural gas purchases, which are passed through to customers at cost.[47] Therefore, the Energy Service segment does not represent a large percentage of NJR's net operating income. As discussed above, net operating income is the more appropriate screening criterion because it better approximates a business segment's contribution to earnings and the corporation's overall financial position. For example, NJR operates a large natural gas distribution system in New Jersey and is generally regarded as a gas distribution company. The Energy Services segment of NJR represents a large percentage of the company's operating revenue but represents a small percentage of net operating income. NJR's regulated operations contribute a larger portion to the company's earnings similar to KEDNY and KEDLI and therefore should be included in the Combined Utility Proxy Group.

Q. Do you believe that the 20 companies in your Combined Utility Proxy Group constitute a sufficiently large proxy group?

A. Yes, I do. The analyses performed in estimating the ROE are more likely to be representative of the subject utility's cost of equity to the extent that the chosen proxy companies are fundamentally comparable to the subject

[46] New Jersey Resource Corporation 2017 Form 10-K, page 74.
[47] New Jersey Resource Corporation 2017 Form 10-K, page 52.

utility. Because all analysts use some form of screening process to arrive at a proxy group, the group, by definition, is not randomly drawn from a larger population. Consequently, there is no reason to place more reliance on the quantitative results of a larger and more dissimilar proxy group simply by virtue of the resulting larger number of observations.

Q. Has the Commission typically relied on similar screening criteria when estimating the ROE?

A. Yes. The Commission has typically relied on screening criteria that are similar to those that I have used to develop my proxy groups. The proxy group that is typically relied on by the Commission is composed of a large group of dividend-paying companies with investment grade bond ratings and regulated revenues of at least 70 percent that are not subject to merger-related or corporate restructuring activities.[48] For the reasons noted above and discussed throughout my Direct Testimony, a proxy group based on these somewhat less selective criteria may be less comparable to the Companies than the two proxy groups I have relied on and therefore may not produce appropriate estimates of the Companies' required ROE.

[48] See, e.g., Case 13-E-0030, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service, Testimony of Craig E. Henry, at 14-16.

Q. Why is it appropriate to include natural gas distribution operations in the screening criteria for the proxy group?

A. Because KEDNY and KEDLI are natural gas distribution companies, a proxy group that recognizes the risks of natural gas distribution operations more closely approximates the risk profiles of KEDLI and KEDNY.

Q. Did you consider additional proxy groups?

A. Yes, I did. Because the Companies are natural gas distribution companies, I also considered a proxy group that is comprised entirely of natural gas distribution companies. In developing that group, I began with the companies that Value Line classifies as "Natural Gas Distribution Companies." That group is currently comprised of 10 domestic U.S. utilities. I applied similar criteria as were used in the development of the Combined Utility Proxy Group:

- I eliminated the companies that are not covered by at least two utility industry equity analysts;
- I eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's because such companies do not have a similar financial risk profile to that of the Companies;
- I eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one

source because such characteristics are incompatible with the DCF model;

- To ensure that the proxy group consists of companies that are primarily regulated utilities, I eliminated companies with less than 70 percent of total operating income derived from regulated utility operations;
- To ensure that the proxy group consists of entities that are primarily natural gas utilities, I eliminated companies that derive less than 50 percent of total operating income from regulated natural gas distribution operations; and
- I eliminated companies known to be party to a merger, acquisition, or other transformational transaction as such activities may have a temporary effect on such companies' stock prices and projections unrelated to the overall cost of capital.[49]

Q. What is the composition of your Natural Gas Proxy Group?

A. My proxy group consists of the companies presented in Figure 8.

[49] For consistency with the Combined Utility Proxy Group, I also eliminated companies that own generation comprising greater than 60 percent of the Company's MWh sales to ultimate customers.

Figure 8: Natural Gas Proxy Group

Company	Ticker
Atmos Energy Corporation	ATO
New Jersey Resources Corporation	NJR
Northwest Natural Gas Company	NWN
ONE Gas, Inc.	OGS
South Jersey Industries, Inc.	SJI
Southwest Gas Corporation	SWX
Spire, Inc.	SR

Q. Is a proxy group of seven companies a reasonable size to estimate the cost of equity?

A. While I recognize that the Natural Gas Proxy Group is somewhat limited in size relative to the Combined Utility Proxy Group, because KEDNY and KEDLI are natural gas distribution utilities, the Natural Gas Proxy Group may be more risk comparable to the Companies than the Combined Utility Proxy Group. Therefore, I have considered the range of results established using both the Natural Gas Proxy Group and the Combined Utility Proxy Group.

VI. COST OF EQUITY ESTIMATION

Q. Please briefly discuss the ROE in the context of the regulated Rate of Return.

A. The rate of return ("ROR") for a regulated utility is based on its weighted average cost of capital, in which the costs of the individual sources of capital are weighted by their respective percentages of total capitalization of the

utility. The ROE included in the ROR is weighted by the percentage of common equity in the regulated utility's ratemaking capital structure.

Q. How is the required ROE determined?

A. While the cost of debt can be directly observed, the cost of equity and the required ROE are market-based and, therefore, must be estimated based on observable market information. The required ROE is determined by using one or more analytical techniques that rely on market data to quantify investor expectations regarding the range of required equity returns. Informed judgment is applied, based on the results of those analyses, to determine where within the range of results the cost of equity for a company falls. As a general proposition, the key consideration in determining the cost of equity is to ensure that the methodologies employed reasonably reflect investors' views of the financial markets, the proxy group companies, and the subject company's risk profile.

Q. What methods did you use to determine the Companies' cost of equity?

A. Consistent with Commission precedent, I used the DCF model and CAPM as the primary approaches. In establishing my recommended ROE, I relied on a multi-stage form of the DCF model, and, consistent with the Commission's stated preference, I used both the traditional form of the CAPM as well as the Zero-Beta form of that model. In both forms of the

CAPM, I incorporated a forward-looking measure of the Market Risk Premium.

Q. Why do you believe it is important to use more than one analytical approach?

A. Because the cost of equity is not directly observable, it must be estimated based on both quantitative and qualitative information. When faced with the task of estimating the cost of equity, analysts and investors are inclined to gather and evaluate as much relevant data as reasonably can be analyzed. As a result, a number of models have been developed to estimate the cost of equity. For that reason, I use multiple approaches to estimate the cost of equity. As a practical matter, however, all of the models available for estimating the cost of equity are subject to limiting assumptions or other methodological constraints. Consequently, many finance texts recommend using multiple approaches when estimating the cost of equity. For example, Copeland, Koller, and Murrin[50] suggest using the CAPM and Arbitrage Pricing Theory model, while Brigham and Gapenski[51] recommend the CAPM, DCF, and "bond yield plus risk premium" approaches.[52]

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[50] Tom Copeland, Tim Koller and Jack Murrin, Valuation: Measuring and Managing the Value of Companies, 3rd Ed. (New York: McKinsey & Company, Inc., 2000), at 214.

[51] Eugene Brigham, Louis Gapenski, Financial Management: Theory and Practice, 7th Ed. (Orlando: Dryden Press, 1994), at 341.

[52] While it has historically been my practice to present the results of a bond yield plus risk premium approach in the context of estimating a reasonable ROE, I have not done so in this case to limit the number of contested issues. The result of such an analysis, however, would suggest more weight should be placed on the CAPM ROE determinations.

Q. How are current market conditions affecting the results of the DCF and CAPM models?

A. As discussed in Section IV, there is concern that current capital market conditions, characterized by historically low Treasury bond yields, are causing utility stocks to be overvalued, thereby reducing the dividend yields in the DCF model. Consequently, the results of the DCF model are understating the forward-looking cost of equity. The CAPM method offers some balance to the sensitivity of the DCF model to low Treasury bond yields. However, low interest rates also impact the CAPM in two ways: (1) if the risk-free rate is based on historical average yields on Treasury bonds, it understates the forward-looking risk-free rate, and (2) if the market risk premium is based on historical returns on large company stocks minus the historical average risk free rate, it understates the forward-looking market risk premium. To adjust for these shortcomings, the risk-free rate in the CAPM analysis should also consider projected yields on Treasury bonds, and the market risk premium should be based on a forward-looking computation of the expected return on the total market less the risk-free rate. Market risk premiums based on long-term historical averages are unresponsive to movements in interest rates and would likely understate the market risk premium and, accordingly, the cost of equity.

Q. What are your conclusions about the results of the DCF and CAPM models?

A. The results of both models have been affected by market conditions and, with traditional data inputs, have a tendency to underestimate the current cost of equity. The DCF model is less reliable in current market conditions because dividend yields for utilities are low and not expected to remain at current levels. As discussed in Section IV, reports from Value Line and Edward Jones demonstrate the expected change in market conditions implies that the ROE calculated using historical market data in the DCF model may understate the forward-looking cost of equity. The results from the CAPM are also affected by the current artificially low yields on Treasury bonds. The use of projected yields on Treasury bonds in the CAPM produces returns that are more reflective of the market conditions that investors expect during the period that the Companies' rates will be in effect. Therefore, properly specified, the CAPM may be a more reliable model in current market conditions than the DCF. Given the sensitivity of each model to market conditions and considering the expectation for changes in those conditions, it is appropriate to equally weight the results of the DCF and CAPM models.

A. DISCOUNTED CASH FLOW MODEL

Q. Please describe the DCF approach.

A. The DCF approach is based on the theory that a stock's current market price represents the present value of all expected future cash flows. In its most general form, the DCF model is expressed as follows:

$$P_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_n}{(1+r)^n} \qquad [1]$$

Where P_0 represents the current market stock price, D_1 … D_n are all expected future dividends, and r is the discount rate, or required ROE. As discussed below, I have not included the constant growth form of the DCF model, but instead have focused on a multi-stage form of the DCF model.

Q. Please generally describe the DCF model you relied on.

A. The multi-stage DCF model is an extension of the constant growth form that enables the analyst to specify growth rates over multiple stages. As with the constant growth form of the DCF model, the multi-stage form defines the cost of equity as the discount rate that sets the current price equal to the discounted value of future cash flows. A multi-stage DCF model addresses the possibility that mean five-year growth rates may not be reasonable in perpetuity and that payout ratios could vary over time.

Q. Please describe the structure of the multi-stage DCF model.

A. The multi-stage DCF model that I have used sets the proxy company's current stock price equal to the present value of future cash flows received

over three time periods. In all three periods, cash flows are equal to the annual dividend payments that stockholders receive. The first period is a short-term growth period that consists of the first five years; the second period is a transition period from the short-term growth rate to the long-term growth rate that occurs over five years (*i.e.*, years 6 through 10); and the third period is a long-term growth period that begins in year 11 and continues in perpetuity. The ROE is then calculated as the rate of return that results from the initial stock investment and the dividend payments over the analytical period.

Q. Has the Commission relied on a multi-stage DCF model in prior cases?

A. Yes, the Commission has relied on a two-stage form of the DCF model in prior cases.[53] The two-stage model that the Commission has relied on and the multi-stage model that I rely on both define the cost of equity as the discount rate that sets the current stock price equal to the discounted value of future cash flows that are expressed as projected dividends. Both models project dividends using growth rates over multiple periods.

[53] See Case 10-E-0362, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc. for Electric Service, Order Establishing Rates for Electric Service, (issued June 17, 2011) ("2011 O&R Rate Order"), at 68-69.

Q. Is the multi-stage form of the DCF model consistent with the intent of the two-stage model relied upon by the Commission?

A. Yes. Both the construction of the multi-stage model and the underlying assumptions are consistent with the two-stage model relied upon by the Commission. The constant growth DCF model assumes the expected growth rate will be constant in perpetuity. The multi-stage forms of the DCF model, including both the two-stage model that the Commission has relied upon and the multi-stage form of the model that is relied on in my analysis, recognize short and long-term growth prospects.

Q. Does the multi-stage form of the DCF model offer improvements over the two-stage model traditionally relied upon by the Commission?

A. Yes. The general form of the two-stage model relied upon by the Commission involves a near-term growth stage based on projected dividends and a long-term growth stage employing an estimated long-term growth rate in dividends.[54] The Commission's application of a two-stage DCF assumes that a company's growth abruptly shifts to a long-run growth state after the initial five-year period. In contrast, the multi-stage model relies on growth rates over three periods, as described above. The multi-

[54] See, e.g., Case 10-E-0362, Case 06-E-1433, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc., for Electric Service, Case 08-E-0539, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service.

stage form of the DCF model provides for a gradual transition to a company's expected long-term growth, whereas the two-stage DCF model assumes the transition from short to long-term growth occurs in one year.

Q. What market data did you use to calculate the current stock price in your DCF model?

A. The stock prices that I relied on in my DCF model are based on the average market closing prices for the proxy companies over the three months ended February 28, 2019

Q. What growth rates did you rely on in the multi-stage DCF model?

A. As shown in Exhibit __ (AEB-1), I began with the current annualized dividend as of February 28, 2019 for each proxy group company. In the first stage of the model, the current annualized dividend is escalated based on the average of the three-to five-year earnings growth estimates reported by First Call, Zacks, and Value Line. For the third stage of the model, I relied on long-term projected growth in Gross Domestic Product ("GDP"). The second stage growth rate is a transition from the first stage growth rate to the long-term growth rate on a geometric average basis.

Q. Why do you believe that earnings growth rates are the appropriate growth rates in the DCF model?

A. Earnings are the fundamental driver of a company's ability to pay dividends; therefore, earnings growth is the appropriate measure of a company's long-term growth. In contrast, changes in a company's dividend payments are based on management decisions related to cash management and other factors. For example, a company may decide to retain certain earnings rather than include those earnings in a dividend issuance. Therefore, dividend growth rates are less likely than earnings growth rates to reflect investor perceptions of a company's growth prospects.

Q. Is there support for the use of analysts' earnings growth estimates in the DCF model?

A. Yes, there is significant academic support for the use of analysts' earnings growth rates. In addition, the majority of the data that are publicly available to investors sets forth analysts' projections of earnings growth rates. Value Line is the only publication I am aware of that provides projected dividend growth rates.

Q. Please summarize the academic research on growth rates and stock valuation.

A. The relationship between various growth rates and stock valuation metrics has been the subject of much academic research. Many published articles specifically support the use of analysts' earnings growth projections in the DCF model in general, as well as for a method of calculating the expected market risk premium. While this article is focused on the calculation of the CAPM, Dr. Robert Harris demonstrates that financial analysts rely on earnings forecasts (referred to in the article as "FAF") and the use of a constant growth DCF formula to estimate the expected market risk premium.[55] Dr. Harris made the following observations:

> […] a growing body of knowledge shows that analysts' earnings forecasts are indeed reflected in stock prices. Such studies typically employ a consensus measure of FAF calculated as a simple average of forecasts by individual analysts.[56]
>
> *****
>
> Given the demonstrated relationship of FAF to equity prices and the direct theoretical appeal of expectational data, it is no surprise that FAF have been used in conjunction with DCF models to estimate equity return requirements.[57]

Dr. Harris's work demonstrates that analysts rely on earnings as the appropriate measure of growth in the DCF model. Professors Carleton and

[55] Robert S. Harris, Using Analysts' Growth Forecasts to Estimate Shareholder Required Rates of Return, Financial Management, Spring 1986, at 66.
[56] *Id*. at 59.
[57] *Id*. at 60.

Vander Weide also performed a study to determine whether projected earnings growth rates are superior to historical measures of growth in the implementation of the DCF model.[58] Although the purpose of that study was to "investigate what growth expectation is embodied in the firm's current stock price,"[59] the authors clearly indicate the importance of earnings projections in the context of the DCF model., concluding that:

> […] our studies affirm the superiority of analysts' forecasts over simple historical growth extrapolations in the stock price formation process. Indirectly, this finding lends support to the use of valuation models whose input includes expected growth rates.[60]

Similarly, Harris and Marston presented "estimates of shareholder required rates of return and risk premia which are derived using forward-looking analysts' growth forecasts."[61] In addition to other findings, Harris and Marston reported that,

> […] in addition to fitting the theoretical requirement of being forward-looking, the utilization of analysts' forecasts in estimating return requirements provides reasonable empirical results that can be useful in practical applications.[62]

The Carleton and Vander Weide study was updated to determine whether the finding that analysts' earnings growth forecasts are relevant in the stock

[58] James H. Vander Weide, Willard T. Carleton, *Investor growth expectations: Analysts vs. history,* The Journal of Portfolio Management, Spring 1988.
[59] *Id*. at 78.
[60] *Id*. at 82.
[61] Robert S. Harris, Felicia C. Marston, *Estimating Shareholder Risk Premia Using Analysts' Growth Forecasts*, Financial Management, Summer 1992.
[62] *Id*. at 63.

valuation process still holds. The results of that updated study continued to demonstrate the importance of analysts' earnings forecasts, including the application of those forecasts to utility companies.[63] Similarly, Brigham, Shome and Vinson noted that "evidence in the current literature indicates that (1) analysts' forecasts are superior to forecasts based solely on time series data; and (2) investors do rely on analysts' forecasts."[64]

Q. What is your opinion of the Commission's historical reliance on dividend per share growth rates during the initial five-year term of its Two-Stage DCF?

A. Sole reliance on Value Line projections of dividend per share growth is not appropriate for several reasons. First, the use of only dividend growth rates ignores the substantial body of academic research demonstrating that earnings growth rates are the most relevant factor in stock price valuation.[65] Second, projections of dividend growth, which would not include growth in retained earnings, only measure a portion of a company's growth. Therefore, earnings growth projections are more complete estimates of total company growth than projected dividend growth rates. Finally, Value Line's 4-6 year projections are not consensus estimates, but reflect the

[63] Advanced Research Center, *Investor Growth Expectations*, Summer, 2004.

[64] The Risk Premium Approach to Measuring a Utility's Cost of Equity, Financial Management, Spring 1985.

[65] The Recommended Decision ("RD") in the GFP indicates that the Telecommunications Group, which included Commission Staff, supported the use of earnings per share growth in the DCF models employed to estimate the ROE (RD at 9).

viewpoint of a single analyst. Therefore, the Commission's models, which have historically relied only on projected dividend per share growth rates from Value Line, reflect the growth expectations of a single analyst in the first stage of the model. In contrast, there are several consensus estimates of projected earnings per share growth rates that are publicly available and widely used by investors, including Zacks Investment Research and Thomson First Call. Each of these consensus forecasts considers the growth expectations for each company based on the expectations of multiple analysts. It is not reasonable to exclude these timely and widely-available sources of information from the analysis when these real-time sources have become the more common data points relied on by investors.

Q. How did you calculate the long-term GDP growth rate?

A. As shown in Exhibit __ (AEB-2), the long-term growth rate of 5.56 percent is based on the real GDP growth rate of 3.22 percent from 1929 through 2018,[66] and a projected inflation rate of 2.27 percent.[67] The projected rate of inflation is based on three measures: (1) the average long-term projected growth rate in the Consumer Price Index ("CPI") of 2.20 percent, as reported by Blue Chip Financial Forecasts;[68] (2) the compound annual growth rate of the CPI for all urban consumers for 2029-2050 of 2.31 percent as projected by the Energy Information Administration ("EIA") in

[66] U.S. Department of Commerce, Bureau of Economic Analysis, National Income and Product Accounts Tables, Table 1.1.6, March 8, 2019.
[67] (1 + 0.0322) x (1 + 0.0227) -1 = 0.0556 or 5.56%.
[68] Blue Chip Financial Forecasts, Vol. 37, No. 12, December 1, 2018, at 14.

the Annual Energy Outlook 2018; and (3) the compound annual growth rate of the GDP chain-type price index for 2029-2050 of 2.29 percent, also reported by the EIA in the Annual Energy Outlook 2018.[69]

Q. Why is the long-term GDP growth rate a reasonable estimate of long-term growth in the multi-stage DCF model?

A. Long-term estimates of GDP growth are commonly used in regulatory proceedings as a proxy for the long-term growth rate in the multi-stage DCF analysis. That application is based on the common theoretical assumption that, over the long-run, all companies in the economy will tend to grow at the same constant rate. That assumption is designed to address the uncertainty associated with estimating individual company growth rates over very long time horizons and is not meant to suggest that company growth rates in the economy will indeed converge in practice over any given period.

Q. Is your calculation of GDP growth consistent with the way in which other analysts' compute estimates of long-term GDP growth?

A. Yes. Investors understand that the U.S. economy goes through cycles of growth and contraction. Therefore, it is appropriate to consider the longest period possible to measure historical real growth in GDP. This view is consistent with Morningstar's explanation about measuring GDP growth:

[69] U.S. Energy Information Administration, Annual Energy Outlook 2018, Table 20.

> Growth in real GDP (with only a few exceptions) has been reasonably stable over time; therefore, its historical performance is a good estimate of expected long-term future performance. By combining the inflation estimate with the real growth rate estimate, a long-term estimate of nominal growth is formed.[70]

Furthermore, Morningstar supports the use of long-term historical data:

> The 87-year period starting with 1926 is representative of what can happen: it includes high and low returns, volatile and quiet markets, war and peace, inflation and deflation, and prosperity and depression. Restricting attention to a shorter historical period underestimates the amount of change that could occur in a long future period. Finally, because historical event-types (not specific events) tend to repeat themselves, long-run capital market return studies can reveal a great deal about the future. Investors probably expect "unusual" events to occur from time to time, and their return expectations reflect this.[71]

Q. How does your estimate of long-term GDP growth compare with investor expectations of long-term utility industry growth rates?

A. The Commission has traditionally relied on Bank of America Merrill Lynch's ("BAML") market return calculations in estimating a company's ROE using the CAPM. Exhibit __ (AEB-3) includes the relevant pages from the BAML *Quantitative Profiles* reports for December 2018 through February 2019. BAML derives the Implied Return using a multi-stage Dividend Discount Model ("DDM"). As shown in Exhibit __ (AEB-3), the Implied Returns for the utility industry were 9.70 percent to 9.90 percent

[70] Ibbotson and Associates, Stocks, Bonds, Bills and Inflation, 1926-2012, 2013 Valuation Yearbook, at 52.

[71] *Id*., at 59.

each month.[72] For those same months, the average dividend yield for the utility industry was 3.30 percent to 3.50 percent.[73] Because the total return consists of capital appreciation (*i.e.*, growth) and dividend yield, that data suggest an expected utility growth rate of approximately 6.35 percent, which is considerably higher than the long-term growth estimate of 5.56 percent used in my multi-stage DCF analysis.

Q. How does your estimate of long-term growth differ from the estimate the Commission has traditionally relied on?

A. The final stage of both the two-stage DCF model that the Commission has relied on and my multi-stage DCF model extends into the future indefinitely. My long-term growth estimate reflects investors' long-term growth expectations for the period from 2029 through 2050. Therefore, the third stage of my multi-stage DCF model reflects investor growth expectations beginning in the first year of the third stage of the model. In contrast, the growth estimate for the two-stage model that the Commission has typically relied on is based on short-term growth rate forecasts. The use of the sustainable growth rate, calculated using Value Line's published projections, provides an estimate of growth four- to six-years in the future. Relying on the sustainable growth rate in perpetuity in the second stage of a two-stage DCF model does not provide a long-run estimate of growth.

[72] Bank of America Merrill Lynch, *Quantitative Profiles,* December 12, 2018, at 58; January 11, 2019, at 57; and February 8, 2019, at 58.
[73] *Id*.

Rather, the use of the sustainable growth rate assumes that the short-term estimate for the four- to six-year period from the Value Line report date is sustained in perpetuity.

In contrast, the long-term growth rate in my DCF analyses reflects both economic forecasts and market-derived projections of inflation over the longest available time period (30 or more years). Those estimates of long-term inflation expectations are combined with the long-term average historical real GDP growth rate to calculate an expected nominal GDP growth rate. Consequently, the long-term growth estimate in my multi-stage DCF model represents investors' and economists' views of nominal long-term GDP growth well beyond the time horizon reflected in the four- to six-year Value Line sustainable growth estimate relied on by the Commission in prior cases.

Q. Does the use of Value Line data to develop the sustainable growth rate address concerns about growth rate bias?

A. No. The sustainable growth rate is the sum of retention growth plus an SV factor,[74] calculated using Value Line data. As such, the sustainable growth rate estimate that the Commission has relied upon is based on a single

[74] Retention growth is the product of the expected earned ROE and the retention ratio (one minus the dividend payout ratio). The SV factor employs an estimate of the market-to-book ratio and the expected expansion rate of outstanding shares of common stock in the future.

analyst's viewpoint of a company's projected four- to six-year growth prospects.

Q. Are there other problems with the use of the sustainable growth rate as an estimate of long-term growth?

A. Yes. Using the sustainable growth rate to estimate the long-term growth of the company uses a very narrowly-defined set of short-term projections based on Value Line data. Specifically, it relies on the following assumptions: (1) projected dividends for year 2; (2) projected dividends for years 4-6; (3) projected earnings for years 4-6; (4) projected book value for year 2; (5) projected book value for years 4-6; (6) current estimate of actual outstanding shares of stock; (7) projected shares of outstanding stock for years 4-6; and (8) current three-month stock price. Each of these assumptions is estimated at most for 6 years into the future. As defined using these assumptions, the sustainable growth rate, which is applied over the long-term in the Commission's two-stage model, does not consider any actual long-term forecasts for the specific company or the economy.

Q. What is your conclusion regarding the methodology typically relied on by the Commission to estimate the sustainable growth rate in the two-stage DCF model?

A. There are several reasons why the Commission's sustainable growth rate should not be relied on in the two-stage DCF model. First, the sustainable growth rate is not a long-term measure of growth and as such should not be

applied in perpetuity in the second stage of the model. Second, the exclusive use of Value Line data, which is a single analyst's viewpoint, to establish the sustainable growth rate assumes that investors do not consider any of the other financial information that is widely available when establishing future dividend expectations. In addition, the sustainable growth rate calculation includes Value Line's ROE projections as an input, implicitly accepting them as reasonable. However, Value Line's ROE projections are often significantly different from the ROE estimates produced by the two-stage DCF model. Finally, the Commission's sustainable growth rate methodology implicitly assumes that investors establish long-term growth expectations based entirely on short-term, company-specific projections. It is unreasonable to conclude that investors would ignore the expectations of long-term macroeconomic growth in establishing the long-term growth estimates for an electric or natural gas distribution utility or any other company.

Q. Have other regulatory Commissions reconsidered the use of the sustainable growth rate in the ROE estimation methodology?

A. Yes. The FERC's long-standing methodology for setting the ROE in utility proceedings was to rely on a single stage DCF model that used two estimates of short-term growth: 1) analysts' estimates of earnings growth, as published by IBES and; 2) the sustainable growth rate, calculated using the (b*r) + (s*v) components that are used by this Commission. The FERC acknowledged that the sustainable growth rate is not a measure of long-term

growth, but is another estimate of short-term growth similar to analysts' earnings projections.

In Opinion No. 531, the FERC determined that it was appropriate to move from a constant growth DCF methodology to a two-stage DCF model for public utility rate cases.[75] In moving to the two-stage DCF, FERC now relies on analysts' estimates of earnings growth in the short-term and a long-term GDP growth rate as the measure of growth in the second stage. The FERC's two-stage model does not rely on a sustainable growth calculation. This was unchanged by the recently proposed methodology which considers the DCF model, along with three other methodologies.[76]

Q. What are the results of your DCF analyses?

A. As shown in Exhibit __ (AEB-1), the multi-stage DCF analysis based on a three-month average stock price and a range of near-term growth rate assumptions produces a mean ROE of 9.20 percent for the Combined Utility Proxy Group and 9.04 percent for the Natural Gas Proxy Group.

[75] Opinion No. 531 147 FERC ¶ 61,234 (June 19, 2014).
[76] FERC, Docket No. EL 11-66-001, et al., Order Directing Briefs, issued October 16, 2018, at para. 32.

Q. Does the multi-stage DCF model discussed above address your concern about low dividend yields?

A. No. While the multi-stage DCF model provides for changes in growth over time, it does not address the low current dividend yields for utility stocks. As discussed in Section IV, currently low dividend yields are causing the DCF model to understate the cost of equity.

Q. What are your conclusions about the results of the DCF model?

A. The results of the DCF model are currently influenced by the low dividend yields on utility stocks due to the low interest rate environment. As discussed previously, one primary assumption of the DCF model is the dividend yield. To the extent these dividend yields are abnormally low and not sustainable, it is important to recognize that the results of the DCF model are understated.

B. CAPITAL ASSET PRICING MODEL

Q. Please briefly describe the Capital Asset Pricing Model.

A. The CAPM is a risk premium approach that estimates the market cost of equity for a given security as a function of a risk-free return plus a risk premium (to compensate investors for the non-diversifiable or "systematic" risk of that security). As shown in Equation [2], the CAPM is defined by four components:

$$k_e = r_f + \beta(r_m - r_f) \qquad [2]$$

where:

k_e = the required market ROE

β = Beta coefficient of an individual security

r_f = the risk-free rate of return

r_m = the required return on the market as a whole

In this specification, the term ($r_m - r_f$) represents the market risk premium. According to the theory underlying the CAPM, investors should be concerned only with systematic or non-diversifiable risk because unsystematic risk can be diversified away. Non-diversifiable risk is measured by the Beta coefficient, which is defined as:

$$\beta = \frac{Covariance(r_e, r_m)}{Variance(r_m)} \quad [3]$$

The variance of the market return, noted in Equation [3], is a measure of the uncertainty of the general market, and the covariance between the return on a specific security and the market reflects the extent to which the return on that security will respond to a given change in the market return.

Q. What risk-free rate did you use in your CAPM model?

A. I used three estimates of the yield on Treasury bonds: (1) the current three-month average yield on 30-year Treasury bonds (3.05 percent);[77] (2) the projected 30-year Treasury yield for 2019-2020 (3.28 percent);[78] and (3) the

[77] Bloomberg Professional.

[78] Aspen Publishers, Blue Chip Financial Forecasts, Vol. 38, No. 3 March 1, 2018, at 2.

projected 30-year Treasury yield for the period 2020-2024 (3.90 percent).[79] In determining the security most relevant to the application of the CAPM, it is important to select the term (or maturity) that best matches the life of the underlying investment. As noted by Morningstar:

> The traditional thinking regarding the time horizon of the chosen Treasury security is that it should match the time horizon of whatever is being valued… Note that the horizon is a function of the investment, not the investor. If an investor plans to hold stock in a company for only five years, the yield on a five-year Treasury note would not be appropriate since the company will continue to exist beyond those five years.[80]

Because utility companies represent long-duration investments, it is appropriate to use yields on long-term Treasury bonds as the risk-free rate component of the CAPM. In my view, the 30-year Treasury bond is the appropriate security for that purpose. Because the cost of capital is intended to be forward-looking, it is appropriate to consider projected measures of the market risk premium and interest rates.

Q. Please describe your estimate of the market risk premium used in your CAPM.

A. The estimated market risk premium is based on the expected return on the S&P 500 Index, less the 30-year Treasury bond yield. The expected return on the S&P 500 Index is calculated using a DCF model for all companies in the index based on market capitalization-weighted growth rates and

[79] Aspen Publishers, Blue Chip Financial Forecasts, Vol. 37, No. 12 December 1, 2018, 14..

[80] Morningstar Inc., Ibbotson SBBI 2013 Valuation Yearbook, at 44.

dividend yields. The market risk premium implied by each of the three Treasury yields discussed above is used in the CAPM analysis.

Q. Is your calculation of the market risk premium consistent with the methodology relied upon in previous cases before the Commission?

A. Yes, it is. The Commission previously has relied upon the calculation of a projected market risk premium, based on the difference between the estimated forward-looking required market return for the S&P 500, as provided by BAML, and the risk-free rate.[81] As a practical matter, that approach is consistent with the Market DCF-derived forward-looking market risk premium estimate discussed above (*see also* Exhibit __ (AEB-5).

Q. What Beta coefficient did you use in your CAPM model?

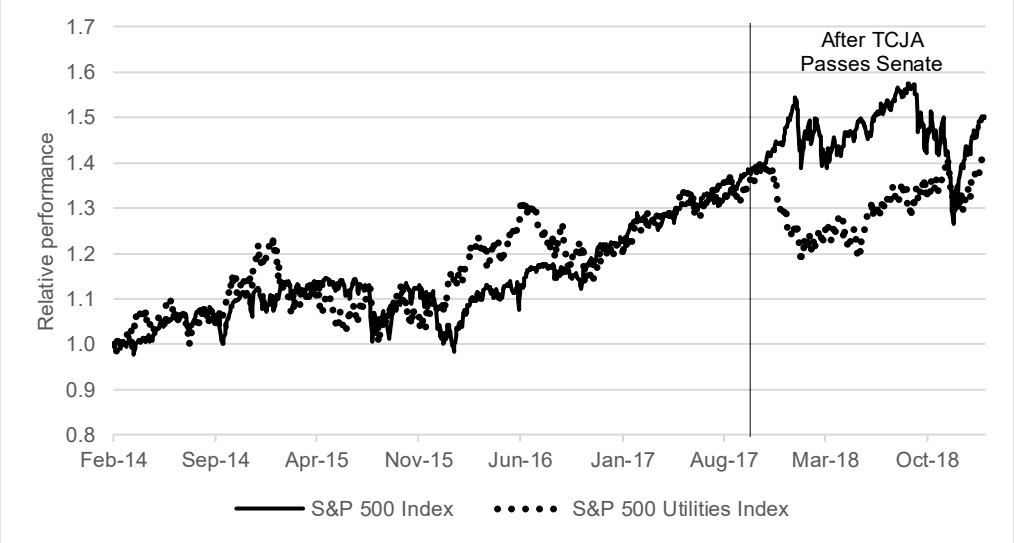
A. I considered the average Beta coefficients for the proxy group companies as reported by Bloomberg and Value Line (*see* Exhibit __ (AEB-6)). The Beta coefficients reported by Bloomberg were calculated using ten years of weekly returns relative to the S&P 500 Index. Value Line's calculation is based on five years of weekly returns relative to the New York Stock Exchange Composite Index.

[81] *See e.g.,* 2011 O&R Rate Order, at 77.

Q. Why did you select a ten-year period to calculate the Beta coefficients from Bloomberg?

A. As I discussed in Section IV, the TCJA has had a significant effect on utility companies. While other industries are able to retain the benefits of a reduced corporate income tax rate, this benefit has largely been passed through to customers by utility companies. This fundamental difference had an effect on investors' view of the utility industry relative to other industries. As shown in Figure 9, after the Senate passed the TCJA on December 2, 2017, utilities significantly deviated from the broader market.

Figure 9: Performance of the Utility Industry Relative to the S&P 500



The TCJA's effect on the utility industry relative to other industries caused a short-term significant shift in the returns on the utility industry relative to the broader market. Over the last three to five years, volatility for the utility industry has been higher than the broader market (as measured by the S&P

500),[82] suggesting higher Beta coefficients for utility companies. However, in short-term calculations of the Beta coefficient, the significant effect of the shift in returns related to the TCJA has outweighed the effect of longer-term measures of relative volatility. As such, to reflect the long-term relationship that suggests utility stocks are less volatile than the broader market (*i.e.* the relative volatility for utility companies has been lower than the S&P 500 over the ten-year measure[83]), I selected a ten-year period to calculate the Beta coefficients from Bloomberg.

Q. Did you consider another form of the CAPM in your analysis?

A. Yes. In prior proceedings, the Commission has relied upon the Zero-Beta CAPM (the form of which is sometimes referred to as the "Empirical CAPM"[84]) in estimating the cost of equity. The Zero-Beta CAPM calculates the product of the adjusted Beta coefficient and the market risk premium and applies a weight of 75.00 percent to that result. The model then applies a 25.00 percent weight to the market risk premium, without any effect from the Beta coefficient. The results of the two calculations are summed, along with the risk-free rate, to produce the Zero-Beta CAPM result, as noted in Equation [4] below:

$$k_e = r_f + 0.75\beta(r_m - r_f) + 0.25(r_m - r_f) \qquad [4]$$

where:

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82 *See,* S&P Dow Jones Indices, Equity, S&P 500 Utilities, February 28, 2019.

83 *Id*.

84 *See e.g.,* Roger A. Morin, New Regulatory Finance, Public Utilities Reports, Inc., 2006, at 189.

k_e = the required market ROE

β = Adjusted Beta coefficient of an individual security

r_f = the risk-free rate of return

r_m = the required return on the market as a whole

In essence, the Zero-Beta form of the CAPM addresses the tendency of the "traditional" CAPM to underestimate the cost of equity for companies with low Beta coefficients such as regulated utilities. In that regard, the Zero-Beta CAPM is not redundant to the use of adjusted Betas; rather, it recognizes the results of academic research indicating that the risk-return relationship is different (in essence, flatter) than estimated by the CAPM, and that the CAPM underestimates the "alpha," or the constant return term.[85]

As with the CAPM, my application of the Zero-Beta CAPM uses the forward-looking market risk premium estimates, the three yields on 30-year Treasury securities noted earlier as the risk-free rate, and the Bloomberg and Value Line Beta coefficients. Exhibit __ (AEB-4) show the results of the CAPM models for the Combined Utility and Natural Gas Proxy Groups. The traditional CAPM model results range from 9.96 percent to 10.81 percent. The Zero-Beta CAPM model results range from 10.92 percent to 11.55 percent. The range established by the traditional CAPM model and

[85] *Id*. at 191.

the Zero-Beta CAPM model results is 9.96 percent to 11.55 percent with a mean of 10.87 percent.

C. Weighted Average Results

Q. Please summarize the results of your analysis and your recommended ROE.

A. As shown in Figure 10 (below), I have presented the DCF and CAPM results weighted equally, and with the RD's proposed 2/3 weighting of the DCF and 1/3 weighting of the CAPM.

Figure 10: Weighted Average Analytical Results

Combined Utility Proxy Group			
	Low	**Mean**	**High**
DCF	8.92%	9.20%	9.67%
Mean CAPM	10.66%	10.72%	10.90%
50%/50% DCF/CAPM	9.79%	9.96%	10.29%
67%/33% DCF/CAPM	9.50%	9.71%	10.08%
Natural Gas Proxy Group			
	Low	**Mean**	**High**
DCF	8.59%	9.04%	10.02%
Mean CAPM	10.89%	10.95%	11.12%
50%/50% DCF/CAPM	9.74%	10.00%	10.57%
67%/33% DCF/CAPM	9.35%	9.68%	10.39%

Q. What was the Commission's reasoning for developing its weighting of the DCF and CAPM methodologies in the RD?

A. At the time of the RD, the Commission did not have a significant amount of experience with the CAPM. The RD noted that the Commission had historically used the CAPM as a check on its DCF results, and was somewhat undecided as to "how far the Commission should go in elevating the status of CAPM."[86] The RD opted for a gradual transition towards the CAPM, ultimately settling on a 1/3 weighting, indicating that "proposals have simply not shown that the CAPM should be raised all at once to parity with the DCF analysis in the setting of returns on equity."[87] To the extent that this was a consideration in the RD's weighting determination, the Commission's 25 years of experience with the CAPM since that time provides a sound basis for altering the weighting of the two ROE methodologies.

Q. Please summarize your conclusion regarding the relative weighting of the CAPM and DCF results.

A. While the RD proposed the 2/3 weighting on the DCF, the weightings and methodologies used to estimate the ROE were left open for additional consideration in future rate proceedings. Since then, the Commission has employed the CAPM as one component of the formula used to develop ROE estimates. There does not appear to be any reason to infer that the

[86] RD, at 27.
[87] *Id*.

Commission has less confidence in the results of the CAPM than those of the DCF. The conditions that warranted the Commission's GFP inquiry and the subsequent RD in the early 1990s exist again today with DCF results considerably lower than those from other models, such as the CAPM, as well as returns authorized in other jurisdictions. Finally, to the extent that dividend yields are low relative to historical levels and could increase as yields on government bonds rise, the DCF model is likely to underestimate the cost of equity. Therefore, it is reasonable to apply equal weighting to the DCF and CAPM methods when determining the ROE for the Companies.

Q. Are the assumptions used in the CAPM less reliable than the assumptions used in the DCF model?

A. Not necessarily. As discussed previously, the CAPM relies on a risk-free rate, Beta and the MRP. The risk-free rate is readily observable and can be projected for the forward-looking period. Beta is estimated using the historical relationship between the risk of the stock and the overall market. Finally, the market risk premium, while not observable, can be estimated for the forward-looking period. My testimony discusses how the dividend yield has been affected by market conditions and therefore, while this assumption may be easy to calculate using historical data, it is not representative of forward-looking market conditions. Therefore, while the CAPM is often criticized as relying on unobservable assumptions, currently

the dividend yield in the DCF model is not reflective of projected market conditions.

VII. REGULATORY AND BUSINESS RISKS

A. RISK ASSESSMENT

Q. Have you performed an analysis of the level of regulatory support that the Companies receive in New York as compared to the proxy group companies?

A. Yes. I conducted an analysis of the regulatory protections that are in place for KEDNY and KEDLI compared with those for the operating utility companies held by the proxy group companies. The results of my analysis are presented in Exhibit ___ (AEB-7). Specifically, I examined the following factors that affect the business risk of KEDNY and KEDLI and the proxy group companies: (1) test year convention; (2) fuel cost recovery; (3) revenue decoupling; and (4) capital cost recovery.

As shown in Exhibit ___ (AEB-7), the majority of the operating companies (*i.e.,* 36 out of 63) in the proxy groups provide service in jurisdictions that allow the use of a fully or partially forecast test year. All of the operating companies held by the proxy groups are allowed to pass through fuel costs and purchased power costs directly to customers, so that the utility does not incur any risk associated with fuel or purchased power costs. It is important to recognize that fuel and purchased power costs typically account for 50 –

60 percent of the total operating costs for a regulated utility. Like KEDNY and KEDLI, 60 percent of the operating utilities held by the proxy group (*i.e.*, 38 out of 63) have revenue decoupling mechanisms or weather normalization adjustment clauses that allow them to break the link between customer usage and revenues. Finally, approximately 56 percent of the operating utilities held by the proxy group (35 out of 63) have capital cost recovery mechanisms that allow them to recover capital investments that are placed into service between rate cases.

Q. Based on these analyses, what is your conclusion regarding the level of regulatory support for KEDNY and KEDLI relative to that of the proxy group companies?

A. My conclusion is that KEDNY and KEDLI have comparable regulatory protection to the proxy group companies. While the Commission has been a leader in implementing mechanisms that reduce the business risk of regulated utilities in New York, many other jurisdictions have taken similar steps in more recent years. A November 2015 report published by the Edison Electric Institute indicates that more and more jurisdictions have moved toward the use of forecast test years since the 2013 survey[88]; fuel

[88] Edison Electric Institute, "Alternative Regulation for Emerging Utility Challenges: 2015 Update," prepared by Pacific Economics Group, November 11, 2015, at 32. (EEI report states: "The ranks of US jurisdictions that allow the use of forward test years have swollen and now encompasses about half of the total. Since our 2013 survey, electric utilities in Pennsylvania have successfully used FTYs and utilities in Arkansas and Indiana have received legislative authorization for their use. Forward test years are the norm in Canadian regulation.")

cost recovery mechanisms have been ubiquitous for many years; revenue decoupling and weather normalization clauses have been approved in many states, especially where declining usage per customer is a concern[89]; and many states have approved capital tracking mechanisms that reduce the regulatory lag associated with significant investments to enhance reliability, service quality and safety.[90]

Q. Are there other risks to the Companies that are specific to New York utility regulation?

A. Yes. In addition to the low equity returns that are typically authorized by the Commission for New York's gas and electric utilities (in 2018 average authorized ROEs in New York were 71 basis points below the national average for gas and electric utilities),[91] New York utilities are subject to strictly-enforced customer service quality, electric reliability, and gas safety measures where the utility is required to achieve predetermined

[89] *Id*., at 21. (EEI report states: "In the electric utility industry, decoupling has been favored in states that strongly support DSM. Since our 2013 survey, decoupling has been adopted for electric utilities in Connecticut, Maine, Minnesota, and Washington state. Decoupling is the most widespread means of relaxing the revenue/usage link for gas distributors. This reflects the fact that gas distributors often experience declining average use and that this has been driven chiefly by external forces.")

[90] *Id*., at 7. (EEI report states: "It can be see that the precedents are numerous and continue to grow. This is the most widely used Altreg tool in the United States. For electric utilities, trackers for emissions controls, generation capacity, advanced metering infrastructure, and general system modernization have been especially common in recent years. Trackers for gas distributors typically address the cost of replacing old case iron and bare steel mains.")

[91] Source: Regulatory Research Associates. 2018 average authorized ROEs for states other than New York (excluding limed issue riders) were 9.61 percent compared to an average authorized ROE of 8.90 percent in New York.

performance benchmarks, or be subject to a negative revenue adjustment for any shortfall.

Q. Please describe the Companies' customer service quantity and gas safety measures.

A. The Companies are subject to a number of customer service quality and gas safety performance metrics for which negative revenue adjustments are incurred for specific levels of non-performance. Figure 11 below summarizes the Companies' potential annual exposure to these negative revenue adjustments.

Figure 11: Summary of Service Quality and Gas Safety Performance Metrics[92]

Performance Metric	Maximum Annual Negative Revenue Adjustment KEDLI	Maximum Annual Negative Revenue Adjustment KEDNY
Customer Service Quality	$9.9 million	$11.7 million
Gas Safety	150 Pre-Tax Base Points	150 Pre-Tax Base Points

Q. Do the negative revenue adjustments associated with these performance metrics differentiate the Companies from the proxy group companies?

A. Yes. They do. The magnitude of the Companies' exposure places the Companies at greater risk than proxy companies on average.

[92] Case 16-G-0058 and 16-G-0059, Joint Proposal dated September 7, 2016.

B. Regulatory Environment

Q. Please explain how the regulatory framework affects investors' risk assessments.

A. The ratemaking process is premised on the principle that, for investors and companies to commit the capital needed to provide safe and reliable utility services, the subject utility must have the opportunity to recover invested capital and the market-required return on such capital. Regulatory commissions recognize that because utility operations are capital intensive, regulatory decisions should enable the utility to attract capital at reasonable terms, which balance the long-term interests of investors and customers. In that respect, the regulatory framework in which a utility operates is one of the most important factors considered in both debt and equity investors' risk assessments.

Because investors have many investment alternatives, even within a given market sector, the Companies' authorized returns must be adequate on a relative basis to ensure their ability to attract capital under a variety of economic and financial market conditions. From the perspective of debt investors, the authorized return should enable the Companies to generate the cash flow needed to meet their near-term financial obligations, make the capital investments needed to maintain and expand their systems, and maintain sufficient levels of liquidity to fund unexpected events. This financial liquidity must be derived not only from internally-generated funds, but also from efficient access to capital markets.

From the perspective of equity investors, the authorized return must be adequate to provide a risk-comparable return on the equity portion of the Companies' capital investments. Because equity investors are the residual claimants on the Companies' cash flows (that is, debt interest must be paid prior to any equity dividends), equity investors are particularly concerned with the regulatory framework in which a utility operates and its effect on future earnings and cash flows.

Q. Please explain how credit rating agencies consider the regulatory framework in establishing a company's credit rating.

A. S&P and Moody's both consider the overall regulatory framework in establishing credit ratings. As shown in Figure 12, Moody's establishes credit ratings based on four key factors:

Figure 12: Moody's Rating Factors

Factor	Weighting
Regulatory Framework	25%
Ability to Recover Costs and Earn Returns	25%
Diversification	10%
Financial Strength	40%
Total	100%

Two of these factors (*i.e.*, regulatory framework and the ability to recover costs and earn returns) are based on the regulatory environment such that

half of Moody's overall assessment of business and financial risk for regulated utilities is based upon the regulatory environment.[93] Moody's further subdivides the first two factors, regulatory framework and the ability to recover costs and earn authorized returns, into sub-factors to help "provide more granularity and transparency on the overall regulatory environment, which is the most important consideration for this sector."[94]

With respect to the regulatory framework, Moody's looks for transparency, predictability, and supportiveness of regulatory commissions.[95] For the second factor, ability to recover costs and earn returns, Moody's evaluates the regulatory elements that directly affect the ability of the utility to generate cash flow and service its debt over time.[96] Moody's views the ability to recover costs on a timely basis and to attract debt and equity capital as crucial credit considerations noting that "[t]he inability to recover costs…has been one of the greatest drivers of financial stress in this sector."[97] This is particularly true as utilities are often cash flow negative due to large capital expenditures, so any lack of timely recovery or sufficiency of rates can strain access to capital markets.

S&P has also identified the regulatory environment as an important factor,

[93] Moody's Investor Service, Rating Methodology, Regulated Electric and Gas Utilities, December 23, 2013, at 6.
[94] *Id.* at 3.
[95] *Id.* at 9-10.
[96] *Id.* at 15.
[97] Id.

stating, "we believe the fundamental regulatory environment in the jurisdictions in which a utility operates often influence credit quality the most."[98]

Q. How does the regulatory environment in which a utility operates affect its access to and cost of capital?

A. The proportion and cost of debt capital available to utility companies are influenced by the rating agencies' assessment of the regulatory environment. Moody's has highlighted the relevance of a stable and predictable regulatory environment to a utility's credit quality, stating that "[b]roadly speaking, the Regulatory Framework is the foundation for how all the decisions that affect utilities are made (including the setting of rates), as well as the predictability and consistency of decision-making provided by that foundation."[99]

Q. Have you conducted any analysis of investors' perceptions of the regulatory framework in which the Companies operate relative to the jurisdictions in which proxy group companies operate?

A. Yes. To assess investors' view of the Companies' regulatory framework, I considered three different rankings: (1) the S&P business and financial rankings; (2) the Regulatory Research Associates ("RRA") ranking of

[98] Standard & Poor's, Assessing U.S. Utility Regulatory Environments, March 11, 2010, at 2.

[99] *Id*.

regulatory jurisdictions; and (3) S&P's ranking of the credit supportiveness of regulatory jurisdictions.

S&P ranks the business profile on a six-tier scale from excellent ("1") to vulnerable ("6"). In addition, S&P ranks financial profile on a similar scale, from minimal ("1") to highly leveraged ("6"). I applied that numeric ranking system to the proxy group companies. As shown in Exhibit __ (AEB-8), both KEDNY's and KEDLI's business profile rankings were ("1"), which is in line with the proxy groups' average jurisdictional ranking that was also excellent ("1.17"). Regarding the financial profile rankings, KEDNY's and KEDLI's rankings were ("4"), again similar to the proxy groups' average jurisdictional ranking which was between intermediate and significant ("3.61").

Q. Please explain how you used the RRA ratings to compare the regulatory jurisdictions of the proxy companies with the Companies' regulatory jurisdiction.

A. RRA assigns a ranking for each regulatory jurisdiction between "Above Average/1" to "Below Average/3," with nine total rankings between these categories. I applied a similar numeric ranking system to the RRA rankings with "Above Average/1" assigned the highest ranking ("9") and "Below Average/3" assigned the lowest ranking ("1"). As shown on Exhibit __ (AEB-9), the New York jurisdictional ranking ("4") was generally

consistent with the proxy groups' average numeric ranking ("5.08") from RRA.

Q. How did you conduct your analysis of the S&P credit supportiveness?

A. For credit supportiveness, S&P classifies each regulatory jurisdiction into five categories that range from "Credit Supportive" to "Most Credit Supportive." My analysis of the credit supportiveness of the regulatory jurisdictions that the proxy companies operate in, as compared with the Companies' regulatory jurisdiction, was similar to the analyses of the S&P business and financial ranking and RRA overall regulatory ranking discussed above. I assigned a numerical ranking to each category, from Most Credit Supportive ("1") to Credit Supportive ("5"). As shown in Exhibit __ (AEB-10), the proxy group average ranking was 2.76, which would be classified between "Highly Credit Supportive" and "Very Credit Supportive", and is slightly above the New York jurisdictional classification of "Very Credit Supportive" ("3"), suggesting investors perceive regulation for the Companies as slightly below average relative to the proxy groups.

Q. What is your conclusion regarding the regulatory framework in New York as compared with the jurisdictions in which the proxy group companies operate?

A. The regulatory framework in which a regulated utility provides service is one of the most important consideration for debt and equity investors.

Based on my analysis, I conclude that New York's regulatory framework has somewhat greater risk than the jurisdictions in which the proxy group companies provide service. While the differences are not significant, my analysis demonstrates that investors perceive regulation for the Companies as slightly below average relative to the proxy group. There is no indication that the business, regulatory and financial risks of the Companies (or other New York utilities) are lower than the industry average. As such, the large differential in the authorized ROE in New York as compared with the nationwide range of returns (71 basis points in 2018) is not supported by the risk assessment.

Q. Have the Companies received any recent ratings actions?

A. Yes. On February 22, 2019 Moody's downgraded the senior unsecured and issuer ratings of KEDNY from A2 to A3. Subsequently, Moody's downgraded KEDLI's senior unsecured and issuer ratings from A2 to A3 on March 29, 2019. Moody's indicated that the rating action on KEDNY "reflects the persistent weakness of KEDNY's key credit metrics and Moody's assessment that there is limited likelihood of a recovery over the medium term."[100] Similarly, in regard to KEDLI, Moody's cited the credit metrics that are likely to fall short of revised guidance for an A2 rating.[101] For both of the Companies, Moody's noted that this rating action takes into

[100] Moody's Investor Services, Ratings Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019.

[101] Moody's Investor Services, Ratings Action: Moody's takes action on certain National Grid subsidiaries and affirms other ratings, March 29, 2019.

account the impact of the Commission's August 2018 order reflecting tax reform, wherein the Commission required a reduction in the Companies' revenue starting in January 2019 from the level authorized in the previous rate plan. In addition, Moody's notes the reduction in revenue allowances in January 2021 due to lower recovery of deferred taxes, increasing expenditures on environmental remediation that will face regulatory lag and expected increased leverage to manage to a 48 percent common equity ratio.[102]

C. CAPITAL EXPENDITURES

Q. Did you consider any other information regarding the Companies' risks relative to the proxy group companies?

A. Yes, I also considered the risk related to the Companies' future capital expenditures as compared with the Combined Proxy Group's capital spending plans.

Q. Please summarize the projected capital expenditure requirements for KEDLI and KEDNY.

A. The combined capital expenditure projections for KEDLI and KEDNY are approximately $4.57 billion for the period from Fiscal Year 2021 through 2023. The Company's program includes significant capital investment in

[102] Moody's Investor Services, Ratings Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019 and Moody's Investor Services, Ratings Action: Moody's takes action on certain National Grid subsidiaries and affirms other ratings, March 29, 2019.

infrastructure projects designed to enhance the safety and reliability of its systems.

Q. Do credit rating agencies recognize the risks associated with significant capital expenditures?

A. Yes, they do. From a credit perspective, the additional pressure on cash flows associated with high levels of capital expenditures exerts corresponding pressure on credit metrics and, therefore, credit ratings. A 2016 S&P report noted:

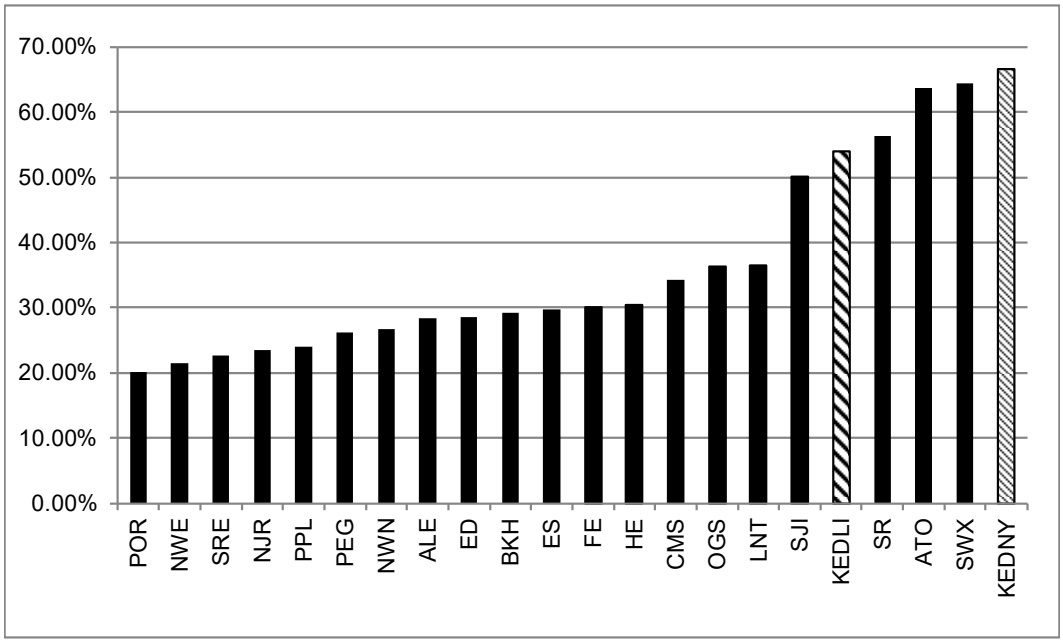
> When applicable, a jurisdiction's willingness to support large capital projects with cash during construction is an important aspect of our analysis. This is especially true when the project represents a major addition to rate base and entails long lead times and technological risks that make it susceptible to construction delays. Broad support for all capital spending is the most credit-sustaining. Support for only specific types of capital spending, such as specific environmental projects or system integrity plans, is less so, but still favorable for creditors. Allowance of a cash return on construction work-in-progress or similar ratemaking methods historically were extraordinary measures for use in unusual circumstances, but when construction costs are rising, cash flow support could be crucial to maintain credit quality through the spending program. Even more favorable are those jurisdictions that present an opportunity for a higher return on capital projects as an incentive to investors.[103]

[103] S&P Global Ratings, Ratings Direct, "Assessing U.S. Investor-Owned Utility Regulatory Environments," August 10, 2016, at 7.

Q. Have you conducted any analysis of the Companies' projected capital expenditures relative to the proxy companies?

A. Yes. I compared the ratio of projected capital expenditures from 2021 through 2023 to net utility plant as of December 31, 2017, for KEDNY and KEDLI with each of the Combined Utility Proxy Group companies. Exhibit __ (AEB-11) shows the ratio of three years of projected capital expenditures to net plant for the proxy groups based on data reported by Value Line. Figure 13 demonstrates that KEDLI's and KEDNY's ratio of projected capital expenditures to net plant are higher than 17 and 20 of the Combined Utility Proxy Group members (there are 20 companies), respectively. Furthermore, as shown in Exhibit __ (AEB-11), KEDLI's planned investment ratio of 53.97 percent and KEDNY's planned investment ratio of 66.57 percent are more than the mean and median of the proxy groups, which suggests that the Companies face greater risk from their construction programs than the typical proxy group company.

Figure 13: Capital Expenditures/Net Plant



Source: Value Line and Company Data

Q. What are your conclusions regarding the effect of the projected capital expenditure plans on the risk profiles of KEDNY and KEDLI and the cost of equity?

A. It is clear that the Companies' capital expenditure requirements as a percentage of net utility plant are higher than the majority of the Combined Utility Group companies. This elevated level of capital expenditures relative to the Combined Utility Group increases the importance of setting a return for KEDNY and KEDLI that is within the range of reasonableness as established by the returns for that group.

VIII. CAPITAL STRUCTURE

Q. Please summarize the Companies' proposed capital structure.

A. As explained in Company Witness Cohen's Direct Testimony, the Companies are requesting a 48 percent to 50 percent equity ratio for ratemaking purposes, which reflects a lower equity ratio than the Companies' actual stand-alone equity ratios. KEDNY's forecast average actual stand-alone equity ratio for the Rate Year and Data Years ranges from 48.90 percent to 50.06 percent[104] and the forecast average actual stand-alone equity ratio for KEDLI for the same period ranges from 48.00 percent to 50.02 percent.[105] The requested equity ratio is consistent with recent Commission precedent regarding the authorized capital structure for utilities.[106]

[104] Direct Testimony of KEDNY Witness Jonathan Cohen, Exhibit __ (JC-1), Schedule 2.
[105] Direct Testimony of KEDLI Witness Jonathan Cohen, at Exhibit __ (JC-1), Schedule 2.
[106] See generally Case 14-E-0493, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc. for Electric Service, Case 14-G-0494, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc. for Gas Service, Case 14-E-0318, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Electric Service, Case 14-G-0319, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Gas Service, and Case 15-E-005, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service.

Q. What is the Commission's policy on determining the authorized equity ratio?

A. The Commission has allowed the use of a stand-alone equity ratio if a utility can demonstrate that the credit rating agencies view that utility's credit on a stand-alone basis independent of its parent.[107]

Q. Do the credit rating agencies view KEDNY and KEDLI's credit on a stand-alone basis?

A. Yes, they do. The credit rating agencies review and assess the credit risk profile of the individual utility on a stand-alone basis, and both KEDNY and KEDLI are rated on their own financial merits and business risk profiles. As noted previously, Moody's recently downgraded the senior unsecured and issuer ratings of KEDNY and KEDLI based on concerns related to key credit metrics based on the Commission's August 2018 order to reduce revenue under the existing rate plan resulting from tax reform. For KEDNY, Moody's also indicated concerns for the additional expenditures for environmental remediation and the expectation of increased leverage to manage to the approved 48 percent equity ratio from an actual equity ratio of 55.6 percent.[108]

[107] Case 14-E-0318, Duah Direct Testimony at 9.
[108] Moody's Investor Service Rating Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019.

Q. Please describe how the Moody's reports for KEDNY and KEDLI demonstrate that Moody's considers the Companies' credit quality on a stand-alone basis.

A. In recent reports, Moody's notes that KEDLI and KEDNY both have strong ring-fencing provisions that protect the stand-alone ratings.

> While there is considerable debt at KEDLI's parent holding companies, the strong ring-fencing provisions applicable to New York utilities reduce the potential for debt to be pushed down to KEDLI. In particular, we view the explicit leverage restriction for KEDLI (to maintain debt /capitalisation ratio at less than 58%) as providing the greatest credit support.
>
> Additional ring-fencing provisions imposed by the NYPSC on KEDLI that we view as credit supportive include: (1) the special preferred share provision that reduces the probability of bankruptcy in a distressed situation, and (2) the requirement for KEDLI to hold an investment-grade rating. [109]

Q. What do you conclude regarding the credit rating agencies' view of the credit quality of KEDNY and KEDLI?

A. Rating agencies are very cognizant of the protective ring-fencing measures that the Commission has established for KEDNY and KEDLI and cite them as the reason why they assess both Companies' credit quality on a stand-alone basis. Because there is factual evidence indicating that the two major credit rating agencies view each of the Companies' credit quality on a stand-

[109] Moody's Investor Services, KeySpan Gas East Corporation: Update following rating downgrade to A3, April 8, 2019 at 5.

alone basis, the stand-alone capital structures proposed in this proceeding are appropriate for the purpose of establishing the ROR on rate base.

Q. What do you conclude regarding the credit rating agencies' view of the credit quality of KEDNY and KEDLI?

A. The recent Moody's report demonstrates concerns regarding the Companies' credit metrics over the medium term. Therefore, it will be important to evaluate the capital structures of the Companies in light of these concerns. As described in Company Witness Cohen's Direct Testimony, the Companies' proposed equity ratios can enhance the Company's cash flow metrics and help mitigate the uncertainty and near-term negative impacts of the TCJA.

Q. Have you conducted any analysis of the Companies' proposed capital structure as compared with the proxy companies?

A. Yes. I have reviewed KEDNY and KEDLI's proposed capital structure as compared with the actual capital structures of the operating companies in the Combined Utility and Natural Gas Proxy Groups for the most recently reported four years. As shown on Exhibit __ (AEB-12), the mean annual equity ratio of the proxy companies over that period is 56.65 percent for the Combined Utility Group and 57.94 percent for the Natural Gas Proxy Group.

Q. What do you conclude from this analysis?

A. The requested 48 to 50 percent equity ratio is conservative considering the equity ratios of the proxy companies and the current business and financial risks of KEDNY and KEDLI, including significant capital investment programs, credit metrics pressures and credit rating agency pressures. This information indicates that the utility operating subsidiaries owned by holding companies with similar business characteristics to KEDNY and KEDLI have for the last four years maintained average common equity ratios more than 8.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding. These higher proxy equity ratios reflect a level of financial risk that is lower than the financial risk implied by the proposed 48.0 percent equity ratio. Therefore, I conclude that the requested equity ratio should be considered a lower bound on the equity ratio relative to the range of reasonableness established by industry practice. As such, it would be reasonable for the Commission to use higher equity ratios for KEDNY and KEDLI closer to those of the proxy group operating companies for ratemaking purposes.

Q. Will the capital structure and ROE authorized in these proceedings affect the Companies' access to capital at reasonable rates?

A. Yes. The level of earnings authorized by the Commission directly affects the Companies' ability to fund their operations with internally generated funds. Both bond investors and rating agencies expect a significant portion

of ongoing capital investments to be financed with internally generated funds.

It also is important to realize that because a utility's investment horizon is very long, investors require the assurance of a sufficiently high return to satisfy the long-run financing requirements of the assets placed into service. Those assurances, which often are measured by the relationship between internally generated cash flows and debt (or interest expense), depend quite heavily on the capital structure. As a consequence, both the ROE and capital structure are very important to debt and equity investors. Furthermore, considering the capital market conditions discussed in Section IV, the authorized ROE and capital structure take on even greater significance.

IX. MULTI-YEAR RATE PLAN

Q. Would a multi-year rate plan impact your ROE recommendation?

A. Yes, it would. The Commission has in many cases approved three-year rate case settlements that often include stay-out premiums. It is my understanding that the Companies will provide four years of forecast data in their rate filings. In keeping with Commission precedent, a stay-out premium would reflect the increased risk faced by the Companies under a multi-year rate plan.

Q. How has New York typically estimated a stay-out premium?

A. The New York approach has typically set the measure of the risk and return trade-off using one half of the yield spread between a one-year and three-year Treasury securities.

Q. Does one half of the yield spread between one-year and three-year Treasuries sufficiently reflect the risk to equity investors inherent in a multi-year stay-out?

A. No. The stay-out premium associated with a multi-year rate plan should not only compensate investors for changes in the level of interest rates or inflation, but also for the potential risk of under-earning that is introduced by "staying out." By staying out of rate cases, the utility may not fully recover material amounts of capital expenditures and may be required to absorb losses due to differences between the cost of service established in the rate plan and actual levels of revenue and expense. The premium should compensate the utility and its investors for these additional risks over and above interest rate risk. In the current market environment, there is additional risk that the authorized ROE for the latter years of a multi-year rate plan will be lower than investors' future requirements as interest rates are expected to increase.

Q. What do you propose as the stay-out premium for a four-year rate plan?

A. The ROE proposed by the Companies of 9.65 percent will not provide the Companies a return commensurate with the return available on investments of similar risk over the term of the multi-year rate plan without an adequate stay-out premium. Consistent with prior cases in which a stay-out premium was included in multi-year rate plans, I recommend that a stay-out premium be included in a multi-year rate plan. I believe 50 basis points is a reasonable, albeit conservative, reflection of the incremental risk to the Company under a multi-year stay-out provision.

X. CONCLUSION AND RECOMMENDATION

Q. What is your conclusion regarding a fair return on book equity for KEDNY and KEDLI?

A. My recommended return on book equity considers the results of the DCF and CAPM models, summarized in Figure 14 (below), and the specific risks to which the Companies are exposed. I conclude that the appropriate ROE for the Companies is within the range of 9.65 percent and 10.55 percent. The Companies are requesting an ROE of 9.65 percent, which is at the low end of the range of reasonableness and is a conservative estimate of the investor-required ROE. Furthermore, if the Commission approves the stay-out period, a premium of 50 basis points should be added to the ROE.

Figure 14: Summary of Analytical Results

Combined Utility Proxy Group			
	Low	**Mean**	**High**
DCF	8.92%	9.20%	9.67%
Mean CAPM	10.66%	10.72%	10.90%
50%/50% DCF/CAPM	9.79%	9.96%	10.29%
67%/33% DCF/CAPM	9.50%	9.71%	10.08%
Natural Gas Proxy Group			
	Low	**Mean**	**High**
DCF	8.59%	9.04%	10.02%
Mean CAPM	10.89%	10.95%	11.12%
50%/50% DCF/CAPM	9.74%	10.00%	10.57%
67%/33% DCF/CAPM	9.35%	9.68%	10.39%

Q. Does this conclude your Direct Testimony?

A. Yes, it does.

Before the Public Service Commission

The Brooklyn Union Gas Company d/b/a National Grid NY

and

KeySpan Gas East Corporation d/b/a National Grid

Rebuttal Testimony

of

Ann E. Bulkley

Case 19-G-0309
Case 19-G-0310

September 18, 2019

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I. INTRODUCTION

Q. Please state your name, affiliation, and business address.

A. My name is Ann E. Bulkley. I am a Senior Vice President of Concentric Energy Advisors, Inc. ("Concentric"), located at 293 Boston Post Road West, Suite 500, Marlborough, Massachusetts 01752.

Q. Are you the same Ann E. Bulkley who provided Direct Testimony in this proceeding?

A. Yes. The terms defined in my Direct Testimony have the same definitions here unless otherwise indicated.

II. PURPOSE AND OVERVIEW OF REBUTTAL TESTIMONY

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my Rebuttal Testimony in these proceedings is to respond to the prepared testimony of the Department of Public Service Staff ("Staff") Finance Panel regarding the appropriate return on equity ("ROE") for KEDNY and KEDLI. My analysis is supported by the data presented in Exhibits__ (AEB-1R) through (AEB-4R).

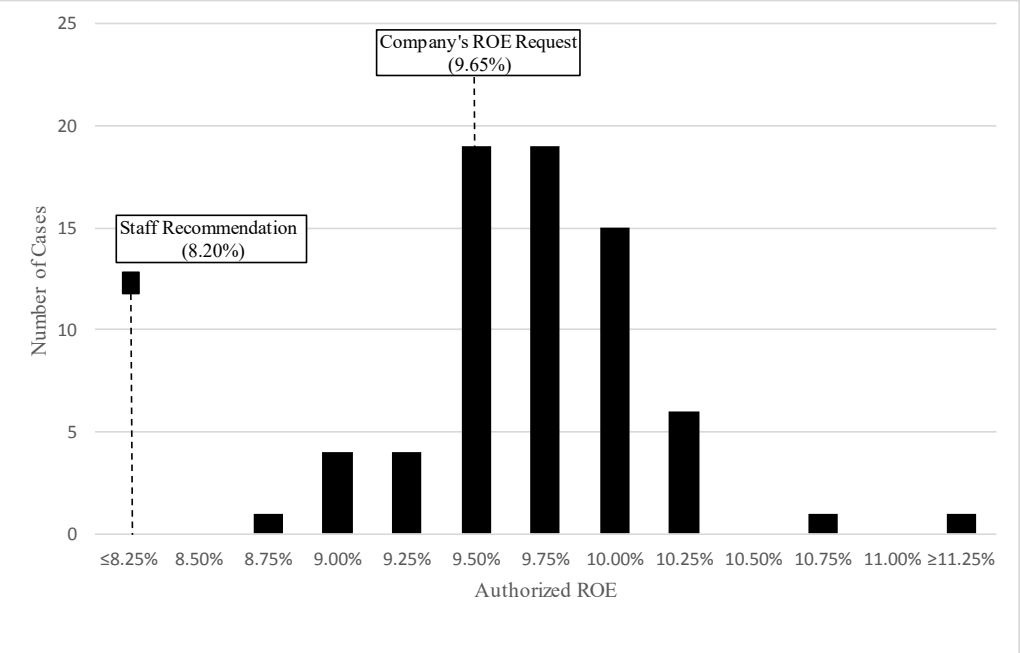
Q. Please summarize your principal concerns with the Staff Finance Panel's recommendations.

A. The Staff Finance Panel's recommended ROE of 8.20 percent is 80 basis points lower than the current authorized ROE for KEDNY and KEDLI of

9.00 percent (December 2016). Staff's recommendation is also well below any authorized ROE for a gas distribution company in the U.S. since at least 1980.[1] Staff has provided no evidence to demonstrate that the Companies' risk profiles are sufficiently different from other natural gas distribution companies to justify a return that is more than 144 basis points lower than the average ROE award for other gas distribution companies over the period from 2017 through 2019. While Staff cites the current low interest rate environment in support of its recommendation, this environment has existed for some time and affects all electric and gas utilities. As shown in Figure 1, over the last three years, the majority of the authorized ROEs have been greater than 9.65 percent.

[1] Source: S&P Global Market Intelligence (Formerly known as SNL Financial).

Figure 1: Authorized ROE for Natural Gas Distribution Utilities
January 2017 through September 2019



Staff applied a two-thirds weighting to the results of the Discounted Cash Flow ("DCF") analysis and a one-third weight to the Capital Asset Pricing Model ("CAPM") analysis, based on the specifications of the 1991 Recommended Decision ("RD") in the Generic Finance Proceeding ("GFP") and other Commission discussions of the ROE estimation methodology. Staff's entire justification for its methodology is that the Commission has used this approach in prior cases. While Staff cites to other cases, it fails to recognize that the Commission has made adjustments to its methodology since the RD, to rely on better and more current information. The Commission's methodology has evolved over time, suggesting that the Commission is open to considering whether the data and methodologies that it has historically relied upon are the best and most relevant given prevailing

market conditions and financial analysis best practices at the time of the case it is deciding.

In contrast to the Commission's demonstrated willingness to change its approach to reflect the circumstances at the time of its decision, Staff has not questioned why the result of its ROE analyses is inconsistent with authorized returns available in New York and other jurisdictions, and whether that difference is, or can be, accounted for by evidence. Rather, Staff has applied the models in a purely mechanistic fashion based on Commission "precedent" without using informed judgment or considering current economic and market conditions, as is appropriate and as the Commission has demonstrated a willingness to do in prior cases.

Furthermore, while Staff acknowledges that credit rating agencies "view unpredictability as an additional risk,"[2] they have not considered how credit rating agencies or equity investors would react to an extraordinarily low authorized ROE for the Companies relative to recently approved ROEs. Such a low ROE is likely to be viewed as punitive by rating agencies and investors and cause rating agencies to question the level of regulatory support being provided by the Commission, with consequences that could be borne by all New York utilities and their customers. As Staff points outs, there is an incremental risk due to the effect of the Companies' moratorium

[2] Prepared Testimony of the Staff Finance Panel, at 92-93.

on processing new natural gas service applications, implying that the Companies may forego growth opportunities in the future.[3] It is not clear how Staff determined that an ROE that is 48 basis points lower than the lowest authorized ROE since at least 1980 would be viewed by the investment community as predictable. More likely, such a result would be viewed as a decrease in the Commission's overall supportiveness, which Moody's identifies as a specific factor that could lead to a downgrade for the Companies.[4] In addition, it would negatively affect the Companies' "Sufficiency of Rates of Returns," subfactor which is currently rated Baa for the Companies.[5] Moreover, predictability is not the only goal of rating agencies and investors, nor is it a requirement in setting the ROE as established by the *Hope* and *Bluefield* standards.

The Staff's recommendation fails to address the concerns of credit rating agencies following tax reform. As discussed in my direct testimony, Moody's recently downgraded the senior unsecured rating and issuer rating of KEDNY and KEDLI, noting the persistent weakness of key credit metrics with limited likelihood of a recovery in the medium term. As discussed in more detail in Section VII, Moody's rationale includes the expectation of reduced revenues for the Companies due to the TCJA.[6]

[3] Prepared Testimony of the Staff Finance Panel, at 75.

[4] *See* Exhibit ___ (SFP-5) at 2 and Exhibit ___ (SFP-6) at 2.

[5] *See* Exhibit ___ (SFP-5) at 8 and Exhibit ___ (SFP-6) at 7.

[6] Moody's Investors Service, Rating Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019 and Moody's Investor Services, Ratings Action:

Staff's recommendation to reduce the ROE by 80 basis points will further weaken the Companies' credit metrics.

The GFP was established in 1991 to address the sensitivity of the DCF model to interest rate fluctuations and to address the unreasonably low returns that are produced by this methodology when interest rates are low. The interest rate environment today is certainly much lower than in 1991 when the Commission decided that the DCF model was not producing reasonable results. While the RD specified conventions for the DCF and CAPM in the GFP, the RD in that case left open the opportunity to consider other methods for "good reason."

The results of Staff's DCF model in this case provide that good reason. Staff's application of the DCF model results in an ROE estimate of 8.22 percent, and its CAPM results in an ROE estimate of 8.04 percent. These estimates are 48 to 66 basis points below the lowest ROEs authorized by the Commission in the past 35 years and 142 to 160 basis points lower than the average authorized ROE for gas distribution companies in all U.S. jurisdictions since January 2017. The assumptions and inputs used to develop Staff's DCF and CAPM models do not result in reasonable estimates of the expected return on equity. Benchmarking Staff's DCF and

Moody's takes action on certain National Grid subsidiaries and affirms other ratings, March 29, 2019.

CAPM results against other observable authorized ROEs supports the position that the parameters of the ROE estimation models in this proceeding must be modified to reflect current market conditions. My rebuttal testimony demonstrates that reasonable adjustments to the models result in returns, using Staff's methodology, and the equal weighting proposed in my Direct Testimony, that are reasonable and consistent with the returns provided on investments of comparable risk.

The GFP provided a framework for the Commission to evaluate whether the financial models used to estimate an appropriate ROE were producing reasonable results under a given set of market conditions. The Commission has since considered its methodology and has made modifications to its preferred approach and assumptions to more reasonably reflect market conditions. Specifically, the Commission has modified the averaging periods relied on for stock prices in the DCF model and the bond yields used in the CAPM from six months to three months to more appropriately reflect market data. In addition, the Commission has changed how it estimates the market risk premium from relying on historical data to projected data, abandoning stale data in favor of data that more accurately reflects market conditions that are relevant in setting the ROE. Based on recent market conditions, that have been acknowledged by other regulatory commissions as anomalous, it would be reasonable for the Commission to again evaluate the relevant market data points to be used in the ROE estimation

methodologies to best estimate the forward-looking cost of equity. For the same reasons, it would be reasonable for the Commission to reconsider the weighting placed on the DCF and CAPM approaches to establish a forward-looking ROE that meets the comparability standards of *Hope* and *Bluefield*.

Q. What are your key conclusions and recommendations regarding the appropriate ROE in this proceeding?

A. My key conclusions are as follows:

- Staff's recommended reduction of the Companies' ROE will likely be viewed as credit negative and may result in further downgrades by Moody's Investor Services.
- Staff's reliance on a proxy group comprised of "electric utility proxy companies" is inconsistent with Staff's prior proxy group screening methodology in other cases and results in a return that does not reflect the business and financial risks of the Companies.
- Staff's mechanical application of Beta coefficients in its CAPM analysis suggests a 120-basis point decline in the cost of equity since the Companies' last cases. Such a significant change warrants further analysis, especially given Staff's DCF analysis suggests there has been no change in the cost of equity since the Companies' last case.[7] Yet Staff provides no further analysis or explanation. As

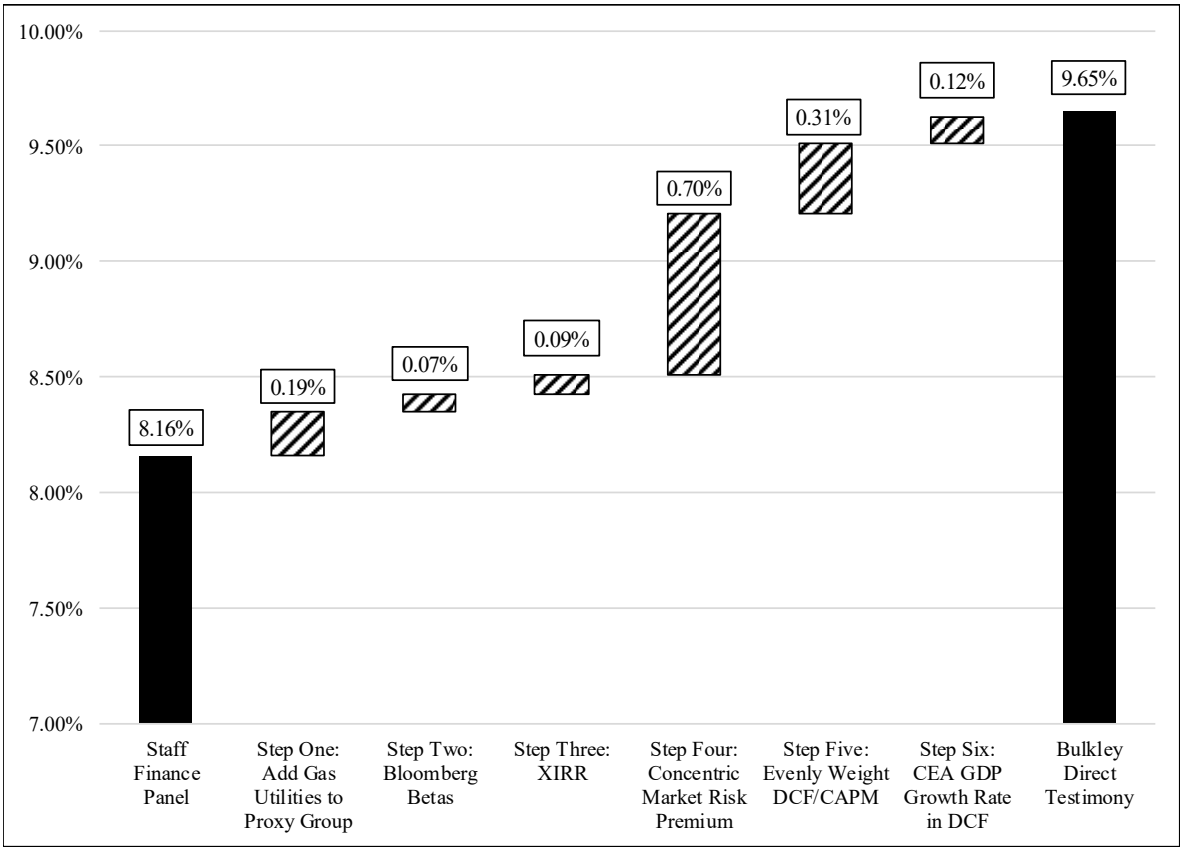
[7] In this case, Staff's DCF result is 8.22 percent. In 16-G-0058 & 16-G-0059 Staff's DCF result was also 8.22 percent.

I described in my Direct Testimony, the performance of utility stocks deviated substantially from the performance of the broader market following the passage of the Tax Cuts and Jobs Act ("TCJA"), while volatility remained high. Staff does not address this fundamental inconsistency and instead relies on analytical results that are below any authorized ROE for a gas distribution company in the U.S. since at least 1980.

- Staff's resistance to use more precise calculation methods that were not available at the time of the GFP is misplaced. As Microsoft Excel has provided advancements that allow for more precise and accurate estimations through the use of the XIRR function, it should be applied in favor of the outdated IRR function.

Section III of my Rebuttal Testimony addresses the technical differences between the analysis presented by Staff and my analysis and recommendation. Reasonable modifications to Staff's proposed assumptions result in significant changes in the resulting ROE. Figure 2 summarizes the effects of modifications made to Staff's DCF model.

Figure 2: Summary of Staff's Adjusted Results[8]



III. RESPONSE TO THE STAFF FINANCE PANEL

Q. What is Staff's ROE recommendation in this proceeding?

A. Staff recommends an ROE of 8.20 percent, based on a two-thirds weighting of its DCF model results of 8.22 percent and a one-third weighting of its average CAPM results of 8.04 percent.[9]

[8] *See* Exhibit ___ (AEB-1R).

[9] Prepared Testimony of the Staff Finance Panel, at 51.

Q. What are the primary areas of disagreement between Staff and you as it relates to the authorized ROE for the Companies?

A. The primary area of disagreement between my analysis and recommendation and that of the Staff is that Staff has dogmatically developed the assumptions used in their ROE formula without considering the effect of market conditions on the models, or the current financial profile of the Companies. As a result, Staff simply runs the models and recommends the results that the model generates. As will be discussed in my testimony, the results of the Staff's model results and ultimate recommendations are not consistent with recently authorized returns in any other jurisdiction and are inconsistent with what the Companies need to maintain their credit ratings, which were downgraded by Moody's earlier this year.

Staff's analysis and criticism of my ROE estimation methodologies are both based on the principle of consistency with prior Commission methodologies. Staff refers multiple times to Commission precedent as the basis for key assumptions in its ROE estimation methodology. The foundation of Staff's criticisms of my methodology is simply that the Commission has not developed the ROE estimation models using the data and methods that I relied on. Staff does not provide any evidence to demonstrate that my assumptions and methods are not those used by investors or are not reasonable. Simply, Staff's position is that my approach

is not what was used in the past. While Staff cites to many instances where it has followed "precedent" with respect to the development of assumptions, Staff fails to acknowledge that there are several instances where the Commission has changed its approach to be responsive to market conditions. This forms the basis of the disagreement between my ROE estimation approach and Staff's methodology.

The specific areas of disagreement related to the calculation of the ROE estimation models are as follows: (1) the composition of the proxy group and the screening criteria used to develop a risk comparable group; (2) the weighting of the DCF and CAPM results; (3) the application of the DCF model and the reasonableness of the results produced by the DCF model under current market conditions; and (4) the application of the CAPM and the reasonableness of making adjustments to the inputs and assumptions used in that model given the current low interest rate environment. The following sections address each of these areas of disagreement.

A. PROXY GROUP COMPOSITION

Q. What is your position with respect to the proxy group composition?

A. An appropriately constructed proxy group consists of companies that are comparable in business and financial risk to the Companies. The importance of selecting a proxy group that is similar in overall financial and business risk to the subject company was endorsed by the United States Court of Appeals for the District of Columbia (the "DC Circuit") in the

Petal Gas Storage decision. The DC Circuit acknowledged that the goal of a proxy group is to rely on companies that possess similar risk to the subject company for the determination of the cost of equity:

> That proxy group arrangements must be risk-appropriate is the common theme in each argument. The principle is well-established. *See Hope Natural Gas Co.*, 320 U.S. at 603 ("[T]he return to the equity owner should be commensurate with returns on investments in other enterprises having corresponding risks."); *CAPP I*, 254 F.3d at 293 ("[A] utility must offer a risk-adjusted expected rate of return sufficient to attract investors."). The principle captures what proxy groups do, namely, provide market-determined stock and dividend figures from public companies comparable to a target company for which those figures are unavailable. *CAPP I*, 254 F.3d at 293–94. Market-determined stock figures reflect a company's risk level and, when combined with dividend values, permit calculation of the "risk-adjusted expected rate of return sufficient to attract investors."
>
> ***
>
> What matters is that the overall proxy group arrangement makes sense in terms of relative risk and, even more importantly, in terms of the statutory command to set "just and reasonable" rates, 15 U.S.C. § 717c, that are "commensurate with returns on investments in other enterprises having corresponding risks" and "sufficient to assure confidence in the financial integrity of the enterprise . . . [and] maintain its credit and . . . attract capital," Hope Natural Gas Co., 320 U.S. at 603.[10]

Consistent with the DC Circuit's decision, I have selected a proxy group of companies with comparable investment risk to the Companies. In contrast,

[10] *Petal Gas Storage v. FERC*, 496 F.3d 695, 699-700 (D.C. Cir. 2007).

Staff applied less stringent screening criteria, which result in a larger, but less comparable proxy group.

Q. Do you believe that Staff's proxy group of only electric utilities is appropriate for purposes of setting the authorized ROE for a gas distribution company?

A. No, I do not. Staff has not included any natural gas distribution-only companies in its proxy group, even though the return on equity is being set for gas distribution companies. The business and operating risks for electric utilities and gas distribution companies are different. I strongly disagree with Staff's opinion that the authorized ROE for the Companies should be established based on a proxy group consisting entirely of electric utilities and excluding gas distributors. From the perspective of investors, there would be no reason to dismiss market data for gas distribution companies when setting the return requirements for a gas distribution company. Rather, in my view, investors would give more weight to market data for companies in the gas distribution industry and less weight to electric utilities.

Q. How would the inclusion of the seven gas distribution companies shown in Figure 8 of your Direct Testimony affect the results of Staff's DCF analysis?

A. As shown in Exhibit___(AEB-1R), adding the seven gas distribution companies from my Direct Testimony to Staff's Exhibit__(SFP-18) produces a mean DCF result of 8.47 percent as compared to the 8.22 percent set forth on Exhibit_(SFP-18).

Q. What justification does Staff offer for relying on a proxy group comprised entirely of electric utilities?

A. Staff contends that my use of a Natural Gas proxy group is not appropriate because the Companies' parent is principally viewed by investors as an electric utility holding company.[11] Staff also argues that its electric utility proxy group should be used because the majority of the holdings companies in the group "have operating subsidiaries that are combination electric and gas utilities" and therefore "reflect the challenges and opportunities facing energy utilities as a whole."[12] This ignores the fact that the Companies are not combination electric and gas utilities, but rather are natural gas-only companies. As such, Staff's reliance on electric utilities, and exclusion of gas-only companies, is inappropriate. Staff further argues a proxy group comprised only of natural gas distribution utilities would not produce a

[11] Prepared Testimony of the Staff Finance Panel, at 53-54.
[12] *Id*., at 54.

sufficiently large proxy group, allowing the analysis to be overly influenced by outlier results.[13] Staff does not, however, explain why natural gas distribution utilities should be excluded from the proxy group.

Q. Do investors view the Companies' parent as an electric utility holding company?

A. No, they do not. As an initial matter, Staff does not appear to have offered any evidence demonstrating that the Companies' parent is viewed by investors as an electric utility holding company. As shown in Exhibit ___ (AEB-2R), approximately 45.00 percent of National Grid USA's regulated operating income is derived from gas operations, and approximately 47.00 percent of National Grid USA's regulated revenue is derived from regulated natural gas operations. In contrast, on average, only 12.00 percent of Staff's proxy group's regulated operating income and regulated revenue is derived from regulated natural gas operations. In fact, more than one-third of Staff's proxy group companies do not have any natural gas distribution operations.

Further, in a credit rating report for the Companies' affiliate, the Massachusetts Electric Company, Moody's characterized the National Grid Plc as "well-diversified," with approximately half of National Grid's US rate base comprised of regulated gas utility investments as of March 31,

[13] *Id*., at 80.

2019.[14] This stands in stark contrast to Staff's testimony that KEDNY and KEDLI's "ultimate parent is largely considered to be an electric utility by investors."[15]

Q. Would your opinion regarding the appropriate return on equity for KEDNY and KEDLI change if the Companies' parent was viewed as an electric utility holding company?

A. No, it would not. Even if the Companies' parent were viewed by investors as an electric utility holding company, that would not be relevant when determining the appropriate return on equity for KEDNY and KEDLI. As part of the Company's merger with National Grid in 2006, extensive ring-fencing provisions were imposed on the Companies. These ring-fencing provisions insulated the Companies from the operations of their corporate parent. For example, the Companies are required to maintain (1) an investment-grade credit rating, (2) a debt to capitalization ratio of less than 56.00 percent (for KEDNY) and 58.00 percent (for KEDLI), and (3) a special preferred share that reduces the likelihood of bankruptcy in a distressed situation.[16] Staff themselves concluded that "the rating agencies still view KEDNY and KEDLI as adequately ring-fenced and thus rate the

[14] Moody's Investors Service Credit Opinion, Massachusetts Electric Company, June 14, 2019.

[15] Prepared Testimony of the Staff Finance Panel, at 54.

[16] *See* Exhibit ___ (SFP-5), at 6, and Exhibit ___ (SFP-6), at 4.

Companies' credit quality substantially on a stand-alone basis, apart from their holding Companies."[17]

The ring-fencing provisions limit the risks to which the Companies are exposed to the parent company's risk, and therefore the Companies' return on equity should be established based on the risk profile of the Companies, and not the risk profile of the parent company. Staff and I agree that debt investors view the Companies substantially on a stand-alone basis because they are insulated from risks associated with National Grid. For example, when summarizing the credit quality of both KEDNY and KEDLI, Moody's begins by discussing the risks associated with gas distribution operations, not electric utility operations.[18] Thus, the Companies' return on equity should be established without considering the operations of the corporate parent.

Given the screening criteria applied by Staff, it is not apparent that there would be any change to the proxy group selection for a vertically integrated electric utility. However, as described in more detail below, the risk profile for a vertically integrated electric utility is meaningfully different from a natural gas distribution company. That demonstrates that Staff has not

[17] Prepared Testimony of the Staff Finance Panel, at 21.
[18] Exhibit ___ (SFP-5), at 1, and Exhibit (SFP-6), at 1.

properly accounted for the risks faced by the Companies, which operate exclusively as natural gas distribution utilities.

Q. Is Staff's testimony in this case regarding the appropriate proxy group consistent with its testimony in other cases?

A. No. In 2016, the Staff Finance Panel offered extensive testimony regarding the appropriate cost of capital for National Fuel Gas Distribution Corp ("NFG") in Case 16-G-0257. Despite the fact that NFG's operations were not as clearly distinct from its parent as the Companies (*e.g.* NFG does not issue its own debt or equity),[19] Staff recognized the importance of including natural gas companies in the proxy group for the gas distribution utility, stating that (emphasis added):

> *The starting point for establishing the proxy group is confirming that all of the companies in the group are in the same specific industry classification as the target company. This is done in order to identify each company as a true peer within that proxy group.* Once the appropriate group of peer companies is established, careful consideration must be given to determine the appropriate screening criteria in order to obtain a group of companies that is large enough without becoming unwieldy. Our objective is to select a representative group of companies whose risks closely match those of [NFG]. A careful balance must be struck between these two potentially conflicting goals. While the objective is to select a group of companies whose risks closely match those of the company being examined, it is also large enough in order that we may have sufficient confidence in its results.[20]

[19] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 14.

[20] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 45-46 [clarification added].

Staff's testimony continued (emphasis added):

> *Ideally, we would have selected a proxy group comprised entirely of regulated gas utilities whose risk profile represents, as closely as possible, the risk characteristics of the individual company being examined.* However, given that the entire Value Line Investment Survey (Value Line) universe of gas utility companies consists of an insufficient number of publicly traded water companies upon which to base our proxy group to produce a reasonable result, we expanded our proxy group to include electric utilities within the Value Line universe that face similar risks and of which investors have similar expectations.[21]

In 2016, the Staff Finance Panel offered testimony regarding the appropriate cost of capital for the Corning Natural Gas Corporation ("Corning") that was almost identical to the testimony quoted above from the NFG case.[22]

Q. What implications does Staff's testimony in the NFG and Corning cases have for KEDNY and KEDLI's cost of capital?

A. As described above, adding the seven gas distribution companies from my Direct Testimony to Staff's Exhibit__(SFP-18) produces a mean DCF result of 8.47 percent as compared to Staff's mean DCF result of 8.22 percent. As shown in Exhibit___(AEB-1R), if the DCF analysis was limited to only the seven gas distribution companies from my Direct Testimony, the mean DCF result would be 9.48 percent. Therefore, Staff has understated its DCF

[21] *Id*., at 46-47.

[22] Case 16-G-0369, Prepared Testimony of the Staff Finance Panel, at 45-47.

results by 126 basis points by failing to select "a proxy group comprised entirely of regulated gas utilities whose risk profile represents, as closely as possible, the risk characteristics of the individual company being examined," consistent with Staff's prior position.[23]

Q. Do you share Staff's concern that there are only seven companies in the Natural Gas Proxy Group?

A. No, I do not. As described in my Direct Testimony, the primary objective of the screening criteria used to select a proxy group is to ensure that the resulting utilities have risk profiles that are as comparable as possible to the Companies.[24] A larger proxy group is not inherently preferable to a smaller proxy group if the larger proxy group is less comparable to the Companies. While I recognize that a proxy group of seven utilities is somewhat limited relative to the Combined Utility Proxy Group, in the NFG and Corning Gas cases, Staff acknowledged that the business and financial risks faced by the Natural Gas Proxy Group are more comparable to these natural gas distribution companies. Considering KEDNY and KEDLI on a stand-alone basis, the natural gas proxy group is also more comparable to the Companies than the Combined Utility Proxy Group. Further, as illustrated in Figure 14 of my Direct Testimony, the average DCF/CAPM results of the Natural Gas Proxy Group are corroborated by the DCF/CAPM results of the Combined

[23] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 46-47.
[24] Direct Testimony of Ann E. Bulkley, at 40.

Utility Proxy Group. This demonstrates that the Natural Gas Proxy Group provides a reasonable indication of the Companies' cost of equity.

Q. What additional concerns does the Staff Finance Panel have with the selection criteria you utilized to select your proxy group?

A. Staff has three additional concerns with my proxy group selection criteria. First, Staff argues that I should have screened based on the percentage of revenue derived from regulated utility operations, rather than screening based on the percentage of operating income derived from regulated utility operations. Second, Staff claims that I misapplied a screening criterion that each company must be covered by more than one equity analyst. Third, Staff believes that I should not have limited my proxy group to transmission and distribution only utilities by excluding companies with owned generation comprising more than 60.00 percent of MWh sales to ultimate customers.

Q. Why is net operating income, rather than revenue, the appropriate screening criterion to utilize for selecting a proxy group for the Companies?

A. As discussed extensively in my Direct Testimony,[25] and expanded upon in response to Information Request DPS 894 (*see* Exhibit__ (AEB-3R)), net operating income better reflects the contribution of each business segment

[25] *Id.*, at 43-45.

to the corporation's earnings. For regulated utilities that collect the cost of purchased gas, fuel, and/or power on a pass-through basis through rates, regulated revenue can fluctuate materially with changes in weather without impacting the corporation's earnings or financial position. In contrast, net operating income, which excludes purchased commodity costs, more closely represents the effect of each business segment on the corporation's overall risk profile. Additionally, as discussed further in response to Information Request DPS-894, multiple other regulatory agencies have accepted the use of an income-based screening criterion.

Q. Why does Staff suggest that revenue is the superior screening criterion?

A. Staff argues that using net operating income, rather than revenue, would inappropriately exclude comparable companies from the proxy group.[26] To illustrate this point, Staff observes that the revenue criterion would result in the exclusion of NextEra Energy, Inc. ("NextEra") from the proxy group. Staff further argues that the revenue criterion better accounts for exposure to risky competitive operations than the net income screening criterion.

Q. Is NextEra comparable to the Companies?

A. No. NextEra, as a vertically integrated electric utility, is materially affected by risks associated with its electric generating portfolio. Consider the

[26] Prepared Testimony of the Staff Finance Panel, at 83-85.

discussion of NextEra's financial data offered by Value Line (*i.e.*, the source that Staff relies on extensively in both its DCF and CAPM analyses):

> This should be another good year for NextEra Energy. The company's utility subsidiary, Florida Power & Light, is benefiting from rate relief and healthy customer growth. A $200 million rate increase took effect after a 1,750-megawatt gas-fired plant began operation at the end of March (ahead of schedule) at a cost of $1.2 billion. The utility is also adding solar capacity, which is recovered through a regulatory mechanism. On the nonregulated side, NextEra Energy Resources continues to add wind, solar, and energy-storage capacity. This subsidiary is also repowering older wind projects. Even so... We cut our 2019 share-earnings estimate by $0.75. First-quarter profits were hurt by mark-to-market accounting charges (which we include in our presentation, even though NextEra excludes these) and subpar production from the company's wind assets.[27]

As illustrated by the Value Line report quoted above, the risks NextEra faces as a vertically integrated electric utility are quite different from the risks faced by the Companies. Investors analyzing NextEra's stock consider the risks associated with company's extensive fleet of electricity generating resources, the impact of weather on the company's generation, and the company's ability to add renewable generation capacity. None of those are risks faced by the Companies, which operate as natural gas distribution utilities.

[27] Value Line Investment Survey, May 17, 2019.

Q. How does the risk profile of your Combined Utility Proxy Group compare to the risk profile of the Companies?

A. As illustrated in Exhibit ___ (AEB-8), Exhibit ___ (AEB-9), and Exhibit ___ (AEB-10), and as discussed in more detail below, the utilities in my combined utility proxy group are, on average, less risky than the Companies. This contradicts Staff's objection that "it is quite possible that using the 70% of operating income criteria could introduce companies into the proxy group that have a higher inherent risk profile than is proper for a regulated utility proxy group."[28]

Q. Are there utilities that are comparable to the Companies that are excluded by the revenue screening criterion?

A. Yes. As described in my Direct Testimony, New Jersey Resources would have been excluded from the Combined Utilities Proxy Group if revenue were used as the screening criterion rather than net operating income. It is difficult to understand why Staff would prefer to include vertically integrated electric utilities, instead of natural gas distribution utilities, in the proxy group used to establish the ROE for a natural gas distribution utility. That fact is especially difficult to understand in light of the extensive ring-fencing provisions insulating the Companies from the operations of the corporate parent.

[28] Prepared Testimony of the Staff Finance Panel, at 84.

Q. Is Staff correct that you inadvertently eliminated three companies that are covered by multiple industry equity analysts?

A. No, Staff has misunderstood the screening criterion. Staff argues that, because Otter Tail, MGE Energy, Inc, and El Paso Electric are followed by multiple equity analysts, each of those companies should have passed my screening criterion.[29] However, the screening criterion does not test whether each company is followed at all by more than one equity analyst. Rather, the screening criterion tests whether the earnings per share growth rates provided by First Call and/or Zacks reflect consensus analyst estimates. As discussed in more detail later in my testimony, consensus estimates that reflect the independent opinions of multiple analysts are more reliable than estimates developed by a single analyst because they are less influenced by idiosyncratic error. The earnings growth rates provided by First Call and Zacks for each of the three utilities identified by Staff do not reflect consensus estimates, and thus each utility was excluded from the Combined Utility Proxy Group. The fact that Staff has identified equity analysts such as the Williams Capital Group and the Maxim Group that cover Otter Tail, MGE Energy, Inc., and El Paso Electric does not mean that there are standardized consensus earnings growth rate estimates available for those utilities.

[29] *Id*., at 85-86.

Q. Is your testimony regarding the importance of relying on consensus estimates similar to Staff's testimony regarding the importance of a sufficiently large proxy group?

A. Yes, it is. Staff explained that relying on a sufficiently large proxy group mitigates the undesirable effects of biased and/or inaccurate growth estimates on the ultimate DCF results.[30] It is important to rely on estimates that reflect a consensus of analysts for that same reason: the more estimates that are reflected, the less the results are influenced by outlier growth rates. For example, Staff takes issue with my use of a 25.50 percent earnings growth rate from Value Line for Northwest Natural Gas Company.[31] However, Staff did not point out that the effect of that growth rate on my mean results is moderated because it is averaged with the considerably lower earnings growth rates projected by First Call and Zacks. Had I relied solely on Value Line, as Staff has, the effect of outlier growth rates would be much more substantial.

B. WEIGHTING OF DCF AND CAPM METHODOLOGIES

Q. Why did the Commission initiate the GFP?

A. As discussed in my Direct Testimony, the GFP was initiated because the Commission recognized that the DCF method was particularly sensitive to interest rate fluctuations and was producing returns far below the returns

[30] *Id*., at 52-53.
[31] *Id*., at 89.

produced by other methodologies.[32] The RD proposed that a two-third/one-third weighting be assigned to the results of the DCF and CAPM analyses, respectively, with less weight given to CAPM because it had only been used to that point as a check on the DCF model.[33] While Staff relies on the GFP to support its decision to place two-thirds weight on the DCF results, Staff does not acknowledge that the RD in the GFP left open the possibility that the weightings and methodologies could be adjusted if necessary to ensure that the results promote regulatory credibility.

Q. Does Staff provide examples of the Commission's thoughts regarding the weighting of the DCF and CAPM results?

A. Yes. Staff cites several cases and provides one specific example of the Commission's discussion of the weighting of the DCF and CAPM results from Case 06-E-1433. As noted by Staff, in that case, the Commission continued to place two-thirds weight on the DCF model results and one-third weight on the CAPM results.

[32] Case 91-M-0509, Proceeding on Motion of the Commission to Consider Financial and Regulatory Policies for New York State Utilities, Recommended Decision, (issued July 19, 1994) ("Generic Finance RD"), at 2.

[33] *Id*., at 27.

Q. What are the important factors to consider in reviewing the Commission's decision in Case 06-E-1433?

A. First, it is important to note that the Commission's decision was issued in October 2007, before the financial crisis of 2007-2009 and the ensuing Great Recession, as well as the Federal Reserve's extended involvement in the financial markets that began following those events and continues today. Therefore, the market data used in Case 06-E-1433 to estimate the projected ROE were not influenced by anomalous market conditions such as have been experienced in recent history and that affect the market data used in the ROE estimation methodologies in these proceedings.

Second, the Commission notes that it changed its calculation of the market return used in the estimation of the market risk premium in the CAPM. The Commission recognized that the use of historical returns published by Ibbotson were stale and less reliable and therefore began relying on projected returns as published by Merrill Lynch. In this same case, the Commission recognized that six-month average stock prices could be "stale."[34] Currently, the Commission's methodology relies on three-month average stock prices. These types of changes demonstrate that the Commission is willing to consider modifications to the ROE estimation

[34] Case 06-E-1433, *Orange and Rockland Utilities*, "Order Setting Permanent Rates, Reconciling Overpayments During Temporary Rate Period, and Establishing Disposition of Property Tax Refunds" (October 18, 2007) at 11.

methodology to include more current and relevant information as market conditions change.

Finally, it is important to note that the Commission decision in Case 06-E-1433 did not state that it would never consider changing the weights on the ROE estimation methodologies. Rather, the Commission's conclusion at that time, nearly ten years ago, was that it was "not ***now*** inclined to deviate from our long-held view that the CAPM should not be entitled to more than one-third of the weight."[35] The Commission explicitly left open the possibility that there would be a point in the future when it would be appropriate to consider such a change. Based on the data I have presented, and the viewpoints provided by other regulatory commissions, it is reasonable to conclude that current market conditions now warrant such a change.

Q. Why do you believe that current conditions in financial markets support reconsideration of the weight placed on the DCF and CAPM Methodologies?

A. When the RD was issued in the GFP in 1991, one of the primary concerns identified by the Commission was that the low interest rate environment was causing the DCF model to understate investors' return requirements.[36]

[35] *Id*., at 15. (Emphasis added.)
[36] 1994 N.Y. PUC Lexis 141, *37.

The Commission also noted that there was nothing sacrosanct about the DCF return on equity analysis.[37] The average daily yield on 30-year Treasury bonds in 1991 was 8.14 percent, whereas the average daily yield on 30-year Treasuries in 2019 has been 2.75 percent.[38] The extraordinarily low interest rate environment today should do nothing to alleviate concerns about how the DCF model is affected by the interest rate environment. On the contrary, if the interest rate environment in 1991 was sufficient reason for the RD in the GFP to conclude that placing one-third weight on the CAPM results was appropriate, then the current interest rate environment should provide sufficient basis for a conclusion that the weighting of the DCF and CAPM methodologies should be modified in these cases.

Q. What are your conclusions regarding the weighting of the DCF and CAPM methodologies?

A. The Companies' proposed ROE of 9.65 percent, which is highly consistent with the weighted average 67%/33% DCF/CAPM mean result for the Combined Utility Proxy Group and the 67%/33% DCF/CAPM mean result for the Natural Gas Proxy Group, is appropriate. While I understand the desire to employ a consistent process to estimate the return on equity, as discussed in the *Hope* decision, "[u]nder the statutory standard of 'just and reasonable,' it is the result reached, not the method employed, which is

[37] *Ibid*.

[38] Source: Bloomberg professional, as of, August 30, 2019.

controlling." Staff's ROE recommendation of 8.20 percent, which is based on a weighting of the DCF and CAPM results that was established 25 years ago, is not comparable to returns available to investors in other jurisdictions for companies with comparable risk.

C. Application of the DCF Model

Q. Please summarize Staff's application of the DCF model.

A. Staff uses a two-stage DCF model that forecasts dividends from 2019 to 2023 using Value Line's estimates of projected dividends in that period, and a "sustainable growth rate" from 2023 forward. Using the DCF methodology, Staff calculates a median ROE for its proxy group of 8.22 percent. Staff applies a 2/3 weight to its DCF model results in deriving its overall ROE recommendation.

Q. What are the differences between your application of the DCF model and Staff's approach?

A. The most significant difference between my application of the DCF model and Staff's approach is the growth rates we use in our respective analyses. I have used a consensus of analysts' EPS growth rates for the proxy group companies as the near-term growth rate, and an estimate of growth in the overall economy for the long-term growth rate. This mitigates the uncertainty associated with forecasting individual companies' growth rates over very long-time horizons. By contrast, Staff uses dividend growth projections from a single source (*i.e.*, Value Line) for their near-term growth

rate, and a "sustainable growth rate" for their long-term growth rate. Staff states that it is "highly unlikely that investors would rely exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections".[39]

Q. Do you agree with Staff's position on earnings growth rates?

A. No, I do not. Staff prefers dividend growth rates rather than earnings growth rates as the near-term growth rate in their DCF analysis. However, as discussed in my Direct Testimony, dividend growth is fundamentally driven by earnings growth.[40] While the model is, indeed, called the "Discounted Cash Flow" model, the cash flows it refers to are those an investor can expect to receive during the time they own the stock. Those cash flows are quarterly dividend payments plus any capital appreciation that occurs between when the stock is purchased and when it is sold. Dividend payments and capital appreciation are both a function of earnings per share ("EPS"), which is ultimately what determines the return on equity to the investor. As discussed in my Direct Testimony, dividends are based on management decisions related to cash management and other factors, and therefore dividend growth rates are less likely to accurately reflect investors' growth expectations than earnings growth rates.[41]

[39] Prepared Testimony of the Staff Finance Panel, at 90.

[40] Direct Testimony of Ann E. Bulkley, at 58.

[41] *Id.,* at 46-47.

I have relied on earnings growth because earnings are the fundamental determinant of a company's ability to pay dividends. As noted by Brigham and Houston:

> Growth in dividends occurs primarily as a result of growth in earnings per share (EPS). Earnings growth, in turn, results from a number of factors, including (1) inflation, (2) the amount of earnings the company retains and invests, and (3) the rate of return the company earns on its equity (ROE).[42]

Investment analysts report predominant reliance on EPS growth projections. In a survey completed by 297 members of the Association for Investment Management and Research, the majority of respondents ranked earnings as the most important variable in valuing a security (more important than cash flow, dividends, or book value).[43]

Academic research also supports the use of EPS growth estimates. A 2002 study in the *Journal of Accounting Research*, examined "the valuation performance of a comprehensive list of value drivers" and found that "forward earnings explain stock prices remarkably well" and were generally superior to other value drivers analyzed.[44] A 2012 study from the journal *Contemporary Accounting Research* found that the sell-side analysts with

[42] Eugene F. Brigham and Joel F. Houston, *Fundamentals of Financial Management*, at 317 (Concise Fourth Edition, Thomson South-Western, 2004).

[43] Block, Stanley B., "A Study of Financial Analysts: Practice and Theory", Financial Analysts Journal (July/August 1999).

[44] Liu, Jing, et al., "Equity Valuation Using Multiples," Journal of Accounting Research, Vol. 40 No. 1, March 2002.

the most accurate stock price targets were those whom the researchers found to have more accurate earnings forecasts.[45]

Q. According to Staff, the Value Line growth rates are produced by a group of analysts, and therefore represent a consensus view. Do you agree?

A. No, I do not. Staff provides a letter from Value Line describing the development of its growth projections, as Exhibit ___(SFP-25). Value Line states that each company it covers is assigned to a lead analyst who is responsible for building the coverage model for that company. Specifically, the letter states, "[e]ach stock in The Value Line Investment Survey is assigned to a specific analyst."[46] The letter goes on to report the quality control procedures applied to the analyst's report. Nowhere in the letter, titled "Quality Control Procedures," does Value Line describe a process whereby multiple independent evaluations are performed and then averaged together to form a consensus view. That is the important element missing from Staff's analysis: Staff relies on a source for growth rate estimates that reflects the opinion of a single analyst.

While I agree that Value Line is a trusted source for investment professionals, it is not the only tool that investors rely on to make decisions.

[45] Gleason, C.A., et al., "Valuation Model Use and the Price Target Performance of Sell-Side Equity Analysts," *Contemporary Accounting Research*.

[46] Exhibit __ (AQ-15).

There are additional data sources readily available that compile the consensus viewpoints of multiple brokerage analysts, and it is reasonable to expect that investors also consider that information. In fact, there have been studies performed comparing Value Line and I/B/E/S analyst earnings forecasts in terms of accuracy, rationality and as proxies for market expectations. In 2001, a study concluded that "I/B/E/S forecasts were superior, as explained by the combination of I/B/E/S's timing advantage and the mitigation of idiosyncratic error through consensus building."[47] Furthermore, the I/B/E/S long-term forecasts were less biased and more accurate.[48]

In addition, Staff's position on the position on the appropriate growth rate is contradictory. Staff criticizes the use of earnings growth rates because "Investors will not just look at expected earnings but also factor in all relevant information when estimating their required return."[49] If investors consider all relevant information, it is inconsistent with that principle to limit the DCF analysis to rely exclusively on a single source of estimated growth rates, as Staff has done. Based on Staff's conclusion, it is reasonable to consider the various sources of consensus growth estimates that I have relied on in addition to the Value Line earnings growth rates.

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47 Ramnath, Sundaresh, Rock, Steven, Shane, Philip, "Value Line and I/B/E/S Earnings Forecasts, November 8, 2001, at 1.

48 *Ibid*.

49 Prepared Testimony of the Staff Finance Panel, at 90.

Q. Do you agree with Staff that reliance on Value Line growth rates lends a degree of predictability to the New York rate setting process for the investment community?

A. No, I do not. The investment community reviews the returns established through the New York rate setting process based on the reasonableness and comparability of the returns to other investment alternatives.

Q. Please summarize your concerns with Staff's sustainable growth rate.

A. I have two main concerns with Staff's sustainable growth rate. First, while Staff asserts that its calculation of the sustainable growth rate represents a measure of long-term growth for the period 2024 and beyond, the inputs to its calculation are, for the most part, shorter-term estimates for the period 2019 and earlier. For example, the "b*r" component of Staff's sustainable growth rate is derived from Value Line forecasts that only extend through 2022. Therefore, Staff's long-term growth rate estimate reflects, at best, one analyst's forecast of only the very early years of the second stage of Staff's DCF model, which theoretically extends into perpetuity. In contrast, my estimate of long-term GDP growth reflects projections over 30 years or more and considers overall measures of economic growth. [50] Second, Staff's sustainable growth rate relies on Value Line's estimate of each proxy company's ROE, as the "r" in the "b * r" component of Staff's growth rate

[50] Direct Testimony of Ann E. Bulkley, at 62-65.

is the expected ROE. This introduces an element of circularity into Staff's calculation, that it fails to reconcile with its results. That is, based on Exhibit __ (SFP-17), the mean and median ROEs assumed in Staff's DCF calculation are 11.14 percent and 10.64 percent, respectively, yet Staff's analysis suggests the proxy group average ROE is 8.22 percent. By that measure, the range of ROEs established in my analysis for my proxy group companies (*i.e.*, 9.65 percent to 10.55 percent) is quite conservative relative to the proxy group ROEs assumed in Staff's sustainable growth estimate.[51]

Q. Have other regulatory commissions abandoned the use of the sustainable growth rate?

A. Yes. In Opinion No. 531, FERC noted that the sustainable "br + sv" growth estimate produces a projection of short-term growth, similar to the IBES growth projections. FERC determined that the use of the IBES projections, without the sustainable growth rate were appropriate in a two-stage DCF model.[52] FERC made no modification to this approach in its recently proposed methodology.[53]

[51] *Id*., at 67.

[52] FERC Opinion 531, at 34, 39.

[53] FERC, Docket No. EL11-66-001, et al., Order Directing Briefs, issued October 16, 2018.

Q. Why is it reasonable to rely on historical averages of real GDP growth to approximate future economic activity?

A. I disagree with Staff that historical averages "are poor indicators of future economic activity."[54] Based on current and recent market conditions, the use of historical real GDP growth is more appropriate than using a current projection of real GDP growth. Economists have reviewed historical growth patterns related to severe financial crises and have concluded that estimates of GDP growth have generally been understated in the decade following severe financial crises. Specifically, the financial crisis and recession that began in 2007 were qualitatively different from most other U.S. economic downturns, which were followed by a rapid return to pre-recession overall output growth levels. In that regard, the current U.S. economic growth situation is similar to that following the two most severe economic events in U.S. history (*i.e.*, the 1929 stock market crash and the 1973 oil shock). Economists who have examined the repercussions of those two historical crises (and similar severe financial crises in other countries) have found that GDP growth rates tended to be lower during the decade following such events.[55] Therefore, it would not be appropriate to assume that current projections of GDP growth are representative of long-term GDP growth starting in 2029 and continuing for the next 200 years.

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[54] Prepared Testimony of the Staff Finance Panel, at 95.

[55] *See,* Reinhart, Carmen M. and Vincent R. Reinhart, "After the Fall," NBER Working Paper 16334, September 2010, in Federal Reserve Bank of Kansas City Economic Policy Symposium Volume, *Macroeconomic Challenges: The Decade Ahead* at Jackson Hole, Wyoming, on August 26-28, 2010, at 2.

Q. Have you performed an analysis of historical GDP growth rates?

A. Yes. I compared the average real GDP growth in the first ten years following the two historical economic crises most comparable to the recent financial crisis (*i.e.*, the 1929 stock market crash and the 1973 oil shock) to the average real GDP growth in the next two decades following each crisis (*i.e.*, eleven to 30 years after the events). I did the same for each of the twentieth-century U.S. recessions for which sufficient data are available. My findings are presented in Figure 3.

Figure 3: Real GDP Growth Rates Following U.S. Economic Downturns[56]

Event	Compound Average Real GDP Growth Rate		
	Decade Following Crisis	Next Two Decades	Difference (Basis Points)
Major Economic Crises			
1929 Stock Market Crash	2.06%	4.72%	266
1973 Oil Shock	2.55%	3.39%	83
Other Recessions			
1937	6.68%	4.15%	-253
1945	3.77%	3.59%	-18
1948	3.79%	3.95%	16
1953	3.60%	3.23%	-37
1957	4.84%	3.13%	-170
1960	4.41%	3.28%	-112
1969	3.57%	3.01%	-56
1980	3.32%	2.45%	-88
1981	3.52%	2.62%	-90

[56] Real GDP data are from the U.S. Bureau of Economic Analysis. The years in which each recession started are from the National Bureau of Economic Research ("NBER"), "US Business Cycle Expansions and Contractions," available at

Figure 3 shows that real GDP growth in the first ten years following the 1929 stock market crash and the 1973 oil shock was substantially lower than real GDP growth in the next two decades following each event. In contrast, eight out of the nine other twentieth century U.S. economic downturns analyzed showed the opposite pattern. In light of the academic research cited above and the findings presented in Figure 3, it is reasonable to believe that current projections of real GDP growth are understated. For that reason, the most reasonable means to forecast long-term GDP growth is to assume a return to long-term historical rates of real GDP growth and to estimate long-term nominal GDP growth based largely on market-based, long-term inflation estimates.

Q. Has Staff relied on historical data in any of their analyses?

A. Yes. For example, in its CAPM analysis, Staff relies exclusively on Value Line Betas that use a 5-year historical estimate of the relative risk of proxy companies and the overall market.

Q. Do you have any additional concerns with the results of Staff's DCF analysis?

A. Yes. One quarter of Staff's proxy group (*i.e.*, seven of the twenty-eight utilities) has a DCF result below 7.00 percent. According to Staff, the risk-

http://www.nber.org/cycles.html. Note that this table excludes the three most recent recessions, which started in 1990, 2001, and 2007 owing to a lack of sufficient data for GDP growth in the following years to calculate comparable long-term GDP growth rates.

free rate as of June 2019 is 2.55 percent.[57] The implied equity risk premium for the proxy group is calculated as the difference between the risk-free rate and the ROE results produced by the two versions of the CAPM developed by Staff. As shown in Exhibit ___ (SFP-20), Staff's traditional CAPM ROE result is 7.63 percent, which implies an equity risk premium of 5.08 percent. Staff's Zero Beta CAPM ROE result is 8.44 percent, which implies an equity risk premium of 5.89 percent. The CAPM analysis provided with my Direct Testimony implies equity risk premiums from 6.37 percent to 8.31 percent, depending on which inputs are reflected.

However, an ROE result of 7.00 percent implies an equity risk premium of 4.45 percent. That is well below the equity risk premiums implied by my analysis, and by Staff's analysis. The fact that one quarter of the utilities in Staff's proxy groups have unreasonably low DCF results calls into question the reliability of that model.

Q. Staff recommends that the Commission reject your use of the "XIRR" function in favor of the "IRR" function. What is Staff's basis for that recommendation?

A. Staff contends that my DCF analysis inappropriately relies on an extended version of Excel's internal rate of return formula. Specifically, Staff takes issue with my use of Microsoft Excel's XIRR function, which can take into account the timing of quarterly dividend payments. Rather, Staff argues in

[57] Exhibit ___ (SFP-20).

favor of the imprecise IRR function, which cannot take into account the timing of quarterly dividend payments when applied to annualized calculations. Staff's basis for rejecting the XIRR function is that (1) the IRR function is suitable for regular cash flows, and (2) the Commission has historically relied on the IRR function.[58]

Q. Please explain the difference between the IRR and XIRR functions.

A. As described in more detail in my response to Information Request DPS-895 (*see* Exhibit__ (AEB-3R)), both the IRR and XIRR functions are Microsoft Excel functions that are used to calculate the internal rate of return for a series of cash flows. The IRR function makes the simplifying assumption that cash flows are available at a set time in each period. In contrast, the XIRR function allows additional inputs as to the timing of cash flows. Specifically, the XIRR function allows the analysis to reflect the assumption that dividends are paid, on average, halfway through the year (*i.e.*, two dividends are paid in the first half of the year, and two dividends are paid in the second half of the year). Staff has provided an analysis demonstrating that the difference in the underlying assumptions between the IRR and XIRR formulas could be as much as 70 basis points.[59] In this case, the difference in the underlying assumptions between the IRR and XIRR formulas has an effect of 11 to 21 basis points on my DCF analysis.

[58] Prepared Testimony of the Staff Finance Panel, at 97-100.
[59] *Id*., at 98.

Q. Why is the XIRR function preferable to the IRR function?

A. The XIRR function allows the analysis to reflect a more nuanced set of assumptions regarding the timing of cash flows. If Staff were correct that the IRR function adequately reflected the timing of cash flows available to utility investors, then the IRR and XIRR functions should return consistent results. The fact that there is a material difference in the results produced by both functions illustrates that Staff's assessment is incorrect.

Q. Please provide an example of why the XIRR function is appropriate, even for regular cash flows.

A. Suppose that there are two dividend paying utilities, identical in every respect but one: utility one, pays a lump-sum dividend once per year; and utility two, pays daily dividends that are equal, on an annualized basis, to the lump-sum dividend paid by utility one. According to Staff's analysis, the result of a discounted cash flow analysis of both utilities would be identical. However, it is a well-recognized financial principle that investors place value the timing in which cash flows are received. While both utilities in this hypothetical pay "regular" dividends, the timing of cash flows matters. The XIRR function allows the analysis to better reflect the timing of cash flows. In contrast, the oversimplified and imprecise IRR function assumes that all dividends are paid at the end of the year. Every utility in my proxy groups, and in Staff's proxy group, pays dividends on a quarterly

basis. Thus, the IRR function, which assumes that investors have to wait longer to receive dividend payments from the utility stock, understates the cost of equity.

Q. When was the XIRR function added to Excel?

A. The XIRR function was not fully integrated into Microsoft Excel until 2007, or more than a decade after the Generic Finance Proceeding ("GFP").[60] As explained in my Direct Testimony, my analysis extends the principles advanced in the Recommended Decision ("RD") in the GFP to reflect best practices in financial analysis today. The XIRR function, which the RD in the GFP would not have contemplated since it was not readily available at the time, is a clear example of a best practice in financial analysis.

Q. What do you conclude about the results of the DCF models under current market conditions?

A. As explained in my Direct Testimony, the DCF models are not producing reasonable results under current market conditions as a result of low dividend yields and high stock valuations.[61] High valuations on utility shares, as suggested by Value Line, could result in an underestimation of the cost of equity using the DCF model, especially if those high valuations are not sustainable in the future. For these reasons, I believe it is appropriate to afford an appropriate weight to the results of other ROE estimation

[60] *See* Microsoft, https://blogs.msdn.microsoft.com/eric_carter/2009/03/19/object-model-changes-for-developers-between-excel-2003-and-excel-2007.

[61] Direct Testimony of Ann E. Bulkley, at 31-32.

methodologies. In particular, the CAPM is more sensitive to changes in interest rates, which is an important risk factor to investors in the current financial market.

D. Application of the CAPM

Q. Please summarize Staff's application of the CAPM.

A. Staff developed its estimate of the risk-free rate of 2.55 percent by using a recent average of 10-year and 30-year Treasury bond yields. For the Beta estimate, Staff used the average Value Line Beta for its proxy group of 0.61. To estimate the market risk premium, Staff subtracted its risk-free rate estimate from an average of the forecast returns for the S&P 500 from Merrill Lynch's May, June and July 2019 Quantitative Profiles. Staff also developed a zero-Beta CAPM using similar inputs as its traditional CAPM analysis, but using a market risk premium based on weighting the Beta times the market risk premium by 0.75 and the market risk premium itself by 0.25. Staff relied on the average of its two CAPM analyses and weighted that result by one-third in the formulation of its overall ROE recommendation.

Q. What are the primary differences between your application of the CAPM and Staff's application of this model?

A. Because the estimation of the ROE is a forward-looking concept, and the ROE that is authorized in these cases will be in effect for some period in the future, my analysis appropriately considers both the recent historical risk-free rate and the projected risk-free rate. The analysis presented in my

Direct Testimony also relies on Bloomberg estimates of Beta that more accurately reflect market conditions. Finally, I estimate the market risk premium based on the difference between the return on large company stocks, as measured by the S&P 500, and the yield on 30-year Treasury bonds.

Q. Why is the use of 30-year Treasury bond yields more appropriate than relying on 10-year and 30-year Treasury bond yields?

A. Staff's rationale for relying on 10-year and 30-year Treasury bond yields in its CAPM analysis is that it is consistent with the approach the Commission has relied on in prior cases, and the use of these securities reflects the expectations of investors who have "intermediate as well as long-term investment horizons.[62] I disagree with Staff's rationale. As noted by Morningstar, "the horizon is a function of the investment, *not the investor*." The use of the 30-year Treasury bond yield as the risk-free rate is consistent with the investment horizon for electric and natural gas utility assets. Taking Staff's rationale to the limits, an investor who plans to hold a position in a utility equity share for only six months would use a certificate of deposit rate to evaluate the potential return. That is clearly not the case for any rational investor considering return requirements.

[62] Prepared Testimony of the Staff Finance Panel, at 114.

Q. Have other regulatory agencies commented on the appropriate securities to use for the risk-free rate in the CAPM?

A. Yes. In Opinion No. 531, the FERC relies on the yield on the 30-year Treasury bond.

> As noted above, the 30-year U.S. Treasury bond yields are a generally accepted proxy for the risk-free rate in a CAPM analysis, and are also considered superior to short- and intermediate-term bonds for this purpose.[63]

FERC made no modification to this approach in its recently proposed ROE methodology.[64]

Q. Why are projected treasury bond yields informative in the CAPM analysis?

A. As discussed in my Direct Testimony, recent market conditions, particularly recent artificially low interest rates, have had an effect on the results of the ROE estimation models. Given these market conditions, I disagree with Staff's position that "current rates are the best indicator of future rates, as they are based on the latest available information to investors."[65] Just as investors have available to them actual and projected growth rates for individual companies, they also have available current and projected yields on Treasury bonds.

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[63] FERC Order 531-B at para 114. Citing also to Roger A. Morin, New Regulatory Finance 151-152 (Public Utility Reports, Inc. 2006) ('the yield on very long-term government bonds, namely, the yield on 30-year Treasury bonds, is the best measure of the risk-free rate for use in the CAPM and Risk Premium methods.").

[64] FERC, Docket No. EL11-66-001, et al., Order Directing Briefs, issued October 16, 2018.

[65] Prepared Testimony of the Staff Finance Panel, at 114.

Furthermore, the use of projected Treasury bond yields is consistent with the use of a forward-looking market risk premium. For this same reason, it is not appropriate to rely only on 3-month average Treasury bond yields, when investors and the market generally are expecting interest rates to increase from the historical lows that have recently been experienced in the market. This is particularly important when the expectation is that interest rates will increase over the period that the ROE that is determined in these cases will be in effect. As discussed previously, consensus forecasts predict interest rates to increase in the near and longer-term. It is reasonable to expect that investors can and do consider this information, as they do other market projections. Therefore, it is reasonable, and consistent with Staff's position regarding the calculation of the market risk premium, to rely on expectations of market conditions to the extent that those expected conditions differ from recent history.

Finally, from a practical perspective, the ROE that is approved by the Commission will be the return that the Companies that will be reflected in rates over the rate period, until such time as another rate proceeding is filed. It is important to reflect investors' expectations of returns over that time period. Therefore, I conclude that it is appropriate to consider consensus estimates of investors' expectation of yields on Treasury bonds in the CAPM analysis.

Q. Does Staff misunderstand your testimony with respect to the use of 30-year Treasury bonds in the CAPM?

A. Yes. Staff has misquoted my testimony on this issue in this case and in several prior cases, including Case 16-G-0369 and Case 16-G-0257. Staff suggests that in my Direct Testimony I conclude that "all utility equity investors have an investment horizon of 30 years."[66] Staff then states that this conclusion is unsubstantiated and erroneous. Staff provides no citation for where this conclusion appears in my Direct Testimony.

Q. What is your testimony with respect to the use of the yield on 30-year Treasury bonds as the risk-free rate?

A. It has been my testimony in each of the cases referenced above that it is important to select the term (or maturity) that best matches the life of the underlying investment.[67] I also cite to Morningstar, which notes that the Treasury security relied on should match the time horizon of what is being valued, and that the time horizon is a function of the investment, not the investor. In each case where Staff has asserted this conclusion from my testimony, I have provided this same clarification in response to Staff's assertion. My testimony does not suggest that equity investors have an

[66] Prepared Testimony of the Staff Finance Panel, at 115. *See* also Case 16-G-0369 Prepared Testimony of Staff Finance Panel, at 95, and Case 16-G-0257 Prepared Testimony of Staff Finance Panel, at 91.

[67] Direct Testimony of Ann E. Bulkley, at 72.

investment horizon of 30 years, nor does my testimony suggest that this would be a relevant factor in determining the appropriate security for the risk-free rate even if investors' time horizons were that long.

Q. Please summarize Staff's concern with your use of Bloomberg's Beta coefficients in the CAPM analysis.

A. Staff contends that the Commission has consistently preferred Value Line's Beta coefficients because they are more accurate, and they reflect the exercise of expert judgment. To support this position, Staff cites two articles analyzing the relative accuracy of Value Line's Beta coefficients compared to other sources.[68]

Q. Why is it reasonable to rely on Bloomberg's Beta coefficients?

A. It is reasonable to consider several measures of market conditions in estimating the ROE. As noted in my Direct Testimony, the Bloomberg Beta coefficient is widely used, and because it is based on a ten-year period as compared to Value Line's five-year period, it mitigates the exogenous effect of the TCJA on utility betas.[69] Neither of the academic studies cited by Staff were published after the Senate passed the TCJA on December 2, 2017. As illustrated by Figure 9 in my Direct Testimony, the performance

[68] Prepared Testimony of the Staff Finance Panel, at 111-112.
[69] Direct Testimony of Ann E. Bulkley, at 74-75.

of utility stocks deviated substantially from the performance of the broader market following the passage of the TCJA.

As Staff explained, Value Line's Beta coefficients are calculated over a five-year period.[70] As shown in Figure 4 below, in the last five years, the performance of the S&P Utilities Index has been significantly less correlated with the performance of the S&P 500 than the long-term historical correlation of the two indices. However, the relative volatility of the two indices is essentially unchanged. This indicates that the relative riskiness of utility stocks is essentially unchanged following the passage of the TCJA, even though utility stocks' Beta coefficients have declined. Thus, CAPM results derived solely from Value Line's Beta coefficients are likely to be understated because they are calculated using the last five years of financial data. In addition, this is inconsistent with Staff's DCF analysis, which suggests there has been no change in the cost of equity since the Companies' last case.[71]

[70] Prepared Testimony of the Staff Finance Panel, at 56.

[71] In this case, Staff's DCF result is 8.22 percent. In 16-G-0058 & 16-G-0059 Staff's DCF result was also 8.22 percent.

Figure 4: Beta Input Analysis

	Correlation (S&P Utilities Index Compared to S&P 500)	Relative Volatility (S&P Utilities Index Compared to S&P 500)
Current Five-Year Value (June 2014 – June 2019)	0.34	1.0304
Long-Term Historical Value (2000 – June 2019)	0.56	1.0318

Q. What effect does the decline in Beta coefficients have on the results of Staff's CAPM analysis?

A. As presented in Figure 5 below, the result of Staff's CAPM analysis in the current case are approximately 120 basis points lower than the result of Staff's CAPM analysis in the Companies' previous rate cases. The nearly 20.00 percent decline in Beta coefficients between those two rate cases is responsible for the vast majority of the decline in the results of Staff's CAPM analysis. The slight decrease in Staff's projected market return is mostly offset by a slight increase in Staff's calculated risk-free rate.

Figure 5: Staff Results Comparison

	Risk-Free Rate	**Beta**	**Market Return**	**CAPM Result**
16-G-0058 & 16-G-0059	2.32%	0.75	11.17%	9.24%
19-G-0309 & 19-G-0310	2.55%	0.61	10.88%	8.04%
Difference	0.23%	-0.14	-0.29%	
Impact on CAPM Result	0.05% Increase	1.08% Decrease	0.22% Decrease	1.20% Decrease

Q. Has Value Line recognized the importance of applying its Beta coefficients in the context of other information?

A. Yes, in Value Line's guidance it states "We propose Beta should be used as one factor in the equity analysis framework. Investors should also look at our Safety rank and Price Stability score when making investment decisions."[72] As shown in Exhibit __ (AEB-4R), since the Companies' last rate case, the proxy companies' Value Line Beta coefficients have decreased by more than twice as much as the proxy companies' Safety rank and Price Stability scores, suggesting that the Beta coefficients may be inconsistent with other risk measures. Consistent with Value Line's guidance, this check of reasonableness suggests other inputs should be considered. As such, I have also considered Bloomberg Beta coefficients in my analysis.

Q. What research did Staff cite regarding Value Line's Beta coefficients?

A. Staff cites a 2011 finance text by H. Kent Baker and Gerald S. Martin titled 'Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice" and an 1983 article by Diana R. Harrington entitled "Whose Beta Is Best?" to suggest that the Value Line estimate of Beta coefficients is superior.[73]

[72] Value Line "Using Beta," Cueter, Andrew J., October 2, 2012, http://www.valueline.com/Tools/Educational_Articles/Stocks/Using_Beta.aspx.

[73] Prepared Testimony of the Staff Finance Panel, at 111-113.

Q. Do those sources demonstrate that Staff's Beta coefficients are superior to those reflected in your CAPM analysis?

A. No, they do not. As an initial matter, I've included Value Line's Beta coefficients in my CAPM analysis. As such, I agree that Value Line Beta coefficients provide relevant information to investors. However, as with all elements of the ROE determination, I believe it is incumbent on the practitioner to review the results of the analyses and exercise judgment as to how to weight those results in the overall ROE determination. As I demonstrated in Figure 9 of my Direct Testimony, the performance of utility stocks deviated substantially from the performance of the broader market following the passage of the TCJA and therefore I also considered Bloomberg Beta coefficients to mitigate the exogenous effect of the TCJA on utility betas. Neither of the academic studies cited by Staff were published after the Senate passed the TCJA on December 2, 2017. In fact, Diana R. Harrington article, "Whose Beta Is Best?" was published only seven months after the introduction of the Bloomberg terminal and therefore does not consider Bloomberg estimates of Beta coefficients.[74]

Further, the conclusions of the articles are consistent with the process I have used in my application of Beta coefficients in my CAPM analysis. Baker and Martin criticize estimates that "involve no input from analysts" and go

[74] McCracken, Harry, "How the Bloomberg Terminal Made History–And Stays Ever Relevant," Fast Company, October 6, 2015, https://www.fastcompany.com/3051883/the-bloomberg-terminal.

on to conclude "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg, which uses the following formula: adjusted beta = historic beta (0.67) +0.33" [75]. However, the smoothing performed by Value Line is similar (adjusted beta = historic beta (0.67) +0.35). And in my application of Bloomberg Beta coefficients, I adjusted the measurement period to mitigate the effects of the TCJA as described above. This is consistent with Baker and Martin's conclusion on Beta, "The objective is to capture investors' view of future risks." As such, I disagree with Staff's conclusion that these sources demonstrate that Value Line Beta coefficients are superior to my application of Bloomberg coefficients that were specified with parameters based on an analysis of the market conditions and my informed judgment.

Q. Please summarize Staff's concern with your estimation of the Market Risk Premium.

A. Staff testifies that the use of the Constant Growth DCF model to develop the market return on the S&P 500 is not appropriate because the companies in the S&P 500 cannot sustain their three-to-five year growth rates in perpetuity.[76]

[75] Exhibit ___ (SFP-33) at 3.

[76] Prepared Testimony of the Staff Finance Panel, at 108-109.

Q. What is your response?

A. The calculation of the MRP is based on the return on the broader market, as measured by S&P 500, less the return on a risk-free instrument. The S&P 500 is an index that includes the largest 500 companies by market capitalization. Over time, the specific companies that are included in the S&P 500 Index will vary, but investor expectations of growth and return overall for the index as a whole may not, based on the selection process involved in the index. Therefore, it is reasonable to assume that the average growth of the index could be sustainable in the long run.

Q. Is there additional support for the reasonableness of the market return you have used to calculate the forward-looking market risk premium?

A. Yes, other alternative sources provide reputable forecasts of market returns that are significantly higher than the estimated market return relied on by Staff. In Figure 6, I provide the return on the S&P 500 calculated using earnings growth projections from Bloomberg Professional, Yahoo! Finance, and Standards and Poor's. The calculated returns for the S&P 500 range from 12.23 percent (Bloomberg Professional) to 14.42 percent (Standard and Poor's). By contrast, the historical market return that Staff relied on is 10.88 percent, which is more than 100 basis points below the market return estimates provided by these reputable sources.

Figure 6: S&P 500 Return Estimates[77]

Source	Estimate Date	Dividend Yield	Growth Estimate	S&P 500 Return
Bloomberg Professional	February 28, 2019	1.97%	10.16%	12.23%
Yahoo! Finance	February 28, 2019	1.97%	11.00%	13.08%
Standard and Poor's	February 28, 2019	1.97%	12.33%	14.42%

Q. What are your conclusions regarding the CAPM analysis?

A. In principle, Staff and I agree that the development of the inputs and assumptions for the CAPM should be done on a forward-looking basis, and that *ex-post* analysis can be flawed in some circumstances. Furthermore, both our CAPM analyses contain some historical data where we believe it to be the best estimate in current market conditions. In particular, I note that recent market conditions should not be considered a proxy for future ROE results. As such, I have developed the assumptions used in the CAPM to be forward-looking to the period that KEDNY's and KEDLI's rates will be in effect. In the instance of Beta, I relied on Bloomberg Beta coefficients in addition to Value Line's Beta coefficients in order to account for the effects of the TCJA on utility Beta coefficients.

[77] Bloomberg and Yahoo! Finance do not report a dividend yield for the S&P 500; therefore, the most recent 12-month average dividend yield as of February 2019 reported in the February 28, 2019 S&P 500 Earnings and Estimate Report was used to calculate the total return.

Based on the specification of the methodology, and the support for this methodology that has been offered by other regulatory agencies as a result of recent market conditions, I believe that it is appropriate to give this form of the CAPM an equal weighting in the estimation of the ROE.

E. CREDIT METRICS

Q. Please describe Staff's analysis regarding the effect its recommended common equity ratio and return on equity would have on the Companies' financial integrity.

A. To assess the effect of its recommended capital structure and return on equity would have on the Companies' financial integrity, Staff analyzed a variety of credit metrics that Moody's and S&P utilize when determining the Companies' credit ratings. The results of that analysis are summarized in Exhibit ___ (SFP-10).

Q. Do you have any concerns with Staff's analysis?

A. Yes, I have two primary concerns with Staff's analysis. First, if Staff's recommended return on equity were adopted, it is likely to undermine investors' perceptions of the supportiveness of the regulatory environment in New York. Second, Moody's has explicitly highlighted concerns around the Companies' Cash Flow from Operation Pre-Working Capital to Debt ("CFO pre-WC/Debt") as a factor that could lead to a downgrade. For KEDNY, Moody's projections for CFO pre-WC/Debt are between 11.2

percent and 13.9 percent.[78] While Moody's projections for KEDLI are somewhat higher,[79] Staff's analysis indicates that an 8.20 percent return on equity would produce a CFO pre-WC/Debt metric of 13.6 percent for KEDLI. Both of these factors would increase the chances of a credit rating downgrade, as explained in more detail below.

Q. Please elaborate on your first concern.

A. Moody's rates the Companies' in accordance with its Regulated Electric and Gas Utilities rating methodology, which consists of four primary factors. The first two primary factors, "Regulatory Framework" and "Ability to Recover Costs and Earn Returns," account for 50.00 percent of Moody's overall rating for the Companies. One of the two sub-factors listed under "Regulatory Framework" is "Consistency and Predictability of Regulation". One of the two sub-factors listed under "Ability to Recover Costs and Earn Returns" is "Sufficiency of Rates and Returns".

Q. How would Staff's recommended return on equity affect those factors?

A. Staff's recommended return on equity of 8.20 percent is likely to be viewed by the credit rating agencies as inconsistent, unpredictable, and insufficient because it is 143 basis points below the year-to-date average authorized return on equity for gas distribution utilities across the U.S., and 50 basis

[78] Exhibit ___ (SFP-5), at 2.
[79] Exhibit ___ (SFP-6), at 2.

points lower than any authorized return on equity since at least 1980. Further, as reflected on Exhibit ___ (AEB-7), Staff's recommended ROE is 80 basis points below the lowest authorized ROE of any operating subsidiary of the utilities in my Combined Utility Proxy Group. An authorized ROE of 8.20 percent would be unprecedented, and likely worsen investors' perception of the supportiveness of the Companies' regulatory environment. The Moody's reports for both KEDNY and KEDLI list "Decrease in the New York Public Services Commission's (NYPSC) overall supportiveness" as a factor that could lead to a credit rating downgrade.[80]

Q. Has Staff offered testimony that corroborates your first concern?

A. Yes, Staff has. Staff states that major credit rating agencies "view unpredictability as an additional risk."[81] For context, Staff offered that testimony while criticizing my choice to use projected earnings growth rates in the DCF model. However, it is difficult to understand how the credit rating agencies would view an 80 basis point reduction in the Company's authorized ROE as "predictable." It is highly unlikely that the investment community would be concerned by the use of First Call and Zacks in the DCF model, but not by an 80 basis point decline in the Companies' authorized ROE.

[80] Exhibit ___ (SFP-5), at 2, and Exhibit ___ (SFP-6), at 2.
[81] Prepared Testimony of the Staff Finance Panel, at 92-93.

Q. Please elaborate on your second concern regarding the CFO pre-WC/Debt metric.

A. Staff states that the CFO pre-WC/Debt metric is the "core coverage ratio used by Moody's to assess the financial health of regulated electric and gas utility companies."[82] According to Staff's analysis, an ROE of 8.20 percent would result in a CFO pre-WC/Debt metric of 13.6 percent for KEDLI.[83]

Q. Has Moody's corroborated Staff's assessment of the importance of the CFO pre-WC/Debt metric?

A. Yes. Moody's has explained that a CFO pre-WC/Debt metric that no longer trends toward the mid-teens in the near-term, and to the high teens over the longer term is a factor that could lead to a credit rating downgrade for KEDNY.[84] Given Moody's projections of a CFO pre-WC/Debt metric between 11.2 percent and 13.9 percent, it is difficult to reconcile how an ROE of 8.20 percent will help KEDNY raise its CFO pre-WC/Debt metric to range sufficient to avoid a downgrade. In addition, Moody's indicated that its "Stable" outlook for KEDLI reflects its expectation that "following the approval of a new rate plan, KEDLI will be able to achieve CFO pre-WC/debt trending towards the high-teens in percentage terms." Moody's

[82] Prepared Testimony of the Staff Finance Panel, at 43.
[83] Exhibit ___ (SFP-6), at 2.
[84] Exhibit ___ (SFP-5), at 2.

further explained that a CFO pre-WC/Debt metric consistently below the high teens is a factor that could lead to a credit rating downgrade for KEDLI.[85]

Q. In light of Staff's assessment that an ROE of 8.20 percent would produce a CFO pre-WC/Debt metric below the high teens, how would Staff's recommended ROE impact KEDLI's credit rating?

A. As described in my Direct Testimony, the TCJA has already put downward pressure on the credit metrics of many utilities and their holding companies.[86] The Companies' are no exception. In fact, earlier this year, Moody's downgraded the senior unsecured rating and issuer ratings of KEDNY and KEDLI, citing weakness in key credit metrics. If it were adopted, Staff's recommended ROE of 8.20 percent, would further erode the Companies' performance on a core Moody's metric, and would likely increase the prospects of another credit rating downgrade.

Q. What are your conclusions regarding the impact Staff's recommended return on equity would have on the Companies' financial integrity?

A. Staff's recommended return on equity is likely to have a negative effect on the Company's financial integrity. It is unpredictable, inconsistent, and insufficient, which would materially worsen investors' perceptions of the

[85] Exhibit ___ (SFP-10), at 3.
[86] Direct Testimony of Ann E. Bulkley, at 31-38.

credit supportiveness of the Companies' regulatory environment. Further, due to the specific factors that Moody's considers when determining the Companies' credit rating, it would particularly strain the credit rating of the Companies.

F. Regulatory Framework and Business Risks

Q. Please summarize Staff's position regarding the Companies' business risks.

A. Staff relies on credit rating agencies' commentary on the risk of the utility segment as a whole to support conclusions regarding the risk of KEDNY and KEDLI. Staff's review of the Companies' business risk is as compared to the overall industry risk for electric utilities and does not provide any evidence as to the relative risk of the Companies and the proxy group.[87] Because the ROE is estimated using the proxy group results as a surrogate for the investor required return, it is important that any analysis of the credit supportiveness and business risk of the Companies be conducted relative to the proxy group.

Q. Is Staff's business risk analysis consistent with the development of its proxy group?

A. No, it is not. Staff supports the use of an electric proxy group on the basis that the parent company of KEDNY and KEDLI has significant electric

[87] *Id*., at 123-133.

operations, and therefore the use of an electric utility proxy group is appropriate, even though KEDNY and KEDLI are natural gas distribution companies. However, in the discussion of business risk, Staff concludes, without an analysis of their specific proxy group, that KEDNY and KEDLI are less risky than most electric utility companies.[88] Based on these differences in risk, it would seem more appropriate in setting the ROE for the Companies to consider the returns of a proxy group that includes natural gas distribution companies that have a more similar risk profile to KEDNY and KEDLI.

Q. Have you conducted any analysis of the risk of your proxy group as compared with KEDNY and KEDLI?

A. Yes. As discussed in my Direct Testimony, I examined two rankings performed by the investment community of U.S. regulatory commissions (one by Regulatory Research Associates, or "RRA", and one by Standard & Poor's, or "S&P"). As noted in my Direct Testimony, RRA accords New York an "Average/1" rating, which is only the fourth best ranking (out of nine).[89] S&P ranks New York in the exact middle of the scale in terms of credit supportiveness, in contrast to the proxy group's overall mean ranking which is slightly above average. This suggests that, while New York may

[88] *Id.,* at 16.

[89] Regulatory Research Associates, https://www.snl.com/web/client?auth=inherit#industry/CommissionProfiles (September 9, 2019).

have a variety of ratemaking policies that lower the risks faced by the Companies, the Companies' regulatory environment is below average when compared to the proxy group. Further, investors' perception of the Companies' regulatory environment is likely to worsen if the Commission authorizes Staff's recommended ROE of 8.20 percent for the Companies', which is 48 basis points lower than the lowest ROE authorized by the Commission in the past 35 years.

Q. Staff notes that revenue decoupling mechanisms reduce revenue risk. Have you considered the effect of the Companies' revenue decoupling mechanisms on the required ROE?

A. Yes, I have. The ROE recommendation is established for a company based on its risk relative to the proxy group. Therefore it is necessary to review the similarity of the Companies' revenue decoupling mechanisms to the alternative rate mechanisms that have been implemented by the proxy companies. Exhibit__(AEB-7) summarizes the alternative rate mechanisms that have been implemented by the proxy companies. As shown in Exhibit __ (AEB-7), and as described in my Direct Testimony, approximately 60.00 percent of the jurisdictions where the proxy companies operate have approved some form of adjustment mechanism (*i.e.*, formula rate plan, revenue decoupling mechanism, straight fixed-variable rate design) to reduce the risk of revenue shortfalls. In addition, the authorized ROEs for the proxy group range from 9.00 percent to 10.50 percent. Therefore, it is

difficult to reconcile Staff's recommendation, which is 80 basis points below the *lowest* authorized return for any of the operating companies of the proxy companies, given the returns available to substantially comparable companies.

Q. Do you agree with Staff's suggestion that the Companies' deferral mechanisms should result in a lower ROE?

A. No, I do not. Since the ROE is established based on the expected returns for a proxy group of comparable companies, the analysis of business risk is necessarily a comparative analysis. The relevant question is not how much of the Company's risk has been reduced, but rather how much risk do the Companies have relative to the proxy group. In that regard, Staff has provided no record evidence. While Staff suggests that most of the Company's revenues and costs are subject to true-ups and deferrals, the recovery of gas and fuel costs are a pass-through cost for nearly all utility companies. Fuel costs are the single largest cost for most utilities. Since nearly all utilities are allowed to pass through purchased power and gas costs directly to customers, there is no risk differential between the Companies and the proxy companies on the recovery of this cost. As noted previously, Staff has not provided any record evidence as to the level of recovery or true-up of the Company's remaining operating costs as compared with the peer group to assess the Company's relative risk profile. As described in the Company's Revenue Requirements Panel Rebuttal

Testimony, Staff has proposed a series of downward-only true-ups for costs associated with full-time equivalents, Service Company IT rent expense, paving, GBE, and credit cards. To the extent any of Staff's proposed downward-only true-ups are implemented, the Companies' will remain exposed to risks associated with any cost overruns or inflation above a certain threshold. Conversely, Staff proposes to eliminate the Company's property tax reconciliation and change KEDNY's city-state construction reconciliation from a 90/10 sharing to an 80/20 sharing. The effect of such asymmetric cost trackers provides little, if any, risk mitigation.

Q. Have the regulatory mechanisms available in New York allowed the Companies to earn their authorized return on equity historically?

A. No. As shown on Exhibit ___ (SFP-5), at 5, KEDNY has under-earned relative to its authorized return on equity in four of the last five years. As shown on Exhibit ___ (SFP-6), at 5, KEDLI has under-earned relative to its authorized return on equity in three of the last five years. While both Companies' achieved returns have recovered in 2018, this sustained historical under-earning casts doubt on Staff's claim that the suite of regulatory mechanisms available in New York insulate the Companies from financial risk.

Q. Did Staff acknowledge the effect of the Companies' moratorium on processing new natural gas service applicants on investors' return requirements?

A. Yes, Staff did. Specifically, Staff stated that this development could "modestly and negatively impact investors return requirements" because it will limit the Companies' future growth opportunities.[90] While Staff argues that the impact of this development may be moderated by adjustments to the design of the Companies' Revenue Decoupling Mechanism, Staff did acknowledge that this is a business risk faced by the Companies.

Q. Did Staff discuss the additional risk faced by the Companies if a multi-year rate plan is approved by the Commission?

A. Yes. Staff appears to agree that investors should be compensated with an ROE premium for the additional risks associated with "staying out" of the rate case process. Specifically, Staff stated:

[90] Prepared Testimony of the Staff Finance Panel, at 75.

> It is important to note, however, that when the Commission evaluates the reasonableness of a multi-year rate plan's ROE, it recognizes the necessity of a premium above the ROE produced using Staff's methodology in recognition of the increased financial and business risks that are inherent when a company is operating within the parameters of a multi-year rate plan. By agreeing not to file for new rates for three years, the Companies reduced their ability to adjust rates in response to changing conditions. Accordingly, locking in rates for an extended period somewhat diminished the Companies' flexibility and thereby increased their level of risk. Additionally, the forecasts of expense items in years two and three of a multi-year rate plan are inherently more inaccurate and thus expose a company to additional operating risk.[91]

While Staff has not quantified the ROE premium that would adequately compensate investors for those incremental risks, I determined that a 50-basis point increase to the Companies' authorized ROE would be appropriate if a multi-year rate plan is approved by the Commission.

Q. What are your conclusions regarding the Companies' business risk relative to the proxy group?

A. As discussed in my Direct Testimony, New York's relative regulatory rankings from a credit perspective, coupled with the heightened level of business risk faced by the Companies due to their capital spending plans, and Moody's concerns about weakened credit metrics demonstrate that the Companies' business risk is above average compared to the proxy

[91] *Id*., at 73-74.

companies.[92] In addition, Staff recognizes that there is an incremental risk due to the effect of the Companies' moratorium on processing new natural gas service.[93] That elevated risk profile increases the importance of setting a return for the Companies that is within the range of reasonableness as established by the returns for the two proxy groups that I used in formulating my return recommendation.[94]

IV. CONCLUSIONS AND RECOMMENDATION

Q. What is your conclusion regarding a fair return on equity for KEDNY and KEDLI?

A. I continue to support a recommended return on equity for the Companies' within the range from 9.65 percent and 10.55 percent based on the analyses my Direct Testimony and the updates proposed to Staff's analyses that are presented in Figure 7, below.

[92] Direct Testimony of Ann E. Bulkley, at 91-94.
[93] Prepared Testimony of the Staff Finance Panel, at 75.
[94] Direct Testimony of Ann E. Bulkley, at 82.

Figure 7: Summary of Staff's Adjusted Results[95]

	Adjustment	Mean DCF Result	Mean Staff CAPM[96]	ROE Result	Incremental Change (bps)
	Staff Estimate	8.22%	8.04%	8.16%	
1	Add gas utilities to Staff Proxy Group	8.47%	8.11%	8.35%	+19 bps
2	Use Value Line and Bloomberg Beta Coefficients	8.47%	8.33%	8.42%	+7 bps
3	Use the more precise XIRR function	8.60%	8.33%	8.51%	+9 bps
4	Apply Market Risk Premium Estimate based on S&P 500 market return	8.60%	10.43%	9.21%	+70 bps
5	Apply GDP Growth Rate for Long-Term Growth in the DCF Model	8.83%	10.43%	9.36%	+15 bps
6	Evenly Weight the DCF/CAPM Analyses	8.83%	10.43%	9.63%	+27 bps
Total Change				9.63%	+147 bps

The Companies' proposed ROE of 9.65 percent is supported by the results of the DCF and CAPM methods, and the specific risks to which the Companies are exposed. In contrast, Staff's recommendation of 8.20 percent is lower than any return that has been authorized by any regulatory

[95] *See* Exhibit ___ (AEB-1R).

[96] CAPM relies on the mean Beta to be consistent with Staff's use of the mean DCF result.

commission and is significantly below the returns authorized by this Commission in recent cases. Therefore, Staff's recommendation does not result in a return that is commensurate with the return on other companies of similar risk. Furthermore, Staff's recommendation is likely to be viewed by rating agencies as unsupportive and could result in a further downgrade of Moody's credit rating for KEDLI, in particular. As a result, Staff's recommended ROE should be rejected.

Q. Does this conclude your Rebuttal Testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Theodore Poe, Jr.

Dated: April 2019

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I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Theodore Poe, Jr., and my business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid"), as the Manager of Gas Load Forecasting and Analysis. I am responsible for supervising the development of the gas load forecasts for National Grid's gas utilities, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies").

Q. Please describe your educational background and professional experience.

A. I graduated from the Massachusetts Institute of Technology in 1978 with a Bachelor of Science in Geology. From 1981 to 1989, I worked as a Research Associate with Jensen Associates, Inc. of Boston where I was responsible for developing a variety of computer forecasting models to

analyze natural gas supply and demand for interstate pipeline and local distribution companies. Upon accepting employment with Boston Gas Company in 1989, I was responsible for modeling and forecasting the natural gas resource requirements of the company's customers and managing its resource-planning process. In 1998 and 1999, I undertook the same responsibility for Essex Gas Company and Colonial Gas Company, respectively. In 2000, I assumed responsibility for modeling and forecasting the natural gas resource requirements of KEDNY and KEDLI. In 2008, I assumed the additional responsibility for modeling and forecasting the natural gas resource requirements of The Narragansett Electric Company and Niagara Mohawk Power Corporation.

Q. Are you a member of any professional organizations?

A. I am a member of the Northeast Gas Association, the New England-Canada Business Council, and the American Meteorological Society.

II. Purpose of Testimony

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to provide historic and forecast customer and customer demand data used by the Companies to prepare the forecast of gas revenues and gas costs for the twelve months ending March 31,

2021 ("Rate Year"). The data includes historic data through February 2018 and forecast data for the Rate Year and the following three years ending March 31, 2022, March 31, 2023, and March 31, 2024 ("Data Year 1," "Data Year 2," and "Data Year 3," respectively). The Companies are including forecast data for Data Year 1, Data Year 2, and Data Year 3 (collectively, the "Data Years") for informational purposes and to facilitate the possibility of a multi-year rate settlement in this proceeding.

Q. Are you sponsoring any exhibits in support of your testimony?

A. Yes, I am sponsoring the following exhibits that were prepared by me or under my direct supervision:

Exhibit __ (TEP-1A) KEDNY Calendar Year Deliveries by Service and Revenue Class for the Period 2013 - 2017

Exhibit __ (TEP-1B) KEDLI Calendar Year Deliveries by Service and Revenue Class for the Period 2013 - 2017

Exhibit __ (TEP-2A) KEDNY Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 - 2017 and the Forecast Period 2018 - 2024

Exhibit __ (TEP-2B) KEDLI Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for

	the Historical Period 2013 - 2017 and the Forecast Period 2018 - 2024
Exhibit __ (TEP-3A)	KEDNY End of Year Customer Count by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-3B)	KEDLI End of Year Customer Count by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-4A)	KEDNY Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-4B)	KEDLI Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-5A)	KEDNY Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-5B)	KEDLI Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-6A)	KEDNY Historical and Projected Economic Indicators Used to Develop the Forecast Models

Exhibit __ (TEP-6B)	KEDLI Historical and Projected Economic Indicators Used to Develop the Forecast Models
Exhibit __ (TEP-7A)	KEDNY Historical Actual Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-7B)	KEDLI Historical Actual Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-8A)	KEDNY Historical Weather Normalized Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-8B)	KEDLI Historical Weather Normalized Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-9A)	KEDNY Historical Actual Customer Counts by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-9B)	KEDLI Historical Actual Customer Counts by Month and by Class for the Period January 2013 through December 2017

Exhibit __ (TEP-10A)	Delivery Forecast Model Specifications – KEDNY – Meter Count
Exhibit __ (TEP-10B)	Delivery Forecast Model Specifications – KEDNY – Use Per Customer
Exhibit __ (TEP-10C)	Delivery Forecast Model Specifications – KEDLI – Meter Count
Exhibit __ (TEP-10D)	Delivery Forecast Model Specifications – KEDLI – Use Per Customer
Exhibit __ (TEP-11)	30-Year Normal Heating Degree Days
Exhibit __ (TEP-12A)	KEDNY Calendar Year Actual Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-12B)	KEDLI Calendar Year Actual Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-13A)	KEDNY Calendar Year Normalized Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-13B)	KEDLI Calendar Year Normalized Use Per Customer by Service and Revenue Class for the Period 2013 - 2017

Exhibit __ (TEP-14A) KEDNY Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024

Exhibit __ (TEP-14B) KEDLI Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024

III. Overview of the Companies' Retail Gas Deliveries

Q. Please provide an overview of the Companies' historic gas deliveries from 2013 through 2017.

A. For KEDNY, actual retail gas deliveries have ranged from 1.484 billion therms to 1.677 billion therms to all customers, excluding non-core (SC 18-5A Non-Core Transport) and electric generation (SC 18 Small Electric Generator Transport and SC 20 Large Electric Generator Transport) (Exhibit ___ (TEP-1A)). "Retail" refers to gas delivered and metered at the customers' burner tips, while "wholesale" refers to gas received and metered flowing into the Companies' distribution systems. The range in volumes of actual retail gas deliveries was primarily driven by fluctuations in the weather, such as the colder-than-normal calendar year 2014 and the warmer-than-normal calendar years of 2016 and 2017. On a weather-normalized basis, KEDNY's retail gas deliveries to all customers,

excluding non-core and electric generation, have grown at a compound annual growth rate of 2.3 percent per annum from 1.480 billion therms in calendar year 2013 to 1.624 billion therms in calendar year 2017 (Exhibit ___ (TEP-8A)). This growth reflects recovery from the 2007-2009 recession and the decrease in domestic natural gas prices. Historic actual billed sales by month and by class for 2013-2017 are provided in Exhibit ___ (TEP-7A) for KEDNY. Historic weather-normalized billed sales by month and by class for 2013-2017 are provided in Exhibit ___ (TEP-8A) for KEDNY.

For KEDLI, actual retail gas deliveries have ranged from 0.912 billion therms to 1.013 billion therms to all customers, excluding small and large electric generation (SC 7 Small Electric Generator Transport and SC 14 Large Electric Generator Transport) (Exhibit ___ (TEP-1B)). Similar to KEDNY, the range in volumes of actual retail gas deliveries for KEDLI was primarily driven by weather fluctuations. On a weather-normalized basis, KEDLI's retail gas deliveries to all customers excluding non-core and electric generation have grown at a compound annual growth rate of 3.3 percent per annum from 0.879 billion therms in calendar year 2013 to 1.000 billion therms in calendar year 2017 (Exhibit ___ (TEP-8B)). Similar to KEDNY, this growth reflects recovery from the 2007-2009

recession and the decrease in domestic natural gas prices. Historic actual billed sales by month and by class for 2013-2017 are provided in Exhibit ___ (TEP-7B) for KEDLI. Historic weather-normalized billed sales by month and by class for 2013-2017 are provided in Exhibit ___ (TEP-8B) for KEDLI.

Q. Please provide an overview of the Companies' historic number of customers from 2013 through 2017, as shown in Exhibit __ (TEP-3A) and Exhibit __ (TEP-3B).

A. As shown in Exhibit ___ (TEP-3A), the total number of KEDNY customers, excluding non-core and electric generation, has grown at a compound average net growth rate of 0.5 percent from 1,229,223 at year-end 2013 to 1,255,103 at year-end 2017, or an average annual addition of 6,470 customers per year. Historic customer counts by month and by class are provided in Exhibit ___ (TEP-9A) for KEDNY.

As shown in Exhibit ___ (TEP-3B), the total number of KEDLI customers, excluding small and large electric generation, has grown at a compound average net growth rate of 1.8 percent from 551,688 at year-end 2013 to 592,733 at year-end 2017, or an average annual addition of 10,261 customers per year. Historic customer counts by month and by class are

provided in Exhibit ___ (TEP-9B) for KEDLI.

IV. Gas Delivery Forecast

Q. Please describe KEDNY's individual service and revenue class distinctions.

A. KEDNY's gas tariff (PSC No. 12) includes the following active service classifications: SC 1, SC 2, SC 3, SC 4, SC 6, SC 7, SC 17, SC 18, SC 20, and SC 21. Certain service classifications contain distinct revenue classes. The revenue classes include Residential Non-Heat, Residential Heat, Commercial Non-Heat, Commercial Heat, and Industrial.

Q. Please describe KEDLI's individual service and revenue class distinctions.

A. KEDLI's gas tariff (PSC No. 1) includes the following active service classifications: SC 1, SC 2, SC 3, SC 4, SC 5, SC 7, SC 9, SC 12, SC 13, SC 14, SC 15, SC 16, and SC 17. Certain service classifications contain distinct revenue classes. The revenue classes include Residential Non-Heat, Residential Heat, Commercial Non-Heat, Commercial Heat, and Industrial.

Q. Please describe how you derived the delivery forecasts for KEDNY and KEDLI.

A. On an annual basis, the Companies prepare their gas load forecasts based on econometric/statistical forecast models developed for residential, commercial/industrial, and multifamily rate groups. The instant forecasts were based on historical billing data from March 2007 through February 2018.

Although the Companies maintain their customer data at the internal rate code level (67 unique rate codes for KEDNY and 52 unique rate codes for KEDLI), the data from these rate codes were aggregated into six rate groups: Residential Heating, Residential Non-Heating, Commercial, Multi-Family, Temperature-Controlled, and Other. By aggregating its data into these six groups, the Companies developed population sets that were large enough for statistical modeling as well as free of variations that are not driven by economic factors. The tables below summarize the service classifications that comprised the rate groups:

KEDNY Rate Groups	
Rate Group	Service Classifications
Residential Heating	SC 1B and SC 17-1B
Residential Non-Heating	SC 1A and SC 17-1A

Commercial	SC 2-1, SC 2-2, SC 17-2-1, and SC 17-2-2
Multi-Family	SC 3, SC 17-3, and SC 18-6M
Temperature-Controlled	SC 6 and SC 18
Other	Other

KEDLI Rate Groups	
Rate Group	Service Classifications
Residential Heating	SC 1B and SC 5-1B
Residential Non-Heating	SC 1A and SC 5-1A
Commercial	SC 2A, SC 2B, SC 5-2A, and SC 5-2B
Multi-Family	SC 3 and SC 5-3
Temperature-Controlled	SC 12 and SC 13
Other	Other

Two different models were developed for each rate group: (i) a model to forecast the number of customers (the number of active, flowing meters), and (ii) a model to forecast use-per-customer. The Companies' volume forecasts are then the product of number of customers multiplied by the use-per-customer. The customer count and use-per-customer models are developed using linear regression techniques, and are selected based on their adjusted R squared after accounting for the effects of autocorrelation and heteroscedasticity.

In the number of customer (meter count) models, the independent variables are a combination of time trends and economic variables, such as population, households, employment, and gas and oil prices. In the

models calculating use-per-customer, the independent variables are a combination of time trends, heating degree days, and economic variables, such as population, housing stock, income, employment, unemployment, gross domestic product, and gas and oil prices.

KEDNY and KEDLI obtained the historic data and forecasts of the independent variables from a leading economic research and forecasting firm as well as energy prices from the U.S. Department of Energy, Energy Information Administration, and used this data to develop the econometric models and generate load forecasts.

Q. Please describe how you derived the customer count forecasts.

A. Because the Companies' economic data is available at the quarterly level, the Companies developed their models at the quarterly level, as well. The Companies obtained the historic and projected economic and demographic data used as the independent variables in the regression equations from Moody's economy.com (publication date: February 2018). The Companies obtained energy price data from the U.S. Department of Energy, Energy Information Administration ("DOE/EIA").

For its meter count (customer count) models, the Companies aggregated their monthly meter count data, which is available by internal rate code to the level of the six rate groups. The Companies then took the meter count at the end of each February as quarter one, the end of May as quarter two, the end of August as quarter three, and the end of November as quarter four.

Linear regression analyses were then performed on the six rate groups for each company to develop its meter count models. After completing the meter count modeling, the results were then allocated to the monthly level using spline functions and to the internal rate code level based on most recent ratios of the various internal rate codes to their rate group.

Summary results of the KEDNY models can be found in Exhibit ___(TEP-10A). Summary results of the KEDLI models can be found in Exhibit ___(TEP-10C). The meter count models for each rate group and the data used to derive them are described in more detail below.

Q. Please describe how you derived the use-per-customer forecasts.

A. The historic monthly use-per-customer values were obtained by dividing the total billed therms for each month by the number of billed customers for the month.

Because the Companies' economic data is available at the quarterly level, the Companies developed their models at the quarterly level, as well. The Companies obtained the historic and projected economic and demographic data used as the independent variables in the regression equations from Moody's economy.com (publication date: February 2018). The Companies obtained energy price data from the DOE/EIA and weather data from the National Oceanic and Atmospheric Administration ("NOAA") and National Weather Service ("NWS").

For its use-per-customer models, each company aggregated its monthly volume data, which is available by internal rate code to the level of its six rate groups and to four quarters per year (the end of each February as quarter one, the end of May as quarter two, the end of August as quarter three, and the end of November as quarter four). To calculate the quarterly use-per-customer, the Companies then divided the quarterly volumes by the corresponding meter counts at the end of each quarter.

Linear regression analyses were then performed on the six rate groups for each company to develop its use-per-customer models. After completing its use-per-customer modeling, quarterly volumes were calculated from the product of quarterly meter count times quarterly use-per-customer. The volume results were then allocated to the internal rate code level based on most recent ratios of the various internal rate codes to their rate group and to the monthly level based on the historical average monthly-to-quarterly percentages.

Summary results of the KEDNY models can be found in Exhibit ___(TEP-10B). Summary results of the KEDLI models can be found in Exhibit ___(TEP-10D). The use-per-customer models for each rate group and the data used to derive them are described in more detail below.

Q. How is the weather variable in the models determined?

A. Weather is a fundamental variable in modeling the natural gas requirements of the Companies' customers. The Companies used gas day (10am – 9am) heating degree days ("HDD") based on temperature observations from the NWS Central Park, New York City, New York weather station in the models for both KEDNY and KEDLI. For the forecasts, as shown in Exhibit ___ (TEP-11), the Companies defined their

normal HDD based on a thirty-year average of the period from July 1988 to June 2018 as 4,535 HDD.

Q. Please explain how the initial forecasts were adjusted for initiatives in the Companies' service territories that could impact the amount of natural gas consumed over the next five years.

A. The Companies' volume forecasts are based on models of meter count and use-per-customer developed for each residential, commercial/industrial, and multifamily rate class. The dependent use-per-customer variable in the Companies' load forecast models is the quotient of historic sales data divided by historic meter count. The Companies' historic sales data includes the impact of actual energy efficiency savings from both the New York State Energy Research and Development Authority ("NYSERDA") and KEDNY/KEDLI sponsored energy efficiency programs. The forecasts include this trend in continuing load reduction based on the historic successes in energy efficiency reductions in load. No further adjustments were made to the forecasts to reflect additional energy efficiency reductions.

As discussed in the direct testimony of the Gas Infrastructure and Operations Panels and Company Witness Elizabeth Arangio, the

Companies have signed a precedent agreement with an upstream pipeline project, known as the Northeast Supply Enhancement project ("NESE"), that will provide access to additional gas supplies necessary to relieve supply constraints in the Companies' service territory. If NESE does not become available by the 2020/21 winter season, the Companies will not be able to prudently add new or additional gas load without jeopardizing the Companies' ability to provide safe, reliable service to its existing firm customers. The Companies' forecasts assume that the NESE project will be approved and in service, and that no moratoriums will be placed on the establishment of new gas services in the Companies' territories during this time period. The Companies' will update their forecasts in Corrections and Updates if the NESE project is not approved.

Q. Has the forecast methodology that you described been used previously by the Companies?

A. Yes. Econometric forecasting has been used by KEDLI and KEDNY for business, gas resource, and infrastructure planning for years, and was used in the Companies' previous rate cases (Cases 16-G-0058 and 16-G-0059). The only change to the Companies' process is to focus on producing retail forecasting at the rate group level and allocating to the rate code level, which reflects discussions with Department of Public Service Staff in the

previous Niagara Mohawk Power Corporation rate case (Case 17-G-0239). Previous forecasts have included regression analysis of individual rate codes, but these analyses often would include marketer movements from sales to and from transportation that could not be attributed to economic or price factors.

Q. Please describe the annual forecast deliveries and the year-end customer count forecasts for KEDNY.

A. As shown in Exhibit ___ (TEP-4A), KEDNY's forecast of normalized retail gas deliveries (excluding non-core and electric generation volumes) is 1.667 billion therms in calendar year 2018, rising to 1.718 billion therms in calendar year 2024. Natural gas deliveries are projected to increase by 8.524 million therms per year or a compound average annual growth rate of 0.50 percent per annum.

As shown in Exhibit ___ (TEP-5A), KEDNY's forecast year-end customer count (excluding non-core and electric generation) is 1,259,420 in calendar year 2018, rising to 1,285,656 in calendar year 2024. Net meter counts are projected to increase by 4,373 meters per year or a compound average annual growth rate of 0.34 percent per annum.

Q. Please describe the annual forecast deliveries and the year-end customer count forecasts for KEDLI.

A. As shown in Exhibit ___ (TEP-4B), KEDLI's forecast of normalized retail gas deliveries (excluding non-core and large electric generation) is 1.060 billion therms in calendar year 2018, rising to 1.172 billion therms in calendar year 2024. Natural gas deliveries are projected to increase by 18.779 million therms per year or a compound average annual growth rate of 1.70 percent per annum.

As shown in Exhibit ___ (TEP-5B), KEDLI's forecast year-end customer count (excluding non-core and electric generation) is 599,441 in calendar year 2018, rising to 629,497 in calendar year 2024. Net meter counts are projected to increase by 5,009 meters per year or a compound average annual growth rate of 0.82 percent per annum.

Q. Please summarize Moody's economy.com economic forecast for downstate New York that was used in the delivery forecast.

A. For KEDNY, population is projected to increase by 0.4 percent per annum over the period from the first quarter of 2018 to the first quarter of 2025, which is a minor decrease from the 0.5 percent per annum growth rate over the period of the first quarter of 2013 to the first quarter of 2018. However,

the number of households is projected to increase by 0.8 percent per annum over the period of the first quarter of 2018 to the first quarter of 2025, which is in line with the 0.8 percent per annum growth rate over the period of the first quarter of 2013 to the first quarter of 2018.

The per capita personal income (in 2009 dollars) for residents in KEDNY's service territory is projected to increase by 1.0 percent per annum over the period from the first quarter of 2018 to the first quarter of 2025, which is a decrease from the 3.3 percent per annum growth rate over the period from the first quarter of 2013 to the first quarter of 2018. The residential natural gas-to-oil price ratio for the first quarter of 2013 (December – February) was 0.36. With lower oil prices, the ratio rose to 0.57 in the first quarter of 2018 and it is projected to decrease to 0.52 by the first quarter of 2025 (Exhibit ___ (TEP-6A))

For KEDLI, population is projected to increase by only 0.03 percent per annum over the period from the first quarter of 2018 to the first quarter of 2025, which is a decrease from the 0.06 percent per annum growth rate over the period from the first quarter of 2013 to the first quarter of 2018. However, the number of households is projected to increase by 0.5 percent per annum over the period from the first quarter of 2018 to the first quarter

of 2025, which is an increase from the 0.3 percent per annum growth rate over the period from the first quarter of 2013 to the first quarter of 2018.

The per capita personal income (in 2009 dollars) for residents in KEDLI's service territory is projected to increase by 0.8 percent per annum over the period from the first quarter of 2018 to the first quarter of 2025, which is a decrease from the 2.2 percent per annum growth rate over the period 2010-2014. The residential natural gas-to-oil price ratio for the first quarter of 2013 (December – February) was 0.38. With lower oil prices, the ratio rose to 0.55 in the first quarter of 2018 and it is projected to decrease to 0.50 by the first quarter of 2025 (Exhibit ___ (TEP-6B))

Q. Please discuss the models used by the Companies to forecast sales and customer counts.

A. As I explained previously, the Companies' load growth forecasts are based on econometric/statistical forecast models developed for each residential, commercial, multifamily, temperature-controlled, and other rate groups and that two different models were developed for each class: (i) a model to forecast the number of customers (meter count) and (ii) a model to forecast use-per-customer.

For the meter count models, the Companies modeled the entire time series of data available for each model. The candidate independent variables are time trends, population, households, employment, and gas and oil prices.

For the use-per-customer models, the Companies divided the time series data for each rate group into two categories – peak season and off-peak season – to reflect periods during which heating customers are using the heating equipment and not using their heating equipment. The use-per-customer during each of those time periods would be different. The breakpoint chosen by the Companies was 600 HDD for the quarter. The candidate independent variables are time trends, heating degree days, population, housing stock, income, employment, unemployment, gross domestic product, and gas and oil prices.

The details of the statistical modeling are provided in Exhibit ___ (TEP-10A) (KEDNY meter count), Exhibit ___ (TEP-10B) (KEDNY use-per-customer), Exhibit ___ (TEP-10C) (KEDLI meter count), and Exhibit ___ (TEP-10D) (KEDLI use-per-customer).

Q. Please explain how KEDNY's rate groups were modeled.

A. Four rate groups (Residential Non-Heating, Residential Heating, Multi-Family, and Commercial) represent 90 percent of the KEDNY retail volumes in calendar year 2017. Two rate groups (Temperature-Controlled and Other) account for the remaining 10 percent.

A. MODELING OF KEDNY RATE GROUPS

1. KEDNY Residential Non-Heating

KEDNY Residential Non-Heating Meter Counts

In KEDNY, SC 1A Residential Non Heat Sales and SC 17-1A Residential Non Heat Transport classes alone accounted for 493,446 customers at year end 2017, or 39.3 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3A)). SC 1A Residential Non Heat Sales and SC 17-1A Residential Non Heat Transport classes combined are projected to decrease to 474,433 customers at year end 2024 (Exhibit ___ (TEP-5A)).

Given that this historical meter count data is highly linear through the years 2007-2018 (*i.e.*, through the "Great Recession" and recovery), the time variable was most appropriate as the data reflects a market not driven by economic factors, but a constrained market where the non-heating

customers are converting at a fixed rate. This can be due to failure rate of their existing non-gas-fired heating equipment over time.

KEDNY Residential Non-Heating Use per Customer

In KEDNY, SC 1A Residential Non Heat Sales normalized use-per-customer was 98 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). SC 17-1A Residential Non Heat Transport normalized use-per-customer was 222 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, SC 1A Residential Non Heat Sales normalized use-per-customer is projected to decrease to 94 therms per year (Exhibit ___ (TEP-14A)). SC 17-1A Residential Non Heat Transport normalized use-per-customer is projected to decrease to 212 therms per year in calendar year 2024 (Exhibit ___ (TEP-14A)).

The peak period KEDNY residential non-heating use-per customer was found to be positively related to billing degree days and negatively related to the residential price of natural gas. The off-peak period KEDNY residential non-heating use-per customer was found to be negatively related to both personal income and the residential price of natural gas.

KEDNY Residential Non-Heating Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4A). In KEDNY, SC 1A Residential Non-Heat Sales and SC 17-1A Residential Non-Heat Transport volumes alone accounted for 0.057 billion therms in calendar year 2017, or 3.7 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-1A)). SC 1A Residential Non-Heat Sales and SC 17-1A Residential Non-Heat Transport volumes are projected to decrease to 0.054 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4A)).

2. KEDNY Residential Heating

KEDNY Residential Heating Meter Counts

In KEDNY, SC 1B Residential Heating and SC 17-1B Residential Heating Transport classes alone accounted for 550,995 customers at year end 2017, or 43.9 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3A)). SC 1B Residential Heating and SC 17-1B Residential Heating Transport classes combined are projected to increase to 595,693 customers at year end 2024 (Exhibit ___ (TEP-5A)).

Given that this historical meter count data is highly linear through the years 2007-2018 (*i.e.*, through the "Great Recession" and recovery), the time variable was most appropriate as the data reflects a market not driven by economic factors, but a constrained market where the non-heating customers are converted to heating customers at a fixed rate and new customers are added to the system at a fixed rate.

KEDNY Residential Heating Use per Customer

In KEDNY, SC 1B Residential Heating normalized use-per-customer was 1,217 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). SC 17-1B Residential Heating Transport normalized use-per-customer was 1,476 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, SC 1B Residential Heating normalized use-per-customer is projected to decrease to 1,163 therms per year (Exhibit TEP-14A). SC 17-1B Residential Heating Transport normalized use-per-customer is projected to rise to 1,597 therms per year in calendar year 2024 (Exhibit ___ (TEP-14A)).

The peak period KEDNY residential heating use-per customer was found to be positively related to billing degree days alone. The off-peak period KEDNY residential heating use-per customer was found to be negatively

related to time.

KEDNY Residential Heating Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4A). In KEDNY, SC 1B Residential Heating and SC 17-1B Residential Heating Transport volumes alone accounted for 0.694 billion therms in calendar year 2017, or 42.8 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8A)). SC 1B Residential Heating and SC 17-1B Residential Heating Transport volumes are projected to increase to 0.736 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4A)).

3. KEDNY Multi-Family

KEDNY Multi-Family Meter Counts

In KEDNY, SC 3 Multi-Family Sales, SC 6M Multi-Family TC Sales, SC 17-3 Multi-Family Transport, and SC 18-6M Multi-Family TC Transport classes accounted for 20,592 customers at year end 2017, or 1.6 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3A)). Combined, these classes are projected to increase to 23,769 customers at year end 2024 (Exhibit ___ (TEP-5A)).

Given that this historical meter count data is highly linear through the years 2007-2018 (*i.e.*, through the "Great Recession" and recovery), the time variable was most appropriate as the data reflects a market not driven by economic factors, but a constrained market where new customers are added to the system at a fixed rate.

KEDNY Multi-Family Use per Customer

In KEDNY, SC 3 Multi-Family Sales use-per-customer was 8,912 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to decrease to 8,616 therms per year (Exhibit ___ (TEP-14A)).

SC 6M Multi-Family TC Sales use-per-customer was 49,319 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to decrease to 37,524 therms per year (Exhibit ___ (TEP-14A)).

SC 17-3 Multi-Family Transport use-per-customer was 31,570 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to decrease slightly to 31,024 therms per year (Exhibit ___ (TEP-14A)).

SC 18-6M Multi-Family TC Transport use-per-customer was 72,461 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to increase to 77,761 therms per year (Exhibit ___ (TEP-14A)).

The peak period KEDNY multi-family use-per customer was found to be positively related to billing degree days alone. The off-peak period KEDNY multi-family use-per customer was found to be positively related to time.

KEDNY Multi-Family Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4A). In KEDNY, SC 3 Multi-Family Sales, SC 6M Multi-Family TC Sales, SC 17-3 Multi-Family Transport, and SC 18-6M Multi-Family TC Transport volumes accounted for 0.435 billion therms in calendar year 2017, or 26.8 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8A)). Combined, these volumes are projected to increase to 0.451 billion

therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4A)).

4. KEDNY Commercial

KEDNY Commercial Meter Counts

In KEDNY, SC 2-1 Non-Residential Non-Heat Sales, SC 2-2 Non Residential Heat Sales, SC 17-2-1 Non Residential Non-Heat Transport, and SC 17-2-2 Non Residential Heat Transport classes alone accounted for 50,536 customers at year end 2017, or 4.0 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3A)). Combined, these classes are projected to increase to 52,306 customers at year end 2024 (Exhibit ___ (TEP-5A)). KEDNY commercial meter counts were modeled as a function of population.

KEDNY Commercial Use per Customer

In KEDNY, SC 2-1 Non-Residential Non-Heat Sales normalized use-per-customer was 5,508 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to increase to 6,125 therms per year (Exhibit ___ (TEP-14A)).

SC 2-2 Non-Residential Heat Sales normalized use-per-customer was 3,793 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For

calendar year 2024, its normalized use-per-customer is projected to increase to 3,913 therms per year (Exhibit ___ (TEP-14A)).

SC 17-2-1 Non Residential Non-Heat Transport normalized use-per-customer was 13,064 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to increase to 15,926 therms per year (Exhibit ___ (TEP-14A)).

SC 17-2-2 Non Residential Heat Transport normalized use-per-customer was 6,339 therms per year in calendar year 2017 (Exhibit ___ (TEP-13A)). For calendar year 2024, its normalized use-per-customer is projected to increase to 7,933 therms per year (Exhibit ___ (TEP-14A)).

The peak period KEDNY commercial use-per customer was found to be positively related to both billing degree days and retail sales. The off-peak period KEDNY commercial use-per customer was found to be positively related to personal income.

<u>KEDNY Commercial Volumes</u>

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery

forecast presented in Exhibit ___ (TEP-4A). In KEDNY, SC 2-1 Non Residential Non-Heat Sales, SC 2-2 Non Residential Heat Sales, SC 17-2-1 Non Residential Non-Heat Transport, and SC 17-2-2 Non Residential Heat Transport volumes accounted for 0.277 billion therms in calendar year 2017, or 17.1 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8A)). Combined, these volumes are projected to increase to 0.322 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4A)).

Q. Please explain how KEDLI's rate groups were modeled.

A. Four rate groups (Residential Non-Heating, Residential Heating, Multi-Family, and Commercial) represent 91 percent of the KEDLI retail volumes in calendar year 2017. Two rate groups (Temperature-Controlled and Other) account for the remaining 9 percent.

B. MODELING OF KEDLI RATE GROUPS

1. KEDLI Residential Non-Heating

KEDLI Residential Non-Heating Meter Counts

In KEDLI, SC 1A Residential General Sales, SC 1A Water Heat Sales, SC 5-1A Residential General, and SC 5-1A Water Heat Transport classes alone accounted for 94,547 customers at year end 2017, or 16.0 percent of

total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3B)). SC 1A Residential General Sales, SC 1A Water Heat Sales, SC 5-1A Residential General, and SC 5-1A Water Heat Transport classes combined are projected to decrease to 71,593 customers at year end 2024 (Exhibit ___ (TEP-5B)).

Given that this historical meter count data is highly linear through the years 2007-2018 (*i.e.*, through the "Great Recession" and recovery), the time variable was most appropriate as the data reflects a market not driven by economic factors, but a constrained market where the non-heating customers are converting at a fixed rate. This can be due to failure rate of their existing non-gas-fired heating equipment over time.

<u>KEDLI Residential Non-Heating Use per Customer</u>

In KEDLI, SC 1A Residential General Sales use-per-customer was 190 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, SC 1A Residential General Sales normalized use-per-customer is projected to decrease to 171 therms per year (Exhibit ___ (TEP-14B)).

SC 1A Residential Water Heat Sales normalized use-per-customer was 458 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, SC 1A Residential Water Heat Sales normalized use-per-customer is projected to decrease to 412 therms per year (Exhibit ___ (TEP-14B)).

SC 5-1A Residential General Transport use-per-customer was 231 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, SC 5-1A Residential General Transport normalized use-per-customer is projected to decrease to 204 therms per year (Exhibit ___ (TEP-14B)).

SC 5-1A Residential Water Heat Transport use-per-customer was 562 therms per year in calendar year 2017 (Exhibit ____ (TEP-13B). For calendar year 2024, SC 5-1A Residential Water Heat Transport normalized use-per-customer is projected to decrease to 485 therms per year (Exhibit ___ (TEP-14B)).

The peak period KEDLI residential non-heating use-per customer was found to be positively related to billing degree days alone. The off-peak

period KEDLI residential non-heating use-per customer was found to be negatively related to time.

KEDLI Residential Non-Heating Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4B). In KEDLI, SC 1A Residential General, SC 1A Water Heat Sales, SC 5-1A Residential General, and SC 5-1AWater Heat Transport volumes alone accounted for 0.027 billion therms in calendar year 2017, or 2.7 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8B)). SC 1A Residential General, SC 1A Water Heat Sales, SC 5-1A Residential General, and SC 5-1A Water Heat Transport volumes are projected to decrease to 0.018 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4B)).

2. KEDLI Residential Heating

KEDLI Residential Heating Meter Counts

In KEDLI, SC 1B Residential Space Heat Sales and SC 5-1B Residential Space Heat Transport classes alone accounted for 425,631 customers at year end 2017, or 72 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3B)). SC 1B Residential Space

Heat Sales and SC 5-1B Residential Space Heat Transport classes combined are projected to increase to 481,316 customers at year end 2024 (Exhibit ___ (TEP-5B)).

Given that this historical meter count data is highly linear through the years 2007-2018 (*i.e.*, through the "Great Recession" and recovery), the time variable was most appropriate as the data reflects a market not driven by economic factors, but a constrained market where the non-heating customers are converted to heating customers at a fixed rate and new customers are added to the system at a fixed rate.

<u>KEDLI Residential Heating Use per Customer</u>

In KEDLI, SC 1B Residential Space Heat Sales normalized use-per-customer was 1,215 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). SC 5-1B Residential Space Heat Transport normalized use-per-customer was 1,440 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, SC 1B Residential Space Heat Sales normalized use-per-customer is projected to decrease to 1,195 therms per year (Exhibit ___ (TEP-14B)). SC 5-1B Residential Space Heat Transport normalized use-per-customer is projected to decrease to 1,386 therms per year in calendar year 2019 (Exhibit ___ (TEP-14B)).

The peak period KEDLI residential heating use-per customer was found to be positively related to billing degree days alone. The off-peak period KEDLI residential heating use-per customer was found to be negatively related to time.

KEDLI Residential Heating Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4B). In KEDLI, SC 1B Residential Space Heat Sales and SC 5-1B Residential Space Heat Transport volumes alone accounted for 0.524 billion therms in calendar year 2017, or 52.4 percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8B)). SC 1B Residential Space Heat Sales and SC 5-1B Residential Space Heat Transport volumes are projected to increase to 0.582 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4B)).

3. KEDLI Multi-Family

KEDLI Multi-Family Meter Counts

In KEDLI, SC 3A Multi-Family Non-Heat Sales, SC 3B Multi-Family Heat Sales, SC 5-3A Multi-Family Non-Heat Transport, and SC 5-3B

Multi-Family Heat Transport classes accounted for 1,661 customers at year end 2017, or 0.3 percent of total customers, excluding non-core and electric generation (Exhibit ___ (TEP-3B)). Combined, these classes are projected to increase to 2,103 customers at year end 2024 (Exhibit ___ (TEP-5B)).

KEDLI multi-family meter count was found to be positively related to households and negatively related to personal income.

KEDLI Multi-Family Use Per Customer

In KEDLI, SC 3A Multi-Family Non-Heat Sales normalized use-per-customer was 8,690 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 9,235 therms per year (Exhibit ___ (TEP-14B)).

SC 3B Multi-Family Heat Sales normalized use-per-customer was 19,668 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to decrease slightly to 19,479 therms per year (Exhibit ___ (TEP-14B)).

SC 5-3A Multi-Family Non-Heat Transport normalized use-per-customer was 17,642 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 18,196 therms per year (Exhibit ___ (TEP-14B)).

SC 5-3B Multi-Family Heat Transport normalized use-per-customer was 41,375 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase slightly to 41,462 therms per year (Exhibit ___ (TEP-14B)).

The peak period KEDLI multi-family use-per-customer was found to be positively related to billing degree days. The off-peak period KEDLI multi-family use-per-customer was found to be negatively related to time.

KEDLI Multi-Family Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4B). In KEDLI, SC 3A Multi-Family Non-Heat Sales, SC 3B Multi-Family Heat Sales, SC 5-3A Multi-Family Non-Heat Transport, and SC 5-3B Multi-Family Heat Transport volumes accounted for 0.052 billion therms in calendar year 2017, or 5.2

percent of total volume, excluding non-core and electric generation (Exhibit ___ (TEP-8B)). Combined, these volumes are projected to increase to 0.065 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4B)).

4. KEDLI Commercial

KEDLI Commercial Meter Counts

In KEDLI, SC 2A Non Residential Water Heat Sales, SC 2A Non Residential General Sales, SC 2B Non Residential Space Heat Sales, SC 5-2A Non Residential Water Heat Transport, SC 5-2A Non Residential General Transport, and SC 5-2B Non Residential Space Heat Transport classes alone accounted for 58,266 customers at year end 2017, or 9.8 percent of total customers excluding non-core and electric generation (Exhibit ___ (TEP-3B)). Combined, these classes are projected to increase to 60,308 customers at year end 2019 (Exhibit ___ (TEP-5B)).

KEDLI commercial meter count was found to be positively related to the time variable.

KEDLI Commercial Use per Customer

In KEDLI, SC 2A Non Residential Water Heat Sales normalized use-per-customer was 1,587 therms per year in calendar year 2017 (Exhibit ___

(TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 2,388 therms per year (Exhibit ___ (TEP-14B)).

SC 2A Non Residential General Sales normalized use-per-customer was 6,088 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 8,385 therms per year (Exhibit ___ (TEP-14B)).

SC 2B Non Residential Space Heat Sales normalized use-per-customer was 3,810 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 4,855 therms per year (Exhibit ___ (TEP-14B)).

SC 5-2A Non Residential Water Heat Transport normalized use-per-customer was 9,609 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 11,365 therms per year (Exhibit ___ (TEP-14B)).

SC 5-2A Non Residential General Transport normalized use-per-customer was 14,210 therms per year in calendar year 2017 (Exhibit ___ (TEP-

13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 19,826 therms per year (Exhibit ___ (TEP-14B)).

SC 5-2B Non Residential Space Heat Transport normalized use-per-customer was 7,937 therms per year in calendar year 2017 (Exhibit ___ (TEP-13B)). For calendar year 2024, its normalized use-per-customer is projected to increase to 10,161 therms per year (Exhibit ___ (TEP-14B)).

The peak period KEDLI commercial use-per-customer was found to be positively related to billing degree days and KEDLI gross domestic product. The off-peak period KEDLI multi-family use-per-customer was found to be positively related to manufacturing employment.

KEDLI Commercial Volumes

The results of the customer count forecasts and the use-per-customer forecasts were then multiplied together to derive the volume delivery forecast presented in Exhibit ___ (TEP-4B). In KEDLI, SC 2A Non Residential Water Heat Sales, SC 2A Non Residential General Sales, SC 2B Non Residential Space Heat Sales, SC 5-2A Non Residential Water Heat Transport, SC 5-2A Non Residential General Transport, and SC 5-2B Non Residential Space Heat Transport volumes accounted for 0.324

billion therms in calendar year 2017, or 32.4 percent of total volume excluding non-core and electric generation (Exhibit ___ (TEP-8B)). Combined, these volumes are projected to increase to 0.437 billion therms in calendar year 2024 on a normalized basis (Exhibit ___ (TEP-4B)).

V. Conclusion

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Supplemental Direct Testimony

of

Theodore Poe, Jr.

Date: June 11, 2019

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I. Introduction

Q. Please state your name and business address.

A. My name is Theodore Poe, Jr. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Theodore Poe, Jr. who previously submitted direct testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies") in these proceedings?

A. Yes, I am. The terms defined in my direct testimony have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of your supplemental direct testimony?

A. The purpose of my supplemental direct testimony is to discuss the impact to the sales forecasts in the Companies' April 2019 rate filings if they are unable to access incremental natural gas supplies from Transcontinental Gas Pipe Line, LLC's Northeast Supply Enhancement ("NESE") project, which would help to relieve current gas supply constraints in the Companies' service territories during the Rate Year and Data Years. In my

direct testimony, I identified certain impacts to the Companies' filings if the NESE project was not placed in service by the winter of 2020/2021. As discussed more fully in the supplemental testimony of Company Witness Elizabeth D. Arangio, recent decisions by the New York State Department of Environmental Conservation ("DEC") and the New Jersey Department of Environmental Protection regarding required water permits create some uncertainty as to whether and when the NESE project may proceed with permitting and construction. While the Companies remain cautiously optimistic the NESE project will go forward, and therefore are not proposing any changes to their April 2019 filings at this time, I address the impact that continued gas supply constraints would have on the Companies' gas sales forecasts in the Rate Year and Data Years.

Q. Does your supplemental direct testimony include any exhibits?

A. Yes. My supplemental direct testimony includes the following exhibits that were prepared or compiled under my direction and supervision:

Exhibit ___ (TEP-1S)	KEDNY Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024 Assuming the Lack of Availability of NESE

Exhibit ___ (TEP-2S)	KEDLI Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024 Assuming the Lack of Availability of NESE
Exhibit ___ (TEP-3S)	KEDNY Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024 Assuming the Lack of Availability of NESE
Exhibit ___ (TEP-4S)	KEDLI Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024 Assuming the Lack of Availability of NESE
Exhibit ___ (TEP-5S)	Comparative Summary of KEDNY and KEDLI Volumes and Customer Counts: Company Initial Filing Forecast versus Lack of Availability of NESE ("No-Growth") Scenario

III. Impact of the Unavailability of the NESE Project

Q. Did your direct testimony discuss the possibility that the NESE project might not be available and that this could affect the Companies' filings?

A. Yes. While the Companies' filings assumed the NESE project would be completed by the winter of 2020/2021, my direct testimony (at 17-18) identified the possibility that the NESE project would not be available as planned and explained that the Companies' gas sales forecasts would be impacted by the resulting gas supply constraints.

Q. Please explain the impact to the Companies' gas sales forecasts if the NESE project is not approved.

A. Tables 1 and 2 below and Exhibit ___ (TEP-5S) summarize the anticipated impacts to KEDNY and KEDLI's gas sales volumes and customer counts if the NESE project is not available.

Table 1 –Sales Volume and Meter Count with NESE Project

		2018	2019	2020	2021	2022	2023
Meter Count	**KEDNY**	1,259,420	1,263,770	1,268,155	1,272,545	1,276,927	1,281,297
	KEDLI	599,441	604,468	609,478	614,484	619,490	624,497
Sales Volume (Therms)	**KEDNY**	1,667M	1,641M	1,656M	1,665M	1,682M	1,700M
	KEDLI	1,059M	1,052M	1,068M	1,089M	1,117M	1,144M

Table 2 –Sales Volume and Meter Count without NESE Project

		2018	2019	2020	2021	2022	2023
Meter Count	**KEDNY**	1,259,420	1,260,976	1,261,396	1,261,509	1,261,509	1,261,509
	KEDLI	599,441	592,757	593,030	593,117	593,117	593,117
Sales Volume (Therms)	**KEDNY**	1,667M	1,635M	1,642M	1,643M	1,643M	1,643M
	KEDLI	1,059M	1,043M	1,045M	1,045M	1,045M	1,045M

As a result of the recent denials (without prejudice) of the water permits for the NESE project, National Grid stopped processing all applications for new or expanded firm gas service in Brooklyn, Queens and on Long Island as of May 15, 2019, thereby curtailing customer load growth in the Companies' service territories. The "no-growth" scenario presented in Exhibit ___ (TEP-5S) assumes that, over the period of May 2019 through April 2020, the Companies will fulfill the residential and commercial gas service requests that were approved (919 customers in KEDNY and 553 in KEDLI) prior to May 15th. No additional customers will be added to the system in the sales forecast. Also, because existing customers will not be able to increase their gas loads, this "no-growth" scenario assumes there will be no changes to the existing uses-by-customer (*i.e.* the normalized March 2019 – February 2020 use-per-customer values will continue into the future).

While the Companies will continue processing applications for non-firm service from qualifying customers to the extent the Companies are able to provide service, the Companies did not adjust their forecasts to reflect additional non-firm customers because there is no conclusive information on the number of customers who might desire such a service in the current environment.

Q. Do the Companies plan to continue to provide updates in these proceedings concerning the status of the NESE project?

A. Yes. The Companies are cautiously optimistic that the NESE project will be approved. As the project's status is uncertain, the Companies will update Staff and other parties on the status of the project throughout these proceedings and will supplement the record as further developments occur.

IV. Conclusion

Q. Does this conclude your supplemental direct testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Theodore Poe, Jr.

Date: July 3, 2019

Q. Please state your name and business address.

A. My name is Theodore Poe, Jr. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Theodore Poe, Jr. who previously submitted direct and supplemental testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes, I am. The terms defined in my direct testimony have the same meanings here.

Q. What is the purpose of your corrections and updates testimony?

A. The purpose of my corrections and updates testimony is to update the Companies' forecasts of projected gas sales for the Rate Year and Data Years based on actual data through February 2019.

Q. Does your testimony also address the Companies' June 11, 2019 supplemental filing?

A. No. The supplemental filing was made to illustrate the potential impacts to the April 30, 2019 filing if KEDNY and KEDLI are unable to obtain incremental natural gas supplies from the NESE project. At this time, the Companies remain optimistic that the NESE project will be approved.

Therefore, all corrections and/or updates discussed herein pertain to the April 30 base rate filing. The Companies will continue to update the parties on the status of the NESE project and implications, if any, to this rate filing.

Q. Does your corrections and updates testimony include any exhibits?

A. Yes. My corrections and updates testimony includes the following exhibits that were prepared or compiled under my direction and supervision:

Exhibit __ (TEP-1ACU)	KEDNY Calendar Year Deliveries by Service and Revenue Class for the Period 2013 – 2017
Exhibit __ (TEP-1BCU)	KEDLI Calendar Year Deliveries by Service and Revenue Class for the Period 2013 – 2017
Exhibit __ (TEP-2ACU)	KEDNY Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
Exhibit __ (TEP-2BCU)	KEDLI Average Annual Calendar Year Use-Per-Customer Based on Total

		Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
	Exhibit __ (TEP-3ACU)	KEDNY End of Year Customer Count by Service and Revenue Class for the Period 2013 – 2017
	Exhibit __ (TEP-3BCU)	KEDLI End of Year Customer Count by Service and Revenue Class for the Period 2013 – 2017
	Exhibit __ (TEP-4ACU)	KEDNY Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
	Exhibit __ (TEP-4BCU)	KEDLI Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
	Exhibit __ (TEP-5ACU)	KEDNY Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024
	Exhibit __ (TEP-5BCU)	KEDLI Forecast End of Year Customer Counts by Service and Revenue Class for the Period 2018 through 2024

Exhibit __ (TEP-6ACU)	KEDNY Historical and Projected Economic Indicators Used to Develop the Forecast Models
Exhibit __ (TEP-6BCU)	KEDLI Historical and Projected Economic Indicators Used to Develop the Forecast Models
Exhibit __ (TEP-7ACU)	KEDNY Historical Actual Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit _ (TEP-7BCU)	KEDLI Historical Actual Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-8ACU)	KEDNY Historical Weather Normalized Billed Sales Data by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-8BCU)	KEDLI Historical Weather Normalized Billed Sales Data by Month and by Class for the Period January 2013 through December 2017

Exhibit __ (TEP-9ACU)	KEDNY Historical Actual Customer Counts by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-9BCU)	KEDLI Historical Actual Customer Counts by Month and by Class for the Period January 2013 through December 2017
Exhibit __ (TEP-10ACU)	Delivery Forecast Model Specifications – KEDNY – Meter Count
Exhibit __ (TEP-10BCU)	Delivery Forecast Model Specifications – KEDNY – Use Per Customer
Exhibit __ (TEP-10CCU)	Delivery Forecast Model Specifications – KEDLI – Meter Count
Exhibit __ (TEP-10DCU)	Delivery Forecast Model Specifications – KEDLI – Use Per Customer
Exhibit __ (TEP-12ACU)	KEDNY Calendar Year Actual Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-12BCU)	KEDLI Calendar Year Actual Use Per Customer by Service and Revenue Class for the Period 2013 - 2017

Exhibit __ (TEP-13ACU)	KEDNY Calendar Year Normalized Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-13BCU)	KEDLI Calendar Year Normalized Use Per Customer by Service and Revenue Class for the Period 2013 - 2017
Exhibit __ (TEP-14ACU)	KEDNY Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-14BCU)	KEDLI Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024

Q. Please describe the changes to the Companies' gas sales forecasts.

A. My direct testimony and exhibits provided forecasts of the Companies' gas sales for the Rate Year and Data Years using actual data through February 2018. The Companies' recently received actual data through February 2019 and used that data to refresh the gas sales forecasts for the Rate Year and Data Years. Historical data, as presented in my corrections and updates exhibits, is now through February 2019. The forecast data in my corrections and updates exhibits now begins in March 2019.

For KEDNY, incorporation of data through February 2019 results in an increase in the Rate Year gas sales forecast from 1,658 million therms to 1,689 million therms and an increase from 1,271,106 customers to 1,274,563 customers (as measured in January of the Rate Year). For KEDLI, incorporation of data through February 2019 results in a slight decrease in the Rate Year gas sales forecast from 1,075 million therms to 1,059 million therms with an increase from 610,304 customers to 613,552 customers (as measured in January of the Rate Year). While there has been little change in the economic forecast between the two gas sales forecasts, the inclusion of the additional year of customer data is the principal driver of these changes. The impacts of the increase in the Companies' gas sales forecasts have been reflected in the corrections and updates testimonies of Elizabeth D. Arangio, the Rate Design Panels, and the Revenue Requirements Panels.

Q. Does this conclude your corrections and updates testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Second Supplemental Testimony

of

Theodore Poe, Jr.

Date: December 13, 2019

Q. Please state your name and business address.

A. My name is Theodore Poe, Jr. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Theodore Poe, Jr. who previously submitted testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes. Previously-defined terms have the same meanings here.

Q. What is the purpose of your second supplemental testimony?

A. The purpose of my second supplemental testimony is to update the Companies' forecast of projected gas sales for the Rate Year and Data Years to reflect the impacts of the recent settlement agreement between the Companies and DPS Staff in Case 19-G-0678 concerning the gas supply constraints in downstate New York and related moratorium on new customer connections.

Q. Does your second supplemental testimony include any exhibits?

A. Yes. My second supplemental testimony includes the following exhibits that were prepared or compiled under my direction and supervision:

Exhibit __ (TEP-2A-SS)	KEDNY Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
Exhibit __ (TEP-2B-SS)	KEDLI Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
Exhibit __ (TEP-4A-SS)	KEDNY Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-4B-SS)	KEDLI Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-14A-SS)	KEDNY Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024

Exhibit __ (TEP-14B-SS)	KEDLI Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-15A-SS)	KEDNY Retail Forecast Spreadsheet, originally presented as Attachment 1 to DPS-068
Exhibit __ (TEP-15B-SS)	KEDLI Retail Forecast Spreadsheet, originally presented as Attachment 2 to DPS-068

Q. Please describe the changes to the Companies' gas sales forecasts.

A. My corrections and updates testimony and exhibits provided forecasts of the Companies' gas sales for the Rate Year and Data Years using data through February 2019 and assumed the availability of capacity supplied by the NESE project by the winter of 2020/2021, which would have enabled new customer connections during the Rate Year and Data Years. While the NESE project is currently projected to be unavailable during the Rate Year and Data Years, the Commission recently approved a settlement agreement ("Settlement") between the Companies and DPS Staff in Case 19-G-0678 wherein the Companies' agreed to lift their gas service moratorium for the next two winters (through September 1, 2021). To enable new customer

connections, the Companies have developed an operational plan that provides for the expanded use of incremental city gate peaking supplies and portable compressed natural gas ("CNG") injections to maintain system pressure during peak conditions, as well as various demand-side and energy efficiency measures to reduce peak demand. The Settlement also provides a framework for identifying a long-term solution to address the current supply constraints.

Because the Companies will continue to add new customers consistent with the growth assumptions underlying my corrections and updates forecast, no updates are required to the forecast of customers to reflect the lifting of the moratorium. My forecast assumes continued sales growth in the Rate Year and Data Years. However, I have updated the Companies' sales forecasts to reflect the anticipated impact of the incremental energy efficiency and demand response programs per the Settlement.

For KEDNY, the updates to the gas sales forecast to reflect the incremental demand response and energy efficiency result in a decrease in the Rate Year from 1,689,417,139 therms to 1,689,187,510 therms compared to the corrections and updates sales forecast. For KEDLI, these updates result in a decrease in the Rate Year from 1,059,164,436 therms to 1,058,941,852

therms compared to the corrections and updates sales forecast. These decreases are attributable to the impact of the incremental energy efficiency and demand response programs that I mentioned previously. Again, there are no changes to the number of customers as new customers and expanded customer connections will continue to be connected at least through September 1, 2021. The impacts of the decreased therms in the Companies' gas sales forecasts have been reflected in the second supplemental testimony of Elizabeth D. Arangio and the Revenue Requirements Panels.

Q. Does this conclude your corrections and updates testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Third Supplemental Testimony

of

Theodore Poe, Jr.

Date: January 29, 2020

Q. Please state your name and business address.

A. My name is Theodore Poe, Jr. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Theodore Poe, Jr. who previously submitted testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes. Previously-defined terms have the same meanings here.

Q. What is the purpose of your third supplemental testimony?

A. The purpose of my third supplemental testimony is to update the Companies' forecast of projected gas sales for the Rate Year and Data Years to reflect the impact of the January 16, 2020 Commission order in Case 18-M-0084 directing implementation of incremental utility energy efficiency and clean heating targets through 2025 (the "EE Order"). The updated forecast uses the forecast provided in my second supplemental testimony, filed on December 13, 2019, as a starting point and then incorporates the projected energy efficiency benefits that will result from the recently implemented targets from the EE Order.

Q. Does your third supplemental testimony include any exhibits?

A. Yes. My third supplemental testimony includes the following exhibits that were prepared or compiled under my direction and supervision:

Exhibit __ (TEP-2A-TS)	KEDNY Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
Exhibit __ (TEP-2B-TS)	KEDLI Average Annual Calendar Year Use-Per-Customer Based on Total Residential Deliveries for the Historical Period 2013 – 2017 and the Forecast Period 2018 – 2024
Exhibit __ (TEP-4A-TS)	KEDNY Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-4B-TS)	KEDLI Forecast Calendar Year Deliveries by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-14A-TS)	KEDNY Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-14B-TS)	KEDLI Forecast Calendar Year Use Per Customer by Service and Revenue Class for the Period 2018 through 2024
Exhibit __ (TEP-15A-TS)	KEDNY Retail Forecast Spreadsheet, originally presented as Attachment 1 to DPS-068
Exhibit __ (TEP-15B-TS)	KEDLI Retail Forecast Spreadsheet, originally presented as Attachment 2 to DPS-068

Q. Please describe the changes to the Companies' gas sales forecasts.

A. My corrections and updates testimony and exhibits provided forecasts of the Companies' gas sales for the Rate Year and Data Years using data through February 2019. My second supplemental testimony and exhibits updated the Companies' sales forecast to reflect the operational plan adopted by the Companies in response to the Settlement between the Companies and DPS Staff in Case 19-G-0678, which provides for new customer connections through September 1, 2021.

The EE Order approved certain energy efficiency targets in an effort to reduce energy consumption and combat climate change. The EE Order directs utilities in New York State, including the Companies, to invest in energy efficiency and clean heat technologies from 2020 through 2025. Once implemented, these technologies are projected to provide reductions in energy usage, baseload, and peak demand in the Companies' service territories. As implementation of the energy efficiency and clean heat technologies will occur in the Rate Year and Data Years, the Companies are updating the forecast to incorporate the energy reductions these technologies are projected to provide. Further discussion on how the technologies from the EE Order will be implemented is provided in the supplemental testimony of the Future of Heat Panel.

For KEDNY, the updates to the gas sales forecast to reflect the incremental demand response and energy efficiency result in a decrease in the Rate Year from 1,689,187,510 therms to 1,687,554,812 therms compared to KEDNY's second supplemental sales forecast. For KEDLI, these updates result in a decrease in the Rate Year from

1,058,941,852 therms to 1,058,168,612 therms compared to KEDLI's second supplemental sales forecast. These decreases are attributable to the impact of the incremental energy efficiency technologies from the EE Order. The impacts of the decreased therms in the Companies' gas sales forecasts have been reflected in the third supplemental testimony of the Companies' Revenue Requirements Panels.

Q. Does this conclude your third supplemental testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Direct Testimony

of

Jonathan Cohen

Dated: April 2019

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I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Jonathan Cohen. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what position?

A. I am employed by National Grid USA Service Company, Inc. ("Service Company"), a subsidiary of National Grid USA ("National Grid"), as Director of US Treasury, Planning, and Strategy (New York).

Q. Please describe your educational background and professional experience.

A. I am a qualified accountant with the Institution for Chartered Accountants in England and Wales and a qualified Treasurer with the Association of Corporate Treasurers. From 2006 to 2012, I worked at BDO LLP, an accounting, tax, audit, and consulting services firm. I worked in audit, specializing in financial services companies, including banks, brokers, and fund managers. I joined National Grid in 2012 as a member of the Corporate Banking Group. In that role, I was responsible for managing the bank relationships, including associated borrowings and revolving credit facilities. I was also responsible for analyzing and monitoring National Grid's credit ratings and managing interactions with its debt investors. I joined National Grid's Debt Capital Markets group in 2015, participating in public bond issuances and managing the associated currency and interest rate risks. I also participated in the sale of significant assets in one of National Grid plc's largest operating companies in the United Kingdom. I began my current position in 2017, where I am

focused on business partnering to determine the efficiency of operations and obtain identified value opportunities. In my current position, I am familiar with the financing activities of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or the "Company").

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to present and support KEDNY's proposed capital structure and overall cost of capital in this proceeding. My testimony provides information for the test year ended December 31, 2018 ("Historic Test Year"), the forecast year ending March 31, 2021 ("Rate Year"), and the years ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (collectively "Data Years"). In addition, I will discuss KEDNY's proposal to continue the currently effective financial protection provisions adopted by the New York State Public Service Commission ("Financial Protections"). Finally, I discuss and support the cost of capital component included in the service company asset recovery charge that is charged to affiliates, such as KEDNY, to recover the costs of shared assets financed and owned by Service Company.

Q. Do you sponsor any exhibits as part of your testimony in this proceeding?

A. Yes. Schedule 1 of Exhibit __ (JC-1) sets forth KEDNY's cost of long-term debt. Schedule 2 contains the projected capitalization and weighted average cost of capital that is proposed to be adopted for KEDNY in this proceeding. Schedule 3 sets forth an analysis of KEDNY's credit metric projections. Schedule 4 sets forth KEDNY's weighted cost of capital for both the Rate Year and for the term of a multi-year rate plan. Schedule 5 sets

forth a forecast sources and uses of funds statement. These exhibits were prepared or compiled under my supervision and direction.

Q. What is the weighted average cost of capital that you are recommending be adopted for KEDNY in this proceeding?

A. KEDNY's proposed Rate Year weighted average cost of capital, as shown on Schedule 4, of Exhibit __ (JC-1), is 6.76 percent. This overall rate of return is based on the following capitalization ratios and cost rates:

Table 1: One-Year Weighted Cost of Capital

	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.57%	4.11%	21.12%
Customer Deposits	0.43%	2.45%	0.01%
Common Equity	48.00%	9.65%	4.63%
Total	100.00%		6.76%

If rates were to be established in a multi-year settlement in this proceeding, the Company proposes the following rates of return, including a stay-out premium on its cost of equity, as shown on Schedule 4 of Exhibit __ (JC-1).

Table 2: Multi-Year Weighted Cost of Capital

	Rate Year (FY21)			Data Year 1 (FY22)			Data Year 2 (FY23)			Data Year 3 (FY24)		
	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.57%	4.11%	2.12%	50.61%	4.11%	2.08%	49.62%	4.11%	2.04%	49.68%	4.16%	2.06%
Customer Deposits	0.43%	2.45%	0.01%	0.39%	2.45%	0.01%	0.38%	2.45%	0.01%	0.32%	2.45%	0.01%
Common Equity	48.00%	10.15%	4.87%	49.00%	10.15%	4.97%	50.00%	10.15%	5.08%	50.00%	10.15%	5.08%
Total	100.00%		7.00%	100.00%		7.06%	100.00%		7.13%	100.00%		7.15%

In calculating the capitalization ratios shown above, all of the goodwill recorded on KEDNY's books was excluded from its total capitalization and common equity balances.

II. Ratemaking Capital Structure

Q. What capital structure does KEDNY propose to use for ratemaking purposes in this proceeding?

A. For a utility that is part of a holding company structure, New York State Public Service Commission ("Commission") precedent supports setting rates using the utility's stand-alone capital structure where the utility can demonstrate that it is financially insulated from its parent and its capital structure is reasonable and in line with its own financial and business risks. KEDNY is proposing for ratemaking purposes to use a capital structure that contains 48 percent common equity for the Rate Year. However, as shown in the table above, should this proceeding result in a settlement where rates are established for multiple years, KEDNY proposes that the capital structure should reflect 49 percent common equity in Data Year 1 and 50 percent common equity in Data Years 2 and 3.

KEDNY believes an increase in common equity in the Data Years is appropriate given the interest of maintaining its credit ratings and achieving continued favorable financing for the benefit of customers. KEDNY's cash flow metrics have been declining over the last few years and it is important that the rates established in this proceeding ensure that those metrics do not continue to decline further. As discussed more fully below, the credit rating agency Moody's Investor Services, Inc. ("Moody's") has reviewed the Company's business and financial forecasts and does not envision cash flow generation to improve in the medium term, citing the impact of the passage of the Tax Cuts and Jobs Act of 2017 ("Tax Reform Act") and environmental remediation spending as two main contributing factors. Higher common equity will improve cash generation during a period when

Moody's sees limited prospects for improvement, reducing risks of credit rating downgrades.

Moody's has recognized that changes in utility capital structures to reflect more equity could act as an offset to negative cash flow impacts. It is important that the rates established in this proceeding ensure that KEDNY's cash flow metrics do not decline to a level that would not support its current A3 rating.

Q. Is KEDNY part of a holding company corporate structure?

A. Yes. KEDNY is indirectly owned by National Grid, which is owned by National Grid plc, the ultimate parent of KEDNY.

Q. Is KEDNY financially insulated from National Grid plc?

A. Yes. KEDNY is financially insulated from National Grid plc, and this insulation is recognized by the financial marketplace. Illustrating this point, in its February 2018 ratings report on KEDNY, Moody's explained that:

> …the strong ring-fencing provisions applicable to New York utilities reduces the potential for debt to be pushed down into KEDNY. In particular, we view the explicit leverage restriction for KEDNY (to maintain a debt to capitalization ratio of less than 56%) as providing the greatest credit support at the current rating level. This provision compares favorably against other New York utilities within the NG group. Additional ring-fencing provisions imposed by the NYSPC for KEDNY that we view as credit supportive include: (1) the "special preferred share" provision that reduces the probability of bankruptcy in a distressed situation and (2) the requirement for KEDNY to hold an investment grade rating.

Similarly, in its September 2018 report on KEDNY, Standard & Poor's ("S&P") stated that:

> We assess the group status of BUG ["Brooklyn Union Gas" aka "KEDNY"] as core to National Grid PLC. At the same time, we assess BUG as insulated, with existing insulation measures supporting a one-notch separation between the subsidiary and its parent. Key insulating measures include:
>
> - Prohibition of dividend distributions if the company does not maintain investment grade credit ratings with at least two of the three major ratings agencies.
> - Prohibition of dividend increases and payments if the company's debt-to-capital ratio exceeds a certain amount.
> - A golden share whose vote is required to file the utility into a voluntary bankruptcy.
> - Clear economic and strategic incentive from the parent to maintain BUG's financial strength.

KEDNY issues debt on a stand-alone basis and has a credit rating separate from National Grid plc and its affiliates. Specifically, while KEDNY has senior unsecured ratings of A3, A-, and A- from Moody's, S&P, and Fitch Ratings, Inc. ("Fitch"), respectively, Moody's, S&P, and Fitch rate National Grid plc as Baa1, BBB+, and BBB+, respectively. Moreover, KEDNY has distinct business risks from National Grid plc. KEDNY operates a gas distribution business in downstate New York, whereas National Grid plc is an international energy company with operations in the United Kingdom and the US, and operates in an array of markets and lines of business.

Q. On average, did KEDNY's capital structure include at least 48 percent common equity in the Historic Test Year?

A. Yes. KEDNY's average capital structure during the Historic Test Year consisted of at least 48 percent common equity.

Q. Please describe KEDNY's current credit ratings.

A. KEDNY has long-term issuer ratings of A3, A-, and BBB+ from Moody's, S&P, and Fitch, respectively. KEDNY's senior unsecured debt ratings are A3, A-, and A- from Moody's, S&P, and Fitch, respectively. In November 2014, Fitch downgraded KEDNY's long-term issuer rating. In October 2015, S&P downgraded KEDNY, lowering its issuer credit and senior unsecured debt ratings one notch from A to A-. In January 2018, Moody's issued a negative ratings outlook for KEDNY alongside several other utilities, including the Company's affiliates KEDLI and Niagara Mohawk Power Corporation ("Niagara Mohawk"), following the enactment of the Tax Reform Act. National Grid's New York utilities were subsequently downgraded to A3 from A2 following Moody's annual reviews in February and March 2019.

Q. Are KEDNY's current credit ratings consistent with the credit rating goals outlined in the "Generic Financing Proceeding" that is often referred to by the Commission in determining the cost of capital in utility rate proceedings?

A. What is referred to as the "Generic Financing Proceeding" – Case 91-M-0509 – resulted in a 1994 Recommended Decision that, although never formally adopted by the Commission, has often been referenced by the Commission when determining the cost of capital to be used in setting rates. The Recommended Decision in the Generic Financing Proceeding proposed that the "A" rating should continue to be the long-term target for utilities and that the Commission should continue to offer utilities ratemaking support for an "A" rating. The Commission recently reaffirmed the importance of strong credit ratings for utilities in

Case 17-M-0815, which considered the impact of the Tax Reform Act on utilities, finding that "strong investment grade credit ratings benefit both utilities and their customers." KEDNY's current ratings are in the "A" rating range, but any further downgrades would move KEDNY into the "Baa/BBB" rating range. KEDNY has limited headroom to maintain an A3 rating under cashflow metric guidance released by Moody's in April 2019, highlighting the importance of credit enhancing provisions like a higher equity percentage in the Company's ratemaking capital structure.

Q. Do customers benefit from "A" credit ratings?

A. Yes. Higher-rated utilities can issue debt at lower costs and on better terms, which benefits customers by lowering the overall rate of return charged to them. The ability of KEDNY to access the debt markets and/or obtain credit on favorable terms may also be affected if KEDNY's rating falls below the "A" range as certain debt investors do not invest in the debt of non-A entities.

Q. What is the impact of the recent downgrade by Moody's?

A. While difficult to quantify, the downgrade from Moody's in February 2019 from A2 to A3 will likely result in modestly higher borrowing costs for customers.

Q. Will KEDNY's proposed capital structure and overall rate of return support its current credit ratings?

A. While I cannot speak for the rating agencies, an analysis of the metrics that the agencies themselves calculate, indicates that if the Commission authorizes the rate relief requested

by KEDNY in this proceeding, then KEDNY will still have tight metrics under the current rating guidelines. In the latest credit report published in April 2019, Moody's noted that a key contributing factor to a further downgrade would be if "CFO pre-WC/debt that no longer appears likely to trend toward the mid-teens in percentage terms in the near term, and to the high teens over the longer term." CFO pre-WC/debt is the ratio of cash flow from operations before consideration of working capital to the Company's debt. There is little headroom in credit metrics for further reduction in cash flow as compared to what is being proposed in the Company's rate filing herein due to KEDNY's planned capital expenditures and the negative cashflow impacts from the Tax Reform Act. The analysis of the effect of the proposed capital structure and rate of return on KEDNY's credit ratings is shown in Schedule 3 of Exhibit __ (JC-1).

Q. Could the Commission's decision in this proceeding adversely impact KEDNY's ability to maintain its current credit ratings?

A. Yes. Failure to authorize a reasonable and achievable return-on-equity ("ROE") or adoption of a revenue requirement that fails to provide support for KEDNY's credit metrics could put it at risk of further credit downgrades to the detriment of customers. Rating agencies indicate that they are particularly focused on cash flow metrics, with the Moody's report covering the downgrade of KEDNY in February 2019, stating:

> Today's rating action reflects the persistent weakness of KEDNY's key credit metrics and Moody's assessment that there is limited likelihood of a recovery over the medium term. KEDNY's ratio of cash from operations before working capital (CFO pre-WC) to debt was 9.8% in the year to March 2018 and is expected to remain weak for the assigned rating.

Moody's also suggests that KEDNY's credit metrics could "come under significant pressure from increasing expenditure on environmental remediation that will not be immediately recovered through revenues," and that regulatory lag for environmental recoveries could result in weaker cash flows and higher leverage for several years. In their latest credit report in March 2019, following the downgrade of KEDNY, Moody's also commented that:

> KEDNY's current rate plan contains an SIR Recovery Surcharge mechanism that would allow it to begin recovering excess SIR expense once cumulative spend exceeds $25 million, a threshold that has not yet been reached. However, recovery in any year is limited to 2.0 percent of revenue, around $30 million-$35 million. KEDNY's next rate plan may provide for immediate recovery of future costs as they are incurred, which would be credit positive. If, however, it instead provides a mechanism similar to the current SIR Recovery Surcharge, there may be a significant lag between pending and eventual recovery. The impact of this lag will depend on the ultimate cost of the projects and the profile of spending, but we estimate that it could reduce KEDNY's CFO pre-WC to debt by 2-3 percentage points in the early 2020s.

As discussed by KEDNY witness Charles F. Willard, the Company is proposing to increase its recovery of SIR costs in the Rate Year and Data Years. Moody's recent pronouncement highlights the critical importance of this proposal to KEDNY's ability to maintain its current credit ratings.

Q. Are there any other factors that could impact KEDNY's credit ratings?

A. Yes. The Tax Reform Act has adversely affected KEDNY's cash flow and played a significant role in the downgrade of KEDNY from A2 to A3 by Moody's in February 2019. The reduction of the maximum corporate tax rate from 35 percent to 21 percent and the new rules regarding bonus depreciation adversely affect KEDNY's cash flow. The reduction in the corporate tax rate will also require KEDNY to return to customers the

difference between taxes that have been collected at the 35 percent tax rate and the new 21 percent tax rate. With regard to the impacts of the Tax Reform Act, Moody's commented in February 2019:

> This assessment takes into account the impact the New York Public Service Commission's order of August 2018, reflecting tax reforms in the United States, which will reduce KEDNY's revenue starting in January 2019 from the level agreed in its current rate plan. Revenue allowances will also be lower from January 2021, compared to previous expectations, due to lower recovery of deferred taxes.

The cash flow issues created by the Tax Reform Act and their resulting potential impact on KEDNY's credit metrics and ratings make it particularly important that the Commission mitigate the downward pressure by authorizing an equity ratio that is sufficient to support KEDNY's current investment grade rating.

Q. Has the Commission acknowledged the challenges to utility creditworthiness created by the Tax Reform Act?

A. Yes. In the Commission's "Order Determining Rate Treatment of Tax Changes" in Case 17-M-0815, the Commission acknowledged the challenges that the Tax Reform Act presented for the credit metrics and cash flow of New York utilities, specifically noting that cash flow concerns arising from the Tax Reform Act resulted in negative ratings outlooks for New York utilities. Additionally, in the Commission's "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan" in Central Hudson's recent rate case (Cases 17-E-0459 and 17-G-0460), the Commission approved the Joint Proposal's use of equity ratios of 49 percent and 50 percent in Rate Years 2 and 3, stating that

> [g]iven the degree of uncertainty regarding the ultimate impact of the Tax Act on the Company's creditworthiness, we find the JP's use of higher common equity ratios in Rate Year 2 and Rate Year 3 to be a responsible and reasonable measure to forestall, or at least diminish, the prospect of higher future borrowing costs attributable to a diminution in Central Hudson's creditworthiness over the next several years.

Q. What have other jurisdictions done to offset the negative cashflow impact of Tax Reform Act?

A. Moody's noted in a June 2018 report that Georgia and Alabama increased authorized equity ratios following passage of the Tax Reform Act, which showed the supportive nature of these jurisdictions. This was noted as a credit positive.

Q. How does a higher equity percentage improve credit metrics?

A. As shown on Schedule 3 of Exhibit __ (JC-1) Moody's and S&P have set cash flow metric expectations for KEDNY. The current rate filing request produces cashflow metrics consistent with these guidelines, albeit without a great deal of headroom. One of the contributing factors of meeting these cashflow guidelines will be the higher equity ratio in years following the Rate Year in the event a multi-year rate plan is authorized.

III. <u>Cost of Debt and Overall Rate of Return</u>

Q. Please outline the planned financing activity reflected in the projected costs of capital for the Rate Year and Data Years.

A. On August 29, 2018, KEDNY filed a financing petition (Case 18-G-0559) requesting multi-year authority to issue new long-term debt securities. On February 8, 2019, the

Commission issued an Order authorizing KEDNY to issue and sell up to $1.4 billion of new long-term debt for the following purposes specified in the Order:

> (a) the acquisition of property, the construction, completion, extension or improvement of [KEDNY's] facilities or the improvement or maintenance of the Company's service within New York State (to the extent such expenditures exceed funds originating from credits to the accumulated provision for depreciation, net salvage and accumulated deferred income taxes); (b) for the discharge or lawful refunding of the Company's obligations; or (c) to reimburse [KEDNY's] treasury for equivalent moneys expended for those purposes prior to March 31, 2019, provided that the Company has kept its accounts and vouchers of such expenditures in such a manner as to enable us to ascertain the amount of moneys so expended and the purposes for which such expenditures were made.

With this financing authority, KEDNY issued $1 billion of long term debt in February 2019 ($550 million in a ten year tenor and $450 million in a thirty year tenor). KEDNY currently plans to issue an additional $400 million of long-term debt in March 2021 and an additional $500 million of long-term debt in March 2023. These issuances will reduce/repay short-term debt, and fund KEDNY's capital expenditure program. The issuance of additional debt will not prevent KEDNY from maintaining the capital structure proposed in this testimony. The projected costs of capital assume that KEDNY will issue new long-term debt as ten-year and thirty-year senior unsecured debt at forecast interest rates of 3.83 percent in March 2021 and 4.46 percent in March 2023, respectively. KEDNY assumes that the costs to issue this debt will be 0.510 percent and 0.935 percent, respectively, of the principal amounts issued and that these costs will be amortized over the lives of the debt.

Q. How were the cost rates of long-term debt for KEDNY shown on Exhibit __ (JC-1), Schedule 4 derived?

A. The long-term debt component of KEDNY's capital structure consists of fixed-rate taxable bonds, all of which support gas investments. Included in the cost of these bonds are the direct coupon expense, the amortization of debt discounts or premiums, and the amortization of issuance costs where applicable.

Q. How were the balances and the cost rates for customer deposits shown on Exhibit __ (JC-1), Schedule 4 determined?

A. KEDNY's forecast balances of customer deposits were assumed to remain equal to the actual monthly balance as of December 31, 2018, the end of the Historic Test Year. In Case 18-M-0611, the Commission set the customer deposits rate for investor-owned utilities to be 2.45 percent, effective January 1, 2019.

Q. What cost rate does KEDNY propose for the common equity component of the capital structure?

A. KEDNY proposes a cost of common equity of 9.65 percent for the Rate Year that is within the range supported by Company Witness Bulkley in her testimony. This proposed cost of equity is at the low end of Ms. Bulkley's recommended range of reasonable returns for her proxy group. If a multi-year settlement is reached, a 10.15 percent cost of common equity should be used to reflect the appropriate stay-out premium recommended by Ms. Bulkley for all years covered by the settlement.

Q. Does KEDNY propose to update its projections of both new debt issuances and cost rates later in this proceeding?

A. Yes. While KEDNY has provided its best forecast of the cost of planned debt issuances and their specific timing, forecasts and assumptions can change over time. KEDNY will endeavor to time its debt issuances to secure the most favorable terms, given capital market conditions, in the interest of minimizing costs for customers. As such, KEDNY will provide updated projections of the cost of new long-term debt during the rate case process and provide the actual terms and costs of any debt issuances completed during the course of the rate case.

IV. Financial Protections

Q. Is KEDNY presently subject to any financial protection provisions adopted by the Commission?

A. Yes. The Commission has previously adopted Financial Protection provisions that generally address the payment of dividends and the components of KEDNY's capital structure.

Q. Does KEDNY believe that the Financial Protections should remain in effect?

A. Yes. The current Financial Protections should continue without modification.

V. Service Company Asset Recovery Charge

Q. What is the service company asset recovery charge?

A. Service Company owns or leases a number of shared assets that are used either by Service Company employees to provide services to affiliates or are used by the affiliates on a shared basis. These are primarily shared office facilities and information technology equipment and software. When Service Company finances and owns the shared assets, it charges the affiliates an asset recovery charge based on a pre-tax return on the asset (net of deferred taxes) and booked depreciation expense. This charge is recovered in KEDNY's rent expense, as set forth in the Revenue Requirements Panel's testimony.

Q. What cost of equity does KEDNY propose to use for the service company asset recovery charge?

A. KEDNY proposes to use the same cost of equity for the service company asset recovery charge as for its cost of equity - *i.e.,* 9.65 percent or 10.15 percent for a four-year rate plan.

Q. What overall weighted average cost of capital does KEDNY propose to use for the service company asset recovery charge and how was it derived?

A. KEDNY proposes to use an assumed 50/50 long-term debt/equity capital structure and the actual cost of long-term debt for the service company asset recovery charge. The table below illustrates the overall weighted average cost of capital for the service company asset recovery charge, assuming KEDNY's requested 9.65 percent cost of equity.

Service Company	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	50.00%	4.2900%	2.15%
Common Equity	50.00%	9.6500%	4.83%
Total	100.00%		6.98%

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Corrections and Updates Testimony

of

Jonathan Cohen

Dated: July 3, 2019

I. Introduction

Q. Please state your name and business address.

A. My name is Jonathan Cohen. My business address is One MetroTech Center, Brooklyn, New York, 11201.

Q. Are you the same Jonathan Cohen who previously submitted prepared direct testimony in this proceeding?

A. Yes. I provided direct testimony as part of KEDNY's April 30, 2019 filing in this proceeding. The defined terms used in my direct testimony have the same meaning in this corrections and updates testimony.

Q. What is the purpose of your corrections and updates testimony?

A. The purpose of my corrections and updates testimony is to correct and/or update KEDNY's capital structure and cost of capital and the Service Company's cost of capital. This updated information is reflected in KEDNY's updated revenue requirement as set forth in the corrections and update testimony of the Revenue Requirements Panel.

Q. Do you sponsor any exhibits as part of your corrections and updates testimony?

A. Yes. Exhibit __ (JC-1CU) sets forth the revised projected weighted average cost of capital resulting from the corrections described below. This exhibit was prepared under my direction and supervision.

Q. Please describe the corrections and updates that are reflected in Exhibit __ (JC-1CU).

A. First, Exhibit __ (JC-1CU), Schedule 2, page 1 sets forth a corrected opening retained earnings balance of $1,253,901,000 as compared to the balance of $1,110,352,000 set forth on Exhibit __ (JC-1), Schedule 2, page 1 of my direct testimony. Second, because of the change in the opening retained earnings balance, an additional $200 million dividend was reflected on the same page of the exhibit in March of 2021 to more properly align the capital structure. Third, Exhibit __ (JC-1CU), Schedule 1, page 5 has been corrected to reflect the correct forecast interest rate -- 4.460 percent -- of the senior unsecured debt to be issued in March 2023. Finally, Exhibit __ (JC-1CU), Schedule 3 corrects a typographical error in the National Grid commentary included in the exhibit.

Q. Do the corrections and updates described above affect the weighted average cost of capital that you are recommending be adopted for KEDNY in this proceeding?

A. Yes. They have a very modest impact. If rates were to be established in a multi-year settlement in this proceeding, the Company's proposed weighted average cost of capital, including a stay-out premium on its cost of equity would be as follows as shown on Schedule 4 of Exhibit __ (JC-1CU).

	Rate Year (FY21)			Data Year 1 (FY22)			Data Year 2 (FY23)			Data Year 3 (FY24)		
	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.58%	4.11%	2.12%	50.61%	4.11%	2.08%	49.62%	4.11%	2.04%	49.67%	4.19%	2.08%
Customer Deposits	0.42%	2.45%	0.01%	0.39%	2.45%	0.01%	0.38%	2.45%	0.01%	0.33%	2.45%	0.01%
Common Equity	48.00%	10.15%	4.87%	49.00%	10.15%	4.97%	50.00%	10.15%	5.08%	50.00%	10.15%	5.08%
Total	100.00%		7.00%	100.00%		7.06%	100.00%		7.13%	100.00%		7.17%

Q. Are you also updating the proposed Service Company capital structure?

A. Yes. The updated Service Company capital structure reflects a reduced cost of long-term debt of 3.7049 percent as compared to the 4.2900 percent included in my direct testimony. The revised Service Company weighted average cost of capital is as follows:

Service Company	Capitalization Rate	Cost Rate	Weighted Cost
Long-Term Debt	50.00%	3.7049%	1.85%
Common Equity	50.00%	9.6500%	4.83%
Total			6.68%

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Jonathan Cohen

Dated: April 2019

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I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Jonathan Cohen. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what position?

A. I am employed by National Grid USA Service Company, Inc. ("Service Company"), a subsidiary of National Grid USA ("National Grid"), as Director of US Treasury, Planning, and Strategy (New York).

Q. Please describe your educational background and professional experience.

A. I am a qualified accountant with the Institution for Chartered Accountants in England and Wales and a qualified Treasurer with the Association of Corporate Treasurers. From 2006 to 2012, I worked at BDO LLP, an accounting, tax, audit, and consulting services firm. I worked in audit, specializing in financial services companies, including banks, brokers, and fund managers. I joined National Grid in 2012 as a member of the Corporate Banking Group. In that role, I was responsible for managing the bank relationships, including associated borrowings and revolving credit facilities. I was also responsible for analyzing and monitoring National Grid's credit ratings and managing its interactions with debt investors. I joined National Grid's Debt Capital Markets group in 2015, participating in public bond issuances and managing the associated currency and interest rate risks. I also participated in the sale of significant assets in one of National Grid plc's largest operating companies in the United Kingdom. I began my current position in 2017, where I am

focused on business partnering to determine the efficiency of operations and obtain identified value opportunities. In my current position, I am familiar with the financing activities of KeySpan Gas East Corporation d/b/a National Grid ("KEDLI" or the "Company").

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to present and support KEDLI's proposed capital structure and overall cost of capital in this proceeding. My testimony provides information for the test year ended December 31, 2018 ("Historic Test Year"), the forecast year ending March 31, 2021 ("Rate Year"), and the years ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (collectively "Data Years"). In addition, I will discuss KEDLI's proposal to continue the currently effective financial protection provisions adopted by the New York State Public Service Commission ("Financial Protections"). Finally, I discuss and support the cost of capital component included in the service company asset recovery charge that is charged to affiliates, such as KEDLI, to recover the costs of shared assets financed and owned by Service Company.

Q. Do you sponsor any exhibits as part of your testimony in this proceeding?

A. Yes. Schedule 1 of Exhibit __ (JC-1) sets forth KEDLI's cost of long-term debt. Schedule 2 contains the projected capitalization and weighted average cost of capital that is proposed to be adopted for KEDLI in this proceeding. Schedule 3 sets forth an analysis of KEDLI's credit metric projections. Schedule 4 sets forth KEDLI's weighted cost of capital for both the Rate Year and the term of a multi-year rate plan. Schedule 5 sets forth a forecast sources

and uses of funds statement. These exhibits were prepared or compiled under my supervision and direction.

Q. What is the weighted average cost of capital that you are recommending be adopted for KEDLI in this proceeding?

A. KEDLI's proposed Rate Year weighted average cost of capital, as shown on Schedule 4 of Exhibit __ (JC-1), is 6.74 percent. This overall rate of return is based on the following capitalization ratios and cost rates:

Table 1: One-Year Weighted Cost of Capital

	One-Year Rate Plan		
	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.37%	4.06%	2.09%
Customer Deposits	0.63%	2.45%	0.02%
Common Equity	48.00%	9.65%	4.63%
Total	100.00%		6.74%

If rates were to be established in a multi-year settlement in this proceeding, the Company proposes the following rates of return, including a stay-out premium on its cost of equity, as shown on Schedule 4 of Exhibit __ (JC-1).

Table 2: Multi-Year Weighted Cost of Capital

	Four-Year Rate Plan (ROE includes stay-out premium, forecasted cost of debt)											
	Rate Year (FY21)			Data Year 1 (FY22)			Data Year 2 (FY23)			Data Year 3 (FY24)		
	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.37%	4.06%	2.09%	50.45%	4.15%	2.09%	49.54%	4.15%	2.06%	49.60%	4.14%	2.05%
Customer Deposits	0.63%	2.45%	0.02%	0.55%	2.45%	0.01%	0.46%	2.45%	0.01%	0.40%	2.45%	0.01%
Common Equity	48.00%	10.15%	4.87%	49.00%	10.15%	4.97%	50.00%	10.15%	5.08%	50.00%	10.15%	5.08%
Total	100.00%		6.98%	100.00%		7.07%	100.00%		7.15%	100.00%		7.14%

In calculating the capitalization ratios shown above, all of the goodwill recorded on KEDLI's books was excluded from its total capitalization and common equity balances.

II. Ratemaking Capital Structure

Q. What capital structure does KEDLI propose to use for ratemaking purposes in this proceeding?

A. For a utility that is part of a holding company structure, New York Public Service Commission ("Commission") precedent supports setting rates using the utility's stand-alone capital structure where the utility can demonstrate that it is financially insulated from its parent and its capital structure is reasonable and in line with its own financial and business risks. KEDLI is proposing for ratemaking purposes to use a capital structure that contains 48 percent common equity for the Rate Year. However, as shown in the table above, should this proceeding result in a settlement where rates are established for multiple years, KEDLI proposes that the capital structure should reflect 49 percent common equity in Data Year 1 and 50 percent common equity in Data Years 2 and 3.

KEDLI believes an increase in common equity in the Data Years is appropriate given the interest of maintaining its credit ratings and achieving continued favorable financing for the benefit of customers. KEDLI's cash flow metrics have declined from March 2016 levels and are expected to decline further according to the credit rating agency Moody's Investor Services, Inc. ("Moody's"). In addition, Moody's does not envision cash flow generation to improve in the medium term, citing the impact of the passage of the Tax Cuts and Jobs Act of 2017 ("Tax Reform Act") as the main contributing factor. Higher common equity will improve cash generation during a period when Moody's sees limited prospects for improvement, reducing risks of further credit rating downgrades.

Moody's has recognized that changes in utility capital structures to reflect more equity could act as an offset to the negative cash flow impact arising from the Tax Reform Act. It is important that the rates established in this proceeding ensure that KEDLI's cash flow metrics do not continue to decline to a level that would not support its current A3 rating.

Q. Is KEDLI part of a holding company corporate structure?

A. Yes. KEDLI is indirectly owned by National Grid, which is owned by National Grid plc, the ultimate parent of KEDLI.

Q. Is KEDLI financially insulated from National Grid plc?

A. Yes. KEDLI is financially insulated from National Grid plc, and this insulation is recognized by the financial marketplace. Illustrating this point, in its February 2018 ratings report on KEDLI, Moody's explained that:

> …the strong ring-fencing provisions applicable to New York utilities reduces the potential for debt to be pushed down into KEDLI. In particular, we view the explicit leverage restriction for KEDLI (to maintain a debt to capitalization ratio of less than 56%) as providing the greatest credit support at the current rating level. This provision compares favorably against other New York utilities within the NG group. Additional ring-fencing provisions imposed by the NYSPC for KEDLI that we view as credit supportive include: (1) the "special preferred share" provision that reduces the probability of bankruptcy in a distressed situation and (2) the requirement for KEDLI to hold an investment grade rating.

Similarly, in its September 2018 report on KEDLI, Standard & Poor's ("S&P") stated that:

> We assess the group status of KGE ["KeySpan Gas East" aka "KEDLI"] as core to National Grid PLC. At the same time, we assess KGE as insulated, with existing insulation measures supporting a one-notch separation between the subsidiary and its parent. Key insulating measures include:

- Prohibition of dividend distributions if the company does not maintain investment grade credit ratings with at least two of the three major ratings agencies.
- Prohibition of dividend increases and payments if the company's debt-to-capital ratio exceeds a certain amount.
- A golden share whose vote is required to file the utility into a voluntary bankruptcy.
- Clear economic and strategic incentive from the parent to maintain KGE's financial strength.

KEDLI issues debt on a stand-alone basis and has a credit rating separate from National Grid plc and its affiliates. Specifically, while KEDLI has senior unsecured ratings of A3, A-, and A- from Moody's, S&P, and Fitch Ratings, Inc. ("Fitch"), respectively, Moody's, S&P, and Fitch rate National Grid plc as Baa1, BBB+, and BBB+, respectively. Moreover, KEDLI has distinct business risks from National Grid plc. KEDLI operates a gas distribution business in downstate New York, whereas National Grid plc is an international energy company with operations in the United Kingdom and the US, and operates in an array of markets and lines of business.

Q. On average, did KEDLI's capital structure include at least 48 percent common equity in the Historic Test Year?

A. Yes. KEDLI's average capital structure during the Historic Test Year consisted of at least 48 percent common equity.

Q. Please describe KEDLI's current credit ratings.

A. KEDLI has long-term issuer ratings of A3, A-, and BBB+ from Moody's, S&P, and Fitch, respectively. KEDLI's senior unsecured debt ratings are A3, A-, and A- from Moody's, S&P, and Fitch, respectively. In November 2014, Fitch downgraded KEDLI's long-term

issuer rating. In October 2015, S&P downgraded KEDLI, lowering its issuer credit and senior unsecured debt ratings one notch from A to A-. In January 2018, Moody's issued a negative ratings outlook for KEDLI alongside several other utilities, including the Company's affiliates KEDNY and Niagara Mohawk Power Corporation ("Niagara Mohawk"), following enactment of the Tax Reform Act. National Grid's New York utilities were subsequently downgraded to A3 from A2 following Moody's annual review in February and March 2019.

Q. Are KEDLI's current credit ratings consistent with the credit rating goals outlined in the "Generic Financing Proceeding" that is often referred to by the Commission in determining the cost of capital in utility rate proceedings?

A. What is referred to as the "Generic Financing Proceeding" – Case 91-M-0509 – resulted in a 1994 Recommended Decision that, although never formally adopted by the Commission, has often been referenced by the Commission when determining the cost of capital to be used in setting rates. The Recommended Decision in the Generic Financing Proceeding proposed that the "A" rating should continue to be the long-term target for utilities and that the Commission should continue to offer utilities ratemaking support for an "A" rating. The Commission recently reaffirmed the importance of strong credit ratings for utilities in Case 17-M-0815, which considered the impact of the Tax Reform Act on utilities, finding that "strong investment grade credit ratings benefit both utilities and their customers."

KEDLI's current ratings are in the "A" rating range, but any further downgrades would move KEDLI into the "Baa/BBB" rating range. KEDLI has limited headroom to maintain

an A3 rating under cashflow metric guidance released by Moody's in April 2019, highlighting the importance of credit enhancing provisions like a higher equity percentage in the Company's ratemaking capital structure.

Q. Do customers benefit from "A" credit ratings?

A. Yes. Higher-rated utilities can issue debt at lower costs and better terms, which benefits customers by lowering the overall rate of return charged to them. The ability of KEDLI to access the debt market and/or obtain credit on favorable terms may also be affected if KEDLI's rating falls below the "A" range as certain debt investors do not invest in the debt of non-"A" rated entities.

Q. What is the impact of the recent downgrade by Moody's?

A. While difficult to quantify, the downgrade from Moody's in March 2019 from A2 to A3 will likely result in modestly higher borrowing costs for customers.

Q. Will KEDLI's proposed capital structure and overall rate of return support its current credit ratings?

A. While I cannot speak for the rating agencies, an analysis of the metrics that the agencies themselves calculate indicates that if the Commission authorizes the rate relief requested by KEDLI in this proceeding, then KEDLI will still have tight metrics under the current rating guidelines. In the latest credit report published in April 2019, Moody's noted that a key contributing factor to a further downgrade would be "CFO pre-WC/debt consistently below the high teens". CFO pre-WC/debt is the ratio of cash flow from operations before

consideration of working capital to the Company's debt. There is little headroom in credit metrics for further reductions in cash flow as compared to what is being proposed in the Company's rate filing herein due to KEDLI's planned capital expenditures and the negative cashflow impacts from the Tax Reform Act. The analysis of the effect of the proposed capital structure and rate of return on KEDLI's credit ratings is shown in Schedule 3 of Exhibit __ (JC-1).

Q. Could the Commission's decision in this proceeding adversely impact KEDLI's ability to maintain its current credit ratings?

A. Yes. Failure to authorize a reasonable and achievable return-on-equity ("ROE") or adoption of a revenue requirement that fails to provide support for KEDLI's credit metrics could put it at risk of credit downgrades to the detriment of customers. Rating agencies indicate that they are particularly focused on cash flow metrics, with the Moody's February 2018 report saying:

> The outlook for KEDLI's rating is negative, reflecting our expectation that financial metrics will be at the bottom of our guidance range following US tax reform, absent significant mitigating actions by the company or additional allowances offered by the NYPSC.

Moody's has also released an updated report in March 2019, stating:

> Under its current rate plan, which began in January 2017, KEDLI's achieved returns have steadily improved to 10.1 percent in the financial year ended March 2018, which is 110 basis points higher than the allowed regulatory returns. Despite this improvement, and as a result of US tax reforms, the decline in KEDLI's revenue starting January 2019 from the level agreed upon in its current rate plan is likely to lead to key credit metrics that will become weakly positioned for its current A2 rating.

Moody's subsequently downgraded KEDLI (in combination with Niagara Mohawk "NIMO") at the end of March 2019, stating:

> The downgrade of the ratings to A3 from A2 reflects Moody's assessment that underlying CFO pre-WC/debt at both KEDLI and NiMo is likely to fall short of Moody's revised guidance for an A2 rating of at least the low-20s, in percentage terms. This assessment takes into account the impact the New York Public Service Commission's order of August 2018, reflecting tax reforms in the United States, which will reduce both companies' revenue starting in January 2019 from the level agreed in its current rate plan. Revenue allowances will also be lower from January 2021, compared to previous expectations, due to the pass-back of excess deferred taxes.

Q. Are there any other factors that could impact KEDLI's credit ratings?

A. Yes. The Tax Reform Act has adversely affected KEDLI's cash flow and played a significant role in the downgrade of KEDLI from A2 to A3 by Moody's in March 2019. The reduction of the maximum corporate tax rate from 35 percent to 21 percent and the new rules regarding bonus depreciation adversely affect KEDLI's cash flow. The reduction in the corporate tax rate will also require KEDLI to return to customers the difference between taxes that have been collected at the 35 percent tax rate and the new 21 percent tax rate. With regard to the impacts of the Tax Reform Act, Moody's commented in February 2018:

> We expect KEDLI's allowed revenues to be reduced following US tax reforms announced in December 2017. Because we did not expect KEDLI to pay significant cash taxes, the reduced revenues will translate directly into lower cash flow from operations pre-working capital (CFO pre-WC). Our negative outlook reflects our expectation that CFO pre-WC to gross debt will remain at the lower end of our mid- to high-teens guidance range as a result of the change.

The cash flow issues created by the Tax Reform Act and their resulting potential impact on KEDLI's credit metrics and ratings make it particularly important that the Commission mitigate the downward pressure by authorizing an equity ratio that is sufficient to support KEDLI's current investment grade rating.

Q. Has the Commission acknowledged the challenges to utility creditworthiness created by the Tax Reform Act?

A. Yes. In the Commission's "Order Determining Rate Treatment of Tax Changes" in Case 17-M-0815, the Commission acknowledged the challenges that the Tax Reform Act presented for the credit metrics and cash flow of New York utilities, specifically noting that cash flow concerns arising from the Tax Reform Act resulted in negative ratings outlooks for KEDLI and other New York utilities.

Additionally, in the Commission's "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan" in Central Hudson's recent rate case (Cases 17-E-0459 and 17-G-0460), the Commission approved the Joint Proposal's use of equity ratios of 49 percent and 50 percent in Rate Years 2 and 3, stating that

> [g]iven the degree of uncertainty regarding the ultimate impact of the Tax Act on the Company's creditworthiness, we find the JP's use of higher common equity ratios in Rate Year 2 and Rate Year 3 to be a responsible and reasonable measure to forestall, or at least diminish, the prospect of higher future borrowing costs attributable to a diminution in Central Hudson's creditworthiness over the next several years.

Q. What have other jurisdictions done to offset the negative cashflow impact of the Tax Reform Act?

A. Moody's noted in a June 2018 report that Georgia and Alabama increased authorized equity ratios following passage of the Tax Reform Act, which showed the supportive nature of these jurisdictions. This was noted as a credit positive.

Q. How does a higher equity percentage improve credit metrics?

A. As shown on Schedule 3 of Exhibit __ (JC-1) Moody's and S&P have set cash flow metric expectations for KEDLI. The current rate filing request produces cashflow metrics consistent with these guidelines, albeit without a great deal of headroom. One of the contributing factors of meeting these cashflow guidelines will be the higher equity ratio in years following the Rate Year in the event a multi-year rate plan is authorized.

III. <u>Cost of Debt and Overall Rate of Return</u>

Q. Please outline the planned financing activity reflected in the projected costs of capital for the Rate Year and Data Years.

A. On August 29, 2018, KEDLI filed a financing petition (Case 18-G-0558) requesting multi-year authority to issue new long-term debt securities. On February 8, 2019, the Commission issued an Order authorizing KEDLI to issue and sell up to $400 million of new long-term debt for the following purposes specified in the Order:

> (a) the acquisition of property, the construction, completion, extension or improvement of [KEDLI's] facilities or the improvement or maintenance of the Company's service within New York State (to the extent such expenditures exceed funds originating from credits to the accumulated provision for depreciation, net salvage and accumulated deferred income taxes); (b) for the

> discharge or lawful refunding of the Company's obligations; or (c) to reimburse [KEDLI's] treasury for equivalent moneys expended for those purposes prior to March 31, 2019, provided that the Company has kept its accounts and vouchers of such expenditures in such a manner as to enable us to ascertain the amount of moneys so expended and the purposes for which such expenditures were made.

With this financing authority, KEDLI intends to issue $400 million of long term debt in September 2021 and $500 million in September 2023. These issuances will reduce/repay short-term debt and fund KEDLI's capital expenditure program. The issuance of additional debt will not prevent KEDLI from maintaining the capital structure proposed in this testimony. The projected costs of capital assume that KEDLI will issue new long-term debt as 30 year and then ten year senior unsecured debt at forecast interest rates of 4.40 percent and 4.06 percent, respectively. KEDLI assumes that the costs to issue this debt will be 0.935 percent and 0.51 percent, respectively, of the principal amounts issued and that these costs will be amortized over the lives of the debt.

Q. How were the cost rates of long-term debt for KEDLI shown on Exhibit __ (JC-1), Schedule 4 derived?

A. The long-term debt component of KEDLI's capital structure consists of fixed-rate taxable bonds. Included in the cost of these bonds are the direct coupon expense, the amortization of debt discounts or premiums, and the amortization of issuance costs where applicable.

Q. How were the balances and the cost rates for customer deposits shown on Exhibit __ (JC-1), Schedule 4 determined?

A. KEDLI's forecast balances of customer deposits were assumed to remain equal to the actual monthly balance as of December 31, 2018, the end of the Historic Test Year. In Case 18-M-0611, the Commission set the customer deposits rate for investor-owned utilities to be 2.45 percent, effective January 1, 2019.

Q. What cost rate does KEDLI propose for the common equity component of the capital structure?

A. KEDLI proposes a cost of common equity of 9.65 percent for the Rate Year that is within the range supported by Company Witness Bulkley in her testimony. This proposed cost of equity is at the low end of Ms. Bulkley's recommended range of reasonable returns for her proxy group. If a multi-year settlement is reached, a 10.15 percent cost of common equity should be used to reflect the appropriate stay-out premium recommended by Ms. Bulkley for all years covered by the settlement.

Q. Does the Company propose to update its projections of both new debt issuances and cost rates later in this proceeding?

A. Yes. While KEDLI has provided its best forecast of the cost of planned debt issuances and their specific timing, forecasts and assumptions can change over time. KEDLI will endeavor to time its debt issuances to secure the most favorable terms, given capital market conditions, in the interest of minimizing costs for customers. As such, KEDLI will provide updated projections of the cost of new long-term debt during the rate case process and

provide the actual terms and costs of any debt issuances completed during the course of the rate case.

IV. Financial Protections

Q. Is KEDLI presently subject to any financial protection provisions adopted by the Commission?

A. Yes. The Commission has previously adopted Financial Protection provisions that generally address the payment of dividends and the components of KEDLI's capital structure.

Q. Does KEDLI believe that the Financial Protections should remain in effect?

A. Yes. The current Financial Protections should continue without modification.

V. Service Company Asset Recovery Charge

Q. What is the service company asset recovery charge?

A. Service Company owns or leases a number of shared assets that are used either by Service Company employees to provide services to affiliates or are used by the affiliates on a shared basis. These are primarily shared office facilities and information technology equipment and software. When Service Company finances and owns the shared assets, it charges the affiliates an asset recovery charge based on a pre-tax return on the asset (net of deferred taxes) and booked depreciation expense. This charge is recovered in the Company's rent expense, as set forth in the Revenue Requirements Panel's testimony.

Q. What cost of equity does KEDLI propose to use for the service company asset recovery charge?

A. KEDLI proposes to use the same cost of equity for the service company asset recovery charge as for its cost of equity - i.e., 9.65 percent or 10.15 percent for a four-year rate plan.

Q. What overall weighted average cost of capital does KEDLI propose to use for the service company asset recovery charge and how was it derived?

A. KEDLI proposes to use an assumed 50/50 long-term debt/equity capital structure and the actual cost of long-term debt for the service company asset recovery charge. The table below illustrates the overall weighted average cost of capital for the service company asset recovery charge, assuming KEDLI's requested 9.65 percent cost of equity.

Service Company	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	50.00%	4.2900%	2.15%
Common Equity	50.00%	9.6500%	4.83%
Total	100.00%		6.98%

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Jonathan Cohen

Dated: July 3, 2019

I. Introduction

Q. Please state your name and business address.

A. My name is Jonathan Cohen. My business address is One MetroTech Center, Brooklyn, New York, 11201.

Q. Are you the same Jonathan Cohen who previously submitted prepared direct testimony in this proceeding?

A. Yes. I provided direct testimony as part of KEDLI's April 30, 2019 filing in this proceeding. The defined terms used in my direct testimony have the same meaning in this corrections and updates testimony.

Q. What is the purpose of your corrections and updates testimony?

A. The purpose of my corrections and updates testimony is to update the Service Company's cost of capital. This updated information is reflected in KEDLI's updated revenue requirement as set forth in the corrections and update testimony of the Revenue Requirements Panel.

Q. Please describe the update to the Service Company's capital structure.

A. The updated Service Company capital structure reflects a reduced cost of long-term debt of 3.7049 percent as compared to the 4.2900 percent included in my

direct testimony. The revised Service Company weighted average cost of capital is as follows:

Service Company	Capitalization Rate	Cost Rate	Weighted Cost
Long-Term Debt	50.00%	3.7049%	1.85%
Common Equity	50.00%	9.6500%	4.83%
Total			6.68%

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Jonathan Cohen

Case 19-G-0309
Case 19-G-0310

September 18, 2019

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Rebuttal Testimony of Jonathan Cohen

I. Introduction

Q. Please state your name, address and business affiliation.

A. My name is Jonathan Cohen. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. Are you the same Jonathan Cohen who previously submitted testimony in these proceedings?

A. Yes. The defined terms in my direct testimony have the same definitions here unless otherwise indicated.

II. Purpose of Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my rebuttal testimony is to address certain cost of capital and capital structure recommendations made by the Department of Public Service Staff ("Staff") Finance Panel and the Staff Information Technology Panel. Specifically, I will address and rebut (i) the Staff Finance Panel's claim that its proposed capital structure and recommended return on equity ("ROE") will support KEDNY and KEDLI's financial integrity; (ii) the Staff Finance Panel's recommendation that the Commission should use the latest 10-year Treasury yield to determine the cost of KEDNY's proposed Rate Year long-term debt issuance rather than a forecast Treasury rate for March 1, 2021; and (iii) the Staff Information Technology Panel's recommendation that the Commission should use the common equity ratio and cost rates for common equity and long-term debt

recommended by the Staff Finance Panel to determine the revenue requirement for National Grid USA Service Company, Inc.'s ("Service Company") rent expenses.

Q. Do other Companies' witnesses address other aspects of Staff's recommendation concerning the Companies' cost of capital?

A. Yes. Companies Witness Ann Bulkley's rebuttal testimony addresses the Staff Finance Panel's recommendations concerning KEDNY and KEDLI's requested returns on equity. Taken together, my rebuttal and Ms. Bulkley's rebuttal demonstrate that the capital structure and overall rate of return for KEDNY and KEDLI proposed in my corrections and updates testimony should be used for ratemaking purposes in these proceedings.

III. Financial Integrity

Q. The Staff Finance Panel asserts (at 33-34) that its recommended common equity ratio, together with its recommended return on common equity, will support the Companies' financial integrity. Do you have any comments concerning this testimony?

A. I am greatly concerned that the Staff Finance Panel's overall recommendation, coupled with the cost disallowance recommendations of other Staff and Intervenor witnesses in these cases, if adopted by the Commission, could adversely affect the Companies' ability to continue to raise capital on commercially reasonable terms. The Staff Finance Panel recommends that the Companies' ROE should be set at 8.2 percent – a rate that is significantly lower than that approved for any regulated gas distribution company in the United States – during a period in which (i) the Companies are attempting to manage an

inability to add incremental interstate pipeline capacity needed to support customer growth; and (ii) at least one witness to this proceeding has submitted testimony that the Companies should shift their focus from seeking growth "to implementing a process of managed decline" to avoid the accumulation of stranded assets.[1] Although I do not share the opinion of those who believe that the Companies are entering into a period of decline, I do believe that, in the current business environment, it is more important than ever that the Commission establish rates that reflect a cost of capital that is commensurate with the returns authorized for similarly situated businesses across the country as well as an overall revenue requirement that affords the Companies a reasonable opportunity to achieve that return. To obtain this result, the Commission must reject the recommendations of the Staff Finance Panel that the Companies' revenue requirement should be based on an ROE of 8.2 percent.

Q. The Staff Finance Panel (at 34) asserts that its recommended common equity ratio and return on equity recommendation, when combined with other quantitative metrics and quantitative factors, produce credit metrics that are consistent with the Companies' current credit ratings. Do you agree?

A. No. Moody's has indicated that it expects CFO Pre-WC/Debt to trend above the mid-teens for KEDLI and the high teens for KEDNY, and that if the metrics trend at or below the mid-teens, the Companies could be at risk of a further downgrade. Based on these indications, and the credit metric results, it appears that Staff's revenue requirement proposal puts KEDLI at a higher risk of a future downgrade. Furthermore, as discussed

[1] *See* Testimony of Robert Wyman at 5.

further by Ms. Bulkley, the low ROE recommended by Staff may adversely impact the credit rating agencies' view of the supportive nature of New York regulation for KEDNY and KEDLI, a key component in the overall credit rating, further increasing the risk of a downgrade for both KEDNY and KEDLI.

IV. Cost of Debt

Q. Please describe the Staff Finance Panel's recommendations regarding the cost of KEDNY's planned issuance of long-term debt in the Rate Year.

A. The Staff Finance Panel recommends (at 48-49) an interest rate of 3.24 percent for KEDNY's planned issuance of $400 million of new long-term debt in the Rate Year, compared to the 3.83 percent rate that I recommended in my direct testimony. The Staff Finance Panel does not agree with the proposed use of forecast Treasury yields to determine the interest rate of the planned Rate Year debt issuance. The Staff Finance Panel's recommended rate of 3.24 percent for this debt issuance is based on the June 2019 yield for 10-year Treasury debt of 2.07 percent, plus KEDNY's proposed credit spread of 1.17 percent. The Staff Finance Panel recommends updating its proposed approach to determining the cost of debt with more current data prior to the Commission's decision.

Q. Do you agree with the Staff Finance Panel's methodology for determining KEDNY's cost of new long-term debt?

A. No, I do not. I agree that the interest rate for the Company's planned Rate Year debt issuance should be updated prior to the Commission's decision; however, the cost of debt

used for ratemaking for the Rate Year should be the forecast cost of debt reflected in Exhibit __ (JC-1), Schedule 1, Page 2, subject to update. KEDNY's forecast interest rate is more reflective of market expectations for the interest rate likely to be available to KEDNY in the Rate Year than the interest rate in effect at the time of a Commission decision in these cases. Although I recognize that the Commission has used actual interest rates at the time it issues its decision to establish long-term debt interest rates in the past, it is nonetheless the case that the forward-looking rates that I have used in these proceedings are widely used in financial markets and fully consistent with the Commission's policy of using a fully forecast rate year to establish rates. There is a reasonable and verifiable link between current interest rates and those forecast for the Rate Year and there is no reason that the Commission should ignore that link in establishing KEDNY's long-term cost of debt.

V. <u>Service Company Asset Recovery Charge</u>

Q. Do you agree with the Staff Information Technology Panel's rate of return recommendation for Service Company?

A. No, I do not. I disagree with the Staff Information Technology Panel's conclusion that the Companies' weighted average cost of capital is the appropriate rate of return for Service Company. The Staff Information Technology Panel ignores the fact that Service Company maintains a capital structure that is separate and distinct from KEDNY and KEDLI. Moreover, Service Company's cost of long-term debt is lower than Niagara Mohawk Power Corporation's cost of long-term debt. Therefore, the proposed rate of return for Service Company is reasonable and conservative.

Q. Do you agree with the ROE component of the Staff Information Technology Panel's rate of return recommendation for Service Company?

A. No, I do not. I agree that the Companies' authorized ROE should be used for the purpose of determining the cost of equity component Service Company's rate of return. As discussed in Ms. Bulkley's rebuttal testimony, however, 8.20 percent is an unreasonable estimate of the Companies' cost of equity. As such, the Staff Information Technology Panel's proposed pre-tax rate of returns of 7.58 percent for KEDNY and 7.54 percent for KEDLI are unreasonable. Therefore, I continue to support the Company's 6.95 percent weighted average cost of capital for Service Company, subject to updated market data prior to the Commission's decision.

VI. Conclusion

Q. Does this conclude your rebuttal testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Supplemental Direct Testimony

of

David B. Doxsee
and
James M. Molloy

Date: January 27, 2020

I. Introduction

Q. Mr. Doxsee, please state your name and business address.

A. My name is David B. Doxsee. My business address is One MetroTech Center, Brooklyn, NY 11201.

Q. Are you the same David B. Doxsee who submitted testimony as part of the Revenue Requirements Panels in these proceedings?

A. Yes. I discuss my credentials in the Panels' testimony.

Q. Mr. Molloy, please state your name and business address.

A. My name is James M. Molloy. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same James M. Molloy who submitted testimony as part of the Revenue Requirements Panels in these proceedings?

A. Yes. I discuss my credentials in the Panels' testimony.

Q. What is the purpose of your supplemental direct testimony?

A. The purpose of our supplemental direct testimony is to provide information to permit us to be included as part of a panel presenting testimony concerning KEDNY and KEDLI's proposed capital structures and overall

costs of capital in these proceedings. KEDNY and KEDLI have previously submitted direct, corrections and updates, and rebuttal testimony concerning these matters from Jonathan Cohen, who serves as National Grid USA Service Company, Inc.'s Director of US Treasury, Planning and Strategy (New York). We are co-sponsoring Mr. Cohen's testimony in this proceeding.

Q. Have you reviewed Mr. Cohen's direct, corrections and updates, and rebuttal testimony as submitted by the Companies in these proceedings?

A. Yes. We are prepared to adopt the testimony previously submitted by Mr. Cohen as part of a panel at the hearing to be held in these proceedings.

Q. Does this conclude your supplemental direct testimony?

A. Yes it does.

A.L.J. COSTELLO: And I just want to make sure, with the companies, d -- did I get all the testimony in that you're --

UNIDENTIFIED SPEAKER: You did, Your Honor.

A.L.J. COSTELLO: Okay. And -- and those affidavits also sponsor the accompanying exhibits with their test -- with their pre-filed testimony. We'll go next to Department of Public Service staff. Let's see.

We have an affidavit by Daniel Gadomski, which we will mark as Exhibit 694. It's a two-page affidavit sworn to on February 25, 2020. And we have an affidavit by John Holst, which will be marked as Exhibit 695. And that's a two-page affidavit sworn to on February 25, 2020. And at this point in the record, the court reporter should insert the following files: Department of Public Service staff, Daniel S. Gadomski, direct testimony, and Department of Public Service staff, John Holst, direct testimony.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter Of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Daniel S. Gadomski
Utility Analyst 1
Office of Market and Regulatory Economics

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Please state your name, employer, and business address.

A. My name is Daniel S. Gadomski. I am employed by the New York State Department of Public Service (Department). My business address is Three Empire State Plaza, Albany, New York 12223.

Q. What is your position at the Department?

A. I am a Utility Analyst 1 economist in the Office of Market and Regulatory Economics.

Q. Please briefly state your educational background and professional experience.

A. I received a Bachelor of Arts Degree in Economics from the State University of New York at Albany in 2014. My current responsibilities at the Department include analyzing compensation and benefits issues in rate case proceedings, analyzing socioeconomic impacts for Article 10 siting of major electric generating facilities, and participating on Department Staff (Staff) teams overseeing utility management audits, including issues related to employee performance targets.

Q. Have you previously provided testimony in proceedings before the New York State Public

Service Commission (Commission)?

A. Yes. I provided testimony on compensation and benefits issues in cases regarding KeySpan Gas East Corporation, d/b/a National Grid (KEDLI), and The Brooklyn Union Gas Company, d/b/a National Grid NY (KEDNY), Cases 16-G-0058 and 16-G-0059; Suez Water New York, Inc., Case 16-W-0130; National Fuel Gas Corporation, Case 16-G-0257; New York American Water Company, Case 16-W-0259; Niagara Mohawk Power Corporation d/b/a National Grid, Cases 17-E-0238 and 17-G-0239; Corning Natural Gas Corporation, Case 16-G-0369; Central Hudson Gas & Electric Corporation, Cases 17-E-0459 and 17-G-0460; Consolidated Edison Company of New York, Inc., Cases 19-E-0065 and 19-G-0066; and the SUEZ Water New York Inc., SUEZ Water Westchester Inc., and SUEZ Water Owego-Nichols Inc., Cases 19-W-0168 and 19-W-0269. Additionally, I have provided testimony in proceedings before the State of New York Board on Electric Generation Siting and the Environment in Cases 16-F-0062 and 16-F-0328, regarding the jobs impact of the Eight Point Wind Facility and the Number Three Wind

Facility, respectively.

Q. What is the purpose of your testimony?

A. I will discuss the reasonableness of KEDNY's and KEDLI's (collectively, the Companies), request for rate recovery of management incentive compensation. In particular, I will discuss the compensation study prepared on the Companies' behalf and describe my review of the Companies' benchmarking of employee compensation. I will also address the annual escalation rate used by the Companies to project management compensation, effective each July 1st, starting July 2019. These issues were addressed in the direct testimony of Maureen P. Heaphy.

Q. In your testimony, will you refer to, or otherwise rely upon, any information produced during the discovery phase of this proceeding?

A. Yes. I will refer to, and have relied upon, the Companies' responses to Staff Information Requests (IRs). These responses are presented in Exhibit__(DSG-1) and are referred to throughout my testimony according to the numerical designation assigned by Staff, for example, DPS-123.

Q. Are you sponsoring any other exhibits?

A. Yes, I am sponsoring two additional exhibits: Exhibit__(DSG-2) contains a diagram of the Commission's two-prong test for recovering incentive pay in rates. Exhibit__(DSG-3) contains the WorldatWork Salary Budget Survey which includes compensation escalation rate forecasts. WorldatWork is a non-profit human resources association and a leading compensation authority for professionals and organizations focused on compensation, benefits, and total rewards.

Background

Q. Please summarize Commission precedent regarding cost recovery of utility incentive compensation programs.

A. The Commission's requirements for incentive compensation cost recovery were set forth in the June 17, 2011 Order Establishing Rates for Electric Service in Case 10-E-0362, a rate proceeding regarding Orange and Rockland Utilities, Inc. (O&R). I will refer to this later in my testimony as the "2011 O&R Rate

Order." In that Order, the Commission explained that a utility must demonstrate that its total level of employee compensation, inclusive of incentive pay, is reasonable relative to peer companies. The Commission explained that utilities should demonstrate the reasonableness of their incentive pay plans by providing a compensation study of similarly-situated companies. However, even if the utility produces a study shown that its compensation is reasonably in line with that of its peers, the Commission also stated that an incentive pay plan will not be deemed reasonable if it includes performance targets that adversely affect customer interests or are inconsistent with Commission policies.

Q. Is a compensation study the only option for utilities to demonstrate the reasonableness of their incentive pay plans?

A. No. The Commission explained further, on page 38 of the 2011 O&R Rate Order, that utility incentive pay plans could be deemed reasonable, and thus eligible for cost recovery, if the utility provides quantifiable or demonstrable

benefits to customers, in a financial sense, or in terms of reliability, environmental impact or customer service.

Q. Did any party seek rehearing of the 2011 O&R Rate Order?

A. Yes, both O&R and the Utility Intervention Unit of the New York Department of State petitioned for rehearing of the 2011 O&R Rate Order.

Q. How did the Commission rule on the rehearing requests?

A. The Commission denied both petitions in its November 21, 2011 Order Denying Petitions for Rehearing and/or Clarification. In the Order on rehearing, the Commission clarified that its objective is to ensure that customers pay no more, and no less, in rates than what is necessary and sufficient to attract and retain employees with the qualifications and motivation to ensure the provision of safe and adequate service. The Commission reiterated its finding that employee compensation plans should align with customer interests, and the utility must provide a quantitative demonstration to support a finding that the cost to ratepayers to fund

incentive compensation is at least matched by the value of the benefits they receive. Furthermore, the Commission explained that a utility could justify recovery of incentive compensation by making a demonstration that total employee compensation levels, inclusive of incentive compensation, are reasonable. The Commission explained that the preferred methodology to demonstrate this would be a comparable total compensation study of similarly situated companies. As long as the plan does not promote employee behavior that would be contrary to customer interests or Commission policies, utilities may be authorized to receive rate recovery of their incentive compensation expenses. I include a diagram of what has been frequently referred to as the Commission's two-prong test in Exhibit__(DSG-2).

Compensation Benchmarking Analysis

Q. Please describe the benchmarking study the Companies used to support their proposal in these rate filings.

A. The Companies hired Willis Towers Watson to

perform a study concerning the overall competitiveness of its management employee compensation and benefits package. The study examined the total employee compensation and benefits package, inclusive of incentive pay, for approximately 92 percent of the management positions at National Grid USA. The Companies' management employees were compared to a benchmarked position derived from a peer group of companies.

Q. What did the Companies conclude from this study?

A. Based on the results of the study, the Companies concluded that KEDNY's and KEDLI's total compensation and benefits programs, including incentive pay, are reasonable and market competitive.

Q. Did you review the Companies' benchmarking study and methodology?

A. Yes. The Companies provided the supporting documents in Exhibit__(MPH-2), and in response to IR DPS-801.

Q. What standard is used to determine whether a utility's benefits and compensation package is market competitive?

A. According to the WorldatWork Handbook of Compensation, Benefits & Total Rewards: A Comprehensive Guide for HR Professionals, which is a widely accepted authority, the value of compensation and benefits paid to a company's employees should fall within plus or minus 10 percent of the market median. This generally accepted rule of thumb referenced in the World at Work Handbook has been used by Staff to evaluate compensation benchmarking results in numerous prior rate cases before the Commission.

Q. Do the Companies assert that their benchmarking study satisfies this standard?

A. Yes. On page 12 of her direct testimony, witness Maureen P. Heaphy states that the study concluded that the value of National Grid USA's overall compensation, benefits, and pension package is slightly below the median value of the peer group's package, but generally within the 10 percent band that is considered the zone of reasonableness.

Q. Did you review the Companies' benchmarking study to determine if it satisfies this standard?

A. Yes. I reviewed the benchmarking study.

Peer Group Used for Benchmarking Analysis

Q. Which companies were included in the peer group used to analyze the competitiveness and reasonableness of the Companies' annual compensation package value?

A. Willis Towers Watson compared the Companies' total compensation and benefits package to the compensation and benefits packages provided selected a peer group consisting of 38 utility and general industry peer companies. The list of the peer group comparison companies was provided in Exhibit__(MPH-2).

Q. How did you review the Companies' peer group.

A. In IR DPS-802, I requested a listing of where the Companies' new hires were previously employed.

Q. Why is this important?

A. Because the goal of a good management compensation program is to attract and retain employees while not paying too much. Thus, it is necessary to evaluate the compensation levels of those firms in competition with the utility in the labor market.

Q. What did you learn from the response to IR DPS-

802?

A. The Companies' response to DPS-802 indicates that the Companies hire employees away from a wide range of general industry and utility industry related companies and many of the companies listed in the peer group are in direct competition with KEDLI and KEDNY in the labor market. Therefore, the span of the peer group companies appears to be reasonably reflective of the Companies' labor market competitors.

Results of Benchmarking Analysis

Q. Do you agree that the Companies' benchmarking study demonstrates that their total compensation is within plus or minus 10 percent of the median total compensation of its similarly situated peers?

A. Yes, for three reasons.

Q. What is the first reason?

A. The position matching was reasonable. The World at Work Handbook of Compensation, Benefits & Total Rewards: A Comprehensive Guide for HR Professionals indicates that the Company should benchmark at least 50% of its positions. The

Companies matched 92 percent of their positions with those of peer companies.

Q. What is the second reason?

A. The Companies' response to IR DPS-935 confirms that the Companies included all of the relevant forms of compensation to its employees in its total compensation benchmarking study.

Q. What is the third reason?

A. Although the Companies reported that that management compensation for KEDLI and KEDNY is 96% of the market average, I calculated that KEDLI and KEDNY compensation was also within 98.5 percent of the market median, thus within the reasonable range of plus or minus 10 percent of the market median.

Performance Targets

Q. Please describe the performance targets in the Companies' Annual Performance Plan.

A. As presented in the response to IR DPS-803 and discussed on page 22 of the direct testimony of Maureen P. Heaphy, the corporate goals of the Annual Performance Plan (the Plan) are built around three key areas of focus – customer,

communities, and people. Within these areas, the key performance metrics are concentrated on safety, reliability, customer ease, customer trust, and controllable costs. The Plan has been structured to establish the following percentages for payouts: (i) Bands C through F – 50 percent based on customers, communities and people objectives and 50 percent based on individual objectives, and (ii) Band B – 40 percent based on individual objectives and 60 percent based on overall financial objectives. The individual goals are designed to enhance employee expertise and performance and focus employees on common goals.

Q. Do the Companies propose to recover the full cost of their incentive pay program?

A. No. As discussed on page 28 of the testimony of Maureen P. Heaphy, the Companies' variable pay expense included in the proposed revenue requirements does not reflect any variable pay component for employees in Band A, or the variable pay component for employees in Band B that is tied to the achievement of overall financial objectives. Attachment 18 of the

response to IR DPS-803 contains the calculation of the incentive dollars amounts allocated to each key area of focus. My review of this response confirms that the portion of the variable pay plan to be included in rates is based on performance targets that will advance the Commission's goals and are not adverse to customers' interests.

Q. Do both prongs of the Commission's incentive pay test require a review of the Company's incentive pay plan performance targets?

A. Yes. Under one prong of the test, the Companies must demonstrate that the compensation program provides either quantifiable or demonstrable benefits to customers in a financial sense, or that it provides customers with quantifiable or demonstrable reliability, environmental, or customer service benefits. Under the alternative prong, the utility should confirm that the incentives will support the provision of safe and adequate service and will have no potential to adversely affect customer interests or to promote results that are inconsistent with Commission policies. Thus, under either prong

of the test, it is necessary to determine how the performance incentive plan goals align with the Commission's policy objectives.

Q. Do the performance targets in the Companies' incentive program promote results that are consistent with Commission policies?

A. Yes. As previously discussed, the portion of incentive compensation for which the Companies requested rate recovery is focused on goals related to safety, reliability, and customer service, as opposed to financial-based goals, which is consistent with Commission policies.

Q. Do the Companies' performance targets promote results that are potentially adverse to ratepayer interests?

A. No. The Companies' incentive compensation program maintains its focus on goals related to safety, reliability, and customer service, which is consistent with Commission policies.

Two-prong Test

Q. Have the Companies satisfied the Commission's two-prong test for including incentive compensation in rates?

A. Yes. The Companies have demonstrated that their total compensation remains within the zone of reasonableness as compared to similarly situated peer companies. The Companies also determined that its incentive compensation performance targets are not adverse to customer interests.

Labor Escalation Factor

Q. What factor did the Companies use to project their management wage increases?

A. According to page 33 of the testimony of Maureen P. Heaphy, the Companies assumed a 3.35 percent management salary increase approved for July 1, 2019 and annual 3.10 percent wage increases in July 1st of each subsequent year.

Q. Are these labor escalation rates reasonable?

A. No. The WorldatWork organization's forecast of "Total Salary Budget Increases," is 3.0 percent for the Actual 2019 median and Projected 2020 median. I provide the WorldatWork forecast of annual United States salary budget increases in Exhibit__(DSG-3). Thus, a 3.0 percent escalation rate would be more reasonable than the escalation rates proposed by the Company.

Q. Do you recommend any adjustments?

A. Yes. I recommend that the Companies' labor escalation factor be adjusted downward to reflect a 3.0% labor escalation rate. I provided this recommendation to the Staff Revenue Requirements Panel. That Panel then reflected this recommendation in Staff's recommended revenue requirements for the Companies.

Q. Does this conclude your testimony at this time?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

John Holst
Utility Analyst
Office of Accounting, Auditing,
and Finance

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Please state your name, employer and business address.

A. My name is John Holst. I am employed by the New York State Department of Public Service (Department) at Three Empire State Plaza in Albany, New York, 12223.

Q. What is your position in the Department?

A. I am employed as a Utility Analyst 2 in the Management and Operations Audit section of the Office of Accounting, Audits, and Finance.

Q. Please describe your educational background and professional experience.

A. I hold a Bachelor of Arts in English from the State University of New York at Stony Brook. I have been employed by the Department as a Utility Analyst since June of 2013. I am responsible for the oversight of management and operations audits, as well as the implementation of the resulting recommendations.

Q. Have you supervised management and operations audits of New York State utilities?

A. Yes. I have served as Project Manager for several operations audits of utility companies. In this role, I have been responsible for

leading the development of the Request for Proposals which defines the scope of each audit, leading the Staff team which selects the independent consultants to perform the audits, overseeing and participating in audit field work, and reviewing and communicating audit findings and recommendations to Staff and the Commission. I have also managed the implementation of audit recommendations as Implementation Manager for two operations audits. In this role, I have been responsible for ensuring that utilities submit timely implementation updates, reviewing the completion of audit recommendations, and reporting implementation status to Staff and the Commission. In addition, I serve as a Subject Matter Expert on the topics of Program and Project Management and Work Management for management and operations audits.

Q. In what cases have you performed these functions?

A. I am the Project Manager and Implementation Manager for Case 13-M-0314, an operations audit which reviewed the accuracy and effectiveness of

certain data provided to the Department by major New York utilities. I am also the Project Manager and Implementation Manager for Case 13-M-0449, an operations audit which examined the adequacy of staffing levels at major New York utilities, and the use of contractors. I am also the Project Manager for Case 17-C-0757, a management and operations audit of Time Warner Cable Information Services. I am a Subject Matter Expert for Case 18-G-0394, a management and operations audit of National Fuel Gas Corporation.

Q. Have you previously testified before the Commission?

A. Yes, I provided testimony in Cases 17-E-0238 and 17-G-0239, in the matter of Niagara Mohawk Power Corporation (NMPC).

Q. Are you sponsoring any exhibits?

A. Yes. I am sponsoring one exhibit, Exhibit___ (JH-1).

Q. Will you refer to any information provided by Brooklyn Union Gas Company (KEDNY) and KeySpan East Gas Corp., (KEDLI) during the discovery phase of this proceeding in your testimony?

A. Yes, I will refer to, and have relied upon, responses to Information Requests (IRs) provided by KEDNY and KEDLI. These responses are contained in Exhibit___(JH-1).

Q. What is the purpose of your testimony in this proceeding?

A. Public Service Law, Section 66(19)(c), requires that "upon the application of a gas or electric corporation for a major change in rates…the Commission shall review that corporation's compliance with the directions and recommendations made previously by the Commission, as a result of the most recently completed management and operations audit." Since KEDNY and KEDLI's rates were last reset in Cases 16-G-0058 and 16-G-0059, the Companies have participated in the implementation of recommendations from three audits which examined various aspects of KEDNY and KEDLI's management and operations. Accordingly, my testimony will address KEDNY and KEDLI's status in relation to three recently completed management and operations audits. These audits include Case 13-G-0009, which was a Comprehensive Management

and Operations Audit of National Grid USA's New York Gas Companies (the gas audit); Case 13-M-0314, which reviewed the Accuracy and Effectiveness of Certain Reliability and Customer Service Systems at all Gas and Combination Gas and Electric Utilities in New York State that Provide Statistics to the Commission on the Services They Provide Customers (the data audit); and Case 13-M-0449, a Focused Operations Audit of the Internal Staffing Levels and the Use of Contractors for Selected Core Utility Functions at Major New York Energy Utilities (the staffing audit). I will also discuss Case 18-M-0195, an ongoing management and operations audit of National Grid's New York electric and gas operations, which includes KEDNY and KEDLI.

Q. Please provide an overview of the management and operations audit process.

A. To initiate a management and operations audit, the Commission issues a Request for Proposals (RFP) for a consultant to conduct an audit. The RFP details the scope of work, deliverables, and expected audit schedule. A consultant is

selected by the Commission to conduct the audit. The chosen consultant, the Department, and the utility or utilities subject to the audit then enter into a multi-party agreement which governs the conduct of the audit. Typically, the consultant develops a work plan, interviews Company personnel, reviews utility documents, performs their analysis, and submits an audit report to Staff. The Commission then releases the report to the public.

Q. What are a utility's responsibilities regarding the outcome of a management and operations audit report?

A. When the Commission issues a management and operations audit report, utilities are required by Public Service Law §66(19)(b) to submit implementation plans for audit recommendations to the Commission. The Commission may then approve, modify, or reject the submitted implementation plan. Upon the Commission's approval or approval with modification, the implementation plan becomes enforceable.

Q. Please discuss Staff's responsibilities following the issuance of a management and

operations audit report.

A. Staff is responsible for monitoring a utility's implementation of audit recommendations. As utilities implement the recommendations, they submit regular implementation updates, as well as evidence of implementation to Staff for review and approval. Accordingly, audit recommendations generally fall into one of three categories of completion status: In Progress, Pending Review (reported as complete by the utility but under review by Staff), and Completed.

Q. What is the background and status of the gas audit?

A. Case 13-G-0009 was a comprehensive management and operations audit of National Grid USA's New York gas companies which examined the gas operations of KEDNY and KEDLI as well as NMPC. The audit was performed by NorthStar Consulting Group. The final audit report was issued on October 2, 2014 and included 31 recommendations for improvement at KEDNY, KEDLI, and NMPC. The Commission directed the implementation of those recommendations in its "Order Approving an

Implementation Plan" issued on May 14, 2015. In accordance with that Order, KEDNY and KEDLI, along with NMPC, filed timely written implementation updates and met with Staff between written updates. The latest implementation update was filed with the Department's Secretary on September 17, 2018. This filing identified 31 recommendations as Implemented, two of which were pending Staff's approval at the time of the filing. On September 28, 2018, the Office of Accounting, Audits and Finance issued a letter to KEDNY, KEDLI and NMPC confirming the completion of the implementation of all audit recommendations from Case 13-G-0009.

Q. What is the background and status of the data audit?

A. Case 13-M-0314 was an operations audit of the effectiveness of the programs and processes used at the utilities to collect data on reliability, safety, and customer service, and the accuracy of the calculations of this data as reported to the Department and the Commission. The audit was performed by Overland Consulting Group. The

final report was issued on April 20, 2016 and included 51 recommendations for improvement at KEDNY as well as 51 recommendations for improvement at KEDLI. The Commission directed the implementation of those recommendations in its "Order Approving Implementation Plans" issued on March 10, 2017. In accordance with that Order, KEDNY and KEDLI, along with NMPC, filed timely written implementation updates and met with staff between written updates. The latest implementation update was filed with the Department's Secretary on March 8, 2018. This filing identified that all recommendations for KEDNY and KEDLI had been implemented. On March 13, 2018, the Office of Accounting, Audits, and Finance issued a letter to KEDNY and KEDLI, as well as NMPC, confirming the completion of all audit recommendations from Case 13-M-0314.

Q. What is the background and status of the staffing audit?

A. Case 13-M-0449 was an operations audit of the internal staffing levels and the use of contractors for selected core utility functions at major New York energy utilities. The audit

was performed by Liberty Consulting Group. The final audit report was issued on February 21, 2017 and included 27 recommendations for improvement at KEDNY, KEDLI and NMPC. The Commission directed the implementation of those recommendations in its "Order Approving Implementation Plans" issued on December 15, 2017. In accordance with that Order, KEDNY, KEDLI, and NMPC filed timely written implementation updates and met with staff between written updates. The latest implementation update was filed with the Department's Secretary on April 15, 2019. The April 15 filing identified that all 27 recommendations had been implemented. On July 30, 2019 the Office of Accounting, Audits, and Finance issued a letter to KEDNY and KEDLI, as well as NMPC, confirming the completion of all audit recommendations from Case 13-M-0449.

Q. Are KEDNY and KEDLI subject to any ongoing management and operations audits?

A. Yes, Case 18-M-0195 is a comprehensive management and operations audit of National Grid USA's New York Electric and Gas Utilities. This

case was initiated on May 17, 2018 by the Commission's "Order Instituting Proceeding and Authorizing Issuance of a Request for Proposals." The audit is ongoing. Per the schedule in the RFP for this audit, a final report is expected in September 2019, and is typically released by the Commission thereafter.

Q. Have KEDNY and KEDLI included costs for conducting future management and operations audits in their revenue requirement?

A. No.

Q. Have KEDNY and KEDLI included implementation costs resulting from any audit recommendation in their rate filing?

A. No.

Q. As a result of your review of the Companies' management and operations audit compliance, do you recommend an adjustment to the Companies' rate request?

A. No.

Q. Do you have any additional recommendations or adjustments to the Companies' proposals for implementation of these recommendations?

A. No

Q. Does this conclude your testimony at this time?

A. Yes, it does.

A.L.J. COSTELLO: And I just want to confirm that that -- that should be everything for staff at this point?

MS. PARKS: Correct, Your Honors.

A.L.J. COSTELLO: Okay. Thank you. And we'll go next to City of New York. I have a two-page affidavit by John H. Lee, sworn to on February 13, 2020. And that will be marked as Exhibit 696. We have an affidavit, two pages, by Elizabeth D. Kelley, sworn to on February 13, 2020. And that will be Exhib -- Exhibit 697. I have a two-page affidavit by Jennifer Roberton, sworn to on February 12, 2020, which will be Exhibit 698. We have an affidavit by Nicholas Patane. I hope I'm pronouncing that correctly. Two pages, sworn to on February 12, 2020, and that will be Exhibit 699. Christine Dianofrio, a two-page affidavit sworn to on February 12, 2020. And that will be Exhibit 700.

We have a two-page affidavit by Natasha Godbe, sworn to on February 12, 2020. That will be Exhibit 701. We have a two-page affidavit by A -- Alexandra Fisher, sworn to on February 12, 2020. And that will be Exhibit 702. We have a two-page affidavit by Lane Kollen, sworn to on February 10,

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas 2020. That will be Exhibit 703. We have a two-page affidavit by Pinar Balci. It's P-I-N-A-R B-A-L-C-I -- sworn to on February 14, 2020. And that will be Exhibit 704. We have a two-page affidavit by Benjamin J. Krakhour, sworn to on February 13, 2020. And that is Exhibit 705. We have a two-page affidavit by David Morkal, M-O-R-K-A-L, sworn to on February 18, 2020. And that is Exhibit 706. We have a two-page affidavit by Ke W -- Wei. And I don't know if I'm pronouncing it correctly, but it is K-E capital W-E-I. Two pages sworn to on February 12, 2020, and that will be Exhibit 707. We have a two-page affidavit by Suzanne Des -- Desroches. It's D-E-S-R-O-C-H-E-S -- sworn to on February 12, 2020, and that will be Exhibit 708. A two-page affidavit by Elissa Velasquez, sworn to on February 14, 2020, which is Exhibit 709. I have a two-page affidavit by Jamie Torres Springer, sworn to on February 13, 2020. And that will be Exhibit 710. We have an affidavit, two pages, by Serga Shabalin, S-H-A-B-A-L-I-N, sworn to on February 24, 2020. And that will be Exhibit 711. And we have a two-page affidavit by Richard A. Bardino, sworn to on February 21, 2020. And that will be Exhibit 712.

I just want to note for the record

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas

that the City of New York indicated that one of its Panel members for the New York City Gas Infrastructure and Safety Panel, specifically, Michael McCutcheon -- it's M-C capital C-U-T-C-H-E-O-N -- the city was unable to locate that witness. He was one of four other pan -- there are four other Panel members whose affidavits we do have. The fifth, they were unable to locate, and so that we're going to admit the testimony without that witness' affidavit.

So at this point in the record, the court reporter should insert the following files: it's City of New York, Energy Efficiency Panel direct testimony; City of New York Low-Income Panel direct testimony; City of New York Alexandra Fisher rebuttal testimony; City of New York Lane Kollen direct testimony; City of New York Gas Infrastructure and Safety Panel direct testimony; City of New York Policy Panel direct testimony; City of New York Rate Design Panel direct testimony; and City of New York Richard Bardino rebuttal testimony. #

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of:

New York City Energy Efficiency Panel

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On Behalf of:

The City of New York

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Cases: 19-G-0309 NYC ENERGY EFFICIENCY PANEL
19-G-0310

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Mr. Lee – My name is John H. Lee. I am the Deputy Director for Green Buildings and Energy Efficiency at the New York City Mayor's Office of Sustainability. My business address is 253 Broadway, 14th floor, New York, New York.

Ms. Kelly – My name is Elizabeth D. Kelly. I am a Senior Policy Advisor at the New York City Mayor's Office of Sustainability. My business address is 253 Broadway, 14th floor, New York, New York.

Ms. Roberton – My name is Jennifer Roberton. I am the Transportation Policy Advisor at the New York City Mayor's Office of Sustainability, or MOS. My business address is 253 Broadway, 14th floor, New York, New York.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. We are submitting this direct testimony before the New York Public Service Commission ("PSC" or the "Commission") on behalf of the City of New York ("City").

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. Lee – I graduated from Rice University with a Bachelor of Architecture Degree and from Harvard University with a Master of Architecture Degree. I am a Leadership in Energy and Environmental Design - Accredited Professional certified by the U.S. Green Building Council. I am a licensed architect.

Ms. Kelly – I graduated from Lake Forest College with a Bachelor of Arts Degree in Urban Studies and French and from Rutgers University with a Master of City and Regional Planning Degree.

Ms. Roberton – I graduated from the University of British Columbia with a Masters of Arts in Planning in 2016. I received an Honours Bachelor of Arts from the University of Toronto. I am an American Institute of Certified Planners Candidate Member, as certified by the American Planning Association.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Mr. Lee – I commenced my public service career with the New York City Department of City Planning as an Urban Designer. From there I moved to the New York City Department of Buildings as a Senior Architect in the codes development divisions. In December 2012, I joined what was then known as the New York City Mayor's Office of Long Term Planning and Sustainability and now known as the Office of Sustainability. I have had positions of increasing responsibility in this Office. In my current capacity, I lead the City's policy and legislative efforts driving the building environment to energy efficiency standards necessary to achieve the City's greenhouse gas reduction goals.

Ms. Kelly – I spent two years organizing community and economic development programs for the North River Commission in Chicago. Subsequently, I was employed by The Community Preservation Corporation for seven years working on sustainability issues. I held positions of increasing responsibility, including

Assistant Vice President for Sustainability Programs. My responsibilities included developing and implementing strategies to leverage private capital in support of environmental sustainability in multifamily affordable housing development projects, engaging with stakeholder and business and community partners, and providing training and technical assistance for sustainability projects. In 2019, I joined the Mayor's Office of Sustainability, where my responsibilities include developing public policies and related programs to reduce greenhouse gas emissions from New York City's public and private buildings and other structures.

Ms. Roberton – I commenced my career in transportation in 2010 as the Education Coordinator at Bikechain, a non-profit mobility access organization based in Toronto, Ontario, Canada. I began working in public service in 2016 as a Program Specialist at the Environment and Energy Division at the City of Toronto. In November 2017, I moved to New York to join MOS as the Transportation Policy Advisor. I lead the Office's transportation portfolio, including the development of a citywide network of electric vehicle, or EV, chargers through policy, infrastructure projects and legislative efforts. A key part of this initiative is the implementation of a $10 million project to install EV fast chargers in every borough, which I manage in partnership with New York City's Department of Transportation, or NYCDOT. I also manage the City's comments to the Public Service Commission on matters regarding vehicle electrification.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Mr. Lee – I have submitted testimony to the Public Service Commission in prior The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corp. d/b/a National Grid ("KEDLI") rate cases. I also recently testified on behalf of the City in the Consolidated Edison Company of New York, Inc. ("Con Edison") 2019 electric and gas rate cases. In addition, I have testified before New York City Council in support of legislation related to energy efficiency in buildings and resiliency. I also frequently give talks at conferences on energy use information, such as benchmarking and energy audits, and energy efficiency issues, revisions to energy codes, and building retrofit measures, as well as NYC Clean Heat and the NYC Retrofit Accelerator.

Ms. Kelly – I have testified before New York City Council in support of legislation related to energy efficiency in buildings and clean energy financing. I also have presented at conferences on topics related to energy. I also testified in the 2019 Con Edison rate cases.

Ms. Roberton – I recently testified on behalf of the City in the 2019 Con Edison rate cases. I also frequently speak at conferences on sustainable transportation, including vehicle electrification, freight efficiency, and mode shift away from private vehicle use, as well as on the City's fast charging program and curbside projects conducted in partnership with Con Edison.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of our testimony is to discuss the City's public policies related to energy efficiency and their interaction with the Companies' various proposals set forth in the direct testimony of the Future of Heat Panel.

Q. PLEASE SUMMARIZE YOUR TESTIMONY AND RECOMMENDATIONS.

A. The City and State of New York have corresponding policies that encourage increasing reliance on energy efficiency to reduce greenhouse gas emissions and the need for fossil-fueled power plants. In this testimony, we recommend changing the structure and scope of energy efficiency programs to target more comprehensive building retrofits, to help to achieve the energy efficiency targets established by the Public Service Commission in its December 13, 2018 Order in Case 18-M-0084, or Energy Efficiency Order. We also make recommendations on the intersection between the Companies' energy efficiency programs and the City's Retrofit Accelerator. We further provide input on the Companies' proposed expanded Demand Response Demonstration Project, as well as their proposed Power-to-Gas ("P2G") program. We then make recommendations on the Companies' efforts to electrify their fleets. Finally, we express concerns about the size and structure of the Companies' shareholder incentive proposals. Any incentives should be linked to superior efforts by the utilities, not business-as-

usual conduct or circumstances outside of the utilities' control, and the Companies should not receive multiple/duplicative incentives for the same outcomes.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A Yes. We are sponsoring Exhibit__(NYCEEP-1), which is a compilation of discovery responses we relied upon in preparing our testimony, some of which we expressly reference herein.

ENERGY EFFICIENCY INCENTIVES FOR RETROFITS

Q. HAVE YOU REVIEWED THE COMPANIES' PROPOSED ENERGY EFFICIENCY PROGRAMS, BUDGETS, AND TARGETS?

A. Yes.

Q. WHAT BUDGETS AND TARGETS DO THE COMPANIES PROPOSE?

A. The Companies propose to maintain their current annual energy efficiency transition implementation plan ("ETIP") budgets through 2020. The Companies also propose using previously-adopted budget amounts for 2020, and the presumptive targets and budgets for 2021-2025, set forth in the Energy Efficiency Order, to determine their proposed budget amounts to achieve incremental energy efficiency savings for the Rate Year and Data Years.

Q. HOW WERE THE BUDGETS AND TARGETS FOR INCREMENTAL ENERGY EFFICIENCY SAVINGS ESTABLISHED?

A. In the Energy Efficiency Order, the Commission stated that the budgets for increased utility energy efficiency efforts were based on the lesser of Commission-authorized or current actual run rates for each utility's existing portfolio. In this

context, run rates for non-low-to-moderate income ("LMI") customers represent the ETIP portfolio level expended funds divided by the portfolio level acquired savings through the fourth quarter of 2017.

Q. ARE THE PROPOSED INCREMENTAL ENERGY EFFICIENCY TARGETS APPROPRIATE?

A. No. The use of historic run rates to establish the Companies' incremental MMBtu savings does not accurately reflect the Companies' actual, recent unit costs to achieve energy efficiency savings. For example, as shown in the Companies' response to City Information Request ("IR") 123, in 2018, KEDNY had an average unit cost of $1.64 per therm of energy efficiency savings, while KEDLI had an average unit cost of $1.61. Over the last two years, the average per-therm unit costs for KEDNY and KEDLI are $1.86 and $1.87, respectively.

In comparison, the Energy Efficiency Order set KEDNY incremental non-LMI MMBtu savings target for 2019-2020 at 231,401 MMBtu, with a total budget of $6,452,620. KEDLI's incremental non-LMI MMBTu savings target for the same two-year period was set at 112,554 MMBtu, with a total budget of $2,674,824. Based on these targets and budgets, the per-therm costs for KEDNY and KEDLI are approximately $2.79 and $2.38, respectively.

Q. HOW SHOULD THE COMPANIES' INCREMENTAL TARGETS BE CALCULATED?

A. The Companies should base their incremental energy efficiency targets on actual recent unit costs. This will encourage the Companies to continue building on recent

successes in their respective energy efficiency programs by continuing to drive down per-unit costs and increasing program effectiveness.

Q. DO YOU HAVE SPECIFIC TARGETS IN MIND?

A. Yes. The Companies should calculate their incremental energy efficiency savings targets by taking their proposed incremental budgets and dividing them by their respective energy efficiency savings unit costs over the last two years. For example, KEDNY proposes a non-LMI budget of $5,972,357 for calendar year 2021. Based on an average unit cost of $1.86 per therm saved over the last two years, the target savings for that budget should be 3,210,944 therms, or approximately 321,100 MMBtu. For the same time period, KEDLI's proposed budget is $2,297,639. Based on a two-year unit cost of $1.87, KEDNY's savings target for 2021 should be 1,228,684 therms, or approximately 122,900 MMBtu.

Q. ARE YOU ADVOCATING FOR THE COMPANIES TO FOCUS PRIMARILY ON 'LEAST COST' OR 'LOW HANGING' ENERGY EFFICIENCY MEASURES?

A. No, in fact later on in our testimony, we advocate for just the opposite – the Companies should be pursuing more comprehensive energy efficiency measures that we contend will maximize energy efficiency savings. We understand, however, that by pushing the Companies to drive down per-unit costs, we may be creating an incentive for the Companies to instead focus on 'low hanging fruit,' or energy efficiency measures that come with low costs and quick paybacks. Again, this is not the intent of our proposal. To avoid this undesirable outcome, the

Commission should adopt the suite of energy efficiency improvements that we discuss below, including payment of upfront energy efficiency incentives, stacking of incentives to increase payment to customers that pursue a combination of efficiency measures at the same time, re-designing incentives to accommodate longer timescales for building owners that undertake comprehensive energy efficiency retrofits, and improving coordination with the City's Retrofit Accelerator.

Moreover, our proposal is consistent with the Energy Efficiency Order, in that we are simply adopting the Companies' actual run rates from the last two years (calendar years 2017 and 2018) and building upon the Companies' demonstrated success in driving down per-unit costs of energy efficiency gains.

Q. PLEASE CONTINUE.

A. Our proposal is also in recognition of the need for incremental energy efficiency gains (among other measures) that has been highlighted by the stalled Northeast Supply Enhancement ("NESE") pipeline. As discussed in the direct testimony of the City's Policy Panel, and as highlighted in the Companies' response to City IR-117, the Companies have not identified any incremental energy efficiency, demand response, or other load management programs that it would plan to implement in the event that the NESE project is ultimately not built. Our proposals here, including our proposal for raising the energy efficiency targets, represent several concrete steps that the Companies can take in the immediate term to improve their energy efficiency and demand response offerings. Certainly, however, we

acknowledge that much more will be needed in the coming years to meet stringent City and State requirements on efficiency, carbon reduction, renewable energy, and the like. This is, in part, why we advocate below for the creation of a collaborative or other similar process, where interested parties can address these broader concerns in a more comprehensive manner.

Q. ARE YOU AWARE OF ANY ISSUES WITH THE CURRENT DESIGN OF THE COMPANIES' ENERGY EFFICIENCY PROGRAM OFFERINGS?

A. Yes. Based on our experience working with building owners and the utilities to implement building retrofits in New York City, we have identified a number of issues with the design of the Companies' energy efficiency program offerings that may be preventing them from achieving the maximum energy efficiency savings possible. First, in many cases, it is difficult to secure the requisite upfront financing needed to install energy efficiency improvements, a problem which is exacerbated if the building owners cannot access utility incentives until after the efficiency measure has been installed. Second, and relatedly, we believe that the Companies can achieve greater savings through their energy efficiency programs by incentivizing building owners to pursue a comprehensive package of energy efficiency programs that, together, have a greater positive impact than if the building owner only pursued individual building improvements.

Q. PLEASE EXPLAIN THE ISSUE OF UPFRONT FINANCING FOR ENERGY EFFICIENCY PROJECTS.

A. Based on our experience working with building owners to implement building retrofits in New York City, and anecdotal evidence from KEDNY and KEDLI energy efficiency program participants, the upfront cost of comprehensive energy efficiency investments can be an impediment for some customers, particularly owners of low income or rent-regulated buildings**.** Some building owners may have limited cash flow or capital reserves to make significant energy efficiency upgrades and thus must seek third party financing to fund the improvements. There are a variety of circumstances that inhibit the ability of some owners to obtain financing or increase financing costs to fund efficiency projects. As a result, these owners are unable to utilize energy efficiency incentive programs and are prevented from proceeding with projects.

Q. WHAT TYPES OF ENERGY EFFICIENCY INCENTIVES DO THE COMPANIES OFFER?

A. Our understanding is that the Companies offer prescriptive, custom, and direct install incentives as part of their energy efficiency programs. For prescriptive measures, incentives can range from $25 to $12,000 per qualifying measure installed. Custom incentives are paid on a per-therm basis, with a maximum cap of up to $100,000 in incentives. Finally, direct install incentives provide free installation of low-cost heating and water saving measures, but no direct payment is made to the customer. Instead, a vendor provides the direct install measure and is paid based on the quantifiable savings achieved thereby.

Q. ARE THESE INCENTIVES PAID UPFRONT?

A. In most cases, no. According to the Companies' response to City IR-122, prescriptive incentives are paid to a customer after the qualifying equipment already has been purchased, and after a detailed verification and documentation process. For custom incentives, customers must complete an application form, schedule a pre-inspection, and submit the required documentation (*e.g.*, cut-sheets, usage data, anticipated project costs, and other technical data). Once the technical review is complete, a Company commitment letter indicating an authorized incentive level is provided to the customer. Once the project is completed, and a post-inspection completed, the incentive is issued. With respect to direct install incentives, the customer must first contract with a qualifying vendor to perform the necessary energy efficiency work.

We do note that, with respect to prescriptive incentives provided via the Companies' online Market Place, qualifying products and quantities can be purchased at a discounted price. In other words, Market Place prescriptive incentives are provided at the point of sale.

Q. DO YOU SEE ANY ISSUES WITH THE INCENTIVE PAYMENT MECHANISM AS DESIGNED?

A. As stated above, the difficulty for some buildings owners to obtain project financing for their energy efficiency investments can prevent otherwise meritorious and beneficial projects from being undertaken, or prevent building owners from undertaking deep retrofits in favor of smaller, less-costly (and likely less beneficial overall) measures. To the extent that the Companies do not provide incentives or

other financial support to a building owner until an energy efficiency project is completed, it can exacerbate this issue.

Q. DO YOU HAVE ANY RECOMMENDATIONS FOR SOLUTIONS?

A. Yes. The Companies should provide funding at the outset for some projects. Project developers then would be able to rely on this funding to obtain financing for the projects, either using the funding as collateral to reduce the financing needed, or as a source of money to repay the financier. To be clear, we propose that the Companies provide a limited amount of funding for this purpose ($1-3 million), at first, to test and verify the concept, and that it be used in partnership with private, 3rd party financing. These efforts will allow for additional private capital to be leveraged in support of larger energy efficiency investments and more comprehensive retrofits.

Q. PLEASE DESCRIBE YOUR PROPOSAL.

A. We propose that KEDNY and KEDLI each segregate a portion of the budgets allotted to their respective multifamily energy efficiency program offerings. The Companies would then encourage building owners and project developers to apply by adjusting the project incentive award timeline to assist the owners in obtaining private financing. Otherwise, the projects would be evaluated and selected in the same manner as currently occurs. Upon selection, the applicable Company would provide a letter verifying the anticipated funding amount the building owner or project developer would receive under the standard program parameters. The verification letter will be contingent on a commitment by a third party financier.

The funding amount could be held in an escrow account by the Companies, the Companies' implementation contractor, or the participating 3rd party lender until the project is complete, then released to the appropriate party. Company funding could be provided subject to repayment if the project is not completed. Under this proposal, the building owner or project developer would be able to use the funds from the Companies to assist in obtaining financing for the project. This approach also should facilitate the deployment of private-market funds for energy efficiency projects and extend the positive impact of incentive dollars much further.

Q. WHAT IS THE PROPOSED TERM FOR THIS PROPOSAL?

A. We propose that it last for the earlier of exhaustion of the funding or three years. Upon its conclusion, the Companies should conduct an evaluation of the performance and default rates to ascertain whether the expected savings are achieved and what the risk of loss is. If this approach results in greater levels of energy efficiency project implementation and corresponding energy savings, and the risk of default, or losses, is minimal, then consideration should be given to broader relaxation of the requirement that funding is provided only after projects are completed.

Q. WHO SHOULD BE THE TARGET OF THIS PROPOSAL?

A. This proposal should target owners of multifamily housing, subsidized housing, limited equity cooperatives, and other low and moderate income residential buildings, who we believe are most likely to benefit from the revised funding structure. These owners often have less cash flow and more difficulty in obtaining

financing for large projects. Providing funding upfront could help them and their tenants obtain the benefits of deep energy efficiency savings that otherwise may be unattainable. The proposal should also target projects where incentive payments are expected to exceed a certain dollar threshold. We propose $50,000 as that threshold.

Q. SHOULD UPFRONT FUNDING BE AVAILABLE TO CUSTOMERS WHO OPT TO INSTALL CUSTOM EFFICIENCY MEASURES?

A. Yes. For customers who choose to install custom energy efficiency measures, the Companies should offer a commitment letter indicating the maximum authorized incentive level. The commitment letters can contain caveats that full payments are not provided until the projects are completed and verified, and that failure to satisfactorily complete the committed projects as proposed will result in the customer being required to reimburse the Companies for any upfront payments provided thereby. The commitment letters (including any conditions) should be designed such that financiers are comfortable with their terms. The combined provision of upfront funding and a commitment letter prior to commencement of projects for purposes of securing financing should increase the number of projects, help leverage private capital in support of energy efficiency, and help achieve the State's, City's, and the Companies' energy efficiency goals

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS ON ENERGY EFFICIENCY PROGRAM DESIGN?

A. Yes. The Companies should also tailor their energy efficiency program offerings to incentivize customers to pursue comprehensive packages of energy efficiency

measures that, together, could result in greater energy savings than discrete measures alone. For example, the Companies could offer higher incentives when related measures are coupled together, such as an increased incentive for new heating and cooling equipment provided a building owner also agrees to make improvements to the building envelope – air sealing, replacing windows and doors, adding insulation, etc.

Combining incentives associated with multiple improvements should encourage or facilitate more comprehensive efficiency measures and produce larger savings. For example, the magnitude of the incentives could increase non-linearly. That is, there is a greater overall value of performing five measures than one or two, so the incentives for multiple measures could be larger than simply adding together the incentives applicable to individual measures if performed discretely. Alternatively, a bonus incentive could be offered when a customer completes multiple measures as part of a comprehensive energy efficiency retrofit that achieves more than 20 percent energy savings.

Q. SHOULD THIS APPROACH FACTOR IN THE TIME IT TAKES TO COMPREHENSIVELY RETROFIT A BUILDING?

A. Yes. Given the expanded nature of this approach, the energy efficiency plan for a customer may need to extend over a period of years. The Companies should design their incentives to accommodate comprehensive upgrades that occur in stages, or over a longer period of time, rather than all at once. Instead of rushing through project scoping to meet prescribed deadlines, customers should be able to

implement projects that maximize savings. This adjustment also would improve coordination with private financing providers, allowing customers to leverage incentive funding to access additional capital to cover the up-front and related costs associated with building improvements not covered by the Companies' funding. Without this adjustment, customers are mostly reliant on self-financing, which further limits the scale and scope of efficiency projects. We recommend this change also be deployed generally.

The above recommendation notwithstanding, we note that we are not experts on the utilities' Technical Resource Manual, but it is our understanding that the Manual does not account for interactivity between measures. As a result, there is no motivation for program implementers or building owners to pursue a combination of comprehensive energy efficiency initiatives.

Q. WHAT ARE YOUR RECOMMENDATIONS REGARDING FUTURE ENERGY EFFICIENCY OFFERINGS?

A. There is precedent for creating a collaborative or other process to occur either as a later part of a rate case or during a rate year to explore issues that are better pursued through discussion than testimony and a hearing. This is such an issue. We recommend that the Commission convene either a technical conference or a collaborative process at which the details of the Companies' energy efficiency programs can be presented and discussed. This forum would allow interested stakeholders to offer and debate proposals for improvements to the Companies' current plans and programs, such as our inclining incentive proposal. Assuming

that consensus can be reached on one or more improvements, they can be presented to the Commission for approval and immediate implementation.

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS ON ENERGY EFFICIENCY?

A. Yes. The City believes it important that the use of the funds approved for the Companies' energy efficiency programs, as well as program performance and the savings actually achieved, be closely monitored. To that end, we recommend that the Companies be required to file a report no less frequently than twice per year containing the following information:

- Details on all expenditures made, broken down by program and including both the reporting period and periods starting with the inception of the program;
- Summary-level information on savings achieved, broken down by program and including both the reporting period and periods starting with the inception of the program; for programs that do not result in direct energy savings, information on the benefits realized from the program should be provided;
- For each program, detailed information on Dth, or MBtu avoided and GHG and peak load reductions; the savings should be broken down by building typology, geographic location – such as zip code, market sector, and technology(ies) deployed;

- Identification of constrained areas and the details we just described for each program that helps to alleviate the constraints;
- Identification of programs and/or specific projects targeted to low income customers, and the details we just described for each program or project, as applicable;
- A discussion of any program changes made over the reporting period and the reasons for such changes;
- A discussion of any program changes that will be made prospectively and the reasons for making the changes.

Additionally, once a year, the Companies should include in their report a discussion of lessons learned and best practices based on its prior experiences. This information would be helpful in designing future programs, and in modifying existing energy efficiency programs to maximize the benefits to customers.

IMPROVED COORDINATION WITH THE RETROFIT ACCELERATOR

Q. PLEASE DESCRIBE THE RETROFIT ACCELERATOR.

A. The City's Retrofit Accelerator has been a growing success and has gained broad acceptance from building owners, developers, contractors, and others. The Retrofit Accelerator was launched in September 2015 and offers a team of energy efficiency advisors to provide independent, customized, advisory services at no cost to building owners to help them complete cost-saving energy efficiency projects. The Retrofit Accelerator team assists building owners and decision-makers in navigating the building retrofit process in order to help participants reduce building

energy consumption and improve the comfort of tenants. This assistance involves identifying financing and incentive options, including those offered by the Companies, and provides guidance for participation in these programs.

Since its launch in 2015, the participants in the Retrofit Accelerator have completed energy efficiency and clean energy projects in thousands of buildings in New York City, with expanded program offerings projected to reach even more buildings throughout the City in the near future. The touchstone of this program is providing free, personalized advisory services to streamline the process of energy efficiency improvements, including helping building decision makers navigate available incentive programs.

Q. IS THERE ANY COLLABORATION BETWEEN THE COMPANIES AND THE RETROFIT ACCELERATOR?

A. Yes. The Retrofit Accelerator website currently offers an "NYC Incentive Map" that helps customers to find programs available in New York City to assist in building retrofits, including energy efficiency rebates and incentives offered by the Companies. For example, the NYC Incentive Map provides a high-level overview of the Companies' Multifamily Program, such as the types of measures covered by the program and the possible incentive amounts, and then provides a convenient link to the dedicated utility website for that program.

The Retrofit Accelerator also has collaborated with the Companies in the past with respect to its 2012 NYC Clean Heat initiative to address the public health hazard presented by heavy heating oil emissions in New York City. As part of the

Clean Heat Initiative, the City worked closely with utilities (including the Companies), contractors, financiers, and others to provide customers the ability to easily work through the process to convert building boilers from heavy fuel oil (*i.e.*, No. 6 fuel oil) to natural gas and No. 2 fuel oil, in order to comply with a regulation that required all use of No. 6 fuel oil to be phased out by 2015. Due in large part to the efforts of the City, the utilities, contractors, and financiers, the Clean Heat Initiative has been very successful.

Q. ARE THERE OPPORTUNITIES TO IMPROVE THE LEVEL OF COORDINATION BETWEEN THE COMPANIES AND THE RETROFIT ACCELERATOR?

A. We believe so. From a customer perspective, having one location in which to obtain information about available programs, apply for incentives, find contractors, and, as necessary, secure financing, would be ideal. In the case of energy efficiency, an approach similar to the coordinated approach taken with the Clean Heat Initiative would help to ensure that consumers are getting all information about available programs and funding.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. There should be greater coordination of messaging, marketing, and outreach between the Companies' energy efficiency programs and the Retrofit Accelerator program. Closer coordination and stronger synergies between the two initiatives will strengthen the positive perception of both programs in the public eye, reduce

duplication of benefits, and result in a clear and unified approach to cut greenhouse gas emissions and improve efficiency.

We also recommend a direct link between the Companies' energy efficiency programs and the Retrofit Accelerator program managers. When customers seek information or assistance via the Retrofit Accelerator, the personnel assisting the customers should be able to contact knowledgeable Company personnel directly and immediately regarding questions about available energy efficiency programs and incentives, and to seamlessly enroll customers in those programs.

We also recognize that the collaborative approach should not be one-sided. That is, the Companies also should work closely with Retrofit Accelerator personnel to identify incremental opportunities for energy efficiency beyond what is provided through the Companies' programs. The Retrofit Accelerator offers services for both new construction and existing buildings, as well as a High Performance Retrofit Track, that are utility and provider agnostic. These services can help drive down city-wide energy demand and increase efficiency for buildings that otherwise might not be eligible for utility programs, or when projects seek assistance completing projects outside the current scope of utility program offerings.

Another process improvement that can be made is closer coordination of customer enrollment. For example, the Retrofit Accelerator currently follows standard industry practice in completing a comprehensive intake survey of all new program participants. To the extent that a Retrofit Accelerator contact is then

transferred to the Companies, there should not be a need for the intake process to be repeated on the Companies' end. We recommend that the Companies' energy efficiency department work with Retrofit Accelerator staff to develop a shared, or common application that can be used by both programs. This will significantly reduce program administrative costs for both parties and best meet customer needs by alleviating the frustration expressed by our shared customers, who are currently being asked to duplicate the intake process.

Lastly, we recommend that the Companies track all customer referrals made to and from the Retrofit Accelerator, including what projects were pursued, and the project status. Consistent and transparent reporting will provide both the Companies and Retrofit Accelerator staff greater insight into where and how customers originate. It also will allow us to quantify the frequency of customer referrals and value of any customer acquisition the Retrofit Accelerator provides. We further recommend that direct customer referrals made by the Retrofit Accelerator to the Companies' programs not be counted as customer acquisitions made by the Companies, and that any associated dollar value be made available to support additional program related outcomes.

EXPANDING DEMAND RESPONSE EFFORTS

Q. WHAT IS THE CITY'S POLICY ON GAS DEMAND RESPONSE?

A. As explained in more detail by the City's Policy Panel, the City generally is supportive on enhanced energy efficiency and demand response efforts. Demand response programs can help to minimize reliance on natural gas by reducing the

need for gas capital expenditures to accommodate growing demand. In turn, such programs can help achieve both the City's and State's clean energy policy goals. In light of the continued uncertainty related to the NESE pipeline project, it is even more crucial now that the Companies have plans in place to effectively manage their current and future gas systems, including through the use of demand response to reduce peak load.

Q. ARE YOU FAMILIAR WITH THE COMPANIES' EXISTING GAS DEMAND RESPONSE DEMONSTRATION ("DRD") PROJECT?

A. Yes. We understand that the City has been an active participant in the DRD project since 2016, and that its experience has generally been positive. Given how well the program has worked to date, we believe that there is an opportunity here to significantly expand on the existing framework of the DRD Project to achieve even greater natural gas consumption and peak load reductions across the downstate National Grid service territories.

Q. HAVE YOU REVIEWED THE COMPANIES' PROPOSAL TO EXPAND THE DRD PROJECT?

A. Yes. Our understanding is that the Companies plan to expand the DRD Project's peak demand reduction targets from 80 to 1,000 Dth per hour of demand reduction for KEDNY, and from 60 to 400 Dth per hour of demand reduction for KEDLI, by 2024. Our understanding is based on the Companies' response to City IR-89 and the information set forth in Schedule 5 of Exhibit__(FOH-1). The Companies also are proposing to include a new penalty mechanism to address program

participants who fail to respond to called curtailment events, or who fail to meet minimum committed reductions.

Q. WHAT ARE YOUR OPINIONS ON THE PROPOSAL?

A. Generally, we support the Companies' decision to expand the DRD Project, particularly given how successful the program has been over the last few years. However, we are concerned that, given recent developments with the NESE pipeline, the proposal does not go far enough to address the lingering concern that growing gas demand will outstrip gas supply in New York City.

Q. DOES THE CITY HAVE ANY SPECIFIC RECOMMENDATIONS ON THE EXPANDED DRD PROJECT?

A. The City's recommendations on the DRD Project are set forth in the Direct Testimony of both the NYC Rate Design Panel and the NYC Policy Panel. Briefly, the Companies should increase the targets of the DRD Project by, among other things, expanding the pool of eligible customers, reducing the requirement for a 10 percent or more committed reduction, increasing customer recruitment, or through other measures. The Companies also should explore the expansion of the DRD Project through a collaborative that includes interested stakeholders

CONCERNS REGARDING THE POWER-TO-GAS PROGRAM

Q. HAVE YOU REVIEWED KEDLI'S PROPOSED P2G DEMONSTRATION PROJECT?

A. Yes.

Q. WHAT IS YOUR OPINION ON THE P2G DEMONSTRATION PROJECT?

A. The City conceptually supports the theory behind the P2G demonstration project, and supports the Companies in evaluating its feasibility. Converting excess renewable energy into pipeline quality natural gas can help to reduce the carbon footprint of the upstream gas system, and can allow the Companies to meet some of their gas system requirements with RNG. This is consistent with both the City's and State's clean energy goals.

We note that in order for P2G to play a meaningful role in decarbonizing the gas system, there first would have to be an excess of renewable energy to convert into methane through this process. The Companies should ensure that, if P2G becomes an integral part of the strategy to increase RNG use, it relies only on excess renewable electricity, and does not otherwise prevent customers from consuming that renewable electricity in the first place. P2G also should not be used as a substitute for comprehensive efforts to reduce overall emissions, gas usage, and peak demand, for example, through expanded energy efficiency efforts.

EXPANDING UTILITY FLEET ELECTRIFICATION

Q. PLEASE DESCRIBE THE CITY'S VEHICLE ELECTRIFICATION POLICY GOALS.

A. The City's goal in the transportation sector is to have an 80 percent sustainable mode share by 2050, meaning that four in five trips in New York City will be by foot, bicycle, or public transit. However, we recognize that sustainable modes may not be accessible for all New Yorkers, which is why for the trips that still need to be taken by vehicles, we want to target zero emission vehicles.

This transportation sector goal is part of a larger goal established by Mayor Bill de Blasio for New York City to reduce citywide greenhouse gas emissions by 80 percent by 2050, based on a 2005 baseline. Transportation accounts for approximately 30 percent of citywide greenhouse gas emissions. The bulk of on road transportation greenhouse gas emissions come from passenger vehicles. According to the *New York City's Roadmap to 80x50* prepared by the Mayor's Office of Sustainability in October 2016, a transition to low carbon fuels and vehicles could account for up to a 13 percent reduction in transportation emissions.

Q. WHAT TYPES OF VEHICLES ARE OF MOST INTEREST TO THE CITY WITH REGARDS TO ITS TRANSPORTATION POLICY?

A. Electric vehicles, or EVs, are of particular interest to the City since they produce no tailpipe emissions and are significantly more energy efficient than their internal combustion engine counterparts. However, the City also sees merit in vehicles that rely on alternatives to fossil fuels, such as renewable natural gas ("RNG").

Q. WHAT IS THE CURRENT COMPOSITION OF THE COMPANIES' FLEET, BY FUEL TYPE?

A. According to the Companies' response to City IR-99, the Companies currently operate a fleet of 275 vehicles that run solely on compressed natural gas ("CNG"), the vast majority of which are located in the Long Island and New York City regions. The Companies also appear to be operating a number of all-electric vehicles as part of their fleets, though most of these vehicles consist of forklifts and small construction vehicles such as backhoes. Importantly, according to their

response to City IR-97, the Companies' fleets currently only include a handful of all-electric or hybrid passenger vehicles.

Q. DOES NATIONAL GRID HAVE ANY PLANS OR INTERNAL POLICIES TO ELECTRIFY ITS PASSENGER VEHICLE FLEET?

A. According to National Grid US's website, and the Companies' response to City IR-98, National Grid has pledged to spend 5 percent or more of its annual acquisition dollars on plug-in EVs and technologies (including charging stations and infrastructure). This 5 percent commitment is cumulative across all of National Grid's businesses. Per the response to City IR-99, the Companies also indicated that National Grid has a general policy to accelerate the process of converting its fleet of passenger cars to all EV vehicles during 2019 and 2020.

Importantly, however, it does not appear that the Companies have any policy to convert their own fleet vehicles to alternative fuels outside of the general policy that broadly covers all National Grid businesses. In response to City IR-124, the Companies stated that they have concrete plans to convert a small number of (*i.e.*, 6) diesel and propane-powered forklifts to electric models, and to collaborate with an established manufacturer on a trial of an all-electric towable backhoe in 2019. With respect to passenger vehicles, the Companies only stated that they expect to continue converting such vehicles to EV, but there are no finalized plans for 2020.

Q. ARE THE COMPANIES' PLANS CONSISTENT WITH THE CITY'S POLICIES?

A. No. In our opinion, the Companies need to do more to accelerate the conversion of their existing fleets of passenger cars to EVs. At present, it does not appear that the Companies have any concrete plans in the near-future to significantly accelerate the conversion of fleet passenger cars to EVs.

Q. WHY IS IT IMPORTANT FOR THE COMPANIES TO ACTIVELY SEEK TO CONVERT THEIR FLEETS TO ALTERNATIVE FUELS?

A. As noted above, in order to achieve an 80 percent reduction in citywide greenhouse gas emissions by 2050, significant in-roads will have to be made with respect to reducing the emissions impact of the transportation sector. This includes electrifying the significant fleet operations that operate in New York City, including the fleets operated by the Companies.

Electrifying the Companies' fleets can produce positive public health benefits. For example, air quality in New York City has been an ongoing concern for many years. Data from the New York City Department of Health and Mental Hygiene shows that areas with large populations of low income residents have 1.7 times the exposure to annual average of fine particulates and over nine times the rate of hospital visits for asthma due to diesel emissions from mobile sources than more affluent areas. Similarly, both gasoline and diesel vehicles contribute to ozone formation that, when inhaled, can have detrimental health impacts such as increasing the likelihood of asthma attacks. By converting their existing gasoline- and diesel-fired vehicles to all-electric, or even to lower emissions fuel types such

as CNG, the Companies can significantly reduce the air quality and health impacts of their vehicles' emissions on the City's population.

Q. WHAT DO YOU RECOMMEND?

A. The Companies should commit to increased EV adoption for their fleets beyond the existing 5 percent goal spread across all National Grid businesses. Specifically, for new light duty fleet purchases, the City recommends that each Company commit at least 25 percent of its annual acquisition budget toward EVs, with the goal of increasing that commitment to 50 percent within five years.

Electric vehicles have a potentially lower overall cost of ownership than their internal combustion engine counterparts (mostly through maintenance and fueling cost reduction). The City has found that its electric fleet vehicles have a maintenance cost of between $204 and $386 annually. Conventional gasoline fleet vehicles, however, cost the City between $922 and $1,805 annually.

In addition, to the extent that acquiring EVs for the Companies' fleets is not feasible (for example, for heavy duty vehicles), the Companies should nonetheless commit to transitioning those vehicles toward CNG- or RNG-fueled vehicles. If the Companies decide to focus on acquiring CNG vehicles in the near term, it should develop plans to transition those vehicles to RNG within 10 years.

EXCESSIVE EARNINGS ADJUSTMENT MECHANISMS

Q. WHAT ARE EARNINGS ADJUSTMENT MECHANISMS ("EAMS")?

A. In our understanding, earnings adjustment mechanisms, or EAMs, are positive-only shareholder incentives designed to align utility incentives with policy goals,

primarily the objectives of Reforming the Energy Vision, or REV, and carbon reduction.

Q. WHAT IS YOUR OPINION OF EAMS?

A. In general, the City is not opposed to providing a utility with shareholder incentives if it performs at an exceptional level. However, incentives need to be designed solely or primarily to reward exceptional performance, not standard or expected behavior, and the amount must be measured and reasonable to avoid unduly burdening customers. As an example, in an Order issued on April 24, 2019 in Cases 16-M-0429 and 14-M-0101, related to interconnection EAMs, the Commission specifically indicated that an EAM should be "an effective measure of exceptional utility performance." We also believe that EAMs should be tied to some demonstrable action taken by the utility, and that utility shareholders should not be rewarded for actions taken solely by third parties, or for circumstances over which the utilities have no control.

We also note that the fact that an EAM, inclusive of its shareholder incentives, may have a cost-benefit ratio in excess of 1.0 is not sufficient to demonstrate exceptional performance or warrant payment of shareholder incentives. The size of the incentives, their impact on customers, and the quality of the performance leading to the outcome must be considered.

Q. HAVE YOU REVIEWED NATIONAL GRID'S EAM PROPOSALS?

A. Yes.

Q. WHAT EAMS DO THE COMPANIES PROPOSE?

A. The Companies propose three EAMs: (i) System Efficiency, split into Peak Reduction and Supply Diversification metrics; (ii) Energy Efficiency, split into Incremental Energy Efficiency and Affordability metrics; and (iii) Carbon Reduction, split into a Full Service metric and a Transportation metric.

Q. DO YOU HAVE ANY GENERAL CONCERNS ABOUT THE PROPOSED EAMS?

A. Yes. We are concerned that, as currently proposed, National Grid's EAMs are excessive in terms of absolute dollar values. Moreover, the City does not believe that actions taken by third parties should form the basis for shareholder incentives, or that utilities should be rewarded when it is unclear if they are taking any actions to achieve specific policy-oriented results. We also are concerned that some of the proposals may be duplicative, and that the Companies potentially could be rewarded multiple times (through different EAMs) for a single occurrence or achievement.

Q. PLEASE EXPLAIN WHY YOU BELIEVE THE COMPANIES' CURRENT EAM PROPOSALS ARE EXCESSIVE.

A. According to the Companies' responses to City IRs 65 and 66, KEDNY and KEDLI could potentially earn at least $25.75 million and $16.42 million, respectively, in 2020 if they achieve the maximum possible award for each of their proposed EAMs. These incentive levels jump to $30.04 million for KEDNY and $19.55 million for KEDLI by 2023. We say "at least" because these amounts do not appear to include any potential earnings associated with the Companies' proposed

Platform Service Revenues, the Non-Pipes Alternatives incentive mechanism, or any positive revenue adjustments proposed by the Companies' Gas Safety Panel, which we do not address in this testimony.

To put this request into perspective, KEDNY could potentially reward its shareholders an amount that is larger than the proposed rate allowance for the utility's low income program during the Rate Year, and KEDLI could reward its shareholders an amount that is triple the Rate Year rate allowance for its low income program. We believe that the Companies' low income customers are more in need of additional benefits than their shareholders.

Q. DO YOU HAVE ANY CONCERNS ABOUT SPECIFIC EAMS?

A. Yes. We have concerns about the proposed Affordability metric, the Carbon Reduction metrics, and the Peak Reduction metric.

Q. PLEASE EXPLAIN YOUR CONCERNS WITH THE AFFORDABILITY METRIC.

A. Under the Affordability metric of the Energy Efficiency EAM, the Companies could earn up to ten 10 basis points for a two percent reduction in the average low-income gas heating customer's bill (or one basis point for every 0.2 percent reduction). In their response to DPS IR-262, the Companies reported that in calendar year 2018, the average bill for a low income heating customer was $125.63 in the KEDNY service territory, and $115.21 in the KEDLI service territory. Thus, for achieving a two percent reduction in those bills ($2.51 and $2.30, respectively),

KEDNY would earn $3.3 million in shareholder incentives and KEDLI would earn $2.1 million in shareholder incentives.

Moreover, according to their response to DPS IR-262, the Companies define "low-income" for the purposes of the Affordability metric as those gas heating customers enrolled in the Energy Affordability Program ("EAP"). In response to DPS IRs 260 and 261, KEDNY reported an average of 62,696 heating customers enrolled in its EAP during calendar year 2018, and KEDLI reported an average of 10,864 heating customers enrolled in its EAP.

Based on the foregoing 2018 figures, as proposed, assuming the Companies achieve the maximum possible savings under this metric (i.e., KEDNY's and KEDLI's EAP heating customers save an average of $2.51 and $2.30 per month, respectively), KEDNY's shareholders would receive $1.75 in incentives for every dollar that an average low-income customer saves during the year, while KEDLI's shareholders would receive $7.00 for every dollar that an average low-income customer saves during the year. These amounts were calculated by multiplying the monthly two percent reduction by 12 to get annual bill savings for an average EAP heating customer, then multiplying annual bill savings by total EAP heating customers to get total estimated annual savings for all low income heating customers, and then dividing total estimated annual savings into the maximum possible shareholder incentive. This disparity in shareholder rewards as compared to the corresponding benefits to low income customers is highly inequitable, and

demonstrates that National Grid's proposed Affordability metric needs substantial reworking.

Q. DOES THE AFFORDABILITY METRIC MEASURE ACTUAL PERFORMANCE BY THE COMPANIES?

A. The answer to this is unclear. The Future of Heat Panel's direct testimony does not specifically elaborate on the precise actions that the Companies intend to take to reduce low-income bills. Rather, the Panel explained on page 106 of its pre-filed Direct Testimony that this metric "incentivizes the Companies for programmatic efforts to reduce low-income customers' energy usage and for animating the market for third parties to likewise provide low-income benefits within the Companies' service territories." As this metric is calculated before any low income discounts are factored in (according to the Companies' response to DPS IR-262), achievement of the metric's targeted bill reduction appear to be independent of the Companies' existing low income programs.

The Future of Heat Panel also did not state whether the Affordability metric would account for factors aside from weather, so it is unclear to us if the Companies' shareholders would be allowed to earn an incentive if average low income customer bills are reduced by, for example, decreases in the cost of natural gas, reductions in usage from ratepayer-funded energy efficiency programs, or actions of third parties unrelated to the Companies' efforts. We note that the Companies have set the minimum baseline for this metric at one basis point for a 0.2 percent reduction of average low income customer bills, or a roughly 25 cent

reduction in average bills, which conceivably could occur as a result of factors outside of the Companies' control.

Q. WHAT IS YOUR RECOMMENDATION?

A. We recommend that the Affordability metric be rejected at this time. As described above, we have numerous concerns with this proposal, including: (i) the size of the shareholder incentive is disproportionate to the benefits derived by low income customers; (ii) the metric can be achieved without any effort by National Grid (*i.e.*, through gas cost reductions or third-party actions); (iii) there is a potential double-count if savings are achieved through ratepayer funded energy efficiency efforts (for which KEDNY and KEDLI are also seeking an EAM); and (iv) it is uncertain how performance will be measured under this metric.

EAMs should be designed to incentive exceptional utility behaviors and outcomes. With the Affordability metric, the Companies could potentially take no action and still receive a windfall, which is borne by customers. This is an unreasonable outcome.

Q. DOES THE CITY SUPPORT OTHER MEASURES TO ASSIST LOW INCOME CUSTOMERS?

A. Yes, absolutely. The City is committed to reducing the energy burden faced by its low income residents. That is precisely why the City's Low Income Panel is proposing to increase the size of the current low income discounts to amounts that are designed to satisfy the Commission's policy of achieving an energy cost burden of no more than six percent of a customer's income. We understand that

the City's Low Income Panel is making several other recommendations to improve the Companies' low income programs, including more frequent file matches with the NYC Human Resources Administration and elimination of reconnection fees for low-income KEDNY customers.

We contend that resources for low income customers are better directed to these efforts, rather than to shareholder incentives. In fact, for KEDNY and KEDLI, there is a combined $5.4 million in proposed shareholder incentives for the Affordability metric. These dollars would be better utilized as part of direct funding increases for existing low income programs.

Q. DO YOU HAVE CONCERNS WITH THE CARBON REDUCTION METRIC?

A. Yes. As proposed, the Carbon Reduction metric is composed of two metrics: (i) the Full-Service CO_{2e} reduction metric; and (ii) the Transportation CO_{2e} reduction metric. We have concerns with both metrics, which we address in turn.

Q. PLEASE EXPLAIN YOUR CONCERN WITH THE FULL-SERVICE CO_{2e} REDUCTION METRICS.

A. We are struggling to differentiate among the Supply Diversification EAM and the Full-Service CO_{2e} Reduction EAM, which appear as if they both could be triggered by the same outcomes. According to the Companies' response to City IR-95, the Supply Diversification EAM measures "the volume of RNG injected into the system," whereas the Full-Service CO_{2e} Reduction EAM "measures the amount of RNG purchased by customers of the Companies." In other words, the Companies

potentially could receive a shareholder incentive for the total supply of RNG that is injected into the gas system, and then again for the amount of that supply purchased by customers through a proposed Green Gas Tariff. This proposal is not reasonable and should be corrected.

Q. SHOULD THE UTILITY BE REWARDED TWICE FOR THE SAME ACHIEVEMENTS?

A. No. EAMs should be designed to reward utilities for achieving exceptional, discrete outcomes, but not in a manner where a single measure (for example, an energy efficiency measure) triggers multiple incentives for the Companies. Accordingly, we recommend rejecting the Full-Service CO_{2e} Reduction EAM.

Q. PLEASE EXPLAIN YOUR CONCERN WITH THE TRANSPORTATION CO_{2E} REDUCTION METRIC.

A. The metric would reward shareholders for achieving certain CO_{2e} reduction targets associated with switching vehicles from diesel or gasoline to CNG/RNG. As discussed herein, we are proposing that the Companies significantly increase their investment in vehicles that run either on electricity or RNG. These investments, if adopted, would result in overlap between ongoing, ratepayer funded efforts and the shareholder incentives proposed under the Transportation CO_{2e} reduction metric.

This proposal also raises a concern about potential double-counting with the Supply Diversification EAM, which measures "the volume of RNG injected into the system." The Future of Heat Panel notes on page 112 of its pre-filed Direct Testimony that, to the extent a fueling "station signs up to use RNG for a portion

of its load, the Companies will also incorporate those savings into the CO_{2e} calculation." In other words, the Companies are proposing to incentivize shareholders for both the injection of RNG into its system and the use of that RNG by a vehicle fueling station.

Q. WHAT DO YOU RECOMMEND?

A. Based on the concerns identified above, we recommend rejecting the full Carbon Reduction metric as proposed. This is not to say, however, that we are opposed to incentives that promote incremental RNG and geothermal deployments by the Companies. While we do not have a specific metric to offer in place of the Carbon Metric reduction, at a minimum any such metric should: (1) reward the utility for superior performance, not business-as-usual behavior; (2) require that the outcomes are driven by utility action, as opposed to actions by third parties; (3) avoid double-counting with other incentives or metrics; and (4) be readily measurable and verifiable.

Q. PLEASE EXPLAIN YOUR CONCERNS WITH THE PEAK REDUCTION METRIC.

A. In the Peak Reduction EAM, the Companies set the baseline for earning an incentive at the forecasted peak reduction under their DRD Project. The Companies already are asking for a combined $2 million for the Project, per Schedule 5 of Exhibit__(FOH-1). If the project achieves its intended purposes – purposes for which National Grid proposes that customers pay for – it would be

entitled to an additional combined $543,000. This amounts to rewarding National Grid for what it is being paid to do.

Q. WHY IS IT PROBLEMATIC TO SET THE BASELINE FOR EARNING AN INCENTIVE AT THE FORECASTED PEAK REDUCTION UNDER THE DRD PROJECT?

A. As we noted above, EAMs should be designed to reward superior performance by a utility, not business-as-usual performance. However, by using the proposed minimum targets of program to establish the minimum threshold for the Companies to earn an EAM on that program, the Companies essentially are rewarding themselves for doing what they would have done anyway.

Q. DO YOU HAVE ANY OTHER COMMENTS ABOUT THE COMPANIES' PROPOSED EAMS?

A. Yes. We note that not all of the proposed EAMs and metrics appear to be rewarding the Companies for business-as-usual performance, or for occurrences out of their control. For example, as explained in the Future of Heat Panel's direct testimony, the Companies are proposing shareholder incentives for incremental MMBtu reductions achieved through their expanded energy efficiency programs, as directed by the Commission in the Energy Efficiency Order. Importantly, the Companies set the minimum threshold for their respective Incremental Energy Efficiency metrics equal to the targets set in the Energy Efficiency Order, and assigned them basis points earnings opportunities of *zero.* In other words, the Companies have appropriately identified the Energy Efficiency Order targets as

baseline targets, and only are proposing to reward themselves for achieving energy efficiency savings above those targets. The Companies should endeavor to design all of their EAMs in such a manner.

Despite our agreement with the *design* of shareholder incentives for the Energy Efficiency EAM, we do not agree with the actual target levels proposed by the Companies for their Energy Efficiency EAM, as discussed in more detail below.

Q. WHAT DO THE COMPANIES PROPOSE FOR THEIR ENERGY EFFICIENCY REDUCTION METRIC?

A. Per their responses to DPS IR-387 and City IR-92, the Companies propose a minimum threshold for their Energy Efficiency metric based on targets set forth in the Energy Efficiency Order, including incremental savings targets. The minimum target is then assigned zero basis points. The maximum target was developed by assuming cost efficiencies equal to each Company's respective lowest average historical run rate from the last four years (adjusted for anomalies and to remove behavioral program results). The mid-point target for the metric is set in between these two points.

Q. DO YOU AGREE WITH THE PROPOSED ENERGY EFFICIENCY METRIC TARGETS?

A. No. As noted above, over the last several years, the unit costs for energy efficiency savings have declined. For example, KEDNY spent an average of $1.64 per therm of energy efficiency savings in 2018, while KEDLI spent an average of $1.61 over

the same period. The Companies should be lauded for improving the effectiveness of their respective energy efficiency programs over time. However, by the same token, new shareholder incentives should encourage the Companies to build upon these successes, not to reward them for continuing business-as-usual. Here, the Companies could each potentially earn 40 basis points for achieving cost efficiencies that they already have achieved in recent years.

Q. HOW SHOULD THE ENERGY EFFICIENCY METRIC TARGETS BE SET?

A. We recommend that the Companies set their minimum targets by assuming cost efficiencies equal to each Company's average historical actual run rate over the last four years. In other words, the current proposed maximum targets should instead be set as the minimum targets. Midpoint and maximum targets should then be set at 10% and 20% savings above the minimum, respectively. This appropriately incentivizes the Companies to strive for superior performance, as opposed to rewarding them for business-as-usual performance.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of:

New York City Low Income Panel:

Nicholas Patané
Christine D'Onofrio
Natasha Godby

On Behalf of:

The City of New York

August 30, 2019

Cases: 19-G-0309 NYC LOW INCOME PANEL
19-G-0310

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION, AND BUSINESS ADDRESS.

A. Mr. Patané – My name is Nicholas Patané. I am an Energy Policy Advisor in the New York City Mayor's Office of Resilience and Mayor's Office of Sustainability. My business address is 253 Broadway, 14th Floor, New York, New York.

Ms. Godby – My name is Natasha Le Godby. I am an Acting Deputy Commissioner for Emergency and Intervention Services for the New York City Human Resources Administration, or HRA. My business address is 150 Greenwich Street, New York, New York.

Ms. D'Onofrio – My name is Christine D'Onofrio. I am the Director of Poverty Research for the New York City Center for Economic Opportunity. My business address is 253 Broadway, 10th Floor, New York, New York.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. The City of New York.

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. Patané – I received a Bachelor's Degree in finance from Georgetown University in 2012 and a Master's Degree in Public Affairs from Princeton University in 2018.

Ms. Godby – I received a Bachelor's Degree in Liberal Arts from CUNY John Jay College of Criminal Justice in 1997 and a Juris Doctorate Degree from the City University of New York in 2002.

Ms. D'Onofrio – I received a Master's Degree and a Ph.D in labor economics from the New School for Social Research.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Mr. Patané – I have worked in the Mayor's Office of Resilience and Mayor's Office of Sustainability for one year. My prior work experience includes two years as a strategist at the New York City Economic Development Corporation and two years of experience as a consultant at Accenture.

Ms. Godby – Upon graduation from law school, I commenced my professional employment as an attorney with the New York City Law Department, where I represented the City in tort actions commenced in New York State Courts. I served as Deputy Public Administrator of Kings County where I managed the estate matters of residents that died intestate. In 2009 I transferred to HRA. I have worked at HRA for 10 years in positions of increasing authority, including Deputy General Counsel, where I provided legal counsel to the Office of Child Support Services and managed attorneys appearing in New York State family court to establish and enforce support orders. I also provided legal counsel to the Investigation, Revenue and Enforcement Administration and managed attorneys appearing in third party liability actions to enforce Medicaid and public assistance liens against personal injury and estate assets of recipients of social services. Now in my present position as Acting Deputy Commissioner of Emergency and Intervention Services, my responsibilities entail managing the Home Energy Assistance Program, or HEAP,

and the Emergency Utility Intervention Program, or EUIP, serving low income New Yorkers at risk for utility disconnections.

Ms. D'Onofrio – I have worked in the Mayor's Office for Economic Opportunity for 11 years. Much of my work involves continuing improvement of the New York City poverty measure, testing the impact of new policies such as the minimum wage or child tax credits on the poverty rate, and advising on the use of this data by multiple agencies. I was instrumental in developing a model that was subsequently adopted by the U.S. Census Bureau as the base for the U.S. Supplemental Poverty Measure. I began my work with the Poverty Research Unit developing a model to simulate tax burden and tax credits using census households as the base population. I have had positions with increasing responsibility and presently am Director of the Poverty Research Unit.

Previously, I was a Senior Legislative Research Analyst for the New York City Council, Division of Finance for over six years. I was responsible for forecasting income tax revenues and modeling the impact of tax changes on families. This work required depth of knowledge about the national and New York City macro economy, income distribution, incidence of taxation and issues of tax equity in the federal, state, and city tax codes. I also was responsible for modeling proposed changes to the City tax code, and I developed the model for the New York City Earned Income Tax Credit. I was part of the team that developed a model to estimate the impact on the city economy and city tax revenues after the World Trade Center attacks in 2001.

I also am a part-time faculty member at the School of Public Engagement at the New School, where I lecture on poverty and the family. I frequently present the results of our poverty research to communities of interest, government agencies, and in academic settings.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Mr. Patané – I previously testified before the Public Service Commission, or Commission, in Cases 19-E-0065 and 19-G-0066, the Consolidated Edison Company of New York, Inc., or Con Edison, rate cases, on behalf of the City. My testimony in those cases similarly focused on low income issues.

Ms. D'Onofrio – I likewise testified in the Con Edison rate cases on behalf of the City.

Ms. Godby - I also testified in the Con Edison rate cases on behalf of the City.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. Our testimony examines the issue of energy cost burden in New York City, and the estimated effects on affordability of an increase in rates proposed by the Brooklyn Union Gas Company d/b/a National Grid NY, or KEDNY, and KeySpan Gas East Corp. d/b/a National Grid, or KEDLI. KEDNY and KEDLI are occasionally referenced in this testimony as National Grid or the Companies. Specifically, we recommend that KEDNY and KEDLI increase the bill discount

amounts provided to low income customers as part of their Energy Affordability Program, or EAP, to reduce the cost burden of New York City's low income residents and realize the State's energy affordability targets. We also offer recommendations on KEDNY's gas service reconnection fees, and on National Grid's existing file-matching program with HRA.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. We offer Exhibit__(LIP-1), which is a compilation of the discovery responses from the Companies that we relied upon in preparing our testimony, some which we expressly reference herein.

Exhibit__(LIP-2) shows the Federal and State poverty thresholds we utilized as part of our energy cost burden analysis.

Exhibit__(LIP-3) shows the results of our energy and gas cost burden analysis.

Exhibit__(LIP-4) is the City's modifications to the National Grid Low Income Workbooks, which were provided by the Companies as Attachment 1 and Attachment 2 to their response to City Interrogatory Request ("IR") 79.

Exhibit__(LIP-5) shows our estimates of the potential bill impacts to other customers if our recommendations for increased low income discounts are adopted.

HRA AND UTILITY LOW INCOME PROGRAMS

Q. WHAT DOES HRA DO?

A. HRA/Department of Social Services is the nation's largest social services agency dedicated to fighting poverty and income inequality by providing New Yorkers in need with essential benefits. HRA serves more than three million New Yorkers

through diverse programs and services such as emergency rental and utility assistance, cash assistance, public health insurance, job placement and training, and emergency/ongoing food assistance, shelter, and housing subsidies.

Q. WHAT TYPE OF ENERGY ASSISTANCE DOES HRA PROVIDE?

A. HRA administers benefits through HEAP, and assists low income homeowners and renters in paying bills for electricity, heating fuel, energy equipment, and related repairs and replacements. HEAP is a federally funded program that provides eligible households with grants to pay regular and emergency energy costs for heating and cooling.

HRA also administers the EUIP, which assists individuals and families who are elderly, blind, disabled, mentally impaired, and/or residing in a neglected or hazardous environment who require financial assistance for their energy bills, primarily in cases where customers are faced with termination of services due to nonpayment, theft of service, or meter tampering. The program also responds to emergencies where the individual is at risk for termination of services and life sustaining medical equipment is present in the household. EUIP is a state-mandated program, and HRA is the designated liaison between these individuals and families and the utility companies.

Q. DOES NATIONAL GRID PROVIDE ANY FUNDING FOR THE HEAP OR EUIP PROGRAMS?

A. No.

Q. DOES NATIONAL GRID ASSIST HRA IN ADMINISTERING THE HEAP OR EUIP PROGRAMS?

A. No, but the Companies provide referrals to HRA and notify HRA of situations covered by the EUIP.

LOW INCOME PROGRAM STRUCTURE

Q. ARE YOU GENERALLY FAMILIAR WITH THE TIERED STRUCTURE OF NATIONAL GRID'S LOW INCOME PROGRAMS?

A. Yes. We understand that Tier 1 includes National Grid customers who receive assistance from HRA, and Tiers 2 and 3 include National Grid customers who receive one or two HEAP adders. Tier 4 includes National Grid customers whose utility bills are paid via direct vouchers by HRA, or for whom HRA guarantees payment to the Companies. The Companies also currently have a Tier 5 for KEDLI, and Tiers 5 and 6 for KEDNY, which they propose to fold into each utility's respective Tier 1 category. We also understand that these tiers were established by the Commission over the objections and concerns expressed by the City and others.

Q. DO YOU HAVE ANY CONCERNS WITH THIS PROGRAM STRUCTURE?

A. Yes. The program structure was designed to achieve an energy cost burden of no more than six percent of a customer's income. Based on the analysis we performed, we can now demonstrate that the program structure falls far short of achieving that goal. Additionally, it is our understanding that the analysis upon which the current program structure was based assumes a family of two persons earning at the

maximum threshold 60 percent of state median income, or 60 SMI. However, many low income customers have earnings below that level, meaning they require more assistance in order to remain within the energy cost burden threshold of six percent of total income spent on energy. The analysis also appears to assume that every low income customer receives a full HEAP discount, and that each such customer pays out-of-pocket for heat. However, HEAP funding each year is finite, and not every eligible person receives a full, or partial, HEAP payment. As discussed below, many low income customers in New York City also do not pay directly for heat and do not receive full HEAP benefits. Further, the receipt of one or two HEAP adders can be unrelated to need, and the tiered approach based on receipt of HEAP adders is unlikely to provide the most assistance to the people with the highest level of need.

Q. WHAT DO YOU RECOMMEND?

A. We strongly urge the Public Service Commission to revisit the low income program structure and revise it to better help achieve its goal of limiting the energy cost burdens of all New Yorkers to six percent of their total income.

ENERGY COST BURDEN ANALYSIS

Q. HAS THE PANEL PERFORMED AN ANALYSIS OF THE ENERGY COST BURDEN ON UTILITY CUSTOMERS WITHIN NEW YORK CITY?

A. Yes. We have performed an analysis of the energy cost burden on customers whose income falls below certain levels. Our analysis and conclusions are summarized in Exhibit__(LIP-1).

Q. PLEASE DESCRIBE YOUR ENERGY COST BURDEN ANALYSIS.

A. We performed a quantitative analysis of the economic burden placed on families from their electric and gas utility bills, with a particular focus on low income New York City residents. We compared data on actual energy cost burdens to the policy the Commission set on page 3 of its May 20, 2016 *Order Adopting Low Income Program Modifications and Directing Utility Filings* in Case 14-M-0565, or Low Income Order, that "an energy burden at or below six percent of household income shall be the target level for all 2.3 million low income households in New York." That is, New Yorkers should pay no more than six percent of their household income for utility expenses. Additionally, based on the Commission's guidance that gas bills should account for no more than half of utility burden, we compared gas costs against a three percent affordability threshold to understand how gas bills impact energy cost burden.[1] We are using the results from that analysis to support our proposal to provide more assistance to those families in New York City whose energy burdens exceed – in some cases significantly – the Commission's affordability targets.

Q. WHO PERFORMED THE ANALYSIS?

[1] *See* Low Income Order at 23 ("Gas service (whether designated as heating or non-heating) is discounted to one-half of a customer's total home energy burden.").

A. Mr. Patané and Ms. D'Onofrio performed the analysis with assistance from others within the New York City Mayor's Office of Sustainability and the Mayor's Office of Economic Opportunity.

Q. ON WHAT DATA IS THE ANALYSIS BASED?

A. The analysis started with the publicly available 2017 Public Use Micro Sample, which is part of the American Community Survey, or ACS. The survey is conducted by the U.S. Census Bureau on an ongoing basis and asks a representative sample of Americans questions related to income, housing, and other topics. The ACS is designed to provide measures of socioeconomic conditions on an annual basis for states and larger localities. It offers a statistically representative sample for New York City. As discussed in its general handbook, the ACS uses a weighting method to ensure that its estimates are consistent with official Census Bureau population estimates by age, sex, race, and Hispanic origin, as well as estimates of total housing units.

The ACS is a standard tool for measuring need. According to the Census Bureau, information from the ACS generates data that helps determine how more than $675 billion in federal and state funds are distributed each year. In questions related to housing, the ACS asks respondents to report the cost incurred for electricity and gas in the previous month and the cost incurred for fuel oil or kerosene in the past 12 months. Because costs are reported as incurred, they are presumed to be net of any direct utility discounts. Also reported in the ACS is

income from wages, transfer payments, self-employment, retirement income, and investment.

Q. HOW DID YOU DEFINE THE TERM "ENERGY COST BURDEN" FOR PURPOSES OF YOUR ANALYSIS?

A. For our analysis, we defined energy cost burden as the sum of (1) reported out-of-pocket monthly natural gas expenses multiplied by 12 plus (2) reported out-of-pocket monthly electricity expenses multiplied by 12 plus (3) reported annual other energy expenses, such as fuel oil or kerosene purchases, minus an estimated HEAP benefit, divided by annual pre-tax income. Tax credits and non-cash subsidies are not included in the income calculation. A family is determined to be energy cost burdened if its energy cost burden is greater than six percent. Six percent is both the State target and a target level used by energy affordability experts, including, for example, Fisher, Sheehan & Colton Public Finance and General Economics, who report a Home Energy Affordability Gap and whose work was relied upon by Department Staff and the Commission in establishing the State's target.

Q. HOW DID YOU DEFINE THE TERM "GAS COST BURDEN" FOR PURPOSES OF YOUR ANALYSIS?

We defined gas cost burden as reported out-of-pocket monthly gas expenses multiplied by 12 minus a HEAP estimate divided by the same pre-tax income used for energy cost burden. A family is considered gas cost burdened if their gas cost burden exceeds three percent (half of the total electric plus gas affordability

target). This is in line with the guidance established by the Commission for implementing the low income utility discounts.

Q. PLEASE DESCRIBE YOUR METHODOLOGY FOR CALCULATING A HEAP ESTIMATE.

A. To calculate a HEAP estimate, we took the average New York City non-emergency HEAP benefit paid for the 2017 federal fiscal year, as reported in the New York State Office of Temporary and Disability Assistance's Temporary and Disability Assistance Statistics released in September 2017. This yields an average HEAP benefit of $46.50. This comparatively low amount reflects the fact that most NYC residents do not receive a full HEAP benefit because many are renters and do not directly pay for their heating bills. We deducted this HEAP estimate from both gas heat and gas non-heat customers; however, we believe this to be a conservative approach that underestimates energy cost burden and gas cost burden for gas heat customers. Since gas non-heat customers typically receive their HEAP benefit directly, we adjust reported gas expenses to reflect this unreported HEAP benefit. However, gas heat customers who are eligible for HEAP typically have their HEAP benefit paid directly to the utility who applies the discount to their bills. Therefore, the gas heat bills reported to the ACS would likely already reflect the HEAP deduction.

By deducting an additional $46.50 from gas heat customers' annual bills, we are likely inflating the amount of HEAP benefits they receive since their reported annual bills should already be net of the full HEAP benefit they receive

which is typically over $300. However, we applied the average HEAP benefit to all customers as a conservative estimate.

Q. HOW DO YOU DETERMINE WHETHER A CUSTOMER IS GAS HEAT OR GAS NON-HEAT?

A. The ACS data collects information on household gas expenses, but does not distinguish the use of that gas. Therefore, to approximate whether a customer is a gas heat or a gas non-heat customer, we applied a monthly bill threshold, above which a customer is likely a gas heat customer and below which a customer is likely a gas non-heat customer. To do this, we created a histogram of reported utility bills and identified that a cutoff of $40 most accurately reflects the number of heat versus non-heat customers reported in KEDNY Exhibit__(RDP-2CU) (this exhibit does not distinguish between low-income and other customers). We believe this is a reasonable cutoff. For a KEDNY residential heating customer to pay a total bill of $40 per month, under National Grid's proposed rates, the customer would have to use only 10 therms per month. This usage level is eight times below average, suggesting that not many residential heating customers are paying bills this low. Similarly, the average KEDNY residential non-heat customer uses 10 therms per month for a projected total bill of $41.69 under proposed rates, which adds additional credence to our cutoff.[2]

[2] *See* KEDNY Exhibit___(RDP-4CU), Schedule 4, pp. 1, 2.

Q. WHAT THRESHOLDS DID YOU USE IN THE ANALYSIS TO CONSIDER A FAMILY AS "LOW INCOME?"

A. We use a threshold of 60 percent of State median income, or 60 SMI, to determine whether a family is low income. 60 SMI is the maximum income eligibility for HEAP, which is the assistance program with the highest income cutoff that enables participation in National Grid's low income programs, and is also the income threshold utilized in the Low Income Order. Low income thresholds, but not National Grid's low income programs, are adjusted for family size.

The thresholds used for SMI are included in Exhibit__(LIP-2). The Federal thresholds identified in the Exhibit are as reported by the Federal Department of Health and Human Services. State thresholds are those used for the 2016-2017 HEAP season. All thresholds used are those from the same year in which the ACS data was reported, thus corresponding to economic conditions and costs in the year the survey was conducted.

The analysis is conducted at the family level. A family is defined as a resource sharing unit composed of related individuals and domestic partners. This is different from a household, since multiple families, subfamilies, or unrelated individuals may reside in the same apartment, home, or other housing unit. Families within a housing unit are defined using the methodology the Mayor's Office of Economic Opportunity developed for the New York City poverty measure using guidelines the National Academy of Sciences suggests, and

comparable to the method the U.S. Census Bureau uses in estimating their Supplemental Poverty Measure.

Q. WHY WAS THE ANALYSIS PERFORMED AT THE FAMILY LEVEL?

A. Poverty levels are defined based on family size and income, not household income. Therefore, to properly evaluate energy costs against the low income thresholds noted in Exhibit__(LIP-2), the family units created for the New York City poverty measure are used. We note that this differs from the Commission's usage of households to determine energy cost burden. In our analysis, more than one family unit in a household would mean that we identify in the census data more than one "resource sharing unit" in that household. For example, a group of family members plus an unrelated couple who are identified as boarders would be considered two families under our analysis, and utility costs would apportioned out to each family according to size. In other words, when there are multiple families in a household, our analysis would show that each is paying less in costs than the combined household and those costs are held relative to each family's income to determine their cost burden. Our understanding is that the vast majority of households in the New York City are single-family households, and so we expect the implication of performing our analysis at the family level (as opposed to the household level) to be minor.

Q. PLEASE EXPLAIN HOW YOU DEFINE THE NATIONAL GRID SERVICE TERRITORY?

A. Our analysis focuses only on New York City customers. For our analysis, we only use responses from respondents living in Staten Island, Brooklyn, and all Queens Community Districts, except Districts 1, 2, 7, 8, and 11, which are assumed to be Con Edison gas customers. Community districts are the most granular geographic division we can make that balances available information with the need for sufficient responses for statistical significance.

We understand that, by only using this subset of data, we essentially are limiting our analysis to the KEDNY service territory, plus the Rockaways portion of the KEDLI service territory.

Q. PLEASE EXPLAIN HOW YOU DEFINE A NATIONAL GRID CUSTOMER?

A. For our analysis, respondents who live in the areas that we defined above and who report directly paying some amount greater than zero for gas are considered National Grid customers.

Q. WHAT DOES THE PANEL MEAN BY DIRECTLY PAYING FOR UTILITY SERVICE?

A. Within New York City, many families with income levels below 60 SMI live in apartment houses or other dwellings in which heat or heat and electricity are provided by the landlord. This includes master-metered buildings. The utility costs for these families may be included in their rent payments or in bills issued by their landlords rather than by Con Edison or National Grid. The numbers presented in our analysis are for families who pay some amount greater than $0

for utility bills. We were not able to differentiate whether those bills are paid directly to Con Edison and National Grid or to some other party, such as energy service companies. Similarly, directly paying for gas implies that the family is paying some amount greater than $0 for gas bills.

Q. PLEASE EXPLAIN THE PROCESS FOR CALCULATING ENERGY COST BURDEN AND GAS COST BURDEN.

A. Our analysis was conducted in R. R is a programming language for statistical computing and data visualization supported by the R Foundation for Statistical Computing. First, we cleaned the data to eliminate group quarters and responses from outside New York City from the base population, and superimposed family units on the census households. Group quarters include facilities like prisons, college dorms, nursing homes, and homeless shelters where costs are typically shared over the institution and are not billed directly to tenants or residents.

Next, we created variables to represent annual utility expenses. For fuel oil, the number reported in the ACS is already annualized. For electric and gas, we multiplied the recorded response by 12.

We then sorted families into income bands, by comparing family income against different income ranges. We then created a variable for all utilities, which is the sum of these values, minus a HEAP estimate. The HEAP estimate we used is the average non-emergency benefit paid to New York City residents in the 2017 HEAP season, or $46.50. Then, we created a variable called “utility burden” which divides the sum of all utilities minus the HEAP estimate, by pre-tax income

for each family. If that ratio is greater than six percent, the family is considered energy cost burdened. Similarly, we created a variable called "gas burden" which is the annualized gas costs minus the estimated HEAP benefit, divided by pre-tax income. If that ratio is greater than three percent, the family is considered gas cost burdened.

Next we applied filters that restrict the data set to National Grid customers. First, we identified respondents whose reported natural gas bill greater than $0. Then we identified respondents who live in the National Grid service territory, as described above.

Following this step, we applied a set of gas discounts and cost increases that reflect hypothetical discount scenarios, and estimated bill increases from the pending rate case. We did this by adding the annualized gas increase or decrease to the annualized gas bill and recalculating whether the gas cost burden is over three percent.

Finally, we generated tables for gas heat and gas non-heat customers that show the different impacts of the hypothetical cost increases or decreases and how these scenarios change the number of families that are gas cost burdened using the 60 SMI threshold.

Q. WHAT ARE YOUR FINDINGS?

A. Schedule 1 of Exhibit__(LIP-3) shows the existing energy and gas cost burden and how the 2017 low income energy cost burden landscape would change under various adjustments to low income customers' utility bills. There are six tables in

Schedule 1. Each table has a similar structure but represents a different look at the data. Table 1 looks at energy burden for all customers. Table 2 looks at gas burden for all customers. Table 3 looks at energy cost burden for non-heat gas customers. Table 4 looks at gas cost burden for non-heat gas customers. Table 5 looks at energy cost burden for heating gas customers. Table 6 looks at gas cost burden for heating gas customers.

As shown in Table 1 of Schedule 1 of Exhibit__(LIP-3), in the National Grid service territory, there are 220,470 families living under 60 SMI who pay more than six percent of their income toward their energy bills. This number represents 18 percent of all customers in the National Grid service territory and 55 percent of all National Grid customers under 60 SMI. Additionally, in the National Grid service territory, there are 148,168 families living under 60 SMI who pay more than three percent of their income toward their gas bills (Table 2). This number represents 12 percent of all customers in the National Grid service territory and 37 percent of all customers in the National Grid service territory under 60 SMI. These results demonstrate that National Grid's existing low income discounts are not sufficient to satisfy the Commission's goal that no low income customer pays more than six percent of income to energy costs, or more than three percent of income to gas costs.

We also found that gas cost burden disproportionately impacts low income families. Only 33 percent of all customers in the National Grid service territory have income levels under 60 SMI, but low income customers represent 68 percent

of gas cost burdened families. Put another way, 37 percent of all low income families are gas cost burdened; in contrast, only 9 percent of non-low-income families are gas cost burdened.

We also see that both energy and gas cost burden are more acute for gas heat customers. As seen in Table 4 of Schedule 1, 15 percent of low income non-heat gas customers are gas cost burdened, whereas over 60 percent of gas heat low income customers are gas cost burdened, as shown in Table 6 of Schedule 1. Said differently, of the 141,168 National Grid low income customers who are gas cost burdened, 22 percent are non-heat gas customers and 78 percent are gas heat customers. Looking to Tables 3 and 5, whereas 42 percent of low income gas non-heat customers are energy cost burdened, nearly 70 percent of low income gas heat customers are energy cost burdened. These findings highlight the need for greater focus on gas heat customers.

In the columns to the right in each table, we see the impacts of various hypothetical incremental discounts, that is, what happens if instead of rates increasing as proposed, monthly bills were discounted by some amount incremental to existing discounts. Looking at Table 6, we see that meeting the state target that no low income family should spend more than 3 percent of their income on gas would require significant discounts. Even with monthly discounts as high as $200 per month – which is not what we are proposing – there are still over 12,000 low income gas heat families who remain gas cost burdened (Table

6) and over 40,000 low income gas heat families who remain energy cost burdened (Table 5).

Q. WILL THE REVENUE REQUIREMENT INCREASE REQUESTED BY NATIONAL GRID IN THESE RATE CASES HAVE ANY EFFECT ON THE GAS COST BURDEN FOR LOW INCOME FAMILIES?

A. Yes. According to the Direct Testimony of National Grid's Shared Services Panel, KEDNY and KEDLI are proposing to maintain the same bill discounts that are in effect as of January 1, 2019, with the exception that Tiers 5 and 6 are being eliminated for both utilities. In other words, the available bill discounts remain the same as currently exist for all customers. However, KEDNY and KEDLI also are requesting rate increases from customers. Therefore, all low income customers participating in the Companies' low income programs would receive a proportionally lower discount than they receive now, and both gas cost burden and energy cost burden would worsen for low income customers.

Q. SHOULD THE RATES PAID BY LOW INCOME CUSTOMERS BE HELD CONSTANT AT THEIR CURRENT LEVELS?

A. No, we are not advocating that. However, by increasing gas delivery rates by 22 percent for an average residential reduced rate non-heat customer and 18 percent for an average residential reduced rate heat customer[3] while holding constant the

[3] *See* KEDNY Exhibit__(RDP-4CU), Sch. 4, pp. 3, 5.

low income discounts, low income customers will not receive the same proportionate benefit as they receive now.

Q. WHAT OTHER IMPACTS OF THE PROPOSED RATE INCREASES ON CUSTOMERS' ENERGY COST BURDENS ARE LIKELY?

A. In addition to low income customers not receiving the same proportionate benefit, the rate increase is likely to result in more customers experiencing an energy cost burden above six percent of their income and a gas cost burden above three percent.

Q. BASED ON YOUR ANALYSIS, HOW MANY MORE CUSTOMERS WILL BECOME GAS COST BURDENED AS A RESULT OF THE PROPOSED RATE INCREASES?

A. Our analysis finds that, unless the low income discounts are increased, 20,000 low income heat gas customers and 7,500 low income non-heat gas customers who are not gas cost burdened today would become gas cost burdened from the proposed rate increase.

Q. HOW DID YOU CALCULATE THIS RESULT?

A. Calculating the exact effect of the increased revenue requirement on customers' energy cost burdens is complicated by the fact that ACS does not track the energy usage of each low income family. Therefore, for the purposes of our analysis, we estimated the impact based on a typical gas non-heat customer and a typical gas heat customer.

In KEDNY Exhibit __ (RDP-4CU) Schedule 4, Pages 3 and 5 of 18, National Grid estimates an average residential KEDNY non-heat reduced rate customer would see a total bill increase of $4.82. Similarly, an average residential KEDNY heat reduced rate customer would see an average increase of $14.57. To estimate the impact of the proposed gas rate increases, we add these amounts to our analysis of gas heat and non-heat customer bills.

It should be noted that this approach will overestimate the impact on KEDLI heat customers and slightly underestimate the effect on KEDLI non-heat customers. As shown in KEDLI Exhibit__(RDP-4CU) Schedule 4, Pages 2 and 4 of 17, the average increase for a residential KEDLI reduced rate non-heat customer is $5.38, and the average increase for a residential KEDLI reduced rate heat customer is $8.49. We would have preferred to add the heat increases for KEDNY and KEDLI separately. However, we cannot maintain statistical significance by splitting KEDLI's New York City territory into a separate analysis. Given there are significantly more New York City KEDNY heat customers than New York City KEDLI heat customers, we believe using the KEDNY increases would be a reasonable approach for the majority of heat customers.

Q. PLEASE CONTINUE.

A. Based on our calculations, it appears that the proposed rate increases would have a significant negative impact on the energy cost burdens and gas cost burdens for low income New York City families. They will both increase the total number of

families that are energy cost burdened and increase the magnitude of the burden for those families who are already cost burdened.

Q. DO YOU BELIEVE THIS OUTCOME IS CONSISTENT WITH THE PUBLIC SERVICE COMMISSION'S INTENT AND GOAL IN ITS LOW INCOME ORDER IN CASE 14-M-0565?

A. No. Using the Public Service Commission's policy as guidance in this case, National Grid's proposal does not maintain the status quo; it makes the situation worse for more New York City families.

Q. BASED ON YOUR ANALYSIS, WHAT ARE YOUR RECOMMENDATIONS TO ADEQUATELY ADDRESS LOW INCOME ENERGY COST BURDEN?

A. We believe that bill discounts should be increased to reduce gas cost burdens for 50 percent of low income heat gas customers. While this is less ambitious than the current state target, we believe this would be a significant improvement over the current practice.

Q. WHAT APPROACH DID YOU USE TO CALCULATE DISCOUNTS NEEDED TO ADDRESS GAS COST BURDENS FOR HALF OF ALL LOW INCOME HEAT GAS CUSTOMERS?

A. To calculate the discounts needed to address gas cost burdens for half of all low income heat gas customers, and to remain consistent with the Generic low income order, we make adjustments to the National Grid Low Income Workbooks for KEDNY and KEDLI as shown in Exhibit__(LIP-4).

We first adjusted median incomes to reflect that the median income for New York City National Grid customers under 60 SMI and over 130 percent of the federal poverty level is $27,808 and median income for Tier 3 customers is $10,921 as found in the ACS data. This fixes a flawed assumption in the workbook structure. The original workbook bases discounts on the high end of the income threshold, that is, it assumes that all low income customers are at 60 SMI when many such customers earn less than 60 SMI. This creates an inherent assumption that many low income customers have more resources at their disposal than they actually do. Basing discounts on the highest income level will only address energy cost burden for families who make income that is equal to the high end threshold. To address gas burden for half of National Grid customers, a median income is required.

Second, we adjust the average bills used in the National Grid Low Income Workbook to reflect the proposed rate increases. Average KEDNY bills should be increased to reflect that gas heat bills would go up $14.57 for low income customers and gas non-heat bills would go up $4.82, and average KEDLI bills should be increased to reflect that KEDLI gas heat bills would go up $8.49 and KEDLI gas non-heat bills would go up $5.38 per Exhibit__(RDP-4CU).

Q. WHAT DISCOUNT LEVELS DID YOU FIND ARE NEEDED TO ADDRESS GAS COST BURDENS FOR HALF OF ALL LOW INCOME HEAT GAS CUSTOMERS?

A. Based on these changes described, we found that the resulting discounts needed to achieve energy affordability for low income customers are much higher than those currently in place. For KEDNY, the required discounts are $59 for Tiers 1, 5, and 6 (compared to $18 today), $79 for Tier 2 customers (compared to $37 today), $99 for Tier 3 customers (compared to $56 today), and $58 for Tier 4 customers (compared to $42 today). For KEDLI this amounts to $52 for Tiers 1 and 5, $72 for Tier 2 customers, $92 for Tier 3 customers, and $51 for Tier 4 customers.

Q. WHAT IS THE ESTIMATED EFFECT OF SUCH DISCOUNTS ON ENERGY COST BURDEN?

A. We estimate that the increased discounts would lift 20,000 low income New York City families out of gas cost burden. Based on our ACS analysis in Table 6 of Schedule 1 of Exhibit__(LIP-3), we found that 190,000 families are low income gas heat customers. Our proposal would solve gas burden for half of this population, *i.e.*, 95,000 families. This equates to 20,000 fewer gas cost burdened families than the 115,000 gas heat low income families who are gas burdened today under existing discounts.

Q. HAS THE COMMISSION ESTABLISHED ANY CONSTRAINTS ON THE ALLOWED SIZE OF UTILITY LOW INCOME BILL DISCOUNT PROGRAMS?

A. Yes. The Commission has placed restrictions that utility bill discounts should not exceed two percent of a utility's revenues, which includes "both total utility revenues and the commodity portion of Energy Service Company revenues

collected through consolidated utility billing to those customers."[4] KEDNY's total reported operating revenues are projected at just over $2 billion in the Rate Year, according to information taken from the Company's Rate Design Panel exhibits.[5] Therefore, that cap dictates that the maximum amount that can be spent on gas low income discounts in KEDNY's territory is approximately $41 million. Similarly, KEDLI's projected total operating revenues are over $1.25 billion in the Rate Year, and the maximum amount that can be spent on gas low income discounts in the KEDLI service territory is approximately $25 million.[6]

We recognize that the final revenue requirements approved by the Commission may be lower than the amounts the Company requested, so these amounts will need to be adjusted correspondingly.

Q. WHAT IS THE AGGREGATE COST OF IMPLEMENTING YOUR PROPOSED DISCOUNT LEVELS?

A. If these proposed needed discounts were implemented, based on anticipated customer usage in KEDNY, the total cost of the discount program would be $59 million per year. For KEDLI, the cost would be $10 million. We understand that this would exceed the threshold of two percent of operating revenues allowable for the KEDNY program; however, we believe such discounts are still justifiable.

Q. PLEASE CONTINUE.

[4] Low Income Order at 30.
[5] KEDNY Exhibit___(RDP-4CU), Sch. 2.
[6] KEDLI Exhibit__(RDP-4CU), Sch. 2.

A. In the generic low income proceeding, the Commission stated that discounts should produce no more than a ten percent energy burden, and that if the ten percent energy burden is reached, the two percent funding limit would need to be reexamined.[7] Our analysis of the ACS data shows that there are currently 145,000 low income National Grid customers who are energy cost burdened over 10 percent of their income. This represent 35 percent of all low income National Grid customers. Moreover, 100,000 low income National Grid customers pay over 5 percent of their income toward their gas bills. The majority of these customers (over 80 percent) are heating gas customers which signals a particularly acute need for relief for heating gas customers. Our analysis of extreme cost burden (*i.e.*, 5 percent gas cost burden and 10 percent energy cost burden) are shown in Schedule 2 of Exhibit__(LIP-3).

Q. IF THERE IS NO RELIEF FROM THE TWO PERCENT CAP, DO YOU HAVE ANY ALTERNATE PROPOSAL FOR DEEPENING THE DISCOUNT LEVELS?

A. Preliminarily, we believe there is strong support for deepening heat discounts as proposed above. As demonstrated by our analysis, existing discount levels fall woefully short of alleviating gas cost burdens for a significant proportion of the low income customer segment. If the Commission is to make meaningful progress towards its ultimate goal of ensuring that no low income customer in the State has

[7] Case 14-M-0565, Proceeding on Motion of the Commission to Examine Programs to Address Energy Affordability for Low Income Utility, Order Granting in Part and Denying in Part Requests for Reconsideration and Petitions for Rehearing (issued February 17, 2017), p. 21.

an energy cost burden greater than six percent (or a gas cost burden greater than three percent), the utility low income programs will have to be significantly increased, perhaps beyond the two percent cap.

That notwithstanding, if the Commission decides that KEDNY and KEDLI must adhere to two percent cap, their discounts should nonetheless be set at levels that will alleviate energy cost burden to the maximum extent possible while avoiding adverse impacts to the most cost burdened families and non-heat gas customers. To that end, as an alternative to our preferred approach, we would recommend that discounts for heating customers should be increased by $20 for all tiers.

In KEDNY, this results in discounts of $38 per month for Tiers 1, 5, and 6; $57 per month for Tier 2; $76 per month for Tier 3; and $62 per month for Tier 4. In KEDLI, this results in the same discounts, except Tier 3 KEDLI customers would receive $77 per month. These discounts would be the same even if National Grid's proposal to eliminate Tiers 5 and 6 in both service territories, and to fold those customers into Tier 1, is adopted.

To be clear, this is not the City's most preferred outcome, as many customers would remain energy cost burdened. That being said, we believe this proposal can significantly improve energy cost burden for low income National Grid customers while still adhering to the two percent cap.

Q. HOW DO THESE DISCOUNTS COMPARE TO DISCOUNTS OFFERED BY UTILITIES WITH SIMILAR CUSTOMERS?

A. Con Edison provides natural gas service to the Bronx, Manhattan, and parts of Queens. Their Low Income Workbook uses the same average monthly bill for a low income gas heat customer, $115, as what is included in KEDNY's Low Income Workbook.[8] Whereas Con Edison provides gas heat discounts of $50.00 per month across all tiers, based on the number of customers across tiers in Exhibit__ (SSP-9), 80 percent of KEDNY customers receive lower benefits than similar Con Edison customers despite having similar energy bills. Furthermore, incomes are not significantly different across service territories. The incomes used in the Con Edison and National Grid Low Income Workbooks are nearly identical, except that Tier 4 incomes in National Grid territory were $220 per year higher. This suggests that National Grid's current discounts are insufficient. Under our proposal, the discounts would be closer to what similarly situated customers in the Con Edison service territory receive.

Q. WHAT WOULD BE THE IMPACT ON OTHER CUSTOMERS OF YOUR ALTERNATE PROPOSAL?

A. Based on the information contained in the Companies' updated Gas Rate Design Panel exhibits regarding the number of customers and operating revenues per service class, we estimate that the cost of increasing low income discounts as proposed would have only a minor effect on other customers. In particular, assuming that costs of the low income programs are allocated to customer classes

[8] *See* Case 19-E-0065 *et al.*, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service, Direct Testimony of the New York City Low Income Panel, Exhibit__(LIP-5) (filed May 24, 2019).

based on operating revenues, we estimate that increasing the discounts as proposed would lead to a monthly increase of $0.28 for a KEDNY residential non-heat customer and $1.22 for a KEDNY residential heat customer. Additionally, we estimate that these discounts would increase monthly KEDLI bills by $0.11 per month for residential non-heat and $0.30 per month for residential heat customers. Our analysis of potential bill impacts on other customers is set forth in Exhibit__(LIP-5). We believe these are modest increases compared to the benefit to low income customers.

Q. ARE YOUR RECOMMENDATIONS CONSISTENT WITH THE COMMISSION'S LOW INCOME ORDER?

A. Yes. In the Low Income Order, the Commission established a default process for setting low income benefit levels that is uniform across all of the utilities. However, the Commission also specifically provided utilities with flexibility to propose alternative means to calculate discounts, so long as they demonstrably accomplished substantially the same results, and left no class of participant underserved. Our recommendations here utilize assumptions that are more indicative of actual income levels of low income households in New York City. As such, they are designed to better accomplish the Commission's policy goal that New York's low income households have an energy burden at or below six percent of household income. Moreover, because our proposed discount levels are based on actual income and energy burden data rather than on an assumption that low income customers earn the maximum threshold of 60 SMI, they would bring more

low income customers up to the Commission's targeted energy burden level of no more than six percent of income. In other words, our proposed increases to current discount levels arguably would make National Grid's EAP *more* inclusive for the thousands of low income households for whom existing discount levels are inadequate.

Q. DO YOU CONSIDER YOUR PROPOSED DISCOUNT CALCULATION METHODOLOGY TO BE A "TOP DOWN" APPROACH?

A. No. To the contrary, we consider our methodology to be a "bottom up" approach. That is, we first analyzed actual income data to develop the appropriate low income household energy burdens, and then calculated the amount of bill discounts needed to achieve an energy burden of no more than six percent of household income. We then extrapolated those increased bill discounts to determine what budgets are necessary for the Companies' respective EAPs, while still remaining within the Commission's established policy that the total budget for a utility's low income program may not exceed two percent of total operating revenues. Importantly, we did not first set each utility's program budget to the maximum two percent cap as a starting point, and then allocate dollars among the different tiers of bill discounts.

RECONNECTION FEES

Q. WHAT ARE RECONNECTION FEES?

A. Reconnection fees are charges that a customer must pay to a utility to reconnect gas service that has been discontinued for non-payment.

Q. WHAT ARE THE COMPANIES' POLICIES REGARDING

RECONNECTION FEES FOR LOW INCOME CUSTOMERS?

A. KEDNY currently charges customers a reconnection fee of $66.13 (plus reimbursement of the Company's City Marshall fees and court filing fees) if service is to be reconnected at the meter or an existing curb valve, and a reconnection fee of $300 if service is to be reconnected at the utility's main by excavation of the street. However, KEDNY also offers a one-time credit of the reconnection fee to low income customers participating in its EAP whose service was disconnected for non-payment.

KEDLI currently charges its customers a $78 reconnection fee, but waives this fee for low income customers.

Q. DO YOU RECOMMEND ANY CHANGES TO THE COMPANIES' RECONNECTION FEE PROGRAMS?

A. Yes. While we appreciate that KEDNY provides low income customers a one-time waiver of the reconnection fee, we propose that KEDNY eliminate their reconnection fees altogether for all low income customers.

Q. WHY SHOULD KEDNY ELIMINATE ITS RECONNECTION FEES FOR LOW INCOME CUSTOMERS?

A. Eliminating reconnection fees for low income customers is appropriate because it will provide them with a better opportunity to remain utility customers and pay their utility bills. In particular, low income customers who are facing termination of service already have demonstrated an inability to pay utility bills. Requiring these customers to pay a fee to reconnect their gas service detracts from the amount

they can put toward paying their past due bills, and creates an additional financial barrier for them to overcome. Customers and the utilities would be much better served if this money was instead put towards direct payment of the customers' utility bills.

Moreover, the amount collected for reconnections is very small when considered in context. According to National Grid's response to City IR-106, on average over the last five years, KEDNY collected approximately $1.3 million in reconnection fees annually from all customers. According to National Grid's response to DPS IR-260, approximately $174,000 of that amount was collected annually from low income customers over the same time period. These amounts, and in particular the amount collected from low income customers, are insignificant when considered in context of the revenue requirement increase that KEDNY is seeking in this rate case, approximately $196 million in the Rate Year. We also understand that each Company's actual annual gas revenue requirements are in the billions of dollars.

Conversely, KEDNY's reconnection fees are substantial, particularly from the perspective of a low income customer. As previously noted, KEDNY charges a reconnection fee of $66.13 (plus fees) for reconnecting service at the meter, which represents nearly 58 percent of an average KEDNY low income heating customer's monthly gas bill. Thus, in context, the loss of these reconnection revenues would be inconsequential for the Companies, but the savings could be material for a customer who already cannot afford his or her utility bill.

Finally, elimination of KEDNY's reconnection fees for low income customers is appropriate to harmonize the utility's practices with those already adopted by KEDLI. This eliminates the possibility of low income customers residing in different portions of New York City (for example, on the Rockaways versus in Brooklyn) being subject to different rules on reconnection fees.

HRA FILE MATCHING PROCESS

Q. PLEASE DESCRIBE THE EXISTING FILE MATCHING PROCESS BETWEEN HRA AND NATIONAL GRID.

A. As part of their 2016 rate cases, National Grid implemented a file matching process with HRA to identify and enroll low income customers in KEDNY's service territory, and in the Rockaways portion of KEDLI's service territory, who are eligible for the Companies' respective EAPs. HRA currently performs the matches and opt-out process between the Companies' records and its records twice a year.

Q. PLEASE DESCRIBE THE OPT-OUT PROCESS.

A. The current matching process was developed to provide notification to HRA's clients of the existence of the National Grid low income programs. Those clients who do not choose to opt out when so notified are considered to have consented to HRA providing their names and limited information necessary for their participation in the programs to National Grid to be enrolled in the low income programs. HRA does not disclose to National Grid which type of benefit(s) any of HRA's clients receive – only that they qualify to be enrolled. Every time a match

is performed, there are some HRA clients who choose to opt-out; HRA does not disclose any information regarding these clients to National Grid.

Q. HAS HRA HISTORICALLY RECEIVED FUNDING FROM NATIONAL GRID FOR ITS ADMINISTRATIVE COSTS ASSOCIATED WITH PERFORMING THE LOW INCOME PROGRAM MATCH?

A. Yes. National Grid currently reimburses HRA for the out-of-pocket costs incurred to perform the matches, up to a cap of $100,000 annually. The reimbursement amount and cap have both been approved by the Commission.

Q. WHAT DO THE COMPANIES PROPOSE IN THIS RATE CASE?

A. The Companies do not propose any changes to both the existing file matching process and the reimbursement mechanism.

Q. WHAT IS YOUR RECOMMENDATION REGARDING THE HRA FILE MATCHING PROCESS?

A. The file matching process and reimbursement mechanism have worked well over the last few years. It is a valuable activity that helps ensure that the City's neediest residents receive additional assistance in paying their utility bills. As such, we recommend that the HRA file matching process and reimbursement mechanism continue, with some modifications. Specifically, we recommend increasing the frequency of the file match, and commensurately increasing the reimbursement cap.

Q. WHAT CAPACITY DOES HRA HAVE TO INCREASE THE NUMBER OF MATCHES PERFORMED EACH YEAR?

A. HRA could increase the match frequency from twice a year to four times a year, or quarterly.

Q. WHAT BENEFITS DO THE PANEL ANTICIPATE AS A RESULT OF MORE FREQUENT MATCHES?

A. We believe that performing the match twice as often will improve the accuracy of customer enrollment in the Companies' low income programs. Performing the match more frequently also will help ensure that customers who are newly-qualified for the Companies' low income programs are properly enrolled and receive the utility low income discount sooner.

While we considered other possible frequencies of the matching process, we believe that a quarterly match strikes a fair and reasonable balance between the desire to continue improving the accuracy and scope of the Companies' low income programs, with the administrative costs and resources that would be required of HRA to perform the matches more than four times a year.

Q. DOES THE PANEL BELIEVE THAT MORE FREQUENT MATCHES WILL INCREASE THE NUMBER OF CUSTOMERS ENROLLED IN THE COMPANY'S LOW INCOME PROGRAMS?

A. Possibly. As stated above, more frequent matches will provide greater opportunities for eligible persons to participate in the Companies' low income programs. However, more frequent matches may not necessarily equate to a sizable increase in participation levels. This is due to perpetual movement of individuals and families coming on and off assistance benefits based on eligibility.

We will not know the exact impact more frequent matches would have on program enrollment until HRA does a full year of quarterly matches.

Q. WHAT IS YOUR PROPOSAL FOR REIMBURSEMENT LEVELS FOR HRA TO PERFORM QUARTERLY MATCHES?

A. As HRA will be performing the match twice as often as it does currently, we recommend that the reimbursement cap be increased by an additional $100,000 per year – for a total of $200,000 per year – to enable HRA to perform a quarterly match.

Q. WHAT IS INCLUDED IN HRA'S ADMINISTRATIVE COSTS RELATED TO PERFORMING THE LOW INCOME MATCH?

A. HRA's administrative costs include the cost of the opt-out process, including the cost of preparing and issuing the letters, envelopes, postage, and computer time to perform the matches. As in any organization, there is a cost associated with each program run on the central processing unit.

Q. WHAT COSTS ARE NOT INCLUDED IN HRA'S REQUEST FOR REIMBURSEMENT?

A. HRA dedicates many hours of its employees' time to this effort. Its employees must refine the programming language, run the match, perform quality control, oversee the mailing process, monitor and catalogue returned mail, respond to questions from clients, process the responses received, and interact with National Grid. This effort also requires management oversight and administrative support.

This time is taken away from these employees' primary job duties and responsibilities to HRA's programs.

HRA continues to be willing to contribute its employees' time for more frequent matches because of the value provided by the National Grid low income programs to its clients. While we have not calculated the total cost being absorbed by HRA to perform quarterly matches, it is likely well in excess of the amount for which reimbursement is sought.

Q. CAN HRA ALLOCATE CITY FUNDS TO THIS EFFORT?

A. No. HRA's role is to administer its own low income programs, not utility low income programs. HRA's funding, whether from federal or state sources, or through the City's budget process, is intended and budgeted to support its own programs. Use of the funds must relate to the programs for which they are allocated. For example, funds for the Supplemental Nutrition Assistance Program could not be used to provide energy-related benefits. For these reasons, HRA does not have any funding available to support National Grid's low income programs.

Q. COULD HRA DELEGATE TO NATIONAL GRID THE RESPONSIBILITY TO CONTACT HRA CLIENTS AND PERFORM THE OPT-OUT STEP OF THE PROCESS TO REDUCE COSTS?

A. No. As established in Case 14-M-0565, information regarding HRA's clients is confidential as a matter of law. Under Federal, State, and local law, there are strict limitations on the information HRA may disclose to a third party about its clients beyond that which is already being provided to National Grid, namely, an indication

of eligibility for the low income program. HRA therefore could not share the preliminary results of the matches with National Grid or have them ask such clients whether they want their information conveyed to the Companies. And, as noted above, at no time does HRA disclose to National Grid which type of benefit(s) any of HRA's clients receive – only that they qualify to be enrolled. If an HRA client opts-out, HRA does not disclose that client's information to National Grid.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Rebuttal Testimony of

Alexandra Fisher

On Behalf of

The City of New York

September 18, 2019

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Alexandra Fisher, Deputy Commissioner, Legal & Regulatory Affairs, New York City Department of Buildings, or "DOB," 280 Broadway, New York, NY 10007.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. The City of New York, or "City."

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND AND PROFESSIONAL EXPERIENCE.

A. I received a Bachelor of Arts in Economics from New York University in 1992 and a Juris Doctor from Brooklyn Law School in 1997.

I have been in my current position since October 2014. In this role, I am responsible for providing legal, policy and general guidance to the DOB Commissioner, senior and other agency staff on a wide range of issues that impact DOB. This includes the analysis of proposed and enacted legislation, code and rule drafting and oversight of licensing operations. I work closely with DOB personnel and other experts to ensure that my guidance is technically sound. Prior to this position, I held legal roles of increasing responsibility in several City agencies including the Office of Administrative Trials and Hearings, the Fire Department, the Department of Correction, and the Office of Payroll Administration. This is my second position with DOB – I also served as Senior Policy Advisor from 2009 to 2012.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Yes. I recently submitted testimony on behalf of the City in the Consolidated Edison Company of New York, Inc. ("Con Edison") 2019 gas rate case, Case 19-G-0066.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. To respond to the pre-filed direct testimony of the New York State Department of Public Service Staff, or "Staff," Pipeline Safety Panel. My testimony is limited to the Panel's recommendations regarding Local Law No. 152 of 2016, or "Local Law 152," and its interaction with the inspection obligations of the Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corp. d/b/a National Grid ("KEDLI," and together with KEDNY, "National Grid" or the "Companies").

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. I offer Exhibit__(AF-1), which contains reports dated April 12, 2016 and November 15, 2016, both which were issued by the New York City Council's Committee on Housing and Buildings, Infrastructure Division, that I relied upon in preparing my testimony; Exhibit__(AF-2), which contains copies of Introduction, or Int., No. 1088-2016, Int. No. 1088-A, and Local Law 152, as well

as the proposed language of 1 RCNY §103-10; and Exhibit__(AF-3), which contains the language of 16 NYCRR §255.3.

LOCAL LAW 152

Q. PLEASE PROVIDE A BRIEF OVERVIEW OF THE DEVELOPMENT OF LOCAL LAW 152.

A. The New York City Council introduced Local Law 152 as Int. No. 1088-2016 on February 24, 2016. The Committee on Housing and Buildings amended the legislation as Int. No. 1088-A on November 15, 2016 and the City Council approved it as amended on November 16, 2016. Mayor Bill de Blasio signed Local Law 152 into law on December 6, 2016; it is codified as Article 318 of Chapter 3 of Title 28 of the Administrative Code of the City of New York. Exhibit__(AF-2) contains copies of Int. No. 1088-2016, Int. No. 1088-A, and Local Law 152.

The City Council introduced Local Law 152 as part of a package of legislation with the goal of improving gas safety, responding to three major gas-related explosions that occurred in New York City during the preceding two years. Each of the three gas-related explosions that precipitated the legislation package tragically resulted in fatalities and injuries, as well as displacement of residents. As noted in the Committee on Housing and Buildings' April 12, 2016 Committee Report, which is attached hereto as Exhibit__(AF-1), the three explosions were the East Harlem Explosion, wherein two buildings at 1644 and 1646 Park Avenue collapsed on March 12, 2014; the East Village Explosion,

wherein one building was damaged and three buildings were destroyed as a result of an explosion at 121 Second Avenue on March 26, 2015; and, the Brooklyn Explosion, in which an explosion occurred at 4206 13th Avenue on October 3, 2015.

Effectively, the objective of Local Law 152 is to improve gas safety and avoid further tragedies by requiring building owners to inspect their buildings' gas piping systems.

Q. PLEASE DESCRIBE DOB'S RULE TO IMPLEMENT LOCAL LAW 152.

A. DOB regularly promulgates rules pursuant to local laws that authorize such rulemaking in accordance with the City Administrative Procedure Act, codified as Section 1041 *et seq.* of Chapter 45 of the New York City Charter. In doing so, the substance of DOB rules must comport with the legislative intent and text of the enabling local law and cannot exceed the limits the City Council imposed. On November 26, 2018, DOB published notice in the City Record of its proposal to add Section 103-10 to Title 1 of the Rules of the City of New York, or "RCNY," pursuant to Local Law 152. DOB held a public hearing and received comments on proposed 1 RCNY §103-10 on January 2, 2019. I note that the rulemaking is ongoing and no final rule has yet been adopted.

Pursuant to Local Law 152, the proposed 1 RCNY §103-10 is intended to promote public safety by requiring periodic inspections of gas piping systems in buildings and the correction of conditions identified by such inspections. Moreover, the proposed rule is intended to effectuate Article 318 of Chapter 3 of

Title 28 of the Administrative Code, which was added pursuant to Local Law 152, by clarifying procedural and substantive requirements associated with the law. In particular, proposed 1 RCNY §103-10 includes provisions that pertain to:

- the periodic inspection cycle schedule and inspection requirements;
- inspector qualifications for performing periodic inspections of gas piping systems;
- the required scope of inspection;
- reporting and certification requirements;
- requirements for reporting unsafe or hazardous conditions; and
- civil penalties for failure to file required certifications.

Q. PLEASE EXPLAIN THE DIFFERENCE BETWEEN THE CITY'S REQUIREMENTS WITH RESPECT TO SERVICE LINE INSPECTIONS AND THE COMPANIES' REQUIREMENTS AS GAS UTILITY COMPANIES.

A. On April 20, 2015, in Case 15-G-0244, entitled Order Establishing Statewide Inspection Schedules and Procedural Requirements, the Public Service Commission ("Commission") adopted a new definition of gas service line, thereby expanding the Commission's jurisdiction and adding two additional inspection requirements to the Operation and Maintenance programs of gas utility companies regulated by the Commission. On pages 1-2 of the Order, the Commission directed that following this change in definition, gas utility companies must complete "leakage surveys and atmospheric corrosion

inspections starting from a building's outside wall up to the outlet of the gas meter."

As provided in 16 NYCRR § 255.3(43), attached as Exhibit__(AF-3), the new definition of gas "service line" reads as follows:

> [T]he piping, including associated metering and pressure reducing appurtenances, that transports gas below grade from a main or transmission line to the outlet of the customer meter or at the connection to a customer's piping, whichever is further downstream where a meter is located within the building; if a meter is located outside the building, the service line will be deemed to terminate at the outset of the building foundation wall.

In contrast, Local Law 152 delineates the scope of gas piping systems inspections as follows: "At each inspection, […] all exposed gas lines from point of entry of gas piping into a building, including building service meters, up to individual tenant spaces shall be inspected […]." The legislative history of Local Law 152 shows that this language originated in Int. No. 1088 and remained unchanged in Int. No. 1088-A, and Mayor de Blasio signed it into law as Local Law 152. DOB's proposed 1 RCNY §103-10 incorporates this scope via reference to §28-318.3.2 of the Administrative Code.

Whereas Local Law 152 requires gas piping inspections of exposed gas lines from the point of entry into a building to individual tenant spaces, the Commission's inspection requirements set forth in 16 NYCRR § 255.3(43) cover piping "from a main or transmission line to the outlet of the customer meter or at

the connection to a customer's piping, whichever is further downstream where a meter is located within the building." Taken together, these two jurisdictional schemes result in three conceivable scenarios.

First, where a meter is located outside a building, National Grid must conduct its inspection through the meter and up to the meter's outlet; the building owner's licensed master plumber would be responsible for all piping from the point of entry from the meter and up to the tenant spaces.

Second, where a meter is located inside a building (for instance, the building's basement), National Grid would be responsible for inspecting all piping from the main/transmission line into the building, through the meter to the meter outlet; the building owner's licensed master plumber would be responsible for inspecting all piping from the point of entry into the building, through the meter and up to the tenant spaces.

Third, where a building contains meters only in individual tenant spaces, National Grid would be required to inspect all piping from the main/transmission line into the building, piping throughout the building, piping into and within individual tenant spaces up through the meter and up to the meter outlet; the building owner's licensed master plumber would be responsible for inspecting all piping from the point of entry into the building, and piping throughout the building up to the tenant spaces.

These three scenarios illustrate, except for buildings with exterior meter locations, that every building will have some length of piping subject to both

Local Law 152 and Commission-required inspections. It should be noted, however, that it is conceivable that there will be some buildings not subject to DOB's jurisdiction, such as Federal buildings, and, therefore may only be subject to Commission-required inspections. Although most buildings could be inspected twice, the inspections will not be "duplicative," as Staff suggests on page 77 of its Pipeline Safety Panel's direct testimony, as the Commission-required inspections do not cover customer piping downstream of the meter outlets. Two separate inspections may bolster safety in that one inspection may reveal conditions that were either missed by the other or arose between inspections.

Q. WHAT IS YOUR RESPONSE TO THE STAFF PIPELINE SAFETY PANEL'S CRITICISM THAT DOB'S RULE DOES NOT ALIGN WITH FEDERAL AND STATE OPERATOR QUALIFIED TRAINING AND QUALIFICATION STANDARDS?

A. Such criticism is misplaced. It appears that the criticism is motivated by an incorrect concern that if Local Law 152 inspections are not performed by operator qualified individuals, such inspections – and, presumably, such inspection data – will not be able to satisfy National Grid's obligations with respect to Commission-required inspections of service lines. Implicit in such criticism is the erroneous presumption that Local Law 152 inspections – paid for by building owners – ought to satisfy the Commission-mandated service line inspections, the responsibility for which the Commission determined properly belongs to National Grid.

The criticism regarding the lack of a requirement in proposed 1 RCNY §103-10 that inspection entities be operator qualified misses obvious foundational characteristics of Local Law 152 – the City is not trying to duplicate the Commission's requirement or relieve gas utility companies from their Commission mandates. Rather, the City is only requiring inspections of gas piping within DOB's jurisdiction and imposing qualification requirements consistent with the City's scope of inspections. The Staff Pipeline Safety Panel fails to appreciate the distinction between the Commission's and the City's jurisdiction.

In sum, it is inappropriate to suggest that the City and DOB ought to incorporate State and Federal operator qualification requirements. The operator is National Grid—not DOB, not the building owner, and not the master plumber retained by the building owner; and the Commission, not the City, is responsible for ensuring that National Grid uses personnel with the requisite operator qualifications.

Q. WHAT IS YOUR RESPONSE TO THE STAFF PIPELINE SAFETY PANEL'S CRITICISM THAT DOB'S PROPOSED RULE DOES NOT EXPLICITLY REQUIRE INSIDE INSPECTIONS FOR NATURAL GAS PIPING IN TENANT SPACES?

A. That criticism is misplaced insofar as it presupposes that DOB has the rulemaking authority to mandate inspections within tenant spaces. DOB's rulemaking authority is limited by the applicable statute or local law. In the context of Local

Law 152, the City Council did not provide authority to DOB to conduct inspections within tenant-owned spaces, and I offer no position on whether the City Council could do so legally.

The Staff Panel's suggestion that DOB's rule should include gas piping within individual tenant spaces suggests that the Panel did not understand Local Law 152, as the suggestion is inconsistent with the City Council's unambiguous intent in specifying—in both Int. No. 1088 and Int. No. 1088-A—that "all exposed gas lines from point of entry of gas piping into a building, including building service meters, up to individual tenant spaces shall be inspected …."

The Staff Pipeline Safety Panel further comments on page 78 of its pre-filed direct testimony that "[i]f the proposed rules do not require inspections within tenant spaces, the inside piping associated with meters inside apartments would not be inspected by the plumber and would require a separate inspection by the Companies." The concern raised here is similarly misplaced – Local Law 152 and proposed 1 RCNY §103-10 are not intended to supplant or facilitate the obligations the Commission imposed on National Grid. Moreover, as noted above, this statement also demonstrates the Panel's lack of understanding of the scope of Local Law 152. In addition, it should also be noted that the attempt to use the Local Law 152 inspection as a means of satisfying National Grid's inspection obligations with respect to the Commission, Staff is, in effect, attempting to disburden National Grid while placing the expense of Commission-imposed inspection duties on the building owner. Such an attempt is especially

misplaced in view of the Commission's recommendations, discussed below, on how National Grid can address its concerns about performing inspections within tenant spaces.

Q. HAS THE COMMISSION ADDRESSED THE INSPECTIONS WITHIN TENANT SPACES?

A. Yes. The Commission addressed the concern about access to tenant spaces at length in its Order Establishing Statewide Inspection Schedules and Procedural Requirements issued on April 20, 2017 in Case 15-G-0244. While the Commission acknowledged that gaining access to inside gas services is a challenge for National Grid in performing the required state inspections, it provided concrete guidance to National Grid on how to surmount this challenge. Specifically, on page 33 of the Order the Commission noted that, "[t]he easiest and most obvious first step for all LDCs is for each to develop safety-related inspection notices to leave at customers' premises, to include as bill inserts, or place on company websites."

Moreover, the Commission highlighted the fact that Public Service Law §65(9) provides National Grid the requisite legal authority to enter, at reasonable times, any building, room, or place supplied with gas for purposes of inspecting meters, pipes, and fittings used for supplying or regulating the supply of gas. Additionally, Public Service Law §65(9) allows National Grid to charge customers one hundred dollars if a customer prevents or hinders the gas utility company's officer/agent from entering the premises to make an inspection.

Lastly, the Commission ruled in the April 20 Order that under certain circumstances National Grid may disconnect service when a customer has paid the one hundred dollar fine but still refuses to provide access.

In sum, the concern the Staff Pipeline Safety Panel raised about Local Law 152 inspections stopping at individual tenant spaces appears out-of-place when viewed in light of the scope of Section 65 of the Public Service Law and the Commission's directives and recommendations to National Grid for gaining access to such spaces. The Commission – not Local Law 152 or DOB's proposed 1 RCNY §103-10 – requires National Grid to inspect individual tenant spaces where such spaces contain meters, and the Commission gave National Grid the means to do so.

Q. WHAT IS YOUR RESPONSE TO THE STAFF PIPELINE SAFETY PANEL'S CRITICISM THAT THE PROPOSED TIMEFRAMES FOR INSPECTION CYCLES IN DOB'S PROPOSED RULE DOES NOT ALIGN WITH NECESSARY FEDERAL AND STATE TIMELINES?

A. As the Panel acknowledged, Local Law 152 specifically provides DOB's commissioner with the discretion to modify the inspection frequency to be consistent with the cycle adopted by the Commission. However, that the proposed rule does not reflect the same cycle now is reasonable given that DOB is establishing the necessary operational processes to receive and track the required certifications of inspections that building owners must file with DOB. While the inspection frequency could be changed in the future, a longer frequency during

the initial rollout cycle allows DOB additional time to establish these operational processes.

Furthermore, from the perspective of building owners, a frequency of one year and three years for business districts and residential districts, respectively, would likely pose significant financial burdens on buildings owners. Tailoring inspection frequencies to align with National Grid's inspection frequency as mandated by the Commission may be more convenient for National Grid, but it could unfairly shift the expense of inspections from National Grid to building owners. Additionally, shorter frequencies of inspections could strain the stock of qualified plumbers available to perform the inspections required by Local Law 152 and the Commission. In view of these considerations, DOB has opted to refrain from exercising its discretion to align the Local Law 152 inspection frequency with the Commission's inspection frequency at this time. Upon completion of at least one full cycle of periodic inspections, DOB may revisit the inspection cycle.

Q. PLEASE DESCRIBE THE WAYS THAT THE CITY IS TRYING TO OPTIMIZE ITS WORK WITH THE GAS UTILITIES ON SERVICE LINE INSPECTIONS.

A. DOB has had numerous meetings with National Grid, Con Edison, and labor association representatives regarding DOB's proposed 1 RCNY §103-10 with the intent of understanding concerns of the gas utility companies and industry professionals. In an attempt to address these concerns, DOB has tried to optimize

1 RCNY §103-10 as it relates to the gas utilities' service line inspections in a number of ways. For instance, in its public comments on 1 RCNY §103-10, submitted together with Con Edison, National Grid urged DOB to change the scheduling of Local Law 152 inspections to an approach where the total number of buildings requiring inspection and associated workload is balanced, which it stated would help balance workforce demands and not overburden building owners who have multiple buildings in a single borough. In response, DOB has explored revising the rule's inspection schedule based on community districts rather than boroughs—and a way that, to the extent possible, evenly distributes amongst National Grid and Con Edison the number of buildings to be inspected each year. DOB scheduled a conference call with National Grid to hear its position on such revisions; however, the call was cancelled by National Grid and never rescheduled. DOB participated in a conference call with Con Edison to discuss revisions to the inspection schedule.

In its public comments, National Grid also urged DOB to change its rule to require the use of utility applications to capture corrosion and leak survey inspection work if the utility chooses to provide such a mobile application. While DOB will not be adding such a requirement to 1 RCNY §103-10, DOB is willing to share with the utilities the data elements of information required within the Local Law 152 inspection reports. DOB's intent here, in sharing such data elements, is to allow the utilities' use of mobile applications in a way that enables them to incorporate information contained in the Local Law 152 inspection

reports into their own Commission-mandated service line inspections. DOB has not received a request for such data elements from National Grid, but has provided such information in response to requests by Con Edison.

Q. DOES THIS CONCLUDE YOUR TESTIMONY AT THIS TIME?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

DIRECT TESTIMONY

AND EXHIBITS

OF

LANE KOLLEN

ON BEHALF OF

THE CITY OF NEW YORK

J. KENNEDY AND ASSOCIATES, INC.

AUGUST 30, 2019

QUALIFICATIONS AND SUMMARY

Q. PLEASE STATE YOUR NAME AND BUSINESS ADDRESS.

A. My name is Lane Kollen. My business address is J. Kennedy and Associates, Inc. ("Kennedy and Associates"), 570 Colonial Park Drive, Suite 305, Roswell, Georgia 30075.

Q. WHAT IS YOUR OCCUPATION AND BY WHOM ARE YOU EMPLOYED?

A. I am a utility rate and planning consultant holding the position of Vice President and Principal with the firm of Kennedy and Associates.

Q. PLEASE DESCRIBE YOUR EDUCATION AND PROFESSIONAL EXPERIENCE.

A. I earned both a Bachelor of Business Administration in Accounting degree and a Master of Business Administration degree from the University of Toledo. I also earned a Master of Arts degree in Theology from Luther Rice University. I am a Certified Public Accountant, with a practice license, Certified Management Accountant, and Chartered Global Management Accountant. I am a member of numerous professional organizations, including the American Institute of Certified Public Accountants and the Society of Depreciation Professionals.

I have been an active participant in the utility industry for more than thirty years, both as an employee and as a consultant. Since 1986, I have been a consultant with J. Kennedy and Associates, Inc., providing services to state government agencies and consumers of utility services in the ratemaking, financial, tax, accounting, and management areas. From 1983 to 1986, I was a consultant with Energy Management Associates, providing services to investor and consumer owned utility companies. From 1976 to 1983,

I was employed by The Toledo Edison Company in a series of positions encompassing accounting, tax, financial, and planning functions.

I have appeared as an expert witness on accounting, depreciation, tax, finance, ratemaking, and planning issues before federal and state regulatory commissions and courts on hundreds of occasions.[1]

Q. ON WHOSE BEHALF ARE YOU TESTIFYING?

A. I am testifying on behalf of the City of New York.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of my testimony is to address the accounting of, and proposed increases in depreciation (amortization) rates for, leak prone pipe ("LPP") plant assets based on depreciation studies performed and presented by Mr. Paul Normand on behalf of Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a Brooklyn Union of L.I. ("KEDLI"; and together with KEDNY, the "Companies").

Q. PLEASE DESCRIBE THE EFFECT OF THE COMPANIES' PROPOSED CHANGES IN DEPRECIATION RATES AND EXPENSE.

A. If adopted, the proposed depreciation (amortization) rates will result in annual LPP depreciation (amortization) expense for KEDNY of $29.258 million, an increase of $27.453 million compared to $1.806 million under present depreciation rates, and annual LPP depreciation expense for KEDLI of $26.555 million, an increase of $24.477 million

[1] My regulatory appearances by jurisdiction and proceeding, as well as the subject matter, are detailed on my Exhibit___(LK-1).

compared to $2.078 million under present depreciation rates, based on plant balances at the study date of December 31, 2018.[2]

Q. PLEASE DESCRIBE THE COMPANIES' PROPOSED CHANGES IN DEPRECIATION RATES.

A. The Companies propose significant increases in the depreciation rates for the LPP plant accounts in order to recover the remaining net book value over 20 years and to recover significant increases in the estimated costs of removal for these assets. The proposed changes in the depreciation rates for the LPP plant accounts ostensibly are due to the Companies' plans to accelerate the retirement and replacement of LPP mains and services. The Companies propose to reclassify the net book value of the LPP plant assets as regulatory assets and then amortize the regulatory assets using the proposed depreciation rates.

KEDNY proposes to increase the average LPP depreciation rates to 34.03% from 2.10%, although the rates vary by plant account.[3] KEDLI proposes to increase the average LPP depreciation rates to 18.64% from 1.46%, although the rates vary by plant account.[4]

KEDNY proposes to shorten the remaining average service lives reflected in the LPP depreciation rates to 20.0 years from 25.8 years. KEDLI proposes to shorten the

[2] For KEDNY, refer to Schedule B of the KEDNY depreciation study. [Exhibit___(PMN-3) attached to Mr. Normand's Direct Testimony. For KEDLI, refer to Schedule B of the KEDLI depreciation study. [Exhibit___(PMN-3) attached to Mr. Normand's Direct Testimony]. I have attached the KEDNY Schedule B as my Exhibit___(LK-2) and KEDLI Schedule B as my Exhibit___(LK-3) for ease of reference.
[3] *Id.*
[4] *Id.*

remaining average service lives reflected in the LPP depreciation rates to 20.0 years from 29.9 years.[5]

KEDNY proposes to increase the cost of removal (net negative salvage) from -60% to -357.3% for account 367.08 *Mains-Steel*, from -38% to -1,968.1% for account 376.05 *Mains-Cast Iron*, from -65% to -357.3% for account 376.09 *Mains-Steel*, from -5% to -111.9% for account 380.02 *Services-Steel*, and from -4% to -248.7% for account 380.05 *Services-Copper*.[6]

KEDLI proposes to increase the cost of removal (net negative salvage) from 0% to -260.8% for account 367.04 *Mains-Cast Iron*, from 0% to -601% for account 367.09 *Mains-Steel*, from -1% to -601% for account 376.08 *Mains-Steel*, from -14% to -260.8% for account 376.10 *Mains-Cast Iron*, from 0% to -38% for account 376.18 *Mains-Plastic*, from 0% to -241.2% for account 380.01 *Services-Steel*, from 0% to -38% for *Services-Plastic*, and from 0% to -30.7% for account 380.03 *Services-Copper*.[7]

Q. PLEASE COMPARE THE MAGNITUDE OF THE COMPANIES' REQUESTS TO THE REMAINING NET BOOK VALUE OF THE LPP PLANT ASSETS.

A. KEDNY seeks to recover $549.060 million through the proposed depreciation rates over the next twenty years compared to the $72.342 million in remaining net book value at December 31, 2018.[8] In other words, KEDNY seeks to recover $476.718 million for costs

[5] For KEDNY and KEDLI, refer to Schedule A column 14 of the depreciation study attached to Mr. Norman's Direct Testimony in each case. [Exhibit___(PMN-2)]. I have attached the KEDNY Schedule A as my Exhibit___(LK-4) and KEDLI Schedule A as my Exhibit___(LK-5) for ease of reference.
[6] Exhibit___(LK-2).
[7] Exhibit___(LK-3).
[8] Exhibit___(LK-4). The plant in-service for all LPP plant assets is $85.974 million (column 1). The accumulated depreciation is $13.632 million (column 12).

of removal in addition to the net book value of the LPP plant assets. KEDLI seeks to recover $489.540 million through the proposed depreciation rates compared to $80.017 million in remaining net book value at December 31, 2018.[9] KEDLI seeks to recover $409.523 million for costs of removal in addition to the net book value of the LPP plant assets.

Q. PLEASE SUMMARIZE YOUR TESTIMONY.

A. I recommend that the Commission reject the Companies proposals to: (1) increase the present depreciation rates and depreciation expense to accelerate recovery of the remaining net book value; and (2) recover potentially hundreds of millions of dollars of highly uncertain and unknown estimated costs of removal. The increases are not necessary and reflect inordinately short recovery periods and excessive estimated costs of removal based on flawed statistical analyses that are inappropriate for this purpose.

The Companies' proposal to shorten the amortization period is unreasonable. Once the net book value is reclassified to a regulatory asset, the remaining lives of the plant assets are irrelevant. There is no compelling reason for the Commission to accelerate the recovery of these costs even if it agrees with the Companies' plan to accelerate the retirement and replacement of the LPP plant assets. The Companies' proposal will compound the costs to customers, who will be required to pay for the cost of the retired LPP assets as well as the cost of the replacement assets.

[9] Exhibit___(LK-5). The plant in-service for all LPP plant assets is $142.470 million (column 1). The accumulated depreciation is $62.453 million (column 12).

The Companies' proposal to significantly increase the estimated costs of removal in the depreciation rates also is unreasonable. There is no need to estimate costs of this potential magnitude. Mr. Normand's estimates for the Companies are unreliable and unreasonable. These estimates sum to almost $900 million over the proposed twenty-year amortization period, nearly six times the net book value of the plant being retired. Mr. Normand's estimates of these costs are not actual estimates developed by the Companies' planning and engineering personnel based on the scope of work, but rather are based on recent costs per mile and per service that vary considerably from year to year.

The better approach is for the Commission to direct the Companies to defer the actual costs of removal to the regulatory assets when the costs are incurred and then depreciate (amortize) the increase in the regulatory assets using the presently approved depreciation (amortization) rates. This approach is equitable to both the Companies and their customers and is fully compensatory to the Companies. It will permit the Companies to recover 100% of their actual costs of removal.

Q. WHAT ARE THE EFFECTS OF YOUR RECOMMENDATIONS?

A. The depreciation expense for KEDNY will be $27.453 million less and for KEDLI will be $24.477 million less than the proposed depreciation expense calculated by Mr. Norman based on LPP plant assets at December 31, 2018.

ACCOUNTING AND RATEMAKING TREATMENT OF LPP PLANT ASSETS RECLASSIFIED TO REGULATORY ASSETS

Q PLEASE DESCRIBE THE TYPICAL ACCOUNTING TO RECLASSIFY THE NET BOOK VALUE OF THE LPP PLANT ASSETS TO REGULATORY ASSETS.

A. The typical accounting consists of a credit to plant in service and an equivalent debit to accumulated depreciation to remove the LPP amounts from the plant and accumulated depreciation accounts. To the extent that the Companies' plant accounts are not fully depreciated, the typical accounting would leave the net book value as a debit balance in the accumulated depreciation accounts. The Companies then would credit the accumulated depreciation and debit regulatory assets for amounts equivalent to the net book value.

Q. PLEASE DESCRIBE THE TYPICAL ACCOUNTING FOR COSTS OF REMOVAL AFTER PLANT COSTS HAVE BEEN RECLASSIFIED TO REGULATORY ASSETS.

A. In such cases, the costs of removal typically are added to the regulatory assets. This is different than if the costs still were recorded in the plant accounts. In that latter circumstance, actual costs of removal are debited to accumulated depreciation and the estimated net negative salvage costs included in the depreciation expense are credited to accumulated depreciation.

Q. PLEASE DESCRIBE THE TYPICAL CALCULATION OF AMORTIZATION EXPENSE.

A. Amortization expense typically is calculated as the amount of the regulatory asset divided by the remaining amortization period (months or years). Alternatively, it could be a dollar amount of amortization expense allowed by the Commission.

Q. DOES IT APPEAR THAT THE COMPANIES PLAN TO FOLLOW THE TYPICAL ACCOUNTING TO RECLASSIFY THE NET BOOK VALUE OF THE LPP PLANT ASSET TO REGULATORY ASSETS AND TO CALCULATE

AMORTIZATION EXPENSE ON A STRAIGHT-LINE BASIS OVER THE REMAINING AMORTIZATION PERIOD?

A. No. It appears that the Companies' plan to maintain the gross plant amounts as the regulatory assets and maintain the accumulated depreciation as accumulated amortization. I conclude this because the Companies calculated their proposed depreciation rates on gross plant, not net plant.[10] I also conclude that the Companies' plan is to continue debiting the actual costs of removal to accumulated amortization and crediting the amortization expense, which includes the estimated costs of removal, to the accumulated amortization. I conclude this because the Companies calculated their proposed depreciation rates on gross plant, not net plant, and embedded the costs of removal into the proposed depreciation rates.[11]

Q. DO YOU AGREE THAT THE ACTUAL COSTS OF REMOVAL SHOULD BE DEBITED TO ACCUMULATED AMORTIZATION?

A. No. The Commission should direct the Company to debit the actual costs of removal to the regulatory assets, not to accumulated amortization, assuming the Commission adopts my recommendation to continue to use the present depreciation rates and does not increase the depreciation (amortization) rates based on Mr. Normand's estimated costs of removal. In this manner, the costs of removal will be properly deferred for recovery and the amortization will commence the month after the costs of removal are deferred.

[10] For KEDNY, refer to Schedule B of the KEDNY depreciation study. [Exhibit___(PMN-3) attached to Mr. Normand's Direct Testimony. For KEDLI, refer to Schedule B of the KEDLI depreciation study. [Exhibit___(PMN-3) attached to Mr. Normand's Direct Testimony].
[11] *Id.*

Q. WHAT IS YOUR RECOMMENDATION?

A. I recommend that the Commission approve the Companies' request to reclassify the LPP plant assets to regulatory assets. I also recommend that the Commission direct the Companies to defer the actual costs of removal when the costs are incurred and to commence amortization using the present depreciation rates. This approach avoids the uncertainty and the need to estimate the future costs of removal using statistical or other methodologies and replace it with the certainty of actual costs when incurred. The Commission retains the authority to revisit the depreciation (amortization) rates in subsequent rate proceedings if necessary and appropriate, but will have the advantage of actual costs that are known and measurable.

AMORTIZATION PERIOD SHOULD NOT BE ACCELERATED

Q. DESCRIBE THE COMPANIES' PROPOSAL TO ACCELERATE THE AMORTIZATION AND RECOVERY OF THE NET BOOK VALUE AND ESTIMATED COSTS OF REMOVAL OVER TWENTY YEARS.

A. Mr. Norman proposes a twenty-year amortization period. He claims that this "is necessary to fully recover all appropriate costs for these facilities within the retirement period proposed."[12]

Q. IS IT NECESSARY OR APPROPRIATE TO RECOVER THESE COSTS "WITHIN THE RETIREMENT PERIOD PROPOSED"?

A. No. The "retirement period" is not a valid basis for setting the amortization period. After the LPP assets are retired, there no longer is any nexus to the prior service lives. The

[12] Direct Testimony of Mr. Paul Normand at 7.

reclassification of the plant costs to regulatory assets severs any relationship with or conceptual underpinning to the prior service lives of the assets. The Companies implicitly acknowledge this fact in their own proposal to use a twenty-year amortization period in the calculation of the depreciation rates. Mr. Norman recommends a twenty-year average service life in the proposed depreciation rates. However, the weighted average remaining service lives for the LPP plant assets is only ten years, assuming that they are retired and replaced uniformly over the next twenty years. The Companies' proposal implicitly recognizes that the amortization of the regulatory assets is no longer tethered to the service lives of the assets, but rather is simply an amortization period that will be determined by the Commission and reflected in the depreciation (amortization) rates.

Q. WHAT ARE THE AVERAGE REMAINING SERVICE LIVES REFLECTED IN THE PRESENT DEPRECIATION RATES?

A. The average remaining service lives for each plant account are shown in the depreciation studies attached to Mr. Norman's testimonies. For KEDNY, they range from 9.5 years to 30.0 years,[13] with an average of 25.8 years. .For KEDLI, they range from 6.4 years to 38.2 years,[14] with an average of 29.9 years.

Q. WHY SHOULD THE COMMISSION MAINTAIN THE SERVICE LIVES REFLECTED IN THE PRESENT DEPRECIATION RATES AND REJECT THE COMPANIES' PROPOSAL TO ACCELERATE THE AMORTIZATION USING A TWENTY-YEAR AMORTIZATION PERIOD?

[13] Exhibit___(LK-4) page 1 column (14).
[14] Exhibit___(LK-5) page 1 column (14).

A. This approach will mitigate the impact on customers of the Companies' proposal to accelerate the LPP plant assets' retirement and replacement. The Company's proposal unnecessarily drives up the revenue requirement and compounds the impact of increased gas rates on customers. The retirement and replacement of the LPP plant assets results in two sets of costs that are included in the revenue requirement in these rate proceedings and in future rate proceedings.

The first set of costs is related to the return on and amortization of the net book value of the retired plant costs plus estimated costs of removal. The second set is related to the return on the net book value and depreciation of the plant in service costs of the replacement plant costs plus estimated costs of removal.

The two sets of costs overlap and both will be included in the revenue requirement until the first set is fully amortized and recovered. Under the Companies' proposal, the recovery of the first set of costs will be accelerated even as the second set of costs is incurred and added to the revenue requirement. In this manner, the Companies' proposal compounds the impact on customers over the shortened recovery period.

Q. WHAT IS YOUR RECOMMENDATION?

A. I recommend that the Commission maintain the present depreciation rates and the service lives reflected in those depreciation rates for the amortization period rather than accelerate the recovery by shortening the amortization period to twenty years.

ESTIMATED COST OF REMOVAL SHOULD NOT BE INCREASED

Q. PLEASE DESCRIBE MR. NORMAND'S STATISTICAL CALCULATION OF THE ESTIMATED COSTS OF REMOVAL.

A. Mr. Normand calculated historical annual percentage costs of removal for each of the plant accounts. He first calculated the costs of removal per mile by dividing the cost of removal by the miles of mains retired for each of the mains plant accounts and the cost of removal per service by dividing the cost of removal by the number of services retired for the services plant accounts. He then calculated the average annual cost of removal per mile and the average cost per service using different numbers of years, i.e., three years, four years, and seven years. He then settled on three-year annual averages for each account that had data. Finally, he calculated the annual cost of removal as a percentage of the gross plant. In some cases, he used the results from a similar account for another account that had no data.[15]

Q. ARE THE RESULTS OF THESE CALCULATIONS RELIABLE AND REASONABLE TO USE AS THE ESTIMATED COSTS OF REMOVAL?

A. No. The results are unreliable and are not reasonable to use as the estimated costs of removal. Mr. Norman's statistical approach may be appropriate for long-lived assets in the context of a normal depreciation study with limited retirements and costs of removal; however, the Companies' plan to retire all their remaining LPP plant assets, is well outside the context of a normal depreciation study.

In this case and considering the magnitude of the retirements and the related costs of removal, the superior approach is not to rely on estimates at all, but instead to defer the actual cost of removal to the regulatory assets. This approach replaces the uncertainty of

[15] Exhibit___(PNM-4) attached to Mr. Norman's Direct Testimony in both cases. I have attached a copy of the KEDNY version of Exhibit___(PNM-4) as my Exhibit___(LK-6) and the KEDLI version of Exhibit___(PNM-4) as my Exhibit___(LK-7) for ease of reference.

using estimates, however determined, with the absolute certainty of using the actual costs when those costs actually are incurred. Estimates are not necessary when actual costs will be available as they are incurred.

Second, the best estimate, if estimates are used, is to have the Companies' planning and engineering personnel estimate the costs in conjunction with the capital costs of the replacement mains and services. Such estimates would be based on the planned scope and timing of the retirement and replacement of the mains and services. In actual experience, the Companies simply allocate the total loaded payroll costs incurred between capital expenditures (plant) and cost of removal. The Companies allocate 70% to capital expenditures (plant) and 30% to cost of removal for mains, and allocate 90% to capital expenditures and 10% to cost of removal for services.[16]

Third, the Companies' actual historical experience has varied significantly and Mr. Norman has offered no evidence that any one historical year or any average of multiple historical years for estimated costs of removal is consistent with the scope or timing of the actual planned mains and services replacements. For example, the cost of removal for account 376.06 *Mains-Cast Iron* for KEDNY ranged from a low of $14,652 per mile to retire 84.6 miles to a high of $2,188,307 to retire 0.8 miles. As another example, the cost of removal for account 376.18 *Mains-Plastic* for KEDLI ranged from a low of $71,364 per mile to retire 0.9 miles of main to a high of $17,807,549 to retire 0.01 miles of main. In none of the prior years did KEDLI retire more than 1.15 miles.[17] There is no evidence that

[16] Response to City of CNY-73. I have attached the narrative portion of this response as my Exhibit___(LK-8).
[17] *Id.*

these historical costs, incurred for different scopes of work, and in many cases, extremely limited retirements and limited activities, provide reliable or reasonable bases to estimate the costs for the retirements of all the remaining LPP plant assets, even if averaged in some manner.

Fourth, the statistical approach to incorporate the costs of removal in the depreciation rates further magnifies the effects of using estimated costs. The estimated costs of removal per mile and per service were multiplied by the miles of main remaining for each account and the services remaining in each account to calculate the estimated dollars for the cost of removal, which Mr. Normand then divided by the gross plant in the account to calculate the percentage for the cost of removal (net negative salvage) component of the depreciation rates for each plant account. These calculations resulted in proposed net negative salvage rates of as much as 1,968% for KEDNY and 601% for KEDLI.[18] In comparison, the current range of net negative salvage rates reflected in the present depreciation rates is 4% to 65% for KEDNY and 0% to 14% for KEDLI.[19]

Finally, the results of Mr. Normand's analyses and proposed depreciation rates simply are not reasonable. The costs of removal resulting from his analyses sum to $476.718 million for KEDNY and to $409.523 million for KEDLI. These costs of removal compare to $72.342 million in remaining net book value for KEDNY and $80.017 million for KEDLI. At a minimum, the Companies should compare these statistical estimated costs of removal to the actual planned costs of removal in conjunction with the replacement of

[18] Exhibit___(LK-2) for KEDNY column (7) and Exhibit___(LK-3) for KEDLI column (7).
[19] *Id.*

the LPP plant assets using the Companies' standard allocation of only 10% to 30% of the loaded payroll costs to costs of removal, with the remainder to the replacement plant. In any event, my recommendation to defer the actual costs of removal to the regulatory assets avoids the need to calculate estimated costs of removal or argue over the methodology.

Q. WHAT IS YOUR RECOMMENDATION?

A. I recommend that the Commission maintain the present depreciation rates and direct the Companies to defer the costs of removal to the regulatory assets for future recovery after the costs actually are incurred and known with certainty. This approach is reasonable and equitable to the Companies and their customers. This approach will provide recovery of the actual incremental costs of removal after they are incurred and are known and certain.

Q. DOES THIS COMPLETE YOUR DIRECT TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of

New York City Gas Infrastructure and Safety Panel

Michael McCutcheon
Ke Wei
Benjamin Krakauer
Pinar Balci, Ph.D.
Chief David Morkal

On Behalf of:

The City of New York

August 30, 2019

Cases: 19-G-0309 NYC GAS INFRASTRUCTURE AND SAFETY PANEL
19-G-0310

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Mr. McCutcheon – My name is Michael McCutcheon. I am currently a Pipeline Coating consultant for Quality Integrated Services, Inc. My business address is 35 North Church Street, Cortland, New York 13045.

Ms. Wei – My name is Ke Wei. I am Assistant Director for Infrastructure for the New York City Mayor's Office of Sustainability and New York City Mayor's Office of Resiliency. My business address is 253 Broadway, 14th floor, New York, New York.

Mr. Krakauer – My name is Benjamin Krakauer. I am the Assistant Commissioner for Strategy & Program Development at New York City Emergency Management, or NYCEM. My business address is 165 Cadman Plaza East, Brooklyn, New York.

Ms. Balci – My name is Pinar Balci. I am the Assistant Commissioner in the Bureau of Environmental Planning and Analysis at the NYC Department of Environmental Protection or DEP. My business address is 59-17 Junction Boulevard, Floor 11, Flushing New York.

Chief Morkal – My name is Battalion Chief David Morkal. I am the Executive Officer to the Chief of Operations for the Fire Department of New York, or FDNY. My business address is 9 Metrotech Center, Brooklyn, New York 11201.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. We are submitting this direct testimony before the New York Public Service Commission ("PSC" or the "Commission") on behalf of the City of New York ("City").

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. McCutcheon – I hold a number of certifications from a variety of professional organizations, including the American Petroleum Institute (Certified Pipeline Inspector) and the American Welding Society (Certified Welding Inspector). I have completed all training necessary to perform audits for Pipeline and Hazardous Materials Safety Administration. In 2004, I completed a certification from the New York State Fire Academy for fire behavior, arson investigation, and fire investigation. I have also completed the NACE International Coating Inspector Program, Level 1.

Ms. Wei – I received a Bachelor of Arts degree in International Relations from Pomona College and a Master in International Affairs with a focus in Energy Management and Policy from Columbia University School of International & Public Affairs.

Mr. Krakauer – I hold a Master's Degree in Public Administration from Binghamton University's College of Community and Public Affairs and a Baccalaureate Degree in Political Science from Binghamton University's Harpur College of Arts and Sciences. Additionally, I am a graduate of the Corning

Community College/Arnot-Ogden Medical Center Emergency Medical Technician-Paramedic Program.

Ms. Balci – I received a bachelor of science from Hacettepe University in Turkey in 1994 and I received a master of science from Hacettepe University in Turkey in 1996. Subsequently, I received a doctorate of philosophy from the University of North Texas in 2002. Following, I received a post-doctorate from Murray State University in Kentucky in 2004.

Chief Morkal – I received a Bachelor of Fine Arts from the University of Evansville in 1981. I am a graduate of the FDNY's Fire Officers Management Institute, Advanced Leadership Course, West Point Combating Terrorism Leadership Program, L-380 "The Point of the Spear", L-381 "Incident Leadership" L-481, and L-580, Staff Ride at Gettysburg National Military Park.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Mr. McCutcheon – I have over 42 years of practical field experience and knowledge construction of liquid and natural gas systems, with experience in all phases of pipeline construction, from large transmission to smaller distribution systems, from wellhead to meter. From 1985 to 2015, I was employed by the NYS Department of Public Service as a Utility Analyst for both regulatory compliance and incident investigation. In this role, I was the lead investigator for 37 incidents, conducting a code and document review for culpability, cause and origin, facility testing, evidence procurement, and root cause analysis. I also conducted inspections for

safety and regulatory compliance of pipeline facilities within New York State, which included manual review, field observation and trailing audits, construction monitoring and review. Since 2016, I have served as an inspector and quality assurance/quality control inspector for several companies, including in my current position as a Pipeline Coating consultant for Quality Integrated Services, Inc.

Ms. Wei – I have been in my current position since January 2016. In this role, I advance and manage regulatory and policy issues and relevant research related to sustainability and resiliency of energy supply and distribution across electricity, natural gas, and fuels. I have spoken at various local, regional, and international conferences and workshops on energy resiliency and sustainability issues. Prior to joining the City, I was a Program Director at Columbia University's Center on Global Energy Policy, including managing their fuel resilience research program. Before that, I was a management consultant with McKinsey & Company.

Mr. Krakauer – My career in emergency services began more than 20 years ago as a volunteer for a local fire department and rescue squad in Suffolk County, New York. Through high school and college, I worked as a Certified First Responder, Emergency Medical Technician, Advanced Emergency Medical Technician – Critical Care, Paramedic, and emergency medical services instructor for a number of departments throughout New York State, including serving as the Chief/Executive Director of a volunteer ambulance service and serving as a Manager and Controlled Substances Officer for a commercial ambulance service that provided coverage to two counties in upstate New York. In 2007, I was

appointed as the Emergency Preparedness Coordinator for Broome County, New York where I focused on all-hazards emergency planning, emergency operations center operations, and public health preparedness. I served as the Incident Commander during Broome County's response to the H1N1 Influenza Pandemic and served as the Planning Section Chief and Deputy Emergency Manager during the American Civic Association mass shooting. In 2010, I joined NYCEM as a Public Warning Specialist, working in Watch Command, sending out emergency messages to the public, and building the City's emergency public notification program, a role which I held for almost two years. I then spent the next six years as the Deputy Director and then Director of NYCEM's Watch Command, where I was responsible for the strategic and operational direction of NYCEM's 24x7 operations center, on-duty emergency managers, and the Notify NYC program. I am currently the Assistant Commissioner for Strategy and Program Development where I oversee intra- and interagency high priority initiatives on behalf of the Commissioner and First Deputy Commissioner, including the citywide mass notification contract, the on-call emergency contracts program, and the interim flood protection measures program. I serve as the leader of one of NYCEM's three on-call teams, consisting of 50 professionals responsible for coordinating the City's response to emergencies and disasters and leading the Emergency Operations Center, or EOC, when activated.

Ms. Balci – I have been with DEP for over seven years. I oversee and manage multiple programs, such as the Green Infrastructure Program, the Municipal

Separate Storm Sewer System Program, integrate planning, and provide City Environmental Quality Review reviews for water and wastewater infrastructure projects. Prior to joining DEP, I worked at the South Florida Water Management District, where I managed the Federal and State Policy and Permitting Division to achieve the goals of the Everglades restoration and capital projects. I also worked as Northern Everglades Implementation Manager overseeing and coordinating all planning, permitting and capital project activities.

I specialize in management, coordination and implementation of water resources and environmental restoration projects. I am familiar with the range of pertinent environmental regulations including the National Environmental Policy Act, the Endangered Species Act, the Clean Water Act and associated requirements.

Chief Morkal – I am a 29-year veteran with the FDNY. I was promoted to Lieutenant the week after the World Trade Center attacks, where I worked in Ladder 18 on the Lower East Side. After being promoted to Captain I was transferred to Engine 23 in Midtown Manhattan. As a Battalion Chief I was assigned to Battalion 4 and worked as the Chief in Charge of Probationary Firefighter School for 2 years. I also worked as the Planning Section Chief for the FDNY Incident Management Team and I am currently one of the Deputy Incident Commanders. I have been activated with the team for Hurricane Gustav in Louisiana, Hurricane Irene in both New York City and Binghamton, New York, Hurricane Sandy in New York City, the snow storm in Buffalo, New York, the East Harlem explosion and collapse, and several wild land fires in the Southwest

with other Incident Management Teams. I have served in my current position as Executive Officer to the Chief of Operations since March 2019.

Q. DO YOU HAVE PRIOR EXPERIENCE TESTIFYING?

A. Mr. McCutcheon – Over the last 30 years, I have made public presentations and been on panels for public information mostly related to damage prevention. I have not previously testified before the PSC in a utility rate case.

Ms. Wei – I speak frequently before international, national and local audiences on the City's energy policy and resilience goals. I also testified in the ongoing Consolidated Edison Company of New York, Inc. electric rate case, Case 19-E-0065, as a member of the New York City Electric Infrastructure Panel.

Mr. Krakauer – I have testified before the New York City Council and a U.S. House of Representatives subcommittee on providing emergency warnings to the public. I also testified on behalf of the City in the ongoing Consolidated Edison Company of New York, Inc. electric rate case, Case 19-E-0065.

Ms. Balci – I do not have prior experience testifying.

Chief Morkal – I have testified before the New York City Council on behalf of the FDNY. I have not previously submitted testimony to the PSC.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. Our testimony will cover five broad areas pertaining to National Grid's gas rate filing. We comment on The Brooklyn Union Gas Company d/b/a National Grid NY's ("KEDNY") and KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I.'s ("KEDLI"; and together with KEDNY, "National Grid" or the "Companies") Gas

Infrastructure and Operations Panel's ("GIOP") pre-filed direct testimonies and their associated exhibits, supplements, and updates. We also comment on National Grid's Gas Safety Panel's pre-filed direct testimony and its associated exhibits, supplements, and updates. Our testimony focuses on the following issues and recommendations:

- National Grid is proposing significant increases in capital spending, nearly tripling from its 2014 spending, with even larger increases in specific programs, namely Leak Prone Pipe replacement and City/State Construction ("CSC"). In light of the increased spending, the programs within the capital budget must be reviewed critically. To protect ratepayers from these significant increases (if the increases are justified), the City recommends two proposals. First, the Commission should continue to require a reconciliation mechanism so that any amounts unspent are returned to customers. Second, the Commission should reject the Companies' proposed changes to the CSC tracker and O&M reconciliation mechanisms.
- National Grid is proposing several storm hardening and system safety projects, including upgrades to the Greenpoint LNG Plant and the proposed installation of remote service shutoff valves. While the City is conceptually supportive of these proposals, they require additional study, greater quality control, or more coordination with the City to improve their effectiveness. Beyond what National Grid has proposed, the City also proposes that

National Grid implement recommendations to harden priority mini-gate and take stations that were contained in the Companies' Storm Hardening Collaborative Report.

- The City is making efforts to offset the deleterious effects of Combined Sewer Overflows (CSOs) through Green Infrastructure Assets, in order to comply with federal clean water act requirements that are enforced by the State. These assets have suffered damage as a result of National Grid actions/inactions. Therefore, the City asks that National Grid coordinate with the City and receive the appropriate training to prevent further damage to these Assets.
- KEDNY and KEDLI propose numerous changes to their array of gas safety-related metrics, including changes to both positive and negative revenue adjustment mechanisms. The City opposes many of these changes as they either seek to reward shareholders for business-as-usual performance or merely reduce the likelihood that the Companies will be subject to negative revenue adjustment without material changes in behavior. Accordingly, the City recommends a series of changes to the Companies' proposed incentive mechanisms to encourage more desirable outcomes.
- Finally, both Companies have a substantial balance of regulatory liabilities. In lieu of using these liabilities to pay for other specific programs, as suggested by National Grid, the full balance of these accounts, with the appropriate carrying charges, should instead be returned to ratepayers.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. We offer Exhibit__(NYCGISP-1), which is a compilation of the discovery responses from National Grid that we relied upon in preparing our testimony, some of which we expressly reference herein.

Exhibit__(NYCGISP-2), which is a compilation of relevant regulations and practice guidelines that we relied upon in preparing our testimony, some of which we expressly reference herein.

Exhibit__(NYCGISP-3), which is a summary of our proposed incentive mechanisms and safety metrics.

CONCERNS WITH THE OVERALL CAPITAL PROGRAM

Q. DO YOU HAVE ANY CONCERNS WITH NATIONAL GRID'S COMPREHENSIVE SUITE OF PROPOSED CAPITAL PROJECTS?

A. According to each Company's GIOP pre-filed testimony, the volume of the Companies' planned capital work continues to increase significantly from 2020 through 2024. The capital work is driven primarily by an aggressive Proactive Main Replacement ("LPP") Program, and KEDNY's CSC Program. The major drivers for the increase in gas capital expenditures in the Rate Year (which are also generally applicable to Data Years ("DY") 1, 2, and 3) include: the LPP Program ($485.3 million), CSC Program ($297.1 million), specific infrastructure upgrades, including KEDNY's Metropolitan Reliability Infrastructure Project and KEDLI's Northwest Nassau Project ($199.2 million, total), and System Integrity Programs ($19.0 million). KEDNY sets forth its major programs on pages 25-53 of its GIOP

testimony, and KEDLI addresses its major programs on pages 25-44 of its GIOP testimony. Both KEDNY's and KEDLI's Exhibit___(GIOP-1CU) detail these major drivers, as well.

Q. WHAT ARE YOUR SPECIFIC CONCERNS WITH THESE PROPOSED INVESTMENTS?

A. We are concerned that this level of capital expenditure is not realistic, and that National Grid may not be able to prudently increase its spending by such amounts while maintaining effective and efficient management, supervision, and cost controls to obtain the desired result of the various programs and projects. We also are concerned as to whether National Grid – as well as its external contractors – has the adequate skilled personnel to competently and safely perform all of the proposed capital work in accordance with all applicable regulations.

Q. WHAT IS THE BASIS FOR YOUR CONCERNS?

A. As mentioned above, the proposed expenditures continue to increase dramatically. The Companies' combined past actual capital spending inclusive of cost of removal, as seen in the response to DPS-84, was $488.7 million in 2014, $600.7 million in 2015, $729.0 million in 2016, $769.7 million in 2017, and $1.04 billion in 2018. In the current proceeding, National Grid is outlining a capital budget proposal of $1.42 billion in the rate year. This is nearly triple the capital spend from 2014.

As the level of spend increases and these programs get larger, it becomes more difficult to manage, requiring more resources, materials and coordination

effort. This places strain on both internal and external resources. Additionally, these programs require new processes and procedures to be effectively managed and remain sustainable for the long term. Thus, it is important that appropriate project controls are in place to ensure prudent use of capital and that programs are managed effectively. We acknowledge that National Grid is proposing to fill approximately 252 and 495 incremental positions by the end of the Rate Year and Data Years, respectively. The Commission should ensure that the Companies have adequate project management staff to handle the proposed increase in capital expenditures effectively.

Q. BEYOND THE MAGNITUDE OF THE COST INCREASES, DO YOU HAVE ANY OTHER CONCERNS?

A. Yes. Recently, the Commission issued its Order Instituting Proceeding and to Show Cause (the "Order to Show Cause") in Case 17-G-0317. The allegations have not been proven; however, they are credible enough for the Commission to require National Grid to answer for them. The sheer number of alleged violations – 1,616 so far, with possibly more to come, according to page 14, fn. 22 of the Order to Show Cause – and the nature of the alleged violations raises legitimate concerns over whether National Grid can safely and prudently manage a capital budget of this size.

Q. WHAT ABOUT THE NATURE OF THE ALLEGED VIOLATIONS RAISES CONCERNS ABOUT PRUDENT SYSTEM MANAGEMENT?

A. These alleged violations, if proven true, are fundamentally safety violations resulting from a material oversight failure by National Grid. Pages 13-14 of the Order to Show Cause concisely summarize the nature of the alleged violations. The New York State Department of Public Service Staff found: (i) that the Queens Pipeline had been installed improperly, "either in very close proximity to, or directly in contact with, various other underground utilities;" (ii) installation of mains and main replacements were done by unqualified contractors; and (iii) joining work was completed by unqualified workers. That these alleged violations arose from unqualified contractors and not National Grid employees does not exculpate the Companies. The consequences of a gas pipeline rupture are the same regardless of whether a contractor or employee violated safety regulations. Moreover, National Grid is responsible for managing its contractors as well as its systems. If National Grid wants to increase the scale and scope of its capital programs at ratepayer expense, it must first demonstrate that it can do so safely and prudently, including identifying gaps in its current project management and oversight approach, and developing strategies to remediate those gaps to ensure that safety concerns are avoided or mitigated.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. In our view, the Main Replacement and Service Replacement Programs should be planned to more thoroughly take advantage of bundling projects, minimizing disruptions to the communities, and ensuring construction is as efficient as possible, thereby reducing the overall program costs. We recommend that the

Companies' risk model for main replacement be reviewed annually to ensure maximum optimization of main work and that the Companies continue to identify opportunities to coordinate with City projects involving street works.

Additionally, the Companies must be able to exhibit that the safety and reliability of their systems is improved at a rate commensurate with the proposed spending, and that the capital programs are prudently managed at a level commensurate with the spending. If that cannot be shown, then the main replacement programs should be decelerated to an appropriate rate. Finally, given the magnitude of the capital budgets the Companies are requesting, we recommend that the Commission require a reconciliation mechanism so that any amounts unspent are returned to customers.

Q. ARE THE COMPANIES PROPOSING ANY CHANGES TO EXISTING RECONCILIATION MECHANISMS?

A. Yes, they are. The Companies propose to continue the downward-only net plant capital tracker that was approved in the 2016 KEDNY and KEDLI Rate Cases. We support this proposal. However, the Companies also have an existing CSC tracker mechanism that currently requires a 90 percent deferral of the difference between forecast and actual CSC spending. The Companies propose to modify this tracking mechanism to a full reconciliation of the difference between forecast and actual spending. KEDNY also proposes to add an O&M tracker "that would fully reconcile the difference between forecast and actual spending in the non-

reimbursable "Support and Protect" O&M expense category" which are certain non-reimbursable CSC O&M expenditures.

Q. DO YOU AGREE WITH THE PROPOSED CHANGES TO THE CSC-RELATED RECONCILIATION MECHANISMS?

A. No. The Companies state that the proposed CSC tracker changes and the new Support and Protect O&M expense tracker is based on the "increasing volatility and scope of unplanned CSC projects…" However, the City's Policy Panel testimony, which is being filed concurrently with this testimony in these cases, is recommending that National Grid designate a high-ranking official to serve as a designated liaison with the City specifically with respect to CSC projects, and that National Grid submit regular reports to the Commission regarding the status of pending CSC projects so that performance can be objectively tracked over time. These recommendations, if adopted, should help address the Companies' stated concerns over increased volatility and scope surrounding CSC. Therefore, the proposed changes to the CSC tracker are not necessary.

In addition, given the size and scope of the CSC program, a full reconciliation mechanism introduces additional, unknown ratepayer exposure. For example, KEDNY has a CSC budget of almost $300 million per year, so even a 10 percent variation between forecasted and actual costs could have significant ratepayer impacts. The Companies should not be insulated from these variations, and doing so would reduce the Companies' incentives for accurate forecasts and effective cost management. Indeed, the Companies themselves recognize the

importance that accurate forecasting plays within the ratesetting process, particularly as it relates CSC costs. In response to DPS-352, KEDNY argues for setting rates using its forecast of actual CSC costs, noting that the "CSC tracker is…not a substitute for setting rates." The Companies' proposal for a full CSC reconciliation mechanism would, however, create just such the substitute that the Companies claim they want to avoid.

For these reasons, the Commission should reject the proposed changes to the Companies' CSC tracker and reject the similar, but related, O&M tracking mechanism.

CONCERNS WITH SPECIFIC CAPITAL PROGRAMS

A. LEAK PRONE PIPE

Q. DO YOU HAVE ANY CONCERNS REGARDING THE COMPANIES' LPP REPLACEMENT PLANS?

A. Yes. The Companies request rate allowances to replace the following miles of LPP in calendar years 2020-2023:

- KEDNY: 70, 75, 80, 85;
- KEDLI: 155, 155, 155, 155.

Minimum removal targets for KEDNY to avoid negative revenue adjustments ("NRAs") are proposed to continue to increase by 5 miles per year. For 2019-2022, minimum targets would be: 60, 65, 70, and 75, with a cumulative minimum of 310 miles for calendar years 2020-2023. Minimum removal targets for KEDLI are held steady at 150 miles per year. The Companies further propose:

- Continuing an existing incentive of two basis points ("BPs") for each full mile of LPP removed above the rate allowance thresholds, capped at 10 BPs.
- Continuing an existing surcharge mechanism (the Gas Safety and Reliability Surcharge, or "GSRS") that allows the Companies to recover costs of retired LPP incremental to the level funded in base rates.

Q. WHAT CONCERNS DO YOU HAVE WITH THESE PROPOSALS?

A. National Grid plans to spend an average of $571.7 million annually on its LPP Program, with an additional $19.7 million annual average on its Service Replacement Program, from 2020 to 2024 (comprising the rate year and data years). These programs are for the replacement of leak prone gas mains and services (defined as small diameter (12" and smaller)) that are: (i) cast and wrought iron; (ii) unprotected steel (non-cathodically protected; (iii) unprotected steel/wrought iron services; and (iv) copper and pre-1974 high-density polyethylene services.

We evaluated the programs and have identified that the combined KEDNY and KEDLI costs associated with these programs have significantly increased from 2014 to the proposed Rate Year spending levels by approximately 645% for the LPP program. The majority of the increases are associated with increased mileage of pipe and services replaced, asphalt and paving of roads and curbs due to the replacement, as well as higher overheads.

Q. PLEASE CONTINUE.

A. As noted above, the Companies' budgets for the LPP Program have reached historic levels, and those levels will only continue to increase under the Companies' proposals. While we agree with the Companies that it is important to continue removing LPP, as doing so is consistent with Commission policy and will reduce leak rates on the Companies' systems, LPP removal efforts must be balanced against customer impacts.

As a result, we support the Companies' requested rate allowances for proactive LPP removal activities, but the requests for an incentive for additional miles above the rate allowance threshold and the request to fund additional miles replaced via the GSRS should be rejected. To the extent that either Company incurs incremental costs to replace additional LPP miles above and beyond the rate allowance threshold, the costs associated with those replacements should be treated similar to other capital projects and deferred for examination and possible recovery in a future rate case.

Moreover, because the Companies' LPP Programs are now designed to replace or retire all LPP within the 20-year horizon called for under Commission policy, there is no need to provide shareholder incentives to the Companies to the extent that either Company exceeds the rate allowance threshold; the Companies do not need an incentive to conform to Commission directives. The existing incentive for each full mile of LPP removed above the rate allowance thresholds should therefore be discontinued.

Q. SHOULD THE COMMISSION CONTINUE THE EXISTING POSITIVE REVENUE ADJUSTMENT IF THE COMPANIES MEET SPECIFIC UNIT COST TARGETS RELATED TO LPP REMOVALS?

A. No. The Companies currently have an incentive mechanism that allows the Companies to earn a positive revenue adjustment ("PRA") of up to 10 BPs if the Companies can meet specific unit cost targets associated with LPP replacements. The Companies propose to continue this incentive mechanism as it currently exists, which means that the unit cost targets embedded in the incentive mechanism would be updated to reflect the unit costs for LPP replacements that are embedded in rates. KEDNY proposes that it start earning incentives for unit costs at or below $1,285/foot, and KEDLI proposes that it start earning incentives for unit costs at or below $306/foot. As a point of comparison, in the current Rate Plan for CY 19, which was set in 2016, KEDNY's incentives begin at $664/foot, and KEDLI's begin at $238/foot. During the current rate plan, the unit costs at which the Companies would begin to earn incentives only increases year-over-year by between 2.5% and 5%.

Q. WHAT ARE YOUR SPECIFIC CONCERNS WITH THIS PROPOSAL?

A. The Companies claim that unit costs for LPP replacements have increased significantly since the last rate case. Thus, for purposes of the LPP unit cost PRA, the Companies argue that the unit cost targets that serve as the starting point for determining whether the Companies have achieved any unit cost savings should also increase under the Companies' proposal.

Under the Companies' proposal, the point at which the Companies would begin earning incentives would be updated to $1,285/foot for KEDNY and $306/foot for KEDLI, respectively, which between CY19 and FY21 represent a 93.5% increase in KEDNY's unit costs and a 28.6% increase in KEDLI's unit costs. While we do not comment on the specific drivers behind the alleged unit cost increases, the proposed threshold resets under the Companies' proposed PRAs nevertheless represent dramatic increases in starting points for both Companies, which raises fundamental questions about the appropriateness of continuing this mechanism 'as-is.' As demonstrated in the charts below, KEDNY in particular has struggled to come anywhere close to earning the productivity incentive under the current rate plan, with actual unit costs in 2017 and 2018 nearly doubling the minimum unit cost incentive thresholds. KEDLI's performance has been better but it, too, has been unable as of yet to achieve any incentives under this mechanism.

KEDNY Historical LPP Removal Productivity Performance (DPS-494)

Year	Minimum incentive threshold (2 BP)	Highest incentive threshold (10 BP)	Actual unit cost (approximate)	Incentive earned (BP)
2017	$618/ft	$580/ft	$630/ft	No (0)
2018	$648/ft	$609/ft	$846/ft	No (0)

Cases: 19-G-0309 NYC GAS INFRASTRUCTURE AND SAFETY PANEL
19-G-0310

KEDLI Historical LPP Removal Productivity Performance (DPS-597)

Year	Minimum incentive threshold (2 BP)	Highest incentive threshold (10 BP)	Actual unit cost (approximate)	Incentive earned (BP)
2017	$223/ft	$209/ft	$250/ft	No (0)
2018	$232/ft	$218/ft	$272/ft	No (0)

Q. IS INCREASING THE THRESHOLD AT WHICH THE COMPANIES MAY EARN AN INCENTIVE IN RATEPAYERS' INTERESTS?

A. No. We certainly support cost-effectiveness, as this PRA seeks to incentivize. However, the Companies' modifications do not incentivize cost-effectiveness. Rather, the modifications requested by the Companies essentially seek to reset the target to move the starting point closer to the Companies' actual per-foot costs to make this incentive easier to achieve. The Companies thus propose to modify this measure to reward business-as-usual behavior instead of rewarding meaningful main replacement cost reductions. Ratepayers, under this proposal, would have to pay for unit costs only a modicum less expensive than the current unit costs and reward the Companies for doing so. Narrowly beating current actual costs that far exceed prior forecasts should not require incentives.

Q. WHAT DO YOU RECOMMEND?

A. The threshold at which the Companies may receive an incentive should remain the same as it is in the current Rate Plan and increase yearly at no more than the same rate than which it increased under the current Rate Plan (e.g., between 2.5-5% per year). As shown in Exhibit___(NYCGISP-3), we are proposing a 3.7% yearly unit cost increase for KEDNY and a 3.5% yearly unit cost increase for KEDLI, which is consistent with the average yearly increases embedded within the current Rate Plan. This adequately rewards cost-effective solutions while ensuring that ratepayers do not reward the Companies for business-as-usual performance. If the Companies contend that an incremental increase to the current unit cost threshold renders the incentive meaningless because it is too difficult to achieve, then this incentive mechanism should be discontinued entirely. LPP unit costs have been rising significantly. The table below, which compiles the responses to DPS-494 and DPS-597, illustrates these increases.

KEDNY and KEDLI LPP Unit Costs ($/ft)

	FY 2015	**FY 2016**	**FY 2017**	**FY 2018**	**FY 2019**
KEDNY	614	478	630	846	1031
KEDLI	227	207	250	272	327

Alternatively, given the significant increases in LPP unit costs in recent years, the prudent course may be to suspend this incentive mechanism for the time being and revisit it once replacement costs have stabilized.

Q. DO YOU HAVE ANY CONCERNS REGARDING THE COMPANIES' REPLACEMENT OF LEAK-PRONE PIPE IN FLOOD-PRONE AREAS?

A. Yes.

Q. WHY IS THE COMPANIES' REPLACEMENT OF LEAK-PRONE MAINS AND SERVICES IN FLOOD PRONE AREAS IMPORTANT?

A. During a flood, the pressure of the flood waters outside of the gas main can be greater than the pressure in the gas main, especially as it pertains to low pressure mains and services. This causes water to enter through any of the leaks (*e.g.*, cracks, holes) present in the pipe. Once there is water intrusion in the mains, it can be a lengthy process to dewater the main, which can require shutting off gas to the affected main and relighting the system.

Q. HOW MUCH LPP DOES EACH COMPANY CURRENTLY HAVE WITHIN THE CURRENT 2015 FEMA FLOODPLAIN?

A. According to the responses to CNY-46 and CNY-47, as of December 31, 2018, KEDNY and KEDLI had the following miles of LPP located within the 2015 FEMA floodplain:

	KEDNY (miles)	**KEDLI (miles)**
Low Pressure	81.8	65.7
High Pressure	13.7	126.9

Q. DO THE COMPANIES SPECIFICALLY TARGET REMOVAL OF LPP WITHIN THE FLOODPLAIN?

A. No. As noted in the response to CNY-48, the Companies have "no distinct timelines for retirement of LPP by type or location." The Companies do, however, consider location within the FEMA's 2015 Preliminary Flood Insurance Rate Maps (PFIRMS) 1% annual chance floodplain (referred to hereafter as the "2015 FEMA 100-year floodplain") and the FEMA 2015 Preliminary Flood Insurance Rate Maps (PFIRMS) 0.2% annual chance floodplain (referred to hereafter as the "2015 FEMA 500-year floodplain") within their main replacement prioritization methodology. In particular, the prioritization methodology includes a 'tie-breaker' mechanism that gives priority to LPP in the 2015 FEMA 100-year or 2015 FEMA 500-year floodplain,[1] in the event that two or more sections of LPP have identical prioritization scores. According to KEDNY's response to CNY-14(i) and CNY-77, KEDNY will remove an additional 2.9 miles of low-pressure LPP in the 2015 FEMA floodplain under its Proactive LPP Program as a "direct result" of this 'tie-breaker' mechanism. KEDLI will remove 13.9 miles, according to its response to CNY-77, but it has not indicated whether this is the direct result of including this 'tie-breaker' mechanism. We understand that, each year, in addition to the LPP replaced pursuant to the Proactive LPP Program, both KEDNY and KEDLI will likely replace several miles of low pressure LPP in the floodplain

[1] Throughout this testimony, when we refer to the 2015 FEMA 100-year and 500-year floodplain, we are referring specifically to FEMA's 2015 PFIRMs 1% annual chance floodplain (100-year floodplain) and FEMA's 2015 PFIRMs 0.2% annual chance floodplain (500-year floodplain).

pursuant to other main replacement programs, including under the CSC Program. These efforts are evidenced in the Companies' response to CNY-76 where, for example, KEDNY indicated that it replaced nearly 24 miles of low pressure LPP in the 2015 FEMA floodplain in 2018, despite having no mechanism in place in 2018 to specifically target LPP in the floodplain.

Q. WHAT IS YOUR RECOMMENDATION?

A. The Companies should proactively improve their efforts to prioritize replacement of leak-prone services and mains in flood prone areas. Initially, the Companies should continue to utilize the tie-breaker mechanism in their main replacement prioritization methodology. However, the Companies' efforts should be bolstered such that each Company is required to achieve total replacement of low pressure LPP in the 2015 FEMA 500-year floodplain by 2030 (*i.e.*, within 10 years). This would require KEDNY and KEDLI to remove or replace approximately 8.2 miles and 6.6 miles per year, respectively, of low pressure LPP in the 2015 FEMA 500-year floodplain – an amount that should be readily achievable without any requirement for incremental funding. According to CNY-76, KEDNY has replaced an average of 10.7 miles per year of low pressure LPP in the 2015 FEMA floodplain[2] from 2014-2018, while KEDLI has replaced an average of 7.9 miles per year over the same timeframe.

[2] National Grid's discovery and testimony on this issue does not specify whether this is within the 2015 FEMA 100-year or 500-year floodplains, but based upon past conversations with National Grid, The City believes it is referring to the 2015 FEMA 500-year floodplain

Q. WOULD YOU SUPPORT AN ACCELERATED EFFORT TO REPLACE LOW PRESSURE LPP IN THE FLOODPLAIN?

Yes, we could support an incentive for accelerated low-pressure LPP replacement in the 2015 FEMA 500-year floodplain and the 2015 FEMA 100-year floodplain. Because of the added risk posed by low-pressure LPP in the floodplain, as discussed herein, an added incentive to hasten floodplain LPP removal could be justified. Thus, we propose a one BP PRA for each full incremental two miles of low-pressure LPP replaced in any given year, but only if the Companies exceed the minimum targets for low-pressure LPP in the floodplain recommended in this testimony, and only if that incremental low-pressure LPP is located within the 2015 FEMA 500-year floodplain (which also encompasses the 100-year floodplain). This incentive would be capped at five BPs. This proposal represents a more appropriate balance between ratepayer and shareholder interests.

B. <u>STORM HARDENING</u>

Q. DO YOU HAVE ANY RECOMMENDATIONS RELATED TO NATIONAL GRID'S STORM HARDENING PROGRAMS?

A. Yes. Many of our recommendations relate specifically to items that were addressed by the Companies in the "2017 Storm Hardening Collaborative Report," which was filed on April 26, 2018 in Cases 16-G-0058 and 16-G-0059. We specifically address four items that were discussed in the 2017 Storm Hardening Collaborative Report: (1) mini-gate and take stations located within the current 2015 FEMA 100-year and 2015 FEMA 500-year floodplains; (2) height of meter

sets in floodplains; (3) the flood study for the Greenpoint LNG plant site; and (4) remote shut-off valves. We also discuss our concerns with the Companies' proposed Expanded Residential Methane Detection program.

Q. WHAT ARE YOUR CONCERNS RELATED TO MINI-GATE AND TAKE STATIONS LOCATION WITHIN THE FUTURE 2050s FLOODPLAIN?

A. The 2017 Storm Hardening Collaborative Report identified 13 mini-gate and take stations that are located within the current 2015 FEMA 100-year and 500-year floodplains, as well as the future 2050s floodplain. Eleven out of these thirteen mini-gate and take stations are located within the current 2015 FEMA 100-year and 500-year floodplains. With regard to these 13 stations, National Grid agreed "[w]here possible and cost effective…[to] consider design changes to the height of telemetry cabinets using [design flood elevations] based upon the City's climate resilience design guidelines methodology for critical facilities with useful lives through 2050s."

Considering the asset class that is at stake – high- and medium-pressure mini-gate and take stations that are located within the current and/or future floodplains – it is extremely important that National Grid storm hardens these stations, including mitigating water intrusion into the underground vaults, elevating telemetry cabinets, and protecting vent poles to ensure that National Grid's control center retains the ability to remotely monitor and communicate with these assets during flooding events. Loss of communications during a flooding event leaves system operators blind to potential pressure build-ups and other issues

that could have the potential for severe neighborhood or system-level consequences. From the City's understanding of this list, we would request that National Grid in particular prioritize the eleven stations that are located in the current 2015 FEMA 100- and 500-year floodplains and mitigate these risks to these gate stations by 2025 at the latest, if not sooner.

Q. HOW IS NATIONAL GRID ADDRESSING THESE 13 MINI-GATE AND TAKE STATIONS?

A. Based upon the Companies' discovery responses, the City understands that one of the stations (Coney Island) from the list of eleven has been storm hardened, including elevating and/or protecting telemetry cabinets due to work done in 2018. Another one (Kings Plaza) is scheduled to be storm hardened in FY 23, and two others should be rehabilitated within a decade. Two new gate stations (Chelsea and Downtown Brooklyn) were constructed with storm hardening guidance in mind. National Grid claims that the other seven stations do not require elevation, and states that other regulator stations in the area can provide remote sensing data indicative of their functioning in the event of a weather event affecting these seven stations. National Grid provided this explanation in CNY-108.

Q. IS NATIONAL GRID PROPERLY ADDRESSING THESE 13 MINI-GATE AND TAKE STATIONS?

A. The City supports the work that National Grid has done to storm harden one existing gate station, as well as to apply storm hardening principles to its two new stations. However, we strongly recommend that the Companies accelerate the

timetable for the rehabilitation at Clifton Gate and Citizens Gate to be completed by 2025, rather than the next decade. With respect to the seven other stations out of the total eleven (that are in the current 2015 FEMA 100- and 500-year floodplains and were identified as vulnerable by the Storm Hardening Collaborative), the City notes that the Companies' response in City-108 is inconsistent with the information that was provided and analyzed during the Collaborative, in particular with respect to Design Flood Elevations of specific assets.

Accordingly, we request that National Grid, per the recommendations contained in its Storm Hardening Collaborative Report, prioritize storm hardening efforts at these stations, including elevation of the cabinets to protect the critical equipment contained in them by 2025. Alternatively, if the Companies have different data than what was shared with the City, showing that these stations are indeed at Design Flood Elevations that do not require additional storm hardening, then the City would request to review this data. Finally, while it is helpful that for some (but not all) of these stations, there are other regulator stations in the area that can provide remote pressure data indicative of the functionality of the station, this is not a blanket substitute for being able to communicate and monitor the performance of these critical assets, especially during a significant coastal storm event.

Q. WHAT ARE YOUR CONCERNS RELATED TO THE HEIGHT OF METER SETS IN THE FLOODPLAIN?

A. The City understands that National Grid installs approximately 500-1,000 new meter sets in the 2015 FEMA floodplains each year. Given the forecasted floodplain expansion in the coming decades, it is critical for National Grid to examine whether (and if so, how) the Company can mitigate flooding risk to these meter sets through common-sense actions like raising the standard height design where feasible. However, in response to CNY-14(j), National Grid reported that it is "not recommending increasing the standard height design for regulators and meter sets for new installations."

Q. WHAT DO YOU RECOMMEND?

A. National Grid should develop and adopt procedures that provide guidance for elevating new meter sets located in the floodplains, for both low-pressure and high-pressure services. We understand that a one-size-fits-all approach may not work due to factors such as different types of housing and building design, and location within the floodplain. We also agree and understand that there will be need for exceptions within any such guidance that accommodates building type, geography, etc. Nevertheless, the development and adoption of guidance procedures related to meter sets in the floodplain is an important step in addressing the risk of flooding. Moreover, as noted in Attachment 1 to CNY-14, adopting such procedures would be in line with the practice of other utility companies that serve customers located within floodplains.

Q. HAS NATIONAL GRID CONDUCTED A FLOOD STUDY FOR THE GREENPOINT LNG PLANT SITE?

A. Yes. Following the completion of the Storm Hardening Collaborative in 2018, KEDNY conducted a desktop flood study of the entire Greenpoint LNG Plant site. At the City's urging, this study was informed by the City's climate projections for future sea level rise, storm surge, and urban precipitation levels. KEDNY considered the City's modeled future 2050s floodplain analysis (a modeled floodplain by the New York City Panel on Climate Change that accounts for how the geographic scope of flooding will increase due to projected sea level rise) in addition to the City's Climate Resilience Design Guidelines,[3] and also considered a flooding scenario occurring during peak demand.

Q. DID THE 2018 GREENPOINT LNG PLANT SITE FLOOD STUDY ADDRESS THE CITY'S CONCERNS?

A. Not entirely, no. The Greenpoint LNG Plant sits along Newtown Creek in Brooklyn, in an area that, according to the City's projections, will become more susceptible to flooding events in the not-too-distant future. Eastern portions of the site are in the current 2015 FEMA 100-year floodplain, while southern and eastern portions of the site are in the current 2015 FEMA 500-year floodplain. However, given its proximity to water, the City's projections show that by both the 2020s and 2050s, significant additional areas of the Plant will be located within the 100-

[3] https://www1.nyc.gov/assets/orr/pdf/NYC_Climate_Resiliency_Design_Guidelines_v3-0.pdf.

year floodplain, with a significant incremental portion of the Plant site falling within the 2020s and 2050s 500-year floodplain as compared to today.[4]

As recognized by National Grid in its pre-filed direct testimony, the Greenpoint LNG Plant is a critical component of the Company's gas supply portfolio and operation network, and can provide a material portion of KEDNY's peak day demand. Loss of this plant during peak demand, perhaps due to a winter flooding event, could therefore have significant reliability and cost implications for National Grid's customers. The criticality of this asset is only magnified given the potential gas supply constraints facing National Grid's service territory. Unfortunately, recent winter storms have highlighted the all-too-real possibility of significant flooding events during the peak gas season. For example, in January and March of 2018, Boston Harbor experienced its first and third highest tides on record since records began nearly a century ago, resulting in inundation of local streets.[5]

Q. WHAT DO YOU RECOMMEND?

A. Given the criticality of the Greenpoint LNG Plant, especially in light of recent permitting decisions that will, at a minimum, delay the NESE pipeline, National

[4] *See* NYC Department of City Planning, NYC Flood Hazard Mapper, *available at* http://www1 nyc.gov/site/planning/data-maps/flood-hazard-mapper.page.

[5] Jeremy C. Flox and John Hilliard, Boston Globe, "Late Friday high tide was milder than the near-record tide from Friday morning" (March 3, 2018) ("Boston Harbor's tide registered at 14.67 feet shortly after 11:15 a m., the third-highest since record-keeping began in 1928, and just a few inches shy of the record of 15.16 feet, which the city set on Jan. 4 during its last major winter storm, according to the National Weather Service."), *available at* https://www.bostonglobe.com/metro/2018/03/03/late-friday-high-tide-was-milder-than-near-record-tide-from-friday-morning/ox46WrIHftityv294LvFwM/story html.

Grid should engage a professional third-party engineering consultant to conduct a more fulsome flood study of the Greenpoint LNG Plant site. This study should include, but not necessarily be limited to, the following components, which were not assessed in the initial National Grid study:

- Hydrological study (*e.g.*, how does the water flow and drain across the site in a flood event, in an intense precipitation event, and in a combined flood and intense precipitation event);
- Geotechnical condition (*e.g.*, assess soil composition to understand how the ground will hold/drain water, including potential contamination impacts to other areas/sites);
- What happens if multiple (not just one) assets are impacted;
- The combined risks at the site due to a flood event with intense precipitation;
- The potential risks of flood mitigation strategies upon drainage;
- The potential risks and physical impacts from wave action;
- The potential risks from high heat and high wind;
- Ground truth elevation data from the 2018 Greenpoint LNG Plant study;
- Ground truth/validate the mitigation strategies that are presented and present potential new recommendations based upon experiences and best practices elsewhere;
- Review potential unintended consequences or secondary effects to the proposed mitigation strategies; and

- Review power distribution across the site, including resiliency/reliability of substation L to support full existing load plus liquefier load, in the event substation M is out of service.

Again, given the criticality of the Greenpoint LNG Plant site, we recommend that this proposed study be completed as soon as possible, and in any event no later than the end of calendar year 2021.

C. ENHANCED RESIDENTIAL METHANE DETECTORS

Q. WHAT IS NATIONAL GRID'S PLAN REGARDING THE ENHANCED RESIDENTIAL METHANE DETECTION PROGRAM?

A. This Program would seek to install commercially available plug-in methane detectors in residential customer homes. It is an expansion of the Residential Methane Detection Pilot Program initiated in the 2016 Rate Cases and, beginning in the Rate Year and continuing for the next three years thereafter, would result in 54,000 detectors installed in KEDNY's service territory each year and 36,000 in KEDLI's each year, for a total planned deployment of 360,000 detectors. National Grid, on page 44 of its Gas Safety Panel testimony, proposes to specifically target "room set meter customers, low income customers, multi-use residences, and customers served by smaller diameter (two and four inch) cast iron piping." The detectors will be accompanied by an educational brochure and distributed in customer-facing interactions, such as service inspections.

Q. HOW DOES NATIONAL GRID PROPOSE TO IMPLEMENT THIS PROGRAM?

A. According to the information provided in the Gas Safety Panel's direct testimony and National Grid's response to CNY-128, the success of the Program is heavily dependent on customers installing and operating the devices, as well as understanding how the devices themselves work and the safety features provided. National Grid will distribute the detectors to customers along with a brochure and the manufacturer's instructions so that the customer may install it. National Grid proposes to send a follow-up letter and brochure, and each device will contain a unique bar code, which National Grid will scan into its system at the time of distribution, so that National Grid can track methane calls from households that have methane detectors. Otherwise, National Grid has proposed no other quality control or customer guidance linked to this program. Under this proposal, National Grid's job effectively ends once the detector is in the customer's hands and does not provide a centralized tracking mechanism for potential leaks.

Q. WHAT IS THE CITY'S ASSESSMENT OF NATIONAL GRID'S PROPOSAL?

A. Conceptually, we support this Program. Methane leaks are dangerous, and improving detection can protect customers, especially the most vulnerable that are being specifically targeted by this Program. Nevertheless, the Program's design and implementation, as proposed by the Companies, raises concerns.

Q. WHAT CONCERNS DOES THE CITY HAVE?

A. Essentially, there is no quality control, limited customer guidance, and no centralized tracking system. These issues could render an otherwise conceptually

helpful program ineffective because it is entirely dependent on the customer to properly install, maintain, and use the device, with little meaningful support from National Grid.

Q. WHAT QUALITY CONTROL ISSUES HAVE YOU IDENTIFIED?

A. National Grid does not know whether a detector works before delivering it to a customer. Considering that National Grid seeks to install 360,000 detectors, even a one percent error rate – not complete failure, but less-than-ideal performance – can place an unacceptable number of customers in jeopardy and reduce the ability of this Program to detect leaks. In addition, because the manufacturer's instructions direct residents to call 911 in the event that the alarm sounds, false positives due to device error or improper installation could place added unnecessary burdens onto the City's emergency response resources. Moreover, per the response to CNY-128, the detectors have a seven-year useful life, but the Companies have no plan to replace the detectors, meaning that the Program will end once the detectors no longer function.

Q. WHAT QUALITY CONTROL MEASURES WOULD HELP ADDRESS THESE CONCERNS?

A. A simple and effective measure would be to test a representative sample of detectors to ensure they work as intended, including the device's detection range and sensitivity to methane leaks not located on the customer's premises or methane from organic or biogenic sources. This also would include testing the end-of-life warning that the manufacturer's instructions indicate will sound after seven years,

per Attachment 3 to CNY-128. Seven years is a long time for a customer to remember that they must eventually replace their detector, and this delay can place customers in jeopardy if they assume that their detector works because no end-of-life alarm sounded.

Q. IS RELYING ON CUSTOMERS TO TEST THEIR OWN DETECTORS SUFFICIENT?

A. No. The brochure and manufacturer's instructions ask that the customer test their own detector weekly to make sure the alarm is sounding properly. However, it is too easy for a customer to forget to test their alarm every week, no matter how colorful the brochure, and it is reasonable to assume many recipients of these detectors will not test on a weekly basis. Considering that this is a ratepayer-funded program, ratepayers should have assurances that National Grid will make an attempt to ensure that the detectors at the heart of their Program work, even if customers forget to test them. Our recommendations in this regard are set forth below.

Q. COULD YOU ELABORATE ON THE LACK OF CUSTOMER GUIDANCE?

A. Yes. Per CNY-128, National Grid gives customers: (i) a brochure that summarizes what the detectors do and how to install them; (ii) the manufacturer's instructions for installation and all associated warnings; and (iii) the detector itself. Additionally, National Grid will follow up with a letter and a second brochure. Aside from this, there is apparently no other customer guidance.

The lack of customer guidance threatens to derail this Program. First, there is no assurance that the customer has actually installed a detector – the Company representative that provides the detector does not perform the installation, nor does that representative observe its installation. Second, no guidance is provided regarding correct installation and maintenance outside of the brochure and manufacturer's instructions. The 12-page set of instructions, provided as Attachment 3 to CNY-128, is itself a barrier. A customer is not likely to read such a document to ensure proper installation. Accordingly, the customer may miss important instructions such as: the manufacturer recommends that several detectors be installed; the detector cannot receive noisy power, such as if another device with a large load is on the same circuit; or that "turbulent air," such as from fans on a hot day, impede the functioning of the device. Customer testing of the device will not indicate to them whether it has been properly installed per this guidance. Moreover, customers have to fill out and place the labels that remind them what to do if the detector sounds. All of these instructions are buried in pages 3 through 5 of the manufacturer's instructions, and missing any one of these important points could render the detector ineffective.

Q. WHAT DO YOU RECOMMEND TO REMEDY THIS ISSUE?

A. We recommend two improvements. First, a Company representative should install the detector and provide concise verbal instructions regarding its use. Having already been tested, this would ensure that a functioning device is placed in the appropriate location. Additionally, it provides another means of informing the

customer of important information without requiring them to read a lengthy document.

Second, twice each year during the four-year deployment schedule, the Companies should test a sample of in-service detectors to ensure proper functioning. This will give the Companies an opportunity to identify and resolve any problems with the detectors, before wide-scale deployment is completed.

Q. WOULD THESE IMPROVEMENTS IMPOSE ADDITIONAL COSTS?

A. We do not believe so. Company installations and sample testing should be able to be accommodated within the budget requested.

Q. DO YOU HAVE ANY ADDITIONAL SUGGESTIONS RELATED TO THE RESIDENTIAL METHANE DETECTORS?

A. Yes. We understand that Con Edison is pursuing a similar, but different, program for residential methane detection, whereby the detectors will have the ability to communicate directly with Con Edison via the AMI meters and infrastructure that Con Edison is currently deploying. We understand that National Grid does not currently have any plans for deploying AMI meters. However, in the long term, there will be a need to monitor gas usage much more closely both from a safety and supply perspective. The Companies should therefore re-examine the potential deployment of AMI and, in doing so, examine whether they can utilize residential methane detectors that have the capability to sync up with AMI, to ensure consistency with Con Edison (and across the City).

D. REMOTE SERVICE SHUTOFF VALVES

Q. WHAT ARE THE COMPANIES' PROPOSALS REGARDING REMOTE SERVICE SHUTOFF VALVES?

A. On pages 85-87 of KEDNY's GIOP testimony and pages 78-80 of KEDLI's, the Companies propose to install remote service shutoff valves at customers located in the FEMA 100-year and 500-year floodplains. Per the response to DPS-561, the valves will be in both the 100-Year and 500-Year floodplain. These valves, National Grid avers, would use a communications network that would allow the valves, once they detect flooding, to selectively stop the flow of gas to areas affected by flooding without impacting other areas. In doing so, the Companies can prevent the negative effects of water infiltrating gas mains, which is discussed elsewhere in this testimony. This program was originally planned to begin in 2017, but lack of a commercially available shutoff valve delayed the program. KEDNY proposes to install 29,000 valves (an expansion from the original program proposal of 25,000 valves), and KEDLI 60,000.

Q. WHAT IS THE CITY'S ASSESSMENT OF THIS PROPOSAL?

A. Our response is twofold. First, we have concerns about how this program will be managed. Second, the data this program would generate presents an excellent opportunity for collaboration with the City. Despite the concerns about program management, given the potential benefits to facilitate faster service restoration and minimizing equipment damage, we recommend approving this proposal, while simultaneously granting the City the ability to leverage the data for its own flood-

monitoring programs. Moreover, we support the placement of valves in both the 100-Year and 500-Year floodplain.

Q. WHAT ARE THE CITY'S CONCERNS REGARDING PROGRAM MANAGEMENT?

A. Fundamentally, this is a nascent, untested technology, and the Program itself has already seen significant delays. Originally, as KEDNY and KEDLI discuss in their GIOP testimonies, they proposed to launch this Program in calendar year 2017. There was no commercially available shutoff mechanism at the time; the valves have only become available as of February 2019. The valves are truly brand new, too. Per CNY-78, the valves have only been tested at the manufacturer's facilities and not in a live situation. Indeed, the Companies have not yet completed their lab testing of the valves. Moreover, per CNY-78(d), no other gas utility is using these valves. Thus, this is a pilot not only for National Grid, but for the gas industry as a whole.

Considering past delays and the untested nature of the valves, it is reasonable to anticipate issues arising that could impact the effectiveness of this program. These include, but are not limited to, equipment failure in real world conditions and improper installations. These concerns are not theoretical. Several years ago, Con Edison installed a new technology, called a Vent Line Protector ("VLP") Device, which was designed to perform a similar function to the remote service shutoff valves; namely, protect gas customers during a flooding event. Over 80 percent (1,837 out of 2,204) of these VLP Devices were initially improperly

installed, and some of the devices were defective, thus requiring remedial action by Con Edison.[6] National Grid should take care to avoid a similar outcome with the remote sensor shutoff valves.

Q. SHOULD THIS PROPOSAL BE IMPLEMENTED DESPITE THESE ISSUES?

A. Yes. Despite our concerns, we agree that this program nevertheless has the potential to offer significant benefits, particularly during flooding events. Because of the storm hardening benefits and the opportunity for collaboration, which is described below, the City supports this proposal with certain enhancements. First, the Companies should implement a plan to methodically and robustly test these devices to ensure they operate as designed, and to gradually deploy such devices in phases. The Companies also need to perform outreach to educate customers on how these remote operated shut off valves will work, and what they are intended to prevent. The Companies should further confirm there will be no "accidental" automatic shutoffs of gas as a result of the operation of these sensors, and that they only will trigger if there is sufficient flooding to cause water intrusion and possible damage to customer equipment.

Q. HOW DOES THIS PROPOSAL PRESENT AN OPPORTUNITY TO SUPPORT THE CITY'S OWN FLOOD MONITORING EFFORTS?

[6] *See* Cases 13-E-0030, et al., Consolidated Edison Company of New York, Inc., Order Adopting Storm Hardening and Resiliency Collaborative Phase Three Report Subject to Modifications (issued January 25, 2016) at 26.

A. The valves activate when they detect flooding, and this information is relayed via a communications network, per the Companies' descriptions in their GIOP testimonies. The Companies propose to install 89,000 valves in total, which provides 89,000 data points on the location and timing of the floods, all in real-time. Furthermore, the Companies will only need to interrupt gas services to the specific areas affected, indicating that this data is quite granular, perhaps even to an individual sensor. This data can do more than just protect customers in the event of flooding – it can be used to track the extent, duration, and frequency of floods. This type of tracking is one of the primary objectives of the "Jamaica Bay Community Flood Watch Project" (the "Flood Watch Project").

Q. COULD YOU ELABORATE ON THE FLOOD WATCH PROJECT?

A. Yes. Due to climate change, the City is experiencing more regular and more extreme tidal flooding, called "sunny day flooding." As sea levels rise, this will only get worse, shifting the flood maps in unexpected ways. The Flood Watch Project seeks to document this sunny day flooding in the Jamaica Bay watershed through citizen science reports so that the City and its citizens can create maps and make predictions about the extent of flooding and its effects. The data the Remote Service Shutoff Valve Program will generate would be a valuable resource to help improve the Flood Watch Project's efforts to help improve forecasts of flooding and coastal hazards.

Q. WHAT DATA WOULD BE REQUIRED TO MAXIMIZE THE EFFECTIVENESS OF THE REMOTE SHUTOFF VALVE PROPOSAL?

A. The City would need access to the following data points for each valve in the City and the Rockaways:

- The latitude and longitude of each valve;
- The elevation of each valve; and
- The date, time, and duration of the flood event.

The City does not request this data in real time; rather, the City requests that National Grid share this data with the City in monthly reports, in a readily-accessible format like Microsoft Excel.

Q. WHY DO YOU SUPPORT VALVE PLACEMENT IN BOTH THE 100-YEAR AND 500-YEAR FLOODPLAINS?

A. Flooding is not just a problem in the 100-Year floodplain; flooding in the 500-Year floodplain has the same similar consequences despite being less frequent, and those customers therefore require the same protection. Importantly, as sea levels rise, there will be increasing likelihood of flooding in both the current 100-year and 500-year floodplains. Data from the valves will help the City better understand what areas are regularly experiencing more flooding due to climate change. Monitoring the growth of the floodplain is vital and can only be done if the valves are there to collect the data.

Q. COULD YOU SUMMARIZE THE CITY'S RECOMMENDATIONS ON THE PROPOSED REMOTE SERVICE SHUTOFF VALVE PROGRAM?

A. Yes.

- Because of the untested nature of the devices and National Grid's issues in getting the device to commercial viability in the past 3 years, the devices should be tested robustly and deployed gradually;
- Because of the protections it provides customers and the data it collects, the Program should go forward, provided that the devices are comprehensively tested, and customers are adequately educated about them;
- National Grid should share the information noted above with the City; and
- The devices should be installed in both the 2015 FEMA 100-year and 500-year floodplains.

CONCERNS ON IMPACTS TO NYC GREEN INFRASTRUCTURE ASSETS

Q. WHAT ARE NYC GREEN INFRASTRUCTURE ASSETS?

A. In New York City, Green Infrastructure, or GI, refers to an array of practices that use or mimic natural systems to manage stormwater runoff. Stormwater is either directed to engineered systems for infiltration or detained at a slower rate before it enters the combined sewer system. Types of Green Infrastructure include:

- Rain Gardens – vegetated or landscaped depressions in the sidewalk designed with an engineered soil that promote infiltration of stormwater runoff into the underlying soil.
- Stormwater Greenstreets – similar to Rain Gardens, but constructed in the roadway and usually larger than a Rain Garden.
- Green Roofs – roofs made up of a top vegetative layer that grows in an engineered soil, which sits on top of a drainage layer.

- Permeable Paving – using a range of materials and techniques, permeable pavers or porous concrete allows water to seep in between paving materials and be absorbed into the ground.

Through Green Infrastructure implementation, DEP is working toward a reduction of 1.67 billion gallons a year in Combined Sewer Overflows ("CSOs") by 2030. CSOs can negatively affect water quality and recreational uses in local waterways. Through the end of 2018, 4,585 Green Infrastructure Assets had been either constructed or were in construction. Approximately 5,500 Assets are planned to enter construction in 2019. A significant portion of these Assets are located in National Grid's service territory in Brooklyn and Queens.

Q. WHAT ARE YOUR CONCERNS RELATED TO GREEN INFRASTRUCTURE ASSETS AS THEY PERTAIN TO NATIONAL GRID?

A. Our primary concern is that NYC Green Infrastructure Program Assets are being damaged by National Grid street work. For example, in north Brooklyn, DEP has recorded damage to 47 Assets from National Grid work crews, totaling approximately $160,000 in damages. The damage is primarily from sediment and cement intrusion into the right-of-way Green Infrastructure Program Assets, which damages the engineered soils and plants located within the Assets. Other impacts include structural damage from National Grid pipe installations either through or adjacent to Green Infrastructure Program Assets. National Grid and its contractors should be following the green infrastructure protection stipulations

required by the DOT Street Opening Permit. It is clear to DEP that National Grid and its contractors are either unaware of these protection stipulations or have not fully adhered to them.

An additional concern about DEP Green Infrastructure Assets is limitations to siting and construction due to interferences with existing National Grid infrastructure. DEP will continue to review these locations with National Grid, but when there is no alternate location for the Asset, DEP retains the right to request National Grid to relocate their infrastructure. National Grid should respond within 10 business days to DEP's request for relocation. In the event that National Grid cannot move its infrastructure, National Grid must provide a specific in-depth justification. In instances where the infrastructure can be moved, it should be relocated in a mutually agreed upon timeframe.

For new National Grid infrastructure or relocation of existing infrastructure, National Grid should consider Green Infrastructure Program implementation, and the infrastructure should be installed with consideration to the future implementation of Green Infrastructure Assets. These considerations include placing natural gas lines six feet instead of three feet from the curb line when feasible, which will allow for DEP to install right-of-way Green Infrastructure Assets.

Q. ARE YOU REQUESTING MONETARY COMPENSATION FROM NATIONAL GRID AS PART OF THIS RATE CASE?

A. Our primary concern is to stop any future damage to Green Infrastructure Assets, and to ensure that all National Grid employees and contractors are properly informed, trained, and follow the protection stipulations required by the DOT Street Opening Permit stipulations. That being said, by submitting this testimony, DEP is not foregoing any right to seek damages for GI assets, and will continue to seek repairs in kind and restitution for damages to GI assets outside of this rate case.

Q. ARE THESE ASSETS VISIBLE TO NATIONAL GRID WORK CREWS?

A. Yes. The GI Program protection stipulations clearly describe the difference between a typical street tree and a DEP GI asset. DEP also provides a presentation on GI at the quarterly Department of Transportation (DOT) Construction Advisory Council Meetings. DEP is preparing additional training materials for National Grid and its contractors to eliminate any remaining confusion. In addition, DEP coordinates with National Grid extensively during the design process, and National Grid is aware of all locations that proceed to construction. Moreover, when GI Assets are in the vicinity of a DOT street opening permit site, the street opening permit includes a notification that indicates that GI Assets are nearby (see below):

BIOSW1	A NYCDEP RIGHT-OF-WAY GREEN INFRASTRUCTURE PRACTICE IS LOCATED ON THIS ROADWAY AND/OR SIDEWALK. PERMITTEE MUST PROTECT ALL NYCDEP GREEN INFRASTRUCTURE DOWNSTREAM OF THE WORK AND WITHIN 5 FEET OF THE WORK AREA. PERMITTEE MUST FOLLOW NYCDEP'S GREEN INFRASTRUCTURE PROTECTION REQUIREMENTS AT http://www.nyc.gov/html/dep/pdf/green_infrastructure/ogi-protection-procedure.pdf PERMITTEE IS RESPONSIBLE FOR RESTORATION OR RECONSTRUCTION OF DAMAGED DEP GREEN INFRASTRUCTURE AS DIRECTED BY DEP.

Finally, DEP maintains a web map that shows locations of built Green Infrastructure Assets, as well as those in construction, and this map can be utilized

by utility personnel to confirm Asset locations, and DEP will include directions to the web map in a forthcoming stipulation update. Despite these existing measures, however, DEP continues to experience issues with damage to Green Infrastructure Assets caused by National Grid work crews.

Q. WHAT DO YOU RECOMMEND TO REDUCE DAMAGE TO GREEN INFRASTRUCTURE ASSETS?

A. Our recommendations are three-fold. First, National Grid should notify DEP a minimum of two weeks prior to street openings, and DEP will send out an inspector during construction to ensure proper GI protections are implemented.

Second, National Grid should provide bi-annual training on identifying and protecting Green Infrastructure Assets to its employees and contractors using DEP-supplied training materials. These training materials are currently in development by DEP. DEP will also offer to provide 'train the trainer' training once per year to National Grid employees and contractors. National Grid should add this training as a requirement of the Companies' third party contracts.

Third, National Grid should identify a central point of contact for Green Infrastructure issues, including construction coordination and damage claims.

CONCERNS WITH PROPOSED GAS SAFETY METRICS AND PERFORMANCE MEASURES

Q. ARE THE COMPANIES SEEKING ANY PERFORMANCE INCENTIVES?

A. Yes. The Companies are proposing to modify the following performance measures: Leak Prone Pipe Replacement, Emergency Response, Leak Management, Damage Prevention, and Gas Regulations Performance.

Q. ARE THE PERFORMANCE INCENTIVES THAT THE COMPANIES SEEK JUSTIFIED?

A. No. Fundamentally, we disagree with the Companies' proposals, which essentially seek to reward them for doing basic prudent utility work. Several of the proposed performance metrics are coupled to shareholder incentives for certain natural gas distribution operations activities that should not be incentivized – emergency response and damage prevention – because these activities are within industry norms of utility operations as required by federal regulations.

The CFR provides minimum pipeline safety regulations governing, in part, the operations of a utility's natural gas distribution system (*See* 49 CFR Part 192). The CFR requires prompt and effective response to and reports of gas odor in or near a building, and to every type of emergency that involves or could involve natural gas or a natural gas pipeline facility. The effectiveness of an operator's emergency response program is a factor considered for risk reduction within the Distribution Integrity Management Program ("DIMP") regulations in 49 CFR Part 192 Subpart P. The DIMP regulations also require an operator to identify and implement measures to address risks to its system and ultimately to people and property. This includes performance of an effective leak management program, which provides that an operator act appropriately to mitigate the hazards

associated with each leak. For non-hazardous leaks, appropriate action may include scheduling the line for repair and continued reduction in leak count.

Other proposed performance metrics are tied to rewarding the Companies for accelerating the implementation of certain capital investment programs or modifying the damage prevention metrics to eliminate components that measure damages that are wholly within each Companies' control. Below is a description of our analysis of each proposed incentive, and an explanation of why such incentive should be rejected by the Commission.

For ease of reference, Exhibit__(NYCGISP-3) includes an overview of our recommendations for the Companies' various gas safety-related incentive mechanisms and safety metrics.

A. LEAK PRONE PIPE INCENTIVES

Q. WHAT ARE THE COMPANIES PROPOSING FOR THEIR LEAK PRONE PIPE RETIREMENT MILEAGE TARGETS AND INCENTIVES?

A. As discussed above, KEDNY is proposing to increase its minimum LPP replacement target by 5 miles per year, starting at its current minimum removal target of 60 miles and accelerating to 75 miles per year by CY 2022, with a cumulative minimum target of 310 miles for CYs 2020-2023. KEDLI proposes to maintain its current 150 miles per year minimum removal target, with a cumulative minimum target of 620 miles for CY2 2020-2023. The Companies propose to maintain the basic structure of this performance metric: each Company will be subject to an NRA should they fail to meet their minimum LPP mileage

target. The LPP mileage targets for the NRA are set well below the levels that will be funded in rates; for both Companies, the NRA is five miles below that which they request in rates. As noted earlier, the Companies also propose to continue the existing PRA of two BPs for each full mile of LPP removed above the level that is funded in rates, up to five miles (10 BPs).

The Companies propose to increase the NRAs for failure to meet their target from the current 8 BPs to a proposed 15 BPs, at the expense of the number of at-risk BPs for the Gas Regulations Performance Measure. The Companies are also proposing to continue its GSRS Mechanism to recover the carrying costs on incremental capital expenditures and O&M expenses associated with the replacement of main above the targets established.

Q. DO YOU OPPOSE THE COMPANIES' PROPOSALS?

A. Yes. First, increasing the NRAs for an incentive that the Companies have had no difficulty meeting at the expense of other NRAs is not appropriate and fails to serve the purpose of the NRA. As there is little likelihood that the Companies will incur their proposed 15 BP penalty, the reallocation does not "more fairly allocate the 150 at risk basis points," as the Companies suggest at page 48 of the Gas Safety Panel, but rather only reduces the likelihood that the Companies will be subject to potential NRAs. Furthermore, as discussed in more detail below, in light of the Order to Show Cause, it is not appropriate at this time to reduce the number of BPs at risk under the Gas Regulations Performance Measure.

Second, the Companies are already proposing increased levels of capital investment and O&M expense. The Companies have not set forth a compelling need to further increase cost burdens on customers by rewarding the Companies for replacing more mains. Moreover, the Companies have not justified that replacing additional mains is even something to be rewarded. Under the Companies' proposals, customers would bear all of the capital and O&M costs for the additional replacements, plus the shareholder incentive.

Q. WHAT DO YOU RECOMMEND?

A. We recommend that the NRAs at risk under the LPP Retirement performance measure not be modified and remain at 8 BPs. In addition, as discussed earlier in this testimony, the PRA for LPP main replacements should be rejected because the Companies have failed to make the case as to why ratepayers should bear such additional costs.

B. EMERGENCY RESPONSE TIME INCENTIVES

Q. WHAT CHANGES ARE THE COMPANIES PROPOSING TO THEIR EMERGENCY RESPONSE SAFETY PERFORMANCE MEASURE?

A. The Companies are proposing a series of positive incentives for their 30 minute emergency response rates. Additionally, the Companies propose to increase the number of at-risk BPs from 12 to 25 at the expense of the number of at-risk BPs for the Gas Regulations Performance Measure.

Q. DO YOU OPPOSE THE PROPOSALS?

A. Yes, they are not acceptable. The Companies should not be rewarded for simply maintaining a safe gas system, which is their fundamental role. Additionally, neither Company has received an NRA under this performance metric. Reallocating BPs to this measure is thus just as inappropriate here as it is for the LPP retirement metric – considering the Companies' past emergency response performance, the Companies' proposal then simply becomes a reduced penalty for violating the Commission's gas safety regulations.

Q. PLEASE PROVIDE THE BASIS FOR YOUR OPPOSITION TO THE COMPANIES' PROPOSED EMERGENCY RESPONSE INCENTIVE PROPOSALS.

A. As a threshold matter, it is the utility's responsibility to provide safe and reliable gas service, including responding to gas leaks in an appropriate time frame to minimize the harmful consequences of gas emergencies to its customers, its employees, the general public, property, and the environment. The Companies are proposing to revise the Emergency Response Safety Performance Measure so that the Company receives an incremental positive incentive for having met a 30 minute response rate to emergency calls.

The expectation to meet a response time target at or below 30 minutes is a current industrywide norm. Prudent operators will continue to strive to improve upon (lower) this norm. Federal regulations (49 CFR Part 192) require an operator to implement an emergency response program that will minimize the hazard

resulting from a gas pipeline (distribution system or transmission system) emergency by providing a prompt and effective response to gas emergencies.

Moreover, CFR §§ 192.605(a), 192.605(b), 192.605(c), and 192.615(a) require an operator to follow procedures for conducting operations, maintenance, and emergency response activities, including procedures for responding promptly to a report of gas odor inside or near a building. The emergency response procedures an operator is required to establish in writing and to follow include emergency plans requiring performance of prompt and effective response to a notice of every type of emergency that involves or could involve gas or a gas pipeline facility (*see* CFR §§ 192.605(c), 192.615(a)(3)).

The presumed expectation for the operator to minimize the risk from the activities of natural gas delivery is also set by the DIMP regulations (49 CFR § 192.1007). The specific contribution of prompt and effective emergency response to risk minimization is the reduction of the consequences of failures that result in gas release or to situations that could imminently result in gas release.

The regulatory requirements to provide a prompt and effective response to gas emergencies, and the inequity of rewarding shareholders for performance that the Companies are required by regulation to do, are the bases for our opposition to the Companies' proposed emergency response incentive proposal.

Q. DO YOU HAVE ANY ADDITIONAL CONCERNS REGARDING THE EMERGENCY RESPONSE TIME INCENTIVES?

A. Yes. The Companies propose a positive revenue adjustment of up to 6 BPs for increasing their 30 minute response rate: 2 BPs for responding to 86% to 88% of calls in 30 minutes, 4 for more than 88% to 90%, and 6 for more than 90%. Neither Company would have met this proposed incentive if it had been in effect during the current rate plan.

We note, however, that Con Edison is proposing an identical incentive mechanism in its ongoing rate case in Case 19-G-0066. According to Con Edison's annual Gas Safety Performance Measures reports for the past five years, Con Edison's actual results for a 30 minute response time have been far better than National Grid's – so much so that Con Edison would have received positive incentives without having to show any improvement, if the metrics proposed by National Grid had been in place in those past five years for Con Edison.[7] Indeed, Con Edison's lowest 30-minute response time in the past five years was in 2014, when it achieved an 87.8% response time. In contrast, KEDNY and KEDLI response times have consistently ranged in the high-70% to low-80% range.

Similar to several of the other gas safety and performance metrics discussed herein, we see no reason why National Grid's performance should lag behind that of Con Edison. Customers living in Brooklyn, Queens, and Staten Island should expect the same level of prompt response to emergencies that

[7] The City also disputed the emergency response time metric in the ongoing Con Edison gas rate case. *See* Case 19-G-0066, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service*, Direct Testimony of New York City Gas Infrastructure Panel (filed May 24, 2019), at pp. 17-19.

customers living in Manhattan and the Bronx have experienced in recent years. Indeed, given the lag in National Grid's recent performance as compared to Con Edison, the Commission should re-visit the minimum required response time for both KEDNY and KEDLI, so that it better aligns with what customers have come to expect in other boroughs.

Q. DO YOU HAVE A PROPOSAL FOR EMERGENCY RESPONSE TIMES FOR KEDNY AND KEDLI?

A. Yes. We propose the following required response times and associated NRAs:

Required Response Time ("RRT")	Calls that must be responded to within RRT	NRA for failure to meet RRT
30 minutes	80%	6 basis points
45 minutes	90%	4 basis points
60 minutes	95%	2 basis points

This proposal recommends a five percent tightening of the 30-minute response time, from 75% to 80%, while maintaining the 45-minute and 60-minute response times and percentages proposed by the Companies. This represents an improvement that should be achievable without the need for additional funding given each Company's recent historical performance. We note that this is still low compared to Con Edison's recent historical performance but nevertheless represents an incremental improvement.

We also recommend that the Commission reject the Companies' proposal to increase the BPs at risk under this performance measure and, instead, retain the total at-risk BPs at 12. Neither Company has had difficulty meeting these metrics in recent years. It would therefore be inappropriate at this time to shift at-risk BPs away from Gas Regulations Performance to Emergency Response Time Performance, especially given the recent Order to Show Cause discussed earlier.

Q. DO YOU HAVE ANY ADDITIONAL COMMENTS OR CONCERNS REGARDING EMERGENCY RESPONSE TIMES?

A. Yes. The City has concerns about how the Companies actually measures their emergency response times for the purpose of tracking this metric.

Q. HOW DO THE COMPANIES CURRENTLY MEASURE EMERGENCY RESPONSE TIME?

A. According to the response to CNY-126, the Companies track and measure emergency response times based on the time difference between when a leak call is first received, and the arrival on site of a "First Responder," that is, a National Grid employee "qualified for emergency response." This First Responder then transmits their arrival time on a mobile application that records an electronically time stamped arrival time.

Q. WHAT ARE YOUR CONCERNS WITH THE EXISTING METHODOLOGY?

A. It is not clear how the Companies designate an employee as a First Responder, or whether that First Responder has any authority to effect actual repairs to a gas leak

without calling for additional assistance to the scene, for example, supervisors or crews with the necessary authority, equipment, and resources to handle leaks.

Q. WHAT DO YOU RECOMMEND TO MITIGATE THIS ISSUE?

A. We recommend that the Commission examine the Companies' emergency response protocols in more detail to determine if changes are warranted, including whether response times should be measured from when a qualified response crew begins remediating a leak, as opposed to when a First Responder first arrives on the scene.

C. GAS LEAK BACKLOG INCENTIVES

Q. WHAT ARE THE COMPANIES' PROPOSALS FOR THEIR LEAK MANAGEMENT PERFORMANCE MEASURES?

A. The Companies have two related leak management performance measures: (1) NRAs for failure to meet a workable leak backlog target and a total leak backlog reduction target; and (2) PRAs for every 50 high-emitting type 3 leaks eliminated beyond their backlog targets. The Companies propose to maintain their workable leak backlog target and continue to reduce their total leaks at the same rate per year as in the current Rate Plan (150 per year for KEDNY and 750 per year for KEDLI). The Companies propose to increase the number of at-risk BPs for their NRAs from 12 to 15 by reducing the number of at risk BPs for their Gas Regulations Performance Measure. Finally, the Companies propose to maintain their PRAs and targets for high-emitting type 3 leaks.

Q. WHAT HAS BEEN NYCEM'S EXPERIENCE WITH LEAKING GAS MAINS?

A. We are aware of at least one gas explosion in New York City – at Lexington Avenue and East 54th Street – that has been attributed to the accumulation of gas in a subterranean structure from a gas main with a small leak. An investigation into that explosion revealed that the leak was classified as type 3 and therefore was not required to be repaired by Con Edison.

Q, WHAT HAPPENED IN THAT INCIDENT?

A. One City employee was seriously injured and two other City employees also were injured.

Q. WHAT ACTIONS, IF ANY, DID NYCEM TAKE AS A RESULT OF THAT INCIDENT?

A. NYCEM formed a working group comprised of City agencies, Con Edison, and National Grid to, among other things, examine the incidences of gas leaks, identify ways to improve communications among and between City agencies and the utilities regarding gas leaks, and recommend ways to enhance public safety.

Q. WAS NATIONAL GRID COOPERATIVE IN THIS EFFORT?

A. Yes. National Grid attended working group meetings, commented on the various versions of the report, and provided the working group with helpful data. We also understand that National Grid has been investigating ways to improve its ability to detect gas leaks during its leak surveys.

Q. WHAT ARE NYCEM'S CONCERNS WITH GAS ACCUMULATIONS IN SUBSURFACE STRUCTURES?

A. If there is a leak in a gas main, the emitted gas will travel to and collect in any voids. Many subsurface structures, such as manholes and service boxes, are not sealed. Gas can travel along pipes, mains, conduits, and ducts into these structures and accumulate. Over time, the potential exists for the gas to reach a concentration level at which it can become explosive in the presence of a flame or heat source. In the incident we discussed, a City crew was repairing potholes using a torch to soften the asphalt.

There are a number of electrical and plumbing activities in which a heat source is in or near one of these subsurface structures or sparks are produced from cutting metal. Based on these concerns, the City would like the Commission to take action to reduce, to the maximum extent possible, the potential for natural gas to accumulate in these structures and potentially cause explosions.

Q. DO YOU HAVE ANY PERSPECTIVE ON HOW OFTEN THESE CONCERNS ARISE WITHIN NEW YORK CITY OVER THE COURSE OF A YEAR?

A. When NYCEM investigated the Lexington Avenue incident, we learned that there were approximately 4,000 active gas leaks of all types and 423 leaks in subterranean structures as of January 2016. Both Con Edison and National Grid have taken action to reduce the number of gas leaks since 2016. Nevertheless, there is still significant room for improvement.

Focusing on National Grid, it reported to the Commission that as of the end of 2018, KEDNY had a total backlog of 2,803 gas leaks and KEDLI had a total backlog of 8,199 gas leaks, which is composed almost entirely of type 3 leaks, as opposed to workable (type 1, 2, or 2A) leaks. This is illustrated in the charts below:

KEDNY Historical Leak Backlog Performance[8]

Year	Workable Leak Backlog Target	Actual Workable Leak Backlog	Total Leak Backlog Target	Actual Total Leak Backlog
2017	≤ 25	8	≤ 3,550	3,118
2018	≤ 25	10	≤ 3,400	2,803

KEDLI Historical Leak Backlog Performance[9]

Year	Workable Leak Backlog Target	Actual Workable Leak Backlog	Total Leak Backlog Target	Actual Total Leak Backlog
2017	≤ 25	0	≤ 10,000	9,232
2018	≤ 25	0	≤ 9,250	8,199

[8] Cases 16-G-0058 *et al.*, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, Gas Safety Performance Metrics Report for Calendar Year 2017 (filed March 30, 2018) (hereinafter "2017 KEDNY Metrics Report"); *id.*, Gas Safety Performance Metrics Report for Calendar Year 2018 (filed March 29, 2019) (hereinafter "2018 KEDNY Metrics Report").z

[9] *Id.*, Gas Safety Performance Metrics Report for Calendar Year 2017 (filed March 30, 2018) (hereinafter "2017 KEDLI Metrics Report"); *id.*, Gas Safety Performance Metrics Report for Calendar Year 2018 (filed March 29, 2019) (hereinafter "2018 KEDLI Metrics Report").

Based on our review of the Companies' gas safety performance reports to the Commission for the past five years and DPS-114, which compiles this information, the leak backlog for each Company was in the thousands of gas leaks each year. Clearly, something needs to be done to change the status quo.

Q. WHAT DO YOU RECOMMEND?

A. We understand that as the Companies replace their most leak-prone pipe, the incidence of gas leaks should decline. However, reliance on the leak-prone pipe replacement program is not sufficient.

The City recommends that the Commission direct the Companies to reset their leak backlog targets to the actual leaks remaining at the end of CY 2018, and to then dramatically reduce their total leak backlog so that their backlogs are negligible within five years. Using actual leak backlog numbers from the end of CY 2018 as the starting point, this would require a reduction in the backlog of about 560 per year for KEDNY and about 1,640 per year for KEDLI.

We also recommend that the Companies continue to reduce their backlog of workable leaks (those that could become hazardous over time) to zero, and then move to repairing all potentially hazardous leaks when they occur. Accordingly, the workable leak backlog target should be set at less than or equal to 10 per calendar year, as opposed to the 25 proposed by the Companies.

Because we are proposing to significantly increase the overall reduction in the leak backlog, the Companies' proposed increase in at-risk BPs for the leak backlog metric is appropriate as it helps motivate the Companies to achieve these

enhanced targets. At the same time, because the Total Leak backlog is so much higher for National Grid than for Con Edison, and because National Grid has successfully mitigated its Workable Leak backlog in the last couple of years, we propose to switch the BPs at risk for the Total Leak and Workable Leak metrics, such that 10 BPs are at risk for failure to meet the Total Leak metric (versus 5 BPs as proposed by the Companies) and 5 BPs are at risk for failure to meet the Workable Leak metric (versus 10 BPs as proposed by the Companies). If our proposal to modify the targets is not accepted, the increase in at-risk BPs should not be approved.

Q. DO YOU BELIEVE THAT IT IS POSSIBLE FOR NATIONAL GRID TO ACHIEVE YOUR PROPOSED LEAK BACKLOG THRESHOLDS USING PROPOSED FUNDING AMOUNTS SET FORTH IN THE COMPANIES' PRE-FILED DIRECT TESTIMONY?

A. Yes. Both Companies have made significant reductions in their total leak backlogs since 2014 – KEDNY has eliminated on average 316 leaks per year, and KEDLI has eliminated on average of 830, with both showing greater reductions in recent years as Proactive LPP replacement programs continue to ramp up. Combined with the synergies from replacing LPP, we believe that both Companies are capable of meeting these targets (particularly if, as proposed, KEDNY's Proactive LPP replacement program continues to ramp up over the next several years).

Moreover, we are proposing to reset the leak backlog targets to the actual leaks remaining at the end of CY 2018 (2,803 for KEDNY; 8,199 for KEDLI). For

KEDNY, at least, the Company should already be well below this target by the time the Rate Year starts. KEDNY reported in response to CNY-38 that its total leak backlog as of June 11, 2019 was 2,467, which is 336 leaks below our proposed reset of 2,803. KEDNY has another seven months before the Rate Year commences to lower its actual backlog even further, which should make these proposed annual reductions that much easier to attain.

We understand that the Company identifies and repairs hundreds of leaks of all types each year. However, it has been doing so for many years, and it still has been able to eliminate a significant portion of its leak backlog as well. It may take five years to fully eliminate the backlog under our proposal, but that elimination should be the goal. Put another way, the goal should not be to simply maintain thousands of minor leaks each year (i.e., the status quo).

To further incentivize eliminating gas leaks, we would also support a PRA that incentivizes the Companies to eliminate the backlog even faster than our proposed timeline. We suggest the following incentive mechanism: one BP for every 10 percent above our proposed yearly leak reduction, capped at five BPs. For example, if KEDNY eliminated 616 leaks instead of 560, it would receive 1 BP; if KEDLI eliminated 2,460 instead of 1,640, it would receive five BPs.

Finally, because we are proposing to materially shorten (and ultimately eliminate) the overall leak backlog, we see no reason to continue the existing PRA for Type 3 'high emitters.' Under our proposal, all leaks will be addressed in a prompt manner, and thus there is no reason to single out a specific category of leak

for special treatment. This has the added advantage of addressing gas leaks in subterranean structures, explained above.

Q. PLEASE DISCUSS THE BENEFITS TO THE PUBLIC OF ELIMINATING GAS LEAKS.

A. There are two primary benefits to doing so. The first is public safety. If gas leaks are fixed as they are identified, there would be far less opportunity for gas to migrate to subsurface structures and accumulate to explosive concentrations. Although gas explosions are rare in New York City, every recent gas explosion has resulted in injuries or death for one or more people. These injuries and deaths are avoidable, and reasonable efforts should be taken to avoid them. The death of a single New York City resident was sufficient cause for the Commission to institute rigorous stray voltage inspection and repair requirements. The Commission should take a similarly aggressive approach with respect to eliminating gas leaks.

Q. WHAT IS THE SECOND PRIMARY BENEFIT?

A. Methane is a very potent greenhouse gas, many times more potent than carbon dioxide. Eliminating gas leaks will reduce fugitive emissions of natural gas into the atmosphere and help to combat global warming. It is our understanding that the State, and the Commission in particular, are taking multiple actions to reduce greenhouse gas emissions, including increasing reliance on renewable and carbon-free resources and seeking substantial expansion of energy efficiency measures. As part of its portfolio of actions, the Commission can, and should, direct National

Grid and the other gas utilities to take steps to reduce and eliminate natural gas emissions from pipe leaks to the maximum extent practicable.

Q. ARE YOU AWARE OF THE PRACTICES OF OTHER UTILITIES REGARDING THEIR GAS LEAK BACKLOG?

A. We are not aware of the details of all utilities, but within New York State we are aware that the practices vary. For example, on May 20, 2019, New York State Electric & Gas Corporation ("NYSEG") and Rochester Gas and Electric Corporation ("RG&E") filed gas rate cases with the Commission in Cases 19-G-0379 and 19-G-0381, respectively. Their filings included pre-filed direct testimony from their Gas Operations Panels. Both Panels testified that the utilities have corporate requirements to repair all gas leaks of all types each year and strive to satisfy these requirements. According to those Panels' exhibits, their leak backlogs are minuscule: across all types of leaks (1, 2A, 2, and 3), NYSEG and RG&E have a three-year average of 10 leaks and 12 leaks, respectively. We understand that the NYSEG and RG&E leak backlog figures may be influenced by the character of their service territories, which may differ from the urban setting that comprises the KEDNY and KEDLI service territories, but the wide differences between NYSEG/RG&E's backlogs and National Grid's are significant enough to warrant mentioning.

We note further that on March 1, 2019, Con Edison reported to the Commission in its report on Gas Safety Performance Measures for 2018, filed in Case 16-G-0061, that as of the end of 2018, it had a total backlog of 283 gas leaks.

This is nearly 10 times lower than KEDNY's leak backlog as of June, 2019. In prepared direct testimony submitted on May 24, 2019 in Case 19-G-0066, the City is similarly recommending that Con Edison reduce its leak backlog to zero.

Q. DO YOU HAVE ANY CONCERNS ABOUT THE MANNER IN WHICH NATIONAL GRID CONDUCTS LEAK SURVEYS?

A. Yes. Following the Lexington Avenue gas explosion discussed above, NYCEM assembled a Natural Gas Interagency Working Group ("Working Group") that was charged with, among other things, improving interagency and utility communication related to active gas leaks that may impact NYC personnel and the general public. As part of that effort, the Working Group identified significant differences in the manner in which NYC's two gas utility companies – Con Edison and National Grid – perform gas leak surveys. Based on Companies' response to CNY-120, National Grid has made improvements in its gas leak survey activities to more closely align with Con Edison's practices. However, significant differences remain between the practices of National Grid and Con Edison, including:

- Con Edison performs monthly mobile 'low speed' surveys of gas mains, which KEDNY performs annually and which KEDLI does not perform at all; and
- Con Edison and KEDLI perform walking surveys of gas service connections in non-business districts every three years, while KEDNY performs these surveys every five years.

Q. WHAT IS YOUR RECOMMENDATION?

A. National Grid's leak survey practices should be aligned both internally and with Con Edison, so that the entire City is assessed in a consistent fashion. For example, KEDNY should conduct monthly mobile 'low speed' leak surveys, and KEDLI should perform similar surveys in the Rockaways portion of its service territory. In addition, KEDNY should perform walking surveys of gas service connections in non-business districts every three years, to align its practices with Con Edison and KEDLI.

D. DAMAGE PREVENTION SAFETY METRICS

Q. WHAT ARE THE COMPANIES' PROPOSALS FOR THEIR DAMAGE PREVENTION SAFETY PERFORMANCE MEASURES?

A. The Companies propose to: (1) increase the total number of at-risk BPs for this measure from 18 to 20 at the expense of the NRAs under the Gas Regulations Performance Measure; (2) remove the distinction between types of damages by only measuring overall damages per 1,000 one-call tickets, which would include third-party damages and no calls in addition to mismarks and company and contractor damages; and (3) add a PRA of up to 6 BPs for reducing overall damages below a certain threshold.

Q. DO YOU OPPOSE THE COMPANY'S PROPOSAL?

A. Yes. We oppose all three changes.

Q. WHY DO YOU OPPOSE THE REALLOCATION OF BASIS POINTS?

A. Increasing the at-risk BPs again is inappropriate – this is a reduction in the penalties for the Gas Regulations Performance Measure, nothing more. We can support increasing the BPs at risk for this performance measure, but not at the expense of other NRAs.

Q. WHY DO YOU OPPOSE THE CONSOLIDATION OF THE METRICS INTO JUST OVERALL DAMAGES?

A. Lumping all damages into overall damages is inappropriate. It eliminates the important purpose served by measuring any damage caused by the Companies not properly labelling their systems (mismarks), or caused by the Companies damaging their own system (company and contractor). These damages are wholly within the Companies' control and represent competent management of their systems to which all ratepayers are entitled. Obfuscating how well the Companies manage their systems would serve no purpose, and overall damages are already reported, so this proposal does not represent an improvement on the performance measures already in place.

Q. DO YOU HAVE CONCERNS ABOUT HOW THE COMPANIES COMPUTED THEIR RECENT PERFORMANCE ON THIS METRIC?

A. Yes. Both Companies may have misreported their 2018 performance on the damage prevention metric. According to the *2018 Pipeline Safety Performance Measures Report* ("Safety Report"), issued June 13, 2019, in Case 19-G-0298, KEDNY and KEDLI were required to omit retransmits or refreshes from their performance calculations since Staff issued its guidance in December 2015, as

noted on pages 9-10 of the Safety Report. However, the Companies did not do so, making year-to-year comparisons difficult, as Staff describes on pages 4 and 5 of the Safety Report. The inclusion of retransmits and refreshes artificially deflated the actual damages per 1,000 tickets. The table below, which compares the Companies' reported performance in each of their 2018 Metrics Reports and Staff's findings in the Safety Report for 2018, exemplifies this discrepancy:

KEDNY and KEDLI 2018 Damage Performance – Company Report vs. Staff Report vs. Targets

Metric (per 1,000 tickets)	KEDNY			KEDLI		
	KEDNY Report	Staff Safety Report	2018 Performance Target	KEDLI Report	Staff Safety Report	2018 Performance Target
Overall Damages	1.17	2.56	≤ 1.87	1.70	2.79	≤ 2.48
Mismark Damages	0.34	0.75	≤ 0.42	0.49	0.79	≤ 0.6
Company and Contractor Damages	0.01	0.03	≤ 0.09	0.09	0.15	≤ 0.13

This discrepancy provides independent grounds for opposing the proposed metric consolidation. The questionable quality of historical information the Companies have provided makes future reporting of all metrics even more important.

Q. SHOULD THE COMPANIES RECEIVE A PRA FOR THIS METRIC?

A. No. The Companies should not be rewarded for minimizing damage to their systems, especially for correctly marking the location of and not damaging their own facilities. This is part of competent system management.

Q. DO THE DISCREPANCIES BETWEEN THE COMPANIES' METRICS REPORTS AND STAFF'S SAFETY REPORT MAKE A REQUEST FOR A PRA INAPPROPRIATE?

A. Yes. The Companies' inclusion of retransmits artificially deflated their numbers, making them appear much lower than what Staff considers them to be. Whatever the case, this discrepancy prevents accurate evaluation of the Companies' historical performance. Until we can accurately measure the Companies' performance relative to their historical performance, there is no basis for shareholder bonuses as there is no way to be assured that there is exceptional performance to reward. Additionally, the inclusion of retransmits calls into question the ability of the Companies to properly compute their metrics. Until there is a consistent record of proper computation, there is again no basis to determine whether shareholder incentives are warranted.

Q. WHAT DO YOU RECOMMEND?

A. Our response is threefold. First, the number of at-risk BPs should not be increased if it reduces those at risk to the Gas Regulations Performance Measure. Our proposal also is to keep 4 BPs at risk for Overall Damages, 10 BPs at risk for Mismarks, and 4 BPs at risk for Company and Contractor.

Second, the current Rate Plan's damage performance categories – total damages, mismarks, and company and contractor – should be maintained, and the thresholds should continue to decrease to encourage better performance. Customers expect continual improvement, so we propose to continue the trend from the current Rate Plan when setting performance metrics. Our proposal can be found in Exhibit__(NYCGISP-3). These targets are based on excluding resent, refreshed, revised, retransmitted, reissued, or relocated one-call tickets, per the Commission's directive in its *Order Approving Implementation Plans*, issued March 10, 2017 in case 13-M-0314, at page 4, footnote 3.

Third, damage prevention, which represents competent management of the Companies' systems, should not be rewarded with PRAs. Considering the Companies' deflation of their damages, there needs to be sufficient time to ensure that the Companies are properly computing their performance and that there is enough data to properly evaluate a PRA. We consider three years or the next rate plan, whichever is later, an appropriate time to develop accurate data that may form the basis for a future request for a PRA.

Q. DO YOU HAVE ANY OTHER CONCERNS?

A. Yes. In addition to raising doubts about KEDNY's and KEDLI's actual performance, the Safety Report suggests that KEDNY and KEDLI performed much worse than they indicated in their 2018 Metrics Reports (and possibly older Metrics Reports). This is illustrated in the table above. The Companies' performance was subject to an NRA, and according to the numbers in the Safety Report, KEDNY potentially failed on all but the company and contractor damages sub-metric in 2018, and KEDLI potentially failed on all three, resulting in a possible NRA of 14 BPs for KEDNY and 18 BPs for KEDLI.[10] The Companies should provide a full accounting of whether they properly computed their performance and whether they should be subject to an NRA.

E. GAS REGULATIONS PERFORMANCE METRICS

Q. WHAT DO THE COMPANIES PROPOSE REGARDING THEIR GAS REGULATIONS PERFORMANCE MEASURES?

A. The Companies propose the following changes to their Gas Regulations Performance Measures: (1) decreasing the total number of BPs at risk from 100 to 75; (2) creating broader "tiers" so that repeated violations of Commission regulations would entail a smaller NRA; and (3) further dividing the types of violations into categories based on whether they are found through Records Audits or Field Audits.

Q. HOW DO THE COMPANIES JUSTIFY THEIR PROPOSED CHANGES?

[10] *See* Case 16-G-0058 *et al.*, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, Joint Proposal (filed September 7, 2016), at page 52; *See id.* at page 100.

A. On page 62 of their Gas Safety Panel, the Companies claim that, because they have improved since 2015, the above changes are appropriate, and on page 48, they describe the proposed reallocation of the BPs at risk as a more fair allocation.

Q. DO YOU FIND THESE CHANGES AND THE COMPANIES' JUSTIFICATION FOR THE CHANGES REASONABLE?

A. No. The Order to Show Cause, which details 1,616 alleged safety-related violations so far, not only calls into question National Grid's ability to safely and prudently manage its system and capital budget, but also the justification for the changes to the Companies' Gas Regulations Performance Measures. The Order to Show Cause highlights a multitude of serious, credible alleged violations of the very same regulations that are the subject of this performance measure. If there has been an improvement since 2015, as National Grid claims, it has not been to a level that could justify reducing the incentive for National Grid to comply with the Commission's safety regulations.

Furthermore, the changes appear to be less about fairness and more about shielding the Companies from the ramifications of their actions. For example, on pages 19-20 of the Order to Show Cause, the Commission notes 499 alleged violations of 16 NYCRR § 255.604, which is one of the "high risk" violations in the current measure, according to page 1 of Appendix 9 to the Joint Proposal in the current rate plan. These allegations, if proven and not subject to a separate penalty proceeding, could impose a significant NRA on National Grid. Combined, if all 1,616 allegations are confirmed, the Companies could be subject to their

entire NRA of 100 BPs. Under the Companies' proposal, going forward the penalty could only be 75 BPs, and perhaps even less. National Grid's changes therefore amount to nothing more than reducing its exposure to penalties.

Accordingly, National Grid's proposed changes to the Gas Regulations Performance Metrics lack justification. The purpose of this performance measure is to incentivize KEDNY and KEDLI to maintain a safe and reliable gas system, and this NRA must remain in place to do so. Indeed, changes to increase National Grid's potential penalty exposure under this measure could be justified in light of the allegations in the Order to Show Cause. For example, National Grid acknowledges on page 62 of its Gas Safety Panel testimony that, as a result of the 2016 KEDNY and KEDLI Rate Cases, there is a cap of ten occurrences per year for non-compliance with each regulation, which it claims helps to alleviate the "overly punitive" nature of the prior metric that allowed for unlimited occurrences. In light of the allegations in the Order to Show Cause – again, which may grow as Staff discovers more – the cap may not deliver the proper incentive.

Our only caveat to our position is that we would support shifting 3 at-risk BPs from the Gas Regulations Performance Measure to the Leak Backlog NRA, provided that our recommendations are adopted to reduce the Workable Leak and Total Leak targets.

RETURN THE HISTORIC NRA BALANCE

Q. WHAT ARE THE COMPANIES' REGULATORY LIABILITIES FOR NEGATIVE REVENUE ADJUSTMENTS IN PRIOR PERIODS?

A. According to the response to CNY-1, KEDNY is carrying a regulatory liability of $21.2 million based on NRAs that were incurred in 2011 and 2013-2017, while KEDLI is carrying a regulatory liability of $13.9 million based on NRAs that were incurred in 2011-2014. As of the date of this testimony, data was not available to determine whether any NRAs have accumulated based on 2018 performance; however, the Order to Show Cause suggests that penalties may be likely.

The Companies propose, at most, to use a small portion of their historical NRA balance to fund two proposed programs: (i) the Residential Methane Detector Program; and (ii) the incremental I&R Control Line Survey and GPS Mapping Program. According to KEDNY Exhibit___(GIOP-8CU) and KEDLI Exhibit___(GIOP-7CU), the total cost for these two programs over the Rate Year and Data Years is $9.08 million and $5.56 million, respectively. Thus, if the Companies' proposals were adopted, KEDNY and KEDLI would retain historic NRA balances of $12.02 million and $8.34 million.

Q. ARE THE COMPANIES' PROPOSALS FOR THE USE OF THESE NRAs APPROPRIATE?

A. No. There is no reason that either Company should be retaining money owed to customers for penalties that were incurred up to eight years ago. Moreover, use of the NRAs should not be conditioned upon, or tied to, specific capital programs.

Q. WHAT ARE YOUR RECOMMENDATIONS?

National Grid should immediately return the full balance of historic NRAs, with interest, to customers. These amounts should be applied as a credit against the

overall revenue requirement so that they are not dependent on use for any specific capital program.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of

New York City Policy Panel

Susanne DesRoches
Elisa Velasquez
Jamie Torres-Springer

On Behalf of:

The City of New York

August 30, 2019

Cases: 19-G-0309 NYC POLICY PANEL
19-G-0310

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Ms. DesRoches – My name is Susanne DesRoches. I am the Deputy Director – Infrastructure and Energy Policy for the New York City Mayor's Office of Sustainability, or MOS, and the New York City Mayor's Office of Resiliency. My business address is 253 Broadway, 14th floor, New York, New York.

Ms. Elisa Velazquez - I have served as Assistant Commissioner and Agency Chief Contracting Officer for the New York City Department of Environmental Protection, or DEP, since 2016.

Mr. Torres-Springer – My name is Jamie Torres-Springer. I am the First Deputy Commissioner of the New York City Department of Design and Construction, or DDC. My business address is 30-30 Thomson Avenue, Long Island City, New York.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. The City of New York, or City.

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Ms. DesRoches – I received a Bachelor's Degree in Industrial Design from the Pratt Institute in 1995 and a Master in Public Administration Degree in Environmental Science and Policy from Columbia University School of International & Public Affairs in 2008.

Ms. Velasquez – I received a Bachelor of Arts Degree in Sociology from Fordham University in 1989 and a Juris Doctor Degree from New York Law School in 1992.

Mr. Torres-Springer – I received a Bachelor of Arts Degree from McGill University in 1988 and a Master's Degree in Public Policy from Harvard University in 2005.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Ms. DesRoches – I have been in my current position since September, 2015. In this role, I am responsible for the City's energy policy and regulatory affairs at the local, state, and federal level and direct the City's efforts to adapt regional infrastructure systems to climate change as outlined in the City's strategic plan, *One New York: The Plan for a Strong and Just City*. I am responsible for the City's analysis and advocacy on a wide range of energy regulatory and legislative matters before the state and federal government. I oversee a team responsible for implementing a cohesive program of resiliency initiatives across several key areas, including energy, telecommunications, water/sewer/waste, and transportation. Prior to joining the City, I served as the Chief for Resilience and Sustainability in the Engineering Department of the Port Authority of New York & New Jersey, where I was charged with incorporating climate resiliency and sustainability into all agency capital projects.

Ms. Velasquez – I am responsible for the procurement activities of the Agency, including competitive sealed bids; requests for proposals; small purchases and purchases; consultant contract negotiations; contract payments; vendor

performance evaluations; establishing MWBE goals and opportunities; and State Prevailing Wage Laws compliance. I also run the fiscal and accounting sections at DEP.

Mr. Torres-Springer – I have served as First Deputy Commissioner of DDC since 2018 and am responsible for a range of project delivery, agency administrative, and intergovernmental relations matters. I helped to lead the agency to its *Strategic Blueprint for Construction Excellence*, released in January of this year, which identifies a series of critical measures to improve on the timing and cost effectiveness of the City's infrastructure and public buildings projects, including among them a number of strategies to improve coordination between DDC and regulated public utilities.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Ms. DesRoches – I have testified before the Public Service Commission in prior rate cases and in the retail access proceeding in Case 12-M-0476. Most recently, I testified in Cases 19-E-0065 and 19-G-0066, which are pending electric and gas rate cases involving Consolidated Edison Company of New York, Inc. ("Con Edison"). Additionally, I have testified before the New York City Council on the City's sustainability initiatives, energy planning, and policy issues. Finally, I speak frequently before international, national, and local audiences on the City's long-

term energy planning and regulatory affairs, infrastructure resiliency, and broader climate change planning.

Ms. Velasquez – I have not testified previously on these issues but have testified as both a fact and expert witness at several vendor criminal trials and grand juries.

Mr. Torres-Springer – I also testified in Cases 19-E-0065 and 19-G-0066, the pending electric and gas rate cases involving Con Edison.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of this testimony is to discuss the City's energy policies, their correlation to State energy policies, and their relationship to these rate cases; the need for Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and Keyspan Gas East Corp. d/b/a Brooklyn Union of L.I. ("KEDLI") (together, "Companies") to develop plans, actions, and activities to be designed to advance the City's and State's policies; the need for balance between the interests of customers and shareholders and more generally the magnitude of the KEDNY and KEDLI rate requests; and the need for better coordination and communication between the Companies and City agencies. We also provide a summary of the City's other pieces of testimony.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. We offer Exhibit__(NYCPP-1), which is a compilation of the discovery responses from the Companies that we relied upon in preparing our testimony, some of which we expressly reference; Exhibit__(NYCPP-2), which contains Volume 7,

entitled "A Livable Climate," or A Livable Climate, of *OneNYC 2050: Building A Strong And Fair City*, a report detailing Mayor de Blasio's updated strategic plan to combat global warming and achieve environmental sustainability, economic equality, and social justice; Exhibit__(NYCPP-3), which contains a copy of the *New York City's Roadmap to 80x50*, or Roadmap to 80x50, a report prepared by MOS and issued in October 2016 to leverage scientific analysis and greenhouse gas emissions modeling to identify the pathways and processes necessary to achieve an 80 percent reduction in greenhouse gas emissions by 2050, as compared to 2005 levels; and Exhibit__(NYCPP-4), which is a copy of the *Citywide Privacy Protection Policies and Protocols* that were adopted earlier this year.

Q. PLEASE SUMMARIZE YOUR TESTIMONY.

A. One of the primary objectives of both the City and State is significantly reducing greenhouse gas emissions through a multi-faceted approach that includes increasing reliance on demand management, renewable gas resources, and expanding energy efficiency programs. At the same time, both the City and State are concerned about maintaining a reliable and resilient gas system, and keeping gas rates affordable, especially for low income and vulnerable residents.

Initially, we discuss the public policy considerations affecting the requested rate increases. To help meet the City's and State's clean energy transition goals while also reducing the need for gas capital expenditures to accommodate growing demand, ambitious demand response and energy efficiency efforts are needed, as discussed in the Public Service Commission's Order issued on December 13, 2018

in Case 18-M-0084, or Energy Efficiency Order. These more ambitious efforts should be included as part of the Companies' core planning activities.

Second, we discuss our concerns with the overall magnitude of the capital expenditures proposed and point to the need to balance the Companies' requests for rate increases and shareholder incentives against costs to consumers. In these rate cases, KEDNY and KEDLI are seeking very substantial increases in their gas revenue requirements. The Companies also are seeking multiple shareholder incentives, some of which would reward them for actions expected of any prudent utility company. The shareholder incentives place incremental burdens on customers over and above the impacts of the revenue requirement increases. In this section, we also address the need to provide low income customers with additional protection from rate increases that they cannot afford to pay.

Third, we address the need to ensure that gas service in New York City remains highly reliable and resilient, while, at the same time, minimizing the potential for excessive stranded costs. In this section, we highlight flaws in the Companies' planning processes that have resulted in a gas moratorium affecting much of their service territories and recommend several actions by the Companies, including the need to develop alternatives to the Northeast Supply Enhancement ("NESE") pipeline, which the Companies continue to simply assume will be constructed. We also highlight problems associated with the Companies' implementation of their gas moratorium.

Fourth, we advocate for increased energy efficiency and demand response measures to be incorporated into system planning, particularly in light of the uncertainty surrounding the NESE pipeline project.

Finally, we request that the Commission require that the Companies institute better, more formalized, high-level communications between them and City agencies, which should result in reduced capital and O&M spending by the Companies.

ENERGY PUBLIC POLICY CONSIDERATIONS

Q. PLEASE PROVIDE A BRIEF DESCRIPTION OF THE CITY'S ENERGY POLICIES.

A. Effective energy policy plays a critical role in driving successful and equitable economic development, public health, sustainability, and resiliency. In that regard, the City actively seeks to balance objectives that include affordable electric and gas rates; a resilient and reliable utility system, especially for critical infrastructure and vulnerable populations; and supporting the transition of energy supply and distribution systems to realize deep greenhouse gas emissions reductions and related sustainability goals. This holistic approach is captured in the Administration's OneNYC strategic plan, which was most recently updated in April, 2019. Exhibits__(NYCPP-2) and __(NYCPP-3) provide details from the OneNYC plan that are pertinent to our testimony.

With respect to gas rates, the City is committed to identifying pathways to reduce the energy cost burden (defined as the sum of electric, gas and/or fuel oil

expenditures relative to pre-tax income) for low income families. The City applauds the Commission's goal of capping energy cost burden for New York State families making less than 200 percent of the Federal Poverty Level at six percent and looks forward to working with the Department of Public Service, other State entities, and stakeholders to achieve this goal. In addition to reducing the economic burdens of New York City and New York State's most vulnerable residents, affordable gas rates are critical to economic development and the continued prosperity of New York City, and in turn, New York State.

Q. WHAT HAS THE CITY DONE TO ADDRESS CLIMATE CHANGE?

A. The City has made a series of significant commitments with respect to deep decarbonization and sustainability that drive our energy policy goals, of which energy efficiency and increasing the deployment of renewables are two key pillars. The City has made unprecedented commitments to achieve its objectives of creating a resilient and low-carbon energy supply, improving air quality, and achieving carbon neutrality by 2050, as set forth in *OneNYC 2050: Building a Strong and Fair City* ("OneNYC").[1] Last month, the New York City Council passed Intro 1253-c, which sets greenhouse gas emissions caps on large buildings, beginning in 2024.

Most recently, Mayor de Blasio announced New York City's Green New Deal, a plan to commit $14 billion that will directly confront income inequality,

[1] *OneNYC 2050: Building a Strong and Fair City,* Vol. 7: "A Livable Climate" (issued April 2019) at 10, available at http://onenyc.cityofnewyork.us/strategies/a-livable-climate/.

create tens of thousands of new jobs, and fight climate change by seeking to further reduce greenhouse gas emissions by nearly 30 percent by 2030.

To better understand the cost, technical feasibility, and appropriate phasing of policies and strategies to meet our 2050 goals, the City is commencing a study on deep decarbonization in coordination with and jointly funded by Con Edison and National Grid that will conclude by spring 2020. The study also will examine potential stranded asset issues resulting from deep decarbonization – particularly for gas infrastructure – and identify steps that can be taken to mitigate the risk of significant stranded assets and stranded costs. We anticipate that the study will inform future regulatory and policy actions.

Further, the City is committed to protecting New York City from the impacts of climate change, focusing on climate resiliency for critical infrastructure and vulnerable populations. To ensure future capital investments that promote resiliency, the City has recently published Climate Resiliency Design Guidelines for use by all City agencies.[2] It has worked with KEDNY and KEDLI for a number of years on issues related to climate change and storm hardening and looks forward to continuing to do so.

Q. WHAT HAS THE CITY DONE TO PROMOTE AND ENCOURAGE ENERGY EFFICIENCY INVESTMENTS?

[2] *Climate Resiliency Design Guidelines – Version* 3.0 (issued March 2019), https://www1.nyc.gov/assets/orr/pdf/NYC_Climate_Resiliency_Design_Guidelines_v3-0.pdf.

A. On the energy efficiency side, the City's *Greener, Greater Buildings Plan* requires large buildings to undertake approaches to energy efficiency, including benchmarking, auditing, retro-commissioning, and lighting upgrades. In 2015, the City released *One City: Built to Last*, which outlined a 10-year action plan to improve the energy efficiency of large buildings by 2025. The City has since launched a series of programs to facilitate implementation of these goals. For example, in 2015, the City launched the NYC Retrofit Accelerator, which provides coordinated outreach and assistance to private building owners and decision-makers so as to accelerate efficiency retrofits and clean energy investments using Local Law 87 data. The NYC Retrofit Accelerator has been very positively received by developers, building owners, contractors, and many others.

In 2019, the City tripled its funding for the second phase of the NYC Retrofit Accelerator, which will be expanded to continue to help building owners implement energy efficiency retrofits and invest in clean energy. The expanded program will reach over 5,000 buildings and hundreds of stakeholders through its targeted outreach campaign for new and existing buildings, on-site trainings, and leadership recognition platform.

The City has led by example by making energy efficiency improvements in, and adding clean distributed energy resources to, municipal buildings. As a result, the City has achieved a 30 percent reduction in greenhouse gas emissions relative to fiscal year 2006 across its own operations and buildings. The City also has

strengthened its building codes and continues to make investments and take actions to make New York City more sustainable and resilient.

Q. PLEASE BRIEFLY DISCUSS THE CITY'S ONENYC INITIATIVE.

A. OneNYC 2050 is Mayor de Blasio's plan for ensuring the future health, welfare and prosperity of New York City. The latest iteration to the plan, *OneNYC 2050: Building A Strong And Fair City*, released on Earth Day 2019, provides an updated and detailed strategy for achieving these goals. It reflects our core strengths as a city – our growing and diverse population, strong economy, and global leadership – and the significant challenges we must address – growing unaffordability, economic insecurity, inequity, and climate change. It imagines a New York City in 2050 that has a vibrant democracy, an inclusive economy, thriving neighborhoods, healthy lives, equity and excellence in education, a livable climate, efficient mobility, and modern infrastructure.

The OneNYC plan sets forth eight goals and 30 initiatives that collectively establish bold targets for a strong, sustainable, and resilient New York City. It includes a plan for achieving carbon neutrality and 100 percent clean electricity, growing the climate economy to create green job opportunities for thousands of New Yorkers, and strengthening communities, buildings, infrastructure, and waterfronts to be more resilient.

Q. WHAT IS THE RELATIONSHIP BETWEEN THE CITY'S POLICIES AND GOALS AND THE STATE'S ENERGY-RELATED POLICIES AND GOALS?

A. The City's and the State's goals are very much in line with each other. Governor Cuomo announced a Statewide Green New Deal earlier this year that includes a commitment to relying on renewables for 70 percent of electricity consumption by 2030, and by 2040 statewide electric demand will generate zero emissions. Moreover, the Governor and the Public Service Commission have established short-term goals for penetration of electric vehicles and energy savings from the installation of heat pumps. Most recently, the State Legislature approved, and the Governor signed, the Climate Leadership and Community Protection Act, or CLCPA, which is sweeping legislation designed to dramatically reduce greenhouse gas emissions.

In sum, any efforts undertaken to achieve the City's goals also would help to achieve the State's goals, and vice-versa. Therefore, the Commission should consider the City's recommendations in these proceedings to be equally applicable to its advancement of the State's policy determinations as they are to advancement of the City's policies.

Q. WHAT SHOULD KEDNY AND KEDLI BE DOING TO ACHIEVE CITY AND STATE POLICY GOALS?

A. Achieving the City's and State's policy goals, reducing the reliance on fossil fuels, and encouraging customers to be more energy efficient will require a coordinated effort among government, the utilities, environmental and community groups, and other stakeholders. We commend the Companies' efforts to address policy goals through their focus on the Future of Heat and their increased spending on energy

efficiency measures. However, as set forth below and in the testimony of other City panels, much more needs to be done.

NEED FOR FAIRNESS AND BALANCED CONSIDERATION OF CUSTOMER AND SHAREHOLDER INTERESTS

Q. IN REVIEWING THE COMPANIES' RATE FILING, HOW SHOULD THE COMMISSION BALANCE THE INTERESTS OF CUSTOMERS AND SHAREHOLDERS?

A. The Companies initially were seeking annual revenue requirement increases of $236.9 million and $49.4 million, respectively. Based on the Corrections and Update Filing ("C&U Filing"), the requested annual revenue requirements are $195.6 million and $61.0 million for KEDNY and KEDLI, respectively.

There is a need to balance the Companies' proposals for their gas systems with the impacts of those proposals on customers. Rates should not be considered an unlimited source of funding for the utilities. The Commission must carefully review and evaluate the Companies' proposals, and it must make adjustments to the requests to ensure that the rates charged to customers are fair and reasonable.

The Companies are planning unprecedented capital spending. We trust that most of the projects included in the Companies' capital plans are, on their face, legitimate utility projects. However, we question whether the Companies have the ability and internal resources to properly manage such a large spending plan and ensure that all expenditures are prudent, and proper practices are in place to prevent waste.

In this regard, the recent Order Instituting Proceeding and to Show Cause, issued by the Commission on July 12, 2019 in Case 17-G-0317 ("Show Cause Proceeding"), is particularly troubling. In that Order, the Commission sets forth detailed allegations by DPS Staff of more than 1,600 violations of Commission regulations by the Companies. These violations indicate a lack of proper safety practices and failed supervision, and resulted in a yet-to-be determined sum of unnecessary spending. Although the extent of the damages resulting from the Companies' operations will be determined in Case 17-G-0317, the detailed allegations suggest that a particularly rigorous review of the Companies' projected costs is required here.

Q. WHAT IS THE POTENTIAL IMPACT OF ADDITIONAL SURCHARGES AND INCENTIVES?

A. In addition to the requested rate increases, the Commission cannot ignore or disregard the additional costs that will be borne by customers related to items that are not included in the revenue requirement. According to the Companies' response to City Information Request 67, or CNY-67, approximately $53.2 million in additional surcharges will be collected from gas customers in 2020. In addition to these surcharges, the Companies together are seeking at least $75.1 million in shareholder incentives, inclusive of the maximum possible Earning Adjustment Mechanisms (or EAM) award, per Appendix B of each Company's Corrected and Updated Revenue Requirement Panel testimony. Thus, the potential bill impacts are larger than the amounts the Companies have stated.

Q. WHAT DO YOU RECOMMEND REGARDING CAPITAL EXPENDITURES?

A. With respect to the capital expenditures in these rate cases, unless the total amounts are substantially reduced, the Commission should require that the amounts allowed to be recovered in rates be subject to reconciliation in the event the Companies do not, or cannot, complete all of the projects they identified. Otherwise, the Companies' shareholders would be unfairly enriched, and the imbalance we spoke of earlier would be exacerbated. Additionally, particularly in light of the Show Cause proceeding, the Commission should hold the Companies responsible for all penalties and additional costs incurred to repair or replace equipment that was installed improperly or work that was not completed in accordance with applicable rules and regulations, Company procedures, and/or good utility practice. These positions are detailed further in the City's Infrastructure and Safety Panel testimony.

Q. EARLIER, YOU TESTIFIED THAT THE COMPANY SHOULD BE PERMITTED TO RECEIVE SHAREHOLDER INCENTIVES FOR SUPERIOR PERFORMANCE. PLEASE EXPAND ON THAT STATEMENT.

A. The City recognizes that sometimes rewards may be needed to induce certain behavior. In the case of utilities, shareholder incentives – both positive and negative – have been used to discourage poor performance and encourage the utilities to do tasks they otherwise may not want to do. At the same time, the

Commission has very broad power to order the utilities to take certain actions or refrain from taking other actions. Violations of Commission Orders can result in monetary penalties of hundreds of thousands of dollars or more. In fact, in response to CNY-1, the Companies identify $13.9 million (KEDLI) and $21.2 million (KEDNY) of penalties that are owed to customers and should be returned.

Accordingly, there is no reason to provide shareholder incentives to the Companies related to any action that the Commission requires them to take or not to take, and for which the Companies would be liable for penalties for failure to comply. An example of this is providing safe and adequate service by expeditiously responding to gas leak and odor calls – there is no reason to reward the Company for the timeliness of such actions, which are required pursuant to good utility practice.

On the other hand, for discretionary actions taken by the Companies, the use of shareholder incentives may be appropriate. In such cases, though, the incentives should reward only superior performance, not business-as-usual conduct. The City Energy Efficiency and Gas Infrastructure and Safety Panels address the Companies' proposed incentives in detail.

Q. HOW SHOULD THE COMPANY'S ENERGY EFFICIENCY PERFORMANCE BE MEASURED?

A. The City's Energy Efficiency Panel addresses this topic in detail. We note that the Companies have been administering energy efficiency programs for many years, and arguably have been doing so more aggressively in recent years. There is now

sufficient information available to determine baseline performance levels. Any shareholder incentives should start at levels above baseline performance, and any increases in the incentives should be commensurate with achievements well above the initial levels.

Q. HOW IS THIS PROPOSAL CONSISTENT WITH THE BALANCE YOU HAVE BEEN EMPHASIZING?

A. Reducing gas usage and gas peak demand benefits all customers because it reduces the need for additional infrastructure. That reduced need, coupled with lower demand and usage, should result in lower commodity and delivery prices for customers. For these reasons, coupling shareholder incentives with achieving high levels of energy savings creates a win-win outcome for shareholders and customers. The recent gas moratorium declared by the Companies only highlights the urgent need to accelerate energy efficiency and demand response measures to better manage their gas systems.

Q. WHAT IS THE IMPACT OF THE REQUESTED RATE INCREASES ON LOW INCOME CUSTOMERS?

A. The City's Low Income Panel presents a recently completed analysis of energy cost burdens on New York City residents. The results of this analysis are striking and should be considered in evaluating the Companies' revenue requirements and shareholder incentive requests.

Q. WHAT DID THE ANALYSIS REVEAL?

A. In 2016, the Commission established that New Yorkers should not pay more than six percent of their income on utility expenses. It then adopted parameters for utility low income programs that it believed would achieve this target. At that time, the City raised numerous concerns about the methodology used and the results obtained. Based upon the results of the City's most recent analysis, as discussed in more detail in the City's Low Income Panel testimony, we continue to believe that the methodology is flawed.

The City's analysis reveals that the discounts established by the Commission are not sufficient to reduce the energy cost burden for low income families to six percent or below. In fact, with the current KEDNY and KEDLI discounts, there are more than 220,470 low income families (under 60 percent of the State Median Income) in the KEDNY service territory and the Rockaways portion of the KEDLI service territory whose energy cost burdens exceed six percent of their pre-tax income. The Companies' proposed rate increases, if adopted without any corresponding changes to existing low income discounts, likely will exacerbate these figures. The City's Low Income Panel has several specific recommendations designed to provide additional protections for low income customers.

Q. HOW DOES THE IMPACT ON LOW INCOME CUSTOMERS RELATE TO THE BALANCING OF CUSTOMER AND SHAREHOLDER INTERESTS?

A. KEDNY and KEDLI are proposing budgets of $25.5 million and $5.5 million, respectively, for gas low income programs in the Rate Year. At the same time, they are seeking more than $45 million and $29 million in shareholder incentives, as well as an increase in its allowed return on equity that has a value, based on the Companies' response to CNY-64, of $35.3 million. In other words, the Companies are seeking benefits for their shareholders that are more than three and a half times larger than the assistance they propose to provide to their customers with the lowest income levels. In our view, the benefits sought for shareholders are not in balance with the benefits that may be received by customers.

Q. WHAT ELSE SHOULD THE COMPANIES DO TO HELP THEIR LOW INCOME CUSTOMERS?

A. As discussed by the City's Low Income Panel, KEDNY and KEDLI should provide deeper discounts to these customers to reduce their energy cost burdens. In addition, reconnection fees are particularly burdensome for low income customers, and should be eliminated altogether. Finally, the Companies should adopt certain improvements to their existing file-matching process with the New York City Human Resources Administration, or HRA, to better identify and enroll eligible low income customers into the utilities' low income programs.

Q. HOW DO YOU RECOMMEND THAT THE COMMISSION ACHIEVE A PROPER BALANCE OF INTERESTS?

A. The Commission has the authority to provide a more equitable and appropriate balance and should do so. For example, our Infrastructure and Safety Panel

recommends the immediate return of certain customer credits being held by the Companies. As discussed by the City's Energy Efficiency and Infrastructure and Safety Panels, the size of the shareholder incentives should be materially reduced, and City Witness Lane Kollen is recommending a significant downward adjustment to the Companies' rate requests based on his analysis and rejection of a change to the depreciation methodology proposed by the Companies. Although the City is not submitting testimony on the appropriate return on equity and capital structure, we understand that Staff and possibly other parties will be doing so and will be proposing both a lower return on equity and a lower equity ratio. We hope that the Commission will consider all of these changes to ensure a proper balance is achieved. Finally, we ask that the relief our Low Income Panel seeks for low-income customers is granted.

MAINTAINING A RELIABLE AND RESILIENT GAS DELIVERY SYSTEM

Q. DOES THE CITY HAVE ANY CONCERNS REGARDING THE RESILIENCE OF THE KEDNY AND KEDLI SYSTEMS?

A. After Hurricane Sandy, the City engaged in a comprehensive assessment of the resilience of utility infrastructure. One weakness that was identified in this assessment pertains to the number of KEDNY and KEDLI assets that are located in or adjacent to the existing 2015 Federal Emergency Management Agency 100-year and 500-year floodplains. Inasmuch as these assets are critical to the gas system, the City believes that the Companies should continue to monitor and invest in actions to ensure the resilience of these assets and consider contingency planning

in the event of asset failure. The City's Gas Infrastructure and Safety Panel makes a number of recommendations designed to address these concerns.

Q. PLEASE COMMENT ON THE COMPANIES' RESPONSE TO THE STALLED NESE PROJECT.

A. On May 15, 2019, the New York State Department of Environmental Conservation ("DEC") refused to grant a water certificate needed for the NESE project to proceed. On May 16, 2019, in response to the DEC rejection, the Companies announced that they would not accept applications for new or additional firm gas service, effectively imposing a gas moratorium. On June 5, 2019, the New Jersey Department of Environmental Protection followed suit and denied permits required by the NESE project.

On June 11, 2019, the Companies filed supplemental testimony to address the consequences of the unavailability of the NESE pipeline. The supplemental testimony states that the loss of the NESE project would result in additional increases to the requested KEDNY and KEDLI revenue requirements of $5.7 million and $15.2 million, respectively. Other than quantifying the impact on revenue requirements, the supplemental testimony did not offer any alternative plans that the Companies would pursue if the NESE pipeline is not built. The Companies maintained a similar approach in its C&U Filing filed on July 3, 2019.

Q. DO YOU HAVE CONCERNS WITH HOW THE COMPANIES HAVE ADDRESSED THE NESE PIPELINE?

A. Yes, very much so. The New York and New Jersey approvals sought by the NESE pipeline sponsors were highlighted as controversial for some time before they were denied. Thus, simply assuming that the approvals would be granted, and not developing alternative plans to address how to continue to provide incremental service to existing and new customers, was and continues to be imprudent. Yet, this appears to be what the Companies have done, and, based on the responses to CNY-118, continue to do.

Q. DOES THE PANEL HAVE ANY ADDITIONAL COMMENTS ON THE COMPANIES' RESPONSES TO THE NESE PROJECT?

A. We believe the Companies should have included alternative plans to the NESE project in their initial rate filing on April 30, 2019, and certainly in either their supplemental or C&U Filing testimony. Imposing a gas moratorium and remaining "cautiously optimistic" about New York and New Jersey regulators reversing their decisions, as the Companies have stated, is not adequate planning.

Q. WHAT SHOULD THE COMMISSION DO?

A. The Commission should require the Companies to develop and file testimony addressing how they will operate their systems, without a moratorium, if the NESE project is not approved. For example, in response to growing demands that led to a moratorium, Con Edison announced that it reached an agreement with Tennessee Gas Pipeline to upgrade compressor stations to provide increased gas capacity to its system. The Companies have not identified any such alternatives, and the

Commission needs to require them to find ways to operate their systems so that they can provide adequate service.

In our testimony and the testimony of other City panels, we highlight alternative demand side programs, such as increased energy efficiency and demand response programs, which could provide load relief for the Companies' system. In short, the Commission should require the Companies to do what they should have done already – provide a combination of supply and demand side measures that meet their customers' needs without solely relying on the NESE project. Viable programs should be funded robustly in order to relieve what is currently a wholly untenable situation. For example, in response to the Con Edison gas moratorium in Westchester County, the Commission, the New York Power Authority, and the New York State Energy Research & Development Authority announced a $250 million Clean Energy Action Plan that included, among other things, $165 million in grants to deploy heat pumps and increase energy efficiency programs. The City would expect an Action Plan in response to the National Grid moratorium in proportion to the scale of the issue affecting National Grid's New York City service territory.

In addition, the Commission should ensure that parties are given a proper and fair opportunity to review, question, analyze, and respond to the Companies' plans showing alternatives when they are filed. To the extent that the timeline for resolution of the gas rate case needs to be extended to allow for this process, the Commission should either ask the Companies to extend the suspension period or

adopt temporary rates so that no portion of the Companies' gas rates are finalized until all aspects of their requests can be properly reviewed.

Q. DO YOU HAVE SPECIFIC CONCERNS ABOUT THE MORATORIUM ITSELF?

A. Yes. It has come to our attention that the Companies' implementation of the moratorium may be unfair to customers. The City recently learned that National Grid has established rigid deadlines for homeowners, building owners, and developers who applied for gas service prior to May 15, 2019 (and whose applications were deemed complete by that date). However, National Grid does not appear to be advising its existing customers and prospective customers of these deadlines. Specifically, for applicants seeking to convert from oil to natural gas, their buildings must be ready to accept gas service and utilize natural gas by November 15, 2019. For applicants seeking gas service for new homes and buildings, the construction must be complete and the premises must be ready to accept gas service and utilize natural gas by May 15, 2020. It appears that National Grid is not publicizing these deadlines.

The homeowners, building owners, and developers who submitted complete gas applications prior to May 15, 2019 have a reasonable basis to believe that they will receive gas service for their homes and buildings. It would be unjust and unfair for National Grid to subsequently deny customer requests for gas connections for failure to meet deadlines of which they are unaware. Notably, Governor Cuomo issued a letter on August 27, 2019, expressing concerns about National Grid's

implementation of the moratorium and directing the Commission to expand its investigation of the moratorium to determine whether allegations of National Grid unfairly delaying or denying service are accurate and, if so, to take appropriate remedial actions.

Q. WHAT SHOULD THE COMMISSION DO WITH RESPECT TO THE MORATORIUM?

A. The Commission should have DPS Staff investigate the underlying causes of the moratorium and determine whether it could have been averted with proper planning. In addition, the Commission should require that the Companies immediately provide notice to customers of how the moratorium is being implemented and the deadlines associated with the moratorium. That effort should include directly notifying each applicant and providing public notice of the deadlines via social media, ads in the New York City daily newspapers, contacts to trade groups and similar organizations, and other means as appropriate. Moreover, the Commission should evaluate the deadlines National Grid has asserted, and consider if there are later, viable deadlines that would provide flexibility for projects already under construction. Finally, the Commission should investigate whether the moratorium is being implemented equitably in order to ensure that customers are being treated fairly and without prejudice.

Q. WHAT ACTIONS SHOULD THE COMPANIES UNDERTAKE WITH RESPECT TO STRANDED COSTS?

A. Given the goals of deep decarbonization and the clean energy transition, the composition of utility infrastructure will change over time. In particular, the need for the Companies' extensive gas infrastructure may diminish. At present, KEDNY and KEDLI are planning to spend approximately $2.3 billion over the next four years to replace leak-prone pipe. As the demand for gas diminishes, the propriety of these investments will need to be re-examined. As for the investments already made and/or being made over the prospective term of this rate case, there is a potential for some to become stranded. We recommend that the Companies, in coordination with interested parties, begin to examine the risk of stranded assets and conduct research and analysis into opportunities to mitigate those risks.

Q. DOES THE CITY HAVE ANY ADDITIONAL COMMENTS ON THE POTENTIAL STRANDED ASSET RISK?

A. There could be a unique opportunity to address a subset of potential stranded asset concerns regarding ongoing leak-prone pipe replacement while advancing the City's and State's beneficial electrification goals. We recommend that the Companies conduct a feasibility study that includes the following elements:

- Identification of neighborhoods with high-concentrations of leak-prone pipe that need to be replaced;
- An assessment of whether it is technically feasible to not replace the leak-prone pipe in certain neighborhoods without affecting other customers, such as at the end of pipeline segments or at the edges of the Company's gas service territory;

- An assessment of the technical feasibility of deploying district energy systems, such as a shared geothermal ground source heat pump system, that could be used to provide heating and cooling needs and domestic hot water for those buildings within identified neighborhoods; this assessment should consider the feasibility of different types of geothermal (e.g., standing column well systems, closed and open loop systems) and other types of district systems to determine best fit (e.g., low carbon high / low-temperature hot water and /or steam loops). The City proposes that such an assessment be conducted in phases, as follows:
  a. Phase One – Retain a consultant to complete a high-level Desktop Study on the feasibility of deploying district energy systems in the Companies' service territories, leveraging existing studies and data. The Desktop Study should align district energy deployment with leak-prone pipe removal to increase financial feasibility and avoid stranded assets;
  b. Phase Two – Based upon the initial Desktop Study, identify ten to twenty sites that indicate high potential and/or represent unique use cases and complete a detailed assessment of those sites to supplement the Phase One Desktop Study, including an assessment of local site characteristics, customer interest, detailed load data, financial feasibility, and more detailed design (with the goal of ranking the sites based on benefits, costs, and feasibility);

c. Phase Three – Conduct a Pilot at several high potential sites, which would be selected to maximize the potential for success and replicability. To the extent possible, the different Pilot sites should test different use cases, and should include the deployment and operation of the geothermal district or another type of district system based upon the assessments in Phases I and II, as well as the establishment of a viable regulatory, ownership, and governance structure; and

d. Phase Four – Assess the scalability of the Pilot to additional locations. As part of this Phase, the consultant should identify what interventions and/or regulatory changes are needed to lead to greater adoption of district systems, as well as the projected benefits and costs of expanding the Pilot.

- Consideration of a regulatory model in which the avoided costs of the leak-prone pipe replacements in the selected neighborhoods can be used to offset the capital costs of a district energy system.

We also recommend that the performance of the feasibility study be preceded by a scoping discussion with interested parties to develop specific details for both the study contents and the manner in which the study is performed.

Q. WHAT OTHER ISSUES SHOULD BE CONSIDERED IF THE FEASIBILITY STUDY CONCLUDES THAT A DISTRICT ENERGY SYSTEM IS VIABLE?

A. There would be a need to examine strategies to incentivize customer behavior to choose a district energy system over other alternatives. This may include providing incentives to support purchases of customer-side equipment and customer interconnection costs. Another issue would be identifying strategies and incentives to encourage every customer on a block to voluntarily convert at the same time in order to eliminate the need for the gas mains. Another key question to be evaluated is whether the total costs of the district energy system would be less than the cost of the leak-prone pipe replacement. As part of this, there could also be value in examining existing depreciation practices for new pipe installed and whether these should be modified to better align with City and State policy goals, including the CLCPA, as this could impact the cost comparison noted above.

Q. IS THIS CONCEPT BEING EXAMINED ELSEWHERE?

A. Yes. A similar evaluation is underway in Massachusetts by the non-profit Home Energy Efficiency Team, which has worked with Massachusetts gas utilities to identify existing leaks on the gas system and is now working with an engineering firm to conduct a GeoMicroDistrict Feasibility Study. That study will explore the potential of replacing natural gas infrastructure with a network of neighborhood-scale district heating systems. Specifically, the study is examining the engineering and economic feasibility and technical potential for implementing and scaling neighborhood-scale thermal energy technologies, and the costs and benefits to both utilities and ratepayers of implementing those systems instead of repairing and replacing existing natural gas infrastructure. In addition, the City is aware of

National Grid's own experience with implementing a shared geothermal pilot in Long Island. There may be learnings from that project that are applicable here.

Q. HOW COULD SUCH A STUDY HELP INFORM THE EFFORTS TO MEET NEW YORK STATE'S AND NEW YORK CITY'S PUBLIC POLICY GOALS?

A. Alternatives such as district energy systems, particularly geothermal district systems, can help the City and State meet long-term decarbonization goals, of which decarbonized heating is a very important and challenging component. Similarly, in addition to looking at renewable gas and alternative low-carbon fuels, there is an ongoing need to evaluate and explore what role gas utilities can play to support this clean energy transition. The use of district energy systems could help reduce a subset of stranded asset risks associated with the new gas infrastructure investments contemplated by the Companies.

Q. ARE THERE ANY CITY RESOURCES THAT COULD SUPPORT SUCH A FEASIBILITY ASSESSMENT?

A. Yes, since 2015, the City has issued one report and promoted two tools to help assess the technical feasibility of building-based geothermal systems and district geothermal systems. Specifically, the City in 2015 published *Geothermal Systems and their Application in New York City*. In 2016, the City assessed the technical feasibility of geothermal systems at the block level for the Roadmap to 80x50 as part of its community energy mapping process. The City is in the process of turning this into an interactive publicly accessible mapping tool. In 2018, the City launched

a Geothermal Pre-feasibility Tool to identify where geothermal heat pump systems may be an option for retrofitting building heating and cooling systems, including an estimated payback period. This tool allows one to search by address to determine the suitability of the location to different geothermal systems, potential capacity, and potential projected payback period.

THE NEED FOR INCREASED ENERGY EFFICIENCY AND DEMAND RESPONSE EFFORTS

Q. HOW SHOULD THE COMPANIES RESPOND TO THE GAS SUPPLY PROBLEMS THAT PROMPTED THEIR MORATORIUMS?

A. The Companies should place more emphasis on gas energy efficiency and demand response as well as move forward with their plans to support renewable gas production. As noted earlier, given the DEC's decision to deny Water Quality Certification for the NESE project, it is critical for existing gas customers to immediately use gas as efficiently as possible to ensure that sufficient gas supplies are available during peak periods. In addition, the Companies should increase their demand response efforts to manage their systems more efficiently. Also, it is critical for the Companies to move forward with alternate sources of gas supply, such as renewable gas. Because new upstream pipelines are not likely to be viable, localized production will become more important to fill the gap between the Companies available gas supplies and projected need. Finally, as noted earlier, as it did in response to the Con Edison moratorium affecting Westchester County, we expect that the State will develop an Action Plan in response to the National Grid

moratorium in proportion to the scale of the issue affecting National Grid's New York City service territory.

Q. WHAT SPECIFIC ENERGY EFFICIENCY MEASURES IS THE CITY RECOMMENDING?

A. The City's Energy Efficiency Panel justifies and recommends a number of measures, including but not limited to:

- Incremental energy efficiency target savings should be increased for both KEDNY and KEDLI, based on the lower per-unit costs for energy efficiency savings that have been achieved in the last two years.
- The Companies should test the concept of providing upfront energy efficiency funding for certain multifamily energy efficiency program offerings.
- Energy efficiency program offerings should be designed to incentivize customers to pursue comprehensive packages of energy efficiency measures (*i.e.*, allow stacking of incentives if several measures are pursued at once).
- Incentives should accommodate comprehensive upgrades that occur in stages, or over a longer period of time, rather than all at once.
- The PSC should establish a collaborative or other process to explore future improvements to the Companies' energy efficiency plans and programs.
- The Companies should provide additional reporting on energy efficiency programs, expenditures, savings, etc.

- There needs to be greater coordination and collaboration between the Companies' energy efficiency programs and the NYC Retrofit Accelerator.
- The Companies' demand response targets should be increased.
- The Companies should increase their efforts to transition their fleet to run on EVs or renewable natural gas.

The Panel also raises a number of concerns regarding EAMs proposed by the Companies to reward their shareholders, namely:

- National Grid's proposed EAMs reward business-as-usual performance, not exceptional performance.
- The proposed EAMs are excessive, both in absolute dollar values and relative to the benefits that customers could receive.
- Many of the proposed EAMs are duplicative of other proposed EAMs or would reward National Grid for occurrences outside their control.

Q. WHAT SHOULD THE COMPANIES DO FROM AN INFORMATION TECHNOLOGY PERSPECTIVE?

A. As the Companies continue to overhaul their information technology systems, including those that are internal as well as customer facing, they should include features and capture data that can support the implementation and tracking of City and State policy goals.

Q. PLEASE DESCRIBE THE FEATURES AND TYPES OF CUSTOMER DATA THAT THE COMPANIES SHOULD INCLUDE IN THEIR SYSTEMS DESIGNS.

A. For the purposes of tracking progress toward meeting broader City and State energy policy goals, we recommend the Companies' information systems include the ability to collect data on:

- The effectiveness of energy efficiency and demand response efforts in managing the gas system;
- Details regarding incentives provided to customers, including measures undertaken, costs of measures, and projected and realized savings; and
- Production, cost and gas quality data associated with any renewable gas projects providing gas to KEDNY and KEDLI, as well as any monetization of Renewable Identification Numbers credits under the federal Renewable Fuels Standard

Q. DO YOU HAVE ANY ADDITIONAL COMMENTS ON THE COMPANIES DATA COLLECTION ACTIVITIES?

A. Yes. We would like to provide some context in which we offer recommendations regarding the data the Companies may collect. The City is committed to protecting the privacy of its residents and visitors so that individuals may access services with confidence, knowing that their personal information will be appropriately and responsibly protected. This is especially important given the rapid advances in technology and the growing volume of virtual data transactions involving such information today. In furtherance of this commitment, the City has established a comprehensive information privacy and security framework, including the *Citywide Privacy Protection Policies and Protocols*, or Policies and Protocols,

which were formally adopted and took effect on January 28, 2019. While the Policies and Protocols govern actions by City agencies and their human services contractors and subcontractors, the City would apply the same standards to safeguarding the data it may seek from the Companies now and in the future.

Moreover, the City believes that these principles, policies, and protocols are worthy of consideration by the Commission as best practices as it considers the City's requests for access to certain utility data in these rate cases and in the broader data access proceedings that are pending before it. Accordingly, we provide the Policies and Protocols with our testimony as Exhibit__(NYCPP-4). This document was issued by the City's Chief Privacy Officer, who heads the Mayor's Office of Information Privacy, with input from the City's Law Department, DoITT, the New York City Cyber Command, Office of Counsel to the Mayor, and the Mayor's Office of the Chief Technology Officer, among others with relevant expertise.

IMPROVEMENTS TO COMMUNICATION AND COORDINATION BETWEEN THE COMPANIES AND NEW YORK CITY AGENCIES

Q. CAN YOU DISCUSS THE NATURE AND EXTENT OF THE INTERACTIONS BETWEEN THE CITY AND THE COMPANIES?

A. The City has ongoing interactions with the Companies on a regular basis across a range of topics, and the Companies require interactions with multiple City agencies many times a day. For example, as discussed above, the Companies work with HRA to ensure that eligible customers are enrolled in the Company's low income programs, and they work nearly daily with HRA to ensure continuation of electric

and/or gas service to low income customers. The Companies must secure permits from the Department of Transportation, or DOT, before performing work on City streets. The Companies must coordinate with the Department of Environmental Protection, or DEP, and DDC, on City infrastructure projects for which the Companies must relocate or protect its utility facilities.

For building-related projects, the Companies may need approvals from the City Planning Commission and/or Department of Buildings. The Companies' provision of utility services to City buildings and facilities entails frequent interactions with the Department of Citywide Administrative Services. Our last example is the frequent discussions between the Companies and the Mayor's Office on a large variety of policy, regulatory, and operational topics. These are just some examples of the extensive interactions that occur.

Q. HAVE YOU HAD DISCUSSIONS WITH CITY AGENCIES REGARDING THEIR INTERACTIONS WITH THE COMPANIES?

A. Yes. In the context of preparing for these rate cases, we have met with many City agencies.

Q. WERE THERE ANY COMMON THEMES IN THE INFORMATION PROVIDED BY THE AGENCIES?

A. Yes. The most prevalent theme was the need for improved communications between the Companies and the agencies, and the need for better coordination of KEDNY and KEDLI projects. Many agencies reported a lack of responsiveness to questions and requests posed by City agencies, inadequate or untimely provision of

information by the Companies to the agencies, and insufficient communication and coordination with regard to overlapping City and utility projects.

Q. WHAT PERCENTAGE OF GAS OPERATIONS BUDGETS COMPRISE CITY/STATE CONSTRUCTION WORK?

A. Based on information provided in the Companies' pre-filed Testimony, a substantial portion of the budgets are dedicated to CSC work; approximately 32% of KEDNY's budget is dedicated to CSC work, for example. As explained below, some of that work is subject to cost sharing with the City.

Q. WHY SHOULD THE COMMISSION INTERCEDE INTO THIS MATTER?

A. These communication problems lead to inefficiencies, and inefficiencies lead to higher utility costs (and higher costs to the City), resulting in upward pressure on rates. The Commission should intercede to help moderate the rate increases sought by the Companies now and in the future. Additionally, as set forth below, the Companies' actions have led to significant delays in the completion of City infrastructure projects, unnecessarily extending disruptions to local residents and businesses and increasing costs borne by City taxpayers.

Q. CAN YOU PROVIDE AN EXAMPLE OF WHERE IMPROVED COMMUNICATIONS COULD HAVE AVOIDED PROBLEMS?

A. Yes. There currently is a backlog in the City's reimbursement of the Companies for CSC projects. The backlog is due in large part to National Grid's failure to submit timely invoices for work performed in 2015-2017. Specifically, National Grid did not submit these invoices to the City until 2017. The City is diligently

processing these historical invoices in an effort to eliminate the backlog, and we have developed a protocol with National Grid to allow for expedited invoice processing (one invoice per week, versus standard processing timeframe of one per month). Nevertheless, if we had a better communications protocol in place in 2015, the backlog would not have developed as it did. The City now meets bi-weekly with National Grid to address invoicing issues and to request supporting documentation to allow for the expedited review.

Q. CAN YOU EXPLAIN THE AUDIT PROCESS FOR CSC REIMBURSABLE EXPENSES?

A. Yes. DEP's audit process is very thorough and can result in longer processing times. However, the process is necessary given that we have found instances in the past where claimed work is not subject to cost sharing, and/or DEP cannot verify that certain work is reimbursable based on the documents provided by National Grid. We have taken steps to improve the processing time, including implementing a "pre-audit" review process with National Grid whereby DEP identifies potential issues and requests more upfront information to support the claimed reimbursement, before the request reaches the formal audit process.

Q. DO YOU HAVE ANY COMMENT ON NATIONAL GRID'S FORECAST OF REIMBURSABLE EXPENSES?

A. Yes. It appears that National Grid's forecast is much higher than the historical average. Per National Grid's response to DPS-84, both Companies spent a combined average of $45.3 million per year on reimbursable CSC work from 2014-

2018. Over the next four years (the Rate Year and all three Data Years), National Grid is projecting an average of $164.1 million/year in reimbursable CSC work. DEP has no way of verifying National Grid's projections.

Q. CAN YOU PROVIDE OTHER EXAMPLES OF THE NEED FOR IMPROVED COMMUNICATIONS?

A. Yes. National Grid delayed the City's reconstruction of Flushing Avenue by 20 months, leading to extended disruptions to the local community, while putting the opening date for a major new grocery business at risk. The cause of the delay was National Grid's schedule for performing test pits and pre-engineering for relocation and support and protection of their facilities. Other notable examples include delays that have occurred at major City infrastructure improvement projects in DUMBO/Vinegar Hill (3 months); on Atlantic Avenue (3 ½ months) and at Coney Island (ongoing 8 month delay). These delays are generally attributable to National Grid not starting work in a timely manner.

Q. PLEASE DISCUSS THE PROBLEMS THAT HAVE ARISEN ON CITY INFRASTRUCTURE PROJECTS BECAUSE OF DELAYS BY THE COMPANIES.

A. As brief background, DDC is the primary construction project manager for the City's in-street infrastructure projects. DDC manages the capital projects undertaken by DEP related to the water and sewer systems, and by DOT related to reconstruction of roadways, sidewalks, and public plazas. Given its role, DDC is

substantially involved in coordinating City infrastructure projects with the City's utilities.

At the outset of a City infrastructure project, DDC sends a 10-day notification letter to the Companies requesting that they provide the expected amount of time it will take to protect or relocate their facilities so that they are not damaged during construction. DDC consistently has challenges getting timely responses on whether National Grid has facilities that conflict with City-planned projects, how National Grid plans to accommodate our projects, and the timeline for supporting and protecting National Grid facilities. Further, National Grid has challenges completing their relocation and protection work within committed timelines due to a limited contractor pool that is certified to do this work.

Q. APPROXIMATELY HOW MANY WORK REQUESTS DOES DDC SUBMIT ANNUALLY TO THE COMPANIES?

A. DDC completes over 90 infrastructure projects each year. The number of work requests provided to the Companies varies by project; there could be as few as one or two requests or 10 or more requests.

Q. HAS DDC ENCOUNTERED ANY PROBLEMS WITH THESE WORK REQUESTS?

A. Yes.

Q. PLEASE DESCRIBE THE PROBLEMS ENCOUNTERED.

A. Coordination, in particular, has been a significant problem. Other problems relate to unexpected design changes made by the Companies and a lack of timely design

and construction responses. On occasion, problems have arisen because of delays by the Companies in retaining contractors to perform the utility protection and relocation work.

Q. APPROXIMATELY HOW LONG DOES IT TAKE THE COMPANIES TO PROCESS DDC WORK REQUESTS?

A. For a typical project, it can take anywhere from three months to a year for the Companies to complete their protection and relocation work. We would note that the time frames vary substantially depending on the size and complexity of each project.

Q. TURNING TO THE COORDINATION ISSUE, HOW DOES DDC RECEIVE NOTIFICATION THAT A REQUEST HAS BEEN PROCESSED BY THE COMPANIES?

A. DDC receives notification from the Public Improvement division of the Companies. The notification letter specifies the time period in which the Companies will complete their work on a project.

Q. DO THE COMPANIES GENERALLY COMPLETE THEIR WORK WITHIN THE STATED TIME FRAMES?

A. Generally, yes. If unforeseen conditions are found during construction, however, the Companies' work can take longer than projected. Recently, DDC has shared updated protocols with National Grid that require the Companies to provide a draft relocation schedule within 90 days after receipt of DDC Mass Mailing #1, or 40 percent design. DDC alerts designated National Grid staff by email and makes

Mass Mailing #1 available through *DDC Anywhere*, an online portal that houses all relevant project design and construction documentation. The Companies also are required to provide an updated final schedule indicating projected completion of utility relocation work within 10 days after receipt of DDC Mass Mailing #2, or 80 percent design. DDC also reserves the right to request schedule updates if there are changes to the expected phasing or completion date.

The schedule submission should include major milestones and phasing information including utility relocation start and finish, both for the overall project area and by street segment. Further, traffic stipulations for relocation work should be provided. This schedule will be used to coordinate relocation efforts and to assist in expediting relocation work.

Q. HAS THE PANEL IDENTIFIED ANY SPECIFIC ISSUES THAT APPEAR TO BE CAUSING DELAYS IN THE COMPANIES' COMPLETION OF PROJECTS IN A TIMELY MANNER?

A. In some cases, the Companies take longer than expected to complete their protection and relocation work. For Gas Cost Sharing projects in particular, National Grid must complete their relocation work in advance of the City commencing its capital projects. Any delay by National Grid has a deleterious cascade effect on dependent projects.

Q. PLEASE DESCRIBE THOSE ISSUES.

A. Significant delays can be caused by inaccurate as-built drawings provided by the Companies, and the discovery of utility facilities within a public right-of-way that

are not shown on any drawings or maps. When such facilities are encountered, the Companies must redesign or reengineer their plans, and they may need to secure additional contractors and/or equipment because of the expanded effort required.

Q. WHAT IS THE EFFECT OF THESE DELAYS ON THE OVERALL PROJECTS?

A. Delays mean additional time and additional costs. Projects take longer than anticipated, which inconveniences the public. Longer timeframes also can translate to higher costs. In this case, higher costs are likely for both Companies' protection and relocation work and for the overall City project – in our view, with better information and better communications and coordination, these delays and incremental costs can be avoided.

Q. WHAT RELIEF IS THE CITY SEEKING FROM THE COMMISSION?

A. The Commission should require the Companies to dedicate executive-level personnel to coordinate with City agencies, and it also should require that processes be put in place for the expeditious provision of information and resolution of disputes. Presently, the Companies have identified a few mid-level or higher personnel who some City officials can call for assistance in resolving discrete issues. The assistance provided by these individuals has been appreciated, but a much more robust, executive-level communications protocol would benefit ratepayers and City taxpayers.

Q. DOES THE PANEL BELIEVE THAT IT WOULD BE USEFUL FOR THE COMPANIES TO HAVE EMPLOYEES SPECIFICALLY ASSIGNED TO THIS MATTER?

A. Yes. Having a dedicated executive point of contact at the Companies to escalate and expedite matters related to City infrastructure projects would improve coordination between the City and the Companies and presumably reduce the number of problems and allow more expeditious resolution of problems that do arise – along with the cost savings resulting from such resolutions.

Q. DO YOU HAVE ANY ADDITIONAL RECOMMENDATIONS REGARDING THIS MATTER?

A. Yes. The Commission should require the Companies to file quarterly or more frequent reports regarding the status of pending City requests so that the Companies' performance can be objectively tracked over time. For example, the Companies could be directed to submit regular reports to DPS Staff on their compliance with submitting the schedules in response to DDC Mass Mailings ##1 and 2, and also track their ability to meet the dates set forth in the schedules, or explain why there is a failure to satisfy those dates. Where extensive delays or inaction are revealed, the Commission can take appropriate action to rectify the problems. The goal would be to reduce unnecessary delays, and associated costs for ratepayers and taxpayers, to the greatest extent possible.

Q. SHOULD THERE BE ANY CONSEQUENCES TIED TO THESE REPORTS?

A. Not at this time. However, if the reports demonstrate a lack of improvement or the existence of recurring problems, there could be a need to impose consequences in the future. If so, these reports could serve as a baseline to measure future performance and set reasonable performance targets.

SUMMARY OF OTHER CITY TESTIMONY

Q. PLEASE PROVIDE A BRIEF OVERVIEW OF THE OTHER TESTIMONY BEING SUBMITTED BY THE CITY.

A. The City is submitting five additional pieces of testimony addressing the following topics:

- The Low Income Panel is addressing the need for changes to the low income programs in order to move closer to achieving the Commission's goal that no utility customer should have an energy cost burden in excess of six percent of that customer's income, as well as elimination of reconnection fees, and improvements to the HRA file matching process.
- The Energy Efficiency Panel is advocating improvements to the Companies' energy efficiency and demand response plans, and highlighting the need for more reasonable shareholder incentives.
- The Infrastructure and Safety Panel is addressing the Companies' proposals for gas capital projects and operations and maintenance activities, as well as the shareholder incentives proposals related to such projects and activities. The Panel also recommends a series of storm hardening-related initiatives based upon the National Grid Storm Hardening Collaborative

that came out of the Companies' 2016 rate cases. The Panel also addresses gas safety and emergency management issues.

- The Rate Design Panel is addressing revenue allocation and rate design issues, including recommendations on how to expand and improve Non-firm service, and highlighting the need for improved communication on billing matters.
- Lane Kollen is addressing the Companies' proposals to accelerate the depreciation of leak prone pipe.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of:

New York City Rate Design Panel:

Richard Baudino
Sergey Shabalin

On Behalf of:

The City of New York

August 30, 2019

Cases: 19-G-0309 NYC RATE DESIGN PANEL
19-G-0310

INTRODUCTION

Q. PLEASE STATE YOUR NAME AND BUSINESS ADDRESS.

A. Mr. Baudino – My name is Richard A. Baudino. My business address is J. Kennedy and Associates, Inc. ("Kennedy and Associates"), 570 Colonial Park Drive, Suite 305, Roswell, Georgia 30075.

Mr. Shabalin – My name is Sergey Shabalin. My business address is DCAS Energy Management, David N. Dinkins Municipal Building, 1 Centre Street, 17th floor, New York, NY 10007.

Q. WHAT IS YOUR OCCUPATION AND BY WHOM ARE YOU EMPLOYED?

A. Mr. Baudino – I am a consultant to Kennedy and Associates.

Mr. Shabalin – I am a Director, Load Management and Utility Affairs for City of New York, Department of Citywide Administrative Services, Division of Energy Management ("DCAS DEM").

Q. PLEASE DESCRIBE YOUR EDUCATION AND PROFESSIONAL EXPERIENCE.

A. Mr. Baudino – I received my Master of Arts degree with a major in Economics and a minor in Statistics from New Mexico State University in 1982. I also received my Bachelor of Arts Degree with majors in Economics and English from New Mexico State in 1979. I began my professional career with the New Mexico Public Service Commission Staff in October 1982 and was employed there as a Utility Economist. During my employment with the Staff, my responsibilities included the analysis of a broad range of issues in the ratemaking field. Areas in which I testified included cost of service, rate of return, rate

design, revenue requirements, analysis of sale/leasebacks of generating plants, utility finance issues, and generating plant phase-ins.

In October 1989, I joined the utility consulting firm of Kennedy and Associates as a Senior Consultant where my duties and responsibilities covered substantially the same areas as those during my tenure with the New Mexico Public Service Commission Staff. I became Manager in July 1992 and was named Director of Consulting in January 1995. Currently, I am a consultant with Kennedy and Associates. Exhibit ____(NYCRDP-1) summarizes my expert testimony experience.

Mr. Shabalin – I received my Master of Science degree with a major in Energy Management from New York Institute of Technology in 2012. I also received my Bachelor of Arts degree with a major in Chemistry from New York University in 2001. I joined DCAS DEM in 2012 as an Energy Billing Analyst focusing on utility billing and tariff matters. I became Director of Billing and Reporting Analytics in 2015. My current role at DEM includes overseeing programs that leverage data-backed operational capabilities of municipal facilities to contribute to the integrity of the City's energy supply (such as demand response and interruptible gas), derive energy and carbon savings (load management) and deployment of real-time monitoring technology across the targeted municipal portfolio. Prior to joining DEM in 2012, I spent time at a Long Island City, New York non-profit Community Environmental Center working on energy audits and energy retrofits in New York's affordable housing stock.

Q. ON WHOSE BEHALF ARE YOU TESTIFYING?

A. We are testifying on behalf of the City of New York.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of our testimony is to address cost and revenue allocation, rate design, and service quality issues for Brooklyn Union Gas Company ("KEDNY") and KeySpan Gas East Corporation ("KEDLI"). In so doing, we will address the Direct Testimony of and certain recommendations made by the Companies' Rate Design Panel ("RDP").

Q. PLEASE SUMMARIZE YOUR CONCLUSIONS AND RECOMMENDATIONS TO THE PUBLIC SERVICE COMMISSION OF THE STATE OF NEW YORK ("COMMISSION").

A. Mr. Baudino: my conclusions and recommendations with respect to cost and revenue allocation and rate design are set forth below. Except where specifically noted, my recommendations relative to KEDNY also apply to KEDLI.

1. The embedded cost of service studies ("ECOSS") that the Companies filed are reasonable, consistent with past practice, and should be accepted by the Commission as the primary guide for revenue allocation.
2. With the exceptions noted in the next bullet regarding the KEDLI TC/IT class, The RDP's proposed interclass revenue allocations for KEDNY and KEDLI appear to be reasonable given the ECOSS results.
3. Although the KEDNY TC/IT class could have its rates and revenue reduced even further, I support the proposed small reduction to that class, as set forth by the KEDNY RDP in its Corrections and Update ("C&U Filing"). Under no circumstances should the KEDNY TC/IT classes receive any rate increase in this

proceeding. With respect to KEDLI, I do not agree with the RDP's proposed increases for the TC/IT service class. This class should receive no increases in this case given its excessive current rate of return.

4. The Commission should approve the RDP's proposed restructuring of the TC/IT classes into two new Non-firm rate classes with Tier 1 and Tier 2 interruption levels. As noted earlier, the KEDNY rates for the new Service Classes ("SC") 18-22 and 22, as set forth in the C&U Filing, should be approved. The KEDLI rates for new Service Classes 18 and 19, as set forth in the C&U Filing also should be set so that these classes receive no increase.

5. The Commission should order the Companies to modify their proposed new Non-firm tariffs to specifically state the interruption temperatures as 15 degrees for Tier 1 and 20 degrees Fahrenheit for Tier 2.

6. In connection with the Companies' proposed new Non-firm service class, customers taking service under those new Non-firm rate schedules and have automatic equipment should be able to elect to take service under Tier 2.

7. Exposing Non-firm customers to System Benefits Charges ("SBC") should be deferred until adequate energy efficiency programs are available to those customers.

8. In light of continuing capacity problems, the Companies should expand the Demand Response Demonstration Project aggressively, beyond what the Companies have proposed.

9. The Non-firm tariffs should be modified so that on-site alternative fuel is not an absolute prerequisite for eligibility.

In addition to the recommendations sponsored primarily by Mr. Baudino, we recommend the following:

10. To the extent that they are not addressed in Cases 16-G-0058 and 16-G-0059, the Commission should adopt the positions advocated by the City in its Comments filed in those dockets, which are attached hereto.

11. The second mandatory test for Non-firm customers should be eliminated.

12. The monetary penalty for so-called "second strike" violations should be reduced or waived if the second interruption notice is issued within five days of the end of a prior event *and* the customer can demonstrate that it could not repair the equipment that prevented it from interrupting. In addition, under these limited circumstances, the violation should not count as a second "strike."

13. Under certain circumstances, monetary penalties otherwise applicable to the New York City Housing Authority for using gas to operate on-site generation facilities to provide electricity to human needs customers during electric emergencies should be waived.

COST AND REVENUE ALLOCATION

Q. HAVE YOU REVIEWED THE ECOSS STUDY PRESENTED BY THE COMPANIES?

A. Yes. The Companies' ECOSS and proposed revenue allocation was presented by the Rate Design Panel. The Company's ECOSS is a traditional study wherein revenue requirements are allocated among customer classes on the basis of cost causation.

Q. WHY IS AN ECOSS IMPORTANT?

A. An ECOSS illustrates the costs a utility incurs to serve each customer class. It is widely accepted that costs should be allocated among customer classes on the basis of cost causation. That principle is perhaps the most universally accepted tenet of allocating costs that cannot be directly assigned to a particular customer class. As such, costs should be allocated to those classes on the basis of how or why those costs are incurred by the utility. The results of such studies are normally used in assigning cost and revenue responsibilities to various customer classes.

Q. DO YOU SUPPORT THE PREMISE THAT COST CAUSATION PRINCIPLES SHOULD GUIDE THE ALLOCATION OF COSTS TO THE CUSTOMER CLASSES?

A. Yes. Rates that are based on consistently applied cost causation principles are fair and reasonable and further the cause of stability, conservation and economic efficiency. Other factors such as simplicity, gradualism, economic development and ease of administration may also be taken into consideration when determining the final allocation of the revenue requirement among classes, but the fundamental starting point and guideline should be the cost of serving each customer class produced by the ECOSS.

Q. PLEASE EXPLAIN THE PURPOSE AND DEVELOPMENT OF A CLASS COST OF SERVICE STUDY.

A. A class cost of service study allocates and assigns the total cost of providing utility service to the classes of customers receiving that service. The development of a class cost of service study consists of three steps: functionalization, classification, and allocation. Functionalization refers to the process by which the Company's investments and expenses are identified and segregated into different cost categories. For KEDNY and KEDLI, these functional categories include supply and storage, transmission, distribution, metering and billing, collections, sales promotions and procurement.

Once functionalization is complete, the utility's costs are classified into categories of demand-related, energy related and customer-related costs in order to facilitate the allocation of costs by applying cost causation principles. In general, all distribution costs are either responsive to increases in natural gas demand or to the number of customers. Demand-related costs are those that are needed to serve the winter peak demands of distribution customers. Demand-related costs are fixed and do not vary with the monthly and annual commodity consumption of the utility's customers. Customer-related costs are associated with the number of customers and are costs that are incurred to connect customers to the system independent of the customer's demand and energy requirements. Primary examples of customer-related costs are investments in meters, services, and a portion of main investment incurred to extend the distribution system to the customer's premises and conform to local and national codes and standards. In addition, such accounting functions as meter reading, bill preparation and revenue accounting are generally considered customer-related costs.

Lastly, the functionalized and classified costs are allocated to the various customer classes based on each class' contribution to the respective cost classifications. In general, demand-related costs are allocated based on each class' contribution to the maximum demand placed on the system by the classes or by the customers within the classes. Customer related costs are allocated based upon the number of customers in each class, weighted to account for the complexity of servicing the needs of the different classes of customers.

Q. HOW DID THE COMPANIES' ECOSS CLASSIFY DISTRIBUTION MAINS?

A. The RDP explained on pages 17 and 18 of its Direct Testimony that it used a minimum-system study to classify distribution mains into customer and demand related costs. Based on this study, 47.08% of the distribution system was classified as customer-related and 52.92% was classified as demand-related. The customer portion of distribution mains was allocated to customer classes based on the number of customers and the demand component was allocated based on class contribution to design-day demand.

Q. IS THE COMPANIES' ALLOCATION OF DISTRIBUTION MAIN COSTS ON A CUSTOMER AND DEMAND BASIS CONSISTENT WITH THE NATIONAL ASSOCIATION OF REGULATORY UTILITY COMMISSIONERS ("NARUC") GAS DISTRIBUTION RATE DESIGN MANUAL DATED JUNE, 1989 ("1989 NARUC MANUAL")?

A. Yes. The NARUC Manuals discuss a number of methodologies and approaches to cost allocation. With respect to the allocation of distribution main costs, the 1989 NARUC Manual states that "[a] portion of the costs associated with the distribution system may be

included as customer cost." (1989 NARUC Manual at 22.) The 1989 NARUC Manual states further that, "[o]ne argument for inclusion of distribution related items in the customer cost classification is the zero or minimum size theory." (*Id.*)

Q. HAS THE COMMISSION RECOGNIZED THE APPROPRIATENESS OF A CUSTOMER COMPONENT IN THE COST OF GAS DISTRIBUTION MAINS?

A. Yes. The Commission has accepted the use of a minimum-system methodology to classify and allocate distribution costs in past National Grid cases.

Q. DO YOU AGREE WITH THE COMPANIES' CLASSIFICATION AND ALLOCATION OF DISTRIBUTION MAINS?

A. Yes. As the RDP pointed out, distribution mains are installed to connect customers to the distribution system and to provide capacity to meet the winter peak. The Companies' ECOSS appropriately utilized a minimum-system study to estimate which portions of distribution mains are demand related and customer related, and then allocated costs to the various customer classes based on that relationship. We recommend that the Commission adopt the Companies' minimum-system methodology for classifying and allocating the cost of distribution mains.

REVENUE ALLOCATION AND RATE DESIGN

Q. WHAT WERE THE RESULTS OF THE ECOSS STUDY AS FILED BY KEDNY?

A. Table 1 below summarizes the class rate of return results of the ECOSS for KEDNY. Table 1 shows the current class rates of return and the relative rates of return at current rates. The class RORs and relative RORs presented in Table 1 were taken from Exhibit ___(RDP-4CU), Schedule 1 filed by the RDP as part of the C&U Filing made on July 3, 2019.

Table 1

KEDNY
Current Class Rates of Return

	Return %	Relative ROR %
SC1A - Res NonHeat	-0.98%	(0.29)
SC-1B Res Heat	4.23%	1.24
SC-1DG Res DG	-6.92%	(2.03)
SC-2-1 Non-Res NonHeat	9.39%	2.75
SC-2-2 Non-Res Heat	2.04%	0.60
SC-3 Multiple Family	1.76%	0.52
SC-4A High Load Factor	11.85%	3.48
SC-4A CNG	6.49%	1.90
SC-4B Year Round AC	16.25%	4.77
SC-6C TC/IT	29.13%	8.54
SC-7 Seasonal	-17.70%	(5.19)
SC-21 DG Sales	-14.30%	(4.19)
Total	3.41%	1.00

Table 1 shows that SC-1A, SC-1DG, SC-7, and SC-21 are actually earning negative returns currently. SC-2-1, SC-4A High Load Factor, SC-6C TC/IT are significantly overearning, which means that the rates these classes are currently paying are significantly greater than the cost based rates that they should be paying.

Q. HOW DID KEDNY PROPOSE TO ALLOCATE ITS REQUESTED REVENUE INCREASE IN THIS PROCEEDING?

A. Table 2 presents the RDP's recommended percentage revenue increases by service class, again from Exhibit ___(RDP-4CU), Schedule 1. This Table also presents the percentage revenue increase and decreases that would be necessary to bring each service class to the overall system average return of 6.76%.

Table 2

KEDNY
Distribution Revenue Allocation

Schedule	Full ECOSS Increase	Company Propsed Incr.
SC1A - Res NonHeat	44.4%	18.6%
SC-1B Res Heat	9.7%	13.1%
SC-1DG Res DG	156.2%	18.6%
SC-2-1 Non-Res NonHeat	-11.7%	3.3%
SC-2-2 Non-Res Heat	24.4%	18.6%
SC-3 Multiple Family	22.6%	18.6%
SC-4A High Load Factor	-16.9%	3.3%
SC-4A CNG	-0.6%	3.3%
SC-4B Year Round AC	-30.3%	3.3%
SC-6C TC/IT	-54.9%	-2.5%
SC-7 Seasonal	146.7%	18.6%
SC-21 DG Sales	312.1%	18.6%
Total	13.8%	13.8%

The RDP's Direct Testimony, pages 26 and 27, presented the methodology it used to assign the revenue increase among KEDNY's service classes. A tolerance band was developed that used a rate of return that is 0.75 times the average at the low end of the band and 1.25 times the average at the high end of the band. The service classes that were within the tolerance band were SC 1B and SC 2-2. Service classes that had class returns above the upper end of the tolerance band were assigned revenue targets five percent over proforma revenue at present rates, with the exception of TC/IT, which was assigned a 2% increase in the Companies' Direct Testimony. Service classes below the lower end of the tolerance band were assigned revenue targets of 28.2% over proforma revenue at present rates.

In the Corrections and Updates Testimony filed by the KEDNY RDP, it explained that in order to update the service class revenue allocation, the revenue allocation from the Direct Testimony was multiplied by 65.8594%, with the exception of the SC 22 TC/IT

service class, which received a 2.5% decrease in order to produce rates no higher than the April 30, 2019 filing.

Q. WHAT IS YOUR CONCLUSION WITH RESPECT TO THE PROPOSED INTERCLASS REVENUE ALLOCATION PROPOSED BY KEDNY IN THIS PROCEEDING?

A. Generally, the revenue allocation proposed by the KEDNY RDP appears reasonable given the ECOSS results and tends to move the service classes toward their cost of service while observing the ratemaking principles of gradualism and avoiding rate shock. The class increases are limited to 1.35 times the system average increase for those classes below their allocated cost of service. For purposes of this case, we support the RDP's proposed rates and revenue allocation for the new SC 18-22 and SC-22 TC/IT classes, although TC/IT customers are providing a very large subsidy to other service classes and could have their rates reduced more significantly based on the results of the Companies' ECOSS.

Q. DID YOU REVIEW KEDLI'S PROPOSED INTERCLASS REVENUE ALLOCATION FOR ITS SERVICE CLASSES?

A. Yes. We reviewed the RDP's proposed revenue allocation for KEDLI. Table 3 below presents the KEDLI distribution revenue allocation in a similar fashion to Table 2 above, with full cost based class rate increases and the RDP's proposed service class increases.

TABLE 3

KEDLI
Distribution Revenue Allocation

Schedule	Full ECOSS Increase	Company Propsed Incr.
SC1A - Res NonHeat	47.9%	12.5%
SC-1B Res Heat	10.6%	6.4%
SC-1DG Res DG	204.6%	12.5%
SC-2-A GS NonHeat	-44.1%	3.9%
SC-2-B GS Heat	5.6%	6.1%
SC-3A Multi Family	-25.1%	3.9%
SC-3B Multi-Family Heat	4.3%	6.1%
SC-9 Wholesale NGV	-81.6%	-36.0%
SC-4-7-12-13 TC/IT	-61.6%	3.9%
SC-15 HLF Service	-21.0%	3.9%
SC-16 Year Round Space Htg.	-30.6%	6.1%
SC-17 DG Sales	26.1%	12.5%
Total	6.2%	6.2%

Note that the TC/IT class receives a 3.9% increase under the RDP proposed revenue allocation despite needing a revenue decrease of 61.6% to achieve cost based rates. Given how far above costs the KEDLI TC/IT customers are based on the ECOSS, and given that these customers could have their rates significantly reduced, we recommend no revenue increase for KEDLI's TC/IT's customers in this case.

Q. ARE YOU RECOMMENDING THAT THE COMMISSION ADOPT THE COMPANIES' PROPOSED REVENUE REQUIREMENTS?

A. No. We have not evaluated the reasonableness of the Companies' proposed revenue requirements.

Q. IF THE COMMISSION APPROVES A LOWER REVENUE REQUIREMENT INCREASE THAN KEDNY OR KEDLI ARE SEEKING IN THIS CASE, HOW SHOULD THIS LOWER INCREASE BE SPREAD AMONG THE COMPANIES' SERVICE CLASSES?

A. If the Commission approves a lower revenue increase for the Companies, then we recommend that the service class increases shown in Table 2 (KEDNY) and Table 3 (KEDLI) be scaled back proportionately, with the exception of the KEDLI TC/IT class, which should remain at zero increase.

Q. PLEASE SUMMARIZE THE RDP'S PROPOSED CHANGES FOR THE TC/IT CLASSES.

A. The KEDNY RDP explained its proposal regarding TC/IT customers beginning on page 32 of its Direct Testimony. Based on the Commission's "Order Approving Tariff Revisions and Directing Further Tariff Filings" issued on February 7, 2019 ("TC/IT Order") in Cases 16-G-0058 and 16-G-0059, the RDP proposed to implement a single service classification (SC 22) with two tiers that includes both TC and IT customers. Tier 1 has a Designated Interruption Temperature of 15 degrees Fahrenheit and Tier 2 has a Designated Interruption Temperature of 20 degrees Fahrenheit. The proposed minimum monthly charge for all SC 22 customers is $375.00. In its C&U Filing, the proposed volumetric per therm rates are $0.3174 for Tier 1 customers and $0.3015 for Tier 2 customers. Tier 2 customers receive a lower rate because, at an interruption temperature of 20 degrees Fahrenheit, they will be interrupted more frequently than Tier 1 customers.

Tier 1 customers would have fully automatic switchover equipment installed so that the Company would be able to remote switch over the customers' equipment to an alternate fuel in the event it calls an interruption. Tier 2 customers would be notified of an interruption through the Company remotely activating an alarm at the semi-automatic Customer sites that would notify customers to manually switch over to their alternate fuel. As written, customers who have automatic switching equipment are limited by KEDNY and KEDLI to Tier 1 service.

Q. DO YOU AGREE WITH THIS LIMITATION?

A. No. Customers with automatic switching equipment are the most reliable source of interruptible capacity, and if they are willing to accept more interruptions in exchange for a lower rate, it is in the Companies' best interest to afford them that option.

Q. ARE THE COMPANIES WILLING TO OFFER THAT OPTION?

A. Apparently, yes. In their response to CNY-54, the Companies stated that Non-Firm customers with automatic switching equipment will automatically be placed into Tier 1, but that if a customer wishes to switch from Tier 1 to Tier 2, it can do so if it formally makes such an election. Please refer to Exhibit ___(NYCRDP-2) for the Company's response to CNY-54.

The Companies did not state in their response how long a Non-firm customer would have to make a formal election to switch to Tier 2 service when new rates are approved in this case. We recommend that customers be given 120 days from Commission approval of the new tariffs to make the election to switch from Tier 1 to Tier 2.

We also recommend that the new TC/IT tariff for both KEDNY and KEDLI include an ongoing provision to allow a customer to switch back from Tier 2 to Tier 1 no later than 120 days prior to the winter heating season. This provision would allow for Tier 2 customers who made the switch from Tier 1 to switch back in the event those customers realize that Tier 2 has too many interruptions. This is a reasonable ongoing provision that would allow TC/IT customers more flexibility and give the Companies adequate prior notice of switching before the heating season begins.

We also recommend that, within 30 days of the Commission's approval of the new tariff, the Companies provide written notice of this option to all customers that otherwise would default into the Tier 1 rate.

Q. DID YOU REVIEW THE COMMISSION'S TC/IT ORDER?

A. Yes.

Q. DID THE COMPANIES FILE PROPOSED TARIFFS FOR TC/IT CUSTOMERS CONSISTENT WITH THE TC/IT ORDER?

A. Subject to our prior comment regarding the option to elect Tier 2 service, yes. The Commission's TC/IT Order ordered the Companies to:

(1) File tariff amendments to modify the penalties for the unauthorized use of gas to the lower of two times the sum of the market price plus the transportation rate, or nine times the applicable sales rate.

(2) File proposed tariff revisions to create and/or revise their current Non-firm service classes to blend the existing TC and IT services and implement two pricing tiers determined by a customer's switching equipment.

(3) File proposed tariff revisions that address the submitted written comments and concerns for the treatment of critical customers in Case 18-G-0565, and Staff's proposed solutions discussed at the IT service technical conference on November 27, 2018.

(4) End the existing moratorium on new TC customers to become effective March 1, 2019.

The Companies' proposed Non-firm tariffs include the ordered language modifying the penalties for the unauthorized use of gas and blends the existing TC/IT classes into one Non-firm class with two pricing tiers based on a customer's switching equipment. The Companies also ended the moratorium on new Non-firm customers. Nevertheless, although the Companies generally complied with the TC/IT Order, in this testimony and in Comments filed in Cases 16-G-058 and 16-G-0059, the City is recommending additional changes to improve the Non-firm service option.

Q. TURNING TO RATE DESIGN, PLEASE COMMENT ON THE RDP'S PROPOSED VOLUMETRIC RATES FOR TIER 1 AND TIER 2 CUSTOMERS FOR KEDNY.

A. The Companies' proposed volumetric rates for Non-firm customers, as set forth in the C&U Filing, are discounted below the corresponding firm service class tail block rates to recognize the lower costs that such customers impose on the system. As noted earlier, based on the ECOSS results, these discounts should be much larger, and the volumetric rates much lower, than the Companies proposed in their C&U Filings. However, given the level of revenue increase in the Companies' filings, we do not oppose the proposed C&U rates

at this time. However, the Commission certainly should not increase Non-firm rates in this proceeding given the huge subsidy being provided by these customers.

Q. ARE TC/IT CUSTOMERS IMPORTANT IN TERMS OF HELPING KEDNY AND KEDLI MANAGE THEIR SYSTEMS DURING PEAK WINTER PERIODS?

A. Yes, and the Commission recognized this fact in its TC/IT Order, page 2, as follows:

> "By offering IT and TC service, the Companies can provide non-firm customers with delivery service during the vast majority of the year, while also ensuring the reliability of the distribution networks when demand for natural gas is highest, generally during cold weather when firm customers use natural gas for heating. Additionally, the Companies design, construct and maintain their distribution networks to serve firm customers when the distribution system experiences peak demand. Thus, the distribution networks do not include the additional infrastructure that would be required to serve TC and IT customers when the system experiences peak demand. This minimizes the cost of the system that must be recovered from firm customers."

The Commission also stated the following on page 14 of the TC/IT Order:

> "Since the Rate Order, it has become apparent that the downstate region is facing a significant and increasing shortage of firm gas transportation capacity. On November 14, 2018, National Grid filed proposed tariff amendments that are intended to provide the Companies flexibility in accepting applications for new or additional gas service if they determine there is insufficient gas supply and/or capacity to serve the prospective customers in the relevant area of their service territories. While the capacity constraints are more acute on Con Edison's gas distribution network, National Grid's load is also growing, which underscores the importance of continuing interruptible services at reasonable terms."

TC/IT customers are an invaluable part of National Grid's system. These customers are even more important now that National Grid is facing serious system constraints during the peak winter season. Continuing to offer TC/IT service at attractive, cost-based rates will be an important part of National Grid's future system planning and will offer the opportunity for the Companies to offer firm service to existing and new customers during a time of constrained capacity.

Q. ON PAGE 57 OF THE KEDNY RDP'S DIRECT TESTIMONY, THE RDP TESTIFIED THAT INTERRUPTIONS WOULD OCCUR FOR TIER 1 AND TIER 2 SC 22 CUSTOMERS AT 15 DEGREES FAHRENHEIT AND 20 DEGREES FAHRENHEIT, RESPECTIVELY. IS THIS CONSISTENT WITH KEDNY'S PROPOSED SC 22 TARIFF?

A. No. Under the Special Provisions section of the proposed SC 22 tariff, the Designated Temperatures are defined as follows:

> "The "Designated Temperature(s)" will be the temperatures set annually prior to the winter season by the Company for each tier. The Company reserves the right to adjust this temperature setting, based on system needs, with a minimum forty-eight (48) hours' notice to Customers. There will be a "Designated Interruption Temperature" for each tier when a Customer's equipment will be switched over to an alternate fuel and a "Designated Resumption Temperature" for each tier when the Customer's equipment may be switched back to natural gas."

This language is similar to the language that defines Designated Temperatures in the Companies' current TC/IT tariffs. It is not consistent with the KEDNY RDP's testimony that explicitly states the interruption temperatures. The KEDLI tariffs have the same problem. Therefore, we recommend that the Commission order the Companies to modify the language in the applicable tariff Special Provisions to explicitly state that the Designated Interruption Temperature and Designated Resumption Temperature are 15 degrees Fahrenheit for Tier 1 and 20 degrees Fahrenheit for Tier 2.

Q. DID THE COMPANIES ADDRESS THE STAFF'S RECOMMENDATIONS IN CASE 18-G-0585 REGARDING THE AVAILABILITY OF ENERGY EFFICIENCY PROGRAMS TO TC/IT CUSTOMERS?

A. Yes, in its KEDNY Direct Testimony on pages 58 and 59, the RDP stated that its proposed tariffs will have all interruptible gas service customers contribute to and be eligible for energy efficiency programs. The RDP's proposal would have all Non-firm customers contribute to the SBC and the service delivery rates for these customers will include Efficiency Transition Implementation Plan ("ETIP") costs.

Q. DID THE RDP SPECIFY WHICH ENERGY EFFICIENCY PROGRAMS WOULD BE AVAILABLE TO TC/IT CUSTOMERS?

A. No. However, the City issued discovery to the Companies seeking more information on this topic (please refer to Exhibit ___(NYCRDP-3), which is National Grid's response to CNY-53). This interrogatory sought information with regard to which energy efficiency programs would be offered to Non-firm customers, when they would be eligible, and the steps that National Grid has taken to assess Non-firm customer interest in the programs, as well as any outreach efforts undertaken by National Grid to attract Non-firm customers to its programs. National Grid's response listed several programs "that would possibly be offered to KEDNY and KEDLI Non-firm customers." The Companies "did not propose a specific start date for Non-firm customers to participate in energy efficiency programs" but stated that they are not opposed to a Rate Year start date. Finally, the Companies stated that they "have not undertaken steps to assess Non-firm customer interest in participating in energy efficiency programs, or planned outreach to attract Non-firm customers."

Q. BASED ON THE RDP'S PROPOSAL AND NATIONAL GRID'S RESPONSE TO CNY-53, WHAT IS YOUR RECOMMENDATION REGARDING NON-FIRM CUSTOMERS PARTICIPATING IN AND BEING CHARGED FOR ENERGY EFFICIENCY PROGRAMS?

A. The Companies' proposal to apply the SBC to Non-firm customers and that ETIP costs be included in their service delivery rates should not be adopted at this time. The Companies' response to CNY-53 indicates that they are not prepared to offer energy efficiency programs to Non-firm customers yet. Therefore, the Companies should not be allowed to charge Non-firm customers for programs that are not being specifically offered to them.

Having said this, based on my conversations with City of New York personnel, the City supports allowing Non-firm customers to participate in, and pay for, energy efficiency programs that are targeted to their needs. If the Companies would respond with programs in which the City's Non-firm accounts could participate, as well as commit to outreach to the City and other Non-firm customers explaining the benefits of energy efficiency programs, then we likely would support charging the SBC and ETIP to Non-firm customers.

Q. HOW SHOULD THE SBC BE APPLIED TO THE TIER 1 AND TIER 2 RATES OF THE NON-FIRM CUSTOMERS, ASSUMING THE COMPANIES OFFER SPECIFIC ENERGY EFFICIENCY PROGRAMS FOR SUCH CUSTOMERS?

A. The total Tier 1 and Tier 2 volumetric rates should not increase over the currently proposed Tier 1 and 2 rates because these rates are already greatly in excess of cost-based rates for Non-firm customers. This means that the if the SBC is added to these customers' rates, there should be a corresponding reduction in the applicable volumetric rates.

Q. WILL THE NEW SC 22 BE SUBJECT TO THE COMPANIES' EARNINGS ADJUSTMENT MECHANISM ("EAM")?

A. No, SC 22 will not be subject to the EAM. National Grid confirmed this in its response to CNY-58, which we have included in Exhibit ___(NYCRDP-4). Currently, TC/IT customers are not subject to the Companies' EAM and it should be made clear that the EAM will not apply to the new TC/IT tariffs proposed by the Companies.

Q. DO YOU HAVE ANY FURTHER RECOMMENDATIONS TO THE COMMISSION?

A. Yes. We recommend that the Demand Response Demonstration Project ("DRD Project") be expanded beyond what the Companies propose.

Q. PLEASE EXPLAIN WHY YOU ARE MAKING THIS RECOMMENDATION TO THE COMMISSION.

A. The Companies' Future of Heat ("FOH") Panel acknowledged that demand response "has the potential to play an important role in the Companies' future of heat strategy." Direct Testimony of the Future of Heat Panel, page 41, lines 7 through 8. The FOH Panel cited two ways in which the Companies are promoting demand response. The first way is the Companies' Non-firm service offerings, which we address earlier and again below. The second is the DRD Project for firm service customers who commit to minimum reductions

when called upon to do so. The FOH Panel presented the very positive results of the DRD Project on pages 43 and 44 of its Direct Testimony. Exhibit ___(FOH-1), Schedule 5 presents the Companies' proposed expansion of the DRD Project.

Q. HAS THE CITY PARTICIPATED IN THE DRD PROJECT?

A. Yes. The City is an active participant in the DRD Project and, in general, is pleased with how the program is run.

Q. WHAT ROLE SHOULD DEMAND RESPONSE PLAY?

A. As noted by the NYC Policy Panel, demand response should play an increased role in managing peak loads, particularly given prevailing capacity constraints. As reported by the FOH Panel, the DRD Project has proven successful and, as a result, the Panel is proposing to expand the program to achieve 1,000 Dt/hour and 400 Dt/hour reductions for KEDNY and KEDLI, respectively, by FY 2024 (Exhibit___FOH-1, Schedule 5).

Q. CAN YOU COMMENT ON THE TARGETS OF THE EXPANDED DRD PROJECT?

A. Yes. Given its success and the increasing need for peak load reductions, the Companies should increase the targets of the DRD Project. This could be achieved by expanding the pool of eligible customers, reducing the requirement for a 10 percent or more committed reduction, increased customer recruitment, or through other measures. Given that the Companies have declared a gas moratorium, expansion of the DRD Project should be explored in a collaborative that includes interested stakeholders.

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS REGARDING NON-FIRM SERVICE OPTIONS?

A. Yes. We recommend expanding Non-firm service eligibility to customers who do not have an alternate fuel source but have backup facilities, are willing to shed load automatically and be remotely interrupted by the Companies, and otherwise comply with the Companies' tariffs. The Non-firm tariffs currently require that a customer **must** have alternate fuel on site to be eligible. It is possible that customers could provide interruptions on demand (i.e., by automatic switching) without alternate fuels. For example, NYCHA is exploring a boiler design that would allow for interruptions on demand by starting up equipment that does not require natural gas or fuel oil. The Companies should be able to accommodate this configuration by providing an electronic or other remote signal to a Non-firm customer that triggers automatic shut-off of a portion or all of the onsite Non-firm gas service, whether or not that shut-off is accompanied by a transition to fuel oil. The primary (and perhaps only) concern from National Grid's standpoint should be whether the gas usage is interrupted at the customer's premises. To the extent the Companies have technological limitations that prevent automatic shutoff of Non-firm gas service in the absence of fuel oil backup, those limitations also need to be addressed and removed to ensure that Non-firm service options are not unnecessarily restrictive. Indeed, expanding the eligibility criteria so that on-site alternate fuel capability is not a prerequisite may spur other innovative ways to provide Non-firm service, without waiving the performance standards otherwise applied to such service.

Q. HOW SHOULD THE EXPANSION OF ELIGIBILITY FOR NON-FIRM SERVICE PROCEED?

A. Expansion of the pool of Non-firm customers could be included in the collaborative we recommended earlier. That collaborative also could address inspection and other rights the Companies may need to ensure that the customer equipment is capable of receiving interruption signals and responding to them, similar to the rights the Companies have to ensure that dual-fuel customers are capable of interrupting when called upon.

Q. WHAT OTHER RATE DESIGN ISSUES DOES THE PANEL WANT TO RAISE?

A. On August 11, 2019, the City submitted Comments in response to the Companies' May 8, 2019 tariff filings ("Filings"). The Filings were made in compliance with the Commission's TC/IT Order issued on February 7, 2019 in Cases 16-G-0058 and 16-G-0059. The City's Comments are attached hereto as Exhibit ___(NYCRDP-5).

In the Comments, the City noted its general support for the consolidation of the TC and IT service classes, as well as the new Non-firm rates set forth in the Filings. However, as noted in the Comments, the new tariffs should be amended to allow Non-Firm customers with automatic-switching equipment to elect the Tier 2 delivery rate, at their option.

The City's Comments also requested that the Commission require that the Companies eliminate the "two-strike" rule, which dictates that customers who do not interrupt twice in any winter period either be transferred to firm service or, in the alternative, have their gas service terminated. In addition, the City's Comments also recommended that all Non-firm customers have the option to remain on that service, subject to paying penalties and filing a remedial plan with the Companies, and that the current criteria for remaining on Non-firm service after "two strikes" be modified. Finally, the Comments requested that the tariff definition of critical care customers be expanded.

To the extent that the Commission determines that changing the "two-strike" rule or the recommendations regarding critical care customers, as set forth in the City Comments, are outside the scope of the Filings, the City requests, for the reasons set forth in Exhibit ___(NYCRDP-5), that such relief be granted in these proceedings.

Q. DOES THE CITY HAVE ANY FURTHER COMMENTS ON THE TARIFFS ADDRESSING NON-FIRM SERVICE?

A. Yes. Given the recent gas moratoriums declared by the Companies, the Companies and interested parties should be looking for ways to incentivize customers to apply for, or stay on, Non-firm service.

Q. PLEASE CONTINUE.

A. Earlier in this testimony, we urge the Commission to require the Companies to commence a collaborative to expand Non-Firm service so that it is not limited exclusively to customers who have on-site alternate fuel capability. Expanding Non-firm service will allow the Companies to more effectively manage their systems in the face of dwindling capacity.

Q. ARE THERE OTHER CHANGES THAT WOULD IMPROVE THE NON-FIRM SERVICE OPTION?

A. Yes. First, the requirement for a second mandatory test is not necessary and is costly to customers, and should be eliminated. Second, the monetary penalty for so-called "second strike" violations should be reduced or waived, and the violation should not count as a "strike," if the second interruption notice is issued within five days of the end of a prior event and the customer can demonstrate that it could not repair the equipment that prevented it from interrupting during the first event. Finally, under certain circumstances

monetary penalties otherwise applicable to the NYCHA for operating on-site generation facilities to provide electricity to human needs customers during electric emergencies should be waived.

Q. PLEASE EXPLAIN WHY THE MANDATORY SECOND TEST SHOULD BE ELIMINATED.

A. The addition of a second, unannounced test at the end of January each year commenced in 2017. The additional test adds costs to customers, increases emissions and creates uncertainty. In addition, it is not clear why it is necessary, particularly for customers with equipment for automatic-switching controlled by the Companies. The Companies typically test their TC/IT customers at the beginning of each winter to ensure that they can respond. Once that is established, the tariff imposes numerous obligations on the TC/IT customer to maintain adequate alternate fuel supplies and to keep any necessary dual-fuel equipment operational. In fact, the Companies have the ability to inspect and test such alternate fuel facilities at any time, and the customer has to grant access to the Companies for inspection and testing purposes. Finally, as discussed in more detail below, there are significant penalties associated with non-compliance, and this alone provides a tremendous incentive for maintaining alternate fuel facilities.

Q. WHAT IS THE RATIONALE FOR REDUCING THE LEVEL OF THE PENALTY UNDER CERTAIN CIRCUMSTANCES?

A. The City appreciates the prior revision that changes the unauthorized use penalty from the higher to the **lesser** of (i) two times the sum of the Market Price for natural gas delivered to the Company's city gate on the day of violation plus the transportation rate the Customer

would be subject to if it were a transportation Customer, or (ii) nine times the applicable sales rate, using nine times the applicable sales rate is excessive. However, as evidenced by the amnesty clause included in the Companies' tariffs, there may be circumstances where the failure to interrupt is not due to a conscious act but, instead, equipment failure. For example, if the first "strike" is due to equipment failure, and despite diligent efforts to fix the problem, the equipment may not be able to be fixed prior to a second event, especially if the second event is in close proximity after the first event. In this case, we would request a threshold of within five days of the end of the first event to qualify as "close proximity." The rationale for this threshold is that the inability to fix the equipment could be due to long lead times or transportation problems, in particular if there is a weekend that falls between the end of the first and the beginning of the second interruption event. If that is the case, the imposition of the penalty is not providing any incentive to change behavior. Accordingly, and as exemplified in concept by the amnesty clause, fairness would dictate that the penalties for the second "strike" should be significantly reduced or waived, and the event should not be considered a "strike." The City acknowledges that the waiver of second strike penalty should require submission of documentation to show a good faith attempt to fix the equipment.

Q. DO YOU HAVE ANY ADDITIONAL RECOMMENDATIONS REGARDING THE APPLICATION OF PENALTIES?

A. Yes. NYCHA currently has or will install natural gas-fired back-up emergency generators on site at 32 of its housing developments in the wake of Hurricane Sandy. Of these, 15 developments are located in National Grid's service territory. This on-site generation is

designed to provide critical electric service to almost 10,000 apartments that house approximately 25,000 of the City's neediest residents during emergencies that interrupt electric service. Although NYCHA requested firm service for these installations, the moratorium has cast doubt on whether they will be able to get that service. Given the summer peaking nature of Con Edison's electric system, outages are far more likely to occur in the summer months rather than the winter months when electricity usage is lower. However, there may be rare instances in which the affected NYCHA apartments experience electric outages during peak gas usage periods in the winter. If the NYCHA facilities with back-up generation are relegated to interruptible gas service and experience an electric outage at the same time that National Grid calls an interruption event for TC/IT customers, then NYCHA would have to make the very difficult choice of either not running its backup generators for the health and safety of its residential customers during an emergency or incurring significant and costly penalties from National Grid. This "lose-lose" situation for NYCHA is the result of the Companies declaring gas moratoriums, and it is not fair that NYCHA residents, including those that suffered significant deleterious impacts during Hurricane Sandy, might have to bear the brunt of decisions wholly outside of their control. Accordingly, penalties imposed on NYCHA should be waived under the unique circumstances set forth here. To address the broader issue of the need for improved comprehensive supply planning, please refer to the City's Policy Panel, which has a number of specific recommendations addressing the gas moratorium.

Q. WHAT IS YOUR RECOMMENDATION?

A. As noted in this testimony, the City is not opposed to the Companies applying reasonable penalties during periods in which interruptions have been called to deter customers from using gas so that they can safely manage their systems. However, in the rare instance that Con Edison experiences an electric outage and NYCHA has to run its back-up generators to protect the health and safety of its human needs customers during a period when National Grid has called for an interruption, it would be appropriate to waive the penalties in this limited circumstance, especially considering that the NYCHA facilities requested firm service. The City is willing to have appropriate NYCHA personnel to sit down with National Grid to establish workable protocols, including establishing the amount of gas needed to maintain critical electric service.

Q. DOES THIS CONCLUDE YOUR DIRECT TESTIMONY?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of the Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**:**	**Case No. 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**:**	**Case No. 19-G-0310**

REBUTTAL TESTIMONY

OF

RICHARD A. BAUDINO

ON BEHALF OF

THE CITY OF NEW YORK

J. KENNEDY AND ASSOCIATES, INC.

SEPTEMBER 18, 2019

Q. PLEASE STATE YOUR NAME AND BUSINESS ADDRESS.

A. My name is Richard A. Baudino. My business address is J. Kennedy and Associates, Inc., 570 Colonial Park Drive, Suite 305, Roswell, Georgia 30075.

Q. DID YOU FILE DIRECT TESTIMONY IN THIS PROCEEDING?

A. Yes. I filed Direct Testimony on behalf of the City of New York, or City, as part of the New York City Rate Design Panel.

Q. WHAT IS THE PURPOSE OF YOUR REBUTTAL TESTIMONY?

A. The purpose of my Rebuttal Testimony is to respond to the Direct Testimony of Danielle Panko, witness for the Utility Intervention Unit ("UIU"). I will also respond to a rate design recommendation contained in the Direct Testimony of the Staff Rates Panel ("SRP"). Finally, I support in concept a recommendation by the Staff Earnings Adjustment Mechanism Panel ("EAM") Panel that proposes to revise the Peak Demand Reduction EAM.

Within my Rebuttal Testimony, I refer to the Brooklyn Union Gas Company and KeySpan Gas East Corporation as KEDNY and KEDLI, respectively. I refer to both KEDNY and KEDLI as either National Grid or the Companies.

UIU WITNESS PANKO

Q. ON PAGE 28, LINES 16 THROUGH 19 OF HER DIRECT TESTIMONY, MS. PANKO RECOMMENDS "USING BOTH THE RESULTS OF THE PRO-FORMA GAS ECOS STUDIES WHICH CLASSIFY DISTRIBUTION MAIN COSTS AS 100% DEMAND-RELATED AND THE RESULTS OF THE ECOS STUDIES PROPOSED BY THE COMPANIES, WHICH USE THE MINIMUM SYSTEM APPROACH." SHOULD THE NEW YORK PUBLIC SERVICE COMMISSION ("COMMISSION") ADOPT THIS RECOMMENDATION?

A. No. The recommendation to utilize an ECOS study that allocates distribution mains based on 100% demand should be rejected by the Commission.

Q. MR. BAUDINO, PLEASE EXPLAIN WHY ALLOCATING DISTRIBUTION MAINS COSTS BASED ON PEAK DEMAND ONLY IS INAPPROPRIATE.

A. The two main functions of distribution mains are to deliver gas during the system winter peak and to connect customers to the system. A properly designed zero-intercept study or minimum size system study recognizes these two functions by classifying main costs into demand-related and customer-related costs, which can then be assigned to customer classes based on their respective contributions to system peak and on the number of customers in each class.

Peak winter demand is one of the primary drivers of the Companies' investment in gas distribution mains. The Companies must have sufficient capacity available on their systems to satisfy the peak winter heating demand, which is caused mainly by heating customers. If the peak winter demand increases, the Companies may need to invest in additional mains to serve the load. During the non-winter months, substantial excess capacity exists on the system. Use of the Companies' distribution system during these months does not cause additional fixed costs to be incurred by National Grid. In a similar manner to peak winter demand, if the number of customers increases, the Companies may need to expand their distribution system investment. Thus, the number of customers connected to the distribution system is another important causative factor in distribution main investment.

Q. PLEASE CONTINUE.

A. The KEDNY and KEDLI Rate Design Panels ("RDP") described the rationale for the

allocation of distribution mains as follows:

> "Distribution mains are installed to connect customers to the distribution system (customer-related) as well as to provide capacity to meet customer requirements on the design day (demand-related). Therefore, the cost of mains is classified as partly customer-related and partly demand-related."[1]

Similarly, the NARUC *Gas Distribution Rate Design Manual* states the following with respect to demand or capacity related costs:

> Demand or capacity costs vary with the quantity or size of plant and equipment. They are related to maximum system requirements which the system is designed to serve during short intervals and do not directly vary with the number of customers or their annual usage. Included in these costs are: the capital costs associated with production, transmission and storage plant and their related expenses; the demand cost of gas; and most of the capital costs and expenses associated with that part of distribution plant not allocated to customer costs, such as the costs associated with distribution mains in excess of the minimum size.[2]

Q. IS THE MINIMUM SIZE DISTRIBUTION SYSTEM METHOD AND ZERO INTERCEPT METHOD THE SAME THING?

A. In concept, yes they are the same. The minimum size system approach identifies the minimum sized distribution main needed to serve customers and then classifies that portion of distribution mains as customer-related. The zero intercept method uses regression analysis to identify the cost of a hypothetical "zero sized" main, the cost of which is necessary to serve customers connected to the system whether or not they place any demand on the system. While there may be subtle differences between the two methods, both recognize that certain distribution main costs should be classified as customer-related and allocated based on the number of customers not on peak demand.

[1] KEDNY Rate Design Panel Direct Testimony at page 17, lines 9 through 12.

[2] *Gas Distribution Rate Design Manual*, NARUC Staff Subcommittee on Gas (June 1989), pp. 23-24.

Q. WHAT IS THE RESULT OF FAILING TO RECOGNIZE A CUSTOMER-RELATED COMPONENT IN THE COST OF DISTRIBUTION MAINS?

A. The result is a significant misallocation of costs that fails to allocate proper cost responsibility to the Company's customer classes. This point can be illustrated with the following simple example. Assume that there is a single industrial customer on KEDNY's system with a peak demand of 300 dekatherms ("Dth"). Further assume that elsewhere on the system, there is a neighborhood of 750 residential customers with a peak demand of 300 Dth. In order to connect all of those residential customers to its system, the Company will have to invest in far more footage of distribution mains for those customers than it would have to invest in for the one industrial customer. That extra investment in distribution mains is due solely to the number of customers on the system, not the peak demands of those customers.

Q. HAS THE COMMISSION RECOGNIZED THE APPROPRIATENESS OF A CUSTOMER COMPONENT IN THE COST OF GAS DISTRIBUTION MAINS?

A. Yes. The Commission has accepted the use of a minimum size system concept to classify and allocate distribution costs in past cases.

Q. CAN YOU PROVIDE A FEW EXAMPLES OF HOW THE COMMISSION HAS ALLOCATED DISTRIBUTION MAIN COSTS IN THE PAST?

A. Yes. In Case 94-G-0885, a National Fuel Gas Distribution Corporation ("NFG") proceeding, the Commission approved both the Administrative Law Judge's approval of NFG's use of the zero intercept method and corresponding rejection of proposals to allocate distribution

costs on the basis of demand.[3] The Commission noted that:

> The Judge also rejected CPB's and PULP's attempt to impeach NFG's cost study. While they criticized the minimum distribution system concept it employed, the Judge found that it makes some sense because clearly no customer can be served without distribution facilities, and the company's approach effectively emphasized the 'minimum' in the minimum distribution system.[4]

The Commission held that "the true interests of residential customers would not be served by over allocating costs to nonresidential customers with competitive alternatives, whose load might thus be lost."[5]

Similarly, in Case 93-G-0162, a Niagara Mohawk proceeding, the Commission declined to sustain objections to the Company's cost studies which allocated minimum system costs and service line costs on a customer basis.[6] The Commission held that the "company's cost estimates are the best presented in the record in these cases."[7]

Q. HAS THE COMMISSION PREVIOUSLY REJECTED A PROPOSAL TO CLASSIFY ALL DISTRIBUTION COSTS AS DEMAND RELATED?

A. Yes. In a Central Hudson Gas & Electric Corporation ("Central Hudson") 2008 rate case, the Commission approved the Administrative Law Judge's recommendation for the continued use of the zero-intercept method and rejected Staff's proposal to allocate

[3] Case 94-G-0885, Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of National Fuel Gas Distribution Corporation for Gas Service, Op. No. 95-16 (issued September 21, 1995) at 49-50, 53; *see also* Recommended Decision (issued June 15, 2005) at 86-93.

[4] Case 94-G-0885, supra, Op. No. 95-16 (issued September 21, 1995) at 49-50.

[5] *Id*. at 53.

[6] Case 93-G-0162, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Gas Service, Op. No. 94-13 (issued May 12, 1994) at 41-45.

distribution costs on the basis of demand. The Commission noted the following:

> Staff proposed to reclassify gas distribution main costs for purposes of the pro forma embedded cost of service study by assigning them entirely to the demand component of rates. Currently, based on the zero-intercept methodology that Central Hudson has used since at least 1990, those costs are classified 55% to the customer component of rates and only 45% to the demand component. Because gas mains constitute 20% of the total cost of gas service, the reclassification results in a very large shift in cost responsibility from residential customers to large gas users. The RD noted that both the existing and proposed methodologies are deemed acceptable by NARUC with no indication that one or the other is superior. It concluded that such a large shift in cost responsibility should not be adopted without compelling evidence that it is necessary to rectify some serious inequity... We have stated repeatedly that we strive to match cost responsibility with cost causation.... At the same time, as we discuss in connection with customer charges and the common cost allocation ratio, we have consistently taken a gradual approach when a sudden, full correction would create unacceptable bill impacts. That situation clearly exists here. Finally, although we find the arguments persuasive as to the assignment of a greater proportion of gas mains costs to the demand component, we are not convinced on this record that no mains costs should be classified as customer related. Accordingly, we direct that for the purpose of setting rates in this case, the allocation of gas mains costs should be 65% demand and 35% customer. This is consistent with the ratio that we adopted for National Grid in approving a Joint Proposal in its recent gas rate case.[8]

Q. DID CENTRAL HUDSON ALSO USE THE ZERO INTERCEPT METHOD IN ITS MORE RECENT RATE CASES?

A. Yes. In its 2014 and 2017 rate cases, Central Hudson used the zero intercept method to support classifying 35% of its distribution mains as a customer-related cost.[9]

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[7] *Id*. at 44.

[8] Cases 08-E-0887, 08-G-0888, 09-M-0004, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Electric Service, et. al., Order Adopting Recommended Decision with Modifications (issued June 2009) at 46-48; *see also*, Recommended Decision at 104-107 (issued April 2009).

[9] Case 14-G-0319, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Gas Service, and Case No. 17-G-0460, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of

Q. DO OTHER NEW YORK UTILITIES ALSO RECOGNIZE A CUSTOMER COMPONENT OF DISTRIBUTION MAINS?

A. Yes. Niagara Mohawk Power Corporation d/b/a National Grid used the zero intercept method in three prior gas rate cases. In those cases, 45.5% of distribution mains were classified as customer-related costs.[10] Finally, the revenue allocations contained in the Joint Proposals approved by the Commission in the last National Grid cases, Cases 16-G-0058 and 16-G-0059, were based on the Companies' ECOS studies, which utilized the minimum distribution system.[11]

Q. DID MS. PANKO PROVIDE THE COMMISSION WITH A REVENUE ALLOCATION PROPOSAL?

A. No, nothing specific. On page 35, Ms. Panko recommended that the revenue allocation process "consider the output, or results, of both of the Companies' filed pro-forma ECOS studies (including the study that allocates distribution main as 100% demand-related)." On page 36, Ms. Panko testified that she provided "illustrative analysis" in her Exhibit ___(DP-1) showing class revenue allocations using boundaries of 0.75 and 1.25 for KEDNY and 0.5 and 1.5 for KEDLI. Ms. Panko concluded her revenue allocation discussion on page 37, and

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Central Hudson Gas & Electric Corporation for Gas Service.

[10] Case 08-G-0609, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Gas Service; Cases 12-E-0201, 12-G-0202, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Electric and Gas Service; Case 17-G-0239, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Electric and Gas Service.

[11] Cases 16-G-0058 et. al, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service, et. al., Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans (issued December 16, 2016) at 38.

recommended that the Commission set specific boundaries around the base delivery revenue change applied to individual service classes to "reflect the magnitude of the delivery increase that is ultimately adopted by the Commission in these proceedings."

I conclude from my review of Ms. Panko's Exhibit ___(DP-1), as well as her discussion of the various boundaries the Commission could use to allocate service class increases, that she did not provide the Commission with any firm guidance or methodology for allocating revenue increases to KEDNY and KEDLI service classes.

STAFF RATES PANEL

Q. BEGINNING ON PAGE 41, LINE 20 THE SRP DISCUSSED ITS APPROACH TO FIRM RATE DESIGN. PLEASE BRIEFLY SUMMARIZE STAFF'S APPROACH.

A. The SRP agreed with National Grid's proposal to gradually reduce the slope of its declining block rate structure. However, because the Staff's overall revenue requirement for KEDNY and KEDLI is significantly lower than National Grid's proposed revenue requirement, the SRP proposed to modify the Companies' proposal in order to "more aggressively flatten the rate blocks."[12] With respect to KEDNY Service Classes 1A, 1B, 2-1, 2-2, 3 and 4B, the SRP recommends holding the minimum charge constant, with the tail and mid-block rates changing, to effect a 20 percent movement towards flat block rates. With respect to KEDLI Service Classes 1A, 1B, 2A, 2B, 3 and 16, the SRP recommends holding the minimum charge constant and increasing the tail block and/or decreasing the mid-block rates to effect a 20 percent movement towards flat block rates. The SRP presented its proposed rates in Exhibit ___(SRP-6).

Q. PLEASE PROVIDE EXAMPLES OF THE COMPONENT RATE INCREASES

[12] Staff RDP Direct Testimony at page 42, lines 9 through 10.

PROPOSED BY THE SRP FOR KEDNY CUSTOMERS.

A. From Exhibit ___(SRP-6), page 1 of 4, the SRP proposed the following rate decreases and increases:

- For SC 1B, a decrease of -8.17% for Block 2 and an increase of 19.56% for Block 3.
- For SC 2-2, a decrease of -5.24% for Block 2, a decrease of -1.31% for Block 3, and an increase of 8.56% for Block 4.

The SRP also showed the following bill decreases and increases for SC 1B in its Exhibit ___(SRP-7):

- Bill reductions (decreases) for monthly consumption of 0 - 100 therms.
- Bill increases above 100 therms, rising to 15.43% for customers using 500 therms per month.

Finally, the SRP showed the following bill decreases and increases for SC 2-2 in its Exhibit ___(SRP-7):

- Bill reductions for customers using up to 900 therms per month.
- Bill increases above 900 therms, rising to 7.86% for customers using 13,000 therms per month.

Q. SHOULD THE COMMISSION FOLLOW THE SRP PROPOSED FIRM RATE DESIGN IN THIS CASE?

A. No. I recommend that the SRP's proposed firm rate design be rejected.

Q. PLEASE EXPLAIN WHY THE SRP'S PROPOSED RATE DESIGN SHOULD BE REJECTED.

A. The SRP's recommendation, which aggressively moves the affected classes toward flat rates, does so more quickly than is advisable at this time. Using SC 1B as an example,

Staff's recommended revenue increase for this class is 0.17%. Higher usage customers in SC 1B, say 140 therms per month, would receive an increase of 2.87% per month, while a customer using 83 therms per month would receive a decrease of -2.70% per month. Exhibit ___(SRP-7) shows a rather wide dispersion of decreases and increases from its rate design proposal. These rate deviations are based on Staff's proposed revenue requirements, which have not been adopted yet.

The SRP's stated rationale for its rate design recommendation is that its lower recommended rate increase for KEDNY and its recommended rate decrease for KEDLI justify an aggressive move to flat rates. However, the SRP did not provide any guidance for how its more aggressive rate design would work at higher revenue increases, should the Commission make such a decision. The Staff merely presented several factors that should be considered if the KEDNY revenue requirement increases substantially, which included (1) potentially reducing the movement toward flat rates to mitigate high bill impact, (2) potentially increasing minimum charges, and (3) allocating the increase to the minimum charge and block rate for SC 1BDG and SC 7.[13]

Finally, it is not apparent to me that the SRP evaluated the impact of its more aggressive rate design on lower income users. It is not always the case that residential customers who use less than the class average amount of therms are lower income. Some residential users may live in housing that is less insulated, and/or they may not have the funds to respond to the SRP's price signal by purchasing more insulation or buying more energy efficient appliances, such as home furnaces, ovens, and so forth. The Commission should consider the effect on lower income residential customers, and higher use commercial

[13] Staff RP Direct Testimony, page 43, lines 11 through 19.

customers for that matter, before it adopts the aggressive rate design proposed by the SRP to flatten rates.

STAFF EAM PANEL

Q. WHAT DOES THE STAFF EAM PANEL SAY ABOUT THE PEAK DEMAND REDUCTION EAM?

A. On pages 21-23, the Staff EAM Panel agrees that a peak demand reduction EAM is needed, but disagrees with the Companies' proposed Peak Demand Reduction metric (in particular, Staff disagrees with basing this metric solely on the results of the Companies' demand response program). The Staff EAM Panel suggests that the metric should be based instead on peak-day usage reductions from firm customers on a per-Heating Degree Day basis, to be measured during the month of January or February. Finally, the Panel suggests reserving up to 10 BPs for this metric, and asks parties to recommend possible metrics and target development strategies as part of rebuttal testimony.

Q. WHAT IS YOUR RESPONSE TO THE PANEL?

A. I support a change to the metric. As noted in the NYC Rate Design Panel direct testimony, reduction of peak demand usage by firm customers is a paramount concern given the Companies' declared moratoriums on new service. However, I do not have specific metrics or targets to suggest at this time, and this effort may be better suited for a collaborative involving all parties.

Q. DOES THIS CONCLUDE YOUR REBUTTAL TESTIMONY?

A. Yes.

A.L.J. COSTELLO: And I just want to confirm with the City of New York that we have all the testimony accounted for.

UNIDENTIFIED SPEAKER: Yes, thank you.

A.L.J. COSTELLO: Okay. Okay. We're now going to deal with the affidavits submitted by the Environmental Defense Fund. We have a two-page affidavit by Joseph C. Von Fisher, sworn to on February 5, 2020. And that will be marked as Exhibit 713. We have a two-page affidavit by Gregory M. Lander, sworn to on February 6, 2020. And that will be Exhibit 714. And we have a two-page affidavit by James Fine, sworn to on February 6, 2020. There is also a one-page California jurat signed by James David Fine on February 6, 2020. And those together will be marked as Exhibit 715.

And at this point in the record, the court reporter should insert the following files: EDF Joseph Von Fisher direct testimony; EDF Gregory Lander direct testimony; EDF James Fine direct testimony. #

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

JOSEPH VON FISCHER

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: August 30, 2019

I. Introduction and Qualifications

Q. Please state your name, title, and business address.

A. My name is Joseph C. von Fischer. My title is Professor and Associate Chair of Biology, Colorado State University ("CSU"). My business address is 100 W. Lake Street, Fort Collins, CO 80523.

Q. On whose behalf are you submitting this testimony in this proceeding?

A. I am submitting this testimony on behalf of the Environmental Defense Fund ("EDF").

Q. Please provide a summary of your education and experience.

A. My primary research expertise is in understanding what processes on the land surface give the atmosphere its greenhouse gas composition. I earned my PhD in Ecology and Evolutionary Biology in 2002 through Cornell University's program in Biogeochemistry and Global Change. From 2002-2003, I conducted post-doctoral training in atmospheric composition at Princeton University. I have been on the faculty at CSU since 2003, where I am now a Professor and Associate Department Chair of Biology.

Over my career, I have won competitive research funding from diverse sources including National Science Foundation, National Aeronautics and Space Administration, United States Geological Survey, the Department of Energy, EDF, NYSEARCH, and Consolidated Edison Company of New York, Inc. ("Con Edison"). This funding has supported my research and allowed me to publish 37 scientific publications, which are routinely cited by other scientists.

Since 2012, I have been collaborating with EDF as the science lead on their "methane mapping" project, where high-sensitivity methane analyzers have been deployed on Google Street View cars. In this collaborative project, we have mapped 15 U.S. cities to date. Each city deployment has been associated with multiple meetings with the utility management team. These conversations have allowed me to deepen my understanding of how distribution systems are managed by utilities, especially with regards to leak surveys, survey technology and the incorporation of survey results into pipeline integrity management. Based on these deployments and the supporting scientific experiments associated with them, I have published 5 scientific papers on methane mapping to date and one industry paper describing how to deploy advanced leak detection to reduce emissions. I describe the results of these papers below.

Based on my training and research experiences, I have developed an expertise in understanding how natural gas leaks in distribution systems can be discovered by high-sensitivity methane analyzer systems, and how information about leak magnitude can be used to prioritize repair and replacement.

I have attached a copy of my curriculum vitae in Exhibit __ (JF-1).

Q. Have you previously filed testimony before regulatory or legislative bodies?

A. Yes, I have testified before the New York Public Service Commission ("Commission") in Case Nos. 16-G-0058, 16-G-0059 and 16-G-0061 and before the Illinois Commerce Commission in Case No. 16-0376. Please refer to Exhibit __ (JF-1) for a detailed listing of my testimonies.

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to provide recommendations to improve The Brooklyn Union Gas Company's d/b/a National Grid NY and KeySpan Gas East Corporation's d/b/a National Grid (collectively, "Companies" or "National Grid") proposed leak repair and pipe replacement activities, in order to advance New York climate policy to reduce natural gas leaks and methane emissions.

Q. Please provide a summary of your testimony and recommendations.

A. I recommend that National Grid deploy Advanced Leak Detection ("ALD") technologies to quantify their distribution system leakage, and to use information about leak size to prioritize pipeline replacement efforts and to repair the largest (i.e., fastest flowing) leaks as quickly as possible. Given that a small number of the largest "super emitter" leaks is typically responsible for the majority of total emissions, a program aimed at finding and rapidly repairing the largest leaks is a relatively easy way to reduce overall system emissions. Because quantifying and reducing emissions is a requirement of New York's recently enacted climate legislation, deployment of ALD technologies would provide National Grid a straightforward path to compliance.

I also note that the current incentive structure for reducing the Companies' leak backlog may be counter-productive. ALD technologies have been shown to help utilities find more leaks in their system than has been possible with traditional technology. Discovering more leaks would likely add leaks to the backlog. If the

incentive structure only rewards the Companies for declining leak backlog counts, the Companies are actually dis-incentivized to use a technology that discovers more leaks. For these reasons, I suggest that the incentive system be re-examined in light of the capabilities of this new technology.

Q. Are you providing any exhibits to your testimony?

A. Yes. I am attaching the following exhibits to my testimony:

- Exhibit __ (JF-1): Curriculum Vitae and List of Expert Testimony
- Exhibit __(JF-2): Pacific Gas and Electric Company Natural Gas Leakage Abatement Report, California Public Utilities Commission Rulemaking 15-01-008 (June 17, 2019)
- Exhibit__(JF-3): Picarro Emissions Quantification Results Final Report in Support of the Methane Leak Surveying Report for the Public Service Electric and Gas Company ("PSE&G") Gas System Modernization Program ("GSMP") II Program
- Exhibit__(JF-4): PSE&G Presentation "Replacement Main Prioritization: A Practical Application of Using Risk and Methane Emissions" (May 2, 2019)
- Exhibit __(JF-5): Kerans, Mike, Picarro Surveyor Leak Detection Study – Sacramento Side-by-Side Study (2012)
- Exhibit__(JF-6): Clark, Timothy, Picarro Surveyor Leak Detection Study – Diablo Side-by-Side Study (2012)

IV. Addressing Methane Leaks from Distribution Networks to Advance New York's Climate Policies

Q. You stated above that one of the purposes of your testimony is to provide recommendations in order to advance New York climate policy to reduce natural gas leaks and methane emissions. Please explain.

A. Natural gas leakage from distribution pipelines is widespread in U.S. cities, including many in New York State. As shown in the chart below, New York has a significant amount of leak prone pipe as compared to other states.

Top Ten States by Miles of Leak Prone Mains[1]

	Unprotected Bare Steel	Cast Iron	Ductile Iron	Copper	**Total Leak Prone Pipe**	Total miles
Grand Total	33,336	22,861	513	13	**56,722**	1,305,025
PA	5,932	2,525	170	2	**8,629**	48,335
NY	4,972	3,175	-	0	**8,147**	49,307
OH	6,197	197	1	1	**6,396**	58,759
NJ	550	3,911	24	1	**4,486**	35,007
TX	3,905	466	-	-	**4,371**	107,799
MA	1,146	2,925	1	0	**4,073**	21,714
CA	3,244	58	-	0	**3,302**	106,806
MI	352	2,389	-	-	**2,742**	59,731
WV	2,546	12	-	-	**2,557**	10,961
IL	28	1,152	205	-	**1,385**	62,168

This leakage is costly from the perspective of ratepayers and the environment. Methane, the principal component of natural gas, is a potent greenhouse gas that,

[1] Pipeline and Hazardous Materials Safety Administration. (2019). 2018 Gas Distribution Annual Report Data. Washington, D.C. Retrieved from https://www.phmsa.dot.gov/data-and-statistics/pipeline/gas-distribution-gas-gathering-gas-transmission-hazardous-liquids

pound for pound, traps 84 times more heat than carbon dioxide over the first 20 years it is released, and is therefore a significant contributor to climate change.[2] As acknowledged by the Companies, retirement of LPP is a "major component" toward reaching methane emission goals, as it "reduces gas losses and fugitive emissions of methane."[3]

Q. Has New York State recognized the importance of prioritizing leak repairs in distribution systems in addressing methane?

A. Yes. Governor Cuomo's May 2017 Methane Reduction Plan directs state agencies, including the Department of Public Service ("DPS"), to develop proposals and policies to inventory emissions and identify strategies for methane capture and elimination. The Plan specifically directs the DPS to "prioritize leak repairs in the distribution system."[4]

Q. Have other recent developments in New York State underscored the importance of addressing methane leaks from distribution networks?

A. Yes. The New York State legislature recently enacted the Climate Leadership and Community Protection Act ("Act" or "CLCPA"), which requires New York to

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[2] Myhre, G.; Shindell, D.; BrÂŐon, F.-M.; Collins, W.; Fuglestvedt, J.; Huang, J.; Koch, D.; Lamarque, J.-F.; Lee, D.; Mendoza, B.; Nakajima, T.; Robock, A.; G. Stephens, T. T.; Zhang, H. In *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; Stocker, T., Qin, D., Plattner, G., Tignor, M., Allen, S., Boschung, J., Nauels, A., Xia, Y., Bex, B., Midgley, B., Eds.; Cambridge University Press, 2013.

[3] *See, e.g.*, Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 29.

[4] Methane Reduction Plan at 6 (May 2017), https://www.dec.ny.gov/docs/administration_pdf/mrpfinal.pdf.

adopt measures to reduce statewide greenhouse gas ("GHG") emissions by 85 percent by 2050, with an additional goal of achieving net zero emissions across all sectors of the economy by that time.[5] The Act acknowledges the short-term climate impact of methane, recognizing that methane causes 84 times as much global warming as the same amount of carbon dioxide.[6] The Act also seeks to ensure that GHG emissions are "real, permanent, quantifiable, verifiable, and enforceable."[7]

Q. Have the Companies made any commitments to reduce methane emissions?

A. Yes, the Companies are establishing a target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035.[8] The Companies acknowledge that retirement of LPP is a major component toward reaching this ambitious goal.[9]

Q. Please explain some of the challenges in addressing methane leakage from distribution pipelines.

A. I perceive two key challenges for addressing methane leakage: 1) there are more leaks in the natural gas system than utilities are aware of, and 2) the current regulatory structure hampers utility initiatives to improve leak detection effectiveness.

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5 https://legislation.nysenate.gov/pdf/bills/2019/S6599.

6 *Id.*, Section 75-0101(2).

7 *Id.*, Section 75-0109(3)(b).

8 Future of Heat Testimony at page 16.

9 Gas Infrastructure and Operations Panel at page 28.

Recent academic findings have demonstrated that observed methane emissions from cities are about twice that reported in the Environmental Protection Agency inventory.[10] In addition, as I describe in a 2018 scientific paper, my team observed that utility crews using traditional technologies were able to locate only 35% of the pipeline leaks that were found using advanced leak detection methods.[11] Interestingly, this result was not unique to our work; two studies by the utility PG&E found a similar fraction of "false negatives" where leaks exist but are not found.[12] Together, these findings suggest that there are many more leaks than those reported in utility inventories. Based on our own published work, and corroborated by the PG&E studies, my team and I estimate that, in areas where utilities are using traditional leak detection technologies, there are 2.4x as many leaks in the system as the utility is aware of.

And, as I explain in detail below, the motivation for utilities to find more leaks is hampered by the regulatory structures for utilities to find and abate leaks. Although there exist well-intended incentives for utilities to reduce the number of leaks in their backlog,[13] an unintended consequence of these policies is that they

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[10] G. Plant *et al.*, Large Fugitive Methane Emissions from Urban Centers Along the US East Coast, https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635.

[11] Weller, Zachary *et al.*, Vehicle Based Methane Surveys for Finding Natural Gas Leaks and Estimating their Size: Validation and Uncertainty, Environmental Science and Technology (2018). If this detection rate is applied at the national scale, then the national inventory for the number of pipeline leaks in natural gas distribution infrastructure would increase by a factor of 2.4. *Id.* at 11925.

[12] Exhibit__(JF-5) and Exhibit__(JF-6).

[13] Gas Safety Panel at pages 53-54.

provide perverse incentives for utilities to not find and record new leaks in their inventories. This could hinder the State's ability to comply with the new climate legislation, which emphasizes the importance of ensuring a complete and accurate accounting of GHG emissions using the best available scientific, technological, and economic information.

Q. Please explain the role advanced leak detection and data analytics could play in addressing these challenges.

A. The idea of ALD is that high-sensitivity methane analyzers can be deployed in survey vehicles to identify where natural gas is leaking from distribution systems. The "advanced" part of this terminology refers in part to the methane sensors using a form of laser-absorption spectroscopy[14] that can more accurately and precisely determine the concentration of methane in the air. In contrast to "traditional technology," the sensors in ALD can quantify much smaller departures from the local background concentration and thus find more leaks. Moreover, ALD uses advanced data analytics that examine the overall neighborhood patterns in the methane concentration to discern when the concentrations are elevated. These sophisticated analyses can reveal subtler leak information than traditional technologies. Finally, ALD deployments can report not just the presence of a leak, but they can also estimate leak size. With leak size information, utilities can

[14] While my testimony refers to ALD, the Companies describe the technology as "CRS" or "CRDS," which is a type of laser absorption spectroscopy. Gas Safety Panel at page 45. While "CRDS" is one component of ALD, ALD also encompasses the advanced analytics or algorithms that utilize the data acquired from the ALD technology to estimate the methane flow rate attributed to a leak indication.

prioritize their repair and replacement efforts of non-hazardous leaks to fix the largest leaks fastest. This prioritization means that big leaks are not emitting methane for as long, thus reducing a utility's annual methane emissions. By simply re-organizing the timing of repair and replacement to fix the biggest leaks first, there is no additional cost.

As other utilities have demonstrated, advanced leak detection technology can be deployed in implementing "Super Emitter" programs, which aim to rapidly detect and repair large grade 3 leaks, leading to significant and rapid GHG emissions reductions.[15] Deploying these methods will also assist the Companies in ensuring a more accurate accounting of GHG emissions, consistent with the new climate legislation.[16]

Q. Could ALD also be used to prioritize pipeline replacement efforts?

A. Yes. Results from ALD yield information about leak location and magnitude. As we illustrated in the publication by Palacios and others for the utility Public Service Electric and Gas Company ("PSE&G"),[17] these leak size and density results can be aggregated by the utility to determine which stretches of pipe have the greatest

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[15] Exhibit __ (JF-2) at page 9.

[16] For example, both Companies provide a high level estimate of potential methane emissions reductions over the next several years based on each respective company's current facilities assuming LPP retirement pursuant to each company's proposed program. Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 30; Gas Infrastructure and Operations Panel, Case 19-G-0310 at page 29.

[17] Palacios, V., S.R. George, J.C. von Fischer and K. Mohlin (2017) "Integrating leak quantification into Natural Gas Utility Operations" white paper in *Public Utilities Fortnightly* https://www.fortnightly.com/white-papers/integrating-leak-quantification-natural-gas-utility-operations.

emissions. After safety considerations have been made, leak magnitude can be used as a metric for prioritizing which pipelines are replaced first.

IV. CSU and EDF Collaboration to Survey Distribution System Leaks

Q. Please provide a brief overview of EDF and CSU's collaborative project to survey methane leaks from distribution systems, and engagement with gas utilities on the use of leak quantification to prioritize pipe replacement and leak repair efforts (the "Project").

A. For the last 7 years, I have collaborated with EDF and Google on a unique project aimed at gathering data on the location and relative size of methane leaks from natural gas distribution systems using three Google Street View ("GSV") cars that are specially equipped with methane sensors.

The goal of the Project is to document patterns in the location and magnitude of natural gas leakage from the distribution systems of different utilities, demonstrate the benefits of using leak quantification technology to prioritize utilities' pipe replacement and leak repair activities, and to create the impetus for, and facilitate, the integration of this technology into utilities' regular leak operations.

The GSV vehicles that collected data as part of this initiative used a Picarro methane concentration analyzer that has a closed-path design with a sampling frequency of one data point every half-second, resulting in samples every 13 feet when driving at 20 miles per hour. These vehicles are also equipped with a Hemispheres Global Positioning System ("GPS") Receiver to document patterns in

methane concentration in space. Drivers were asked to drive every public street in a study area at least twice to ensure accuracy. Re-driving helped confirm the persistence of elevated methane readings and characterize their emission rate more accurately.

Data collected in urban systems and during controlled release experiments informed our development of a data analysis algorithm to interpret data collected through this Project. We conducted a number of "controlled release" studies at CSU by releasing methane at known rates and then driving through the resulting plume with the GSV car. We analyzed this data for patterns in plume size and shape to develop an algorithm that predicts leak size from measured patterns in methane concentration. We also validated the leak rate estimates using comparisons with a number of more costly and labor-intensive approaches and found that the method we used had more than enough precision and accuracy to be used effectively in utility leak reduction efforts.

Q. In how many cities has EDF/CSU mapped methane leaks from natural gas distribution pipelines as part of the Project?

A. To date, EDF and CSU have published methane maps for leaks from natural gas distribution systems in 13 cities: Boston, MA; Burlington, VT; Chicago, IL; Dallas, TX; Elizabethtown, NJ; Indianapolis, IN; Jacksonville, FL; Los Angeles, CA; New York City (Staten Island), NY; Syracuse, NY; Birmingham, AL; Mesa, AZ; and

Pittsburgh, PA.[18] In addition, we have conducted regional mapping in collaboration with the New Jersey utility PSE&G[19] and with the New York utility Con Edison.[20]

Q. Please summarize key findings and implications for utilities' pipeline replacement and leak repair programs.

A. In general, we found more natural gas leaks (expressed as leak indications per mile of roadway) in systems where federal data indicates high fraction of older, leak-prone materials in the distribution systems. That is, a high abundance of leaks per mile correlates with a higher percentage of leak-prone mains on the relevant utility system.

Scientific studies from across the natural gas supply chain have found that the majority of methane emissions come from a few larger leaks, and data gathered through this Project is consistent with that broader trend.[21] A 2014 Stanford University study noted, based on many independent experiments, that a small number of large leaks are responsible for a large fraction of the leakage from natural gas systems.[22] Likewise, across the 6100 natural gas leak indications found in the

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18 https://www.edf.org/climate/methanemaps.

19 https://www.edf.org/climate/methanemaps/pseg-collaboration.

20 https://www.edf.org/climate/methanemaps/con-edison.

21 Adam R. Brandt, Garvin A. Heath, and Daniel Cooley, Methane Leaks from Natural Gas Systems Follow Extreme Distributions, *Environmental Science & Technology* 2016 *50* (22), 12512-12520DOI: 10.1021/acs.est.6b04303, https://pubs.acs.org/doi/abs/10.1021/acs.est.6b04303.

22 A. R. Brandt et al., "Methane Leaks from North American Natural Gas Systems", *Science*, Vol. 343, 14 February 2011, available at http://www.novim.org/images/pdf/ScienceMethane.02.14.14.pdf.

15 urban areas we have mapped to date, we found that the larges 16% of those leaks are responsible for 50% of the emissions. Thus, efforts aimed to find and fix those largest super-emitter leaks can have a disproportionate and cost-effective impact on the overall emissions.

Only by measuring leak flow rate (i.e. the volume of methane emissions released from a given leak per unit time) can utilities use this information, alongside safety information of course, to prioritize the largest leaks for repair or the leakiest pipe segments for replacement. I summarize our findings in the table below.

City	Miles surveyed	Leaks found	Leaks per mile	Total Emissions (L/min)	Emissions per mile
Birmingham, AL	345	168	0.49	592	1.7
Boston, MA	2,607	2,349	0.90	6,142	2.4
Burlington, VT	113	13	0.12	18	0.2
Chicago, IL	1,037	402	0.39	834	0.8
New York, NY*	200	380	1.90	1,029	5.1
Dallas, TX	705	414	0.59	1,226	1.7
Elizabethtown, NJ*	212	224	1.06	655	3.1
Indianapolis, IN	767	3	0.00	6	0.0
Jacksonville, FL	820	81	0.10	160	0.2
Los Angeles, CA	1,674	354	0.21	1,104	0.7

Mesa, AZ	164	1	0.01	1	0.0
Newark, NJ*	813	313	0.38	781	1.0
Pittsburgh, PA	321	235	0.73	553	1.7
Staten Island, NY	990	976	0.99	2,621	2.6
Syracuse, NY	493	212	0.43	539	1.1
SUM or MEAN	11,262	6,125	0.54	16,262	1.5

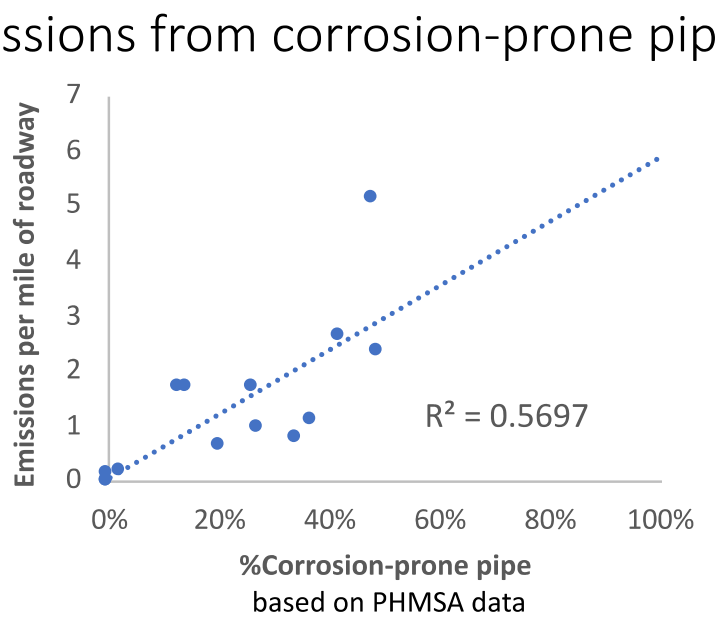
Our work shows that cities differ sharply in the number of leaks observed, depending, in part, on whether, and the extent to which, they had conducted accelerated pipeline replacement programs ("APRPs"). In general, cities that are known to have relatively newer, less leak prone distribution systems were found to have fewer leaks than cities with older, more leak prone systems. For instance, Burlington, VT and Indianapolis, IN registered 11 and 5 leaks, respectively, while we found hundreds or thousands of leaks in cities known to have older, more leak prone gas infrastructure such as Boston, MA, Staten Island, NY and Syracuse, NY.

For a more direct comparison, these leak counts should be normalized to the length of distribution lines or to the gas flow rate beneath the survey area, but appropriate data and/or infrastructure information were not in place to make this estimate. As a surrogate for this analysis, I calculated the frequency of leaks per unique km of roadway surveyed. This normalization did not alter our broad conclusions from an analysis of the raw data; there only were 6 leaks per 100 km

in Burlington and 0.4 per 100 km in Indianapolis, while Boston had 56 per 100 km, Staten Island had 63 per 100 km and Syracuse had 28 per 100 km.

In all cities, low leak flow rates were far more common (83% to 100% of cities' leaks) than medium (0% to 17%) or high (0% to 2%).

We estimated the average leak flow rates for each city as the sum of estimated leak flow rates per 100 km of roadway surveyed. When multiplied by the length of roadways in each city, we conservatively estimate that these leaks from Boston amount to at least 1300 tons methane yr^{-1}, 1000 tons methane yr^{-1} in Staten Island and 300 tons methane yr^{-1} in Syracuse.



The figure above shows the relationship between the percentage of corrosion-prone pipe and the estimated emissions per mile of corrosion-prone pipe. The corrosion-prone pipe data are drawn from the federal PHMSA database for the year when the surveys were conducted. Only 13 urban areas are shown because we

omitted results from New York, NY and Elizabethtown, NJ where the surveys were not a random sample of the area but instead a targeted survey of known areas with high abundance of leak-prone pipe.[23]

Many sectors of the natural gas supply system have a "fat tail problem" where large, infrequent leaks cause a significant proportion of the total system leakage.[24] The magnitude of the potential for leak reduction is stark: cities that were known to have relatively newer, less leak prone systems due to the implementation of APRP programs show an average 95% reduction in the normalized system leakage rate as compared to other cities.

Q. Have the results of this Project been published in peer reviewed scientific journals?

A. Yes, some of the results have been published, and more publications are underway. A description of the methodology was published in Environmental Science and Technology in 2017,[25] and a validation of false positives and leak size estimate was published in Environmental Science and Technology in 2018.[26] The approach for

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[23] This analysis has not been subject to peer review and was performed to illustrate the trends observed in the Project.

[24] A. R. Brandt et al., "Methane Leaks from North American Natural Gas Systems," *Science*, Vol. 343, 14 February 2011, available at http://www.novim.org/images/pdf/ScienceMethane.02.14.14.pdf.

[25] Joseph C. von Fischer *et al.*, Rapid Vehicle-Based Identification of Location and Magnitude of Urban Natural Gas Pipeline Leaks, Environmental Science & Technology (2017), https://pubs.acs.org/doi/abs/10.1021/acs.est.6b06095/.

[26] Zachary D. Weller *et al.*, Vehicle-Based Methane Surveys for Finding natural Gas Leaks and Estimating Their Size: Validation and Uncertainty, Environmental Science & Technology (2018), https://pubs.acs.org/doi/full/10.1021/acs.est.8b03135.

using leak size information for prioritizing pipeline repair and replacement was published in Public Utilities Fortnightly in 2017.[27] In 2018, we published a statistical study in the journal Environmetrics where we evaluate how greater survey effort finds more leaks.[28] A 2019 publication reports on updates to our data analysis algorithm and makes the algorithm publicly available.[29] Finally, in a second 2019 publication, we show how the GSV vehicle deployment could be used to estimate methane emissions from other sources, in this case fertilizer plants.[30]

I summarize each of these studies here:

- Von Fischer and others (2017): This is the first scientific paper to show that natural gas leak indications can be detected using ALD approaches. We outlined the methodology and described early findings in the first five cities we had mapped.

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[27] Virginia Palacios *et al.*, Integrating Leak Quantification into Natural Gas Utility Operations, Public Utilities Fortnightly (May 2017), https://www.fortnightly.com/white-papers/integrating-leak-quantification-natural-gas-utility-operations.

[28] Zachary D. Weller *et al.*, A calibration capture-recapture model for interfering natural gas leak population characteristics using data from Google Street View cars, Environmetrics (January 2018), https://doi.org/10.1002/env.2519/.

[29] Weller, Z.D., D. Yang* and J.C. von Fischer (2019) "An open source algorithm to detect natural gas leaks from mobile methane survey data." *PLoS ONE* 14(2): e0212287 https://doi.org/10.1371/journal.pone.0212287.

[30] Zhou, X., F.H. Passow, J. Rudek, J.C. von Fischer, S.P. Hamburg, and J.D. Albertson (2019) "Estimation of methane emissions from the ammonia fertilizer industry using a mobile sensing approach" *Elementa: Science of the Anthropocene.* 7(1) p19. https://www.elementascience.org/article/10.1525/elementa.358/.

- Weller and others (2018): we establish the efficacy of mobile methane surveys for managing local natural gas distribution systems by evaluating their ability to detect and locate natural gas leaks and quantify their emissions. In two cities, we found that three-quarters of leak indications from mobile surveys corresponded to natural gas leaks, but local distribution companies' field crews did not find most of these leaks, indicating that the national methane activity factor for leaks in local NG distribution pipelines is underestimated by a factor of 2.4. We found the median distance between mobile-estimated leak locations and actual leak locations was 19 meters. A comparison of emission quantification methods (mobile-based, surface enclosure, and tracer ratio) found that the mobile method overestimated leak magnitude for the smallest leaks but accurately estimated size for the largest leaks that are responsible for the majority of total emissions. Across leak sizes, mobile methods adequately rank relative emission rates for repair prioritization, and they are easily deployed and offer efficient spatial coverage.
- Palacios and others (2017): We describe how natural gas utilities can incorporate leak flow rate data into existing pipeline replacement and leak repair prioritization frameworks to more rapidly and efficiently reduce leakage on their system. Leak distributions typically demonstrate a "fat-tail," where a few, large leaks are responsible for the majority of lost gas volumes. Through ranking and ordering leak flow rate data, utilities can

identify a subset of the largest leaks to repair or the leakiest pipelines to replace, and capture more gas per dollar spent on leak repair or pipeline replacement. This benefits ratepayers, who pay for the cost of lost gas, and also carries broader environmental and societal benefits.

- Weller and others (2018): While new mobile monitoring technology has revolutionized our ability to measure pollutant levels over large regions, statistical methods for making inferences from data collected by these mobile systems are still being developed. In this paper, we introduce a new "capture–recapture" model to answer key questions from data collected by mobile monitoring systems. We apply our new method to characterize populations of natural gas leaks in urban areas using data collected by atmospheric methane analyzers placed on GSV cars. The new calibration capture–recapture ("CCR") model combines data from controlled methane release experiments and data collected from mobile air monitors to enable inference for several natural gas leak population characteristics, including the number of undetected leaks and the total methane output rate in a surveyed region. Our methodology is a novel application of capture–recapture modeling. The CCR model addresses challenges associated with using a capture–recapture model to analyze data collected by a mobile monitoring system such as a variable sampling effort. The CCR model provides a new framework for inferring the total number of leaks in natural gas distribution systems and offers critical insights for informing intelligent

infrastructure repair policy that is both cost effective and environmentally friendly.

- Zhou and others (2019): In this study, we estimate methane emissions from an important downstream consumer of natural gas, the ammonia fertilizer industry, which commonly uses natural gas as a feedstock and a fuel for the production of ammonia and other upgraded products. Using a GSV car equipped with a high-precision methane analyzer, we adopted a mobile sensing approach to measure methane mixing ratios along public roads that are downwind of the ammonia fertilizer plants. Useful data were collected from six plants, which represent >25% of the total number of U.S NG-based ammonia fertilizer plants, and use >20% of the total NG consumption by this industry. Based on the measured data, a source characterization model was applied to estimate the methane emission rates from the upwind plants. Assuming that the estimates are representative of emissions during normal operations of a plant, we calculated the NG loss rate (i.e. the ratio between NG emission rate and NG throughput). If the sampled plants are representative of the U.S. ammonia fertilizer industry, the industrial-averaged NG loss rate (± standard deviation) is estimated to be 0.34% (±0.20%), and the total methane emissions (± standard deviation) from this industry are estimated to be 29 (±18) Gigagram per year (Gg CH_4/yr) in 2015–2016. This is significantly higher than the reported methane emissions of 0.2 Gg CH_4/yr from the U.S. EPA's Facility Level Information

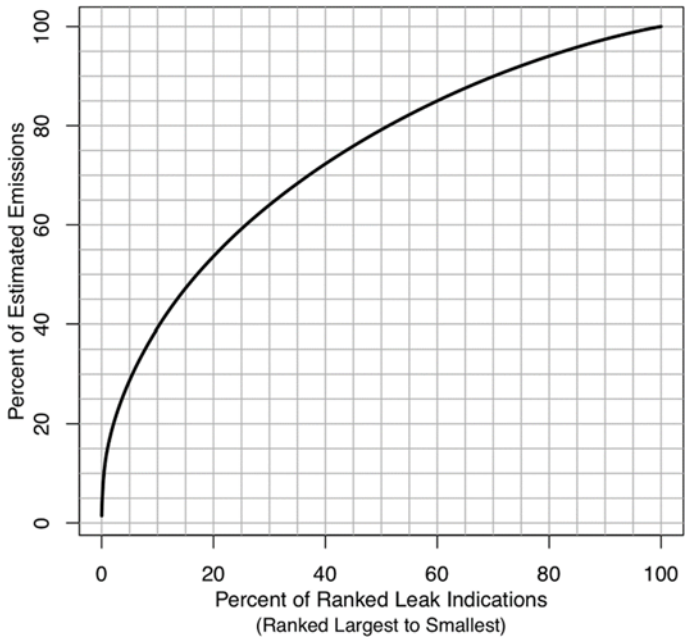
on Greenhouse Gas Tools (FLIGHT). This study begins to fill an important knowledge gap in quantifying methane emissions along the NG value chain, and demonstrates the capability of mobile sensing for characterizing emissions.

- Weller and others (2019) While survey platforms for ALD are commercially available, the associated data processing software largely constitute a black box due to their proprietary nature. In this paper we describe our updated, algorithm for identifying leak indications using data from mobile CH_4 surveys, and provide an under-the-hood look at the choices and challenges associated with data analysis. We also describe how our algorithm has evolved over time, and the data-driven insights that have prompted these changes. Applying our algorithm to data collected in 15 urban areas produced more than 6100 leak indications and estimates of the leaks' size. We use these results to characterize the distribution of leak sizes in local natural gas distribution systems. Mobile surveys are already an effective and necessary tool for managing natural gas distribution systems, but improvements in the technology and software will continue to increase its value.

Q. How can the use of advanced leak detection technology and leak quantification methodologies in leak prioritization ensure that ratepayer funding is deployed efficiently?

A. I have used a graphical tool called a "cumulative emissions curve" to illustrate the impact of eliminating the largest system leaks. A cumulative emission curve is shown below for the >6,100 leak indications that our Project has identified in the 15 urban areas where we have surveyed. To make this graph, we rank the leaks from largest to smallest, and then sum their emissions. The line on the graph thus illustrates how much of total emissions are caused by some percentage of the largest emissions. In the graph below, we can draw some direct conclusions about the role of the largest leaks in the system. For example:

- 20% of emissions come from the largest 2.5% of leaks
- 50% of emissions come from the largest 16% of leaks



Because a small fraction of the leaks are responsible for a large fraction of emissions, any effort that targets those large leaks for immediate repair will have a beneficially disproportionate effect on reducing overall system leakage.

V. Integration of Advanced Leak Detection into Utility Operations

Q. Have other utilities employed advanced leak detection, data analytics and quantification?

A. Yes. New Jersey's largest utility, PSE&G, received New Jersey Board of Public Utilities approval to implement a three year, $905 million pipe replacement program,[31] where PSE&G was required to consider data on the volume of methane emissions leaked from its pipes, in conjunction with other relevant factors, to identify those that are most in need of replacement.[32] Compared to a business as usual scenario, by using leak flow rates for prioritization, PSE&G achieved an 83% reduction of methane emissions early on by replacing one-third fewer miles of gas lines.[33]

As part of the second phase of its gas system modernization program in BPU Docket No. GR17070776, PSE&G committed to contract with a third party

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[31] *Decision and Order of the New Jersey Board of Public Utilities In The Matter Of Public Service Electric And Gas Company for Approval of a Gas System Modernization Program and Associated Cost Recovery Mechanism*, Docket No. GR15030272, November 16, 2015, retrieved from http://www.nj.gov/bpu/pdf/boardorders/2015/20151120/11-16-15-2F.pdf.

[32] Further information about this analysis can be accessed at https://www.edf.org/climate/methanemaps/pseg-collaboration.

[33] *Id.*

vendor to conduct a leak survey in 2018 on 280 miles of leak prone pipeline grids.[34] Leak survey data will be used to generate an "Estimated Flow Rate per Mile (Liter/min/mile)."[35] PSE&G will then develop a ranking threshold which will be used to prioritize grids for replacement in subsequent program years.[36]

Q. Has PSE&G shared the results from its Gas System Modernization Program?

A. Yes. The Methane Leak Surveying Report on PSE&G's Gas System Modernization Program II explains that "[w]ith methane maps and their aggregated emissions data…it is possible to make accurate, surgical construction decisions at the grid or individual pipeline section level as desired."[37] The report concludes:

> This variability shows the power of the methane mapping technique for providing additional granularity that can be used to maximize methane emissions reductions and/or maximize remediation of the maximum number of belowground leaks through changes to construction priorities based on these methane maps and associated data.[38]

A recent PSE&G presentation also concludes that its use of advanced leak detection technology is "better for the environment, [provides] less chance of non-

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[34] *In the Matter of the Petition of Public Service Electric and Gas Company for Approval of the Next Phase of the Gas System Modernization Program and Associated Cost Recovery Mechanism*, BPU Docket No. GR17070776, Stipulation of Settlement and Agreement at P 24 (April 18, 2018). The BPU approved this settlement in a June 1, 2018 order.

[35] *Id*.

[36] *Id.*

[37] Exhibit __(JF-3) at page 8.

[38] *Id.* at page 11.

hazardous leaks getting worse, and [results in] fewer potential customer calls/complaints."[39]

Q. Are you aware of other utilities' efforts to integrate advanced leak detection and data analytics into their operations?

A. Yes. Elizabethtown Gas Company ("Elizabethtown") recently became the first New Jersey utility to officially adopt and integrate advanced leak detection techniques into its operations as it implements its $300 million Infrastructure Investment Program ("IIP"). Pursuant to the settlement agreement in this case, Elizabethtown will integrate advanced leak detection technology information and methane emission flow rates in prioritizing its IIP projects.[40] The Peoples' Gas Light and Coke Company ("PGL") in Chicago, Illinois agreed to conduct a pilot program in which "leak flow rate data, collected by a contracted service provider or PGL using advanced leak detection and quantification technology, will be considered in prioritizing leak-prone pipe ("LPP") replacement under the [System Modernization Program]."[41] In addition, Peoples Gas of Pittsburgh committed to using advanced leak detection and data analytics to cut methane emissions from its

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[39] Exhibit __(JF-4) at slide 26.

[40] *In the Matter of the Petition of Elizabethtown Gas Company to Implement an Infrastructure Investment Program ("IIP") and Associated Recovery Mechanism Pursuant to N.J.S.A. 48:2-21 and N.J.A.C. 14:3-2A*, BPU Docket No. GR18101197, Stipulation of Settlement and Agreement at P 11 (May 30, 2019). The New Jersey Board of Public Utilities approved this settlement in a June 12, 2019 order.

[41] *Illinois Commerce Commission On its Own Motion v. The Peoples Gas Light and Coke Company*, ICC No. Docket 16-0376 at page 77 (January 10, 2018 Final Order).

distribution system by 50%.[42] After mapping and measuring the leaks from its underground pipes, Peoples will use the data to prioritize upgrades to achieve the greatest climate benefits.

Q. Have utilities in New York also explored the benefits of advanced leak detection and data analytics?

A. Yes, in 2016, Consolidated Edison Company of New York Inc. ("Con Edison"), EDF, and CSU conducted a pilot program to survey Con Edison's backlog of Type 3 leaks and characterize the leaks by size (i.e. flow rate). The leak flow rate data provided by EDF allowed Con Edison to repair the largest emitting non-hazardous leaks, representing an estimated reduction in nearly twice the amount of methane emissions compared to a business-as-usual scenario. EDF estimated that Con Edison reduced about 30% of the total emissions from surveyed areas. These results and an interactive map showing the leak locations are available on EDF's website.[43]

Q. Has Con Edison acknowledged the benefits of advanced leak detection?

A. Yes. The Company has stated that "[d]etection of gas leaks through state-of-the-art technology is critical to our comprehensive approach to risk management and

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[42] Environmental Defense Fund. January 8, 2019. Peoples Gas, EDF Unveil New Commitment to Help Protect the Climate by Cutting Methane Emissions From Pittsburgh Utility System in Half. Retrieved from: https://www.edf.org/media/peoples-gas-edf-unveil-new-commitment-help-protect-climate-cutting-methane-emissions.

[43] Environmental Defense Fund. Innovative collaboration fixes non-hazardous leaks faster. Retrieved from: https://www.edf.org/climate/methanemaps/con-edison Accessed on: May 20, 2019.

commitment to public safety."[44] Con Edison also notes that "through enhanced leak detection, we can identify, respond and remediate leaks more rapidly, reducing risk, keeping the public safe, and protecting the environment by reducing emissions of methane, a greenhouse gas."[45]

VI. The Companies' Leak Prone Pipe Repair and Replacement Efforts

Q. Please describe the pilot projects that EDF is conducting in collaboration with National Grid.

A. EDF and National Grid have collaborated on a range of pilot projects, in order to advance the use of new technological solutions, including advanced leak detection and quantification methods, to build on National Grid's efforts to enhance its system modernization and maintenance efforts. These projects are aimed at providing additional data and perspective around underground infrastructure condition, leakiness and program need, to help National Grid and the Commission balance the need for pipe replacement/repair efforts with the associated costs.

Q. Do the Companies propose to continue to explore the use of ALD technology to enhance leak program repair?

A. Yes, as explained by the Gas Safety Panel:

> During CY 2019, the Companies are continuing to assist Colorado State with its next phase of analysis of the 2018 CRD testing results. The Companies are providing access to and guidance on their gas system GIS mapping to enable the University to conduct geospatial analysis. The Companies will continue to

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[44] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company for Gas Service*, Case No. 19-G-0066, Direct Testimony of Gas Policy Panel at pages 15-16.

[45] *Id.* at page 16.

support these efforts and will consider the results and possible expanded application of the CRDS technology.[46]

Q. Do you have any recommendations regarding how the Companies should integrate advanced leak detection technology into their operations?

A. The Companies should conduct regular, comprehensive surveys of their infrastructure using ALD technology, especially in areas with high density of leak-prone pipe. It would serve the Companies well to set an accurate GHG baseline so that they can identify and quantify the largest sources of methane emissions in their service territories and receive regulatory credit for methane reductions from abating the largest leaks. The surveys should gather information on leak size so that the "super emitters" or leaks of exceptionally large size can be quickly found and repaired.

Q. Please explain what you mean by "super emitters."

A. Super emitters are the largest leaks on a utility's system, measuring 10 standard cubic feet per hour or greater. A "Super Emitter" Program identifies the highest emitting leaks for targeted replacement to satisfy emission reduction requirements.[47]

Q. Are you aware of any utilities that have implemented such a program?

A. Yes, Pacific Gas and Electric Company's ("PG&E") conducted its first "Super Emitter" leak survey in August 2018, where PG&E targeted the largest leaks and

[46] Gas Safety Panel, Case 19-G-0309 at pages 45-46.

[47] https://naturalgas.picarro.com/case_studies/super_emitters.

aimed to repair them promptly no matter the leak grade.[48] In 2018, PG&E found 220 "Super Emitter" leaks. The impact of the program was estimated at a reduction of 90 Mscf and PG&E expects emissions to continue to decline in the upcoming reporting years.[49]

Q. How would implementation of a Super Emitter program assist the Companies in complying with New York's new climate legislation?

A. In light of the CLCPA, finding and abating super-emitters is low hanging fruit that benefits ratepayers and advances the goals of the Act as well as safety. Because earlier reductions provide enhanced benefit in comparison to later in the terms of the Act, I recommend that the Companies pursue this type of program and receive regulatory benefit under the Act for doing so. If the Companies do not employ such a program now, as EDF witness Ferit Ucar observes, detection of large methane emissions at a later date "could cast doubt on earlier-demonstrated achievements of GHG reductions and hamper the Companies' ability to have future emissions trajectories deemed to be in compliance with the requirements of the Act."[50]

VII. Leak Backlog Incentive

Q. Please explain the current incentives the Companies receive for eliminating the highest volume Type 3 leaks.

[48] Exhibit __ (JF-2) at page 3.

[49] *Id.* at page 9.

[50] Ucar Direct Testimony at page 17.

A. The current leak backlog performance metrics and incentives were established in the Companies' 2016 rate cases. The metrics set annual leak backlog targets for workable leaks (Types 1, 2, and 2A) and total leaks and imposed Negative Revenue Adjustments for failure to meet targets.[51] The Companies are incentivized to repair Type 3 leaks and are permitted to earn a positive revenue adjustment of one basis point for each 50 such leaks eliminated, up to a maximum of five basis points per year.[52]

Q. Please explain the Companies' proposal for the leak backlog metrics and incentives.

A. The Companies' proposed metrics and incentives are presented in the tables below:

KEDNY Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	3,100*	5 basis points
	CY 21	2,950 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	2,800 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	2,650 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 150 reduction from CY19 target

[51] Gas Safety Panel at pages 53-54.

[52] *Id.* at page 55. The Companies must achieve both annual backlog reduction targets to qualify for this incentive. *Id.*

KEDLI Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	7,750*	5 basis points
	CY 21	7,000 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	6,250 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	5,500 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 750 reduction from CY19 target

KEDNY & KEDLI # High Emitting Type 3 Leaks Repaired	Positive Revenue Adjustment	Total Leak Target Reduction Subsequent Year
0- 49	None	0
50-99	1 basis point	50
100-149	2 basis points	100
150-199	3 basis points	150
200-249	4 basis points	200
>=250	5 basis points	250

Q. Do the current incentive mechanisms incentivize the Companies to deploy modern technology such as advanced leak technology that would find and quantify more leaks?

A. No. As I noted above, advanced leak detection technology and analytics are typically able to find many more existing and ongoing leaks than traditional technologies. At the same time, the current incentive structure rewards the utility primarily for reporting a lower number of leaks in their backlog at the end of the year. For these reasons, National Grid is currently disincentivized to use ALD, which could be expected to find more leaks. At the time of the Companies' 2016 rate case, ALD was not an established technology. Since that time, ALD has become a scientifically and commercially established technology, and it has been adopted by other utilities to find leaks and quantify their emissions.

Q. Do you have a recommendation to resolve this disconnect?

A. Yes. The current incentive structure should be revisited, as it ultimately serves to discourage the Companies from finding more leaks on its system and serves as an obstacle to achieving New York State climate and methane abatement goals and policies.

Q. Please further explain how your recommendation would work in practice.

A. I recommend that the Companies first complete a methane leak survey of its entire service territory using advanced leak detection technology. Using the information gathered from this initial survey, the Companies could then establish a system-wide baseline leak flow rate. Next, a volumetric leak reduction target could be

established within the current leak abatement incentive. In order to receive its annual maximum positive incentive, the Companies would be required to achieve a 50% reduction over three to five years which, according to our data, would require abatement of approximately the largest ca. 20% of leaks in its non-hazardous leak inventory. This objective could be met through a combination of pipeline replacement and Type 3 leak repairs, allowing the utility to optimize its approach to leak mitigation through pipeline replacements when necessary.

Q. Please explain why you are recommending a three to five year range for the Companies to achieve a 50% reduction in emissions reductions.

A. I am recommending a range of three to five years for two reasons: first, the current, estimated rate of advanced leak detection technology surveying is such that a system-wide survey could take several months to complete; and second, in recognition of the time that may be required for the Companies to integrate leak flow rate information into its geospatial information systems, and prioritization systems for leak repair and pipeline replacement. I believe it is possible for the Companies to achieve 50% emissions reductions within a three-year period, but also that some flexibility in the timeline is warranted to ensure that integration of the technology and information derived from its use is well thought out and positions the Companies to make efficient prioritization decisions over the long term.

Q. How does the three to five year range impact your recommendations regarding the Companies' incentive structures?

A. I recommend that the Companies use their proposed incentive structure based on the current leak backlog until a system-wide leak survey using advanced leak detection technology has been completed, and the Companies have developed plans for prioritizing leak repairs and pipeline replacements based on leak flow rate, after taking safety into consideration. After the Companies have made their prioritization plans available to the Commission, the new incentive structure based on percentage emission reductions should come into effect.

Q. Does this conclude your testimony?

A. Yes.

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

GREGORY LANDER

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: August 30, 2019

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Gregory M. Lander. My business address is 83 Pine Street, Suite 101, West Peabody, MA 01960.

Q. What is your occupation and by whom are you employed?

A. I am President of Skipping Stone, LLC ("Skipping Stone").

Q. Please state your educational background and experience.

A. I graduated from Hampshire College in Amherst, Massachusetts, in 1977, with a Bachelor of Arts degree. In 1981, I began my career in the energy business at Citizens Energy Corporation in Boston, Massachusetts ("Citizens Energy"). I became involved in the natural gas business of Citizens Energy in 1983. Between 1983 and 1989, I served as Manager, Vice President, President and Chairman of Citizens Gas Supply Corporation (a subsidiary of Citizens Energy).

I started and ran an energy consulting firm, Landmark Associates, from 1989 to 1993, during which time I consulted on numerous pipeline open access matters, a number of Federal Energy Regulatory Commission ("FERC") Order No. 636 rate cases, pipeline certificate cases, fuel supply and gas transportation issues for independent power generation projects, international arbitration cases involving renegotiation of pipeline gas supply contracts, and natural gas market information requirements cases (FERC Order Nos. 587 *et seq*.). In 1993, I founded TransCapacity LP, a software and natural gas information services company.

Since 1994, I have also been a Services Segment board member of the Gas Industry Standards Board ("GISB") and its successor organization, the North American Energy Standards Board ("NAESB"). During the period 1994 to 2002, I served as a Chairman of the Business Practices Subcommittee, the Interpretations Committee, the Triage Committee, and several GISB/NAESB Task Forces. I am currently a Board Member of NAESB and have served continuously in that capacity since 1997.

Skipping Stone, Inc. acquired TransCapacity in 1999, and since that time I have headed up Skipping Stone's Energy Logistics practice, where my specialization has been interstate pipeline capacity issues, information, research, pricing, acquisition due diligence and planning. In 2001, Skipping Stone launched CapacityCenter.com, a pipeline capacity information service. In 2004, Skipping Stone was acquired by Commerce Energy Group, a national retail energy services provider. In 2005, I was appointed President of Skipping Stone, which operated as a wholly owned subsidiary of Commerce Energy Group. In 2008, I purchased substantially all of the assets of Skipping Stone and now operate essentially the same business as before the Commerce Energy transaction as Skipping Stone, LLC.

From 1984 to present, I have maintained a deep familiarity with a wide range of pipeline transportation issues, beginning with access to pipeline capacity to make competitive sales, resolution of the pipeline take-or-pay contracting regime, pipeline affiliate marketer concerns, restructuring of the pipelines from merchants

to transporters and thereafter, and definitions of what constituted a pipeline capacity "right" for the purposes of formulating the then newly commenced capacity release and capacity rights trading business process. I continue to be involved in nearly all facets of the capacity information and trading business as part of my duties at Skipping Stone. In addition, I have been the lead principal on all 50+ pipeline and storage mergers and acquisitions transactions as well as all pipeline and storage facility expansion projects for which Skipping Stone has been retained by potential purchasers and project sponsors to provide economic due diligence consulting and market analysis. In addition, I have testified before, participated in or assisted with proceedings before, state public utilities commissions and/or their staffs in the states of Maine, Massachusetts, Missouri, Virginia, South Carolina, California, Rhode Island, New York and New Jersey with respect to infrastructure matters, integrated resource plans, and fuel cost recovery proceedings. Please refer to Exhibit __ (GL-1), which contains my current CV.

Q. Have you previously filed testimony before regulatory commissions?

A. I have filed testimony before the New York Public Service Commission ("Commission"), Massachusetts Department of Public Utilities, the Maine Public Utilities Commission, the Virginia State Corporation Commission, the Missouri Public Service Commission, the California Public Utilities Commission, and the South Carolina Public Service Commission. I have also filed testimony in several

FERC proceedings. Please refer to Exhibit __ (GL-1), which contains a full list of case names and docket numbers in which I have participated as a witness.

Q. On whose behalf are you testifying in this proceeding?

A. I am submitting testimony on behalf of the Environmental Defense Fund ("EDF").

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to provide recommendations regarding how the gas supply planning process of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, "Companies") could be improved.

Q. Please provide a summary of your testimony and recommendations.

A. My testimony first details the changed market conditions and new developments that compel the need for revisions to the Companies' gas supply planning process. I then provide an overview of the current gas supply planning process in New York and compare this process to other states. To improve the Companies' gas supply planning process, I propose three recommendations. I first suggest that the Companies should be subject to an annual gas supply process, with discovery rights for intervenors and technical conferences as needed. I next suggest that the Companies should be required to submit a long-range plan in a format, which would become the basis for evaluating future cost recovery. The long-range plan would set forth projections of demand and resources to meet that demand,

identifying the "all-in" cost for each resource. Finally, I suggest additional categories of information the Companies should submit in order to assist the Commission in its review of gas supply issues. Taken together, I explain how these refinements will help identify potential issues well in advance of experiencing demand/supply mismatches; manage the fixed costs commitments made by the Companies; and provide a more thorough framework for the Companies to consider alternatives that would have lower all-in costs to customers.

Q. Was your testimony prepared by you or under your supervision?

A Yes.

Q. Are you sponsoring any exhibits?

A. Yes, I am sponsoring the following exhibits:

Exhibit __ (GL-1): CV and List of Expert Testimony of Gregory M. Lander

Exhibit __ (GL-2): Winter Supply Review Data Request; Case No. 18-M-0272

Exhibit __ (GL-3): Rhode Island Joint Memorandum; Docket 4816

Exhibit __ (GL-4): Con Edison Response to EDF 6-48; Case No. 19-G-0066 *et al.*

Exhibit __ (GL-5): Con Edison Response to EDF 6-50; Case No. 19-G-0066 *et al.*

III. Need for Revisions

Q. Have there been changes in market conditions over the last decade that have affected natural gas utilities?

A. Yes. Over the past decade, two major trends have marked the evolution of the natural gas business. These two major trends, viewed generally, have had and will

continue to have profound impacts on gas distribution utilities, especially in the northeastern United States. The first of these trends is the explosive growth in natural gas supply from the Marcellus and Utica shales in the regions of northeastern and southwestern Pennsylvania, southeastern Ohio and northern West Virginia. The second is the dramatic increase in gas-fired electric generation, especially in Ohio, Pennsylvania, New Jersey, New York and New England. The second of these trends, driven by plentiful, low-priced natural gas has placed strains on the legacy pipeline infrastructure serving the same region. The dramatic growth in gas-fired generation has had a particular impact on the legacy pipeline infrastructure especially during winter periods when both domestic gas fueled heating needs and electric demands have placed simultaneous demands on pipelines. In the past, Local Distribution Companies ("LDCs") like the Companies were the primary users of pipelines during the winter. Now, LDCs and generators seek to use the same pipelines at the same hours of the day. As a result, the flexibility that the LDCs used to have with respect to hourly takes is being much more frequently restricted by the pipelines to the LDCs' specified contract rights. At the same time, because the loads of electric generators can be as large as (or larger) than many of the LDCs' take stations for whole swaths of their distribution territories, these combined demands are a new and growing factor that the industry has to cope with. Thus, LDC capacity planning that previously may have viewed its planning criteria as a daily load, now has to increasingly plan for peak hours of many more days in the winter than its "design

day." While in the past LDCs' on-system peaking resources may have been used, or planned to be used, during three to twenty days per winter, now may need to be called upon throughout the winter to meet hourly demands that are in excess of the LDCs' hourly contract rights. The significance of this is that even though the hours of use of on-system resources may increase, when looked at from an annual utilization basis, may not exceed 3% to 6% of 8760 hours per year (i.e., 250 to 500 hours per year). Given such low load factors and given the cost of incremental pipeline capacity, it simply may not make sense to commit to long-term capacity contracts in light of significant costs, the State's climate policies, and the Commission's findings on non-pipeline alternatives.[1]

Q. Please explain the consequences of this market evolution.

A. In addition to the tightening of pipeline operations which have reduced LDCs' previous flexibility to take gas at a greater hourly rate than their strict contract entitlements, producers and marketers have subscribed to large amounts of capacity from supply areas both to and past large metropolitan demand centers, and, in many cases, all the way from the supply areas to the U.S. Gulf coast to supply Liquefied Natural Gas ("LNG") export facilities. This increase in pipeline capacity is a resource, which, if it can flow into (as opposed to past), LDC distribution locations can be contracted for by LDCs based upon a price which

[1] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Gas Service*, Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan, Case No. 17-G-0460 *et al.* at page 75 (June 14, 2018) (finding that that non-pipeline alternatives should be "explored as a universal practice as an alternative to traditional investments.").

makes it in the supplier's interest to supply the LDC instead of taking the gas to export. Such arrangements would have the effect of reducing the need to build additional capacity from supply areas to market, and instead provide and plan for the capacity to exist along the "last mile" (i.e., into and from the LDC take stations). These combined facts—LDC short duration needs coupled with existing capacity moving gas past the LDC demand centers—present an opportunity to view capacity planning in a different way than ever before. Thus, a portfolio approach, a combination of on-system resources to meet needle hourly peaks plus contracted supply otherwise bound for export, have the effect of making LDC capacity planning a multi-dimensional exercise.

Q. Has National Grid acknowledged the need for refinements to its gas supply planning in other jurisdictions?

A. Yes. In a February 20, 2019 Joint Memorandum submitted to the Rhode Island Public Utilities Commission, National Grid stated:

> In the past, the [Long Range Plan] filings were not controversial and tended to raise few complicated issues. But now, the Company needs to plan in a way that assures adequate capacity and delivery security under supply contracts, the magnitude and implications of which have grown substantially. As a result, the current framework and template for the Company's long-range planning is no longer sufficient for an appropriate regulatory review.[2]

Q. Would you agree that these same concerns are relevant to National Grid's New York utilities?

A. Yes, the two trends I detailed above have profound impacts on gas distribution utilities, both in the northeastern United States and in New York State.

[2] Exhibit__(GL-3) at page 2.

Q. Has the Commission recognized the need to review and assess policies, programs, and regulations in light of these changes?

A. Yes. In a January 28, 2019 Statement, Chair John Rhodes directed DPS Staff to "develop a report to review the changing market conditions that gave rise to Con Edison's decision [to institute a moratorium] and to develop recommendations to make certain that utilities across the state are able to meet customer needs in a manner consistent with the State's energy goals."[3] The Statement also acknowledges the need for utilities to help consumers conserve energy and to solicit for natural gas supply alternatives. My recommendations set forth below will help the Commission and the Companies satisfy these objectives.

Q. Have there been any other recent developments in New York that compel utilities to revisit their gas supply planning?

A. Yes. On July 18, 2019, Governor Cuomo signed into law the Climate Leadership and Community Protection Act.[4] The Act requires a reduction in statewide greenhouse gas emissions by 85 percent by 2050 from 1990 levels, with an additional goal of achieving net zero emissions across all sectors of the economy by that time. The Act provides that in issuing approvals and decisions, *all* State agencies, offices, authorities, and divisions "shall consider whether such decisions

[3] *In the Matter of Staff Investigation into a Moratorium on New Natural Gas Services in the Consolidated Edison Company of New York, Inc. Service Territory*, Case No. 19-G-0080, Statement from Public Service Commission Chair John B. Rhodes on Consolidated Edison's Decision to Stop Accepting New Gas Customers on a Temporary Basis in Westchester County (January 28, 2019).

[4] New York State Climate Leadership and Community Protection Act (CLCPA), https://legislation.nysenate.gov/pdf/bills/2019/S6599.

are inconsistent with or will interfere with the attainment of the statewide greenhouse gas emissions limits established in Article 75 of the Environmental Conservation Law…."[5]

Q. Please explain the impact of this new legislation on gas supply planning.

A. This Act underscores the importance of thoughtful, transparent, and robust planning. As New York proceeds forward on its path outlined in the legislation there is a need for a transparent demonstration of the true demands of the system and the all-in costs of meeting that demand with various resources, being mindful not to lock-in greenhouse gas emissions from unnecessary long-lived and possibly stranded infrastructure.

IV. Suggested Improvements to Gas Supply Planning

Q. Please provide an overview of the current gas supply planning process in New York.

A. It is my understanding that the Commission opens a docket every year in order to examine each gas utility's supply plans. Each utility provides information regarding their portfolio and purchasing strategies, anticipated portfolio changes, gas supply portfolio details, among other issues.

Q. Have you reviewed the Companies' gas supply submittal in Case No. 18-M-0272?

A. Yes, I have reviewed the Companies' July 16, 2018 response to Staff's Winter Supply Review Data Request, attached to my testimony as Exhibit __ (GL-2), and

[5] CLCPA, Section 7(2).

the various updates and reports submitted by the Companies in that same proceeding. In addition, I have reviewed the testimony of Elizabeth Arangio submitted in these proceedings.

Q. Do you have any observations regarding these annual gas supply review proceedings?

A. Yes. Although the Commission has broad discretion to review the long-range programs of gas utilities,[6] the Commission historically has not made any formal public findings regarding the sufficiency of each gas utility's supply plan in these proceedings. Rather, it seems to be just a process between the utility and DPS Staff. Moreover, they are not formatted in such a manner as would make it possible to compare what is set forth in these planning documents and what customers ultimately will be asked to pay. Intervenors historically have not had discovery rights in these proceedings and there are no transcribed, on the record, technical conferences, let alone a formal evidentiary hearing.

Q. How does this process in New York compare to other state commissions' review of gas utility supply plans, of which you are aware?

A. At present, based upon my knowledge, there is no "perfect" state process. However, there are several states which require submissions of data, which if combined, come close to the type of transparency that could and should be the hallmark of an ideal process. For instance, in North Carolina, Piedmont Natural

[6] Public Service Law § 5(2) (the Commission has broad discretion to "encourage all persons and corporations subject to its jurisdiction to formulate and carry out long-range programs, individually or cooperatively, for the performance of their public service responsibilities with economy, efficiency, and care for the public safety….").

Gas files historic and projected load duration curves and against such curves presents its "resource stack" of pipeline capacity and on-system supplementals (e.g., LNG and CNG) to demonstrate its resource sufficiency.[7] In Massachusetts, the utilities present design day demand (net of conservation and energy efficiency) and against that they present their contractual[8] and on-system resource stack and identify surplus or deficit conditions with respect to that matching of forecasted demand to contracted resources.[9] In addition, many utilities that I am aware of also conduct post-winter analyses of peak hourly takes at the take station level and compare these experienced takes to their system model's predictions of takes under the experienced degree day conditions. In many cases this "post-action" analysis coupled with future demand growth projections identifies potential hourly demand versus hourly contracted resource deficits.

In Rhode Island, the Division of Public Utilities and Carriers (i.e., the Staff) and Narragansett Electric Company (a National Grid company) have proposed to

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7 Annual Review of Gas Costs Pursuant to G.S. 62-133.4(c) and Commission Rule R1-17(k)(6), North Carolina Utilities Commission Docket No. G-9, Sub 727, Testimony and Exhibits of Gennifer Raney on behalf of Piedmont Natural Gas Company, Inc. (August 1, 2018), https://starw1.ncuc.net/NCUC/ViewFile.aspx?Id=feb95b8f-afe1-4fab-8040-edd252c431a3.

8 Here, contractual includes both utility held pipeline capacity and Delivered Service contracts, the description of the service provided, and the daily quantity available under the contract.

9 *See, e.g.*, Boston Gas Company d/b/a National Grid Long-Range Resource and Requirements Plan (November 1, 2018), https://fileservice.eea.comacloud.net/FileService.Api/file/FileRoom/10008562.

improve the information available in the gas supply planning process.[10] One such improvement is the presentation of the effective per unit fixed costs based upon the planned load factor utilization of the individual transportation resource paths. Adding to this effective (i.e., load factor sensitive) fixed cost, the modeled cost of gas, generates an "all-in" cost presentation. The revised proposed protocol, set forth in a February 20, 2019 Joint Memorandum, is attached to my testimony as Exhibit__(GL-3).

Q. Do you have any recommendations to improve the Companies' gas supply planning process?

A. Yes, my recommendations fall into three general categories. First, I propose changes to the process by which gas supply issues are addressed. These changes would augment and supplement the current annual Staff review of gas supply plans. Second, I propose changes to how the Commission should review and consider gas supply information. In essence, each Company would be required to submit a long range plan, which would form the basis for future cost recovery in a gas cost reconciliation proceeding. Third, I propose changes to the types of information that each Company should submit to better inform gas supply decision making.

Q. Please explain your first category of changes to the process by which gas supply issues are addressed.

[10] At present, in part because the protocol is pending before the Rhode Island Public Utilities Commission, the information is not available outside of the Division and the Company.

A. First of all, the process should be an annual process. It should be an open process with intervenors having discovery rights, and to the extent it is more administratively efficient to have a series of technical conferences (following discovery), then those should be transcribed. If needed, an evidentiary hearing should be established. As the Commission has recently acknowledged, "[g]as supply constraint solutions will need to involve greater visibility of the distribution planning process to stakeholders and local communities, to enable joint problem solving."[11]

Q. Please explain your second category of changes to how the Commission should review and consider gas supply information.

A. As a starting place, each Company should submit a long-range plan, which would set forth projections of demand, by peak hour by Company operational "division" and by day by "division." Against that demand, the resources to meet that demand should be set based upon the contracts and the on-system supply capabilities of each Company. Each Company should then identify the cost of each resource (fixed cost and projected[12] or known[13] variable costs) and the projected load

[11] *Petition of Consolidated Edison Company of New York, Inc. for Approval of the Smart Solutions for Natural Gas Customers Program*, Case No. 17-G-0606 at 35, Order Approving with Modification the Non-Pipeline Solutions Portfolio (February 7, 2019).

[12] Here, "projected" would be in those cases where future costs are based upon market indices and projections of those future values.

[13] Here, "known" would exclude instances where negotiations have not been completed or where revealing "known" would impact negotiations for similarly situated transactions.

factor utilization of the resources so that all-in costs can be reviewed and alternatives that might result in lower all-in cost be evaluated.

Q. Do the Companies provide any of this information already?

A. In Ms. Arangio's testimony, this type of presentation was provided for certain peak period resources. My recommendation is that it be done for all transportation paths based upon modeled utilizations of the capacity paths. In addition, it would be helpful to provide this presentation for design day, normal winter weather and design winter weather.

Q. Once each Company submits a long range plan, what would follow?

A. An agreed-upon long range plan would become the basis for both the annual (and/or semi-annual) gas cost reconciliation proceedings; and, for rate case revenue requirement development. In the gas cost reconciliation proceedings, the long range plan would provide the baseline. Differences between the baseline and the actuals in the gas cost reconciliation proceeding would be evaluated as "variances from plan." This is in contrast to the Commission having to review the gas cost reconciliation proceedings with information, resources, structures, and costs that may not have been revealed previously.

Q. Please explain what you mean by "all-in cost."

A. The "all-in cost" is determined by looking at the annual facilities' fixed costs plus commodity/O&M cost per unit of demand met taking into account the load factor of the annual demand to be met. Below I provide an example of "all-in" cost calculations.

	Annual Facilities' / Fixed Costs	Annual O&M / Commodity Costs	Peak Hour Demand (Dth/Hr)	Annual Incremental Demand Met	All-in Cost ($/Dth)
Ex. 1	$5,000,000	$1,800,000	1,000	150,000	$45.33
Ex. 2	$15,768,000	$420,000	1,000	150,000	$107.92

Ex. 1 Assumptions: Annual Cost of CNG Facility is $5 MM; CNG $/Dth $12;
Ex. 2 Assumptions: Annual Cost of New build PL Capacity at $1.80/Dthd; $/Dth $2.80;

Common Asssumptions: 1,000 Dth/Hr (24,000 Dthd); and 150 Hours/Yr Equivalent Full use.

Q. Are you saying that the figures above are representative of actual costs that the Companies might incur?

A. Not necessarily. This is an illustrative example demonstrating how each Company would calculate the "all in" cost of various alternatives.

Q. Please explain your third category of changes as to the types of information that the Companies should submit.

A. I recommend the Companies file the material they file today in the gas supply planning dockets reviewed by Staff, a long range plan as outlined above, and the following, available to intervenors:[14]

1) Historic (i.e., actual) daily winter period demand curves for the prior 5 years by class along with the prior demand forecasts for the same periods;

2) Historic (i.e., actual) daily non-winter period demand curves for the prior 5 years by class along with the prior demand forecasts for the same periods;

[14] At present, the Companies provide this material to Staff but it is redacted when posted to the Commission's website.

3) Historic (i.e,, actual) system winter period demand curves, (hourly and daily) for each of the Company's take stations for the prior five years along with the demand forecasts for the same periods;

4) Historic (i.e., actual) system non-winter period demand curves, (hourly and daily) for each of the Company's take stations for the prior five years along with the demand forecasts for the same periods;

5) The historic (i.e., actual) resource stacks of the Company employed to meet those historic demand curves;[15]

6) The Company's forecasted winter period system demand duration curves for the next five years;

7) The Company's forecasted non-winter period system demand duration curves for the next five years;

8) The Company's forecasted winter period demand curves, (hourly and daily) for each of the Company's take stations;

9) The Company's forecasted non-winter period demand curves, (hourly and daily) for each of the Company's take stations;

10) The resource stacks (including separate presentation of their respective fixed and projected variable costs and projected load factor utilization) the

[15] Note that for the hourly resources by take station, the Company should indicate a) its hourly contract rights, b) any overrun services it received to meet hourly demands; and c) Company-operated facilities employed to meet demands not otherwise met by contract rights or overruns service(s). In addition, the Company should note the extent to which delivered services (either to the Company or for the benefit of the Company's transportation customers) contributed to meeting demands by take station.

Company has under contract to meet the Company's forecasted forward period demand curves;[16] and

11) For those forecasted demands not met by existing contract rights plus Company-operated facilities, the Company should identify all potential resources (including non-pipeline solutions) under consideration and each such resource's forecasted all-in cost (as defined above) and provide the detailed analysis and assumptions used for the build-up of such resources' all-in costs presented by the Company. In addition, the Company should identify potential non-pipeline solutions not under consideration for each forecasted period, and the detailed analysis performed as to why the particular potential non-pipeline solutions are not under consideration for the subject period(s).

Q. How would this information assist the Commission?

A. The historic information would be used to evaluate previously forecasted demands to actual realized demands and the resources used to meet those actual demands. The forecasted demand information would identify gaps, if any, between forecasted demands and resources. The presentation of potential resources, their timing, all-in costs, and capabilities would assist the Commission in both understanding the available alternatives and the trade-offs involved with each. I recommend that the culmination of this planning, presentation and

[16] Note that for the hourly resources, by take station, the Company should indicate then-existing contract rights and Company-operated facilities' hourly (and total) capabilities to be employed to meet forecasted demands not met by take station contract rights. In addition, the Company should note the extent to which delivered services (either to the Company or for the benefit of the Company's transportation customers) contributed to meeting demands by take station.

consensus building process be Commission approval, based on a traditional review of reasonableness.

Lastly, the consensus resource stack on contracts, costs and capabilities would form the baseline against which subsequent gas reconciliation proceedings are conducted and against which variances from plan are identified and worked through.

Q. Would these refinements lead to benefits?

A. A more robust planning process would lead to multiple benefits, including: (1) identifying potential issues well in advance of experiencing demand/supply mismatches; (2) a way to manage and contain the fixed cost commitments made by the Companies, which will protect ratepayers against unreasonable financial risk and protect the Companies against prudency risk from after-the-fact regulatory challenges; and (3) a more thorough framework for the Companies to consider alternatives that would have lower all-in costs to customers.

V. New York Facilities System

Q. Please provide an overview of the New York Facilities System.

A. As Ms. Arangio explains:

> KEDNY and KEDLI are parties to an agreement with Consolidated Edison Company of New York, Inc. ("Con Edison") concerning the ownership and operation of the New York Facilities System ("NYFS"), the high pressure gas transmission system serving the three downstate New York distribution companies. The agreement permits the parties to contract for the transportation and receipt of gas from various interstate pipelines that interconnect with the NYFS, including Transcontinental Gas Pipeline Company LLC ("Transco"), Texas Eastern Transmission LP ("Texas Eastern"), Iroquois Gas Transmission System LP

("Iroquois"), and Tennessee Gas Pipeline Company ("Tennessee").[17]

Q. How do the Companies and Con Edison plan for the needs of the NYFS?

A. My understanding is that the Companies and Con Edison prepare joint hydraulic models of the NYFS,[18] file an updated appendix annually with the Commission, and discuss the modeling results from time-to-time with Staff at their request.[19]

Q. Do you have any recommendations to improve the planning process for the NYFS?

A. Yes. I recommend that the Commission undertake a proceeding where the planning assumptions of the NYFS are formalized and integrated into the more general, downstate energy planning process for the NYFS members. The Commission may even find that it appropriate to view and regulate the NYFS as a functioning independent not-for-profit gas grid operator. This would entail formally identifying which facilities of the downstate gas utilities be contained within such organization, defining cost allocation and cost recovery mechanisms for the organization and finally, defining the goals and charges of such organization in the planning, operating, and optimal execution of the State's energy and climate policies.

Q. Does this conclude your testimony?

A. Yes.

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[17] Arangio Testimony at page 5.

[18] Exhibit___(GL-4).

[19] Exhibit___(GL-5).

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

JAMES FINE

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: February 6, 2020

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is James Fine. My business address is 123 Mission Street, 28th Floor, San Francisco, CA 94105.

Q. What is your occupation and by whom are you employed?

A. I am a Senior Economist in the Energy Program at Environmental Defense Fund ("EDF").

Q. Please describe your responsibilities in that position.

A. As a member of EDF's Energy Program, I conduct research and analysis to inform EDF's strategy and support its advocacy. My work includes resource adequacy assessment, long-term procurement planning, energy efficiency, demand response (DR), renewable energy, residential tariff design, and smart grid deployment. Since 2009, I have worked with the California Public Utilities Commission (Commission or CPUC), California Air Resources Board (CARB), California Independent System Operator (CAISO), California Energy Commission (CEC), California investor-owned utilities (IOUs) and many other stakeholders on clean energy policies.

Q. Please describe your educational and professional background.

A. I received my B.S. in Economics from the University of Pennsylvania Wharton School in 1989, and my Ph.D. from the University of California Berkeley, Energy and Resources Group, in 2003. I have over 20 years of experience working in the field of environmental economics, including energy, with the last ten years focusing

primarily on clean energy policy. I consulted with M.Cubed and Envair from 1994 to 2007 and was an assistant and adjunct professor at the University of San Francisco. I have provided expert testimony in several CPUC proceedings in California and elsewhere, including residential electricity rate reform, vehicle-grid integration and incentive programs for distributed clean energy resources. In addition, I have a significant amount of experience with EDF's work in other states, such as Illinois, Ohio, Texas, New Jersey and New York.

Q. Have you previously filed testimony before the New York Public Service Commission?

A. No.

Q. On whose behalf are you testifying in this proceeding?

A. I am submitting testimony on behalf of EDF.

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to make recommendations concerning how The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid (collectively, "Companies") can meet the evidentiary burden that the Commission will need to require in order to approve a rate plan after the effective date of the New York Climate Leadership and Community Protection Act ("CLCPA" or the "Act"). In addition, my recommendations provide a structural approach whereby the Companies can quantify and potentially receive credit for successful outcomes from their efforts to

reduce emissions under the Act. Ferit Ucar, who was previously my fellow economist at EDF, previously filed a version of this testimony in August 2019 in this proceeding, on behalf of EDF. I am filing this substantively identical testimony in place of Mr. Ucar's testimony because he is no longer an EDF employee.

Q. Please provide a summary of your testimony and recommendations.

A. My testimony describes the greenhouse gas ("GHG") emission reduction goals that have been adopted by New York State through the CLCPA; one of the mechanisms through which agencies such as the Commission are required to ensure that the State remains on track to achieve those goals; and the evidentiary showing that utilities, when they are seeking to obtain the Commission's approval of a rate plan, will need to make in order for the Commission to make the determinations of consistency and non-interference with the CLCPA that the Act requires in connection with any such approval. It also describes progress that has been made in a different setting, the EAM Collaborative arising out of Consolidated Edison's 2016 rate case, in considering what information is relevant for a similar analysis and what obstacles need to be overcome in order for that information to be obtained in a timely fashion. Finally, I describe potential opportunities for the Companies to reap benefits from exercising climate leadership in developing accurate emissions accounting tools in parallel with their leading GHG reduction activities.

Q. Was your testimony prepared by you or under your supervision?

A Yes.

Q. Are you sponsoring any exhibits?

A. Yes, I am sponsoring the following exhibits:

Exhibit __ (JF-1): CV of James Fine

III. New York's Climate Goals

Q. Please describe the policies relevant to GHG emissions that were in effect in the Companies' service territories and relevant to the Companies' operations at the time that the Companies filed their initial rate case on April 30, 2019 and corrections and updates on July 3, 2019.

A. As the Companies' Future of Heat Panel noted in its testimony, prior to the filing of the rate case, Governor Cuomo had adopted goals including 40% GHG emissions reductions (from 1990 levels) from the energy sector by 2030, the longer-term goal of decreasing total carbon emissions 80% by 2050, as well as a requirement that 100% of electricity consumed in New York be carbon neutral by 2040.[1] In addition, in May 2017 New York adopted a Methane Reduction Plan, which identifies 25 actions that are expected to "reduce methane emissions in alignment with New York State's commitment to reduce greenhouse gas emissions"[2] and, among other things, calls upon the New York Department of Environmental Conservation ("DEC") and the New York Department of Public Service ("DPS") to "[e]valuate and identify best technology and methods to identify leaks in each portion of the

[1] Future of Heat Panel testimony at page 7.

[2] N.Y. Methane Reduction Plan at 4 (May 2017), https://www.dec.ny.gov/docs/administration_pdf/mrpfinal.pdf.

system."[3] Further, in April 2019, shortly before the filing of the rate cases, the City of New York passed Intro 1253, which requires buildings over 25,000 square feet to meet building emissions limits that are in alignment with the City's own goals of 40% reductions by 2030 and 80% reductions by 2050 (from 2005 levels). This new City law, which is now known as Local Law 97, applies to the Companies' customers in the portion of the Companies' service territories that is in New York City.[4]

Q. Have any significant developments involving New York State climate policy occurred since the Companies' filed their initial rate cases on April 30, 2019 and corrections and updates on July 3, 2019?

A. Yes. On July 18, 2019, Governor Cuomo signed into law the Climate Leadership and Community Protection Act.[5]

Q. Please provide an overview of the key provisions of the Act.

A. The Act requires New York to adopt measures to reduce statewide GHG emissions by 85 percent by 2050 from 1990 levels, with an additional goal of achieving net zero emissions across all sectors of the economy by that time (the remaining 15 percent can come from carbon offsets).[6] The Act ensures that the short-term climate impact of methane, which causes 84 times as much global warming as the same

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3 *Id.* at 5.

4 New York City Local Law 97 (2019).

5 New York State Climate Leadership and Community Protection Act (CLCPA), https://legislation.nysenate.gov/pdf/bills/2019/S6599.

6 CLCPA, https://legislation.nysenate.gov/pdf/bills/2019/S6599.

amount of carbon dioxide over a twenty-year horizon,[7] will be recognized in GHG reporting.[8] The Act also recognizes that GHG emissions attributable to New York include "greenhouse gases produced outside of the state that are associated with the generation of electricity imported into the state and the extraction and transmission of fossil fuels imported into the state,"[9] and seeks to ensure that GHG emissions reductions are "real, permanent, quantifiable, verifiable, and enforceable."[10]

IV. CLCPA Implementation

Q. Does the Act define a role for the Commission in ensuring that the State meets its ambitious GHG limits?

A. Yes. The Act establishes a new Section 66-p of the Public Service Law, which requires the Commission to establish a program to require that (a) a minimum of 70% of the statewide electric generation secured by jurisdictional load serving entities to meet the electrical energy requirements of all end-use customers in New York State in 2030 be generated by renewable energy systems, and (b) that by the year 2040 the statewide electrical demand system will be zero emissions. The new Section 66-p will further require that, beginning in 2024 and again every two years thereafter the Commission conduct a comprehensive review of these programs,

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7 IPCC, 2014: Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. *See* Box 3.2, Table 1.

8 Environmental Conservation Law § 75-0101(2) and §75-0105.

9 Environmental Conservation Law § 75-0101(13).

10 Environmental Conservation Law § 75-0109(3)(b).

including progress toward meeting targets. Section 66-p further provides that, no later than July 1, 2024, the Commission shall establish programs to require procurement of at least 9 gigawatts (GW) of offshore wind electricity by 2035 and 6 GW of photovoltaic solar generation by 2035, and to support 3 GW of statewide storage capacity by 2030. Section 66-p also directs the Commission to specifically provide a minimum level of benefit for low-income customers and disadvantaged communities. In addition to these specific provisions applicable to the Commission, the Act provides that in issuing approvals and decisions, *all* State agencies, offices, authorities, and divisions "shall consider whether such decisions are inconsistent with or will interfere with the attainment of the statewide greenhouse gas emissions limits established in Article 75 of the Environmental Conservation Law…."[11]

Q. What information would be necessary for the Commission to assess whether approving utility proposals would be inconsistent with and/or would interfere with compliance with statewide GHG emissions limits?

A. First of all, the Commission would need a comprehensive current baseline of GHG emissions in each regulated utility service territory. Once the baseline is established, the Commission would need information and data to be able to assess actual progress being made in each utility service territory towards GHG emissions reductions, ideally on an annual basis to assess consistency with the Act. Finally, the Commission would need to be provided with estimated GHG impacts from the utility proposals as well as projections of GHG emissions with an assessment of

[11] CLCPA, Section 7(2).

variability and uncertainty to be able evaluate whether proposals interfere with compliance with future statewide GHG emissions limits as required by the Act.

V. Role of Utilities in facilitating CLCPA Implementation

Q. Do you have any recommendations regarding how the Commission should make a determination of consistency with and non-interference with State GHG goals?

A. The Commission's ability to make this determination is premised upon receiving sufficient information and analyses from the proponent of the request. In the case of a utility rate case, the proponent would generally be the utility.

Q. Are utilities well positioned to play a role in CLCPA implementation?

A. Yes. The decisions of utilities are central to the achievability of the State's climate goals. In fact, in this case, the Companies acknowledged their role in achieving the then-applicable State climate goals, citing to the State's goal of decreasing carbon emissions 80 percent by 2050, and National Grid's "Northeast 80x50 Pathway." The Companies state that their "gas networks will play an integral role in meeting these ambitious goals and delivering the lower-carbon economy of the future."[12]

Q. In your opinion, have gas utilities to date provided sufficient means and tools to make such a demonstration?

[12] Future of Heat Panel, Case 19-G-0309 at page 7.

A. No. As my colleague, Naim Jonathan Peress, has previously asserted to the Commission, gas utilities have generally not provided the tools or the means to assess and weigh climate impacts—good, bad, or indifferent.[13]

Q. What information or tools should regulated utilities provide to assist the Commission to assess consistency with and non-interference with CLCPA?

A. Regulated utilities will need to develop a comprehensive GHG assessment methodology, provide the data, do the monitoring and knowledge management needed to organize, update, and augment the data, and then plug in the data in empirical tools and metrics to assess the GHG impacts of their operations. Ideally, these data and tools could be used to inform utility decision making such as around supply arrangements and infrastructure investment as well as to assist the Commission in making the determinations required under the Act. The tools that are used to assess current and forecasted GHG emissions should make use of the "best available economic models, emissions estimation techniques and other scientific methods," as contemplated by sections of the Act addressing methodological considerations.[14] These tools could be used to establish a current baseline, and after such baseline is established, measurement of GHG emissions in subsequent years would allow assessments of progress toward GHG goals and the

[13] *In the Matter of Staff Investigation into a Moratorium on New Natural Gas Services in the Consolidated Edison Company of New York, Inc. Service Territory*, Case No. 19-G0080, Public Statement Hearing Transcript at page 128 (February 13, 2019) (statement of N. Jonathan Peress on behalf of EDF).

[14] *See* Environmental Conservation Law, § 75-0103(14)(b).

accuracy of projections, and to inform any adjustments needed for future projections of GHG emissions.

Q. Can you describe the appropriate scope and content of a GHG assessment tool that would provide the Commission with the information it would need to determine consistency with and non-interference with the CLCPA?

A. An appropriate GHG assessment methodology would be a holistic analysis of GHG emissions within a service territory, taking into consideration all emissions (or at a minimum all energy-related emissions) within the geographical footprint of the service territory, not solely those emissions that are directly attributable to the utility's own system. Such a holistic methodology would be appropriate primarily because of the nature of the GHG emissions limits established in the Act, which pertain to economy-wide emissions, including upstream emissions.

Q. Is it unreasonable to judge a request by a utility that only owns and operates one energy system by such a holistic GHG metric?

A. No. It might be unreasonable to judge a utility's own climate impact by a holistic metric, but that is not the nature of this exercise. The purpose of this metric is to give the Commission the foundation it needs to find that approving the utility's proposal will not handicap achievement of statewide targets. No methodology focused solely on emissions attributable to a specific utility's system would allow the Commission to make such a determination as long-term statewide GHG outcomes arise as a result of many factors and actions of multiple actors, all of which are interdependent. For example, utility actions and practices affect the

actions and practices of customers, suppliers, and other parties, shaping market outcomes that do not arise directly from the utility's own system. Gas utility supply decisions may indirectly affect the level of GHG emissions associated with natural gas extraction taking place in various locations; their treatment of natural gas customers that are electric generators may affect the rapidity with which the electric generation fleet is converted to lower-emitting or non-emitting generation; and their practices with respect to line extensions and non-pipeline solutions may affect the speed of electrification of building heat. There are also second-order effects; if natural gas utilities influence the GHG emissions intensity of the electric generation fleet serving their service territory, they will also, indirectly, influence the GHG emissions intensity of newly electrified applications, such as transportation and electric building heat. Knowing that their rate plans will be evaluated on a holistic basis with respect to all these indirect market outcomes aligns the natural gas utilities' business goals with the state's emissions limits and reduces the effect of incentives they might otherwise face to engage in activities that could be expected to hinder achievement of the GHG emissions limits.

Q. As a practical matter, what steps do gas utilities need to take to enable this holistic evaluation?

A. Gas utilities need to work with state and local entities to gather the requisite, economy-wide information that will make a holistic GHG emissions assessment feasible. This will require measuring emissions associated with the Company's own operations, including greenhouse gas emissions resulting from the combustion of

natural gas the Company delivers to customers as well as an accurate assessment of methane loss from the Company's own distribution system. It will require the Company to obtain reliable information about the upstream emissions associated with the natural gas it delivers, as well as information about greenhouse gas emissions in other sectors of the economy. In its fully developed form, this may require gathering information from private parties that are not regulated by the Commission, such as fuel suppliers, and accounting for methane loss at customer premises. Using this economy-wide information, a current status of emissions attributable to the Company's geographical footprint can be established.

After establishing a current, holistic GHG baseline for its service territory, the utility would need to provide projections of GHG emissions for future years based on projected economy-wide energy consumption and pertinent emissions factors. Projections of energy consumption for each fuel should take into account the impacts of energy efficiency, electrification, population growth, and other relevant factors. Projections of emission factors for natural gas should take into account the estimated impacts from renewable natural gas and biogas and changes in methane leakage rate from the natural gas system. Projections of emissions factors for electricity should take into account the changing GHG emissions-intensity of the generation fleet, as required by the CLCPA. Moreover, since any future projections would contain inherent uncertainty and variability, these projections should be accompanied by an uncertainty and sensitivity analyses in order to provide the Commission with necessary information to evaluate risks.

Q. Would the establishment of a baseline and a means to assess progress provide a useful operational tool for the Companies?

A. Yes. At the present time, the Companies have at their disposal various means of reducing GHG emissions in their geographical footprints. The sooner the Companies begin quantifying overall emissions and planning to support the dramatic reductions required under the Act, the more their business practices and the resulting market outcomes can be expected to align with the goals of Act. Comparatively early compliance with the goals of the Act would be advantageous for the Companies because they could reasonably expect the Commission to look more favorably upon the proposals of Companies that take such initiative, and, aided by such suitable tools, the Companies could work with the Commission to develop a system of incentives for emissions reductions that take place ahead of the schedule required under the Act. Conversely, the later the Companies establish a defensible baseline, the harder it will be for them to obtain full recognition for their early leading-edge work to reduce economy-wide GHG emissions.

Q. Would the establishment of incentives that reward regulated utilities for climate leadership where GHG reductions are achieved faster than required by law be appropriate?

A. Yes. Because climate change is a result of the cumulative GHG emissions that are present in the atmosphere and there is limited time to reduce them before the point

at which climate change becomes irreversible and catastrophic,[15] such earlier-than-mandated GHG emissions reductions may have more value than emissions reductions in later years. The New York's Legislature recognized the vital importance of climate leadership in the recitals to the Act, finding and declaring as follows:

> "Action undertaken by New York to reduce greenhouse emissions will have an impact on global greenhouse gas emissions and the rate of climate change. In addition, such action will encourage other jurisdictions to implement complementary greenhouse gas reduction strategies and provide an example of how such strategies can be implemented. It will also advance the development of green technologies and sustainable practices within the private sector, which can have far-reaching impacts such as reduction in the cost of renewable energy components, and the creation of jobs and tax revenues in New York."[16]

The N.Y. Legislature further found and declared:

> "By exercising a global leadership role on greenhouse gas mitigation and climate change adaptation, New York will position its economy, technology centers, financial institutions, and businesses to benefit from national and international efforts to address climate change."[17]

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[15] *See generally* 2018 IPCC Special Report, Global Warming of 1.5°C, *available at* http://www.ipcc.ch/report/sr15/

[16] CLCPA, Section 1(3).

[17] CLCPA, Section 1(12).

In light of the potentially high value of early GHG reductions, the commensurately high cost of delay, and New York's stated intent to exercise leadership in the struggle to mitigate climate change, incentives for regulated companies that reward climate leadership would be appropriate.

Q. For the years 2014 through 2018, the Companies provide estimated methane emissions reductions associated with the replacement of leak prone pipe.[18] Do these estimates provide an accurate assessment of methane loss from the Company's system?

A. No. The Companies' emission reduction estimates are based on the methodology set forth in the Environmental Protection Agency's ("EPA") regulations.[19] However, recent academic findings have demonstrated that observed methane emissions are about twice that reported in the EPA inventory.[20] In practice, the only reliable way to get an accurate assessment of methane emissions is to measure them. The need for accurate GHG accounting in the context of the CLCPA provides another reason why the Companies should promptly integrate Advanced Leak Detection technology into their business practices system-wide. A comprehensive and accurate measurement of methane emissions is crucial to establishing an accurate GHG baseline for the Companies and tracking their progress toward reducing emissions.

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[18] Future of Heat Panel, Case 19-G-0309 at pages 14-15.

[19] *Id.* at page 15 (citing 40 C.F.R. § 98, Subpart W).

[20] G. Plant *et al.*, Large Fugitive Methane Emissions from Urban Centers Along the US East Coast, https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635.

Q. How would the creation and establishment of ongoing, comprehensive, and accurate measurement of methane emissions assist the Companies in efforts to meet the goals of the CLCPA?

A. As further described in the testimony of Joe von Fisher, such a measurement system can provide more accurate and useful tools in the Companies' leak prioritization efforts. Combining advanced leak detection technology with traditional leak surveys offers utilities unique insight into their systems that is not possible using only traditional leak survey methods. Deploying these methods will also assist the Companies in ensuring a more accurate accounting of GHG emissions, consistent with the Act.[21] Moreover, as stated previously, it is in the best interest of the Companies to identify and quantify the largest sources of methane emissions in their service territories now to set an accurate GHG baseline because detection of such large sources at a comparatively late date could cast doubt on earlier-demonstrated achievements of GHG reductions and hamper the Companies' ability to have future emissions trajectories deemed to be in compliance with the requirements of the Act.

[21] For example, both Companies provide a high level estimate of potential methane emissions reductions over the next several years based on each respective company's current facilities assuming LPP retirement pursuant to each company's proposed program. Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 30; Gas Infrastructure and Operations Panel, Case 19-G-0310 at page 29.

VI. Prior Efforts to Develop this Holistic Perspective

Q. Is there any precedent for aligning a utility's business model with holistic GHG outcomes?

A. I am not aware of any full deployment of such an approach, but EDF previously introduced a similar concept in the Consolidated Edison collaborative that followed that utility's 2016 rate case, where EDF proposed it as the basis of an Earnings Adjustment Mechanism ("EAM"). In the collaborative, parties considered various ways to use an EAM to align Consolidated Edison's earnings opportunities with favorable GHG outcomes. Most of the discussion in the collaborative concerned a metric that would add up specifically identified ("targeted") emissions-reducing activities and reward Consolidated Edison for sufficient levels of such activities. EDF encouraged parties to consider a "broad" metric, as an alternative to this targeted approach. In the EAM context, the term "broad" provides a useful contrast to "targeted," but other than when referring to the EAM opportunities considered in the Consolidated Edison collaborative, the term "holistic" provides a better description of the nature of the analysis that I am recommending.

Q. Please describe what you meant by a "broad" metric in the EAM context.

A. In the context of an EAM, a "broad" GHG metric would assess measured, holistic, economy-wide GHG emissions outcomes in the utility service territory and reward the utility for the achievement of GHG emissions reductions above and beyond the historic trend and/or business-as-usual projections. This would automatically recognize GHG reductions of all types, including emissions reductions that result

from improved utilization of utility energy, emissions reductions that result from improved price signals, emissions reductions that result from reduced carbon intensity of the utility's energy resources themselves (for example, substituting RNG/biogas for conventional natural gas), emissions reductions that result from substituting usage of energy provided by a utility for more carbon-intensive practices not involving the utility at all (for example, electrification of vehicles, substituting natural gas for higher-emitting delivered fuels, or electrifying heating applications that would otherwise have been fossil fueled), as well as emissions reductions from non-energy sources (for example, capturing fugitive methane emissions from waste to produce biogas).

Q. Why would a "broad" metric be advantageous for evaluating greenhouse gas outcomes in an EAM?

A. While an EAM based on a "broad" metric would be more challenging than a targeted metric to design as the impact of exogenous factors like weather and the state of the economy would have to be isolated – and the proper time horizon might be longer than a typical rate case settlement plan – a "broad" metric could overcome the various limitations associated with a narrowly-defined targeted GHG reductions metric. Compared to the targeted approach, a "broad" metric would be much simpler for the utility and its regulator to calculate because it would have fewer elements that would have to be calculated solely for purposes of computing the metric. The shortcomings of a narrow approach to the EAM are also relevant to my reasons for recommending a holistic, emissions-measurement-based assessment in

connection with CLCPA consistency and non-interference. First, the narrow approach relies on "deemed" values and does not assess measured GHG emissions reductions, which could deviate materially from the values assumed before the fact. Additionally, such an approach fails to provide the utility with any incentive for ensuring that short-term actions are taken in a manner that is consistent with optimal long-term GHG performance. The narrow metric also misses GHG emissions reduction opportunities beyond the targeted set of activities, and does not account for interactive effects among activities, whether favorable or unfavorable with respect to emissions outcomes. It also fails to account for market effects (indirect effects that result from the activities that the utility is tracking) as well as any activities that are not necessarily known to the utility. For all these reasons, a metric based on a set of pre-selected activities, however carefully designed, in effect predetermines the utility's actions, and so fails to provide a significant incentive for innovation, including innovation that is needed to achieve the requirements of the CLCPA.

Q. Did the Collaborative recommend adoption of a "broad" GHG metric as the basis of an EAM?

A. No, largely for reasons of feasibility. As the Commission observed in the order approving the metrics proposed by the Collaborative participants in a Report, "[t]he Report…endorses the Broad GHG Reduction metric as a scorecard metric without associated financial incentives. The Broad GHG Reduction metric measures GHG reduction in New York City without attributing those reductions to specific

technologies or market activities in the Consolidated Edison service territory based on New York City's annual GHG Emissions Inventory. The Report notes that while the parties to the Collaborative agree that the Broad GHG Reduction metric has merit, this metric was not selected as an EAM for 2019 due to the complexities of developing both a service territory-wide inventory [footnote omitted] and targets to meaningfully measure achievement."[22]

Q. Please describe the complexities of developing a holistic, "broad" metric, which made it infeasible to deploy as an EAM in the short term.

A. The most immediate barriers were very concrete. Although a comprehensive GHG inventory exists for New York City, which is most of Consolidated Edison's service territory, there is no such inventory for Westchester. Additionally, as the 2018 Report from the Consolidated Edison Collaborative noted, the New York City GHG Inventory, which could provide the backbone for such a metric, generally lags too far behind to be useful for an EAM on the typical EAM schedule. Other complexities would come into play when establishing targets and earnings opportunities for a holistic EAM. Compared to a targeted GHG emissions reduction EAM, a broad GHG emissions EAM would constitute a far more significant departure from conventional performance-based regulation, and would require thoughtful consideration of how utility practices can contribute to large-scale

[22] Case 16-E-0060, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service*, Order Approving Recommendations of the 2018 Outcome-Based Earnings Adjustment Mechanisms Collaborative Report (April 25, 2019) at 4-5.

market outcomes. But in any event, a means to quantify and assess conformity with the Act is necessary, and developing those structures sooner (e.g., beginning immediately) will improve the Companies' ability to manage the full range of emissions reduction levers available to them and facilitate early emission reductions, advancing the goals of the Act.

Q. What is the current status of the "broad" metric as it relates to Consolidated Edison?

A. The 2018 Report of the EAM Collaborative contemplated that it would be reported as a Scorecard, and very basic data on GHG emissions from the stationary energy and transportation sectors in New York City for the Calendar Year 2017 were included in the 2018 EAM report, which was filed in April 2019.[23] However, I am unaware of any efforts to fill methodological gaps that make it infeasible to include the portion of the service territory that is in Westchester or to coordinate with the City of New York to expedite the collection and availability of pertinent data to address the time lag.

Q. Is this holistic approach to greenhouse gas emissions appropriate where, as here, the utility under consideration is a gas-only utility, rather than a combination utility (as Consolidated Edison is in most of its service territory)?

[23] Case 16-E-0060, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service*, Consolidated Edison 2018 Energy Efficiency Earnings Adjustment Mechanism Achievement Report (April 1, 2019) at 11.

A. Yes. First, the holistic approach accurately aligns with the exercise of giving the Commission the information it needs to determine whether the utility's proposed business plans are consistent with and will not interfere with achievement of the state's statutory GHG emissions limits. By contrast, any evidentiary showing limited to the operation of the natural gas system would provide little or no basis for making any such determination. Moreover, even gas utilities have an opportunity to contribute – favorably or unfavorably – to GHG emissions outcomes from sectors other than their own (including vehicular emissions and emissions from heating buildings that are currently heated using delivered fuels). A holistic evaluation of service territory emissions properly gives the utility a reason to be mindful of such opportunities to influence those other sectors when developing long-term plans, investing in infrastructure, and generally operating its business. That said, it goes without saying that, compared with a combination utility, a gas-only utility has fewer levers to pull that affect economy-wide GHG emissions. That may be an important consideration when setting targeted levels of economy-wide GHG emissions reductions for purposes of providing a utility with an incentive in the form of additional earnings opportunity. However, that consideration is not particularly relevant where, as here, the question is not whether the utility should be entitled to additional earnings, but instead concerns consistency (or lack thereof) with State GHG emissions limits that are themselves not specific to the natural gas system.

Q. How does your prior thinking about the "broad metric," and the Consolidated Edison collaborative experience, inform your recommendations about the holistic approach to providing evidence for a determination of consistency and non-interference in the context of the CLCPA?

A. My prior experience suggests that although the reasons for taking a holistic approach are compelling, the practical barriers to doing so are substantial. There is no time to lose in developing detailed, accurate, and timely GHG emissions inventories for individual service territories, to assess consistency with goals when the relevant deadlines are upon us and to provide a current baseline and a holistic way to measure progress in reducing GHG emissions, and for the Companies to identify and implement practices that facilitate such progress and be recognized for their achievements in this area by their regulator. These inventories have to be performed on a recurring basis and cannot be excessively lagging. Although the utilities themselves will need to provide this information to the Commission to support Commission approval of their plans, the utilities are not, presently, especially well situated to create economy-wide emissions inventories. They will need to work with local and state government entities to create these inventories as well as with electric utilities in their gas service area. The gas utilities should start with their systems and get accurate emissions inventories associated with delivery and consumption of natural gas in their service territory while beginning to work with local and state government entities and electric utilities on the other components of the inventory. The CLCPA directs the Department of

Environmental Conservation to issue a statewide GHG emissions report within two years and each year thereafter.[24] The Act also provides guidance on how to develop GHG inventories for this report.[25] In developing GHG emissions data for their geographic footprints, utilities should follow the same guidance to ensure that their methodologies are consistent with those being used to develop the statewide greenhouse gas emissions inventory and report.

Q. Does this conclude your testimony?

A. Yes.

[24] *See* Environmental Conservation Law § 75-0105(1).

[25] *See* Environmental Conservation Law § 75-0105.

A.L.J. COSTELLO: And that is it for the Environmental Defense Fund. And I just want to make sure that we have everybody.

MS. MURPHY: Yes. Thank you, Your Honor.

A.L.J. COSTELLO: Okay. Okay. We'll move next to the Long Island Power Authority. We have one affidavit. It's a two-page affidavit by Rick Shansky, sworn to on February 4, 2020. And that will be Exhibit 716. And at this point the court reporter should insert the file marked LIPA Rick Shansky, direct testimony. #

BEFORE THE NEW YORK STATE
DEPARTMENT OF PUBLIC SERVICE

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.	Case Number: 19-G-0310

DIRECT TESTIMONY OF

RICK SHANSKY

LONG ISLAND POWER AUTHORITY

Dated: August 30, 2019

Q. PLEASE STATE YOUR NAME, TITLE AND BUSINESS ADDRESS.

A. Rick Shansky, Vice President of Operations Oversight, Long Island Power Authority ("Authority"), 333 Earle Ovington Boulevard, Suite 403, Uniondale, New York 11553.

Q. WHAT ARE YOUR RESPONSIBILITIES AT THE AUTHORITY?

A. I direct the Authority's oversight of its primary contractor PSEG Long Island ("PSEG-LI"), as well as its affiliate that performs day-to-day power and fuel procurement. I am also responsible for managing the Authority's participation in wholesale power markets.

Q. PLEASE SUMMARIZE YOUR EDUCATIONAL BACKGROUND AND PROFESSIONAL EXPERIENCE.

A. I have a Bachelor of Science in Electrical Engineering from Rensselaer Polytechnic Institute and a Master of Science in Energy Management from NY Institute of Technology. I am licensed as a Professional Engineer in the State of New York. I have more than 35 years of experience in the electric utility industry, and previously held positions at Consolidated Edison Company of New York ("Con Edison") and the Long Island Lighting Company ("LILCO") in the areas of energy management, resource planning, fuel and purchased power, and generation planning. I joined the Authority in

2008, and held management positions in the Power Markets Department before assuming my current function in 2014.

Q. HAVE YOU PREVIOUSLY TESTIFIED IN RATE PROCEEDINGS IN NEW YORK STATE?

A. Yes. I testified for the Authority in Department of Public Service Matter 15-00262. Prior to that, I testified as a witness for Con Edison in Public Service Commission ("PSC") Cases 07-S-1315, 05-S-1376, 03-S-1672, 99-S-1621, and 94-E-0334.

Q. ARE YOU SPONSORING ANY EXHIBITS IN THIS MATTER?

A. Yes. I am sponsoring Exhibit __(RS-1), which is KEDLI's response to Information Request LIPA-1.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The Authority is requesting that KeySpan Gas East Corp. d/b/a National Grid ("KEDLI") eliminate its Value Added Charge ("VAC") and Minimum Bill Obligation ("MBO") elements from its charges for services pursuant to SC-14, transportation service to electric generators.

Q. PLEASE EXPLAIN THE AUTHORITY'S INTEREST IN THIS MATTER.

A. The Authority provides electric service to 1.1 million customers in Nassau and Suffolk Counties and the Rockaway Peninsula. Prior to 1998, LILCO was a

combined electric and gas utility that owned electric generating plants on Long Island, and also provided gas service to those plants. The Authority is the successor in ownership to LILCO's electric system. National Grid Generation, LLC is the successor in ownership to the legacy generating plants, and KEDLI is the gas local distribution company ("LDC") on Long Island. The Authority is responsible for providing the fuel used by many of the electric generators under contract to it, and all of the natural gas used by those plants is delivered by KEDLI. As a public power utility, the Authority recovers all its costs from its customers without margin or profit. What the Authority pays to KEDLI for local gas transportation service is a part of the cost of electric service to the Authority's customers.

The Authority is the largest gas transportation customer of KEDLI on Long Island. For the twelve months ending May 2019, KEDLI delivered 53 million dths of interruptible gas on behalf of the Authority at a total charge of $19 million.[1] Only 54% of that charge represented the direct cost of transportation service and balancing. The remaining 46% was comprised of VAC and MBO.

[1] Includes service to Northport at a negotiated rate as of January 2019.

As evidenced by KEDLI responses to questions 2 and 3 in Exhibit __ (RS-1), the cost justification for those charges is currently unsupportable.

Q. HAS THIS MATTER NOT BEEN ADDRESSED PREVIOUSLY BY THE COMMISSION?

A. Yes, the matter has been addressed, but not resolved. There is a pending generic case that has not seen any activity for quite some time. Prior to the start of the generic inquiry, there were predecessor proceedings as well.

In December 2016, the Commission adopted the terms of the Joint Proposal submitted in KEDLI's last rate case (Cases 16-G-0058 et al.) that established a three-year rate plan, and provided for a two-phase collaborative to consider power generation matters. Phase Two of the power generation collaborative culminated in the Phase Two Report.[2] The Phase Two Report, submitted by KEDLI, stated, in pertinent part, that "further actions related to value-based and cost-based rate design for power generators should be tabled until Staff releases its White Paper in the Generic Generator Proceeding, at which point the collaborative parties will reconvene to determine how best to

[2] Cases 16-G-0058 et al., *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, Report of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid Concerning Phase Two of The Power Generation Collaborative (Jan. 9, 2018).

proceed . . . ."

As the Authority explained in its comments on the Phase Two Report, the Authority's ratepayers are unduly burdened by the VAC.[3] The Authority recommended that the Commission relieve that burden by immediately eliminating the VAC for power generators. The Commission declined to adopt the Authority's recommendation, stating that such a change may be feasible only in concert with changes to other aspects of rate design for natural gas service to electric generators.[4]

Q. HAVE THERE BEEN OTHER PROCEDURAL ATTEMPTS TO ADDRESS VAC FOR ELECTRIC POWER GENERATORS?

A. Yes. In July 2015, the Commission instituted a proceeding to re-examine the VAC.[5] In November 2015, parties submitted comments demonstrating that the Commission should remove the VAC from LDC tariffs. After reviewing the comments, Staff determined "that a single component of delivery service

[3] Cases 16-G-0058 et al., *supra*, LIPA Comments (Apr. 23, 2018), at 3.

[4] Cases 16-G-0058 et al., *supra*, Order Adopting Tariff Amendments Related to the Power Generation Collaborative Phase Two Report (June 18, 2019), at 7–8.

[5] *See generally* Case 15-G-0469, *Proceeding on Motion of the Commission to Review the Appropriateness of the Value Added Charge for Natural Gas Utilities of New York State*.

to electric generators should not be evaluated in a vacuum," and that all the components of the rate need to be analyzed.[6]

As Staff noted in the Report, most generators taking gas delivery service from the LDCs have not been exposed to the basic rate because they have taken service under negotiated contracts.

Q. WHY ARE YOU REQUESTNG ACTION ON VAC AND MBO NOW?

A. While it is understandable that the Commission would like to examine the whole picture, that examination has lagged and electricity customers are continuing to pay excessive rates for interruptible local transportation service. Moreover, assessing the true cost of such service and making any required adjustments to the components of SC-14 rates is best accomplished in individual utility rate cases. With respect to VAC, it appears that the Authority is one of the few LDC customers that actually pays the VAC, while most others are exempted from it by contractual arrangements with their LDC. As demonstrated by the comments in Case 17-G-0011, the Authority is one of a handful of generation customers that is charged a VAC at all. For example, Central Hudson Gas and Electric Corp. reported that while the VAC is in its

[6] Case 17-G-0011, *In the Matter of a Review of Tariff Provisions Regarding Natural Gas Service to Electric Generators*, Staff Initial Findings Report and Additional Questions (Jan. 19, 2017).

tariff, it does not provide transportation service to any generator under the tariff. Further, its individually negotiated contracts do not charge the VAC. Similarly, Con Edison and Orange and Rockland advocated for the elimination of the VAC. Neither of these companies has ever charged the VAC. In sum, the Authority appears to be one of a handful of customers that is subject to the VAC. The fact that the Authority is a public power entity and does not operate for profit is all the more reason to remove the VAC. Notably, in the pending but quiescent generic case, no party argued that the VAC should be retained. Even KEDLI proposed that the VAC should be simplified.

While the VAC was originally conceived as a means to share a portion of a generator's profits from increasing wholesale electricity prices with gas ratepayers, the Authority does not earn profits from its sale of electricity in the wholesale market, but instead passes on all such benefits and costs to its customers. Thus, the VAC serves merely as a tax on the Authority's customers—many of whom receive no corresponding benefit in their gas bills because they do not have access to KEDLI's natural gas service. The VAC is also an element in the Authority's bids for electricity sold to the New York Independent System Operator ("NYISO"), and can therefore influence the wholesale price of electricity. As the VAC by design is unrelated to the cost

of gas transportation service, it has the potential to introduce a distortion into the wholesale electricity markets. For all of the above reasons, the Commission should eliminate the VAC from KEDLI's rates without further delay. In the event the Commission declines to eliminate the VAC entirely, it should exempt non-profit customers from it.

Q. WHY SHOULD MBO BE ADDRESSED IN THIS PROCEEDING?

A. MBO associated with interruptible service is a significant portion of the total charges paid by the Authority for that service. In the 12 months ending May 2019, the MBO was over $7 million. In theory, MBO is meant to recover fixed costs associated with low volume use of KEDLI's system. Yet, KEDLI has not demonstrated that it incurs any fixed costs to support interruptible service, inasmuch as KEDLI's system design excludes service to interruptible generation loads.[7] This rate case is the best forum for assessing the fixed cost of interruptible service, if any, and determining the proper replacement for MBO.

Q. DOES THIS COMPLETE YOUR PRE-FILED DIRECT TESTIMONY?

A. Yes.

[7] *See, e.g.*, Exhibit __ (RS-1), question 1. Arguably, KEDLI incurs variable operation and maintenance expenses to support interruptible transportation service, though such costs have not been presented in KEDLI's direct case.

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas

A.L.J. COSTELLO: We'll now move onto the Sane Energy Project. We have a two-page affidavit by -- and I'm -- apologize if I don't say your names correctly -- Lee Diesche. Is that --

MS. DIESCHE: Diesche.

A.L.J. COSTELLO: Diesche? Okay. Lee Diesche. Two-page affidavit marked -- I'm sorry, sworn to February 19, 2020. And that will be marked as Exhibit 717. And we have a two-page affidavit by Kim Fraczek.

MS. FRACZEK: Fraczek.

A.L.J. COSTELLO: Fraczek -- I apologize. Sworn to on February 19, 2020. And that will be marked as Exhibit 718. And at this point in the record, the court reporter should insert the following files: Sane Energy, Kim Fraczek and Lee Diesche, direct testimony; Sane Energy, Kim Fraczek and Lee Diesche, supplemental testimony. #

BEFORE THE NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations Of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	**Case 19-G-0310**

DIRECT TESTIMONY OF KIM FRACZEK AND LEE ZIESCHE ON BEHALF OF SANE ENERGY PROJECT

August 30, 2019

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III. PROCEDURAL RECOMMENDATION: PUBLIC PARTICIPATION

I. INTRODUCTIONS AND BACKGROUND

Q.Please state your name and business address.

A. Kim Fraczek and Lee Ziesche. 232 E 11th Street, New York, NY 10003

Q. What is your occupation?

A:(FRACZEK) Director of Sane Energy Project
(ZIESCHE) Community Engagement Coordinator of Sane Energy Project

Q.Have you previously testified before the Commission?

A:Yes. We recently provided direct testimony for the Con Edison rate case. We have also provided written comments and attended multiple PSC public hearings on numerous PSC cases and have helped New Yorkers understand the public commenting processes and submit thousands of comments on numerous cases before the PSC on energy issues.

Q. What is Sane Energy Project

A. Sane Energy Project is a non-profit organization representing 8,500 New Yorkers. We are based in New York City, and work with communities across New York State to stop the buildout of fracked gas infrastructure and hasten the just transition to renewable energy. Since 2011, we have worked directly with scientists, health professionals, and impacted communities across the state to raise awareness about the health and climate impacts of fracked gas infrastructure, and we work with communities to identify and remove barriers to community-owned renewable energy. We've worked with thousands of New Yorkers to prepare and provide public comment and participate in the regulatory process for over a dozen fracked gas infrastructure projects across New York.

Q.Please summarize your background.

A:(FRACZEK) I have been an employee of Sane Energy Project since 2014. Prior to this, was a volunteer coordinator for Sane Energy Project since its inception in 2011. My background is 12 years in corporate creative direction and digital rapid prototyping leading teams of designers in New York, USA, Tokyo, Japan, Hong Kong and Visakhapatnam, India in the fashion industry, to produce products for the American Market. I began volunteering my time to remedy water quality issues with people in Pennsylvania, where my family lives, and built a non-profit organization to focus on this work full-time. I have traveled extensively in Pennsylvania and

New York to organize with communities, scientists, medical doctors and elected officials to change laws that are lenient toward the fossil fuel industry that compromises human health, wildlife health and air and water quality detriment.

(ZIESCHE) I have a bachelors in Telecommunications from the University of Florida School of Journalism with a minor in International Development and Humanitarian Assistance. I am currently in post-production for my full length documentary directorial debut. For three and a half years I was employed as the Grassroots Coordinator for International WOW Company's award winning GASLAND films. In that time I traveled around the world to communities being impacted by fracking and fracked gas infrastructure with a focus on communities in Pennsylvania and New York and often organized events with leading experts on the health and climate impacts of shale gas development. Since 2017 I've been working as the Community Engagement Coordinator for Sane Energy Project. During that time I've continued to travel to communities in New York and Pennsylvania being impacted by fracked gas infrastructure and produced over a dozen short documentary films telling their stories.

Q: What is your purpose in submitting this testimony?

A: The purpose of the testimony is to address the failure of the Company to present a proposal that takes into account the unique climate and health impacts of fracked gas and the growing public demand to get off fossil fuels. The Company is proposing to spend millions of ratepayer dollars on Storm Hardening while at the same time investing hundreds of millions of dollars on fossil fuel infrastructure that will only make future storms stronger by contributing to climate change.

II. GAS ISSUES

A. Introduction and Background

Q: What will you address relating to the Company's business activities?

A. We have reviewed the Company's policy overview of its gas business activities, and this testimony addresses four issues relating to these activities.

First, we oppose the Company's proposals to spend hundreds of millions of ratepayer dollars on gas capital investments to replace Leak Prone Pipeline (LPP) instead of retire it and convert customers from oil to gas because these proposals conflict with state and city climate goals when the unique climate impacts of fracked gas are taken into account.

Second, we oppose the Company's proposal to invest hundreds of million of ratepayer dollars to extend the life of LNG and CNG facilities.

Third, we'd like to address that National Grid's dependence on the Williams NESE Pipeline shows a deep disregard for public demand around energy and incorrect calculations of fracked gas greenhouse emissions.

Fourth, taking into account the unique health impacts of shale gas we request funds be used for radon testing to insure the safety of the Company's service.

Q. What is the overarching context in which the Company's gas business activities must be evaluated?

A.Shale gas production in the Appalachia region has increased rapidly since 2012 meaning an increasing amount of the gas supply coming into New York is extracted from the Marcellus Shale using high volume hydraulic fracking. Nationally, by 2016, 69%[1] of all wells drilled in this country were fracked and that number has continued to rise. The proposed Williams NESE Pipeline that much of the Company's rate case proposal supports would be transporting fracked gas from Pennsylvania. And in her Direct Testimony[2] Elizabeth D. Arangio confirmed more of the Company's gas will be fracked gas saying "as supplies from the Marcellus shale region became abundant and readily accessible, the Companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada." Because of the extreme method used to develop gas from the Marcellus Shale, fracked gas has greater environmental impacts for the climate and on human health[3] than conventional gas and must be evaluated differently.

B. Gas Proposals Conflict with City & State Greenhouse Gas Reduction Goals

Q.What kind of gas capital expenses is the Company proposing?

A.The company is proposing to replace and expand Leak Prone Pipeline (LPP) and other pipeline through its Proactive Main Replacement Program, Gas System Reinforcement Program

[1] See EIA *Hydraulically fractured horizontal wells account for most new oil and natural gas wells* https://www.eia.gov/todayinenergy/detail.php?id=33972

[2] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)

[3] The *Compendium of Scientific, Medical, and Media Findings Demonstrating Risks and Harms of Fracking (*the Compendium*)* is a fully referenced compilation of the evidence outlining the risks and harms of fracking. https://concernedhealthny.org/compendium/

and Metropolitan Reliability Infrastructure (MRI) Program. In its Future of Heat Panel testimony the company says it will replace 225 miles per year for a total of 930 miles over four years. In its Future of Heat Panel testimony the Company is also proposing to increase gas demand by converting oil customers to fracked gas through its Customer Connections Programs adding 7,000 new customers. The Company claims both these programs will help them meet New York State greenhouse gas reduction goals and their own methane reduction goals of 60% by 2035.

Q.What are New York State greenhouse gas reduction goals?

A:The New York State Energy plan sets a 40% greenhouse gas reduction target for 2030 and 80% reduction by 2050. The recently passed and signed into law Climate Leadership and Community Protection Act (CLCPA) also requires 40 percent emissions reduction by 2030 and an increased requirement of 85 percent emissions reductions by 2050.

Q:What are the city's goals greenhouse gas emission reduction goals?

A: Intro 1253 passed by the New York City Council requires buildings over 25,000 square feet to reduce greenhouse gas emissions by 40% by 2030 and 80% by 2050.

Q: Can the Company expand gas capacity and meet state and city greenhouse gas emissions reduction goals?

A.No. When the full life cycle of methane leakage is taken into account, gas expansion goes against city and state goals to reduce greenhouse gas emissions. Methane, the main component of gas is a powerful greenhouse gas that is 86-106 times more potent a warming agent than C02 for the first 20 years it's in the atmosphere. FLIR footage has shown high leakage rates of methane at the wellhead and all along the transportation route at compressor and metering stations. Cornell scientist Robert Howarth estimates that when looking at upstream and downstream emissions from shale gas, leakage rates range between 3.8% and 12%[4]. Even if upstream emissions were not taken into account Howarth estimates the downstream emissions associated with storing gas and delivering shale gas to market to be 2.5%. With leakage rates like this it is impossible to meet city and state goals to reduce greenhouse gas emissions and is more likely taking us in the wrong direction by increasing emissions.

[4] See Robert W Howarth *Methane emissions and climatic warming risk from hydraulic fracturing and shale gas development: implications for policy* https://www.eeb.cornell.edu/howarth/publications/f_EECT-61539-perspectives-on-air-emissions-of-methane-and-climatic-warmin_100815_27470.pdf

Q.How are the Company's assertions about meeting greenhouse gas emission goals inadequate?

A. When it comes to oil to gas conversions through the Customer Connections Program, the company primarily bases its claims of reducing greenhouse gas emissions on the fact that when burned, gas emits less CO2 than oil, but ignores upstream and downstream methane leakage. In doing so they are presenting an incomplete and misleading representation of greenhouse gas emissions associated with their projects and their environmental impact. Additional when talking about reducing greenhouse gas emissions the Company proposes projects to reduce methane leaks[5] within their system while at the same time expanding gas capacity and extending the overall life use of pipelines through replacements. Expanding and extending gas capacity will lead to more fracking and fracking infrastructure needed to transport the gas, thus increasing overall methane emissions.

Q: Does the Company's project proposals to limit methane address the issue?

A: No. While identifying and fixing leaks is important, the technologies proposed to identify leaks do not address the upstream emissions especially when paired with pipeline and infrastructure enhancements that will overall increase gas use.

Q:What should the Company do instead of investing in gas capital project to replace pipeline and convert customers?

A:The Company should not invest any new ratepayer funds in expanding gas capacity and extending the life of pipelines and should instead invest in gas demand reduction and beneficial electrification. The company should look at retiring LPP instead of replacing it and convert customers to non-gas alternatives.

C. LNG & CNG Reliability

Q:What is the company proposing?

A: The company is proposing the following projects as investments to create LNG and CNG reliability: Compressed Natural Gas Special Projects, LNG Blanket, LNG – Special Projects,LNG TANK 1 and Tank 2 Upgrade Projects, LNG - Vaporizers 7 & 8 Replacement and LNG - Salt Water Pump House.

[5] Future of Heat Panel

Q:Do you support this investment? Why or why not?

A: No. As stated in above testimony the increased dependency and expansion of fracked gas goes against New York policy to reduce greenhouse gas emissions. We need to plan for shutting these facilities down rather than investing hundreds of millions of dollars on infrastructure that will increase our reliance on dirty fossil fuels.

Q: In addition to climate impacts are there other reasons the Company should plan on closing LNG & CNG facilities instead of investing in them?

A: Dangerous to surrounding communities and workers. LNG is highly volatile and explosive, and notably New York City saw a disaster that killed 40 workers[6].

Q: What do you recommend?

A: The Company should not invest any new funds that would extend the use of LNG and CNG facilities and should instead invest in gas demand reduction.

D.Williams NESE Pipeline

Q:Does the Company propose projects connected to the approval of the Williams Northeast Supply Enhancement Pipeline?

A: Yes. According to the Company the proposed Marine Park Regulator Station Project is required to "utilize the increased capacity from the Transco Northeast Supply Enhancement project."[7] The Company also claims in its testimony that the pipeline is critical to meet increase demand in its service territory and it would suspend its customer connections programs if the pipeline is not approved[8].

Q:Do you support investments connected to the Williams Northeast Supply Enhancement Pipeline? Why or why not?

A: No. As mentioned in above testimony, climate science and city and state law require a dramatic reduction of greenhouse gases and it's impossible to lower greenhouse gas emissions and expand fracked gas infrastructure at the same time. Additionally New Yorkers, including many of the Company's ratepayers, have overwhelming called on New York State to reject

[6] https://www.silive.com/news/2013/02/40_years_ago_today_staten_isla.html
[7] Exhibit ___ (GIOP-5) Page 90 of 120
[8] Gas Infrastructure and Operations Panel - Pages 118 & 119

Williams Transco's application to build the Northeast Supply Enhancement (NESE) project. In their temporary denial of the project in May the New York Department of Environmental Conversation said 14,000 public comments were submitted on the project and over 90% were opposed to the pipeline. Since then New Yorkers have submitted an additional 25,000 public comments opposing NESE on the pipeline's reapplication and a recent poll[9] found the more New Yorkers learn about the pipeline, the more likely they are to oppose, and in fact by a margin of nearly 5-to-1, New Yorkers would rather have the state expand clean energy solutions over dirty fossil fuels. The Company's continued push for NESE project goes against what New Yorkers ratepayers are demanding and against city and state climate goals. Additionally, according to a study[10] done by PSE Health Energy for Earthworks the NESE project would cause an estimated 7.8 million metric tons of CO2 equivalent per year (in their mid-range modelling scenario). New York City as a whole produces about 50 million metric tons of CO2 equivalent. This pipeline alone would produce so much climate pollution that it would be equivalent to roughly 15% of New York City's greenhouse gas emissions taken as a whole or roughly about 4% of the entire State of New York's greenhouse gas emissions. The Company's claims that the NESE project and related investments in gas expansion would lower greenhouse gas emissions and meet state and city goals is simply false.

D: Need for Radon Testing

Q: What about the Company's proposal that fracked gas expansion has "environmental benefits" is inaccurate?

A: The Company's claim their Customer Gas Connections provides "environmental benefits"[11] and reduces local emission fails to address the unique health threats that come with gas fracked from the Marcellus Shale. The Marcellus Shale in northeastern Pennsylvania has higher than average levels of uranium and thorium. These elements break down to form radium-226. Signs of radon in gas & other fracked gas chemicals in our homes, have been on the rise since 2004[12] when high-volume hydraulic fractured drilling began. Radon is the leading cause of lung cancer in non-smokers in the United States[13].

Q What is your recommendation?

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[9] https://www.nrdc.org/experts/rich-schrader/new-poll-new-yorkers-reject-williams-pipeline
[10] https://www.psehealthyenergy.org/wp-content/uploads/2018/02/NY.Pipelines-1.pdf
[11] Exhibit ___ (GIOP-5) Page 1 of 120
[12] Increased Levels of Radon in Pennsylvania Homes Correspond to Onset of Fracking Johns Hopkins Bloomberg School of Public Health https://www.jhsph.edu/news/news-releases/2015/increased-levels-of-radon-in-pennsylvania-homes-correspond-to-onset-of-fracking.html
[13] https://www.cancer.org/cancer/cancer-causes/radiation-exposure/radon.html

In order for the Company to comply with its requirement to provide safe service we recommend the Company provide EPA scientific radon monitoring[14] in all gas supplied to customers. Additionally, we recommend meeting with New York elected officials who drafted the 2014 Radon Bill[15] to monitor and mitigate radon in our gas supply.

III. PROCEDURAL RECOMMENDATION: PUBLIC PARTICIPATION

Q:What barriers have you experienced in participating in this proceeding?

A: As two part-time employees working for an organization, it has been very difficult for Sane Energy Project to have the capacity to draft our own testimony and financially impossible for us to hire our own experts. Additionally the public we work with has had no opportunity to bring evidence to the table or ask questions of the Company.

Q:What is your procedural recommendation?

A: We recommend Intervenor Funds available for the public and since it is our rate payer dollars National Grid is proposing to spend, the PSC should require utilities to fund the public to hire our own experts. We also recommend multiple public hearings in locations where energy burdens are greatest earlier in the process so members of the public can provide evidence before settlement negotiations begin. We also recommend discovery be available to all members of the public not just individuals who have the capacity to intervene as a party. The climate crisis and unwanted expansion of fracked gas infrastructure have made New Yorkers more aware of and wanting to shape the energy system of New York than ever before.

[14] https://www.epa.gov/radon
[15] https://nyassembly.gov/leg/?default_fld=&bn=A06863&term=2013&Summary=Y&Text=Y

BEFORE THE NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

Case 19-G-0309

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations Of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

Case 19-G-0310

SUPPLEMENTAL TESTIMONY OF KIM FRACZEK AND LEE ZIESCHE ON BEHALF OF SANE ENERGY PROJECT

January 29, 2020

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C. LNG & CNG
D. Clove Lakes Transmission Loop Project
E. Riverhead Transmission and Southeast Suffolk Projects
F. Leak Prone Pipe
G. New Research On Radon Threat

I. INTRODUCTIONS AND BACKGROUND

Q.Please state your name and business address.

A. Kim Fraczek and Lee Ziesche. 140 2nd Avenue, New York, NY 10003

Q. What is your occupation?

A:(FRACZEK) Director of Sane Energy Project
(ZIESCHE) Community Engagement Coordinator of Sane Energy Project

Q.Have you previously testified before the Commission?

A:Yes. We recently provided direct testimony for the Con Edison rate case. We have also provided written comments and attended multiple PSC public hearings on numerous PSC cases and have helped New Yorkers understand the public commenting processes and submit thousands of comments on numerous cases before the PSC on energy issues.

Q. What is Sane Energy Project

A. Sane Energy Project is a non-profit organization representing 8,500 New Yorkers. We are based in New York City, and work with communities across New York State to stop the buildout of fracked gas infrastructure and hasten the just transition to renewable energy. Since 2011, we have worked directly with scientists, health professionals, and impacted communities across the state to raise awareness about the health and climate impacts of fracked gas infrastructure, and we work with communities to identify and remove barriers to community-owned renewable energy. We've worked with thousands of New Yorkers to prepare and provide public comment and participate in the regulatory process for over a dozen fracked gas infrastructure projects across New York.

Q.Please summarize your background.

A:(FRACZEK) I have been an employee of Sane Energy Project since 2014. Prior to this, was a volunteer coordinator for Sane Energy Project since its inception in 2011. My background is 12 years in corporate creative direction and digital rapid prototyping leading teams of designers in New York, USA, Tokyo, Japan, Hong Kong and Visakhapatnam, India in the fashion industry, to produce products for the American Market. I began volunteering my time to remedy water quality issues with people in Pennsylvania, where my family lives, and built a non-profit

organization to focus on this work full-time. I have traveled extensively in Pennsylvania and New York to organize with communities, scientists, medical doctors and elected officials to change laws that are lenient toward the fossil fuel industry that compromises human health, wildlife health and air and water quality detriment.

(ZIESCHE) I have a bachelors in Telecommunications from the University of Florida School of Journalism with a minor in International Development and Humanitarian Assistance. I am currently in post-production for my full length documentary directorial debut. For three and a half years I was employed as the Grassroots Coordinator for International WOW Company's award winning GASLAND films. In that time I traveled around the world to communities being impacted by fracking and fracked gas infrastructure with a focus on communities in Pennsylvania and New York and often organized events with leading experts on the health and climate impacts of shale gas development. Since 2017 I've been working as the Community Engagement Coordinator for Sane Energy Project. During that time I've continued to travel to communities in New York and Pennsylvania being impacted by fracked gas infrastructure and produced over a dozen short documentary films telling their stories.

Q: What is your purpose in submitting this testimony?

A: The purpose of the testimony is to address the Company's Second Supplemental Testimony as it relates to Case 19-G-0678 and recent developments regarding implementation of state climate law and research on radioactivity in gas.

Q: Does this testimony contain exhibit?

Yes.

Exhibit_Sane 1
Exhibit_Sane 2
Exhibit_Sane 3

II. GAS ISSUES

A. Introduction and Background

Q: What will you address relating to the Company's business activities?

A. We have reviewed the Company's Second Supplemental Testimony, and this testimony addresses six issues relating to the Company's business activities.

First, we oppose the Company's proposal to accelerate construction of the Metropolitan Reliability Infrastructure (MRI) project and oppose the Company's request to raise rates to finish the project.

Second, we oppose the Company's proposal to expand the use of portable compressed natural gas (CNG) and any new investments to extend the life of CNG and liquified natural gas (LNG) facilities.

Third, we oppose ratepayer dollars being spent on the Clove Lakes Transmission Loop Project.

Fourth, we oppose ratepayer dollars being spent on the Riverhead Transmission and Southeast Suffolk Phase 2 projects.

Fifth, despite a small reduction in cost we still oppose the Company's proposals to spend hundreds of millions of ratepayer dollars on gas capital investments to replace Leak Prone Pipeline (LPP).

Sixth, taking into account new groundbreaking reporting in Rolling Stone we request the Company reconsider their position not to use funds for radon testing to ensure the safety of the Company's service.

Q. What is the overarching context in which the Company's gas business activities must be evaluated?

A.Shale gas production in the Appalachia region has increased rapidly since 2012 meaning an increasing amount of the gas supply coming into New York is extracted from the Marcellus Shale using high volume hydraulic fracking. Nationally, by 2016, 69%[1] of all wells drilled in this country were fracked and that number has continued to rise. The proposed Williams NESE Pipeline that much of the Company's rate case proposal supports would be transporting fracked gas from Pennsylvania. And in her Direct Testimony[2] Elizabeth D. Arangio confirmed more of

[1] See EIA *Hydraulically fractured horizontal wells account for most new oil and natural gas wells* https://www.eia.gov/todayinenergy/detail.php?id=33972

[2] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)

the Company's gas will be fracked gas saying "as supplies from the Marcellus shale region became abundant and readily accessible, the Companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada." Because of the extreme method used to develop gas from the Marcellus Shale, fracked gas has greater environmental impacts for the climate and on human health[3] than conventional gas and must be evaluated differently.

B. Metropolitan Reliability Infrastructure

Q.What did the Company propose regarding the MRI project in their Second Supplemental Testimony?

A: In their Second Supplemental Testimony the Company asserts "KEDNY's MRI Reliability project must be accelerated by one year to provide the hydraulic system support that is critical for enabling new customer connections over the next few winters".[4] The Company is proposing to spend an additional 184,670,417 ratepayer dollars to finish the pipeline.

Q.Do you support this proposal? Why or why not?

A: No we do not support this proposal because any investment to expand or extend the life of fracked gas infrastructure goes against climate science as well as city and state climate law to dramatically reduce greenhouse gas emissions. In response to SANE IR-4 the Company says "An assessment of greenhouse gas emissions was not done for this project" [5] and when asked if upstream methane emissions were taken into account the Company said "National Grid cannot control or measure upstream emissions on facilities it does not own and/or operate" yet the Climate Leadership and Community Protection Act (CLCPA) which Governor Andrew Cuomo signed into law on July 18, 2019 requires State agencies to take into account the full lifecycle of methane emissions which the Company confirms they are not doing.

As Sane Energy highlighted in our Direct Testimony "Cornell scientist Robert Howarth estimates that when looking at upstream and downstream emissions from shale gas, leakage rates range between 3.8% and 12%[6]. Even if upstream emissions were not taken into account Howarth

[3] The *Compendium of Scientific, Medical, and Media Findings Demonstrating Risks and Harms of Fracking (*the Compendium*)* is a fully referenced compilation of the evidence outlining the risks and harms of fracking. https://concernedhealthny.org/compendium/

[4] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

[5] Exhibit_Sane 1

[6] See Robert W Howarth *Methane emissions and climatic warming risk from hydraulic fracturing and shale gas development: implications for policy*

estimates the downstream emissions associated with storing gas and delivering shale gas to market to be 2.5%. With leakage rates like this it is impossible to meet city and state goals to reduce greenhouse gas emissions and is more likely taking us in the wrong direction by increasing emissions." [7]

Q: Have there been any developments since August that make Howarth's research more relevant to this rate case? If so, what?

A: Yes, Robert Howarth has been named to the Climate Action Council which will advise on the implementation of the CLCPA.

Q. Have local elected officials made statements on the MRI project? If so, have they been in support or opposition?

Yes, State Senator Julia Salazar, Assemblymember Joe Lentol, City Council Member Antonio Reynoso, City Council Member Rafael Espinal, all local elected officials who have the pipeline within their district, as well as citywide elected officials Comptroller Scott Stringer and Public Advocate Jumaane Williams have made statements in opposition to the pipeline.

Q. Have community members made statements on the MRI project? If so, have they been in support or opposition?

As reported in the BKLYNR, Brooklyn Daily Eagle and Brooklyn Paper community members expressed opposition to the pipeline on January 14 at a Brooklyn Community Board -1 meeting. Community members specifically expressed safety and climate concerns as well as the Company's failure to do outreach on the project as reasons for opposition[8]. BKLYNR reported that "audience members were altogether dissatisfied with the responses from National Grid....They wanted to know why they weren't being heard, and why National Grid refused to give the community a platform in the decision process for this kind of project."[9] and Brooklyn

https://www.eeb.cornell.edu/howarth/publications/f_EECT-61539-perspectives-on-air-emissions-of-methane-and-climatic-warmin_100815_27470.pdf

[7] Sane Energy Direct Testimony
[8] Exhibit_Sane 2
[9] 'Public Utilities Under Public Control!' — North Brooklyn National Grid Pipeline Not Popular With Locals https://bklyner.com/north-brooklyn-pipeline/

Paper reported "How can you say you communicated anything to this community," said Antonia Ortiz. "No one knew about this until they had to get their car because they were digging a hole." [10]

Q:Has the public beyond the local community commented on the MRI project? And if so what have they said?

A: Yes, as of January 29, 2020 1,827 comments have been submitted to the PSC and a majority oppose the expansion of fracked gas infrastructure. Additionally over 3,800 New Yorkers have signed a petition[11] to Governor Cuomo and Commissioner John Rhodes calling on them to reject the Company's proposal to spend almost 185 million ratepayer dollars to complete the MRI project.

Q: What do you recommend?
We recommend that the public and local elected officials' call for an immediate halt to construction be listened to and no additional funds should be approved for the MRI project.

C. LNG & CNG

Q:What is the company proposing regarding CNG?

A: In their Second Supplemental Testimony the Company claims "For the winter 2020/2021, to meet increased gas demand from new and existing customers, the Glenwood CNG facility will be expanded to double its capacity. Two additional CNG injection facilities (a third and a fourth CNG facility) will be required in KEDLI's service territory at locations to be determined. At this point, five potential properties that are owned by National Grid in KEDLI's service territory are being considered. The capital costs for each new facility are estimated at approximately $1 million in FY 2020 and $10 million in FY 2021….Finally, the Companies require a CNG Filling Station to be located on Staten Island where trucks will be filled with CNG for transportation to both KEDLI's and KEDNY's CNG injection sites. The total cost for the project is approximately $30 million, and the project is expected to be in service in March of 2022." [12]

Q:Do you support this investment? Why or why not?

[10] RESIDENTS FUME AS NAT GRID BUILDS NORTH BROOKLYN PIPELINE https://www.brooklynpaper.com/residents-fume-as-nat-grid-builds-north-brooklyn-pipeline/
[11] https://actionnetwork.org/petitions/tell-governor-cuomopublic-service-commission-listen-to-the-clcpa-reject-all-of-national-grids-fracked-gas-pipelines-and-proposals?source=direct_link&
[12] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

A: No, we do not support the investments to expand CNG or build new CNG facilities because as stated in above testimony investments in fracked gas infrastructure go against climate science and city and state climate law. We recommend the Company invest more in gas demand reduction and energy efficiency.

Q:What is the company proposing regarding LNG?

In their Second Supplemental Testimony the Company claims "interim solutions to the capacity constraints involve additional reliance on the Companies' LNG capabilities, including updates to the Companies' existing LNG Tank Upgrade programs and to KEDNY's LNG Truck Load/Unload Station Program. Additionally, new investments are required for KEDNY's Greenpoint LNG site to add: (i) a new portable LNG facility and (ii) two additional vaporizers."[13]

Q:Do you support these investments? Why or why not?

No, we don't support these investments. While no investments in fracked gas infrastructure comply with city and state climate law, studies have found LNG increases emissions over regular gas because it is more intensive than the supply chain for pipeline gas, because of the extra processing steps such as processing, refrigeration, transportation, storage and handling. In order to meet our emission reduction benchmarks, we should be retiring LNG not investing in it.

Q: In addition to climate impacts are there other reasons the Company should plan on closing LNG & CNG facilities instead of investing in them?

A: Dangerous to surrounding communities and workers. LNG is highly volatile and explosive, and notably New York City saw a disaster that killed 40 workers[14].

Q: What do you recommend?

A: The Company should not invest any new funds that would expand the use of or extend the life of LNG facilities and should instead invest in gas demand reduction and energy efficiency.

D. Clove Lakes Transmission Loop Project

Q:What is the company proposing?

[13] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[14] https://www.silive.com/news/2013/02/40_years_ago_today_staten_isla.html

A: In their Second Supplemental Testimony the Company asserts "the Clove Lake Transmission Loop Project has been added to the FY 2021 and FY 2022 capital plans to enable KEDNY to conduct a feasibility study and engineering analyses. Looping of the Clove Lakes Transmission Main with a new 30-inch transmission main across the borough of Staten Island will facilitate KEDNY's ability to take more gas through the TETCO Goethals Take Station and ultimately move this gas across KEDNY's system to supply constrained areas." [15]

Q:Do you support this investment? Why or why not?

A: No we do not support the Company's proposal to invest 2 million ratepayer dollars to conduct a feasibility study for a pipeline expansion. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law. The Company asserts the project "is not expected to be in service until FY 2025, at the earliest." In order to meet emission reduction goals of the CLCPA and Climate Mobilization Act New York will be reducing gas use leaving a pipeline coming into service in 2025 a stranded asset.

E. Riverhead Transmission and Southeast Suffolk projects

Q:What is the company proposing?

A: In it Second Supplemental Testimony the Company asserts "KEDLI's updates include adding back the Riverhead Transmission and Southeast Suffolk Phase 2 projects that were removed in the rebuttal testimony. These projects are required to support new customer connections in eastern Long Island. Moreover, to enable new customer connections, KEDLI has accelerated Phase 1 of the Southeast Suffolk project to ensure it will be in service in FY 2022. Accordingly, Exhibit __ (GIOP-1SS) reflects a shift of $5 million of the project forecast from FY 2022 to FY 2021." [16]

Q: What is the Riverhead Transmission project?

A: According to Direct Testimony filed by the Company in April "this project will install 8200 feet of 12" 350 psig transmission main on Mill Rd from County Rd 58 to Middle Rd, then on Middle Rd to Joise Ave." [17]

[15] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[16] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[17] Gas Infrastructure and Operations Panel

Q:Do you support this investment? Why or why not?

A: No we do not support the Company's proposal to invest over 3.5 million ratepayer dollars to install new transmission main. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law. The Company asserts the project "will be constructed in FY 2024."[18] In order to meet emission reduction goals of the CLCPA and Climate Mobilization Act New York will be reducing gas use leaving a pipeline being constructed in 2024 a stranded asset.

Q: What is the Southeast Suffolk project?

A:According to Direct Testimony filed by the Company in April the project will "install approximately 8,500 feet of 16-inch, 350 psig transmission main, a new regulator station, and 5,800 feet of 12-inch steel distribution main. The Company's capital investment plan includes approximately $20 million in the Rate Year, $21.6 million in Data Year 1, $0.1 million in Data Year 2, and $0.3 million in Data Year 3."[19]

Q:Do you support this investment? Why or why not?

A:No we do not support the Company's proposal to invest over 42.2 million ratepayer dollars to install a new transmission main, a new regulator station and new distribution main. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law.

F. Leak Prone Pipe

Q:What was the Company's update regarding LPP?

A:In it Second Supplemental Testimony the Company asserts "KEDLI renegotiated its agreement with a contractor that provides services for the Proactive Main Replacement (LPP) Program. A copy of the renegotiated agreement was provided in the Company's response to IR No. DPS-1043.The agreement results in reduction of the program's unit costs from $306 per foot to $304 per foot, as shown in Exhibit (GIOP-4SS). Revising the cost per foot reduces KEDLI's capital expense forecast by $1.929 million in the Rate Year, as shown in Exhibit (GIOP-1SS)." [20]

[18] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[19] Gas Infrastructure and Operations Panel
[20] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

Q: Does the reduction in cost change your position on LPP?

A:No. We still argue that the Company should not invest any new ratepayer funds in extending the life of pipelines.The Company should be planning on retiring LPP instead of replacing it.

G: Need for Radon Testing

Q: What was Sane Energy's recommendation on radon testing in testimony filed in August?

In Direct Testimony Sane asserted "In order for the Company to comply with its requirement to provide safe service we recommend the Company provide EPA scientific radon monitoring[21] in all gas supplied to customers. Additionally, we recommend meeting with New York elected officials who drafted the 2014 Radon Bill[22] to monitor and mitigate radon in our gas supply." [23]

Q:Why did Sane make that recommendation?

A: We made that recommendation because the Company asserted in Direct Testimony[24] that more of their gas supply is coming from the Marcellus Shale in northeastern Pennsylvania which has higher than average levels of uranium and thorium. These elements break down to form radium-226. Signs of radon in gas & other fracked gas chemicals in our homes, have been on the rise since 2004[25]when high-volume hydraulic fractured drilling began. Radon is the leading cause of lung cancer in non-smokers in the United States[26].

Q: Have there been any significant developments regarding radioactivity in gas since Sane Energy Project filed testimony in August? If so, what are the developments?

A: Yes. A groundbreaking, 20-month investigation on oil and gas radioactivity by reporter Justin Nobel was published in Rolling Stone on January 21, 2020[27].

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[21] https://www.epa.gov/radon
[22] https://nyassembly.gov/leg/?default_fld=&bn=A06863&term=2013&Summary=Y&Text=Y
[23] Sane Direct Testimony
[24] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)
[25] Increased Levels of Radon in Pennsylvania Homes Correspond to Onset of Fracking Johns Hopkins Bloomberg School of Public Health
https://www.jhsph.edu/news/news-releases/2015/increased-levels-of-radon-in-pennsylvania-homes-correspond-to-onset-of-fracking.html
[26] https://www.cancer.org/cancer/cancer-causes/radiation-exposure/radon.html
[27] America's Radioactive Secret
https://www.rollingstone.com/politics/politics-features/oil-gas-fracking-radioactive-investigation-937389/

Q:What did the investigation find?

A: Nobel's reporting found that major fossil fuel companies have known for decades that gas has high levels of naturally occurring radioactivity and that radioactivity builds up within their infrastructure. Nobel wrote "In the early 1970s, Exxon learned radioactivity was building up in pumps and compressors at most of its gas plants. "Almost all materials of interest and use to the petroleum industry contain measurable quantities of radionuclides," states a never-publicly released 1982 report by the American Petroleum Institute, the industry's principal trade group, passed to *Rolling Stone* by a former state regulator...Despite dire climate warnings, the U.S. oil-and-gas industry is in the midst of an epic boom, what a 2018 Department of Energy paper calls an "oil-and-gas production renaissance." Pipelines, power plants, and shipping terminals are being developed across the nation at a dizzying pace. But in the excitement of this boom there is little mention of the pipes, pumps, and filters in these plants that will become coated with radioactivity."

Q: Is the Company testing for radon?

A: No. In response to Sane IR-3 when asked if testing for radon the Company said "where gas is introduced into the National Grid system, gas monitoring instruments are needed to monitor odorant levels, BTU, composition, hydrates, and hydrocarbon dew point ("HCDP")." and went further denying the radioactivity of the Marcellus Shale saying "the Companies do not accept the conclusion asserted in the question that "Marcellus Shale has been proven to have high levels of radon and hydraulic fracking uses dozens of carcinogenic chemicals". [28]

Q: What do you recommend?

A: We recommend the Company do radon testing in accordance with bill A1020[29] which has been introduced in the New York State Assembly.

[28] Exhibit_Sane 3
[29] https://www.nysenate.gov/legislation/bills/2019/A1020

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas

A.L.J. COSTELLO: We have a two-page affidavit on behalf of the New York State Laborers' Organizing Fund, signed by John Hutchings, and sworn to on February 10, 2000 -- it says 2019. It should be 2020. The commit -- the notary stamp is -- no. It should be 2020. It should -- needs -- would be -- need to be corrected. But that's Exhibit 719. And at this point, the court reporter should insert the fi -- file entitled New York State Laborers' John Hutchings direct testimony. #

BEFORE THE STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	Case 19-G-0309
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	Case 19-G-0310

DIRECT TESTIMONY

OF

JOHN HUTCHINGS

ON BEHALF OF THE NEW YORK STATE LABORERS ORGANIZING FUND

August 30, 2019

Q: Please state your name, affiliation, and address.

A: My name is John Hutchings. I am the Director of the New York State Laborers Organizing Fund ("NYSLOF"). My business address is 42-40 Bell Boulevard, Bayside, NY 11361.

Q: What is the New York State Laborers Organizing Fund?

A: The New York State Laborers Organizing Fund represents over 40,000 members around the state. We are affiliated with the Laborers International Union of North America ("LIUNA"), one of the largest labor unions in the United States representing a half million workers across the United States and Canada. In New York, our members primarily work in the construction industry, and are organized into more than 24 local unions and five district councils. Our members build and maintain vital energy infrastructure across a diverse portfolio, including natural gas transmission facilities, renewable generation assets, and nuclear facilities. Thousands of our members work for contractors employed by utilities on accelerated pipe replacement projects which are critical to the safety and reliability of New York's energy assets. Hundreds of our members work for construction contractors on pipe construction activities for The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies" or "National Grid"). In this testimony, I refer to NYSLOF and LIUNA interchangeably.

Q: Please summarize your background and professional experience.

A: I have been involved in the construction industry for over 30 years. I began my career in the construction industry as a Laborer in Binghamton, NY in 1985. I was a Union Trustee for

Laborers' Local 7 for over 10 years. I have served as the Director of the New York State Laborers' Organizing Fund for over 10 years

Q: What are your duties and responsibilities in your current position?

As Director, I supervise and manage a statewide staff of 21 organizers, researchers, and policy analysts as we work to increase union market share in the construction industry and service our local unions. I am responsible for conceptualizing and executing statewide campaigns and supporting local organizing efforts.

Q: What is the purpose of your testimony?

A: The purpose of my testimony is to describe the work LIUNA's members perform on National Grid's system, and recommend conditions that NYSLOF asks that the Commission consider imposing on any approval of the proposed rate increase. The conditions are intended to mitigate undue risks in contracting practices, ensure National Grid contractors only employ highly qualified and reputable labor resources, and to enhance transparency of National Grid's use of contractors, as well as workplace safety and reliability.

In addition, my testimony promotes the importance of the expansion of natural gas infrastructure discussed by the Companies in this case.

Q: Please explain why the subjects you plan to address are relevant to this proceeding.

A: Outside contractor employees play a significant role in National Grid's provision of safe and reliable gas services to the public, including by performing essential construction work on the Companies' gas distribution lines. My understanding is that outside contractor expense is included in National Grid rates.

The expansion of natural gas infrastructure, such as the construction of the Transcontinental Gas Pipe Line, LLC's Northeast Supply Enhancement ("NESE") project, is vitally important for affordable gas rates for consumers, for supporting quality construction pipeline jobs for hundreds of workers, and facilitating economic growth in National Grid's service territory. Though the approval of NESE is outside the scope of the State of New York Public Service Commission ("Commission"), it is our view that the Commission can be a leader in advocating for expanded energy infrastructure and approve the Companies capital expenditures should NESE proceed.

Q: Please describe the extent of the Companies' use of outside, contractor employees in the conduct of its operations?

A: National Grid's use of contractor employees is vital to its conduct of utility operations, including modernizing its infrastructure, installing gas services, and providing for safe and reliable services. In 2019, KEDNY projects a maximum of 40 main crews and 40 service crews. KEDLI estimates a maximum of 37 main crews and 53 service crews. In 2020, maximum crew levels will increase 29 percent over 2019 levels. See Exhibit No. NYSLOF-1 to this testimony. Assuming there are five workers on each construction crew, there are approximately 850 contractor workers employed on National Grid's system during peak construction.

Q: Do NYSLOF members work for contractors who perform work on the National Grid system?

A: Yes. NYSLOF does not represent in-house workers, only the employees of certain construction contractors. National Grid uses a number of contractors who are signatory to collective bargaining agreements with LIUNA, including Hallen Construction Company, Inc., Bond Brothers Inc., and Henkels & McCoy.

Q: Is National Grid proposing to recover expenditures paid to contractors in this rate case?

A: Yes. Construction contractors perform the bulk of pipe replacement activities, including projects related to mandated programs such as proactive main replacements of Leak Prone Pipe ("LPP"), large diameter main rehabilitation, and reactive and proactive service replacements. Accordingly, contractor costs comprise the bulk of the Companies' LPP costs. For KEDNY, 74 percent of LPP unit costs are contractor expenditures. For KEDLI, 65 percent of LPP unit costs are contractor-related. See Exhibit No. NYSLOF-2.

Furthermore, though Vendor Costs include all vendors and not just construction contractors, Vendor Costs at the Companies comprise the majority of total costs. At KEDNY, for example, payments to one of its construction contractors was greater than KEDNY's own labor costs and comprised 37% of total costs.

Q: What is the relevance of these data?

A: These data show two things. First, that National Grid is reliant on outside contractors to perform vital safety and reliability services on its system. Second, that expenditures paid by National Grid to outside contractors comprise a significant portion of costs the Companies are proposing to recover in rates.

Q: How does National Grid procure construction services from outside contractors for Significant Capital Program projects?

A: According to National Grid's Sourcing and Qualifying Contractors Process Flowchart, the process consists of the following steps: 1) Identify/Source Potential Contractors and Add to Bid List; 2) Non-Standard Bid Event, 3) Framework/Non-Framework Contract, 4) Understand Available Resources and Identify Future Resource Needs; and 5) Make Ready Approved

Contractors. See Exhibit No. NYSLOF-3. National Grid's Contractor Resource Team assesses contractors based on experience, approach, qualification of staff, resource availability, equipment, and safety culture. See Exhibit No. NYSLOF-4. Bids from Contractors are evaluated on four types of criteria: Technical Capability, Resource Capability, Experience, and Commercial/Price. Each category contain specific elements which are weighted to produce a final bid score. For example, under Resource Capability, the number of available crews a contractor comprises 50% of the total score for that category. See Exhibit No. NYSLOF-5.

Q: What are the safety metrics used by National Grid to evaluate contractors?

A: National Grid uses ISNetworld to track Contractor Evaluation Criteria. The Criteria include insurance requirements; acknowledgement of the Companies' Contractor Safety Form; workplace safety statics that include OSHA citations, fatalities, Total Recordable Incident Rate ("TRIR"), Days Away, Restricted, or Transferred ("DART") Rate; a health and safety questionnaire used during the New Vendor Onboarding Application Form; and a review of the Contractor's written health and safety program. See Exhibit No. NYSLOF-6. National Grid also evaluates contractors on various performance metrics related to safety, performance, and compliance. See Exhibit No. NYSLOF-7. National Grid's N-1402 National Grid Contractor Safety Requirements details contractor safety requirements that are unique to operations and various functional groups. Contractors must be in compliance with these policies as a contractual condition. See Exhibit No. NYSLOF-8.

Q: What safety risks exist for Laborers performing work related to natural gas transmission and distribution pipeline?

A: Working on natural gas facilities poses many significant risks, both to individual worker safety and overall system reliability. Working with natural gas is extremely dangerous and, when installing or maintaining gas lines, it is critical to minimize the risk of line ruptures or punctures, which can cause accidental ignition or explosion. Even absent ignition, the high pressure in the pipes could cause life-threatening damage if there were a rupture. Laborers perform the work of digging out a line when it is necessary to access the line for repair, maintenance, or replacement. While operators using mechanized equipment dig up the area of within two feet of an existing transmission lines, Laborers dig the rest by hand. Given the associated risks, this job requires painstaking accuracy. Along with rupture, ignition risks, there are concerns when trenching about the potential for "cave ins." This can happen when Laborers or Welders are in the ditch or if Laborers or Welders are walking along the ditch above ground along the edge of the ditch. To minimize this risk, Laborers employ safeguards, such as using creating shielding or shoring around a trench. There are also risks associated with working in extreme climates, as Laborers are frequently required to do. Laborers must constantly be aware of their surroundings, and these concerns may be heightened when working in poor weather conditions. Working with and directing heavy equipment and machinery presents other safety risks. Laborers are responsible for acting as the Operator's eyes because the Operator can't see everything going on around him. For instance, the Laborer must be alert to unexpected things like a child or dog coming near the heavy machinery, and be prepared to signal the Operator to stop immediately. Of course, a Laborer must also be alert to the possibility that the Operator could hit a line that had not been located correctly. Acting as the Operator's eyes requires having good communication skills and a well-established relationship of trust with the Operator. The reason that the latter is important is that each party needs to be able to read the other's signals. Accidents are more likely to occur if

hand signals are used incorrectly or there is bad communication with the operator, or a Laborer is not anticipating properly what the Operator is going to do. In addition, because most of this work happens in the street, there is always the constant danger of workers being injured by the surrounding traffic.

Q: What role does proper training play in ensuring workplace safety and the successful completion of assigned tasks?

A: It plays a very significant role. The need for proper training should be obvious—the work these employees are performing is dangerous, and, where done improperly, life-threatening for both workers and customers. In recent years, there have been a number of high-profile incidents that have highlighted the problems that can occur at gas distribution utilities. Pipeline incidents caused by excavation damage can result in fatalities and injuries, as well as significant costs, property damages, environmental damages, and unintentional fire or explosions.

LIUNA is dedicated to making sure its workforce is the most qualified and safest in the industry. LIUNA annually invests more than $100 million in training workers in construction and environmental skills curriculum. Over the past half-century, LIUNA has developed its own training programs, which are currently used by contractor employers across the United States. Through its training program, LIUNA has worked towards offering a standardized course of high-quality training.

In response to new Operator Qualification Regulations, approved under DOT 49 CFR Parts 192 and 195, LIUNA Training developed Pipeline Safety Worker and Instructor training leading to Pipeline Operator Qualification (OQ).

Thousands of our members are OQ trained at our training facilities. There are at least five LIUNA training centers that train for Northeast Gas evaluations.

It is important to note that LIUNA views Operator Qualification as a baseline minimal standard. Our training goes beyond OQ. For example, LIUNA offers a 64-hour Pipeline Technology Class. Major topics include: employee obligations, chain of command on a pipeline, general hazards and precautions, trench and excavation safety, soil testing and identification, potholing and underground utility lines, the duties of the various crews (fence, ditch, clearing, etc.), cleaning and inspecting pipes, and fabrication yards. Several of the topics require students to perform tasks or demonstrate skills being taught. Special emphasis is placed on following proper techniques, developing safe work habits, and recognizing unsafe work conditions.

LIUNA offers a 12-hour Trench and Excavation Safety Class. Major topics are designed for workers who work in trenches and excavations. Students learn about OSHA's standards for building protective systems in trenches and excavations, and other regulations to ensure safety on a job site. Students also learn how to install different kinds of protective systems, with special emphasis placed on following proper techniques and developing safe work habits.

Q: How does National Grid ensure the qualification of its contractor personnel?

A: The Northeast Gas Association ("NGA") maintains contractor operator qualifications ("OQ") in its OQ database system. NGA tracks contractor qualification, and the Companies verify such qualifications. See Exhibit No. NYSLOF-9.

Q: In your opinion, how do National Grid's contract procurement process, contractor evaluation process, and safety and health standards compare to other owners of construction projects?

A: In my opinion, National Grid has a very thorough contract procurement process, and contractor evaluation system. The criteria used to evaluate contractor performance is very detailed and comprehensive when compared to other owners of construction projects in New York State. The feedback from LIUNA members on National Grid projects is also very positive. These workers feel supported to raise any health and safety concern that may arise on the job. The only recommendation we have is that National Grid adopt terms and conditions in their construction contracts that are similar to terms utilized by Consolidated Edison ("ConEd") to encourage higher labor standards with the goals of reducing construction labor risks, ensuring National Grid has the qualified labor resources required to meet the increase in pipe replacement activities, and improving workplace—and customer—safety.

Q: What is the recommendation that you would encourage the Commission to impose on National Grid?

A: ConEd's "Standard Terms and Conditions for Construction Contracts" includes a provision on Labor which states that only union labor from building trades locals and affiliated with the Building and Construction Trades Council of Greater New York can be employed. I have attached a copy of ConEd's "Standard Terms and Conditions for Construction Contracts" to this testimony. See Exhibit No. NYSLOF-10.

Q: Why is NYSLOF recommending National Grid adopt this recommendation?

A: The use of union labor from construction trade locals affiliated with the Building and Construction Trades Council of Greater New York provide benefits to ratepayers. First, this condition ensures that only reputable and qualified labor resources are employed on National Grid projects. Mitigating undue labor risks is especially critical given the increase in direct

capital expenditures proposed by the Companies in this case. KEDNY's total direct capital expenditures are forecasted to grow by 27 percent in 2024 compared to the historic test year. KEDNY's total direct capital expenditures are forecasted to grow by 52 percent in 2024 compared to the historic test year. Furthermore, increased LPP targets in future years require that contractors have the highly trained and qualified labor sources available to perform the work. KEDNY has 1,458 miles of LPP remaining in its inventory. ConEd leads the nation in the ratio of repaired leaks to miles in part because its contractors only employ reputable union labor. See Exhibit No. NYSLOF-11.

The issue of New York State utilities' employment of contractors is not new to the Commission. In Case No. 13-M-0449, *Focused Operations Audit of the Internal Staffing Levels and the Use of Contractors for Selected Core Utility Functions at Major New York Energy Utilities*, the Operations Audit of Staffing Levels at the Major New York State Energy Utilities" ("Staffing Audit") published in February of 2017 confirmed the following: 1) Contractor employees are a critical component of utilities' labor mix, 2) The importance of contractor employees has grown as utilities accelerate pipe replacement activities, 3) Utilities need to plan for how they will meet the need to grow contractor resources in order to meet ambitious pipe replacement targets, and 4) Growing competition by utilities and other energy projects for skilled gas workers will strain this finite labor pool. By adopting this condition, the Commission can ensure that National Grid has the qualified and reputable labor resources it needs to meet pipe replacement targets.

Q: Please explain the importance of NESE to the construction industry, and expanded natural gas infrastructure.

A: The last topic I will touch upon briefly is the need for proper focus and attention on infrastructure. The experience of our Local affiliates across New York State is that constraints on natural gas and opposition to pipelines not only deny qualified construction laborers an opportunity to work, it challenges broader economic growth and expansion. In this rate case, residential ratepayers will experience a greater rate increase if NESE is not built compared to if the pipeline is constructed. Furthermore, a delay or denial of NESE will reduce KEDNY's contractor labor force by one to three crews, and reduce KEDLI's contractor workforce by approximately 13 crews due to the reduction in capital programs. See Exhibit No. NYSLOF-12.

NYSLOF recommends that the Commission approve the Companies' capital expenditures should NESE proceed.

Q: Does this conclude your testimony?

A: Yes.

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas

A.L.J. COSTELLO: We have a two-page affidavit by Robert Mark Bob Wyman, sworn to on February 12, 2020. And that will be marked as Exhibit 720. The court reporter should insert Bob Wyman direct testimony and Bob Wyman rebuttal testimony. #

BEFORE THE
NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	)))))	Case Number: 19-G-0309
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.	))))	Case Number: 19-G-0310

DIRECT TESTIMONY TESTIMONY OF BOB WYMAN

August 30, 2019

Introduction

Q. What is your name?

A. I am commonly known as Bob Wyman. My formal name is Robert Mark Wyman.

Q. Mr. Wyman, have you previously testified before the Commission?

A. Yes. I presented testimony during the recent Central Hudson, Orange & Rockland, and Consolidated Edison rate cases. I have also given verbal testimony at a variety of PSC public hearings, have helped others prepare testimony for a variety of PSC proceedings, and am an active party in a variety of PSC matters.

Q. Do you always participate in PSC matters as an individual, representing only yourself?

A. No. In addition to participating in PSC proceedings as an individual, I have, from time to time, assisted others in preparing for these proceedings or have spoken for them. For instance, I have several times assisted NY-GEO[1] in developing testimony and/or evidence. I also represented NY-GEO as a participant in ConEd's "Peak Gas Collaborative" which had been established as a result of the prior ConEd rate case (16-G-0061).[2].

Q. What is your background?

A. I am a New York City-based advocate for renewable energy and Beneficial Electrification with a focus on geothermal heat pumps. I am a member of the New York Geothermal Energy Organization, (NY-GEO)[3], the Renewable Thermal Alliance[4], and the International Ground Source Heat Pump Association (IGSHPA)[5] on whose Advo-

[1] https://ny-geo.org/

[2] 16-G-0061, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service.

[3] https://ny-geo.org/

[4] https://cbey.yale.edu/programs-research/renewable-thermal-alliance

[5] https://igshpa.org/

cacy Committee I serve. I am a founder and shareholder, but not an employee, of Dandelion Energy[6], the Google X[7] spin-off company that currently installs geothermal heat pump systems in New York State.

Although most of my career was spent in the computer software business, during which I worked for a variety of companies, including Digital Equipment Corporation, Microsoft, Google, Medio Multimedia, and Pubsub Concepts,[8] I have been advocating on energy related issues for at least 45 years, having first given public testimony in hearings concerning Pres. Nixon's "Project Independence"[9] in 1973.

[6] https://dandelionenergy.com/
[7] https://x.company/
[8] See my LinkedIn profile for more detail: https://www.linkedin.com/in/bobwyman/
[9] https://en.wikipedia.org/wiki/Project_Independence

Purpose of Testimony

Q. What is your purpose in submitting this testimony?

A. I wish to comment on several aspects of the proposals made by National Grid ("The Company") in the current cases and to advocate for several modifications or additions to those proposals. My testimony is intended to ensure a more complete record of the relevant issues and to advocate for policies that will be more fair, more in the public interest, and more likely than those proposed by the Company to conform to existing State and New York City laws and policies.

Some of the issues that I wish to discuss include:

- The Company's proposals concerning
  - Gas expansion
  - Depreciation
  - Excess Deferred Income Tax (EDFIT).
  - Planning a "Process of Managed De-Capitalization"

Gas Expansion and Gas Capital Investment

Q. Should the Company's proposed gas expansion projects be accepted?

A. No. This is undoubtedly the last rate case during which the Company may even hope that it has the opportunity to receive approval for either gas expansion or for any substantial gas capital investment, other that required for safety and reliability. For all the reasons that will prevent the Company, and any other gas utility in New York State, from obtaining approval for gas expansion in future rate cases, the Company's proposals in this rate case should be, at least, severely limited — if not rejected.

Q. Why are future approvals for gas expansion unlikely?

A. A continuously growing public awareness of the full climate, health and economic costs of burning fossil fuels, including gas, has already risen to such a level that public authorities, both at the State and City level, have been compelled to enact laws, policies and regulations that require a dramatic reduction in the future use of these fuels. The goal of these laws is to ensure cleaner, healthier air and water, a reduction in climate impacts, and the adoption of more efficient and economical methods for accomplishing those things currently done with fossil fuels such as gas. The result is an unprecedented regulatory and market environment for the Company.

In the past, the Company has always been able to assume broad government and public support for increasing its customer base, increasing its capacity to serve, and for making long-term gas capital investments. In the past, the Company has also always been able to assume that the assets they invested in would be used at, or close to, full capacity over all or most of their technical lives. However, our new laws and policies now demand that the Company's focus shift from seeking perpetual growth to implementing a process of managed decline. The law, policies and public opinion now require that the Company focus on reducing the number of customers they serve, reducing the quantity of gas they supply, and crafting a process of managed decapitalization in order to avoid the accumulation of stranded assets as we find the useful life of assets ending much earlier than was typical in the past..

Q. Why are you so sure that gas sales must fall in the future?

A. New York State and New York City have both established aggressive non-optional goals limits on carbon emissions. State law requires an 85% reduction in emissions by 2050 and New York City law requires an 80% reduction. New York City's limits are defined in Local Law 97-2019 and are expressed using calendar year 2005 as the base

year. In addition to the 80% reduction by 2050, the City also requires a 30% reduction by 2030.

Although State law requires dramatic emissions reductions by 2050, one might argue that it is possible that emissions reductions might be uneven throughout the state. Thus, KEDNY or KEDLI would be able to reduce emissions by less than 80% if emissions from sources in other parts of the state are reduced by some higher amount. However, KEDNY's emissions are controlled by both State and New York City law thus, reductions elsewhere in the State will have no impact on KEDNY's requirement to comply with City law. Given that only KEDNY and ConEd provide natural gas in New York City, it is useful to look at the impact of New York City law on KEDNY's prospects in the future. In New York City, KEDNY will only be able to reduce its sales reductions if it can shift the burden of reduction to ConEd...

According to the New York City Greenhouse Gas Inventory,[10] in 2005, the City's carbon emissions from all sources was 61,062,452 tCO2e/year. Thus, the City's law requires that carbon emissions in 2030 be reduced to 42,743,716 tCO2e/year and by 2050 to no more than 12,212,490 tCO2e/year. However, **the inventory of NYC's 2017 emissions reports that carbon emissions attributable to natural gas use in buildings[11] were 17,639,988 tCO2e or 144% of the 2050 city-wide, all-sources limit and 41% of the 2030 limit.** (It should be recognized that natural gas use in New York City has increased since 2017. Thus, when the Inventory is updated for 2019, it will show that emissions from natural gas are even greater than 144% of the 2050 limit.) Given that carbon emissions from natural gas already dramatically exceed the City's limit on carbon emissions from all sources, it is clear that emissions from natural gas must

[10] https://on.nyc.gov/ghginventory

[11] "Stationary Source" not including power generation.

be reduced. If we assume that in 2050, natural gas will be the only source of carbon emissions in the City (a fantastic assumption) it would be necessary to reduce natural gas use by 31% in order to avoid exceeding the limits established by City law. Of course, if we assume, as we must, that there will be other sources of carbon emissions, it is clear that natural gas use must be reduced by more than 31% before 2050.

Q. Have carbon emissions from buildings been declining in New York City?

A. Yes, as shown in Figure 1, emissions from buildings for all energy sources other than natural gas have decreased by 21% since 2005. Emissions from oil use are down 53%, emissions from electricity use are down 45% and emissions from steam are down 58%. However, the 35% increase in emissions from natural gas has limited the reduction in this sector's total emissions.

Q. Will further reductions in emissions from other sources or by improving energy use efficiency allow avoidance of reductions in natural gas emissions?

A. No. As stated above, natural gas emissions are already at a level equal to or higher than 144% of the citywide, all-sources limit on carbon emissions for 2050. Thus, even if emissions from all other sources in buildings, as well as all other sources in all sectors (including transportation) were completely eliminated, it would still be necessary to dramatically reduce emissions from natural gas. Similarly, improvements in energy efficiency, such as higher efficiency furnaces, window treatments, or increased insulation would not allow us to avoid dramatic reductions in natural gas use. Such improvements may reduce the amount of emissions from non-natural gas sources and thus might minimize the total reduction required, but emissions from natural gas must still be reduced by at least 30%.

Q. Is it necessary for gas use to be reduced 80% by 2050 in order to meet the requirements of Local Law 97-2019?

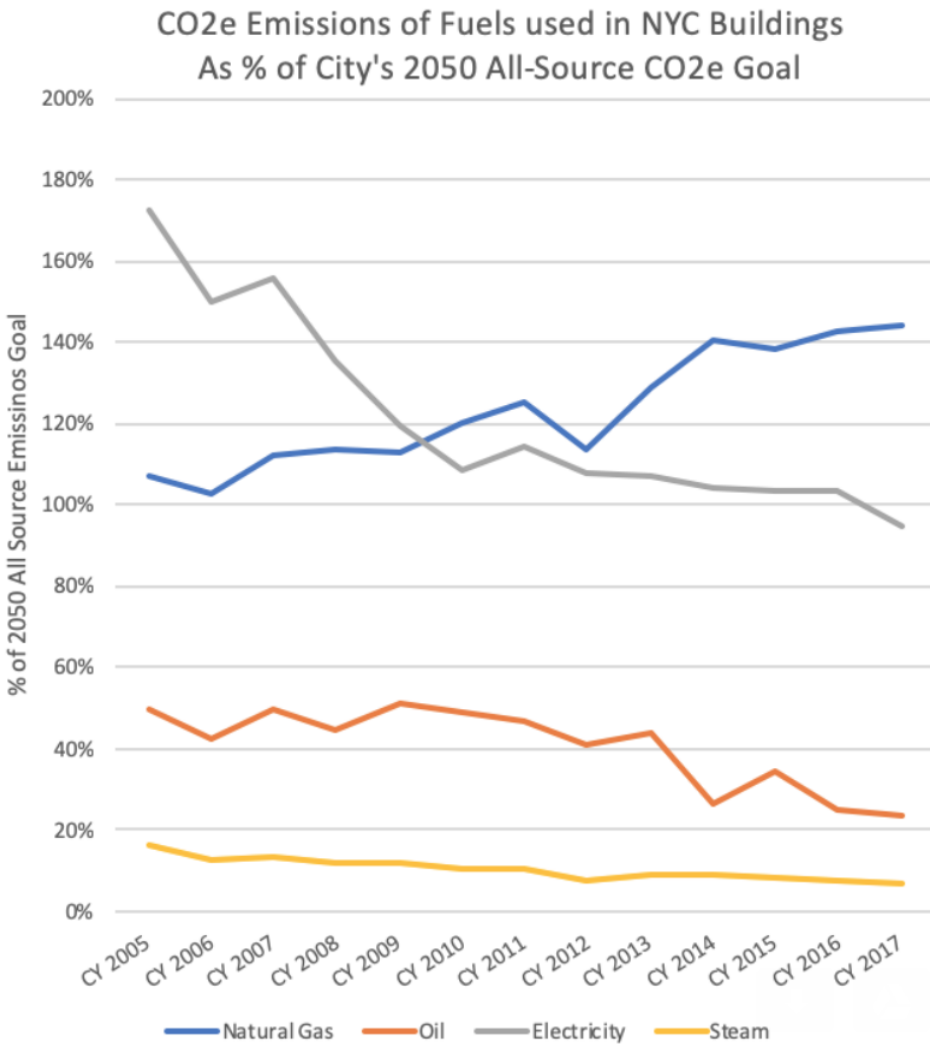


Figure 1: Emissions from Buildings in New York City (Source: NYC Emissions Inventory)

A. No. Although doing so would be desirable, in order for the City's citywide, all-source limits to be met it is only necessary to reduce emissions from natural gas by some quantity larger greater than 30% of the City's natural gas consumption in 2017. How much greater than 30% would be determined by the emissions produced by others sources and by how much natural gas use has risen since 2017. A 30% reduction would be sufficient only if natural gas were to be the only source of carbon emissions in 2050 and if gas expansion had ended in 2017. It is likely that a reduction of between

50% and 80% would be necessary to allow the City's limits to be met.

Q. Given the current law and policies, why is the Company proposing gas capital investment?

A. Because it is all they know to do... The Company has an obligation not only to the public but also to its shareholders. Those shareholders require that the Company be able to show that it has the ability to produce long term profit streams that will benefit its shareholders. However, given the current regulatory requirements, the only means the Company has to produce revenues is to invest in capital assets, from which it earns an allowed return on equity.

Gas utilities are not, in fact, in the "business of selling gas." At least, that isn't how they make money. Gas utilities sell gas to their customers at, or near, that commodity's cost. Additionally, even though they spend a great deal to operate and manage their large and complex transmission and distribution systems, those O&M expenses are paid by ratepayers at cost – without a profit. The only way a gas utility makes money in its regulated business is from their allowed return on the equity they invest in capital assets. Thus, in order to assure shareholders that there will be future profits, the Company must always be able to demonstrate a plan and ability to ensure that shareholder's capital is, and will remain, fully invested in capital assets that deliver a reasonable return on equity. In the past, this has not been a challenge since it has always been possible to obtain approval for additional and ever-growing capital investments. As old assets were retired or depreciated the Company was always able to invest in new gas assets to replace the old. But, times have changed. Continued investment in a large, expensive system whose use is required to decline is simply no longer reasonable.

Given today's laws, policies, etc., it is clear that continued investment in gas assets is not in the public interest, however, such investments are necessary to convince shareholders to maintain their investments in the Company. The conflict between the interests of the Company and those of shareholders appears to be significant. However, we should not be surprised if the Company seeks to prioritize shareholders' interests.

Q. If continued investment is in shareholders' interest, what policies would be most in the public interest?

A. To serve the public interest, and to conform to the law, the Company should:

- Establish a formal, managed moratorium or permanent end to gas expansion
- Establish a year-by-year "glide-path" for reducing gas consumption that conforms to the various legally expressed limits on greenhouse gas emissions in 2030, 2050, etc.
- Create programs, incentives, etc. to ensure that gas consumption stays at or below the levels defined by the glide-path over its full extent.
- Accelerate depreciation and thus revenue-recovery from ratepayers for assets whose useful lives will be shorter than was historically typical.
- The Company should petition the Commission and the Legislature to eliminate the "100 foot rule" which currently forces the Company to provide service on-demand, no matter how much doing so may be against the public interest.

Q. What do you mean by "managed moratorium?"

A. Unlike the current, informal moratorium during which the Company has simply refused to process new applications for service, the Company should do what was done by ConEd in Westchester County and announce a deadline for the acceptance of new applications for gas service. This will allow those who have projects under construc-

tion or in design, but have not yet applied for gas service, to determine whether or not they will be able to receive gas service in the future. The Company should require that applicants demonstrate a capacity to accept service within one year of having that service approved.

Once the initial collection of applications have been accepted, the Company should accept applications only from those customers who can prove that they have special needs or no viable alternative to gas service – given the nature of their usage requirements. Those customers should then only be granted service if it can be shown that sufficient demand has been relieved in order to ensure that the new gas consumption will leave total consumption below that which is required by the "glide-path."

Q. How should be glide-path be defined?

A. The precise shape of the glide-path should be proposed by the Company, in cooperation with parties, after a detailed study of the effectiveness of potential incentives, legal mandates, workforce development, etc. A simple start might be to assume a straight-line decline over the thirty years remaining between now and 2050. Clearly, various parties will advocate for alterations to such a curve, probably to assume greater declines in future years and less decline in near-term years. The actual glide-path should be the result of extensive study and stakeholder discussion.

Q. Why must cost recovery be accelerated?

A. An example might best show the need: The Company currently estimates that services, the connections between customers and mains, will have an average service life of between 55 and 60 years, depending on the type of pipe used. However, it is clear that between now and 2050, only thirty years from now, many customers will terminate gas service and switch to cleaner, cheaper, more sustainable alternatives, such as heat pumps. As those customers switch away from gas, their services will be re-

tired before expected. (Note: "Services" only have one customer. They are not shared assets.) Thus, the actual Average Service Life for these abandoned assets will be less than that which was historically common. In order to ensure that stranded assets (including un-depreciated assets not in use) do not accumulate, it will be necessary to reduce Average Service Life expectations in order to avoid an inter-generational cost-shift to the ever-declining number of future ratepayers.

The amount by which cost recovery is accelerated should be determined upon careful consideration of the glide-path for use decline. I recommend that the Company institute an open depreciation study which will include participation from public stakeholders to determine the best cost recovery processes and rates.

Q. Accelerated depreciation will increase rates. How can we reduce that impact?

A. Given the recent tax law changes, it has become clear that previous rate payers paid too much for deferred Federal Income Taxes. Given that we now know that public policy and law will result in Average Service Lives which are shorter than originally expected, it is clear that previous ratepayers under-paid for asset depreciation. Given that we have both over-payments and under-payments, it would make sense to use the over-payments by previous ratepayers to balance out, as much as possible, the under-payments by those same ratepayers. Thus, the Excess Deferred Federal Income Tax, which might otherwise be returned to ratepayers as lower rates, should be used instead to reduce the depreciation deficiencies which exist now and which will grow in the future.

Q. If cost recovery is accelerated, won't this create windfall profits for the Shareholders?

A. No. Accelerated depreciation does not create a profit for shareholders. It simply converts the shareholders share of asset values into an equivalent share of capital. If anything, one might argue the shareholders would prefer that cost-recovery not be ac-

celerated since doing so reduces the amount of equity which is subject to the allowed return on equity. However, failing to accelerate cost-recovery will result in stranded assets. Thus, the shareholders are better off getting their capital returned to them in a form that allows it to be reinvested in assets that are more likely to deliver acceptable returns over time.

Q. If the amount of equity invested in gas assets is reduced, how do the shareholders get their required return?

A. The Company will either have to find ways to invest the equity in non-gas assets or return the equity to its shareholders through dividends or share buy-backs.

Q. Other than gas assets, what could the Company invest in?

A. As noted before, the company isn't really in the business of selling gas. They make their money by investing in pipes in the ground. While the gas-pipe business is going away, the pipes-in-the-ground business need not. What needs to change is what is in the pipes. The Company should: **Pump heat, not gas.**

It is clear that district geothermal heating schemes have the potential to serve much of the need once served by gas space and water heating. Such systems are quite common in Europe and the Company has recently operated pilots exploring the technology. As has been shown, the key capital investment in such systems is that which is made in the ground loops and distribution pipes.

Like a gas system, a geothermal district heating system is heavily invested in pipes-in-the-ground. It should also be noted that the pipes required for district heating systems will have Average Service Lives which are similar to, if not longer than, the Average Service Lives of the gas pipes that the Company has traditionally invested in but no longer should. Unlike gas pipes, the installation of district heating systems fits

well within existing and anticipated law and policy.

Q. This all sounds complicated. Why can't we wait to deal with these issues later?

A. The longer we allow the accumulation of gas asset to continue, the more expensive it will be for ratepayers to support the gas system in the future. The key problem here is that once a new gas capital investment is made, its costs are essentially fixed over the remaining life of the asset. For instance, if the Company invests in a new service connection with a sixty-year life, it will recover one sixtieth of its up-front cost each year, plus the cost of financing the remaining un-recovered amount. Then, given that the Company uses volumetric rates to recover its costs of delivery, the fixed costs are spread over the number of units of gas sold each year.

Q. Explain why fixed costs are such a problem when sales decline.

A. As long as the number of units of gas sold remains unchanged, then the cost-recovered per unit will stay essentially the same each year until costs are fully recovered. However, if sales decline, then each year the fixed asset costs will be spread over a smaller number of units of gas than during the year before. That means that after a sales decline, the per-unit cost recovery will increase.

For example, if gas consumption were to decline by 80% between now and 2050, then the per-unit cost recovery for an asset installed in 2019 but still in use in 2050 would be 500% greater than it was in 2019.[12] This is because, in 2050, the fixed costs would be spread over just 20% of the number of units sold in 2019. If the decline in sales by 2050 is only 50%, then the per-unit cost would increase by a still substantial 200%.

Q. What is the likely result of the per-unit cost increases that will be driven by reduced gas sales?

[12] Compute the increase by dividing one by one minus the percentage decline. In this case, the formula would be: $\frac{1}{(1-.80)} = \frac{1}{.2} = 5$

A. The result will be a gas **"death-spiral."**

As gas' per-unit costs increase, alternatives to gas will become more competitive and those ratepayers who have the capital or credit needed to invest in alternatives to gas, such as heat pumps, will abandon gas and find shelter with the cheaper alternatives. However, by abandoning gas, they will drive sales even lower and that will increase the per-unit costs to the point where even more ratepayers decide to abandon gas. The cycle then repeats. As the per-unit costs increase, gas sales will decrease even faster, forcing the per-unit costs increases to accelerate. This is referred to as a "death-spiral."

In time, we will find that gas will become known as the "poor-man's fuel" since the remaining gas users will increasingly be those who can't afford to abandon it — no matter how much they might wish to. Low and moderate income ratepayers, who have the most limited financial resources and the greatest need for more affordable energy, will end up carrying the burden of the stranded assets left behind by those who are financially more fortunate.

Q. Does that conclude your direct testimony in this case?

A. Yes.

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REBUTTAL TESTIMONY OF BOB WYMAN

September 18, 2019

Introduction

Q. What is your name?

A. I am commonly known as Bob Wyman. My formal name is Robert Mark Wyman.

Q. Are you the same Bob Wyman who submitted direct testimony in this case?

A. Yes.

Q. What is your purpose in submitting this testimony?

A. I wish to comment on several aspects of the direct testimony submitted by other parties in this case. My rebuttal testimony is intended to ensure a more complete record of the relevant issues and to advocate for policies that will be more fair, more in the public interest, and more likely than those proposed by the Company to conform to existing State and New York City laws and policies.

The issues that I wish to discuss include:

- Staff opposition to the proposed "Expanded Geothermal Demonstration Pilot Program"
- New York City's testimony concerning potential stranded asset risk.

Staff Opposition To Expanded Geothermal Demonstration Program

Q. When and why was the"Geothermal Demonstration Pilot Program" begun?

A. Accepting "Staff's recommendation that the Companies [KEDNY, KEDLI] should be required to explore the use of geothermal technologies to improve sustainability and overall system efficiency by displacing peak gas consumption rather than by address-

ing those needs through the addition of pipeline capacity,"[1] the Commission's December 16, 2018 order directed that KEDLI should "implement a pilot program to demonstrate geothermal heating and cooling as an alternative to either new or existing firm or interruptible gas service."

As ordered, KEDLI initiated a Geothermal Demonstration Project "to test shared-loop ground-source heat pump ("GSHP") systems as a cost-effective heating and cooling alternative to natural gas infrastructure in the Glenwood Village community of Riverhead, New York."[2].

Q. Has the Geothermal Demonstration Project been successful?

A. The Companies testify that the project was, in fact, a success and has proven that "geothermal technology has the potential to play an important role in the suite of NPAs called upon to sustainably meet customer heating requirements."[3] Based on an analysis of program results, the company estimates that "each geothermal customer could realize an average annual energy cost savings of between $1,000 and $1,500 and nearly 6.75 metric tons of CO_2 emission reductions from the displacement of approximately 800 gallons of heating oil."[4] The Companies have not estimated the potential savings when natural gas is the alternative fuel.

Q. Please summarize the Companies' proposal for an Expanded Geothermal Demonstration Project.

A. The Companies propose to expand the existing program to include both KEDNY and KEDLI. They propose to own and operate the installed equipment and to charge participating customers a fixed monthly fee of approximately $440 per year to recover

[1] CASE 16-G-0058, *ORDER ADOPTING TERMS OF JOINT PROPOSAL AND ESTABLISHING GAS RATE PLANS*, Dec 16, 2016. pp. 121-122

[2] The Future of Heat Panel, pp. 50-57

[3] The Future of Heat Panel, pp. 52.

[4] The Future Of Heat Panel, pp. 53.

the costs of engineering, design and installation.

Q. Why does Staff oppose expansion of the Geothermal Demonstration Project?

A. In it's testimony,[5] Staff provides three reasons for opposing expansion:

1. "The existing Geothermal Demonstration Project is just entering the data analysis phase and has not yet been proven to be cost-effective."
2. "KEDNY and KEDLI have not demonstrated that utility ownership of a geothermal system would not adversely affect third-party market participants that may be able to provide additional affordable alternatives."
3. "[T]he Companies have not performed a solicitation, such as for a non-pipeline alternative, seeking third-party developers to demonstrate that there is a need for utility ownership."

Q. Is Staff's first objection, concerning un-proven cost-effectiveness compelling?

A. No. The existing Geothermal Demonstration Project, because of its small size and limited operational experience, cannot be expected to have already produced compelling cost-effectiveness data.

It should be recognized that the existing demonstration has only explored the utility of shared geothermal loops in one of a very large number of housing stock types.[6] Even if this one project had been shown to be either cost-effective or not cost-effective, the results from such a limited demonstration would not actually be useful in assessing the wisdom of expanding such efforts to other types of housing stock.

In order to usefully determine if, where, and when shared loop geothermal systems are cost-effective, it will be necessary to pilot such systems in a wide-variety of con-

[5] DPS Staff, *Staff Efficiency & Sustainability Panel*, pp. 52

[6] i.e. The first pilot was installed for a cluster of manufactured homes. While it is important to study this type of housing, such housing comprises only a small percentage of the housing which exists in the Companies' service territory.

ditions and over periods of time long enough to ensure that a good sampling of operational results during a variety of weather patterns can be had.

Ideally, the Company would, as part of their proposed expansion, commit to ensuring that additional demonstrations would be sited in such a way as to explore cost-effectiveness in a variety of situations not yet explored. Some candidates might include:

- Multi-family housing
- Row houses in dense urban environments
- Mixed-use systems which combine commercial heating/cooling loads with residential heating/cooling loads.
- Systems that rely on river, lake, ocean, sewage or waste water as a heat sink/source rather than ground water
- Buildings currently using gas or clusters of buildings which include one or more currently using gas.

While Staff's concern for cost-effectiveness is undoubtedly well-intentioned, restricting expansion of this program at this time prevents the gathering of precisely the cost-effectiveness information that Staff rightly and properly seeks.

Q. Is Staff's second objection, concerning potential interference with the competitive market compelling?

A. Yes and No. Certainly, we should all be concerned whenever a regulated utility seeks to expand its scope into a market which might otherwise be served by others. However, in this case, it is clear that there are currently virtually no "third-party market partipants" whose efforts might be "adversely affected" by an expansion of the existing Geothermal Demonstration Program. Thus, it is too early for this concern to

command our actions in this case.

In fact, the experience and learning that would come from an expanded program might provide the knowledge needed in order to convince third-parties to get into this market. The more successes the Companies accumulate and the more evidence they provide that shared geothermal systems cost-effectively address market needs and that such systems can be profitably owned and operated, the more likely we are to need to address the concern expressed by Staff. On the other hand, a clear demonstration by the Companies that investment in such systems is not prudent, the more likely we are to find that third-parties don't waste their efforts entering a market that holds no promise. Alternatively, the Companies' inability to make such systems work might identify specific issues that others would then address using innovative methods.

Q. How should we address Staff's concern for interference in competitive markets?

A. While there are virtually no third-party participants in the market for shared geothermal systems, there is a small but growing number of companies that currently provide geothermal systems for individual buildings or for collections of adjacent buildings (e.g. Campus-wide systems) having a single owner. The entrance of the Companies into this market for single-owner buildings could have dramatically negative impacts. Thus, It would be very appropriate, in fact essential, to prohibit the Companies from competing with, or interfering with, these existing market participants. The Companies should be restricted to providing shared systems which serve buildings owned by multiple owners.

Even though there are few third-parties in the shared geothermal market today, it is possible that the Company might use means available to it as a regulated utility

to offer systems at rates unachievable by other potential market participants. This would tend to discourage potential third-parties from considering market entrance. (For instance, if the Companies were allowed to recover the costs of such projects through the ratebase shared by all of its other customers, it might be able to amortize costs over periods of time longer than is possible for others while also allocating costs to program non-participants.) Thus, the Companies should be required to ensure that the costs associated with installing systems in the demonstration program will be recovered exclusively from participants in the demonstration program.

Additionally, it would be reasonable to clearly state that approval of an expansion of the demonstration program in this rate case will not imply that future rate cases might authorize continued growth of the program into something other than a demonstration. During each future rate case, careful consideration should be given to determining if continued market participation by the Companies is likely to provide an unwanted interference with the development of a competitive market for shared geothermal systems. In fact, the Companies should be warned that if, at some future time, it is found to be undesirable for them to continue providing the services installed during demonstrations, they may be required to sell-off to third-party market participants any previously installed systems.

Q. Is Staff's third objection, concerning the lack of a solicitation compelling?

A. Somewhat. While the benefits of open solicitations need not be enumerated, it should be recognized that one currently un-stated goal of these demonstrations should be to assist the Companies, the Legislature, and the Public Service Commission in determining whether or not is practical and desirable for utilities to engage more fully in the shared geothermal market. This is particularly important given the expected future decline in natural gas use and the potential reduction in opportuni-

ties for the Companies to either invest in natural gas infrastructure or to continue the employment of those currently on their staffs.

Some argue, as I do, that the Companies should be permitted to shift their focus from the expansion of their natural gas infrastructure to a focus on district heating systems, preferably shared geothermal systems. In brief, while they may stay in the business of providing and operating "Pipes in the Ground," they should learn to **Pump Heat, Not Gas** and become "Thermal Utilities." This transition is necessary, in part, since if the Companies cannot find an alternative to natural gas infrastructure expansion, then they can only be expected to do all in their power to thwart efforts to reduce the consumption of natural gas and thus compromise achievement of the goals established by the Climate Leadership and Community Protection Act as well as NYC Local Law 97-2019. At the same time that we seek a managed process of decapitalization of the Companies' gas assets, we should also ensure that they have a means to profitably deploy their investors' equity through a recapitalization based on district heating assets. Balancing decapitalization with recapitalization is necessary to ensure that all stakeholders interests are best served.

Given that shared geothermal systems are not simply "things we want installed," but also potentially core elements of the Companies' future business model, we should ensure that the Companies are not limited in their ability to retrain existing employees[7] or contractors in order to install the demonstration systems. Any experience they gain in doing so will inform future decisions not only of the Company but of the Commission, Staff and parties to future rate cases.

[7] Ideally, the various unions whose members are employed by the Companies, or their contractors, would become active participants in planning, developing and implementing such retraining programs. Similar unions in Europe have accumulated extensive experience in providing properly trained labor for such projects. One hopes that some knowledge transfer could be facilitated.

Q. What should happen once the demonstrations deliver useful data?

A. Given that district heating systems and shared geothermal systems have proven to be successful and popular in Europe, in areas not that different the urban and near-urban sections of the Companies' territory, it is expected that the demonstration pilots will provide convincing evidence that such systems can provide a viable alternative to gas and oil based systems here. At that point the Commission will need to answer some important questions, including:

- Should utilities be allowed to install shared geothermal systems in non-demonstration programs?
- Should third-party providers of shared geothermal systems be regulated?

I suggest that the answer to both of these question is "Yes." I also recognize that these are not questions that can be resolved in this rate case.

Q. Why should we allow utilities to invest in shared geothermal and district heating systems?

A. As stated elsewhere, I believe that we must provide a pathway to future profitability for the existing gas utilities who have served our interests in the past. As we require that utilities reduce their investment in gas expansion, we should provide them with an alternative investment that will allow them to provide their shareholders with confidence that their investments will continue to afford reasonable returns in the future. We should also show a pathway to a future business model that will avoid the high cost of capital and low credit ratings that are typical of businesses in decline and which would create significant burdens for any remaining gas customers.

Q. Why should the shared geothermal and district heating markets be regulated?

A. Historically, it has been found that the management and development of distribution systems serving multiple customers, such as those for water, gas, electricity, tele-

phone calls, cable TV, and steam, present complex issues that require resolution by a neutral third party dedicated to the public interest. Thus, these "public services" are now regulated by the Public Service Commission. In time, I'm confident that we'll find that district heating systems, including shared geothermal systems, will present much the same kind of issue that has justified regulation of other shared distribution systems in the past. Given this, I suggest that it would be appropriate to begin now the discussion of how such shared assets should be regulated and when such regulation should begin.

While I suggest that PSC regulation of shared systems will probably be found to be necessary, I do not propose that such regulation should be considered appropriate for systems installed for the benefit of individual owners.

New York City's testimony concerning stranded asset risk

Q. To what testimony do you refer?

A. The "New York City Policy Panel" testimony, starting on page 25 and continuing until page 31.

Q. Do you agree with the City's proposals concerning stranded asset risk?

A. Yes. But, I think the City understates that significance of the stranded asset risk. They say:

> "Given the goals of deep decarbonization and the clean energy transition, the composition of utility infrastructure will change over time. In particular, the need for the Companies' extensive gas infrastructure *may* diminish." (emphasis added.)

While I recognize that the City may have sought to be "gentle" in its illumination of

the issue, I believe the reality is that, given current law and policy, it is a certainty that the need for the Companies' extensive gas infrastructure *will* diminish.

We do stakeholders no service by sugar coating the implications of current law or shying away from stating clearly the consequences of our actions to ensure increased energy efficiency and a reduction of greenhouse gas emissions.

Q. What is it about their proposal that you support?

A. I strongly support the City's call for studies of stranded asset risks and for their proposing the preparation of:

> "An assessment of whether it is technically feasible to not replace the leak-prone pipe in certain neighborhoods without affecting other customers, such as at the end of pipeline segments or at the edges of the Company's gas service territory;"

While the City has provided a number of useful suggestions for similar projects to be studied, I suggest that the list should be expanded to include a study of the Non-Essential Gas Asset Retirement program recently described by Central Hudson.[8]

That program's goals are summarized as follows:

> 1. Non-Essential Gas Asset Retirement
>
> Central Hudson's TMA [Transporation Mode Alternative] projects are designed to strategically retire high-cost natural gas infrastructure by seeking high efficiency electric alternatives to replace natural gas end uses in customer's homes. For each identified area, performance will be measured based on the retirement of all prescribed gas services and mains. The conversion approach will vary by customer based on existing baseline condi-

[8] Cases 17-E-0459, 17-G-0460, *Central Hudson Gas & Electric Corporation's Non-Tariff Implementation Plan & Compliance Filing for Non-Pipe Alternatives: Three Transportation Mode Alternatives,* June 21, 2019

tion, but will primarily focus on high efficiency cold climate air-source heat pumps and electric heat pump water heaters. Other appliance types will be considered based on the existing natural gas end uses for each home.

CHGE may, in pursuing this program, provide important learning that could inform the Companies in pursuing similar efforts. Of course, it would also make sense for the Companies to consider geothermal systems, shared or not, as alternatives to the air-source systems which are the focus of CHGE's program.

Q. Does that conclude your rebuttal testimony in this case?

A. Yes.

A.L.J. COSTELLO: And we have a two-page affidavit on behalf of State New York Real Estate Services, signed by Jared Rodriguez, sworn to on February 10, 2020. And that will be marked as Exhibit 721. And the court reporter should insert the following files at -- at this point in the record: State New York Jared Rodriguez direct testimony and State New York Jared Rodriguez rebuttal testimony. All of these files should be inserted as if the testimony were orally given here today. #

TESTIMONY OF JARED RODRIGUEZ

Q. Would the witness please state his name and business address?

A. Jared Rodriguez and my business address is 40 West 57th Street, Floor 23, New York, New York 10019.

Q. Mr. Rodriguez, by whom are you employed and in what capacity?

A. I am employed by Estates NY Real Estate Services LLC ("Estates"), which I will also sometimes refer to as "LeFrak," as Assistant Director, Utilities Conservation Engineering.

Q. How long have you been employed by LeFrak?

A. I have been employed by LeFrak for 7 years.

Q. Please briefly describe your educational and business experience.

A. I attended Tufts University and received a Bachelor of Science of Civil Engineering in 2008, and New York University, where I studied real estate development, sustainable development and finance, receiving a Master of Science degree in 2012. I founded an energy and sustainability consulting firm where I assisted clients with their plans for renewable energy planning, development and finance.

Q. Please generally describe your current responsibilities.

A. My responsibilities as Assistant Director, Utilities Conservation Engineering include identifying ways to foster energy efficiency throughout the residential and commercial real estate portfolio in existing buildings and in new construction, planning capital projects related to

energy-related buildings systems and developing a procurement strategy for utilities of all types.

Q. Of what local and regional energy bodies are you a member?

A. Member, Sustainability Committee, Real Estate Board of New York (REBNY); Member, 80x50 Buildings Partnership, Urban Green Council; Member, NYC 2030 District Multifamily Retrofit Working Group; Member, and the Carbon Neutral Buildings Roadmap Technical Advisory Group, NYSERDA.

Q. For which intervening party are you submitting testimony in Rate Case 16-G-0059?

A. I am testifying on behalf of Estates NY Real Estate Services LLC.

Q. Have you previously submitted testimony on behalf of Estates or any affiliate or subsidiary of Estates before the New York State Public Service Commission ("Commission")?

A. Yes, I testified on behalf of Estates in Case No. 16-G-0059.

Q. Please describe Estates and provide an overview of its properties.

A. Estates is the owner/manager of a very large real estate portfolio in the New York metropolitan region and in South Florida, Washington State and California. The firm owns and manages over 18,000 residential apartment units and over 20 million square feet of commercial office and retail space.

Q. Is Estates a customer of National Grid (The Brooklyn Union Gas Company dba National Grid NY) which is also referred to as the "Company" in this testimony?

A. Yes, the Estates properties include 38 buildings ("Buildings") which are, in the aggregate, one of the National Grid's largest gas customers taking Firm Service.

Q. What is the total estimated annual average gas usage of the Buildings under the Firm Service Classification?

A. 5,557,710 therms, which is the average of the most recent 3 years of annual usage.

PURPOSE OF TESTIMONY

Q. What is the purpose of your testimony in this National Grid rate case?

A. The purpose of my testimony is as follows:

1. To demonstrate the value of natural gas demand response to the Company, the Company's ratepayers and to all the citizens of New York State;

2. To lend support for the furtherance and expansion of the geothermal Non-Pipe Alternative ("NPA");

3. To support efforts to decarbonize the New York City and State energy system via electrification of building equipment which contributes to peak gas demand and capacity constraints;

4. To lend support for the furtherance of the efforts by National Grid to introduce and develop more sources of biomethane and renewable natural Gas ("RNG") and Hydrogen Gas via a Green Gas Tariff and other efforts; and

5. To make recommendations regarding the SC-3 rate class which serves many low income and family customers in New York City.

DISCUSSION

Q. Please state give a general overview of recent energy/environmental legislation and other agency efforts in New York State.

A. Recent legislation includes the Climate Leadership and Community Protection Act, NYISO carbon pricing, NYC Local Law 97, and additional efforts to modify NYSERDA or utility-managed incentive programs to further buildings energy decarbonization and efficiency.

Q. How do you think these initiatives will affect Estates moving forward?

A. My expectation is that these initiatives will require large capital (or other) expenditures at properties to modify or replace building equipment to comply with regulations. Owners of properties from smaller multi-family residential to large commercial have substantial sunk costs related to natural gas infrastructure and heating plant equipment; which burns

natural gas as a primary fuel. It is important to consider and calculate the full lifecycle costs that Customers will face as decarbonization efforts progress. In addition, customers unable to comply with NYC Local Law 97 because of the age of the building or other factors may be burdened with additional costs such as purchasing renewable energy credits (RECs) so as to achieve deductions from the required annual building emissions.

Q. Are there any solutions which would continue to use natural gas or biomethane?

A. Yes. Decarbonizing the natural gas supply is a possible solution to avoiding the substantial waste associated with the abandonment of any portion of the natural gas infrastructure (which infrastructure has been increasingly adopted in recent years by Customers) which would be a desirable outcome of decarbonization efforts. Such solutions would include the use of renewables such as agricultural or diary industry-sourced gaseous fuels like biomethane or hydrogen as discussed below.

Q. Do you favor supplementing gas network decarbonization by heating electrification efforts?

A. Heating electrification should be given serious consideration. Electrification would assist in avoiding expanding the natural gas network with its capacity constraints. One favorable method of managing decarbonization by electrification is expanded use of electric domestic hot

water equipment. Specifically, natural gas demand for the sole purpose of producing domestic hot water leads to network capacity constraints during periods of cold weather. This is caused by the peaking nature of domestic hot water demand in the morning coinciding with likely high natural gas demand response periods.

Q. How would you respond to an argument that carbon-producing fossil fuels including natural gas are used to generate electricity which would produce additional pollution thereby reducing or neutralizing decarbonization via conversation of gas-fired hot water equipment to electric power? As a corollary, how would you respond to an argument that such electrification may well necessitate the construction of additional generation plants and an expansion of the electric gird to accommodate the additional usage and demand that would follow electrification?

A. Taking these arguments one at a time, atmospheric pollution by carbon-producing combustion is more easily addressed in a baseline electric power plant than at individual gas burner tips. Also, baseload electric generation plants in NYS include sources that do not use fossil fuel such as hydro, nuclear and utility scale wind and solar. As to the possible necessity for the addition of electric utility generation, transmission and distribution plant, I note that these would address peak usage and my intent in this testimony is that the combination of initiatives and technologies that I am recommending

in this testimony (when expanded to include electricity - such as electric demand response) would reduce peak usage of both gas and electricity such that, for this reason, any significant expansion of electric utility plant would be unnecessary.

Q. Do you see other alternatives to natural gas?

A. Yes, the conversion of equipment to utilize hydrogen gas, including blended forms, remains an alternative as I discuss later in my testimony.

Q. Is gas demand response an effective means of helping the State and City to reach its goals?

A. Yes, the Company's pilot gas demand response program should be significantly expanded to help address the capacity issues and enable the State, City and the Company to reach their respective goals for maximizing the use of the system and reducing the overall amount of fossil fuels consumed.

Q. How will these goals be achieved by the use of gas demand response?

A. Gas demand response allows the Company to maximize its gas network to bring additional commodity gas to its Customers, thereby offsetting carbon intensive liquid fossil fuels and reducing carbon emissions. A gas demand response program incentivizes Customers to limit usage at times of capacity constraints via direct payments to participating

Customers which incentivize them to limit gas use generally and especially during such periods. This, in effect, monetizes timely responses during system constraint periods. Customers may utilize electrification and gas-to-electric switching as a means of achieving demand response goals, which not only helps to alleviate capacity limitation issues but also reduces carbon emissions. Peak period gas consumption, most likely driven by domestic hot water demand, appears to me to be an appropriate target for electrification in the near term.

Q Are there approaches or technologies, in addition to electrification, that you believe would be helpful to achieving demand response goals?

A. Load shifting is another approach to reducing demand. An example of load shifting technology is being implemented by companies such as RadiatorLabs, which modify radiator enclosures to store energy and release it on demand. It is possible to avoid natural gas consumption during peak gas periods with RadiatorLabs' technology. Pairing this technology with advanced electrification technologies can yield significant reductions in gas capacity constraints. RadiatorLabs' technology appears to be a "bridge" technology that is able to yield significant reductions in natural gas peak load now, while the building sector shifts to low-carbon heating technologies.

Q. Do you have recommendations to the Parties in this Proceeding regarding gas demand response?

A. We urge the Parties to consider methods of enhancing or incentivizing gas demand response via electrification as a means of achieving decarbonization goals, reducing overall costs to gas ratepayers, maximizing gas network efficiency and utilization, improving local air quality, and fostering a market for the development of new methods or technologies. We also urge the Company to expand its existing pilot gas demand response program and ultimately make the program permanent.

Q. How does the development and implementation of geothermal NPA heating help Estates and the overall energy marketplace in terms of energy efficiency, carbon reduction and resiliency?

A. Geothermal technology offers highly energy efficient production of domestic hot water and space heating. The technology, particularly shared loop geothermal energy systems, will likely play a key role in reinforcing the Company's ability to meeting decarbonization goals via heating electrification and reducing peak gas capacity. Without addressing potential legal steps necessary for the Company to engage a line of business producing and selling geothermal energy, the Company may be able to develop alternative revenue sources with an affiliated thermal company selling thermal energy as a means of offsetting possible losses incurred by

an overall reduction in fossil-based natural gas consumption. Alternatively, or concurrently, the production and sale of geothermal energy would also provide private businesses with opportunities resulting in job creation in New York State. Geothermal technology helps Estates NY meet regulatory requirements in that it provides a cost-effective low- to zero-carbon source of heat and remains the only likely technologically feasible alternative for the wholesale electrification of large multifamily and commercial buildings.

Q. Can you provide an explanation of a thermal company?

A. A thermal company is a company that builds, maintains and manages shared infrastructure (which would include locations in public rights of way and on private property) that provides a heating and cooling medium for water source heat pumps or similar technologies. A thermal company's distribution facility is most likely comprised of a horizontal water pipe network, heat exchangers, pumps and pumping stations, and physical connections to customers and heat sources or sinks. The thermal company, would provide a platform for sharing thermal energy across various customers, and might offer the possibility of establishing a bi-direction thermal distribution system similar to the bi-directionality of the electric distribution network envisioned by New York States Renewing the Energy Vision (REV) plan.

Q. What other benefits might a thermal company provide?

A. A thermal company may also provide a platform for new business ventures that provide heat sources or sinks via vertical bore fields (geothermal), thermal storage media, or via industrial waste. The New York City Department of Environmental Protection ("DEP") recently established an Office of Energy and Resource Recovery, which may facilitate recovery of energy resources via thermal energy extraction from the sanitary sewer network or production of Renewable Natural Gas from DEP-managed wastewater treatment facilities. If any applicable legal hurdles can be surmounted and this can be accomplished by the Company, such a model would be complementary to the Company's goals as expressed in the Future of Heat Panel's direct testimony, the Company's natural gas business, the State of New York's decarbonization goals and the goals of the Department of Public Service. I strongly encourage the Commission to look into establishing a regulatory framework for regulated monopolies to establish thermal company affiliates. Such a framework, either in conjunction with utility affiliates or independently, would provide new private business opportunities and hold the possibility of significantly reducing cost risks for Customers in light of the State's decarbonization goals.

Q. How would the development of more biomethane and RNG help the City and State reach their stated goals?

A. Renewable natural gas and hydrogen blending help New York City and State achieve decarbonization goals by moving natural gas toward a regime harmonized with the Renewing the Energy Vision (REV) framework. We strongly recommend that the Commission recognize the need for a gas REV framework to foster gas network bi-directionality, and an increase in local biomethane, RNG and hydrogen suppliers and sources, and to recognize the gas network's potential as a large-scale and seasonal energy storage and delivery system. Rethinking the gas network is vital to the State and City achieving its goals. Providing a pathway to, and a regulatory framework for, the Company forming a thermal company affiliate is desirable to in connection with natural gas being folded into REV.

Q. Please describe a typical SC-3 tenant and how increases in the SC-3 rate impact these families?

A. Estates NY's building stock is primarily comprised of rent stabilized apartments regulated by the New York State Department of Homes and Community Renewal "(DHCR"). Therefore, a high proportion of our tenants are low- and moderate-income residents. Higher natural gas and electric costs naturally burden each property, creating upward pressure on rents for those residents who do not reside in rent stabilized apartments. The Company's rate increases have typically outpaced rent increases,

reducing funds necessary to maintain these buildings and provide resident amenities.

Q Please discuss the current SC-3 rate and the potential problems with its level.

A. The Company is proposing an increase in base distribution rates of over 30% in the tail block for the SC-3 customer class. The overall increase in proposed bills to the typical SC-3 customer is pegged at 11%. (Exhibit RDP, Schedule 4 Page 8 of 18.). It is important for the Parties in this case to realize that this massive increase in base rates is hidden by low commodity costs which dampen the overall impact of the rate increase. There is no question that the natural gas market has been very favorable to the consumer over the past ten years. On the other hand, to expect that commodity costs will continue to stay low going forward and yet seek the proposed increases in customer bills is inconsistent. It is therefore important to call out the real increase here for the SC-3 customer will be in the range of 25% on a typical bill. This increase is on top of a greater than 25% increase in base rates agreed upon in the last rate case. This means that base rates will have increased by over 50% in less than four (4) years for this rate class, which primarily consists of low- and average-income family customers in New York City.

Q. What could happen to the overall bill impact if natural gas prices were to return to 10 year or even twenty-year averages?

A. If gas prices were to rise to even five-year averages, the average bill would then increase by over $1.00 per MMBtu as a result of the purchased gas passthrough mechanism which shifts all gas cost risk to the customer. Should this return to more historically normal gas prices occur, the overall bill impacts of this rate increase would be much larger.

Q. What is your recommendation to the Commission on the level of Rate SC-3?

A. The Commission should not be misled by the current environment of low gas prices in thinking that the rate increase sought by the Company is likely to be limited to only a 10% increase in rates. The Customers will bear ALL the risk associated with potentially higher gas commodity costs and, given the uncertainty of deliverability into New York during peak periods as well as the uncertainty associated with the price of the commodity at the wellhead, any projection of similar gas cost going forward is, at best, speculative. The Commission must understand that the increase in base rates for this class of customer is exorbitant unless the goal of the Commission and other New York State agencies is to raise the cost of natural gas delivery in order to shift customer behavior and enhance decarbonization goals. If this is an accurate assessment of the

future of heat and/or gas distribution in the state, the Commission must make provisions for Customers to feasibly and economically convert buildings to electrified heat or other low-carbon alternatives.

Q. What other concerns do you have related to Company's provision of natural gas delivery to your facilities?

A. If the goal of the Commission and New York State is to reduce the size of the Company's Customer pool, the Commission must recognize that, should this occur, it would necessarily increase the cost to remaining Customers via the Company's regulated recovery of fixed costs related to maintaining and/or expanding natural gas delivery capacity in the KEDNY service territory. If this conclusion is accurate, I urge the Commission to immediately cease approval of additional fixed cost acquisition by the Company and provide the Company access to additional Customers via new business models such as shared loop geothermal or thermal company ventures. Increasing or maintaining the size of the Customer pool would provide necessary price stability for those Customers that remain connected to the natural gas distribution network.

Q. Does that conclude your testimony?

A. Yes

REBUTTAL TESTIMONY OF JARED RODRIGUEZ

Q. Would the witness please state his name and business address?

A. Jared Rodriguez and my business address is 40 West 57th Street, Floor 23, New York, New York 10019.

Q. Are you the same Jared Rodriguez who filed Direct Testimony on behalf of Estates NY in this National Grid Rate Case?

A. Yes.

Q. For what reasons are you filing Rebuttal Testimony in this case?

A, I believe that it is important that the Commission, in this rate case, fully take into consideration the many viable opportunities to improve the long-term energy future of New York State in this proceeding. Specifically, Estates believes that the Future of Heat Panel has laid out three particular strategies/programs with significant merit that could economically and effectively lower GHG emissions across the service territory.

Q. What are these three programs?

A. Expanded Gas Demand Response ("DR"), creation of a renewable natural gas ("RNG") tariff - "Green Gas Tariff" - and development and expansion of the geothermal shared loop program – the Geothermal Demonstration Project. Each of these programs has the potential to lower GHG emissions, provide economic benefits to the Company's customer and more efficiently utilize the Company's current plant in service. I detailed the reasons for our support of these programs in my initial testimony. I would like to add that

the Commission should endeavor to have its goals and directives substantially align with the long range plans of New York State. In that regard, it is very important for the Commission to give the Company the tools necessary to decarbonize and shift to low-carbon business models like shared loop geothermal, otherwise the objectives of the Climate Leadership and Community Protection Act of 2019 will not be met.

Q, Upon review of the direct testimony of the other participants in this case what were the major points of agreement that you could find?

A. I was happy to see that Staff, in the Prepared Direct Testimony of the Staff Efficiency and Sustainability Panel ("DT E&S Panel") in Cases 19-G-0309 & 19-G-0310 (Pgs. 40-48) and the Staff Policy Panel ("DT Policy Panel") (Pg. 26, Lines 6-8) regarding the expansion of Companies' two DR programs, support the Company's proposals in principle for Demand Response.

Q. What concerns do you have regarding such Staff testimony?

A. I am concerned that, while the Staff supports the above two DR programs in principle, Staff did not fully support them by allowing the expenses for the necessary FTE's to support these DR programs to be collected in rates (DT E&S Panel Pg. 48, Lines 1-4, 14-16).

Q. Did you also have concerns regarding Staff's testimony regarding the proposed expanded Geothermal Demonstration Project?

A. Yes. In its testimony, Staff expresses a negative view towards the Geothermal Demonstration Project and neither supports that project nor the addition of a proposed incremental FTE (DT E&S Panel Pg. 52, Lines 18-24; DT Policy Panel, Pg., 25, Lines 15-19 and Pg. 26, Lines 11-13).

Q We believe that a failure to fully staff or support these programs in the rates to be implemented in this rate case will cause these program to be expanded too slowly or not at all at a time when New York State needs clear and decisive action with regard to rapid decarbonization as well as reliable natural gas deliverability right now.

Q. Would you like to elaborate on the concerns you have expressed?

A. Yes, as to my concern that Staff's direct testimony was not supportive of National Grid's proposal on the development of geothermal shared loops, this technology is viable right now and has been demonstrated across the country as well as in New York City. We need to further develop new business models to meet our GHG goals as a state. The current reliance solely on electrification via air-source heat pumps is myopic and we should be pursuing all forms of thermal energy to reduce our reliance on fossil fuels to heat our homes and businesses. To accomplish the worthy goal of bringing this technology to the fore, we can see no better partner than National Grid, the Company with the most experience of delivering thermal energy underground.

Q. Should the Company be granted regulated monopoly status with regard to implementing geothermal shared loop systems?

A. Yes. When geothermal loops are shared by separate private entities for the purpose of accessing remote thermal "sinks or sources" like geothermal well fields or industrial waste heat, and when these shared loops are located in public rights of way, the Company should function as a regulated monopoly in a similar fashion to its natural gas delivery regulated monopoly status. Granting a shared loop thermal energy business model to the Company provides a number of significant benefits to the service territory and to the people of New York State. These benefits include but are not limited to: (1) acceleration of the electrification of heat; (2) reduction of risk to the Company and to rate payers related to the acquisition of stranded assets in the form of natural gas infrastructure; (3) radical reduction of financial, regulatory and logistical barriers to end-user owners of geothermal heating and cooling systems; and (4) provides a platform for market transmission and the development of new economic activities by private sector actors as thermal energy providers. These concepts expand on my previous testimony as it relates to the development of a regulatory framework for thermal energy within the context of New York State's Renewing the Energy Vision. Over the medium and long-term, granting regulated monopoly status to the Company for the purpose of developing and maintaining shared thermal energy infrastructure (i.e. geothermal shared loops)

is the most logical next step in decarbonizing energy for the benefit of the environment.

Q. Does that conclude your testimony?

A. Yes.

A.L.J. COSTELLO: And I just want to make sure I -- is there any party who had testimony or affidavits that I did not address? No? Okay. Good. Do you want to talk about anything we have to do with exhibits?

A.L.J. LEARY: Okay. I'm going to complain, sorry. There are a couple things on the exhibit list that really need to be addressed. And I'm going to point out the parties that need to address them. And by and large, I mean, you know, almost 700 exhibits, this was actually a good job by everybody.

KEDNY and KEDLI, the companies, Exhibits 335 and 336, 350 and 351, 355, 366, 367 -- that's it -- there is inadequate information in the description of those exhibits for us. So for example, they will say quote, "Relevant IR responses." Many of the parties properly included which IRs and gave some kind of description. You need to fill those in to tell us what they actually are. Quote, "Relevant portions of company IR responses" isn't -- isn't really helpful for us to be able to have ingress and egress from this exhibit list, and that's what we're looking for.

So do you need me to repeat those numbers?

MR. O'BRIEN: No, we have them.

A.L.J. LEARY: Great. Okay. DPS staff, 413, 416, 424, 475, 476, 477, and 478, 492, 510, 516, 518, 520. Same -- I'm sorry. Not done. Oh, yeah, that's it. Same comment. I need some description in those. I'll sug -- my suggestion as to how we deal with this will come at the close. I just want to get through.

New York City, Exhibits 536, 540, 548. This looks like it's DPS staff and the company. At the end of the exhibit list, we have multiple parties. It would be Exhibits 596, 597, and 599. Again, these are generally relied upon responses to information requests that are not identified on this exhibit list. And we -- we are not going to have the time to go back and try to piece this together. So I'm going to ask the companies to -- or the company, and New York City, and DPS staff to advise us what these IOR responses are that need to be identified.

This is what I'm going to suggest, a simple email to Judge Costello and I with the exhibit number.

You're the entity offering the exhibit, the description completed -- I'm sorry. The exhibit number, the DMM item number, which should be right on this list already. Actually, I'm -- I'm going to take that back. You don't need to do the DMM item niter -- number. And you don't need to do the pre-filed designation number. You just need to deal with the description.

So give me the exhibit number. Give me the revised description that identifies what we're talking about on those exhibits. And we will fill in the -- and complete the exhibit list, including any of the exhibits that were offered newly during the testimony of the hearing.

And that would begin, I believe, with DPS Exhibit 599. And go through Exhibit 721 that Judge Costello just covered. Any additional exhibits being proffered by any party?

MR. GOODRICH: So --

A.L.J. LEARY: I'm sorry, go ahead, Mr. Goodrich.

MR. GOODRICH: Oh, I'm sorry. I thought that was a question.

A.L.J. LEARY: No, no. It is a

question. It is a question. I was just going to propose. I wasn't asking, because we know that there are folks thinking about this. What we were going to suggest is a protocol to make that happen in the event that any party would like to offer something. So we want to keep it as informal as possible, but we need some basis to -- to rule on whether it should go in or not and to have other parties be given the opportunity to object. So the protocol -- what -- what were you going to --

MR. GOODRICH: Well, I was going to say, there are a number of IR response that various parties have stipulated to, while there were a couple that we sort of put in -- in -- en -- en masse. There are some that like, for example, we circulated the -- the responses to a few IRs provided by Maureen Hefei this morning. And the company had noted -- had -- had -- had her lay the foundation for it in -- in their affidavit from her. But I don't believe they've been given any exhibit numbers. And I don't know if there's others. I think there are others from other parties. Those are the ones that we were dealing with.

A.L.J. LEARY: I know UIU has the

potential for offering something. Okay.

MR. GOODRICH: But I mean, these are ones that are currently sort of being offered.

A.L.J. LEARY: Stipulated.

MR. GOODRICH: But we haven't given them exhibit numbers, I don't think.

A.L.J. COSTELLO: Do you which -- which documents, specifically?

MR. GOODRICH: So we provided them this morning. It's -- if you can pull up the email or -- you have it. It's IR responses to an -- certain IRs that Ms. Hefei responded to. He's going to provide the -- the numbers. And the company's affidavit for Ms. Hefei included the -- some language to the effect of that -- that she provided those responses.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: Well, we'll mark that as Exhibit 722.

MR. GOODRICH: Well, there's --

MR. FORST: Yeah, so if -- if you refer to the email, there's four actual responses. That'd be DPS 1018, DPS 1052, and DPS 1084, which also has a confidential supplement, which is another

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document. While those have been shared during discovery, they are referred to in Ms. Hefei's affidavit in paragraph three as being sponsored by the company's witness, Ms. Hefei.

MR. GOODRICH: So I guess what we would ask is if we could e -- identify an exhibit for each of those four.

A.L.J. COSTELLO: So for DPS 1018, it would be Exhibit 722. DPS 1052 would be 723. 1084 would be 724, and the confidential material too, 1084, will be 725. Does that work?

MR. FORST: That works for us, Your Honors, thank you.

A.L.J. LEARY: I would suggest in correcting those parts that I -- or those exhibits that I identified, that at the end of that email -- you can do this by email -- you just reiterate those four exhibit numbers just so that it squares with what we have here. So we're up to exhibit number 725, which will be DPS IR 1085, confidential.

A.L.J. COSTELLO: So 1084?

MR. FORST: Just for clarity, that's DPS --

A.L.J. LEARY: Oh, I'm sorry, 1084.

MR. FORST: For supplemental, which is a confidential document. Thank you.

A.L.J. LEARY: But you're also putting in the redacted?

MR. FORST: So there, the company did not provide a redacted response document. The response in the attachments itself are deemed confidential.

A.L.J. LEARY: Okay. So you're designating 1084 as one exhibit number and the attachment?

MR. FORST: No. So it's a little bit complex. Response DPS 1084, which has provided Exhibit number 724, that's fine. The company then offered a supplemental response with attachments, which then they claimed confidentiality over. And that is being submitted as Exhibit 725. So it's not --

A.L.J. LEARY: Got it.

MR. FORST: -- same response or a redacted confidential version. It's a supplemental response.

A.L.J. LEARY: It's the initial response and the supplemental response.

MR. FORST: That's correct, Your Honor.

A.L.J. LEARY: So let's call it that. But just put those at the end of your clarifying these other exhibits.

MR. FORST: Understood.

A.L.J. COSTELLO: I just want to -- we have to admit them. So I'm going to --

A.L.J. LEARY: Does anybody else have any other exhibits that, like DPS staff, they'd like to offer at this time --

MR. DECICCO: Y -- Your Honors, we have --

A.L.J. LEARY: -- by stipulation?

MR. DECICCO: -- we have the five exhibits that we've agreed with UIU to -- to update form our Rates Design Panel. So we would request five exhibit numbers, and I can tell you what -- what we intend to update.

A.L.J. LEARY: Okay. Go ahead, shoot.

MR. DECICCO: It's Exhibit RDP3, which is the embedded cost of service study.

A.L.J. COSTELLO: Okay. That will be Exhibit 726.

MR. DECICCO: It's RDP4, which is the rev -- revenue allocation rate design and bill impacts.

A.L.J. COSTELLO: That will be Exhibit 728.

MR. DECICCO: RDP6, which is the merchant function charge update.

A.L.J. COSTELLO: That will be Exhibit 728.

MR. DECICCO: RDP12, which is an embedded cost of service study using 100% demand allocation for mains.

A.L.J. COSTELLO: That will be Exhibit 729.

MR. DECICCO: And RDP16, which is work papers associated with the above e -- exhibits.

A.L.J. COSTELLO: And that will be Exhibit 730.

MR. FORST: Just for my sanity, those are the ones that are being updated to reflect the third supplemental?

MR. DECICCO: Correct.

MR. FORST: Okay.

A.L.J. LEARY: So again, I would

appreciate if you would note in your email that corrects the exhibits, I identified as not having sufficient descriptor language, that you put at the end of that email these new exhibits that we've assigned today, Exhibits 726 through 729 with the description.

A.L.J. COSTELLO: 730.

A.L.J. LEARY: I'm sorry, 730, the work papers. Whoops -- with the description, the date. Just go back to the exhibit Excel spreadsheet and utilize those columns that are there for these new exhibits today. Anyone else?

MS. MURPHY: Your Honors, this is Erin Murphy on behalf of EDF. I -- this might be a bit more complicated than just an exhibit. But I appreciated Judge Leary raising earlier the question about the 1990 emissions numbers. And Patrick with the company circulated that spreadsheet. So we would be interested in getting that into the record, and I'm not sure if we need to do like an affidavit from the company or if we can just number that as an exhibit as is.

A.L.J. LEARY: It was subject to the Panel's testimony. I'm -- I'm going to ask if you're

waiving any cross examination associated with that testimony. I mean, this may open up a Pandora's box. And the answer's no. Sorry.

A.L.J. COSTELLO: The answer's yes, you're waiving. And even if you're not, the answer's no, you're not getting cross. Re -- the record should reflect my own laughter at this. Okay. So what's the company have to say about this?

MR. O'BRIEN: We can stipulate to admitting it.

A.L.J. LEARY: All right. That's 731?

A.L.J. COSTELLO: 731.

A.L.J. LEARY: Let's give that 731. I'm going to ask EDF to provide an email that would contain the information noted on the Excel spreadsheet. Under DMM number, you're not going to have that. Do we get this into DMM somehow? I don't think we need to, but you don't need a DMM number.

A.L.J. COSTELLO: We don't need the number. We can deal with that.

A.L.J. LEARY: We'll deal with that. Good. What else? That's it. Any other party have exhibits that -- UIU, I think we took care of this through the company offering those exhibits.

MS. O'HARE: Yeah, and just to clarify, I believe the first couple of exhibits, the company was talking about where it talked about on the 19th about the bill impacts issue that wasn't specifically to a UIU request. And it was RDP12-R that's being updated pursuant to the UIU request.

A.L.J. LEARY: Okay. The company will take care of that in the description, I assume. Anyone else?

A.L.J. COSTELLO: L -- let me --

MS. O'HARE: And just to note for the record, we do have a discovery request that we sent out yesterday, and we're reviewing that. And we might potentially request that that be added to the record, but we were still deciding.

A.L.J. LEARY: We're going to cover that next.

MS. O'HARE: Okay. Thank you.

A.L.J. COSTELLO: Yeah, let me just admit what we have and --

A.L.J. LEARY: Yeah.

A.L.J. COSTELLO: -- and just find out. All right. So we have pre-filed exhibits we had marked one -- numbers one through 598. There were not

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MR. GOODRICH: I do have one.

A.L.J. COSTELLO: Okay.

MR. GOODRICH: To hearing Exhibit 666. I'll let that go.

A.L.J. LEARY: We -- I -- actually, I -- I think that we gave that a number, but we never -- during the hearing, this was your -- what you prepared Mr. Molloy?

MR. MOLLOY: Yes.

A.L.J. LEARY: I think that's right.

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: I -- I -- the company hasn't offered it. And are -- they're stipulating now.

A.L.J. COSTELLO: I think -- yeah, I think we marked it, but it was --

MR. MALONEY: We -- we marked it. And we --.

MR. GOODRICH: So 666 will not be admitted.

A.L.J. COSTELLO: It will not be admitted.

Okay. Anything further with respect to objections?

MR. GOODRICH: For -- forgive me for prolonging this. I just have one question. As I was looking through the -- our list of the -- the exhibits done at the hearing, 613 and 614 are stated as appendix to KEDNY surcharge, and appendix to KEDLI surcharge. And this is our own attempt to write down what it was. And I know that copies were not provided at the hearing, and I cannot for the life of me figure out what that is.

A.L.J. LEARY: 613 and 614?

MR. O'BRIEN: Yeah. So it -- it was -- it's attached to Chuck Willard's (phonetic spelling) direct testimony.

MR. GOODRICH: Okay. So it was attachments to his testimony.

MR. O'BRIEN: Yeah, it just didn't make it into the exhibit list.

MR. GOODRICH: That's fine then. I was

just like I don't know what this is.

A.L.J. LEARY: Willard testimony?

MR. O'BRIEN: Yes, the Willard direct testimony, both for KEDNY and KEDLI.

MR. GOODRICH: So they were like appendices to the testimony.

A.L.J. COSTELLO: There were -- yeah, appen -- appendice -- appendix one.

MR. GOODRICH: Okay.

A.L.J. COSTELLO: And for both companies. Thank you.

A.L.J. LEARY: I'm going to ask the company to put that in your email so that -- I mean, I have it here. But I just want to make sure that I'm hearing from you that this is what is the appendix to the Willard testimony. I think it's data April 30, 2019. Although, that could be his testimony. I don't know when this appendix came in.

A.L.J. COSTELLO: Okay. Any further objections with respect to hearing exhibits? Okay. Not hearing any, we're going to move all of the exhibits identified at the hearing into evidence at the -- into the evidentiary record at this point.

A.L.J. LEARY: Okay. Boy, this table

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of contents, I think I want to go over the schedule again just so the parties know. The initial post-hearing briefs, I have a March 23, 2020 date. Does that sound right to everybody?

MR. FORST: That's correct, Your Honors.

A.L.J. LEARY: Yeah. We did move the reply back by a couple of days to April 3, 2020. Okay? I just want to reiterate our request that any citations give the jump page, as we call it, and any of the orders of the commission including those adopting joint proposals, you need to have the case number, the date the order is issued, and the jump page. Because in many of the generic proceedings, we have multiple orders, and don't make the mistake. Because again, Judge -- Judge Costello and I are not going to have the time to try to figure out what you're talking about. So watch your citations.

What did we decide on further additions to the exhibit list? Circulate to the parties and then see if there's an objection?

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: All right. So any further additions beyond the discussion today of

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exhibits should be the subject of a -- an email to us copied to the parties proffering the exhibit for admission and providing some basis or rationale as to why this will inform the commission's decision in how it relates to a material issue in the proceeding. So give us some basis and a basis on which anybody objecting can respond.

Responses to the admission of additional exhibits, we're going to have to shorten the time. This is a motion, but because we want to wrap-up everything that's in this record quickly, I would ask the parties to respond within -- what do you think, two days?

A.L.J. COSTELLO: Well, why don't we wait to see if we can set it? But it's going to be abbreviated.

A.L.J. LEARY: You want to set it if there's an objection?

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: So tell them to notify us first if there's an objection?

A.L.J. COSTELLO: Yeah. I would say within in two days, assuming they notify us. And then --

A.L.J. LEARY: Okay. Two days to object. And then we will rule on whether it should be admitted or not. Everybody understand that protocol? It's a little less formal. Yes?

MR. LOUGHNEY: Just a question, Your Honor. The party proffering an exhibit after the hearing is closed, do they have to explain why it was not proffered during the hearing? It's kind -- you could -- you could come forward with any kind of document at this point, right?

A.L.J. LEARY: I think you're right. And I -- we're not going to look upon anything that could have been offered. This is material that's new. This is information that's new. This is something that -- outstanding recovery requests, for example. So that's not going to be looked upon favorably. And I'm glad you asked that question. It's just coming in with some new study that nobody on any Panel, witness Panel or any testimony, has been elicited about it. I can't say how we're going to rule. But you can tell my -- by -- by my attitude how we will review it.

Let's see. Okay. Any other questions on that?

MR. FORST: Your Honors, is there an

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ultimate deadline for the submission of new exhibits possibly in advance of the brief being filed?

A.L.J. LEARY: You have a proposal?

MR. FORST: My co-counsel is indicating that St. Patrick's Day would be an appropriate date.

MR. GOODRICH: That's like -- well, actually, that might be a Saturday, but it's -- I'm not sure what day of the week it is. But like that's -- that would be like five days before the brief is due. So that would give enough time, I would hope, for people to object, and then -- and then a ruling on whether it's going to be allowed.

MR. FORST: I think ultimately it's --

A.L.J. LEARY: It's a Tuesday, and it's fine. Is that okay with everybody? Anybody want to do something different?

MS. KASON: I'm sorry, what is that date?

A.L.J. LEARY: That's a Tuesday, March 17th. Isn't it amazing how a woman named Leary would know exactly when St. Patrick's Day is, which I never celebrate? I always forget. Anything else on that? So there's a deadline of March 17 for offering new

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MS. MURPHY: So similarly -- this is Erin Murphy for EDF. Is -- is there a date when discovery is closing?

A.L.J. LEARY: We haven't -- generally, I -- I understand that unless we have motion for good cause shown, the Office of Hearings does generally not close discovery because of things that happen even in briefing. Unfortunately, though, anyone who is served with discovery can bring to our attention that this is outside the scope, or whatever you want to say about it, and seek some kind of protection from responding.

I will say that discovery, you know, post-hearing discovery is -- is ver -- has -- has got to be very narrow, and you have got to present some nexus between -- related to why you're looking for it and the material issues in the hearing.

A.L.J. COSTELLO: And why -- why it wasn't requested earlier.

A.L.J. LEARY: And why it wasn't requested earlier. So we don't want to open up a lot of back and forth between the parties. That, we can tell you right now, is discouraged. That having been

19-G-0309/19-G-0310 - 2-25-20 - Brooklyn Union Gas said, discovery is not closed as long as you have a legitimate reason based upon new information why you need this -- this information from another party. Does that help? Okay.

A.L.J. COSTELLO: And -- and the later it gets, the less likely that anything that you obtain in discovery would be admitted into the record. Because it would be prejudicial at some point for people to be responding. You know, we're -- so just keep that in mind as well.

A.L.J. LEARY: Okay. Moving -- anyone else on the issue of discovery. Finally, the briefing schedule. I did that. The table of contents to the brief has taken on a life of its own, and I'm blaming the company for this. Because -- please let the record reflect, I'm joking. There are many, many, many issues in this case.

The uniform table of contents is not designed to capture every single issue. What it's designed to do is to organize the issues into discrete areas, which the companies outline of disputed issues does, to a certain degree. Unfortunately, it doesn't quite match where we generally go with recommended decisions and commiss -

- and the commission goes with the orders.

So this is what I'm going to do, because several of the parties commented on the table of contents. And thank you for providing those com -- comments. I'm going to provide to you what is our working draft that you've had now of the companies' disputed issues and how they're framed. In bold on that document will be roughly the subject headings for the uniform table of contents, but it will disclose to everyone what someone could raise in the briefing on these many, many different issues, many, many different projects, many, many different adjustments, many different whatever.

So this is a one-year rate case, and that is a live issue about what's appropriate in a one-year rate case. I want to make that clear to everyone. And hopefully, in bolding this information -- and you will see what will not be in the uniform table of contents, but will be in that particular area of the brief, you will be able to decide what you're going to discuss in your briefs and what you're not.

But I'm still not finished yet. And I was here until almost 8:00 last night. So I still

need to get this to Judge Costello and get some impact -- feedback. And -- and I'm actually looking to a second person in -- in the Office of Hearings, who has a lot of rate case experience, to give me some feedback on this. And the -- they've been actively giving feedback already, but just so you know, I'm not going to wait to -- to get this out to you. I'm going to get you the full whatever seven pages it now is that lists lots of -- lots of issues.

Unfortunately, I looked at it today, and there are some issues in two places, and that kind of thing. And sometimes they belong in two places. So I am looking for feedback, at least in this next round, by Friday of where you want to see that issue discussed. What is the most appropriate location for a particular issue to be discussed?

We put it in a particular place. It doesn't mean it has to go in that place. And this is the -- the -- the real dilemma with these litigated rate cases, is what's the structure of, you know, the brief? All of you have written briefs that have, you know, three, four, five point headings and discrete things covered in each point heading. This is a little bit different.

So I will get it out to you as soon as I get some green light from Judge Costello and Chief Judge Lecakes. And please be studios in getting me something back, suggestions, and to all the parties, by Friday. We'll still work with that date, and then we'll close of business Friday hopefully have a final document. And you're off and running on the briefs. Anything else? Anybody have any questions on that, Mr. Berkley?

MR. BERKLEY: Yes, Your Honors. What is the deadline now for looking over the transcripts?

A.L.J. LEARY: We did not said that -- set that. I'm very glad you raised that. We have posted -- what are we up to? Day six was posted yesterday.

A.L.J. COSTELLO: I'm not sure if it was day six, but --

A.L.J. LEARY: I'm pretty sure day six was yesterday, and there'll be more today. If your briefing is due on the 23rd, there will be various deadlines, depending on how long you've had the transcript or had it available. You've been here. So it's not reasonable for us to impose a deadline right now. But let's just impose a deadline for the first,

let's say, five, which are available to you now, and have been available. And let's do some sequential dates, say -- what do you think? Yeah, this has got to get done right away. So can we set next week, which goes from March 2 to March 6, as the sequential dates for reviewing the transcripts and correcting?

We're really looking for the sponsoring party, meaning the parties sponsoring the testimony, primarily to correct it. But on cross examination, I understand that there are things that you will want to make sure are accurate as well. So let's start with March 2, and we will go with -- what did we -- yeah, how many do we have? We had nine?

MR. GOODRICH: Well, there's nine days.

A.L.J. LEARY: No, I know. But what do we have up now?

A.L.J. COSTELLO: I will let you know.

A.L.J. LEARY: What have we posted to date?

A.L.J. COSTELLO: I'm going to let you know in a minute. Let me get on the system.

We have volumes one through four. That's it.

A.L.J. LEARY: All right. We have volumes one through four. Let's -- let's have --

A.L.J. COSTELLO: And we'll have more by -- by the end of the week. We'll have more of them.

A.L.J. LEARY: Yeah, there will be a lot more posted hopefully by the end of this week. But let's just do, on a preliminary basis, March 2 for volume one, March 3, volume two -- I'm sorry. I'm saying volume one and it's the first hearing day for March 2. Second hearing day, March 3, third hearing day, March 4, fourth hearing day, March 5, fifth hearing day, March 6. Do you want to do two a day? So we'll have four additional. Hmm.

A.L.J. COSTELLO: We'll have five additional. And we have confidential.

A.L.J. LEARY: Okay. We're going to have to double up on this, unfortunately. But I, as I sit here, cannot tell you who's going to be overburdened, because they had, you know, three witness panels. So I would like to hear from the -- from DPS and the company. Nobody else needs to weigh in here, because they're the two that are going to be most imposed upon by the corrections. And I believe

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one of the -- or two of the Panels will actually be reading their own testimony to assist you. I'd like to hear from both parties on how you want to split it up between March 2 and March 6.

MR. GOODRICH: So a couple questions. One, I heard reference to the confidential transcript. But if I understand correctly, since we haven't done any live confidential te -- te -- testimony, everything in that transcript should really simply be --

A.L.J. LEARY: Corrected.

MR. GOODRICH: -- pre-filed. So there really shouldn't be any corrections there other than flipping through and make suring -- make -- making sure what's supposed to be there is there, right?

A.L.J. COSTELLO: That's correct.

MR. GOODRICH: Okay. So that's -- I don't see that as an issue. And am I understanding then that what you're proposing is you would like us -- you would like to have all nine days reviewed by March 6, is that what you're saying?

A.L.J. LEARY: I mean, it depends on what we can get up by the end of this week.

MR. GOODRICH: Yeah.

A.L.J. LEARY: And it's going to depend on the court reporting service. But answer, yes, we may have to go into the next week.

MR. GOODRICH: Okay. I would -- I would say that I'm -- I would say that I would be fine going into the next week. I would also say that I think that if we are allowed some flexibility if we find that the day three is the one that's going to require a lot of man hours that we can send an -- send a note before close of business on March 4 and say, this is the one that's going to take more time. Can we -- can we talk about -- you know, like as long as there's some flexibility if we run into issues, I'm fine with, you know, coming up with dates and -- and aiming for them.

A.L.J. LEARY: I think that's a very good idea. Anybody have a problem with that? And that would be applicable to the company as well?

MR. O'BRIEN: Yeah. I -- I -- as long as we have, like -- like Mr. Goodrich said, as long as we have flexibility, that -- that works for us.

A.L.J. LEARY: Okay. So let's use March 2 through March 6 to get the majority of the work done. It's not like you can't draft a brief that

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leaves this blank until you get the corrected transcript. So hopefully by the following week, we will put into DMM those final corrected transcripts. But just remember, this is -- this is really a key labor intensive next step that is often overlooked. And it -- it really -- the -- the integrity of the record is really important here, as it is in any hearing, or trial, or whatever.

Does that sound good to everybody? Okay. Anything else? Anybody else want to raise any issues?

I want to say on the record, you know, for the benefit of your respective constituents, what a pleasure it was to preside. And Judge Costello, he can speak for himself, although I am speaking for him as well. We were really impressed with the preparation, the courtesy, the Panels included. And I want to con -- ask you to convey to the Panels how well they did. And even for a litigated case that you don’t always look forward to, this really was a pleasure. You all did a fantastic job, and make sure you show this part of the transcript to your clients, or whoever’s paying the bill, so that they know what went on here for three weeks. And it was -- it was

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pretty tedious at times. Although, things moved very quickly. They really did. And I -- I very much appreciated it. Anything to add?

A.L.J. COSTELLO: No, I agree. And thank you, everybody. I think we're going to go off the record.

A.L.J. LEARY: Thank you.

(Off the record 1:04 p.m.)

STATE OF NEW YORK

I, JANET WALLRAVIN, do hereby certify that the foregoing was reported by me, in the cause, at the time and place, as stated in the caption hereto, at Page 1 hereof; that the foregoing typewritten transcription consisting of pages 4073 through 5685, is a true record of all proceedings had at the hearing.

IN WITNESS WHEREOF, I have hereunto subscribed my name, this the 3rd day of March, 2020.

JANET WALLRAVIN, Reporter